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Steven G. Krantz

# Harmonic and Complex Analysis in Several Variables

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Steven G. Krantz

# Harmonic and Complex Analysis in Several Variables

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*To the memory of Walter Rudin*

# Preface

The subject of harmonic analysis dates to the days of Bernoulli and Euler, but it was first formalized by Joseph Fourier in the early nineteenth century. In the intervening 200 years we have seen the subject grow in many directions. The Fourier transform, Fourier analysis on locally compact abelian groups, group representations, Fourier integral operators, and many other fields find their genesis in the early work of Fourier.

It is safe to say that harmonic analysts are always seeking new venues in which to ply their craft. One of the newest of these is the context of the function theory of several complex variables. There were a few rather tentative early papers of Alberto Calderón and Antoni Zygmund in the 1950s. Explorations of nonlocal solvability of partial differential equations and of subelliptic estimates for the $\overline{\partial}$-Neumann problem and of properties of plurisubharmonic functions certainly anticipated some of the key ideas that would become more explicit later. Papers of Lewy, Kohn, and Hörmander describe some of these results.

One of the notable developments in the modern theory is the work of Korányi in 1969. He showed that the study of the boundary behavior of holomorphic functions will be dramatically different in the several complex variable context (as opposed to the one complex variable context). Shortly thereafter, in 1972, Elias M. Stein wrote his seminal tract on the boundary behavior of holomorphic functions. This was the first coherent treatment of several aspects of the harmonic analysis of several complex variables—including maximal functions, area functions, the boundary behavior of Hardy space functions, and other topics. Stein introduced a number of important and original techniques that are still used today.

One of the notable features of Stein's book mentioned in the last paragraph is its terseness. It is *very hard work* to read that book, and to fill in all the details. In the meantime, there have been many research papers about the Bergman and Szegő kernels, area integrals, boundary behavior of holomorphic and meromorphic functions, singular integrals, and many other aspects of harmonic analysis in the several complex variables context. There have been some books on the Bergman theory, at least one in the several complex variable arena; but there has been no attempt to address the harmonic analysis of several complex variables taken as a coherent whole.

That is the purpose of the current book. Assuming a very basic background in the complex analysis of one and several complex variables, together with the basics of real and functional analysis, we develop the harmonic analysis of several complex variables from first principles. Every effort has been made to render this book as self-contained as possible. On the one hand, the Bibliography is encyclopedic. On the other hand, we do not want the reader to be frequently rushing off to the library to seek additional information and background.

This book is written with the student in mind. There are many examples, copious explanations, and exercise sets at the end of each chapter. Each chapter begins with a Prologue, to introduce the reader to the subject matter that is about to be presented. Each section begins with a Capsule, to give a spirited launch to that unit. Each major result (theorem or proposition) is preceded by a Prelude to help put the idea in context. The text is also sprinkled with Exercises for the Reader; these provide encouragement for the neophyte to pick up his/her pencil and get to work. Finally, we indulge in the conceit of repeating frequently used material, just to make many chapters more self-contained.

This author has been involved in the development of the harmonic analysis of several complex variables almost since its inception. He has contributed a number of key ideas to the subject. So he feels well qualified to engage in this exposition. Our goal is to plant a flag for the subject and to lay the foundation for future work.

It is a pleasure to thank my teacher E. M. Stein for teaching me the foundations of this subject. The publisher engaged several insightful reviewers who contributed many useful suggestions and corrections. My thanks to them. Jerry Folland read a great part of the book and contributed many insights, corrections, and suggestions. His edits were both insightful and incisive. I give him my sincere thanks.

My many collaborators and students (too numerous to mention by name) have also taught me a great deal. I also thank my Editor Elizabeth Loew for her unwavering encouragement and support. I of course assume all responsibilities for any errors or inaccuracies in the text. I look forward to hearing from readers of the book.

St. Louis, MO, USA Steven G. Krantz

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# Chapter 1
# Introduction and Review

**Prologue:** This is a book about the harmonic analysis of functions of several complex variables. Here "harmonic analysis" means the decomposition and agglomeration of functions. In practice what harmonic analysis has become over the years is the study of integral operators and their actions on function spaces. This is thanks to Poisson summation and related ideas (see [KRA2, Ch. 1]). That is the point of view that we take in the present book.

In particular, we shall study the Poisson integral operator, the Poisson–Szegő operator, the Szegő operator, the Bergman operator, the $\overline{\partial}$-Neumann operator, and several others. Along the way, a number of important partial differential operators will come into the picture. These include the Laplacian, the $\overline{\partial}$ operator, the Monge–Ampère operator, the de Rham operator, and others.

It is well to begin our studies with a review of some of the classical ideas in the context of one complex variable. This will give our studies both context and a reference point. That is what we shall do in the next section.

## 1.1 Harmonic Analysis on the Disc in the Complex Plane

**Capsule:** The idea of studying the boundary limits of holomorphic functions on the disc goes back at least to the studies by Pierre Fatou in 1906. It is remarkable to look at his original papers to see how quickly and thoroughly he had digested the very new ideas of Lebesgue about measure theory and adroitly applied them to his new insights about the boundary behavior of functions on the disc. In point of fact, Fatou's approach to these questions was by way of Fourier series—which was a hot topic at the time. In today's context we take a more function-theoretic view of the matter. This is appropriate for philosophical reasons, and also because we later want to generalize the ideas to several complex variables.

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Let

$$D = \{z \in \mathbb{C} : |z| < 1\}.$$

We think of the boundary of the disc as $\partial D = \{z \in \mathbb{C} : |z| = 1\}$ and also as $\mathbb{T} = \{e^{i\theta} : 0 \leq \theta < 2\pi\}$. It is natural to identify the latter realization of the boundary with the interval $[0, 2\pi)$ and we do so without further comment (see [KAT, Ch. 1] for a detailed discussion of this matter). We equip this interval with Lebesgue measure as usual. The interaction of the Fourier analysis of the boundary $\partial D$ with the function theory of the interior $D$ gives rise to a rich theory. That will be the focus of the present section. This section is largely expository, and many proofs are omitted. Complete treatments may be found in [ZYG3, Ch. 7], [KRA1, Ch. 8], and [KRA2, Ch. 5].

If $f \in L^1([0, 2\pi))$, then we define (for $j \in \mathbb{Z}$)

$$\widehat{f}(j) = \frac{1}{2\pi} \int_0^{2\pi} f(t) e^{-ijt}\, dt.$$

Here the functions $t \mapsto e^{ijt}$ are the *characters* of the circle group $\mathbb{T}$. See [KRA2, Ch. 1] for a derivation of the characters of the circle group. It is natural, from the point of view of Hilbert space, to consider the series

$$Sf(t) \sim \sum_{j=-\infty}^{\infty} \widehat{f}(j) e^{ijt}.$$

Here the symbol $\sim$ means "is associated with." We ask whether $Sf$ converges to $f$ in some sense (either pointwise or in norm or in some other wise). Note that we use the symbol $\sim$ in this last display because we don't want to a priori suggest anything about the convergence of the formally given series.

It would be easy to write an entire book about the convergence of Fourier series (see the classic tome [ZYG3, Ch. 2, 3, 4], the modern classic [STE2, Ch. 3], and the more accessible reference [KRA2, Ch. 1]). Some basic theorems are these:

**Theorem 1.1.1 (Dini)** *Let $f$ be an integrable function on $[0, 2\pi)$. Let $P$ lie in the interior of this interval and assume that $f$ is differentiable at $P$. Then $Sf(P)$ converges to $f(P)$.*

**Theorem 1.1.2 (M. Riesz)** *Let $f$ be an $L^p$ function on $[0, 2\pi)$, $1 < p < \infty$. Then $Sf$ converges to $f$ in $L^p$ norm.*

**Theorem 1.1.3 (Kolmogorov)** *There exists an $L^1$ function $f$ on the interval $[0, 2\pi)$ so that $Sf(x)$ fails to converge to $f(x)$ at all $x$.*

**Theorem 1.1.4 (Carleson)** *Let $f$ be an $L^2$ function on $[0, 2\pi)$. Then $Sf(x)$ converges to $f(x)$ for almost every $x$.*

**Theorem 1.1.5 (Hunt, Sjölin)** *Let $f$ be an $L^p$ function on $[0, 2\pi)$, $p > 1$. Then $Sf(x)$ converges to $f(x)$ for almost every $x$.*

If $f$ is an integrable function on the boundary $\partial D$ of the unit disc, then we may extend $f$ to a harmonic function $F$ on $D$ with the Poisson integral formula

$$F(re^{i\theta}) = \frac{1}{2\pi} \int_0^{2\pi} \frac{(1-r^2)}{1 - 2r\cos(\theta - t) + r^2} \cdot f(e^{it})\, dt\,. \tag{1.1.6}$$

It can be shown that

$$\lim_{r \to 1^-} F(re^{i\theta}) = f(e^{i\theta})$$

for almost every $\theta$ in $[0, 2\pi)$.

By way of justification of formula (1.1.6), it is worth noting that the harmonic continuation of $e^{ij\theta}$ to the disc is $r^j e^{ij\theta} = z^j$ when $j \geq 0$ and is $r^{|j|} e^{ij\theta} = \bar{z}^{-j}$ when $j < 0$. And it is straightforward to calculate that

$$\sum_{j=0}^{\infty} r^j e^{ij\theta} + \sum_{j=-1}^{-\infty} r^{|j|} e^{ij\theta} = \frac{1 - r^2}{1 - 2r\cos\theta + r^2}\,.$$

Of course the disc is also equipped with the Cauchy integral formula, having kernel $1/(\zeta - z)$. The Cauchy integral

$$Cf(z) = \frac{1}{2\pi i} \oint \frac{f(\zeta)}{\zeta - z}\, d\zeta \tag{1.1.7}$$

of the function $f(e^{i\theta}) = e^{ij\theta}$ for $j \geq 0$ is in fact $z^j$. The Cauchy integral of the function $f(e^{i\theta}) = e^{ij\theta}$ for $j < 0$ is 0.

The Cauchy integral (1.1.7) is equivalent to the Szegő integral

$$\mathcal{S}f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(e^{i\theta}) \cdot \frac{1}{1 - ze^{-i\theta}}\, d\theta\,.$$

The Szegő integral operator is actually the Hilbert space projection of $L^2(\partial D)$ to the space of $L^2$ functions on $\partial D$ which are boundary traces of holomorphic functions. See [KRA2, §1.5] for the details.

Another useful integral operator is the Bergman operator. On the disc $D$ it is given by

$$\mathcal{K}f(z) = \frac{1}{\pi} \iint_D \frac{f(\zeta)}{(1 - z \cdot \bar{\zeta})^2}\, dA\,.$$

The operator $\mathcal{K}$ is the Hilbert space projection from $L^2(D)$ to the intersection of $L^2(D)$ with the space of holomorphic functions. This operator, and the kernel

$$K(z,\zeta) = \frac{1}{(1 - z \cdot \overline{\zeta})^2},$$

have important invariance properties under conformal mappings. The kernel $K$ also gives rise to the *Bergman metric*, which was the first nontrivial Kähler metric. See [KRA1, §1.4] for the details.

### 1.1.1 The Boundary Behavior of Holomorphic Functions

Throughout this section we let $D \subseteq \mathbb{C}$ denote the unit disc. Let $0 < p < \infty$. We define

$$H^p(D) = \left\{ f \text{ holomorphic on } D : \sup_{0<r<1} \left[ \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^p \, d\theta \right]^{1/p} \equiv \|f\|_{H^p} < \infty \right\}$$

Also define

$$H^\infty(D) = \left\{ f \text{ holomorphic on } D : \sup_D |f| \equiv \|f\|_{H^\infty} < \infty \right\}.$$

The fundamental result in the subject of $H^p$, or *Hardy*, spaces (and also the fundamental result of this section) is that, if $f \in H^p(D)$ for $0 < p \leq \infty$, then the limit

$$\lim_{r \to 1^-} f(re^{i\theta}) \equiv \widetilde{f}(e^{i\theta})$$

exists for almost every $\theta \in [0, 2\pi)$. For $1 \leq p \leq \infty$, the function $f$ can be recovered from $\widetilde{f}$ by way of the Cauchy or Poisson integral formulas; for $p < 1$ this "recovery" process is more subtle and must proceed by way of distributions. Once this pointwise boundary limit result is established, then an enormous and rich mathematical structure unfolds (see Y. Katznelson [KAT, Ch. 3], K. Hoffman [HOF, Ch. 3–4], S. Krantz [KRA1, §8.1]).

In fact—see below—we shall prove a result that is distinctly stronger than the radial convergence just described. We shall instead prove a theorem about *nontangential* convergence (Fatou himself also proved such a theorem). This is a qualitatively stronger result, and also much more useful. For example, the notion of nontangential convergence is a conformal invariant.

Recall from the last subsection that the Poisson kernel for the disc is

$$P_r(e^{i\theta}) = \frac{1}{2\pi}\frac{1-r^2}{1-2r\cos\theta+r^2}.$$

Let, for $0 < p < \infty$,

$$\mathbf{h}^p(D) = \left\{ f \text{ harmonic on } D : \sup_{0<r<1}\left[\frac{1}{2\pi}\int_0^{2\pi} |f(re^{i\theta})|^p\, d\theta\right]^{1/p} \equiv \|f\|_{\mathbf{h}^p} < \infty \right\}$$

and

$$\mathbf{h}^\infty(D) = \left\{ f \text{ harmonic on } D : \sup_D |f| \equiv \|f\|_{\mathbf{h}^\infty} < \infty \right\}.$$

Obviously $H^p(D) \subsetneqq \mathbf{h}^p(D)$ for $0 < p \leq \infty$. Throughout this section, arithmetic and measure theory on $[0, 2\pi)$ (equivalently on $\partial D$) is done by identifying $[0, 2\pi)$ with $\mathbb{R}/2\pi\mathbb{Z}$. See [KAT, Ch. 1] for more on this identification procedure.

**Prelude** The next proposition establishes right away—and with a nice, soft, argument—that a boundary function exists. Once we have that function in our hands, then we can begin to analyze it.

**Proposition 1.1.6** *Let $1 < p \leq \infty$ and $f \in \mathbf{h}^p(D)$. Then there is an $\widetilde{f} \in L^p(\partial D)$ such that*

$$f(re^{i\theta}) = \int_0^{2\pi} \widetilde{f}(e^{i\psi}) P_r(e^{i(\theta-\psi)})\, d\psi.$$

**Proof** Define $f_r(e^{i\theta}) = f(re^{i\theta}), 0 < r < 1$. Then $\{f_r\}_{0<r<1}$ is a bounded subset of $(L^{p'}(\partial D))^*, p' = p/(p-1)$. By the Banach-Alaoglu theorem (see W. Rudin [RUW2, p. 68]), there is a subsequence $f_{r_j}$ that converges weak-$*$ to some $\widetilde{f}$ in $(L^{p'}(\partial D))^* = L^p(\partial D)$. For any $0 < r < 1$, let $r < r_j < 1$. Then

$$f(re^{i\theta}) = f_{r_j}\left((r/r_j)e^{i\theta}\right) = \int_0^{2\pi} f_{r_j}(e^{i\psi}) P_{r/r_j}\left(e^{i(\theta-\psi)}\right) d\psi$$

because $f_{r_j} \in C(\overline{D})$. Now $P_{r/r_j} \in C(\partial D) \subseteq L^{p'}(\partial D)$. Thus the righthand side of the last equation is

$$\int_0^{2\pi} f_{r_j}(e^{i\psi}) P_r(e^{i(\theta-\psi)}) d\psi + \int_0^{2\pi} f_{r_j}(e^{i\psi})\left[P_{r/r_j}(e^{i(\theta-\psi)}) - P_r(e^{i(\theta-\psi)})\right] d\psi.$$

As $j \to \infty$, the second integral vanishes (because the expression in brackets converges uniformly to 0) and the first integral tends to

$$\int_0^{2\pi} \widetilde{f}(e^{i\psi}) P_r(e^{i(\theta-\psi)}) d\psi.$$

This is the desired result. □

**Remark** It is easy to see that the proof breaks down for $p = 1$ since $L^1$ is not the dual of any Banach space. This breakdown is not merely ostensible: the harmonic function

$$f(re^{i\theta}) \equiv P_r(e^{i\theta})$$

satisfies

$$\sup_{0<r<1} \int_0^{2\pi} |f(re^{i\theta})| d\theta < \infty,$$

but the Dirac $\delta$ mass is the only measure of which $f$ is the Poisson integral. □

**Exercise for the Reader** If $f \in \mathbf{h}^1$ then there is a Borel measure $\mu_f$ on $\partial D$ such that $f(re^{i\theta}) = P_r(\mu_f)(e^{i\theta})$. Here one calculates the Poisson integral of a measure just as one calculates the Poisson integral of a function.

**Prelude** This next proposition is a sort of converse to Proposition 1.1.6.

**Proposition 1.1.7** *Let $f \in L^p(\partial D)$, $1 \leq p < \infty$. Then $\lim_{r\to 1^-} P_r f = f$ in the $L^p$ norm.*

**Remark** The result is false for $p = \infty$ if $f$ is discontinuous. The correct analogue in the uniform case is that if $f \in C(\partial D)$ then $P_r f \to f$ uniformly.

As an exercise, consider a Borel measure $\mu$ on $\partial D$. Show that its Poisson integral converges in the weak-$*$ topology to $\mu$. □

**Proof of Proposition 1.1.7** If $f \in C(\partial D)$, then the result is clear by the solution of the Dirichlet problem (or, more easily, by simple properties of the Poisson kernel). If $f \in L^p(\partial D)$ is arbitrary, let $\epsilon > 0$ and choose $g \in C(\partial D)$ such that $\|f - g\|_{L^p} < \epsilon$. Then

$$\begin{aligned} \|P_r f - f\|_{L^p} &\leq \|P_r(f-g)\|_{L^p} + \|P_r g - g\|_{L^p} + \|g - f\|_{L^p} \\ &\leq \|P_r\|_{L^1} \|f - g\|_{L^p} + \|P_r g - g\|_{L^p} + \epsilon \\ &\leq \epsilon + o(1) + \epsilon \end{aligned}$$

as $r \to 1^-$. □

**Prelude** The next covering lemma is a basic version of the celebrated Wiener covering lemma that is so widely used in harmonic analysis. In fact there are a great many different covering lemmas. And it is known that a number of central results in the subject are essentially logically equivalent to certain covering lemmas.

**Proposition 1.1.8** *Let $K \subseteq \partial D$ be compact, and let $\{I_\alpha\}_{\alpha\in A}$ be a covering of $K$ by open intervals. Then there is a subcovering $\{I_{\alpha_j}\}_{j=1}^M$ such that every point of $K$ is contained in at least one but not more than two of the $I_{\alpha_j}$'s. [We call such a subcover a cover of "valence two."]*

**Proof** Since $K$ is compact, $\{I_\alpha\}_{\alpha\in A}$ has a finite subcollection $\{I_{\alpha_j}\}_{j=1}^J$ that still covers $K$. Now, in sequence, discard any $I_{\alpha_j}$ that is contained in the union of the other intervals. □

**Definition 1.1.9** If $f \in L^1(\partial D)$, let

$$Mf(\theta) = \sup_{R>0} \frac{1}{2R} \int_{-R}^{R} |f(e^{i(\theta-\psi)})|\, d\psi \,.$$

The function $Mf$ is called the *Hardy–Littlewood maximal function* of $f$.

**Remark** Thanks in part to the influence of E. M. Stein (see [STE2, Ch. 2]), the Hardy–Littlewood maximal function now plays a prominent role in the study of linear harmonic analysis. Many important operators are dominated by the Hardy–Littlewood maximal function or variants thereof. Certainly our treatment of the boundary behavior of holomorphic functions—both in one and several complex variables—will reflect this significant role. □

**Definition 1.1.10** Let $(X, \mu)$ be a measure space and $f : X \to \mathbb{C}$ be measurable. We say that $f$ is weak $p$, $0 < p < \infty$, (or sometimes of weak-type $p$) if $\mu\{x : |f(x)| > \lambda\} \leq C/\lambda^p$, all $0 < \lambda < \infty$. The space weak-type $\infty$ is defined to be $L^\infty$.

**Lemma 1.1.11 (Chebycheff's Inequality)** *If $f \in L^p(X, d\mu)$, then $f$ is weak-type $p$, $1 \leq p < \infty$.*

**Proof** Let $\lambda > 0$. Then

$$\mu\{x : |f(x)| > \lambda\} \leq \int_{\{x:|f(x)|>\lambda\}} |f(x)|^p/\lambda^p d\mu(x) \leq \lambda^{-p}\|f\|_{L^p}^p. \qquad \square$$

**Exercise for the Reader** There exist functions that are of weak-type $p$ (with respect to Lebesgue measure on the line, for example) but not in $L^p$, $1 \leq p < \infty$.

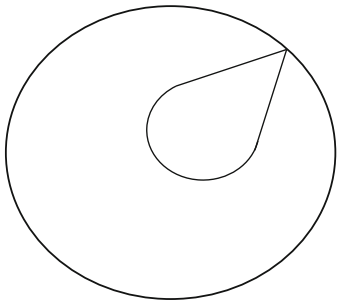
**Definition 1.1.12** An operator $T : L^p(X, d\mu) \to \{\text{measurable functions}\}$ is said to be of *weak-type* $(p, p)$, $0 < p < \infty$, if

$$\mu\{x : |Tf(x)| > \lambda\} \leq C\|f\|_{L^p}^p/\lambda^p\,, \qquad \text{all} \quad f \in L^p, \quad \lambda > 0.$$

**Prelude** Theorem 1.1.15, combined with Proposition 1.1.13 below, will be the key to proving our result about boundary limits. This establishes an important paradigm for studying such problems.

**Proposition 1.1.13** *The operator $M$ is of weak-type* $(1, 1)$.

**Figure 1.1** A Stolz region.



**Proof** Let $\lambda > 0$. Set $S_\lambda = \{\theta : |Mf(e^{i\theta})| > \lambda\}$. Let $K \subseteq S_\lambda$ be a compact subset with $2m(K) \geq m(S_\lambda)$. For each $k \in K$, there is an open interval $I_k \ni k$ with $|I_k|^{-1} \int_{I_k} |f(e^{i\psi})| d\psi > \lambda$. Then $\{I_k\}_{k\in K}$ is an open cover of $K$. By Proposition 1.1.8, there is a subcover $\{I_{k_j}\}_{j=1}^M$ of $K$ of valence not exceeding 2. Then

$$\begin{aligned} m(S_\lambda) \leq 2m(K) &\leq 2m\left(\bigcup_{j=1}^M I_{k_j}\right) \leq 2\sum_{j=1}^M m(I_{k_j}) \\ &\leq \sum_{j=1}^M \frac{2}{\lambda} \int_{I_{k_j}} |f(e^{i\psi})| d\psi \\ &\leq \frac{4}{\lambda} \|f\|_{L^1} . \end{aligned}$$

□

**Definition 1.1.14** If $e^{i\theta} \in \partial D, 1 < \alpha < \infty$, then define the *Stolz region* (or *nontangential approach region* or *cone*) with vertex $e^{i\theta}$ and aperture $\alpha$ to be

$$\Gamma_\alpha(e^{i\theta}) = \{z \in D : |z - e^{i\theta}| < \alpha(1 - |z|)\}.$$

See Figure 1.1.

**Prelude** The next result is the heart of the matter. It shows how a maximal function that is obviously closely related to the question of nontangential boundary values is majorized by the Hardy–Littlewood maximal function. This, together with our already proven estimates for the maximal function, will be the key to our proof of Fatou's theorem.

**Theorem 1.1.15** *If $e^{i\theta} \in \partial D, 1 < \alpha < \infty$, then there is a constant $C_\alpha > 0$ such that if $f \in L^1(\partial D)$, then*

$$\sup_{re^{i\phi} \in \Gamma_\alpha(e^{i\theta})} |P_r f(e^{i\phi})| \leq C_\alpha Mf(e^{i\theta}).$$

**Proof** For $re^{i\phi} \in \Gamma_\alpha(e^{i\theta})$, we have

$$|\theta - \phi| \le 2\alpha(1-r).$$

Therefore, for $1/\alpha \le r < 1$, we obtain

$$\begin{aligned}
|P_r f(e^{i\phi})| &= \left| \frac{1}{2\pi} \int_0^{2\pi} f(e^{i(\phi-\psi)}) \frac{1-r^2}{1-2r\cos\psi + r^2} d\psi \right| \\
&= \left| \frac{1}{2\pi} \int_0^{2\pi} f(e^{i(\phi-\psi)}) \frac{1-r^2}{(1-r)^2 + 2r(1-\cos\psi)} d\psi \right| \\
&\le \frac{4}{2\pi} \sum_{j=0}^{\log_2(\pi/\alpha(1-r))} \int_{S_j} |f(e^{i(\phi-\psi)})| \frac{1-r^2}{(1-r)^2 + 2r(2^{j-1}\alpha(1-r))^2} d\psi \\
&\quad + \frac{1}{2\pi} \int_{|\psi|<\alpha(1-r)} |f(e^{i(\phi-\psi)})| \frac{1-r^2}{(1-r)^2} d\psi,
\end{aligned}$$

where $S_j = \{\psi : 2^j\alpha(1-r) \le |\psi| < 2^{j+1}\alpha(1-r)\}$. Now this is

$$\begin{aligned}
&\le \frac{4\alpha}{4\pi\alpha^2} \sum_{j=0}^{\infty} \frac{1}{2^{2j-2}(1-r)} \int_{|\psi|<(2+2^{j+1})\alpha(1-r)} |f(e^{i(\theta-\psi)})|\, d\psi \\
&\quad + \frac{2}{2\pi} \frac{1}{1-r} \int_{|\psi|<3\alpha(1-r)} |f(e^{i(\theta-\psi)})|\, d\psi \\
&\le \frac{32}{\pi} \sum_{j=0}^{\infty} 2^{-j} \left[ \frac{1}{2\alpha(2+2^{j+1})(1-r)} \int_{|\psi|<(2+2^{j+1})\alpha(1-r)} |f(e^{i(\theta-\psi)})| d\psi \right] \\
&\qquad + \frac{6\alpha}{\pi} \frac{1}{2\cdot 3\alpha(1-r)} \int_{|\psi|<3\alpha(1-r)} |f(e^{i(\theta-\psi)})|\, d\psi \\
&\le \frac{32}{\pi} \cdot \sum_{j=0}^{\infty} 2^{-j} Mf(\theta) + \frac{6\alpha}{\pi} Mf(\theta) \\
&\le \frac{64}{\pi} Mf(\theta) + \frac{6\alpha}{\pi} Mf(\theta).
\end{aligned}$$

If $0 < r \le 1/\alpha$ then

$$\begin{aligned}
|P_r f(\phi)| &\le \frac{1}{2\pi} \int_0^{2\pi} |f(e^{i(\phi-\psi)})| (2\alpha/(\alpha-1)) d\psi \\
&\le \frac{2\alpha}{\alpha-1} Mf(\theta).
\end{aligned}$$

□

**Prelude** The next theorem is our "toy" version of the main result (Theorem 1.1.26) below. It is only a preliminary result, because it is stated just for $1 < p \leq \infty$. This result is just the right one for harmonic functions, but not for holomorphic functions. One of the main things that we learn in this section is the difference between harmonic and holomorphic functions. Later, in the several variable context, we shall learn even more profound differences.

**Theorem 1.1.16** *Let $f \in \mathbf{h}^p(D)$ and $1 < p \leq \infty$. Let $\widetilde{f}$ be as in Proposition 1.1.6 and $1 < \alpha < \infty$. Then*

$$\lim_{\Gamma_\alpha(e^{i\theta})\ni z\to e^{i\theta}} f(z) = \widetilde{f}(e^{i\theta}), \quad a.e.\ e^{i\theta} \in \partial D.$$

**Proof** It suffices to handle the case $p < \infty$ and $f$ real-valued. If $\epsilon > 0$ then choose $g \in C(\partial D)$ real-valued so that $\|\widetilde{f} - g\|_{L^p(\partial D)} < \epsilon^2$. We know by the theory of the Dirichlet problem (or by elementary properties of the Poisson kernel) that

$$\lim_{\Gamma_\alpha(e^{i\theta})\ni z\to e^{i\theta}} g(z) = g(e^{i\theta}), \qquad \text{all} \quad e^{i\theta} \in \partial D, \tag{1.1.16.1}$$

where $g(re^{i\theta}) \equiv P_r g(e^{i\theta})$. Therefore

$$\begin{aligned}
&m\{e^{i\theta} : \limsup_{\Gamma_\alpha(e^{i\theta})\ni z\to e^{i\theta}} |f(z) - \widetilde{f}(e^{i\theta})| > \epsilon\} \\
&\leq m\{e^{i\theta} : \limsup_{\Gamma_\alpha(e^{i\theta})\ni z\to e^{i\theta}} |f(z) - g(z)| > \epsilon/3\} \\
&\qquad + m\{e^{i\theta} : \limsup_{\Gamma_\alpha(e^{i\theta})\ni z\to e^{i\theta}} |g(z) - g(e^{i\theta})| > \epsilon/3\} \\
&\qquad + m\{e^{i\theta} : |g(e^{i\theta}) - \widetilde{f}(e^{i\theta})| > \epsilon/3\} \\
&\leq \{e^{i\theta} : C_\alpha M(\widetilde{f} - g) > \epsilon/3\} + 0 + (3\|g - \widetilde{f}\|_{L^p}/\epsilon)^p.
\end{aligned}$$

In the last estimate we used Lemma 1.1.11 and Proposition 1.1.13. Now the last line is majorized by

$$\begin{aligned}
C'_\alpha \|\widetilde{f} - g\|_{L^1}/(\epsilon/3) + 3^p\epsilon^p &\leq C''_\alpha \|\widetilde{f} - g\|_{L^p}/(\epsilon/3) + 3^p\epsilon^p \\
&\leq C'''_\alpha \epsilon.
\end{aligned}$$

It follows that

$$\lim_{\Gamma_\alpha(e^{i\theta})\ni z\to e^{i\theta}} f(z) = \widetilde{f}(e^{i\theta}), \qquad a.e. \quad e^{i\theta} \in \partial D. \qquad \square$$

The informal statement of Theorem 1.1.16 is that $f$ has nontangential boundary limits almost everywhere. Thanks to work of Littlewood and Rudin, among others, this result is known to be sharp (that is to say, there is no theorem with a geometrically broader approach region).

Theorem 1.1.16 contains essentially all that can be said about the boundary behavior of harmonic functions. Why are holomorphic functions better? And what do we mean by "better?" What we mean is that $H^p$ functions have boundary limits even for $0 < p \leq 1$. The classical way of addressing this matter (which will, in the long run, be seen to be misleading) is to use Blaschke factorization:

**Definition 1.1.17** If $a \in \mathbb{C}$, $|a| < 1$, then the *Blaschke factor* at $a$ is

$$B_a(z) = \frac{z - a}{1 - \overline{a}z}.$$

It is elementary to verify that $B_a$ is holomorphic on a neighborhood of $\overline{D}$ and that $|B_a(e^{i\theta})| = 1$ for all $0 \leq \theta < 2\pi$.

**Lemma 1.1.18** *If $0 < r < 1$ and $f$ is holomorphic on a neighborhood of $\overline{D}(0, r)$, let $p_1, \ldots, p_k$ be the zeros of $f$ (listed with multiplicity) in $D(0, r)$. Assume that $f(0) \neq 0$ and that $f(re^{it}) \neq 0$, all $t$. Then*

$$\log |f(0)| + \log \prod_{j=1}^{k} r |p_j|^{-1} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{it})| dt.$$

**Proof** The function

$$F(z) = \frac{f(z)}{\prod_{j=1}^{k} B_{p_j/r}(z/r)}$$

is holomorphic and nonvanishing on a neighborhood of $\overline{D}(0, r)$, hence $\log |F|$ is harmonic on a neighborhood of $\overline{D}(0, r)$. Thus

$$\begin{aligned} \log |F(0)| &= \frac{1}{2\pi} \int_0^{2\pi} \log |F(re^{it})| dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{it})| dt \end{aligned}$$

or

$$\log |f(0)| + \log \prod_{j=1}^{k} \frac{r}{|p_j|} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(re^{it})| dt. \qquad \square$$

Notice that, by the continuity of the integral, Lemma 1.1.18 holds even if $f$ has zeros on $\{re^{it}\}$.

**Corollary 1.1.19** *If $f$ is holomorphic in a neighborhood of $\overline{D}(0,r)$ then*

$$\log|f(0)| \leq \frac{1}{2\pi}\int_0^{2\pi} \log|f(re^{it})|dt.$$

**Proof** The term $\log\prod_{j=1}^{k}|r/p_j|$ in Lemma 1.1.18 is positive. □

**Corollary 1.1.20** *If $f$ is holomorphic on $D, f(0) \neq 0$, and $\{p_1, p_2, \dots\}$ are the zeros of $f$ counting multiplicities, then*

$$\log|f(0)| + \log\prod_{j=1}^{\infty}\frac{1}{|p_j|} \leq \sup_{0<r<1}\frac{1}{2\pi}\int_0^{2\pi}\log^+|f(re^{it})|dt.$$

**Proof** Apply Lemma 1.1.18, letting $r \to 1^-$. □

**Prelude** This next is a key fact about Hardy space functions on the disc. There is no known analogue in several complex variables (however see the significant partial results in [VAR]).

**Corollary 1.1.21** *If $f \in H^p(D), 0 < p \leq \infty$, and $\{p_1, p_2, \dots\}$ are the zeros of $f$ counting multiplicities, then $\sum_{j=1}^{\infty}(1-|p_j|) < \infty$.*

**Proof** Since $f$ vanishes to finite order $k$ at 0, we may replace $f$ by $f(z)/z^k$ and assume that $f(0) \neq 0$. It follows from Corollary 1.1.20 that

$$\log\prod_{j=1}^{\infty}\left\{\frac{1}{|p_j|}\right\} < \infty$$

or $\prod(1/|p_j|)$ converges; hence $\prod|p_j|$ converges. So $\sum_j(1-|p_j) < \infty$. □

**Prelude** This next is a companion result to the preceding one.

**Proposition 1.1.22** *If $\{p_1, p_2, \dots\} \subseteq D$ satisfy $\sum_j(1-|p_j|) < \infty, p_j \neq 0$ for all $j$, then*

$$\prod_{j=1}^{\infty}\frac{-\overline{p}_j}{|p_j|}B_{p_j}(z)$$

*converges normally on $D$.*

**Proof** Restrict attention to $|z| \leq r < 1$. Then the assertion that the infinite product converges uniformly on this disc is equivalent with the assertion that

$$\sum_j \left| 1 + \frac{\overline{p}_j}{|p_j|} B_{p_j}(z) \right|$$

converges uniformly. But

$$\begin{aligned} \left| 1 + \frac{\overline{p}_j}{|p_j|} B_{p_j}(z) \right| &= \left| \frac{|p_j| - |p_j| \overline{p}_j z + \overline{p}_j z - |p_j|^2}{|p_j|(1 - z\overline{p}_j)} \right| \\ &= \left| \frac{(|p_j| + z\overline{p}_j)(1 - |p_j|)}{|p_j|(1 - z\overline{p}_j)} \right| \\ &\leq \frac{(1 + r)(1 - |p_j|)}{1 - r}, \end{aligned}$$

so the convergence is uniform. □

**Definition 1.1.23** Let $0 < p \leq \infty$ and $f \in H^p(D)$. Let $\{p_1, p_2, \dots\}$ be the zeros of $f$ counted according to multiplicities. Let

$$B(z) = \prod_{j=1}^{\infty} \frac{-\overline{p}_j}{|p_j|} B_{p_j}(z)$$

(where each $p_j = 0$ is understood to give rise to a factor of $z$). Then $B$ is a well-defined holomorphic function on $D$ by Proposition 1.1.22. Let $F(z) = f(z)/B(z)$. By the Riemann removable singularities theorem, $F$ is a well-defined, non-vanishing, holomorphic function on $D$. The representation $f = F \cdot B$ is called the *canonical factorization* of $f$.

**Exercise for the Reader** All the assertions of Definition 1.1.23 hold for $f \in N(D)$, the Nevanlinna class. Here we say that $f \in N$ provided that

$$\sup_{0<r<1} \frac{1}{2\pi} \int_0^{2\pi} \log(1 + |f(re^{it})|)\, dt < \infty\,.$$

Note that $H^p \subsetneqq N$ for every $p > 0$.

**Proposition 1.1.24** *Let $f \in H^p(D)$, $0 < p \leq \infty$, and let $f = F \cdot B$ be its canonical factorization. Then $F \in H^p(D)$ and $\| F \|_{H^p(D)} = \| f \|_{H^p(D)}$.*

**Proof** Trivially, $|F| = |f/B| \geq |f|$, so $\| F \|_{H^p} \geq \| f \|_{H^p}$. If $N = 1, 2, \dots$, let

$$B_N(z) = \prod_{j=1}^{N} \frac{-\overline{p}_j}{|p_j|} B_{p_j}(z)$$

(where the factors corresponding to $p_j = 0$ are just $z$).

Let $F_N = f/B_N$. Since $|B_N(e^{it})| = 1$, all $t$, it holds that $\|F_N\|_{H^p} = \|f\|_{H^p}$. If $0 < r < 1$ then

$$\int_0^{2\pi} |F(re^{it})|^p dt^{1/p} = \lim_{N\to\infty} \int_0^{2\pi} |F_N(re^{it})|^p dt^{1/p}$$
$$\leq \lim_{N\to\infty} \|F_N\|_{H^p} = \|f\|_{H^p}.$$

Therefore $\|F\|_{H^p} \leq \|f\|_{H^p}$. □

**Corollary 1.1.25** *If $\{p_1, p_2, \dots\}$ is a sequence of points in $D$ satisfying $\sum_j (1 - |p_j|) < \infty$ and if $B(z) = \prod_j (-\overline{p}_j/|p_j|) B_{p_j}(z)$ is the corresponding Blaschke product, then*

$$\lim_{r\to 1^-} B(re^{it})$$

*exists and has modulus* 1 *almost everywhere.*

**Proof** The conclusion that the limit exists follows from Theorem 1.1.16 and the fact that $B \in H^\infty$. For the other assertion, note that the canonical factorization for $B$ is $B = 1 \cdot B$. Therefore, by Proposition 1.1.24,

$$\int |\widetilde{B}(e^{it})|^2 dt^{1/2} = \|B\|_{H^2} = \|1\|_{H^2} = 1$$

hence $|\widetilde{B}(e^{it})| = 1$ almost everywhere. □

**Prelude** The next theorem is (a generalization of) the big result of Fatou from 1906. It really set a course of study, both in one and several complex variables, for the twentieth century. Fatou's theorem had no precedent in the history of mathematics. It was both original and profound. And it contributed to the rich symbiosis between the Fourier analysis of the circle and the complex analysis of the disc.

**Theorem 1.1.26** *If $f \in H^p(D), 0 < p \leq \infty$, and $1 < \alpha < \infty$, then*

$$\lim_{\Gamma_\alpha(D) \ni z \to e^{i\theta}} f(z)$$

*exists for almost every $e^{i\theta} \in \partial D$ and equals $\widetilde{f}(e^{i\theta})$. Also $\widetilde{f} \in L^p(\partial D)$ and*

$$\|\widetilde{f}\|_{L^p} = \|f\|_{H^p} \equiv \sup_{0<r<1} \left( \int_0^{2\pi} |f(re^{i\theta})|^p d\theta \right)^{1/p}.$$

**Proof** By Definition 1.1.23, write $f = B \cdot F$ where $F$ has no zeros and $B$ is a Blaschke product. Then $F^{p/2}$ is a well-defined $H^2$ function and thus has the appropriate boundary values almost everywhere. *A fortiori*, $F$ has nontangential boundary limits

almost everywhere. Since $B \in H^\infty, B$ has nontangential boundary limits almost everywhere. It follows that $f$ does as well. The final assertion follows from the corresponding fact for $H^2$ functions (exercise). □

## Exercises

1. Prove the following result of Carathéodory: If $f$ is holomorphic on the disc $D \subseteq \mathbb{C}$ and $f$ has non-tangential limit 0 on a set $E \subseteq \partial D$ of positive one-dimensional measure, then $f \equiv 0$. [*Hint:* Let $E_0 \subseteq E$ be closed and have positive measure. Let $\Omega = \cup_{\zeta \in E_0} \Gamma_1(\zeta)$. Map $\Omega$ conformally to $D$. This mapping extends to the boundary in such a way that sets of zero measure are mapped to sets of zero measure. Now invoke the standard boundary uniquess theorem for functions continuous on $\bar{D}$ and holomorphic in $D$.] See [KOO, p. 59] for details.
2. Prove a special case of the *Marcinkiewicz interpolation theorem*:

   - If $(X, \mu), (Y, \nu)$ are measure spaces and $T$ maps

   $$L^p(X, \mu) \to \{\text{measurable functions on } Y\},$$

   we say that $T$ is *sublinear* if $|T(f + g)(x)| \leq |Tf(x)| + |Tg(x)|$ for all $f, g$.
   - Let $T$ be a sublinear operator as in the first bullet point. If $1 \leq p_0 < p_1 \leq \infty$ and $T$ is of weak-types $(p_0, p_0)$ and $(p_1, p_1)$ then prove that $T$ maps $L^p$ to $L^p$, all $p_0 < p < p_1$. [*Hint:* Let $f$ be in $L^p$ and break it into two pieces, one of which is weak-type $p_0$ and the other of which is weak-type $p_1$.]

   In fact the full statement of the theorem is this:

   **Theorem** Suppose that $T$ is a sublinear operator of weak type $(p_j, q_j)$, where $1 \leq p_j \leq q_j \leq \infty, j = 0, 1$ and $q_0 \neq q_1$. Then $T$ is of strong type $(p_t, q_t)$ provided that

   $$\frac{1}{p_t} = \frac{1-t}{p_0} + \frac{t}{p_1} \qquad \text{and} \qquad \frac{1}{q_t} = \frac{1-t}{q_0} + \frac{t}{q_1},$$

   $0 < t < 1$.
3. Let $E \subseteq \partial D$ be an $F_\sigma$ of first category and $\phi$ a continuous function on $D$. F. Bagemihl and W. Seidel [BAG] proved that there exists a holomorphic function $f$ on $D$ such that $\lim_{r \to 1^-} |\phi(r\zeta) - f(r\zeta)| = 0$ for all $\zeta \in E$. We omit the rather technical proof.

   Give an example to show that the Bagemihl–Seidel theorem cannot hold if "radial convergence" is replaced by non-tangential convergence. Conversely, give an example to show that the result of Exercise 1 cannot hold if non-tangential convergence is replaced by radial convergence (feel free to use Bagemihl–Seidel here).

**4.** Prove the following classical result of Hardy and Littlewood (see [GOL, p. 411]).

**Theorem** Let $f : D \to \mathbb{C}$ be holomorphic and bounded. Let $0 < \alpha < 1$. Then $f$ is Lipschitz $\alpha$ if and only if there is an $\mathbb{N} \ni k > \alpha$ and a $C_k > 0$ such that

$$|f^{(k)}(z)| \leq C_k(1 - |z|)^{\alpha - k}, \qquad \text{all} \quad z \in D. \tag{$*$}$$

Prove that $(*)$ holds for one $k > \alpha$ if and only if it holds for all $k > \alpha$.

**5.** This problem is philosophically related to the preceding one. The classical result, once again, is due to G. H. Hardy and J. E. Littlewood [HAR1, HAR2].

**Theorem** Let $f : D \to \mathbb{C}$ be holomorphic. Let $0 < p < 1$. If $f' \in H^p(D)$, then $f \in H^q(D)$ where $q = p/(1-p)$. If $f' \in H^1$, then $f \in A(D)$; indeed, $f|_{\partial D}$ is absolutely continuous.

Here $A(D)$ is the space of functions which are continuous on $\overline{D}$ and holomorphic on $D$.

It is even possible to prove results for $p > 1$: if $f' \in H^p, 1 < p \leq \infty$, then $f$ is Lipschitz $\alpha$, where $\alpha = 1 - 1/p$. Although the reader should have no difficulty verifying the assertion for $p = \infty$, he/she may encounter trouble with $0 < p < \infty$. Nevertheless, for $1 < p < \infty$, the problem is accessible if we use some Sobolev space ideas.

We here briefly recall the concept of Sobolev space. If $s \geq 0$ is an integer, then we say that a function $f \in L^2$ is in the Sobolev space $W^s$ if the weak derivative $(\partial^\alpha / \partial x^\alpha) f \in L^2$ for every multi-index $\alpha$ satisfying $|\alpha| \leq s$. If $s \in \mathbb{R}$ is not an integer, and we even allow here $s$ to be negative, then we define $W^s$ to be those functions satisfying

$$\int |\widehat{f}(\xi)|^2 (1 + |\xi|^2)^{s/2} \, d\xi < \infty, .$$

The reader may easily use Plancherel's theorem to check that these two definitions are equivalent when $s$ is a nonnegative integer. It is also common to use the notation $H^s$ for the Sobolev spaces. One will encounter the Sobolev space norm written as $\| \ \|_{W^s}$ or $\| \ \|_{H^s}$ or even more briefly as $\| \ \|_s$.

Identify $f$ with its boundary function $\widetilde{f}$. Prove that the hypotheses imply that $(d/d\theta)\widetilde{f}(e^{i\theta}) \in L^p([0, 2\pi])$ in the sense of distributions. Prove that, for $p > 1$, this implies that $\widetilde{f}$ is in Lipschitz $1 - 1/p$ on $\partial D$. Hence $f$ is in Lipschitz $1 - 1/p$ on $\overline{D}$.

When $p = 1$, matters are even simpler.

**6.** A function $f$ is said to be in *BMO* (functions of bounded mean oscillation) if

$$\sup \frac{1}{|B|} \int |f - f_B| \, dV < \infty \, .$$

Here $B$ are balls in Euclidean space and $f_B$ is the average of $f$ over $B$.

If $f \in H^p(D), 0 < p < \infty$, let $\widetilde{f}$ denote the boundary function. Define

$$BMOA(D) = \{f \in H^2(D) : \widetilde{f} \in BMO(\partial D)\}.$$

Prove that this space is unchanged if we replace the exponent 2 by any $p \geq 1$. Use Fefferman's theorem (see Exercise 6) to show that

$$\left(H^1(D)\right)^* = BMOA(D).$$

**7.** If $f \in L^1(\partial D)$ then $Pf(re^{i\theta})$ is a harmonic function on $D$ with almost everywhere radial limit $f$ on $\partial D$. Let $Qf(re^{i\theta})$ be the unique harmonic conjugate to $Pf$ on $D$ that vanishes at 0. Let $F(re^{i\theta}) = Pf(re^{i\theta}) + iQf(re^{i\theta})$. Then $F$ is holomorphic on $D$. If $f$ is real, $f \geq 0$, then $G(z) \equiv e^{-F(z)} \in H^\infty(D)$. Therefore $G$ has almost everywhere radial boundary limits on $\partial D$. Hence $Qf(re^{i\theta})$ has almost everywhere radial boundary limits on $\partial D$. Call this boundary function $Hf(e^{i\theta})$ (the *Hilbert transform* of $f$). Extend $H$ to all of $L^1$ by linearity. If $f \in L^1(\partial D)$ then $Hf$ need not be in $L^1$ (consider $f_j \in L^1(\partial D)$, $\|f_j\|_{L^1} = 1$, $f_j$ tending weak-$*$ to the Dirac $\delta$ mass at 1). Define

$$H^1_{\text{Re}}(\partial D) = \{f \in L^1(\partial D) : Hf \in L^1(\partial D)\}.$$

Norm the space by

$$\|f\|_{H^1_{\text{Re}}} = \|f\|_{L^1(\partial D)} + \|Hf\|_{L^1(\partial D)}.$$

Then the map

$$\Phi : H^1_{\text{Re}}(\partial D) \to H^1(D)/i\mathbb{R}$$

given by

$$f \mapsto Pf(re^{i\theta}) + iQf(re^{i\theta})$$

is a surjective Banach space isomorphism. It is a deep theorem (see C. Fefferman and E. M. Stein [FES]) that $(H^1_{\text{Re}})^* = BMO$.

**8.** Construct an example of a bounded analytic function $f$ on the disc, $f$ not identically zero, such that for almost every $P$ in the boundary of the disc there is a tangential sequence $\{z_j\}$ in $D$ approaching $P$ along which $f$ tends to 0. Here a sequence in $D$ is "tangential" if it escapes every non-tangential approach region. Consider variants of this result: can you replace the sequence $\{z_j\}$ by a curve at each point? If you replace "bounded" by $H^p$, then can you strengthen the example? A good reference for this sort of result is I. Priwalow [PWW] or [COL, Ch. 2, 5].

**9.** Prove that any $H^1$ function on the disc is the product of two $H^2$ functions. Generalize this result to other $p$.

**10.** Prove that if an $H^p$ function $f$ has boundary limit zero on a set of positive boundary measure then $f \equiv 0$.

**11.** Prove that, if $f \in H^2(D)$, then

$$\iint_D |f(z)|^2 \, dA(z) < \infty \,,$$

so that $f$ is in the Bergman space. Prove that the converse is not true.

**12.** Say that $f$ is in the Nevanlinna class if

$$\sup_{0<r<1} \int_0^{2\pi} \log(1 + |f(re^{i\theta}|) \, d\theta < \infty \,.$$

What would be a suitable norm on this space? Show that each $H^p$ is a subspace of the Nevanlinna class. Which of the results of this chapter can you generalize to the Nevanlinna class?

# Chapter 2
# Boundary Behavior

**Prologue:** In the papers [ZYG1] and [CALZ1, CALZ2], Calderón and Zygmund began to explore the boundary behavior of holomorphic functions on domains in $\mathbb{C}^n$. Another pioneering paper in this vein is [AIZ].

Aĭzenberg as well as Calderón and Zygmund establish the existence of nontangential boundary limits for holomorphic functions with a growth estimate on certain domains (including the unit ball) in $\mathbb{C}^n$. These are the same boundary limit theorems that one can prove for harmonic functions in that context. It is now understood that holomorphic functions of several complex variables are special. The real part of a holomorphic function in that context is harmonic, to be sure. But it is in fact *pluriharmonic*, and that is a much more special property. It turns out that holomorphic functions of several complex variables have boundary limits through much broader approach regions than nontangential (see [DIK, Ch. 3] for a thorough and authoritative treatment of this phenomenon). These regions are called *admissible approach regions*, and have been the subject of intense study. We shall treat them in detail in what follows.

## 2.1 The Modern Era

**Capsule:** In this section we begin to lay the groundwork for the study of the boundary behavior of holomorphic functions on domains in several complex variables. There will be several new ingredients, including admissible approach regions and the Poisson–Szegő kernel. And there will be new types of arguments. These will include harmonic majorants and some interesting function-theoretic arguments.

It was Lev Aĭzenberg in 1959 [AIZ] who established that Hardy space functions on fairly general domains in $\mathbb{C}^n$ have nontangential boundary limits. This is an interesting result, but does not really explore new territory since it is analogous to (and proved in a similar manner to) the one-dimensional result. Aĭzenberg's result followed on earlier work of Calderón and Zygmund.

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The groundbreaking result, from our point of view, in the harmonic analysis of several complex variables was the work in [KOR1, KOR2] by Korányi. In these papers, Korányi studies the boundary behavior of Hardy space functions $f$ in $H^2$ of the unit ball $B$. [As usual, $B = \{z = (z_1, z_2, \ldots, z_n) \in \mathbb{C}^n : |z_1|^2 + |z_2|^2 + \cdots + |z_n|^2 < 1\}$.] In dimension 2, write

$$B = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 < 1\}.$$

Set, for $0 < p < \infty$,

$$H^p(B) = \left\{ f \text{ holomorphic on } B : \sup_{0<r<1} \left( \int_{\partial B} |f(r\zeta)|^p \, d\sigma(\zeta) \right)^{1/p} \equiv \|f\|_{H^p(B)} < \infty \right\},$$

where $d\sigma$ is rotationally invariant area measure on $\partial B$. Let $H^\infty(B)$ be the space of bounded holomorphic functions on the ball, equipped with the obvious norm.

Let $P \in \partial B$, $\alpha > 1$, and set

$$\mathcal{A}_\alpha(P) = \{z \in B : |1 - z \cdot \overline{P}| < \alpha(1 - |z|)\}.$$

This "admissible approach region" at $P \in \partial B$ should be compared and contrasted with the more classical nontangential approach region

$$\Gamma_\alpha(P) = \{z \in B : |z - P| < \alpha(1 - |z|)\}.$$

This region $\Gamma_\alpha$ is completely analogous to the Stolz angles or cones that we studied in the one-variable context on the unit disc in $\mathbb{C}$.

The approach region $\mathcal{A}_\alpha(P)$ allows for parabolic approach in certain directions. Both these approach regions "touch" the boundary at $P$. However, they give different geometric means for approaching $P$. See Figure 2.1.

In fact it is worth noting, and we make much of this in what follows, that the classical approach region $\Gamma_\alpha$ is modeled on the isotropic ball $\beta_1(P, r) = \{z \in \partial\Omega : |z - P| < r\}$ while the new approach region $\mathcal{A}_\alpha$ is modeled on the non-isotropic ball $\beta_2(P, r) = \{z \in \partial\Omega : |1 - z \cdot P| < r\}$. By this we mean the following. The set of

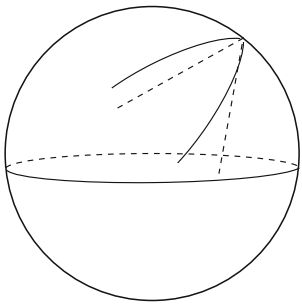


**Figure 2.1** An admissible approach region.

points in $\Gamma_\alpha$ that is distance $\delta$ from the boundary of the ball is, roughly speaking, a classical isotropic Euclidean ball of radius $\alpha\delta$. By contrast, the set of points in $\mathcal{A}_\alpha$ that is distance $\delta$ from the boundary of the ball is, roughly speaking, a nonisotropic ball of radius $\alpha\delta$.

We certainly understand the shape of $\beta_1$: It is a round ball. As for $\beta_2$, notice the following. For simplicity we take the dimension be $n = 2$ and we let $P = (1, 0) \in \partial B$. Now if $z \in \beta_2(P, r)$ then certainly

$$|1 - z_1| < r\,.$$

So the ball has extent $r$ in the $z_1$, or complex normal, direction. But now we calculate that

$$|z_2|^2 < 1 - |z_1|^2 = (1 - |z_1|)(1 + |z_1|) \leq 2(1 - |z_1|) \leq 2|1 - z_1| < 2r\,.$$

Hence $|z_2| < \sqrt{2}\sqrt{r}$. So the ball $\beta_2$ has extent $\sim \sqrt{r}$ in the $z_2$ or complex tangential direction. We see explicitly then that the ball $\beta_2$ has a nonisotropic shape, and therefore so does $\mathcal{A}_\alpha$.

**Theorem 2.1.1** *Let $f \in H^2(B)$. Then, for $\sigma$-almost every $\zeta \in \partial B$, we have that the limit*

$$\widetilde{f}(\zeta) \equiv \lim_{\Gamma_\alpha(\zeta) \ni z \to \zeta} f(z)$$

*exists. We also have that the limit*

$$\widetilde{f}(\zeta) \equiv \lim_{\mathcal{A}_\alpha(\zeta) \ni z \to \zeta} f(z)$$

*exists.*

The first of these results, as has been indicated above, was proved by Aĭzenberg and by Calderón and Zygmund. The second is due to Adam Korányi [KOR1, KOR2].

It had been a matter of commonly held belief, going back to J. E. Littlewood, Walter Rudin, and others, that nontangential approach was the optimal approach for a Fatou-type theorem on the boundary limits of Hardy space functions. Numerous well-known examples illustrated the point (see [DIK, Ch. 1, 2] for details). So Korányi's result came as something of a shock. How did he establish this remarkable theorem?

The methodology is fundamental and important. As illustrated in Chapter 8 of [KRA1], the key fact used to prove the classical result (on the disc) about nontangential boundary behavior is that the Poisson integral is bounded above by the Hardy–Littlewood maximal function (see Theorem 1.1.15 in the one-variable context):

**Theorem 2.1.2** *If $\zeta \in \partial D$, $1 < \alpha < \infty$, then there is a constant $C_\alpha > 0$ such that, if $f \in L^1(\partial D)$, then*

$$\sup_{re^{i\phi} \in \Gamma_\alpha(\zeta)} |P_r f(e^{i\phi})| \leq C_\alpha M f(\zeta) .$$

Here $M$ is the Hardy–Littlewood maximal function and $P_r f$ is the Poisson integral. Korányi's strategy was to take advantage of the multivariable complex-analytic setting by using a different kernel. He could have used the Poisson kernel for the unit ball in $\mathbb{R}^{2n} \approx \mathbb{C}^n$, but that would have only given him nontangential convergence—nothing new. He instead used the Poisson–Szegő kernel, which has a different sort of singularity than the Poisson kernel. And the Poisson–Szegő integral is in turn dominated by a *different* maximal function. Let us now consider the details.

As explained in Section 1.5 of [KRA1], the Szegő kernel for the unit ball in $\mathbb{C}^n$ is given by

$$S(z, \zeta) = \frac{(n-1)!}{2\pi^n} \frac{1}{(1 - z \cdot \overline{\zeta})^n} .$$

Here $c_n$ is a constant that depends only on the complex dimension of the ambient space. This kernel is the canonical reproducing kernel for the Hardy space $H^2(B)$.

Now a classical construction of Hua [HUA, Ch. 3] gives rise to a new reproducing kernel which is positive. Namely, we set

$$\mathcal{P}(z, \zeta) = \frac{|S(z, \zeta)|^2}{S(z, z)} .$$

It is a straightforward calculation to see that integration against $\mathcal{P}(z, \zeta)$ reproduces elements of $C(\overline{B}) \cap \mathcal{O}(B)$ (that, is functions holomorphic on $B$ that extend to be continuous on $\overline{B}$)—see [KRA1, Ch. 8], [KRA18, Ch. 1, 3] for more on these matters. The explicit formula for $\mathcal{P}$ on the ball is

$$\mathcal{P}(z, \zeta) = \frac{(n-1)!}{2\pi^n} \frac{(1 - |z|^2)^n}{|1 - z \cdot \overline{\zeta}|^{2n}} .$$

We see that the singularity of the Poisson–Szegő kernel is the nonisotropic expression that we encountered in the definition of $\mathcal{A}_\alpha(P)$. In short, the classical Hardy–Littlewood maximal operator based on round balls fits the classical Poisson kernel because the singularity of that kernel is isotropic—in other words, it is *round*. By contrast, a new Hardy–Littlewood type maximal operator (which we shall define in a moment) based on nonisotropic balls will fit the Poisson–Szegő kernel.

In fact let us compare the two maximal functions. Let $P \in \partial\Omega$. If $r > 0$ then let

$$\beta_1(P, r) = \{\zeta \in \partial B : |\zeta - P| < r\} .$$

This is of course a standard, isotropic Euclidean ball intersected with $\partial B$. Also let

$$\beta_2(P,r) = \{\zeta \in \partial B : |1 - \zeta \cdot \overline{P}| < r\}.$$

We can plainly see that the balls $\beta_2$ are modeled on the nonisotropic geometry that we have seen before in the definition of $\mathcal{A}_\alpha(P)$. We can define a maximal function

$$M_1 f(P) = \sup_{r>0} \frac{1}{\sigma(\beta_1(P,r))} \int_{\beta_1(P,r)} |f(\zeta)|\, d\sigma(\zeta).$$

Here $d\sigma$ is $2n-1$-dimensional Hausdorff measure on $\partial B$. Alternatively, we can say that $d\sigma$ is rotationally invariant area measure.

Likewise, we define

$$M_2 f(P) = \sup_{r>0} \frac{1}{\sigma(\beta_2(P,r))} \int_{\beta_2(P,r)} |f(\zeta)|\, d\sigma(\zeta).$$

The difference between the two maximal functions is the balls that are used.

Now the main point is that $\partial B$, equipped with area measure $d\sigma$ and the balls $\beta_1(P,r)$, is a space of homogeneous type (see below) in the sense of Coifman and Weiss [COG1, Ch. 2]. See the discussion of this concept below. And also $\partial B$, equipped with area measure $d\sigma$ and the balls $\beta_2(P,r)$, is a space of homogeneous type. On a space of homogeneous type, it is automatic that the corresponding maximal function, as defined above, is of weak-type (1,1) and strong type $(p,p)$ for $1 < p \leq \infty$. It is a matter of direct estimation (as in Proposition 1.1.15 above and Section 8.1 of [KRA1]) to see that the Poisson integral is majorized by $M_1$ and the Poisson–Szegő integral is majorized by $M_2$. The rest of the Fatou theorem is standard and well-known machinery.

### *2.1.1 Spaces of Homogeneous Type*

These are fundamental ideas of K. T. Smith [SMI] and L. Hörmander [HOR5] which were later developed by R.R. Coifman and Guido Weiss [COG1] into a coherent theory.

**Definition 2.1.3** We call a locally compact Hausdorff space $X$ a *space of homogeneous type* if it is equipped with a collection of open balls $B(x,r)$. We assume that these balls, for fixed $x$, form a neighborhood basis for the topology at $x$, and also that $B(x,r) \subseteq B(x,s)$ when $r < s$. We also hypothesize the existence of a Borel regular measure $\mu$, together with positive constants $C_1, C_2$, such that

**(3.3.1) The Positivity Property:** $0 < \mu(B(x,r)) < \infty$ for $x \in X$ and $r > 0$;

**(3.3.2) The Doubling Property:** $\mu(B(x,2r)) \leq C_1 \mu(B(x,r))$ for $x \in X$ and $r > 0$;

**(3.3.3) The Enveloping Property:** If $B(x,r) \cap B(y,s) \neq \emptyset$ and $r \geq s$, then $B(x, C_2 r) \supseteq B(y,s)$.

We frequently use the notation $(X, \mu)$ to denote a space of homogeneous type. In some contexts a space of homogeneous type is equipped with a metric as well (and the balls are defined in terms of the metric), but we opt for greater generality here.

We can now define the Hardy–Littlewood maximal function on $L^1(X,\mu)$. If $f \in L^1(X,\mu)$, then define

$$Mf(x) \equiv \sup_{R>0} \frac{1}{\mu(B(x,R))} \int_{B(x,r)} |f(t)| d\mu(t).$$

By way of studying this new maximal operator, we need a covering lemma.

**Prelude** As we predicted in the last chapter, the fundamental covering lemma in harmonic analysis is that of Wiener. We now state and prove this result in full generality. A broad treatment of different types of covering lemmas appears in [DEG].

**Lemma 2.1.4 (Wiener)** *Let $K \subseteq X$ be a compact set. Let $\{B_j\}$ be a covering of $X$ by open balls. Then there is pairwise disjoint subcollection $\{B_{j_k}\}$ so that the dilated balls $\{C_2 B_{j_k}\}$ still cover $K$.*

**Proof** Since $K$ is compact, we may as well suppose at the outset that $\{B_j\}$ is finite.

Now select $B_{j_1}$ to be the largest ball in the collection (if there is more than one such, then just pick one). Next, select $B_{j_2}$ to be the largest of the remaining balls that is disjoint from $B_{j_1}$ (again, if there is more than one then just pick one). Choose $B_{j_3}$ to be the largest of the remaining balls that is pairwise disjoint from *both* $B_{j_1}$ and $B_{j_2}$. Continue in this fashion. Since the collection of balls is finite, the process will stop.

We end up with a pairwise disjoint collection of balls $\{B_{j_k}\}$ as specified in the lemma. And we claim that their $C_2$ dilations (the constant $C_2$ coming from the definition of space of homogeneous type) cover $K$ (here the $C_2$ dilation of $B(P,r)$ is simple $B(P, C_2 r)$). To see this, we will in fact show that the $C_2$ dilations of the $\{B_{j_k}\}$ cover all the original $B_j$. So pick a ball $B_j$. Let $B_{j_k}$ be the first ball from the subcollection that intersects $B_j$. Then, by the way that we chose the $B_{j_k}$, it must be that the radius of $B_{j_k}$ is at least that of $B_j$. But then the enveloping property tells us that $C_2 B_{j_k}$ contains $B_j$. So we see that *every* ball $B_j$ is covered by $\{C_2 B_{j_k}\}$. Hence $K$ is covered by $\{C_2 B_{j_k}\}$. That is what we wished to prove. □

Now the exact same proof that we gave in Section 1.1 tells us that the Hardy–Littlewood maximal operator $M$ is of weak-type $(1,1)$. And the maximal operator is trivially bounded on $L^\infty$. By Marcinkiewicz interpolation (see [STW2, Ch. 5]), it follows that $M$ is bounded on $L^p$ for $1 < p < \infty$.

## 2.2 Estimates for the Poisson Kernel

**Capsule:** While the Cauchy kernel is the same for every domain in the complex plane, and the Bochner–Martinelli kernel is the same for every domain in complex space, such is not the case for many of the most important kernels. In particular, the Poisson kernels for different domains are different. But what *is* true is that one can obtain estimates for the Poisson kernel of a smoothly bounded domain that depend only on the diameter of the domain and the $C^2$ norm of a defining function. We discuss some of these estimates in the present section.

One of the key ideas in our treatment of the boundary behavior of holomorphic functions in the context of one complex variables was the majorization of the Poisson integral by the Hardy–Littlewood maximal function. In this estimation, the explicit form of the Poisson kernel for the disc was exploited. If we wish to use a similar argument to study the boundary behavior of harmonic and holomorphic functions on general domains in $\mathbb{R}^N$ and $\mathbb{C}^n$, then we must again estimate the Poisson integral by a maximal function.

However there is no hope of obtaining an explicit formula for the Poisson kernel of an arbitrary smoothly bounded domain. In this section we shall instead study some rather sharp estimates that turn out to be as good as an explicit formula. Interestingly, there is scant treatment of this matter in print, but see [KRA1, Ch. 8] and [KRA13] as well as [KER4].

The following point needs to be clarified. We have already noted that, if we are to obtain new results about admissible boundary values for holomorphic functions, then we must use a new kernel—namely the Poisson–Szegő kernel. And we shall. Nevertheless, the classical Poisson kernel is a useful tool for passing back and forth between the boundary of a domain and its interior. So we shall make good use of it as well. That is why we spend time in this section studying and estimating the Poisson kernel.

It is possible in some cases (but not all cases) to obtain estimates on the Poisson–Szegő kernel. But we shall use some function-theoretic tricks to avoid the need for such estimates.

Recall (see Section 1.3 of [KRA1]) that the Poisson kernel for a $C^2$ domain $\Omega \subseteq \mathbb{R}^N$ is given by $P(x, y) = -\nu_y G(x, y), x \in \Omega, y \in \partial\Omega$. Here $\nu_y$ is the unit outward normal vector field to $\partial\Omega$ at $y$, and $G(x, y)$ is the Green's function for $\Omega$. Recall that, for $N > 2$, we have $G(x, y) = c_n|x - y|^{-N+2} - F_x(y)$, where $F$ depends in a $C^{2-\epsilon}$ fashion on $x$ and $y$ jointly and $F$ is harmonic in $y$. It is known that $G$ is $C^{2-\epsilon}$ on $\overline{\Omega}\times\overline{\Omega}\backslash\{\text{diagonal}\}$ and $G(x, y) = G(y, x)$. It follows that $P(x, y)$ behaves qualitatively like $|x-y|^{-N+1}$. [These observations persist in $\mathbb{R}^2$ by a slightly different argument.] The results below will make this intuitive observation more concrete and precise.

We begin with a geometric fact:

**GEOMETRIC FACT:** Let $\Omega \subset\subset \mathbb{R}^N$ have $C^2$ boundary. There are numbers $r, \widetilde{r} > 0$ such that, for each $y \in \partial\Omega$, there are balls $B(c_y, r) \equiv B_y \subseteq \Omega$ and $B(\widetilde{c}_y, \widetilde{r}) \equiv \widetilde{B}_y \subseteq {}^c\overline{\Omega}$ that satisfy

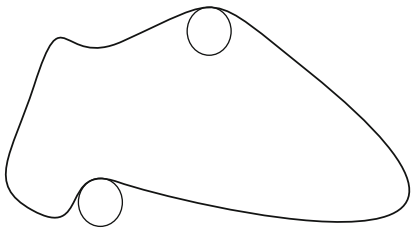


**Figure 2.2** Osculating balls.

(i) $\overline{B}(\widetilde{c_y}, \widetilde{r}) \cap \overline{\Omega} = \{y\}$;
(ii) $\overline{B}(c_y, r) \cap \overline{{}^c\Omega} = \{y\}$.

Let us indicate why these balls exist. Fix $P \in \partial\Omega$. Applying the implicit function theorem to the mapping

$$\begin{aligned} \partial\Omega \times (-1, 1) &\to \mathbb{R}^N \\ (\zeta, t) &\mapsto \zeta + t\nu_\zeta, \end{aligned}$$

at the point $(P, 0)$, we find a neighborhood $U_P$ of the point $P$ on which the mapping is one-to-one. By the compactness of the boundary, there is thus a neighborhood $U$ of $\partial\Omega$ such that each $x \in U$ has a unique nearest point in $\partial\Omega$ (such a neighborhood $U$ is called a *tubular neighborhood of the boundary*—see [HIR, Ch. 4]). It further follows that there is an $\epsilon > 0$ such that, if $\zeta_1, \zeta_2$ are distinct points of $\partial\Omega$, then $I_1 = \{\zeta_1 + t\nu_{\zeta_1} : |t| < 2\epsilon\}$ and $I_2 = \{\zeta_2 + t\nu_{\zeta_2} : |t| < 2\epsilon\}$ are disjoint sets (that is, the normal bundle is locally trivial in a natural way). From this it follows that, if $y \in \partial\Omega$, then we may take $c_y = y - \epsilon\nu_y, \widetilde{c_y} = y + \epsilon\nu_y$, and $r = \widetilde{r} = \epsilon$.

We may assume in what follows that $r = \widetilde{r} < \operatorname{diam}\Omega/2$. See Figure 2.2.

We now consider estimates for $P_\Omega(x, y)$. For the rest of this section, $x$ will be an element of $\Omega$ and $y$ will be an element of $\partial\Omega$. We shall not provide the complete detailed proofs here, but instead refer the reader to [KRA1, §8.2]. The key idea is to compare the Green's function and Poisson kernel for the given domain $\Omega$ with the corresponding functions for the internally and externally tangent balls $B(c_y, r)$ and $B(\widetilde{c_y}, \widetilde{r})$ (and also for the complements of these balls).

**Proposition 2.2.1** *Let $\Omega \subseteq \mathbb{R}^N$ be a domain with $C^2$ boundary. Let $P \equiv P_\Omega : \Omega \times \partial\Omega \to \mathbb{R}$ be its Poisson kernel. Then, for each $x \in \Omega$, there is a positive constant $C_x$ such that*

$$0 < C_x \leq P(x, y) \leq \frac{C}{|x - y|^{N-1}} \leq \frac{C}{\delta(x)^{N-1}}.$$

*Here $\delta(x) = dist\,(x, \partial\Omega)$.*

**Proof** Proof omitted. See [KRA1, p. 332].

**Exercises for the Reader** Use the explicit formula for the Poisson kernel of $B(0,R) \subseteq \mathbb{R}^N$ to prove that, if $u$ is a non-negative harmonic function on $B(0,R)$ and $\xi \in B(0,R)$, then:

**(i)** $u(\xi) \leq 2(R/d(\xi))^{N-1} \cdot u(0)$;
**(ii)** $u(\xi) \geq [d(\xi)/(2^N R)] \cdot u(0)$, where $d(\xi) = \text{dist}\,(\xi, {}^cB(0,R))$

These first two inequalities are essentially the Harnack inequalities. Apply inequality **(ii)** to $G_\Omega(x,\cdot)$ on a ball $B_y^*$ internally tangent to $\partial\Omega$ at $y \in \partial\Omega$ with $x \notin B_y^*$ to obtain

**(iii)** $G_\Omega(x,t) \geq C_x[d(t)]G_\Omega(x,c_y)\,, \quad t \in B_y^*\,, \quad x \in \Omega;$

Conclude that

**(iv)** $-\frac{\partial}{\partial \nu_y} G_\Omega(x,y) \geq C_x G_\Omega(x,c_y) > 0,$

so that

**(v)** $P_\Omega(x,y) \geq C'_x > 0.$

If $x \in \Omega\,, \delta(x) < r$, let $\pi x \in \partial\Omega$ be the nearest point to $x$. Apply **(ii)** to $P_\Omega(\cdot,y)$ on $B_{\pi x}$ to obtain

**(vi)** $P_\Omega(x,y) \geq C\delta(x) \cdot P_\Omega(c_{\pi x}, y)$ provided $\delta(x) < r$;

Conclude from **(vi)** and the fact that $\{c_{\pi x}\} \subset\subset \Omega$ that

**(vii)** $P_\Omega(x,y) \geq C'\delta(x)$ for all $x \in \Omega\,.$

For the case $\delta(x) < r$ and $y = \pi x$, we may compare $P_\Omega(x,y)$ with $P_{B_y}(x,y)$ to obtain

**(viii)** $P_\Omega(x,y) \geq c|x-y|^{-N+1}$ when $y = \pi x$.

With the results of the exercise in hand, it is now easy to prove the next proposition. We omit the details, but refer the reader to [KRA1, §8.2].

**Proposition 2.2.2** *If $\Omega \subset\subset \mathbb{R}^n$ is a domain with $C^2$ boundary, then*

$$P_\Omega(x,y) \geq C \cdot \frac{\delta(x)}{|x-y|^N}.$$

The next estimate for $P_\Omega$ (Proposition 2.2.6) is the most useful and the deepest. In order to prove it we must first consider an auxiliary function to which we will compare it.

**Lemma 2.2.3** *Let $B \subseteq \mathbb{R}^N$ be the unit ball. Let $\widetilde{B} = B((0,\ldots,0,1),1)$ and $T = B \cap {}^c\widetilde{B}$. Let $b = \partial B \setminus \overline{\widetilde{B}}$. There is a non-negative harmonic function $H$ on a neighborhood of $\overline{T}$ and a $c > 0$ such that*

(2.2.3.1) $H(0) = 0$;
(2.2.3.2) $H(x) > 0$ *if* $x \in \overline{T} \setminus \{0\}$;
(2.2.3.3) $H \geq 1$ *on* $b$;
(2.2.3.4) $H(0,\ldots,0,-t) \leq ct$ *, all* $0 \leq t \leq 1$.

**Lemma 2.2.4** *There is a constant $C = C(\Omega, N)$ such that $G_\Omega(x,y) \le C|x-y|^{-N+2}$.*

**Lemma 2.2.5** *We have*

$$G_\Omega(x,y) \le C(\Omega, N) \cdot \frac{\delta(y)}{|x-y|^{N-1}}, \tag{2.2.5.1}$$

$$G_\Omega(x,y) \le C(\Omega, N) \cdot \frac{\delta(x)}{|x-y|^{N-1}}. \tag{2.2.5.2}$$

The final estimate on $P_\Omega(x,y)$ is as follows.

**Prelude** This next result is really the climax of this section. It shows that, for all practical purposes, the Poisson kernel on a smoothly bounded domain in $\mathbb{R}^N$ behaves just like the Poisson kernel on the disc. In particular, this proposition will enable us to establish suitable maximal function estimates. The proof of Proposition 2.2.6 is highly nontrivial, and we refer the reader to [KRA1, §8.2] for the details.

**Proposition 2.2.6** *If $\Omega \subset\subset \mathbb{R}^N$ is a domain with $C^2$ boundary, then*

$$P_\Omega(x,y) \le C\frac{\delta(x)}{|x-y|^N}.$$

## 2.3 Subharmonicity and Boundary Values

**Capsule:** As previously noted, there is no concept of Blaschke product in several complex variables, and in particular no concept of the canonical factorization. So we must come up with new tools in order to tame $H^p$ functions on domains in $\mathbb{C}^n$. This will include harmonic majorants, admissible approach regions, and new types of maximal function estimates for holomorphic functions on admissible regions. This will all involve some new complex function theory, and new ways to think about boundary values. It is an exciting and challenging adventure.

Let $\Omega \subseteq \mathbb{C}^n$ be a domain and $f : \Omega \to \mathbb{R}$ a function. The function $f$ is said to have a *harmonic majorant* if there is a (necessarily) non-negative harmonic $u$ on $\Omega$ with $|f| \le u$. We are interested in harmonic majorants for subharmonic functions. Note that the property of having a harmonic majorant is a potential-theoretic substitute for the property of being bounded.

**Proposition 2.3.1** *If $f : B \to \mathbb{R}^+$ is subharmonic, then $f$ has a harmonic majorant if and only if*

$$\sup_{0<r<1} \int_{\partial B} f(r\zeta) d\sigma(\zeta) < \infty.$$

**Proof** Let $u$ be a harmonic majorant for $f$. Then

$$\int_{\partial B} f(r\zeta)d\sigma(\zeta) \leq \int_{\partial B} u(r\zeta)d\sigma(\zeta) = \omega_{N-1} \cdot u(0) \equiv C < \infty,$$

as claimed.

Conversely, if $f$ satisfies

$$\sup_r \int_{\partial B} f(r\zeta)d\sigma(\zeta) < \infty,$$

then the functions $f_r : \partial B \to \mathbb{C}$ given by $f_r(\zeta) = f(r\zeta)$ form a bounded subset of $L^1(\partial B) \subseteq \mathcal{M}(\partial B)$. Let $\widetilde{f} \in \mathcal{M}(\partial B)$ be a weak-$*$ accumulation point of the functions $f_r$. Then $F(r\zeta) \equiv P\widetilde{f}(r\zeta)$ is harmonic on $B$, and for any $x \in B$ and $1 > r > |x|$ we have

$$\begin{aligned} 0 \leq f(x) &\leq \int P(x/r, \zeta) f_r(\zeta) d\sigma(\zeta) \\ &\to \int P(x, \zeta) d\widetilde{f}(\zeta) = F(x) \qquad \text{as} \quad r \to 1^- \end{aligned}$$

□

A consequence of Proposition 2.3.1 is that not all subharmonic functions have harmonic majorants. For example, the subharmonic function $f(\zeta) = e^{1/|\zeta-1|^2}$ has no harmonic majorant on the unit disc in $\mathbb{C}$. Harmonic majorants play a significant role in the theory of boundary behavior of harmonic and holomorphic functions. Proposition 2.3.1 suggests why growth conditions may, therefore, play a role. The fact that $f$ harmonic implies $|f|^p$ subharmonic only for $p \geq 1$ whereas $f$ holomorphic implies $|f|^p$ subharmonic for $p > 0$ (exercise) suggests that we may expect different behavior for harmonic and for holomorphic functions.

By way of putting these remarks in perspective and generalizing Proposition 2.3.1, we consider the space of harmonic functions $\mathbf{h}^p(\Omega)$ (resp. the space of holomorphic functions $H^p(\Omega)$), with $\Omega$ any smoothly bounded domain in $\mathbb{R}^N$ (resp. $\mathbb{C}^n$). These are defined below. First we require some preliminary groundwork.

A tool that will be used repeatedly in what follows is the concept of defining function. Let $\Omega \subseteq \mathbb{C}^n$ be a domain. A *defining function* for $\Omega$ is a $C^1$ function $\rho : \mathbb{C}^n \to \mathbb{R}$ such that

$$\Omega = \{z \in \mathbb{C}^n : \rho(z) < 0\}$$

and $\nabla\rho \neq 0$ on $\partial\Omega$. Clearly only a domain with $C^1$ boundary can have such a defining function. In case the defining function is $C^k$, then the boundary of $\Omega$ will be $C^k$ according to any other geometrically natural definition (see Appendix I in [KRA1] for a detailed consideration of these matters).

Let $\Omega \subset\subset \mathbb{R}^N$ be a bounded domain with $C^2$ boundary. Let $\phi : \mathbb{R} \to [0,1]$ be a $C^\infty$ function supported in $[-2,2]$ with $\phi \equiv 1$ on $[-1,1]$. Then we see, with $\delta_\Omega(x) \equiv \text{dist}\,(x, \partial\Omega)$ and $\epsilon_0 > 0$ sufficiently small, that

$$\rho(x) = \begin{cases} -\phi(|x|/\epsilon_0)\delta_\Omega(|x|) - (1 - \phi(|x|/\epsilon_0)) & \text{if } x \in \overline{\Omega} \\ \phi(|x|/\epsilon_0)\delta_\Omega(|x|) + (1 - \phi(|x|/\epsilon_0)) & \text{if } x \notin \overline{\Omega} \end{cases}$$

is a $C^2$ defining function for $\Omega$ (see [KRPA4] for information about the smoothness of $\delta_\Omega$). The implicit function theorem implies that, if $0 < \epsilon < \epsilon_0$, then $\partial\Omega_\epsilon \equiv \{x \in \Omega : \rho(x) = -\epsilon\}$ is a $C^2$ manifold that bounds $\Omega_\epsilon \equiv \{x \in \Omega : \rho_\epsilon(x) \equiv \rho(x) + \epsilon < 0\}$. Now let $d\sigma_\epsilon$ denote area measure on $\partial\Omega_\epsilon$. Then it is natural to let

$$\mathbf{h}^p(\Omega) = \{f \text{ harmonic on } \Omega : \sup_{0<\epsilon<\epsilon_0} \int_{\partial\Omega_\epsilon} |f(\zeta)|^p d\sigma_\epsilon(\zeta)^{1/p}$$

$$\equiv \|f\|_{\mathbf{h}^p(\Omega)} < \infty\}, \qquad 0 < p < \infty,$$

$$\mathbf{h}^\infty(\Omega) = \{f \text{ harmonic on } \Omega : \sup_{x\in\Omega} |f(x)| \equiv \|f\|_{\mathbf{h}^\infty} < \infty\}.$$

In case $\Omega$ is a subdomain of *complex space* $\mathbb{C}^n$, we define

$$H^p(\Omega) = \mathbf{h}^p(\Omega) \cap \{\text{holomorphic functions}\}, \quad 0 < p \leq \infty.$$

The next lemma serves to free the definitions from their somewhat artificial dependence on $\delta_\Omega$ and $\rho$.

**Lemma 2.3.2 (Stein [STE1, Ch. 1])** *Let $\rho_1, \rho_2$ be two $C^2$ defining functions for a domain $\Omega \subseteq \mathbb{R}^N$. For $\epsilon > 0$ small and $j = 1, 2$, let*

$$\Omega_\epsilon^j = \{x \in \Omega : \rho_j(x) < -\epsilon\},$$
$$\partial\Omega_\epsilon^j = \{x \in \Omega : \rho_j(x) = -\epsilon\}.$$

*Let $\sigma_\epsilon^j$ be area measure on $\partial\Omega_\epsilon^j$. Then, for $f$ harmonic on $\Omega$, we have*

$$\sup_{\epsilon>0} \int_{\partial\Omega_\epsilon^1} |f(\zeta)|^p d\sigma_\epsilon^1(\zeta) < \infty$$

*if and only if*

$$\sup_{\epsilon>0} \int_{\partial\Omega_\epsilon^2} |f(\zeta)|^p d\sigma_\epsilon^2(\zeta) < \infty.$$

*(Note: Since $f$ is bounded on compact sets, equivalently the supremum is only of interest as $\epsilon \to 0$, there is no ambiguity in this assertion.)*

**Proof** By definition of defining function, $\operatorname{grad}\rho_i \neq 0$ on $\partial\Omega$. Since $\partial\Omega$ is compact, we may choose $\epsilon_0 > 0$ so small that there is a constant $\lambda, 0 < \lambda < 1$, with $0 < \lambda \leq |\operatorname{grad}\rho_i(x)| < 1/\lambda$ whenever $x \in \Omega, \delta_\Omega(x) < \epsilon_0$. If $0 < \epsilon < \epsilon_0$, then notice that for $x \in \partial\Omega_\epsilon^2$ we have

$$B(x, \lambda\epsilon/2) \subseteq \Omega$$

and, what is stronger,

$$B(x, \lambda\epsilon/2) \subseteq \left\{t : -3\epsilon/\lambda^2 < \rho_1(t) < -\lambda^2 \cdot \epsilon/3\right\} \equiv S(\epsilon). \tag{2.3.2.1}$$

Therefore

$$|f(x)|^p \leq \frac{1}{V(B(x, \lambda\epsilon/2))} \int_{B(x,\lambda\epsilon/2)} |f(t)|^p dV(t).$$

As a result,

$$\begin{aligned}
\int_{\partial\Omega_\epsilon^2} |f(x)|^p d\sigma_\epsilon^2 &\leq C\epsilon^{-N} \int_{\partial\Omega_\epsilon^2} \int_{B(x,\lambda\epsilon/2)} |f(t)|^p dV(t) d\sigma_\epsilon^2(x) \\
&= C\epsilon^{-N} \int_{\mathbb{R}^N} \int_{\partial\Omega_\epsilon^2} \chi_{B(x,\lambda\epsilon/2)}(t) |f(t)|^p d\sigma_\epsilon^2(x) dV(t) \\
&\leq C\epsilon^{-N} \int_{S(\epsilon)} |f(t)|^p \int_{\partial\Omega_\epsilon^2 \cap B(t,\lambda\epsilon/2)} d\sigma_\epsilon^2(x) dV(t) \\
&\leq C'\epsilon^{-N}\epsilon^{N-1} \int_{S(\epsilon)} |f(t)|^p dV(t) \\
&\leq C'' \sup_\epsilon \int_{\partial\Omega_\epsilon^1} |f(t)|^p d\sigma_\epsilon^1(t).
\end{aligned}$$

Of course the reverse inequality follows by symmetry. □

One technical difficulty that we face on an arbitrary (nonconvex) $\Omega$ is that the device of considering the dilated functions $f_r(\zeta) = f(r\zeta)$ as harmonic functions on $\overline{\Omega}$ is no longer available. However this notion is an unnecessary crutch, and it is well to be rid of it. As a substitute, we cover $\Omega$ by finitely many smoothly bounded domains $\Omega_1, \ldots, \Omega_k$ with the following properties:

(2.3.3) $\Omega = \cup_j \Omega_j$;

(2.3.4) For each $j$, the set $\partial\Omega \cap \partial\Omega_j$ is a smooth $(N-1)$-dimensional manifold with boundary;

(2.3.5) There is an $\epsilon_0 > 0$ and a vector $\nu_j$ transversal to $\partial\Omega \cap \partial\Omega_j$ and pointing out of $\Omega$ such that $\Omega_j - \epsilon\nu_j \equiv \{z - \epsilon\nu_j : z \in \Omega_j\} \subset\subset \Omega$, all $0 < \epsilon < \epsilon_0$.

We leave the detailed verification of the existence of the sets $\Omega_j$ satisfying (2.3.3)–(2.3.5) as an exercise. See Figure 2.3 for an illustration of these ideas.

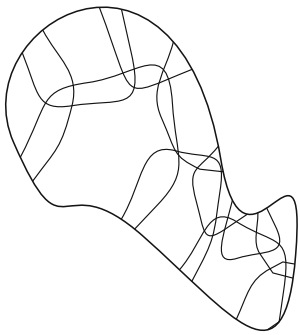


**Figure 2.3** Geometrically advantageous subregions.

For a general $C^2$ bounded domain, the substitute for dilation will be to fix $j \in \{1, \ldots, k\}$ and (locally) consider the translated functions $f_\epsilon(x) = f(x - \epsilon\nu_j), f_\epsilon : \Omega_j \to \mathbb{C}$, as $\epsilon \to 0^+$.

**Prelude** The next result ties together all of our key ideas. It establishes that an $\mathbf{h}^p$ function has a $p$th power modulus with harmonic majorant (and conversely) and also that such a function is the Poisson integral of an $L^p$ function on the boundary. As we know from our studies in Chapter 1, these are the first steps in getting our hands on the boundary behavior of harmonic and holomorphic functions. Further developments will be seen in the later sections.

**Theorem 2.3.6** *Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with $C^2$ boundary and $f$ harmonic on $\Omega$. Let $1 \leq p < \infty$. The following are equivalent:*

**(2.3.6.1)** $f \in \mathbf{h}^p(\Omega)$;

**(2.3.6.2)** *If $p > 1$ then there is an $\widetilde{f} \in L^p(\partial\Omega)$ such that $f(x) = \int_{\partial\Omega} P(x, y)\widetilde{f}(y)d\sigma(y)$ [resp. if $p = 1$ then there is a $\mu \in \mathcal{M}(\partial\Omega)$ such that $f(x) = \int_{\partial\Omega} P(x, y)d\mu(y)$]. Moreover, $\|f\|_{\mathbf{h}^p} \cong \|\widetilde{f}\|_{L^p}$.*

**(2.3.6.3)** *$|f|^p$ has a harmonic majorant on $\Omega$.*

**Proof (2.3.6.2) $\Rightarrow$ (2.3.6.3)** If $p > 1$, let

$$h(x) = \int_{\partial\Omega} P(x, y)|\widetilde{f}(y)|^p d\sigma(y).$$

Then, treating $P(x, \cdot)d\sigma$ as a positive measure of total mass 1, we have

$$\begin{aligned} |f(x)|^p &= \left| \int_{\partial\Omega} \widetilde{f}(y)P(x, y)d\sigma(y) \right|^p \\ &\overset{\text{(Jensen)}}{\leq} \int_{\partial\Omega} |\widetilde{f}(y)|^p P(x, y)d\sigma(y) \equiv h(x). \end{aligned}$$

The proof for $p = 1$ is similar.

**(2.3.6.3) $\Rightarrow$ (2.3.6.1)** If $\epsilon > 0$ is small, $x_0 \in \Omega$ is fixed, and $G_\Omega$ is the Green's function for $\Omega$, then $G_\Omega(x_0, \cdot)$ has non-vanishing gradient near $\partial\Omega$ (use Hopf's lemma). Therefore

$$\widetilde{\Omega}_\epsilon \equiv \{x \in \Omega : -G_\Omega(x, \cdot) < -\epsilon\}$$

are well-defined domains for $\epsilon$ small. Moreover (*check the proof*), the Poisson kernel for $\widetilde{\Omega}_\epsilon$ is $P_\epsilon(x, y) = -\nu_y^\epsilon G_\Omega(x, y)$. Here $\nu_y^\epsilon$ is the normal to $\partial\widetilde{\Omega}_\epsilon$ at $y \in \partial\widetilde{\Omega}_\epsilon$. Assume that $\epsilon > 0$ is so small that $x_0 \in \widetilde{\Omega}_\epsilon$. So if $h$ is the harmonic majorant for $|f|^p$ then

$$h(x_0) = \int_{\partial\widetilde{\Omega}_\epsilon} -\nu_y^\epsilon G_\Omega(x_0, y) h(y) d\sigma(y). \tag{2.3.6.4}$$

Let $\pi_\epsilon : \partial\widetilde{\Omega}_\epsilon \to \partial\Omega$ be normal projection for $\epsilon$ small. Then

$$-\nu_y^\epsilon G_\Omega(x_0, \pi_\epsilon^{-1}(\,\cdot\,)) \to -\nu_y G_\Omega(x_0, \,\cdot\,)$$

uniformly on $\partial\Omega$ as $\epsilon \to 0^+$. By 2.2.1, $-\nu_y G_\Omega(x_0, \,\cdot\,) \geq c_{x_0} > 0$ for some constant $c_{x_0}$. Thus $-\nu_y^\epsilon G_\Omega(x_0, \pi_\epsilon^{-1}(\cdot))$ are all bounded below by $c_{x_0}/2$ if $\epsilon$ is small enough. As a result, (2.3.6.4) yields

$$\int_{\partial\widetilde{\Omega}_\epsilon} h(y) d\sigma(y) \leq 2h(x_0)/c_{x_0}$$

for $\epsilon > 0$ small. In conclusion,

$$\int_{\partial\widetilde{\Omega}_\epsilon} |f(y)|^p d\sigma(y) \leq 2h(x_0)/c_{x_0}.$$

**(2.3.6.1)** $\Rightarrow$ **(2.3.6.2)** Let $\Omega_j$ be as in (2.3.3) through (2.3.5). Fix $j$. Define on $\Omega_j$ the functions $f_\epsilon(x) = f(x - \epsilon\nu_j), 0 < \epsilon < \epsilon_0$. Then the hypothesis and (a small modification of) Lemma 2.3.2 show that $\{f_\epsilon\}$ forms a bounded subset of $L^p(\partial\Omega_j)$. If $p > 1$, let $\widetilde{f}_j \in L^p(\partial\Omega_j)$ be a weak-$*$ accumulation point (for the case $p = 1$, replace $\widetilde{f}_j$ by a Borel measure $\widetilde{\mu}_j$). The crucial observation at this point is that $f$ is the Poisson integral of $\widetilde{f}_j$ on $\Omega_j$. Therefore $f$ on $\Omega_j$ is completely determined by $\widetilde{f}_j$ and conversely (see also the exercises at the end of the section). A moment's reflection now shows that $\widetilde{f}_j = \widetilde{f}_k$ almost everywhere $[d\sigma]$ in $\partial\Omega_j \cap \partial\Omega_k \cap \partial\Omega$ so that $\widetilde{f} \equiv \widetilde{f}_j$ on $\partial\Omega_j \cap \partial\Omega$ is well-defined. By appealing to a partition of unity on $\partial\Omega$ that is subordinate to the open cover induced by the (relative) interiors of the sets $\partial\Omega_j \cap \partial\Omega$, we see that $f_\epsilon = f \circ \pi_\epsilon^{-1}$ converges weak-$*$ to $\widetilde{f}$ on $\partial\Omega$ when $p > 1$ (resp. $f_\epsilon \to \widetilde{\mu}$ weak-$*$ when $p = 1$).

Referring to the proof of **(2.3.6.3)** $\Rightarrow$ **(2.3.6.1)** we write, for $x_0 \in \Omega$ fixed,

$$\begin{aligned} f(x_0) &= \int_{\partial\Omega_\epsilon} -\nu_y^\epsilon G_\Omega(x_0, y) f(y) d\sigma_\epsilon(y) \\ &= \int_{\partial\Omega} -\nu_y^\epsilon G_\Omega\left(x_0, \pi_\epsilon^{-1}(y)\right) f\left(\pi_\epsilon^{-1}(y)\right) \mathcal{J}^\epsilon(y) d\sigma(y), \end{aligned}$$

where $\mathcal{J}^\epsilon$ is the Jacobian of the mapping $\pi_\epsilon^{-1} : \partial\Omega \to \partial\Omega_\epsilon$. The fact that $\partial\Omega$ is $C^2$ combined with previous observations implies that the last line tends to

$$\int_{\partial\Omega} -\nu_y G_\Omega(x_0, y)\widetilde{f}(y)d\sigma(y) = \int_{\partial\Omega} P_\Omega(x_0, y)\widetilde{f}(y)d\sigma(y)$$

$$\left(\text{resp.} \int_{\partial\Omega} -\nu_y G_\Omega(x_0, y)d\widetilde{\mu}(y) = \int_{\partial\Omega} P_\Omega(x_0, y)d\widetilde{\mu}(y)\right)$$

as $\epsilon \to 0^+$. □

**Exercises for the Reader**

**1.** Prove the last statement in Theorem 2.3.6.

**2.** Imitate the proof of Theorem 2.3.6 to show that if $u$ is continuous and *subharmonic* on $\Omega$ and if

$$\sup_\epsilon \int_{\partial\Omega_\epsilon} |u(\zeta)|^p d\sigma(\zeta) < \infty, \qquad p \geq 1,$$

then $u$ has a harmonic majorant $h$. If $p > 1$, then $h$ is the Poisson integral of an $L^p$ function $\widetilde{h}$ on $\partial\Omega$. If $p = 1$, then $h$ is the Poisson integral of a Borel measure $\widetilde{\mu}$ on $\partial\Omega$.

**3.** Let $\Omega \subseteq \mathbb{R}^N$ be a domain with $C^2$ boundary and let $\rho$ be a $C^2$ defining function for $\Omega$. Define $\Omega_\epsilon = \{x \in \Omega : \rho(x) < -\epsilon\}, 0 < \epsilon < \epsilon_0$. Let $\partial\Omega_\epsilon$ and $d\sigma_\epsilon$ be as usual. Let $\pi_\epsilon : \partial\Omega_\epsilon \to \partial\Omega$ be orthogonal projection. Let $f \in L^p(\partial\Omega), 1 \leq p < \infty$. Define

$$F(x) = \int_{\partial\Omega} P_\Omega(x, y)f(y)d\sigma(y).$$

**a.** Prove that $\int_{\partial\Omega} P_\Omega(x, y)d\sigma(y) = 1$, any $x \in \Omega$.

**b.** There is a $C > 0$ such that, for any $y \in \partial\Omega$,

$$\int_{\partial\Omega_\epsilon} P_\Omega(x, y)d\sigma_\epsilon(x) \leq C, \qquad \text{any} \quad 0 < \epsilon < \epsilon_0;$$

**c.** There is a $C' > 0$ such that

$$\int_{\partial\Omega_\epsilon} |F(x)|^p d\sigma_\epsilon \leq C', \qquad \text{any} \quad 0 < \epsilon < \epsilon_0;$$

**d.** If $\phi \in C(\Omega)$ satisfies $\|\phi - f\|_{L^p(\partial\Omega)} < \eta$ and

$$G(x) \equiv \int_{\partial\Omega} P_\Omega(x, y)\,(\phi(y) - f(y))\,d\sigma(y),$$

then

$$\int_{\partial\Omega_\epsilon} |G(x)|^p d\sigma_\epsilon(x) \leq C'\eta\,, \qquad \text{any} \quad 0 < \epsilon < \epsilon_0.$$

**e.** With $\phi$ as in part **d.** and $\Phi(x) = \int_{\partial\Omega} P_\Omega(x,y)\phi(y)d\sigma(y)$, then $(\Phi|_{\partial\Omega_\epsilon}) \circ \pi_\epsilon^{-1} \to \phi$ uniformly on $\partial\Omega$.

**f.** Show that $F \circ \pi_\epsilon^{-1} \to f$ in the $L^p(\partial\Omega)$ norm.

## 2.4 Pointwise Convergence for Harmonic Functions on Domains in $\mathbb{R}^N$

**Capsule:** We next turn to the question of pointwise boundary convergence for $\mathbf{h}^p$ functions on a smoothly bounded domain in $\mathbb{R}^N$, $1 < p < \infty$. Just as in the one variable situation, this will be a preliminary step to studying pointwise boundary convergence for $H^p$ functions on a smoothly bounded domain in $\mathbb{C}^n$. Many of the steps and arguments in the present section will have a ring of familiarity (although the details will be rather more recondite). Again we reference [AIZ] for historical background.

Let $\Omega \subseteq \mathbb{R}^N$ be a domain with $C^2$ boundary. For $P \in \partial\Omega, \alpha > 1$, we define

$$\Gamma_\alpha(P) = \{x \in \Omega : |x - P| < \alpha\delta_\Omega(x)\}.$$

This is the $N$-dimensional analogue of the Stolz region considered in Chapter 1. See Figure 2.4. Our theorem is as follows:

**Theorem 2.4.1** *Let $\Omega \subset\subset \mathbb{R}^N$ be a domain with $C^2$ boundary. Let $\alpha > 1$. If $1 < p \leq \infty$ and $f \in \mathbf{h}^p$, then*

$$\lim_{\Gamma_\alpha(P)\ni x\to P} f(x) \equiv \widetilde{f}(P) \qquad \textit{exists for almost every } P \in \partial\Omega\,.$$

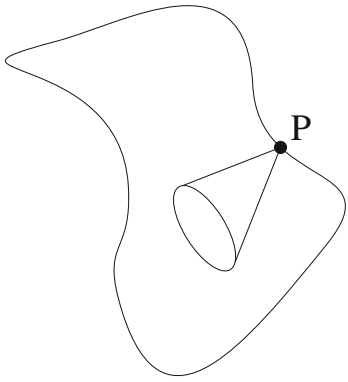



**Figure 2.4** The $N$-dimensional Stolz region.

*Moreover,*

$$\|\widetilde{f}\|_{L^p(\partial\Omega)} \cong \|f\|_{\mathbf{h}^p(\Omega)}.$$

**Proof** We may as well assume that $p < \infty$. We already know from (2.3.6.2) that there exists an $\widetilde{f} \in L^p(\partial\Omega)$, $\|\widetilde{f}\|_{L^p} \cong \|f\|_{\mathbf{h}^p}$, such that $f = P\widetilde{f}$. It remains to show that $\widetilde{f}$ satisfies the conclusions of the present theorem. This will follow just as in the arguments in Chapter 1 as soon as we prove two things: First,

$$\sup_{x\in\Gamma_\alpha(P)} |P\widetilde{f}(x)| \le C_\alpha M_1\widetilde{f}(P), \tag{2.4.1.1}$$

where

$$M_1\widetilde{f}(P) \equiv \sup_{R>0} \frac{1}{\sigma(B(P,R)\cap\partial\Omega)} \int_{B(P,R)\cap\partial\Omega} |f(t)|d\sigma(t).$$

Second,

$$\sigma\{y \in \partial\Omega : M_1\widetilde{f}(y) > \lambda\} \le C\frac{\|\widetilde{f}\|_{L^1(\partial\Omega)}}{\lambda}, \qquad \text{all} \quad \lambda > 0. \tag{2.4.1.2}$$

Now (2.4.1.1) is proved just as in Chapter 1. It is necessary to use the estimate given in Proposition 2.2.6. On the other hand, (2.4.1.2) is not obvious; we supply a proof in the paragraphs that follow. □

**Lemma 2.4.2 (Wiener)** *Let $K \subseteq \mathbb{R}^N$ be a compact set that is covered by the open balls $\{B_\alpha\}_{\alpha\in A}$, $B_\alpha = B(c_\alpha, r_\alpha)$. There is a subcover $B_{\alpha_1}, B_{\alpha_2}, \ldots, B_{\alpha_m}$, consisting of pairwise disjoint balls, such that*

$$\bigcup_{j=1}^{m} B(c_{\alpha_j}, 3r_{\alpha_j}) \supseteq K.$$

**Proof** We already proved a more general version of this result in the section on spaces of homogeneous type. We shall not repeat the details. □

**Corollary 2.4.3** *Let $K \subseteq \partial\Omega$ be compact, and let $\{B_\alpha \cap \partial\Omega\}_{\alpha\in A}$, $B_\alpha = B(c_\alpha, r_\alpha)$, be an open covering of $K$ by balls with centers in $\partial\Omega$. Then there is a pairwise disjoint subcover $B_{\alpha_1}, B_{\alpha_2}, \ldots, B_{a_m}$ such that $\cup_j \{B(c_\alpha, 3r_\alpha) \cap \partial\Omega\} \supseteq K$.*

**Proof** The set $K$ is a compact subset of $\mathbb{R}^N$ that is covered by $\{B_\alpha\}$. Apply the preceding Lemma 2.4.2 and restrict to $\partial\Omega$. □

**Lemma 2.4.4** *If $f \in L^1(\partial\Omega)$, then*

$$\sigma\{x \in \partial\Omega : M_1 f(x) > \lambda\} \le C\frac{\|f\|_{L^1}}{\lambda},$$

*all $\lambda > 0$.*

**Proof** Let $S_\lambda = \{x \in \partial\Omega : M_1 f(x) > \lambda\}$. Let $K$ be a compact subset of $S_\lambda$. It suffices to estimate $\sigma(K)$. Now for each $x \in K$, there is a ball $B_x$ centered at $x$ such that

$$\frac{1}{\sigma(B_x \cap \partial\Omega)} \int_{B_x \cap \partial\Omega} |f(t)| d\sigma(t) > \lambda. \tag{2.4.4.1}$$

The balls $\{\partial\Omega \cap B_x\}_{x\in K}$ cover $K$. Choose, by Corollary 2.4.2, disjoint balls $B_{x_1}, B_{x_2}, \ldots, B_{x_m}$ such that $\{\partial\Omega \cap 3B_{x_j}\}$ cover $K$, where $3B_{x_j}$ represents the threefold dilate of $B_{x_j}$ (with the same center). Then

$$\begin{aligned}\sigma(K) &\leq \sum_{j=1}^{m} \sigma(3B_{x_j} \cap \partial\Omega) \\ &\leq C(N,\Omega) \sum_{j=1}^{m} \sigma(B_{x_j} \cap \partial\Omega),\end{aligned}$$

where the constant $C$ will depend on the curvature of $\partial\Omega$. But (2.4.4.1) implies that the last line is majorized by

$$C(N,\partial\Omega) \sum_j \frac{\int_{B_{x_j}\cap\partial\Omega} |f(t)| d\sigma(t)}{\lambda} \leq C(N,\partial\Omega) \frac{\|f\|_{L^1}}{\lambda}.$$

This completes the proof of the theorem. □

## 2.5 Boundary Values of Holomorphic Functions in $\mathbb{C}^n$

**Capsule:** This section and the next present the culmination of our efforts. It presents, with proof, Stein's result (which generalizes Korányi's pioneering work) about admissible boundary limits on domains in $\mathbb{C}^n$. It should be noted that Stein's theorem is sharp for strongly pseudoconvex domains, but not for more general domains (such as finite type domains in $\mathbb{C}^n$). This point is explained more fully in [DIK, Ch. 1].

Everything in Section 2.4 applies *a fortiori* to domains $\Omega \subseteq \mathbb{C}^n$. However, on the basis of our experience in the classical case, we expect $H^p(\Omega)$ functions to also have pointwise boundary values for $0 < p \leq 1$. That this is indeed the case is established in this section by two different arguments.

First, if $\Omega \subset\subset \mathbb{C}^n$ is a $C^2$ domain and $f \in H^p(\Omega)$, we shall prove through an application of Fubini's theorem (adapted from a paper of Lempert [LEM2]) that $f$ has pointwise boundary limits in a rather special sense at $\sigma$-almost every $\zeta \in \partial\Omega$. This argument is self-contained. After that, we derive some more powerful results using a much broader perspective. We shall not further develop the second

methodology in this book, but we present an introduction to it for background purposes. Further details may be found in E. M. Stein [STE1, Ch. 2] and C. Fefferman and E. M. Stein [FES]. The logical progression of ideas in this chapter will proceed from the first approach based on Lempert [LEM2].

It should be noted that the ideas presented here were anticipated in the notable paper [HOR5] and a bit later in [STE6]. The first of these papers, which is not cited sufficiently frequently, is remarkably prescient.

**Proposition 2.5.1** *Let $\Omega \subset\subset \mathbb{C}^n$ have $C^2$ boundary. Let $0 < p < \infty$ and $f \in H^p(\Omega)$. Write $\Omega = \cup_{j=1}^k \Omega_j$ as in (2.3.3) through (2.3.5), and let $\nu_1, \ldots, \nu_k$ be the associated outward normal vectors. Then, for each $j \in \{1, \ldots, k\}$, it holds that*

$$\lim_{\epsilon \to 0^+} f(\zeta - \epsilon \nu_j) \equiv \widetilde{f}_j(\zeta)$$

*exists for $\sigma$-almost every $\zeta \in \partial\Omega_j \cap \partial\Omega$.*

**Proof** We may suppose that $p < \infty$. Fix $1 \leq j \leq k$. Assume for convenience that $\nu_j = \nu_P = (1 + i0, 0, \ldots, 0), P \in \partial\Omega_j$, and that $P = 0$. If $z \in \mathbb{C}^n$, write $z = (z_1, \ldots, z_n) = (z_1, z')$. We may assume that $\Omega_j = \cup_{|z'|<1}\{(z_1, z') : z_1 \in \mathcal{D}_{z'}\}$, where $\mathcal{D}_{z'} \subseteq \mathbb{C}$ is a diffeomorph of $D \subseteq \mathbb{C}$ with $C^2$ boundary. For each $|z'| < 1, k \in \mathbb{N}, 0 < 1/k < \epsilon_0$, let $b^k_{z'} = (\mathcal{D}_{z'} \times \{z'\}) \cap \{z \in \Omega_j : \text{dist}\,(z, \partial\Omega_j) = 1/k\}$. Define $B^k = \cup_{|z'|<1} b^k_{z'}$. Now we see that

$$\sup_k \int_{B^k} |f(\zeta)|^p d\sigma_k \leq C_0 < \infty\,, \tag{2.5.1.1}$$

where $\sigma_k$ is surface measure on $B^k$. Formula (2.5.1.1) may be rewritten as

$$\sup_k \int_{|\zeta'|<1} \int_{b^k_{\zeta'}} |f(\zeta_1, \zeta')|^p d\widetilde{\sigma}_k(\zeta_1) dV_{2n-2}(\zeta') \leq C_0, \tag{2.5.1.2}$$

where $\widetilde{\sigma}_k$ is surface measure on $b^k_{\zeta'}$. If $M > 0, k \geq k_0 > 1/\epsilon_0$, we define

$$S^M_k = \left\{ \zeta' : |\zeta'| < 1, \int_{b^k_{\zeta'}} |f(\zeta_1, \zeta')|^p d\widetilde{\sigma}_k(\zeta_1) > M \right\}. \tag{2.5.1.3}$$

Then (2.5.1.2), (2.5.1.3), and Chebycheff's inequality together yield

$$V_{2n-1}(S^M_k) \leq \frac{C_0}{M}\,, \qquad \text{all} \quad k.$$

Now let

$$S^M = \left\{ \zeta' : |\zeta'| < 1, \int_{b^k_{\zeta'}} |f(\zeta_1, \zeta')|^p d\widetilde{\sigma}_k(\zeta_1) \leq M \right.$$
$$\left. \text{for only finitely many } k \right\}$$
$$= \bigcup_{\ell=k_0}^{\infty} \bigcap_{k=\ell}^{\infty} S^M_k .$$

Then $V_{2n-1}(S^M) \leq C_0/M$. Since $M$ may be made arbitrarily large, we conclude that for $V_{2n-2}$ almost every $\zeta' \in D^{n-1}(0,1)$, there exist $k_1 < k_2 < \cdots$ such that

$$\int_{b^{k_m}_{\zeta'}} |f(\zeta_1, \zeta')|^p d\widetilde{\sigma}_{k_m}(\zeta_1) = O(1) \qquad \text{as} \quad m \to \infty.$$

It follows that the functions $f(\cdot, \zeta') \in H^p(\mathcal{D}_{\zeta'})$ for $V_{2n-2}$ almost every $\zeta' \in D^{n-1}(0,1)$. Now we have the desired result just by invoking the 1-dimensional theorem and Fubini's theorem. □

At this point we *could* prove that $f$ tends nontangentially to the function $\widetilde{f}$ constructed in Proposition 2.5.1. However, we do not do so because a much stronger result is proved in the next section. The remainder of the present section consists of a digression to introduce the reader to the second point of view mentioned in the introduction. This second point of view involves far-reaching ideas arising from the "real variable" school of complex analysis. This methodology provides a more natural, and much more profound, approach to the study of boundary behavior.

The results that we present are due primarily to E. M. Stein [STE1]. To present Stein's ideas, we first need an auxiliary result of A. P. Calderón, (see Stein [STE2, Ch. 3]), and K. O. Widman [WID]. Although it is well within the scope of this book to prove this auxiliary result on a half space, a complete proof on smoothly bounded domains would entail a number of tedious ancillary ideas (such as the maximum principle for second-order elliptic operators). Hence we only state the needed result and refer the reader to the literature for details.

Let $\Omega \subset\subset \mathbb{R}^N$ be a domain with $C^2$ boundary and let $u : \Omega \to \mathbb{C}$ be harmonic. If $P \in \partial\Omega$ then we say that *f is nontangentially bounded* at $P$ if, for each $\alpha > 1$, it holds that

$$\sup_{x \in \Gamma_\alpha(P)} |f(x)| < \infty .$$

Now the result is as follows:

**Prelude** The next theorem, due to A. P. Calderón, has been tremendously influential in modern harmonic analysis. It proves the equivalence of two fundamental ideas.

**Theorem 2.5.2** *Let $\Omega \subset\subset \mathbb{R}^N$ have $C^2$ boundary. Let $u : \Omega \to \mathbb{C}$ be harmonic. Let $E \subseteq \partial\Omega$ be a set of positive $\sigma$-measure. Suppose that $u$ is nontangentially bounded at $\sigma$-almost every $P \in E$. Then $u$ has nontangential limits at $\sigma$-almost every $P \in E$.*

**Remarks**

(i) Obviously, if $u$ has a nontangential limit at $P \in \partial\Omega$, then $u$ is nontangentially bounded at $P$.

(ii) Theorem 2.5.2 has no analogue for radial boundedness and radial limits, even on the disc. For let $E \subseteq \partial D$ be a $F_\sigma$ set of first category and measure $2\pi$. Let $f : D \to \mathbb{C}$ be given by $f(z) = \sin\left(\frac{1}{1-|z|}\right)$. Then $f$ is continuous, bounded by 1 on $D$, and does not possess a radial limit at any point of $\partial D$. Of course $f$ is *not* holomorphic. But by a theorem of Bagemihl and Seidel [BAG], there is a holomorphic (!) function $u$ on $D$ with

$$\lim_{r\to 1^-} |u(re^{i\theta}) - f(re^{i\theta})| = 0$$

for every $e^{i\theta} \in E$. □

**Prelude** The next theorem is the poor man's version of a boundary limit theorem in several complex variables. Once we have the machinery of admissible approach regions in place, we can prove something much stronger.

**Theorem 2.5.3** *Let $\Omega \subset\subset \mathbb{C}^n$ have $C^2$ boundary. Let $f \in H^p(\Omega), 0 < p \leq \infty$. Then $f$ has nontangential boundary limit at almost every $P \in \partial\Omega$.*

**Proof** We may assume that $p < \infty$. The function $|f|^{p/2}$ is subharmonic and uniformly square integrable over $\partial\Omega_\epsilon, 0 < \epsilon < \epsilon_0$. So $|f|^{p/2}$ has a harmonic majorant $h \in \mathbf{h}^2(\Omega)$. Since $h \in \mathbf{h}^2$, it follows that $h$ has almost everywhere nontangential boundary limits. Therefore $h$ is nontangentially bounded at almost every point of $\partial\Omega$. As a result, $|f|^{p/2}$, and therefore $|f|$ itself, is nontangentially bounded at almost every point of $\partial\Omega$. So, by 2.5.2, $f$ has a nontangential boundary limit $\widetilde{f}$ defined at almost every point of $\partial\Omega$. □

We conclude this section with a recasting of the ideas in the proof of 2.5.3 to make more explicit the role of the maximal function. We first need two lemmas.

**Lemma 2.5.4** *Let $\{X, \mu\}$ be a measure space with $\mu \geq 0$. Let $f \geq 0$ on $X$ be measurable and $0 < p < \infty$. Then*

$$\int_X f(x)^p d\mu(x) = \int_0^\infty p s^{p-1} \mu_f(s) ds = -\int_0^\infty s^p d\mu_f(s),$$

*where $\mu_f(s) \equiv \mu\{x : f(x) \geq s\}, 0 \leq s < \infty$.*

**Proof** We have

$$\begin{aligned}\int_X f(x)^p d\mu(x) &= \int_X \int_0^{f(x)^p} dt d\mu(x)\\ &\overset{\text{(Fubini)}}{=} \int_0^\infty \int_{\{x:f(x)^p \geq t\}} d\mu(x) dt\\ &\overset{(t=s^p)}{=} \int_0^\infty p s^{p-1} \mu\{x : f(x) \geq s\} ds\\ &\overset{\text{(parts)}}{=} -\int_0^\infty s^p d\mu_f(x)\end{aligned}$$

□

**Prelude** The proof of the next result is an instance of the Marcinkiewicz interpolation theorem (see [STW2, Ch. 5]). In turn, Marcinkiewicz is a case of the so-called "real interpolation method" that is generally attributed to Lions and Peetre (see [LIP1, LIP2]). It is an important and powerful technique.

**Lemma 2.5.5** *The maximal operator $M_1$ is bounded from $L^2(\partial\Omega)$ to $L^2(\partial\Omega)$.*

**Proof** We know that $M_1$ maps $L^\infty$ to $L^\infty$ (trivially) and is weakly bounded on $L^1$. After normalizing by a constant, we may suppose that

$$\|M_1 f\|_{L^\infty} \leq \frac{1}{3}\|f\|_{L^\infty}, \qquad \text{all} \quad f \in L^\infty(\partial\Omega). \tag{2.5.5.1}$$

Also we may assume that

$$\sigma\{\zeta \in \partial\Omega : |M_1 f(\zeta)| > \lambda\} \leq \frac{\|f\|_{L^1}}{\lambda}, \qquad \text{all} \quad \lambda > 0, \quad f \in L^1(\partial\Omega). \tag{2.5.5.2}$$

If $f \in L^2(\partial\Omega)$, $\|f\|_{L^2} = 1, 0 < t < \infty$, we write

$$f(\zeta) = f(\zeta) \cdot \chi_{\{\zeta:|f(\zeta)|<t\}} + f(\zeta) \cdot \chi_{\{\zeta:|f(\zeta)|\geq t\}} \equiv f_1^t(\zeta) + f_2^t(\zeta).$$

Then

$$\begin{aligned}\|M_1 f\|_{L^2}^2 &= 2\int_0^\infty \sigma_{M_1 f}(\lambda)\lambda d\lambda\\ &\leq \int_0^\infty \sigma_{M_1 f_2^\lambda}(\lambda/2)\lambda d\lambda + \int_0^\infty \sigma_{M_1 f_1^\lambda}(\lambda/2)\lambda d\lambda\\ &\equiv I + II.\end{aligned}$$

Now

$$\begin{aligned} I &\leq 2\int_0^\infty (\|f_2^\lambda\|_{L^1}/\lambda)\lambda d\lambda \\ &= 2\int_0^\infty \int_\lambda^\infty \sigma_f(t)dtd\lambda \\ &= 2\int_0^\infty \sigma_f(t)\int_0^t d\lambda dt \\ &= 2\int_0^\infty \sigma_f(t)\cdot t dt \\ &= \|f\|_{L^2}^2. \end{aligned}$$

Furthermore, notice that since $|f_1^\lambda(\zeta)| < \lambda$ for all $\zeta$, it follows that $|M_1 f_1^\lambda(\zeta)| \leq \lambda/3$. Hence $\sigma_{M_1 f_1^\lambda}(\lambda/2) \equiv 0$. Therefore $II = 0$. That finishes the proof. □

Let us now examine more closely the interplay between harmonic majorants and maximal functions to obtain a quantitative version of the nontangential boundedness. Let notation be as in Theorem 2.5.3. Let $h$ be the least harmonic majorant for $|f|^{p/2}$ (assuming that $p < \infty$). Notice that the dominated convergence theorem (or Fatou's lemma) implies that $\widetilde{f} \in L^p(\partial\Omega)$. (*Exercise:* For a rigorous proof, you must either consider the Jacobian of $\pi_\epsilon^{-1}$ or restrict attention to the $\Omega_j$'s as in (2.3.3)–(2.3.5). Now, for $\alpha > 1$ fixed and $P \in \partial\Omega$, we have

$$f_1^{*,\alpha}(P)^{p/2} \equiv \sup_{x\in\Gamma_\alpha(P)} |f(x)|^{p/2} \leq \sup_{x\in\Gamma_\alpha(P)} |h(x)| \leq C_\alpha M_1\widetilde{h}(P).$$

Therefore

$$\begin{aligned} \|f_1^{*,\alpha}\|_{L^p(\partial\Omega)}^p &\leq C_\alpha \|M_1\widetilde{h}\|_{L^2(\partial\Omega)}^2 \\ &\leq C\cdot C_\alpha \|\widetilde{h}\|_{L^2(\partial\Omega)}^2 \\ &\leq C'_\alpha \sup_\epsilon \int_{\partial\Omega_\epsilon} |f(\zeta)|^p d\sigma(\zeta) \\ &\leq C''_\alpha \int_{\partial\Omega} |\widetilde{f}(\zeta)|^p d\sigma(\zeta) \\ &= C''_\alpha \|\widetilde{f}\|_{L^p(\partial\Omega)}^p. \end{aligned} \tag{2.5.6}$$

Inequality (2.5.6) is valid for $0 < p < \infty$. It is central to the so-called real variable theory of $H^p$ spaces. For instance, one has:

**Theorem 2.5.7** *Let $u$ be a real harmonic function on $D \subseteq \mathbb{C}$. Let $0 < p < \infty$. Then $u$ is the real part of an $f \in H^p(D)$ if and only if, for some $\alpha > 1$,*

$$\|u_1^{*,\alpha}\|_{L^p(\partial D)} < \infty.$$

*Under these circumstances, $\|u_1^{*,\alpha}\|_{L^p(\partial\Omega)} \cong \|f\|_{H^p(D)}$.*

Theorem 2.5.7 was originally proved by methods of Brownian motion (D. Burkholder, R. Gundy, and M. Silverstein [BUR]). C. Fefferman and E. M. Stein [FES] gave a real-variable proof and extended the result in an appropriate sense to $\mathbb{R}^N$. The situation in several complex variables is rather more complicated (see J. Garnett and R. Latter [GARNL], S. G. Krantz and D. Ma [KRM], and S. G. Krantz and S.-Y. Li [KRL2]).

This ends our digression about the real-variable aspects of boundary behavior of harmonic and holomorphic functions. In the next section, we again pick up the thread begun in Proposition 2.5.1 and prove a theorem that is strictly stronger. This requires a new notion of convergence.

## 2.6 Admissible Convergence

Let $B \subseteq \mathbb{C}^n$ be the unit ball. The Poisson kernel for the ball has the form

$$P(z,\zeta) = \frac{\Gamma(n)}{2\pi^n} \cdot \frac{1-|z|^2}{|z-\zeta|^{2n}},$$

whereas the Poisson–Szegő kernel has the form

$$\mathcal{P}(z,\zeta) = \frac{(n-1)!}{2\pi^n} \cdot \frac{(1-|z|^2)^n}{|1-z\cdot\overline{\zeta}|^{2n}}.$$

As we know, an analysis of the convergence properties of these kernels entails dominating them by appropriate maximal functions. The maximal function involves the use of certain balls, and the shape of the ball should be compatible with the singularity of the kernel. That is why, when we study the *real analysis* of the Poisson kernel, we consider balls of the form

$$\beta_1(\zeta, r) = \{\xi \in \partial B : |\xi - \zeta| < r\}, \quad \zeta \in \partial B, \quad r > 0.$$

In studying the *complex analysis* of the Poisson–Szegő kernel (equivalently, the Szegő kernel), it is appropriate to use the balls

$$\beta_2(\zeta, r) = \{\xi \in \partial B : |1 - \xi\cdot\overline{\zeta}| < r\}, \quad \zeta \in \partial B, \quad r > 0.$$

These new nonisotropic balls are fundamentally different from the classical (or *isotropic*) balls $\beta_1$, as we shall now see. Assume without loss of generality that $\zeta = \mathbf{1} = (1, 0, \ldots, 0)$. Write $z' = (z_2, \ldots, z_n)$. Then

$$\beta_2(\mathbf{1}, r) = \{\xi \in \partial B : |1 - \xi_1| < r\}.$$

Notice that, for $\xi \in \partial B$,

$$\begin{aligned} |\xi'|^2 &= 1 - |\xi_1|^2 \\ &= (1 - |\xi_1|)(1 + |\xi_1|) \\ &\leq 2|1 - \xi_1| \end{aligned}$$

hence

$$\beta_2(\mathbf{1}, r) \subseteq \{\xi \in \partial B : |1 - \xi_1| < r, |\xi'| < \sqrt{2r}\}.$$

A similar computation shows that

$$\begin{aligned} \beta_2(\mathbf{1}, r) &= \{\xi \in \partial B : |1 - \xi_1| < r, |\xi'| = \sqrt{1 - |\xi_1|^2}\} \\ &\supseteq \partial B \cap \{\xi : |\operatorname{Im} \xi_1| < r/2, 1 - r/2 < \operatorname{Re} \xi_1 < 1, |\xi'| < \sqrt{r}\}. \end{aligned}$$

In short, the balls we are considering have dimension $\sim r$ in the complex space containing $\nu_1$ and dimension $\sim \sqrt{r}$ in the orthogonal complement (see Figure 2.5). The word "nonisotropic" means that we have different geometric behavior in different directions, and this is why we call the new balls nonisotropic.

In the classical setup, we considered *cones* modeled on the balls $\beta_1$:

$$\Gamma_\alpha(P) = \{z \in B : |z - P| < \alpha(1 - |z|)\}, \quad P \in \partial B, \quad \alpha > 1.$$

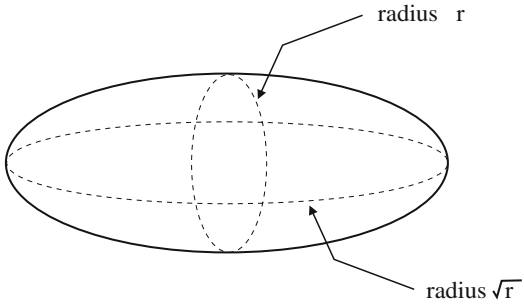



**Figure 2.5** A nonisotropic ball.

In the new situation we consider *admissible regions* modeled on the balls $\beta_2$:

$$\mathcal{A}_\alpha(P) = \{z \in B : |1 - z \cdot \overline{P}| < \alpha(1 - |z|)\}.$$

Our new theorem about boundary limits of $H^p$ functions is as follows:

**Prelude** The next result is the fundamental theorem of Korányi from 1969 (see [KOR1, KOR2]). It was a profound insight, and really took the world by storm.

**Theorem 2.6.1** *Let $f \in H^p(B), 0 < p \leq \infty$. Here $B \subseteq \mathbb{C}^n$. Let $\alpha > 1$. Then the limit*

$$\lim_{\mathcal{A}_\alpha(P) \ni z \to P} f(z) \equiv \widetilde{f}(P)$$

*exists for $\sigma$-almost every $P \in \partial B$.*

Since the Poisson–Szegő kernel is known explicitly on the ball, then for $p \geq 1$ the proof is deceptively straightforward: One defines, for $P \in \partial B$ and $f \in L^1(\partial B)$,

$$M_2 f(P) = \sup_{r>0} \frac{1}{\sigma(\beta_2(P, r))} \int_{\beta_2(P,r)} |f(\zeta)| d\sigma(\zeta).$$

Also set $f(z) = \int_{\partial B} \mathcal{P}(z, \zeta) f(\zeta)\, d\sigma(\zeta)$ for $z \in B$. Then, by explicit computation,

$$f_2^{*,\alpha}(P) \equiv \sup_{z \in \mathcal{A}_\alpha(P)} |f(z)| \leq C_\alpha M_2 f(P), \qquad \text{all} \quad f \in L^1(\partial B).$$

This crucial fact, together with appropriate estimates on the operator $M_2$, enables one to complete the proof along classical lines for $p \geq 1$. For $p < 1$, matters are more subtle.

We forego the details of the preceding argument on $B$ and instead develop the machinery for proving an analogue of Theorem 2.6.1 on an arbitrary $C^2$ bounded domain in $\mathbb{C}^n$. In this generality, there is no hope of obtaining an explicit formula for the Poisson–Szegő kernel; indeed, there are no known techniques for obtaining estimates for this kernel on arbitrary domains (however see C. Fefferman [FEF1], Boutet de Monvel and Sjöstrand [BOUTS], and A. Nagel, J. P. Rosay, E. M. Stein, and S. Wainger [NARSW] for estimates on strongly pseudoconvex domains and on domains of finite type in $\mathbb{C}^2$). Therefore we must develop more geometric methods which do not rely on information about kernels. The results that we present were proved on the ball and on bounded symmetric domains by A. Korányi [KOR1, KOR2]. Many of these ideas were also developed independently in Gong Sheng [SHE]. All of the principal ideas for arbitrary $\Omega$ are due to E. M. Stein [STE1, Ch. 3].

Our tasks, then, are as follows: (1) to define the balls $\beta_2$ on the boundary of an arbitrary $\Omega$; (2) to define admissible convergence regions $\mathcal{A}_\alpha$; (3) to obtain appropriate estimates for the corresponding maximal function; and (4) to couple the

maximal estimates, together with the fact that "radial" boundary values are already known to exist (see Proposition 2.5.1) to obtain the admissible convergence result.

If $z, w$ are vectors in $\mathbb{C}^n$, we continue to write $z \cdot \overline{w}$ to denote $\sum_j z_j \overline{w}_j$. (*Warning:* It is also common in the literature to use the notation $z \cdot w = \sum_j z_j \overline{w}_j$.) Also, for $\Omega \subseteq \mathbb{C}^n$ a domain with $C^2$ boundary, $P \in \partial\Omega$, we let $\nu_P$ be the Euclidean unit outward normal at $P$. Let $\mathbb{C}\nu_P$ denote the complex line generated by $\nu_P$: $\mathbb{C}\nu_P = \{\zeta \nu_P : \zeta \in \mathbb{C}\}$. This latter is commonly referred to as the complex normal space at the boundary point $P$.

By dimension considerations, if $T_P(\partial\Omega)$ is the $(2n-1)$-dimensional real tangent space to $\partial\Omega$ at $P$, then $\ell = \mathbb{C}\nu_P \cap T_P(\partial\Omega)$ is a (one-dimensional) real line. Let

$$\begin{aligned} \mathcal{T}_P(\partial\Omega) &= \{z \in \mathbb{C}^n : z \cdot \overline{\nu}_P = 0\} \\ &= \{z \in \mathbb{C}^n : z \cdot \overline{w} = 0 \ \forall\, w \in \mathbb{C}\nu_P\}. \end{aligned}$$

*A fortiori*, $\mathcal{T}_P(\partial\Omega) \subsetneqq T_P(\partial\Omega)$. If $\alpha \in \mathcal{T}_P(\partial\Omega)$, then $i\alpha \in \mathcal{T}_P(\partial\Omega)$. Therefore $\mathcal{T}_P(\partial\Omega)$ may be thought of as an $(n-1)$-dimensional complex subspace of $T_P(\partial\Omega)$. Clearly, $\mathcal{T}_P(\partial\Omega)$ is the complex subspace of $T_P(\partial\Omega)$ of maximal dimension. It contains all complex subspaces of $T_P(\partial\Omega)$. We may think of $\mathcal{T}_P(\partial\Omega)$ as the real orthogonal complement in $T_P(\partial\Omega)$ of $\ell$.

Now let us examine the matter from another point of view. The complex structure is nothing other than a linear operator $J$ on $\mathbb{R}^{2n}$ that assigns to $(x_1, x_2, \ldots, x_{2n-1}, x_{2n})$ the vector $(-x_2, x_1, -x_4, x_3, \ldots, -x_{2n}, x_{2n-1})$ (think of multiplication by $i$). With this in mind, we have that $J : \mathcal{T}_P(\partial\Omega) \to \mathcal{T}_P(\partial\Omega)$ both injectively and surjectively. Notice that $J\nu_P \in T_P(\partial\Omega)$ while $J(J\nu_P) = -\nu_P \notin T_P(\partial\Omega)$. We call $\mathbb{C}\nu_P$ the *complex normal* space to $\partial\Omega$ at $P$ and $\mathcal{T}_P(\partial\Omega)$ the *complex tangent space* to $\partial\Omega$ at $P$. Let $\mathcal{N}_P = \mathbb{C}\nu_P$. Then we have $\mathcal{N}_P \perp \mathcal{T}_P$ and

$$\begin{aligned} \mathbb{C}^n &= \mathcal{N}_P \oplus_{\mathbb{C}} \mathcal{T}_P \\ T_P &= \mathbb{R}J\nu_P \oplus_{\mathbb{R}} \mathcal{T}_P. \end{aligned}$$

**EXAMPLE 2.6.2** Let $\Omega = B \subseteq \mathbb{C}^n$ be the unit ball and $P = \mathbf{1} = (1, 0, 0, \ldots, 0) \in \partial\Omega$. Then $\mathbb{C}\nu_P = \{(z_1, 0, \ldots, 0) : z_1 \in \mathbb{C}\}$ and $\mathcal{T}_P = \{(0, z_2, z_3, \ldots, z_n) = (0, z') : z' \in \mathbb{C}^{n-1}\}$. □

The next definition is best understood in light of the foregoing discussion and the definition of $\beta_2(P, r)$ in the boundary of the unit ball $B$. Let $\Omega \subset\subset \mathbb{C}^n$ have $C^2$ boundary. For $P \in \partial\Omega$, let $\pi_P : \mathbb{C}^n \to \mathcal{N}_P$ be (real or complex) orthogonal projection.

**Definition 2.6.3** If $P \in \partial\Omega$ let

$$\begin{aligned} \beta_1(P, r) &= \{\zeta \in \partial\Omega : |\zeta - P| < r\}; \\ \beta_2(P, r) &= \{\zeta \in \partial\Omega : |\pi_P(\zeta - P)| < r, |\zeta - P| < r^{1/2}\}. \end{aligned}$$

**Exercise for the Reader** The ball $\beta_2(P, r)$ has diameter $\sim \sqrt{r}$ in the $(2n-2)$ complex tangential directions and diameter $\sim r$ in the one (complex normal) direction. Therefore $\sigma(\beta_2(P, r) \approx (\sqrt{r})^{2n-2} \cdot r \approx Cr^n$. By contrast, $\sigma(\beta_1(P, r) \approx Cr^{2n-1}$.

If $z \in \Omega, P \in \partial\Omega$, we let

$$\delta_P(z) = \min\{\text{dist}\,(z, \partial\Omega), \text{dist}\,(z, T_P(\Omega))\}.$$

Notice that,if $\Omega$ is convex, then $\delta_P(z) = \delta_\Omega(z)$.

**Definition 2.6.4** If $P \in \partial\Omega, \alpha > 1$, let

$$\mathcal{A}_\alpha = \{z \in \Omega : |(z-P) \cdot \overline{v}_P| < \alpha\delta_P(z), |z-P|^2 < \alpha\delta_P(z)\}.$$

We call $\mathcal{A}_\alpha(P)$ the *admissible approach region of aperture* $\alpha$ *at* $P$.
Notice that $\delta_P$ is used because near non-convex boundary points we still want $\mathcal{A}_\alpha$ to have the fundamental geometric shape of (paraboloid $\times$ cone) as shown in Figure 2.6.[1]

**Definition 2.6.5** If $f \in L^1(\partial\Omega)$, $P \in \partial\Omega$, and $r > 0$, then we define

$$M_j f(P) = \sup_{r>0} \sigma(\beta_j(P, r))^{-1} \int_{\beta_j(P,r)} |f(\zeta)| d\sigma(\zeta), \qquad j = 1, 2.$$

**Definition 2.6.6** If $f \in C(\Omega), P \in \partial\Omega$, then we define

$$f_2^{*,\alpha}(P) = \sup_{z \in \mathcal{A}_\alpha(P)} |f(z)|.$$

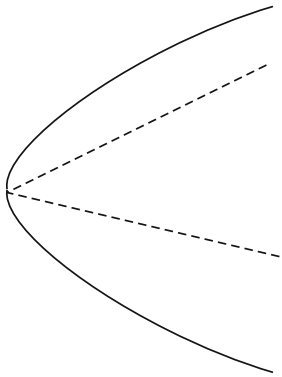


**Figure 2.6** An admissible approach region.

[1] A geometrically more natural way to define the admissible approach regions is in the language of invariant metrics—see [KRA7].

The first step of our program is to prove an estimate for $M_2$. This will require a covering lemma (indeed, it is known that weak type estimates for operators like $M_j$ are logically equivalent to covering lemmas—see A. Cordoba and R. Fefferman [COR]). It is natural for us at this point to exploit the paradigm of a space of homogeneous type.

In particular, it needs to be checked that the balls $\beta_2$ satisfy the axioms for such a space. The first two axioms are straightforward to check, especially if one exploits the compactness of the boundary. It is generally the enveloping property that is the most challenging. We address that now.

Suppose that $\beta_2(x,r) \cap \beta_2(y,s) \neq \emptyset$ with $r \geq s$. Thus there is a point $a \in \beta_2(x,r) \cap \beta_2(y,s)$. We may assume that $r = s$ by the doubling property. We thus have $|x-a| \leq r^{1/2}$, $|y-a| \leq r^{1/2}$, hence $|x-y| \leq 2r^{1/2}$. Let the constant $M \geq 2$ be chosen so that $\|\pi_x - \pi_y\| \leq M|x-y|$. (We must use here the fact that the boundary is $C^2$.) We claim that $\beta_2(x,(3+4M)r) \supseteq \beta_2(y,r)$. To see this, let $v \in \beta_2(y,r)$.

The easy half of the estimate is

$$|x-v| \leq |x-y| + |y-v| \leq 2r^{1/2} + r^{1/2} = 3r^{1/2}.$$

Also

$$\pi_x(x-v) = \pi_x(x-a) + \pi_y(a-v) + \{\pi_x - \pi_y\}(a-v).$$

Therefore

$$\begin{aligned} |\pi_x(x-v)| &\leq r + 2r + \|\pi_x - \pi_y\||a-v| \\ &\leq 3r + M|x-y| \cdot |a-v| \\ &\leq 3r + M2r^{1/2}(|a-y| + |y-v|) \\ &\leq (3+4M)r. \end{aligned}$$

This proves the enveloping property. Thus we have the following:

**Corollary 2.6.7** *If $f \in L^1(\partial\Omega)$, then*

$$\sigma\{\zeta \in \omega : M_2 f(\zeta) > \lambda\} \leq C\frac{\|f\|_{L^1(\partial\Omega)}}{\lambda}, \qquad \text{all} \quad \lambda > 0.$$

**Proof** By the general theory of spaces of homogeneous type, once we have verified the axioms then we know that we have a Wiener-type covering theorem. And therefore we immediately get the weak-type (1,1) estimate. □

**Corollary 2.6.8** *The operator $M_2$ maps $L^2(\partial\Omega)$ to $L^2(\partial\Omega)$ boundedly.*

**Proof** Exercise. Note that $M_2$ is weak-type $(1,1)$ and strong type $(\infty,\infty)$ and apply Marcinkiewicz interpolation. □

**Remark** In fact there is a general principal at work here. If a linear operator $T$ is bounded from $L^\infty$ to $L^\infty$ and is also weak-type $(1,1)$ then it is bounded on $L^p$, $1 < p < \infty$. Of course $M_2$ is *not* linear. Instead it is *sublinear*: $M_2(f+g)(x) \leq M_2 f(x) + M_2 g(x)$. This is sufficient for the result stated.

The results mentioned in the last paragraph are instances of "interpolation" theorems for operators. For a more thorough discussion of this topic, see E. M. Stein and G. Weiss [STW2, Ch. 5] or J. Bergh and J. Löfström [BERL, Ch. 3]. □

**Prelude** The next lemma is the heart of the matter: it is the technical device that allows us to estimate the behavior of a holomorphic function in the interior (in particular, on an admissible approach region) in terms of a maximal function on the boundary. The argument comes from Stein [STE1, Ch. 3] and Barker [BAR].

**Lemma 2.6.9** *Let $u \in C(\overline{\Omega})$ be non-negative and plurisubharmonic on $\Omega$. Define $f = u|_{\partial\Omega}$. Then*

$$u_2^{*,\alpha}(P) \leq C_\alpha M_2(M_1 f)(P)$$

*for all $P \in \partial\Omega$ and any $\alpha > 1$.*

**Proof** After rotating and translating coordinates, we may suppose that $P = 0$ and $\nu_P = (1 + i0, 0, \ldots, 0)$. Let $\alpha' > \alpha$. Then there is a small positive constant $k$ such that if $z = (x_1 + iy_1, z_2, \ldots, z_n) \in \mathcal{A}_\alpha(P)$ then $\mathcal{D}(z) = D(z_1, -kx_1) \times D^{n-1}((z_2, \ldots, z_n), \sqrt{-kx_1}) \subseteq \mathcal{A}_{\alpha'}(P)$ (see Figure 2.7).

We restrict attention to $z \in \Omega$ so close to $P = 0$ that the projection along $\nu_P$ given by

$$z = (x_1 + iy_1, \ldots, x_n + iy_n) \to (\widetilde{x}_1 + i\widetilde{y}_1, x_2 + iy_2, \ldots, x_n + iy_n) \equiv \widetilde{z} \in \partial\Omega$$

makes sense. [Observe that points $z$ that are far from $P = 0$ are trivial to control using our estimates on the Poisson kernel.]

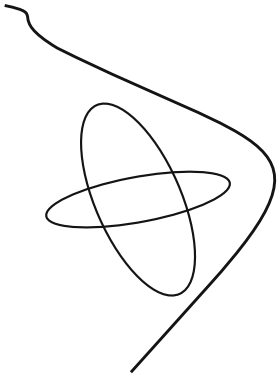


**Figure 2.7** The polydisc that describes the shape of the domain near a boundary point.

The projection of $\mathcal{D}(z)$ along $\nu_P$ into the boundary lies in a ball of the form $\beta_2(\widetilde{z}, Kx_1)$—*this observation is crucial.*

Notice that the subharmonicity of $u$ implies that $u(z) \leq Pf(z)$. Also there is a $\beta > 1$ such that $z \in \mathcal{A}_{\alpha'}(0) \Rightarrow z \in \Gamma_\beta(\widetilde{z})$. Therefore the standard argument leading up to (2.4.1.1) yields that

$$|u(z)| \leq |Pf(z)| \leq C_\alpha M_1 f(\widetilde{z}). \tag{2.6.9.1}$$

Now we bring the complex analysis into play. For we may exploit the plurisubharmonicity of $|u|$ on $\mathcal{D}(z)$ by invoking the sub-averaging property of subharmonic functions in each dimension in succession. Thus

$$\begin{aligned} |u(z)| &\leq \left(\pi|kx_1|^2\right)^{-1} \cdot \left(\pi(\sqrt{-kx_1})^2\right)^{-(n-1)} \int_{\mathcal{D}(z)} |u(\zeta)| dV(\zeta) \\ &= Cx_1^{-n-1} \int_{\mathcal{D}(z)} |u(\zeta)| dV(\zeta). \end{aligned}$$

Notice that, if $z \in \mathcal{A}_\alpha(P)$, then each $\zeta$ in the last integrand is in $\mathcal{A}_{\alpha'}(P)$. Thus the last line is

$$\begin{aligned} &\leq C' x_1^{-n-1} \int_{\mathcal{D}(z)} M_1 f(\widetilde{\zeta}) dV(\zeta) \\ &\leq C'' x_1^{-n-1} \cdot x_1 \int_{\beta_2(\widetilde{z}, Kx_1)} M_1 f(t) d\sigma(t) \\ &\leq C''' x_1^{-n} \int_{\beta_2(0, K'x_1)} M_1 f(t) d\sigma(t) \\ &\leq C'''' \left(\sigma\left(\beta_2(0, K'x_1)\right)\right)^{-1} \int_{\beta_2(0,K'x_1)} M_1 f(t) d\sigma(t) \\ &\leq C'''' M_2(M_1 f)(0). \end{aligned}$$

□

Now we may prove our main result:

**Prelude** The next theorem is the crowning result of our efforts in this chapter. This is the main theorem of the classic work [STE1, Ch. 3]. This result trumps the pioneering Korányi theorem, because Korányi worked only on the unit ball. Stein's theorem is true on *any smoothly bounded domain*. His proof introduced a number of new techniques that have proved to be of lasting value.

But it should be noted that his theorem is sharp (in terms of the shape of the approach region) only on a strongly pseudoconvex domain. The paper that makes this point clear—by way of the Lindelöf principle—is [CIK]. The paper [KRA14] also explores these ideas. And the forthcoming tract [DIK] will tell the full story.

**Theorem 2.6.10** *Let $0 < p \leq \infty$. Let $\alpha > 1$. If $\Omega \subset\subset \mathbb{C}^n$ has $C^2$ boundary and $f \in H^p(\Omega)$, then for $\sigma$-almost every $P \in \partial\Omega$ we have*

$$\lim_{\mathcal{A}_\alpha(P)\ni z\to P} f(z)$$

*exists.*

**Proof** We already know that the limit exists almost everywhere *in the special sense of* Proposition 2.5.1. Call the limit function $\widetilde{f}$. We need only consider the case $p < \infty$. Let $\Omega = \cup_{j=1}^k \Omega_j$ as usual. It suffices to concentrate on $\Omega_1$. Let $\nu = \nu_1$ be the outward normal given by (2.3.5). Then, by Proposition 2.5.1, the Lebesgue dominated convergence theorem implies that, for $\partial\widetilde{\Omega}_1 \equiv \partial\Omega \cap \partial\Omega_1$,

$$\lim_{\epsilon\to 0} \int_{\partial\widetilde{\Omega}_1} |f(\zeta - \epsilon\nu) - \widetilde{f}(\zeta)|^p d\sigma(\zeta) = 0. \tag{2.6.10.1}$$

For each $j, k \in \mathbb{N}$, consider the function $f_{j,k} : \Omega_1 \to \mathbb{C}$ given by

$$f_{j,k}(z) = |f(z - \nu/j) - f(z - \nu/k)|^{p/2}.$$

Then $f_{j,k} \in C(\overline{\Omega}_1)$ and is plurisubharmonic on $\Omega_1$. Therefore a trivial variant of Lemma 2.6.9 yields

$$\begin{aligned}\int_{\partial\widetilde{\Omega}_1} |(f_{j,k})_2^{*,\alpha}(\zeta)|^2 d\sigma(\zeta) &\leq C_\alpha \int_{\partial\widetilde{\Omega}_1} |M_2(M_1 f_{j,k}(\zeta))|^2 d\sigma(\zeta)\\ &\leq C'_\alpha \int_{\partial\widetilde{\Omega}_1} |M_1 f_{j,k}(\zeta)|^2 d\sigma(\zeta)\\ &\leq C''_\alpha \int_{\partial\widetilde{\Omega}_1} |f_{j,k}(\zeta)|^2 d\sigma(\zeta),\end{aligned}$$

where we have used Corollary 2.6.8, Lemma 2.6.9, and the *Proof* of Lemma 2.5.5. Now let $j \to \infty$ and apply (2.6.10.1) to obtain

$$\int_{\partial\widetilde{\Omega}_1} \sup_{z\in\mathcal{A}_\alpha(\zeta)} |f(z) - f(z - \nu/k)|^p d\sigma(\zeta) \leq C''_\alpha \int_{\partial\widetilde{\Omega}_1} |\widetilde{f}(\zeta) - f(\zeta - \nu/k)|^p d\sigma(\zeta). \tag{2.6.10.2}$$

Let $\epsilon > 0$. Then

$$\begin{aligned}&\sigma\{\zeta \in \partial\widetilde{\Omega}_1 : \limsup_{\mathcal{A}_\alpha(\zeta)\ni z\to\zeta} |f(z) - \widetilde{f}(\zeta)| > \epsilon\}\\ &\leq \sigma\{\zeta \in \partial\widetilde{\Omega}_1 : \limsup_{\mathcal{A}_\alpha(\zeta)\ni z\to\zeta} |f(z) - f(z - \nu/k)| > \epsilon/3\}\end{aligned}$$

$$+\sigma\{\zeta \in \partial\widetilde{\Omega}_1 : \limsup_{\mathcal{A}_\alpha(\zeta)\ni z\to\zeta} |f(z-\nu/k) - f(\zeta-\nu/k)| > \epsilon/3\}$$

$$+\sigma\{\zeta \in \partial\widetilde{\Omega}_1 : \limsup_{\mathcal{A}_\alpha(\zeta)\ni z\to\zeta} |f(\zeta-\nu/k) - \widetilde{f}(\zeta)| > \epsilon/3\}$$

$$\leq C\int_{\partial\widetilde{\Omega}_1} \sup_{z\in\mathcal{A}_\alpha(\zeta)} |f(z) - f(z-\nu/k)|^p d\sigma(\zeta)/\epsilon^p$$

$$+0$$

$$+C\int_{\partial\widetilde{\Omega}_1} |\widetilde{f}(\zeta) - f(\zeta-\nu/k)|^p d\sigma(\zeta)/\epsilon^p$$

where we have used (the proof of) Chebycheff's inequality. By (2.6.10.2), the last line does not exceed

$$C'\int_{\partial\widetilde{\Omega}_1} |\widetilde{f}(\zeta) - f(\zeta-\nu/k)|^p d\sigma(\zeta)/\epsilon^p.$$

Now (2.6.10.1) implies that, as $k \to \infty$, this last quantity tends to 0. Since $\epsilon > 0$ was arbitrary, we conclude that

$$\limsup_{\mathcal{A}_\alpha(\zeta)\ni z\to\zeta} |f(z) - \widetilde{f}(\zeta)| = 0$$

almost everywhere. □

The theorem says that $f$ has "admissible limits" at $\sigma$-almost every boundary point of $\Omega$. An inspection of the arguments in the present section suggests that Theorem 2.6.10 is best possible only for strongly pseudoconvex domains. At the boundary point $(1,0)$ of the domain $\{(z_1,z_2) : |z_1|^2 + |z_2|^{2m} < 1\}$, the natural interior polydiscs to study are of the form

$$\{(1-\delta+\xi_1,\xi_2) : |\xi_1| < c\cdot\delta, |\xi_2| < c\cdot\delta^{1/2m}\}.$$

This observation, together with an examination of the proof of 2.6.10, suggests that the aperture in complex tangential directions of the approach regions should vary from boundary point to boundary point—and this aperture should depend on the Levi geometry of the point. A theory of boundary behavior for $H^p$ functions taking these observations into account, for a special class of domains in $\mathbb{C}^2$, is enunciated in A. Nagel, E. M. Stein, and S. Wainger [NASW1]. A more general paradigm for theories of boundary behavior of holomorphic functions is developed in S. G. Krantz [KRA7]. Related ideas also appear in S. G. Krantz [KRA14]. See also the forthcoming book [DIK].

## Exercises

**1.** A $k$-dimensional real manifold $M \subseteq \mathbb{C}^n$ is called *totally real* if, whenever $P \in M$ and $w \in T_P(M)$ then $Jw \notin T_P M$ (here $J$ is the complex structure tensor). In other words, $M$ is totally real if no tangent space $T_P(M)$ has a non-trivial complex subspace. Prove that if $U \subseteq \mathbb{C}^n$ is an open set, $M \subseteq U$ is a totally real submanifold of real dimension $n, f : U \to \mathbb{C}$ is holomorphic, and $f|_M = 0$, then $f \equiv 0$ on $U$. (*Hint:* First consider the special case $M = \{(x_1+i0, \ldots, x_n+i0)\} \subseteq B(0,1) = U$. Treat the general case by considering the power series expansion of $f$.)

**2.** Refer to Exercise 1 for terminology. Prove that there are no totally real manifolds in $\mathbb{C}^n$ of dimension $k > n$.

**3.** Let $U \subseteq \mathbb{C}^n$ be open. Let $K_j : U \times U \to \mathbb{C}, j = 1, 2$, satisfy $K_j(\cdot, \zeta)$ is holomorphic for each $\zeta$ and $K_j(z, \cdot)$ is conjugate holomorphic for each $z$. If $K_1(z, z) = K_2(z, z)$, all $z \in U$, then prove that $K_1 \equiv K_2$.

**4.** Let $B \subseteq \mathbb{C}^2$ be the unit ball and $f : B \to \mathbb{C}$ a bounded holomorphic function. Let $\gamma : [0, 1] \to \partial B$ be a $C^2$ *complex normal* curve, that is, $\gamma'(t) \notin \mathcal{T}_{\gamma((t)}$ for all $t \in [0, 1]$. Then $f$ has radial limits almost everywhere (with respect to one-dimensional measure) along $\gamma$. Complete the following outline to prove this result of A. Nagel and W. Rudin [NARU]:

**(a)** Restrict $\gamma$ to a subarc if necessary and reparametrize so that $\gamma : [0, 1] \to \partial B$ and $i\delta\gamma'(t) + \gamma(t) \in B$, all $0 \leq t \leq 1, 0 < \delta$ small.

**(b)** Define $\phi(x + iy) = f(\gamma(x) + iy\gamma'(x)), 0 \leq x \leq 1, 0 < y < \delta$. Then $\alpha = \partial\phi/\partial\overline{z}$ is bounded.

**(c)** There is a $u$ on $U = \{x + iy : 0 < x < 1, 0 < y < \delta\}$ such that $\overline{\partial} u = \alpha, u \in \text{Lip}_{\epsilon}$, some $\epsilon > 0$.

**(d)** The function $\phi - u$ is in $H^\infty(U)$. Therefore $\phi - u$ has boundary values almost everywhere on $\partial U \cap \{y = 0\}$. Thus $\phi$ has the same property.

**5.** Complete the following outline to construct an example (N. Sibony [SIB1]) of a pseudoconvex domain $\Omega \subsetneq D^2(0, 1)$ with the property that

**(i)** $\Omega$ equals the interior of $\bar{\Omega}$,

**(ii)** $\bar{\Omega} \neq \bar{D}^2(0, 1)$,

**(iii)** every bounded holomorphic function on $\Omega$ analytically continues to $D^2(0, 1)$.

**Outline of Construction**

**(a)** Let $\{a_j\} \subseteq D$ have no accumulation points in $D$ but have every point of $\partial D$ as an accumulation point. Define $u(z) = \sum_j \frac{1}{j^2} \log |(z - a_j)/2|$.

**(b)** Let $v(z) = \exp u(z)$. Then $v$ is continuous and subharmonic on $D$ and $0 \leq v < 1$ on $D$.

**(c)** Let $\Omega = \{(z,w) \in \mathbb{C}^2 : |z| < 1, |w| < \exp(-v(z))\}$. Then $\Omega$ satisfies properties (i) and (ii) above and $\Omega$ is pseudoconvex. [*Note:* The function $(z,w) \to |w| \exp v(z)$ is plurisubharmonic.]

**(d)** Let $f$ be bounded and holomorphic on $\Omega$; say that $\|f\|_{L^\infty} \leq 1$. Then $f$ has a power series expansion

$$f(z,w) = \sum_{j \geq 0} f_j(z) w^j,$$

where

$$f_j(z) = (j!)^{-1} \frac{\partial^j f}{\partial w^j}(z,0).$$

Each $f_j$ is holomorphic on $D$.

**(e)** Write, for $0 < r < 1$,

$$f_j(z) = \frac{1}{2\pi i} \int \frac{f\left(z, re^{i\theta} \exp(-v(z))\right)}{r^j e^{ij\theta} \exp(-jv(z))} i d\theta.$$

Conclude that $|f_j(z)| \leq |\exp jv(z)|$, in particular, $|f_j(a_\ell)| \leq 1$, all $j$ and $\ell$.

**(f)** Each $f_j \in H^\infty(D)$. If $\widetilde{f_j}$ is the boundary limit function of $f_j$ then $|\widetilde{f_j}(e^{i\theta})| \leq 1$ almost everywhere hence $|f_j| \leq 1$.

**(g)** The series representation in (d) converges normally for points $(z,w) \in D^2(0,1)$.

**6.** Show that the standard rigid imbedding of the 2-torus into $\mathbb{C}^2$ (where we think of the torus as the product of two circles, and each circle is the unit circle in $\mathbb{C}$) renders (the imbedded) $\mathbb{T}^2$ as a totally real submanifold. That is, the tangent space at any $P$ in the boundary of the imbedded torus never contains a complex line in $\mathbb{C}^2$.

**7.** Let $M \subseteq \mathbb{C}^2$ be a smooth, real, $k$-dimensional regularly imbedded submanifold, $1 \leq k \leq 4$. If $k \geq 3$, then $M$ cannot be totally real. If $k = 2$ then, near a point $P \in M$, we may choose coordinates such that $M$ is locally given by $\{(z, r(z)) : z \in U\}$, some open $U \subseteq \mathbb{C}$. Show then that $M$ is totally real at $(z, r(z))$ if and only if $(\partial r/\partial \bar{z})(z) \neq 0$.

An abstract version of these ideas is as follows. Let $M \subseteq \mathbb{C}^2$ be a smooth real hypersurface. Let $\mathbb{P}_1$ be the projective space of one-dimensional complex lines in $\mathbb{C}^2$. Let $\psi : M \to \mathbb{C}^2 \times \mathbb{P}_1$ be given by $\psi(z) = (z, \mathcal{T}_z(M))$. Prove that $\psi(M)$ is totally real in $\mathbb{C}^2 \times \mathbb{P}_1$ at the point $\psi(z)$ if and only if $z$ is a point of strong Levi pseudoconvexity in $M$. The same result is true in $\mathbb{C}^n$ but is a little more difficult to prove (see S. Webster [WEB2]).

**8.** This problem provides a glimpse of some of the ideas of Lelong.

**(a)** Examine the proof of the Poisson integral formula to see that it also yields the *Riesz decomposition* for a subharmonic function: if $\Omega \subset\subset \mathbb{R}^N$ has $C^2$

boundary, $u \in C^2(\bar{\Omega})$, and $u$ is subharmonic on $\Omega$, then

$$u(x) = \int_{\partial\Omega} P_\Omega(x,y)u(y)d\sigma(y) - \int_\Omega (\Delta u(y))G_\Omega(x,y)dV(y).$$

Use a limiting argument to extend this formula to all subharmonic functions which are in $C(\bar{\Omega})$.

**(b)** Let $f \in C^2(\bar{D}), D$ the disc in $\mathbb{C}$. Let $f|_D$ be holomorphic. Assume that the zeros of $f$ are $\{\alpha_1, \ldots, \alpha_k\} \subseteq D$. Let $\delta_k$ be the unit Dirac mass at $\alpha_k$. Then $\Delta \log|f| = C \cdot \sum_k \delta_k$ in the sense that

$$\int_D \phi(z)\Delta \log|f(z)|dV(z) = C \cdot \sum \phi(\alpha_k), \qquad \text{all} \quad \phi \in C_c^\infty(D).$$

Use Green's formula to prove this result.

**(c)** Use a limiting argument to extend (b) to elements of the Nevanlinna class.

**(d)** Lelong has proved that if $f$ is holomorphic on $\Omega \subset\subset \mathbb{C}^n$ then $\omega_0 = \Delta \log|f|$ is a non-negative measure supported on $\mathcal{N} \equiv \{z \in \Omega : f(z) = 0\}$. If $E \subseteq \Omega$ is Borel, then

$$\mathcal{H}^{2n-2}(E \cap \mathcal{N}) = C \cdot \int_\Omega E\, d\omega_0.$$

Taking **(a)** through **(d)** for granted, we now prove that if $\Omega \subset\subset \mathbb{C}^n$ has $C^2$ boundary and $f$ is in the Nevanlinna class on $\Omega$, then the zero set of $f$ satisfies an analogue of the Blaschke condition on $\Omega$.

**(e)** Let $\rho$ be a defining function for $\Omega$ and, for $0 < \epsilon < \epsilon_0$, let $\Omega_\epsilon, \rho_\epsilon, d\sigma_\epsilon$ be as in the text. Let $P_\epsilon$ be the Poisson kernel for $\Omega_\epsilon$ and $G_\epsilon$ the Green's function for $\Omega_\epsilon$. Fix an $f$ in the Nevanlinna class on $\Omega$. Let $\Delta \log|f| = \omega_0$. Let $\Gamma$ be the fundamental solution for the Laplacian and define

$$V_0(x) = \int \Gamma(x-y)d\omega_0(y).$$

Then $V_0$ is subharmonic.

**(f)** Let $H(z) = \log|f(z)| - V_0(z)$. Then $H$ is harmonic on $\Omega$. It follows that, for $0 < \epsilon < \epsilon_0, z_0 \in \Omega \setminus \{f(z) = 0\}$, we have

$$H(z_0) = \int_{\partial\Omega_\epsilon} P_\epsilon(z_0,\zeta)H(\zeta)d\sigma_\epsilon(\zeta),$$

$$\int_{\Omega_\epsilon} G_\epsilon(z_0,\zeta)d\omega_0(\zeta) + V_0(z_0) = \int_{\partial\Omega_\epsilon} P_\epsilon(z_0,\zeta)V_0(\zeta)d\sigma_\epsilon(\zeta).$$

**(g)** Conclude from (f) that

$$\int_{\Omega_\epsilon} G_\epsilon(z_0,\zeta)d\omega_0(\zeta) + \log|f(z_0)| = \int_{\partial\Omega_\epsilon} P_\epsilon(z_0,\zeta)\log|f(\zeta)|d\sigma_\epsilon(\zeta).$$

**(h)** Let $\epsilon \to 0$ to obtain

$$\int_{\Omega} G(z_0,\zeta)d\omega_0(\zeta) + \log|f(z_0)| \leq C_{z_0}\|f\|_N.$$

**(i)** Conclude from (h) that

$$\int_{\Omega} \delta_\Omega(\zeta)d\omega_0(\zeta) < \infty.$$

This is the Blaschke condition.

**(j)** G. M. Henkin [HEN2] and H. Skoda [SKO] have proved that if $\Omega$ is strongly pseudoconvex with smooth boundary (and satisfies a mild topological condition), if $\mathcal{N} \subseteq \Omega$ is the zero set of *some* holomorphic function, and if $\mathcal{N}$ satisfies the Blaschke condition in (i), then $\mathcal{N}$ is the zero set of a function in the Nevanlinna class.

**(k)** Notice that some sort of topological condition on $\mathcal{N}$ is necessary. For any compact set in $\Omega$ satisfies the Blaschke condition trivially, but it could never be the zero set of a holomorphic function.

**9.** Let $\Omega \subset\subset \mathbb{C}^n$ be a domain with $C^2$ boundary. Let $f \in H^p(\Omega), 0 < p < \infty$. Suppose that $f$ has radial boundary limit 0 on a set $E \subseteq \partial\Omega$ of positive $\sigma$ measure. Then $f \equiv 0$. (*Hint:* Apply the Riesz decomposition to the subharmonic function $\log^+|f|$.) Does the result prevail if we merely assume that $f \in \mathbf{h}^p$? (See S. G. Krantz [KRA9] for more on this matter.)

**10.** Let $\Omega_1, \Omega_2$ be bounded domains in $\mathbb{C}^n$ with $C^2$ boundary. Let $\phi : \Omega_1 \to \Omega_2$ be a biholomorphic map which extends $C^2$ to $\bar{\Omega}_1$ and such that $\phi^{-1}$ extends $C^2$ to $\bar{\Omega}_2$. Prove that there is a $C > 0$ such that if $P_1 \in \partial\Omega_1$ and $\mathcal{A}_\alpha(P_1)$ is an admissible region in $\Omega_1$ then

$$\mathcal{A}_{\alpha/C}(\phi(P_1)) \subseteq \phi(\mathcal{A}_\alpha(P_1)) \subseteq \mathcal{A}_{C\alpha}(\phi(P_1)).$$

Which of the hypotheses on $\phi$ are really needed here?

**11.** Consider the mapping $\phi : S^3 \to \mathbb{C}^3$ given by $(z,w) \mapsto (z,w,f(z,w))$. Here $S^3$ is the 3-sphere. Prove that the image of this mapping is a totally real manifold if and only if

$$w\frac{\partial f}{\partial \bar{z}} - z\frac{\partial f}{\partial \bar{w}} \neq 0$$

for all $(z,w) \in S^3$.

**12.** Let $\Omega \subseteq \mathbb{C}^n$ be a domain. Let $J$ be the real linear operator on $\mathbb{R}^{2n}$ that gives the complex structure. Let $P \in \partial\Omega$. Let $z = (z_1, \ldots, z_n) = (x_1 + iy_1, \ldots, x_n + iy_n) = (x_1, y_1, \ldots, x_n, y_n)$ be an element of $\mathbb{C}^n \cong \mathbb{R}^{2n}$. The following are equivalent:

(i) $w \in \mathcal{T}_P(\Omega)$;
(ii) $Jw \in \mathcal{T}_P(\Omega)$;
(iii) $Jw \perp \nu_P$ and $w \perp \nu_P$.

**13.** With notation as in the previous exercise, let $A = \sum_j a_j(z)\partial/\partial z_j, B = \sum_j b_j(z)\partial/\partial z_j$ satisfy $A\rho|_{\partial\Omega} = 0$, $B\rho|_{\partial\Omega} = 0$, where $\rho$ is any defining function for $\Omega$. Then the vector field $[A, B]$ has the same property. (However note that $[A, \overline{B}]$ does not annihilate $\rho$ on $\partial\Omega$ if $\Omega$ is the ball, for instance.) Therefore the holomorphic part of $\mathcal{T}_P$ is integrable (see G. B. Folland and J. J. Kohn [FOK, Ch. 1, 4]).

**14.** If $\Omega = B \subseteq \mathbb{C}^2, P = (x_1 + iy_1, x_2 + iy_2) \approx (x_1, y_1, x_2, y_2) \in \partial B$, then $\nu_P = (x_1, y_1, x_2, y_2)$ and $J\nu_P = (-y_1, x_1, -y_2, x_2)$. Also $\mathcal{T}_P$ is spanned over $\mathbb{R}$ by $(y_2, x_2, -y_1, -x_1)$ and $(-x_2, y_2, x_1, -y_1)$.

# Chapter 3
# The Heisenberg Group

**Prologue:** In many ways the heart of modern harmonic analysis is the theory of singular integrals. Most any question in the classical theory of Fourier series can be rendered as a problem about the Hilbert transform (which is the only singular integral in dimension one) or the maximal Hilbert transform. In higher dimensions there are many singular integrals. See [KRA2, Ch. 5, 6] for background on these ideas.

It is natural to wonder what are the right singular integrals for the study of the function theory of several complex variables, and how such integrals might be characterized. A full answer to this question is not known at this time. But, at least in some contexts, there are fairly complete answers.

In fact the best understood venue for Fourier analysis and singular integrals in several complex variables is the boundary of the unit ball $B$ in $\mathbb{C}^n$. And the reason for this is fairly straightforward. Just as the unit disc in the plane may be identified (via the Cayley transform) with the upper half plane, and the boundary of the disc identified with the line, so the ball in $\mathbb{C}^n$ may be identified (by a generalized Cayley transform) with a "Siegel upper half space" and the boundary of the ball identified with its parabolic boundary. That boundary, in turn, may be mapped in a natural fashion to the Heisenberg group. Thus the boundary may be equipped with both translations (in the Heisenberg group structure) and (nonisotropic) dilations. So some Fourier analysis, and certainly a theory of singular integrals, is feasible. We shall provide some of the details of these ideas in this and the next chapter.

Analysis on the Heisenberg group is fascinating because the group is topologically Euclidean but analytically non-Euclidean. Many of the most basic ideas of analysis must be developed again from scratch. Even the venerable triangle inequality, the concept of dilation, and the method of polar coordinates, must be re-thought. One of the main points of our work will be to define, and then to prove estimates for, singular integrals on the Heisenberg group.

One of the really big ideas here is that the index of the critical singularity—the singularity for a singular integral kernel—will not be the same as the topological dimension of the space (recall that, on $\mathbb{R}^N$, the critical index is $N$). Thus we must develop the concept of "homogeneous dimension." It is also the case that the Fourier transform—while it certainly exists on the Heisenberg group—is not nearly as useful a tool as it was in classical Euclidean analysis. The papers [GEL1, GEL2, GEL3] and [GELS] provide some basis for analysis using the Fourier transform on the Heisenberg group. While this theory is rich and promising, it has not borne the sort of fruit that classical Euclidean Fourier theory has.

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## 3.1 Prolegomena

While the Heisenberg group may have figured implicitly in earlier studies of representation-theoretic questions, it was the paper [FOS1] that put the group on the map for harmonic analysts. The motivation for that paper—rather technical estimates for the $\overline{\partial}_b$ equation—was rather recondite, but the paper also served to set up all the analytic machinery for further studies of analysis on the Heisenberg group. Folland and Stein were particularly interested in the contact geometry of the Heisenberg group, and how it models the contact geometry of a strongly pseudoconvex point. But the part of their work that has had a lasting influence was the setting of the foundations of analysis on the group.

In particular, singular integrals, nonisotropic dilations, triangle inequalities, measures, integral operators, homogeneous dimension, commutators, and other artifacts of hard analysis were treated precisely and in detail in [FOS1]. The theory of singular integrals in this new setting had been developed in the earlier papers [KNS, COG1], and [KOV]. But Folland and Stein [FOS1] gave this work a new context and meaning. Many new techniques were introduced in this paper, including a particularly incisive analysis of fractional integral operators. Later works, including [ROST, FOS2, Ch. 1–3], and [KRA24], built on the foundations laid in [FOS1].

As noted elsewhere in this book, harmonic analysis has grown over the years from studies of the circle group and the real line to studies on $\mathbb{R}^N$, spaces of homogeneous type, and various Lie groups. But the Heisenberg group has proven to be a particularly fertile ground for new developments in our subject. And we expect much additional progress in the future.

## 3.2 The Upper Half Plane in $\mathbb{C}$

**Capsule:** It almost seems like a step backward to now revert to the classical upper half plane. But we shall understand the Heisenberg group as a Lie group that acts on the boundary of the Siegel upper half space. And that is a direct generalization of the classical upper half plane. So our first task is to understand that half plane in a new light, and with somewhat new language. We shall in particular analyze the group of holomorphic self-maps of the upper half plane, and perform the Iwasawa decomposition for that group (thought of as a Lie group). This will yield a new way to think about translations, dilations, and Möbius transformations.

As usual, we let $U = \{\zeta \in \mathbb{C} : \operatorname{Im}\zeta > 0\}$ be the upper half plane. Of course the unit disc $D$ is conformally equivalent to $U$ by way of the Cayley map

$$\begin{aligned} c : D &\longrightarrow U \\ \zeta &\longmapsto i \cdot \frac{1-\zeta}{1+\zeta}. \end{aligned}$$

If $\Omega$ is any planar domain then we let $\mathrm{Aut}\,(\Omega)$ denote the group of conformal self-maps of $\Omega$, with composition as the binary operation. We call this the *automorphism group* of $\Omega$. We equip the automorphism group with a topology by using uniform convergence on compact sets (equivalently, the compact-open topology). We sometimes refer to convergence in the automorphism group as "normal convergence." Then the automorphism groups of $D$ and $U$ are canonically isomorphic (both algebraically and topologically) by way of

$$\mathrm{Aut}\,(D) \ni \varphi \longmapsto c \circ \varphi \circ c^{-1} \in \mathrm{Aut}\,(U)\,.$$

We shall use the Cayley map and the upper half plane $U$ to help us to understand the automorphism group of the disc. We shall denote automorphisms of the upper half plane with a tilde ($\widetilde{\ }$) and those of the disc without a tilde. This usage will be clear from context.

We now wish to understand $\mathrm{Aut}\,(D)$ by way of the so-called *Iwasawa decomposition* of the group. This is a decomposition of the form

$$\mathrm{Aut}\,(D) = K \cdot A \cdot N\,,$$

where $K$ is compact, $A$ is abelian, and $N$ is nilpotent. We shall use the Iwasawa decomposition as a guide to our thoughts, but we shall not prove it here (see [HEL, Ch. 6] for the chapter and verse on this topic). We also shall not worry for the moment what "nilpotent" means.[1] The concept will be explained in the next section. In the present context, the "nilpotent" piece is actually abelian, and that is a considerably stronger condition.

Now let $K$ be the group of those automorphisms of $D$ that fix the origin. By Schwarz's lemma, the elements of $K$ are simply the rotations of the disc. Hence $K$ is certainly a compact group, for it may be canonically identified (both algebraically and topologically) with the unit circle. Alternatively, it is easy to see (using Montel's theorem) that any sequence of rotations contains a subsequence that converges to another rotation.

To understand the abelian part $A$ of the group, it is best to work with the unbounded realization $U$ (that is, the upper half plane). Then consider the group of dilations

$$\widetilde{\alpha}_\delta(\xi) = \delta\xi$$

for $\delta > 0$. This is clearly a subgroup of the full automorphism group of $U$, and it is certainly abelian. Let us examine the group that it corresponds to in the automorphism group action on $D$. Obviously we wish to consider

$$\alpha_\delta(\zeta) = c^{-1} \circ \widetilde{\alpha}_\delta \circ c \in \mathrm{Aut}\,(D)\,. \tag{3.2.1}$$

[1] Suffice it to say that a group is nilpotent of step $m$ if all $m$th order commutators in the group equal the identity—and if $m$ is the least such integer. In particular, an abelian group is nilpotent of step 1.

We know that $c^{-1}(\xi) = [i - \xi]/[i + \xi]$. So we may calculate the quantity in (3.2.1) to find that

$$\alpha_\delta(\zeta) = \frac{(1-\delta) + \zeta(1+\delta)}{(1+\delta) + \zeta(1-\delta)} = \frac{\zeta + (1-\delta)/(1+\delta)}{1 + \zeta(1-\delta)/(1+\delta)} \tag{3.2.2}$$

This is the "dilation group action" on the disc. The far righthand side of (3.2.2) shows that the mapping is a Möbius transformation of the disc. Clearly the dilations are much easier to understand, and the abelian nature of the subgroup more transparent, when we examine the action on the upper half plane $U$.

Next we look at the nilpotent piece, which in the present instance is in fact abelian. We again find it most convenient to examine the group action on the upper half plane $U$. This subgroup is the translations:

$$\widetilde{\tau}_a(\xi) = \xi + a\,,$$

where $a \in \mathbb{R}$. Then the corresponding automorphism on the disc is

$$\tau_a(\zeta) = c^{-1} \circ \widetilde{\tau}_a \circ c \in \text{ Aut}\,(D)\,.$$

With some tedious calculation we find that

$$\tau_a(\zeta) = \frac{-a + \zeta(2i-a)}{(2i+a) + a\zeta} = \frac{2i-a}{2i+a} \cdot \frac{\zeta + a/(a-2i)}{1 + [a/(a+2i)]\zeta}\,.$$

This is plainly a Möbius transformation followed by a rotation. Again, the "translation" nature of the automorphism group elements is much clearer in the group action on the unbounded realization $U$, and the abelian property of the group is also much clearer.

Notice that the group of translations acts simply transitively on the boundary of $U$. For if $\widetilde{\tau}_a$ is a translation then $\widetilde{\tau}_a(0) = a + i0 \equiv z \in \partial U$. Conversely, if $z = a + i0 \in \partial U$ then $\widetilde{\tau}_a(0) = a + i0 = z$. So we may identify the translation group with the boundary, and vice versa. This simple observation is the key to Fourier analysis in this setting.

We will use these elementary calculations of the Iwasawa decomposition in one complex dimension as an inspiration for our more sophisticated calculations on the ball in $\mathbb{C}^n$ that we carry out in the next two sections.

## 3.3 The Significance of the Heisenberg Group

**Capsule:** In this section and the next we begin to familiarize ourselves with the unit ball in $\mathbb{C}^n$. We examine the automorphism group of the ball, and we detail its Iwasawa decomposition. The subgroup of the automorphism group that plays the role of "translations" in

the classical setting turns out to be the Heisenberg group. There are certainly other ways to discover the Heisenberg group, but this one turns out to be most natural for us. It is important to understand that the Heisenberg group acts simply transitively on the boundary; this makes possible the identification of the group with the boundary. One may alternatively do harmonic analysis on the boundary of the ball by exploiting the unitary group action. We shall not explore that approach here, but see [FOD].

Complex analysis and Fourier analysis on the unit disc $D = \{\zeta \in \mathbb{C} : |\zeta| < 1\}$ work well together because there is a group—namely the group of *rotations*—that acts naturally on $\partial D$. Complex analysis and Fourier analysis on the upper half plane $U = \{\zeta \in \mathbb{C} : \operatorname{Im}\zeta > 0\}$ are symbiotic because there is a group—namely the group of *translations*—that acts naturally on $\partial U$.[2] We also might note that the group of dilations $\zeta \to \delta\zeta$ acts naturally on $U$ for $\delta > 0$. One of the main points here is that the disc $D$ and the upper half plane $U$ are conformally equivalent. The *Cayley transform*

$$c : D \to U$$

is given explicitly by

$$c(\zeta) = i \cdot \frac{1-\zeta}{1+\zeta}\,.$$

Notice that $c$ is both one-to-one and onto. Its inverse is given by

$$c^{-1}(\mu) = \frac{i-\mu}{i+\mu}\,.$$

We would like a similar situation to obtain for the domain the unit ball $B = \{(z_1, \ldots, z_n) \in \mathbb{C}^n : |z_1|^2 + \cdots |z_n|^2 < 1\}$. It turns out that, in this situation, the unbounded realization[3] of the domain $B$ is given by

$$\mathcal{U} = \{(w_1, \ldots, w_n) \in \mathbb{C}^n : \operatorname{Im} w_1 > \sum_{j=2}^{n} |w_j|^2\}\,.$$

It is convenient to write $w' \equiv (w_2, \ldots, w_n)$. We refer to our domain $\mathcal{U}$ as the *Siegel upper half space*, and we write its defining equation as $\operatorname{Im} w_1 > |w'|^2$.

[2] It must be noted, however, that the rotations on the disc and the translations on the upper half plane do not "correspond" in any natural way; certainly the Cayley transform does not map the one group to the other. This anomaly is explored in the fine text [HOF, Ch. 3, 8].

[3] We use here the classical terminology of Siegel upper half spaces. Such an upper half space is defined with an inequality using a quadratic form. The resulting space is unbounded. However, when the quadratic form is positive definite then the domain has a bounded realization—that is to say, it is biholomorphically equivalent to a bounded domain. See [KAN, Ch. 1, 7] for details of this theory.

Now the mapping that shows $B$ and $\mathcal{U}$ to be biholomorphically equivalent is given by

$$\begin{aligned} \Phi : B &\longrightarrow \mathcal{U} \\ (z_1, \ldots, z_n) &\longmapsto \left( i \cdot \frac{1 - z_1}{1 + z_1}, \frac{z_2}{1 + z_1}, \ldots, \frac{z_n}{1 + z_1} \right) . \end{aligned}$$

We leave it to the reader to perform the calculations to verify that $\Phi$ maps $B$ to $\mathcal{U}$ in a holomorphic, one-to-one, and onto fashion. The inverse of the mapping $\Phi$ may also be calculated explicitly.

Just as in one dimension, if $\Omega \subseteq \mathbb{C}^n$ is any domain, we let $\mathrm{Aut}\,(\Omega)$ denote the collection of biholomorphic self-mappings of $\Omega$. This set forms a group when equipped with the binary operation of composition of mappings. In fact it is a topological group with the topology of uniform convergence on compact sets (which is the same as the compact-open topology). Further, it can be shown that, at least when $\Omega$ is a bounded domain, $\mathrm{Aut}\,(\Omega)$ is a real Lie group (never a complex Lie group, except for the trivial case when the automorphism group has dimension zero—see [KOB, Ch. 4–5]). We shall not make much use of this last fact, but it a helpful touchstone in our discussions. There is a natural isomorphism between $\mathrm{Aut}\,(B)$ and $\mathrm{Aut}\,(\mathcal{U})$ given by

$$\mathrm{Aut}\,(B) \ni \varphi \longmapsto \Phi \circ \varphi \circ \Phi^{-1} \in \mathrm{Aut}\,(\mathcal{U}) . \tag{3.3.1}$$

It turns out that we can understand the automorphism group of $B$ more completely by passing to the automorphism group of $\mathcal{U}$. We used this technique in the last section to understand the automorphism group of the disc. We shall again indulge in that conceit right now. We shall use, as we did in the more elementary setting of the disc, the idea of the Iwasawa decomposition $G = KAN$.

The compact part $K$ of $\mathrm{Aut}\,(B)$ is the collection of all automorphisms of $B$ that fix the origin. It is easy to prove, using a multivariable version of the Schwarz lemma (see [RUW3, Ch. 8]), that each of these automorphisms is a unitary rotation. This means that the mapping is given by an $n \times n$ matrix whose rows form a Hermitian orthonormal basis for $\mathbb{C}^n$ and which has determinant 1. Of course this group is compact because it is a closed, bounded subset of Euclidean space.

Implicit in our discussion here is a fundamental idea of H. Cartan: If $\Omega$ is any bounded domain and $\{\varphi_j\}$ a sequence of automorphisms of $\Omega$, and if the $\varphi_j$ converge uniformly on compact subsets of $\Omega$, then the limit mapping $\varphi_0$ is either itself an automorphism, or else it maps the entire domain $\Omega$ to a point in the boundary. The proof of this result (which we omit, but see [NARU]) is a clever combination of Hurwitz's principle and the open mapping theorem. In any event, since the mappings in the last paragraph all fix the origin, then it is clear that the limit mapping cannot map the entire domain into the boundary. Hence, by Cartan, the limit mapping must itself be an automorphism.

One may utilize the isomorphism (3.3.1)—this is an explicit and elementary calculation—to see that the subgroup of Aut $(\mathcal{U})$ that corresponds to $K$ is $\widetilde{K}$ which is the subgroup of automorphisms of $\mathcal{U}$ that fixes the point $(i, 0, \ldots, 0)$. Although $\widetilde{K}$ is *a priori* a compact Lie group, one may also verify this property by a direct argument as in the last two paragraphs.

Thus we have disposed of the compact piece of the automorphism group of the unit ball. Now let us look at the abelian piece $A$. For this part, it is most convenient to begin our analysis on $\mathcal{U}$. Let us consider the group of dilations, which consists of the nonisotropic mappings

$$\widetilde{\alpha}_\delta : \mathcal{U} \longrightarrow \mathcal{U}$$

given by

$$\widetilde{\alpha}_\delta(w_1, \ldots, w_n) = (\delta^2 w_1, \delta w_2, \delta w_3, \ldots, \delta w_n)$$

for any $\delta > 0$. Check for yourself that $\widetilde{\alpha}_\delta$ maps $\mathcal{U}$ to $\mathcal{U}$. We call these mappings nonisotropic (meaning "acts differently in different directions") because they treat the $w_1$ variable differently from the $w_2, \ldots, w_n$ variables. The group is clearly abelian. It corresponds, under the mapping $\Phi$, to the group of mappings on $B$ given by

$$\alpha_\delta(z_1, \ldots, z_n) = \Phi^{-1} \circ \widetilde{\alpha}_\delta \circ \Phi(z)\,. \tag{3.3.2}$$

Now it is immediate to calculate that

$$\Phi^{-1}(w) = \left( \frac{i - w_1}{i + w_1}, \frac{2iw_2}{i + w_1}, \ldots, \frac{2iw_n}{i + w_n} \right).$$

Of course it is just a tedious algebra exercise to determine $\alpha_\delta$. The answer is

$$\alpha_\delta(z) = \left( \frac{(1 - \delta^2) + z_1(1 + \delta^2)}{(1 + \delta^2) + z_1(1 - \delta^2)}\,,\ \frac{2\delta z_2}{(1 + \delta^2) + z_1(1 - \delta^2)}\,,\right.$$
$$\left. \cdots\,,\ \frac{2\delta z_n}{(1 + \delta^2) + z_1(1 - \delta^2)} \right).$$

One may verify directly that $z \in B$ if and only if $\alpha_\delta(z) \in B$.

It is plain that the dilations are much easier to understand on the unbounded realization $\mathcal{U}$. And the group structure, in particular the abelian nature of the group, is also much more transparent in that context.

We shall find that the "nilpotent" piece of the automorphism group is also much easier to apprehend in the context of the unbounded realization. We shall explore that subgroup in the next section.

## 3.4 The Heisenberg Group Action on $\mathcal{U}$

**Capsule:** Here we study the action of the Heisenberg group on the Siegel upper half space and on its boundary. We find that the upper half space decomposes into level sets that are "parallel" to the boundary (similar to the horizontal level lines in the classical upper half plane), and that the Heisenberg group acts on each of these. We set up the basics for the convolution structure on the Heisenberg group.

If $G$ is a group and $g, h \in G$, then we define a *second order commutator* of $g$ and $h$ to be the expression $\lambda(g,h) \equiv ghg^{-1}h^{-1}$. We call this "second order" because it involves two group elements. [Clearly, if the group $G$ is abelian, then this expression will always equal the identity; otherwise not.] If $g, h, k \in G$ then a *third order commutator* is an expression of the form $\lambda(\lambda(g,h),k)$. Of course higher order commutators are defined inductively.[4]

Let $m$ be a nonnegative integer. We say that the group $G$ is *nilpotent of order m* (or step $m$) if all commutators of order $(m+1)$ in $G$ are equal to the identity, and if $m$ is the least such integer. Clearly an abelian group is nilpotent of order 1. It turns out that the collection of "translations" on $\partial\mathcal{U}$ is a nilpotent group of order 2. In fact that group can be *identified* in a natural way with $\partial\mathcal{U}$ (in much the same way that the ordinary left-right translations of the boundary of the classical upper half plane $U$ can be identified with $\partial U$). We now present the details of this idea.

The *Heisenberg group* of order $n-1$, denoted $\mathbb{H}^{n-1}$, is an algebraic structure that we impose on $\mathbb{C}^{n-1} \times \mathbb{R}$. Let $(\zeta, t)$ and $(\xi, s)$ be elements of $\mathbb{C}^{n-1} \times \mathbb{R}$. Then the binary Heisenberg group operation is given by

$$(\zeta, t) \cdot (\xi, s) = (\zeta + \xi, t + s + 2\,\mathrm{Im}\,(\zeta \cdot \overline{\xi}))\,.$$

It is clear, because of the Hermitian inner product $\zeta \cdot \overline{\xi} \equiv \zeta_1\overline{\xi}_1 + \cdots + \zeta_{n-1}\overline{\xi}_{n-1}$, that this group operation is non-abelian (although in a fairly subtle fashion). Later on we shall have a convenient means to verify the nilpotence, so we defer that question for now.

Now an element of $\partial\mathcal{U}$ has the form $(\mathrm{Re}\, w_1 + i|(w_2, \ldots, w_n)|^2, w_2, \ldots, w_n) = (\mathrm{Re}\, w_1 + i|w'|^2, w')$, where $w' = (w_2, \ldots, w_n)$. We identify this boundary element with the Heisenberg group element $(w', \ \mathrm{Re}\, w_1)$, and we call the corresponding mapping $\Psi$. In other words,

$$\Psi : \partial\mathcal{U} \to \mathbb{H}^{n-1}$$

$$(\mathrm{Re}\, w_1 + i|w'|^2, w') \mapsto (w', \ \mathrm{Re}\, w_1)\,.$$

[4] In some sense it is more natural to consider commutators in the Lie algebra of the group. By way of the exponential map and the Campbell–Baker–Hausdorff formula (see [SER, Ch. 4]), the two different points of view are equivalent. We shall describe some of the Lie algebra approach in the material below. For now, the definition of commutators in the context of the group is a quick-and-dirty way to get at the idea we need to develop right now.

Now we can specify how the Heisenberg group acts on $\partial\mathcal{U}$. If $w = (w_1, w') \in \partial\mathcal{U}$ and $g = (z', t) \in \mathbb{H}^{n-1}$, then we have the action

$$g[w] = \Psi^{-1}[g \cdot \Psi(w)] = \Psi^{-1}[g \cdot (w',\ \mathrm{Re}\, w_1)] = \Psi^{-1}[(z', t) \cdot (w', \mathrm{Re}\, w_1)]\,.$$

More generally, if $w \in \mathcal{U}$ is *any element* then we write

$$\begin{aligned} w &= (w_1, w_2, \ldots, w_n) = (w_1, w') \\ &= \Big( (\,\mathrm{Re}\, w_1 + i|w'|^2) + i(\mathrm{Im}\, w_1 - |w'|^2), w_2, \ldots, w_n \Big) \\ &= \Big( \mathrm{Re}\, w_1 + i|w'|^2, w' \Big) + \Big( i(\,\mathrm{Im}\, w_1 - |w'|^2), 0, \ldots, 0 \Big)\,. \end{aligned}$$

The first expression in parentheses is an element of $\partial\mathcal{U}$. It is convenient to let $\rho(w) = \mathrm{Im}\, w_1 - |w'|^2$. We think of $\rho$ as a "height function." In short, we are expressing an arbitrary element $w \in \mathcal{U}$ as an element in the boundary plus a translation "up" to a certain height in the $i$ direction of the first variable.

Now we let $g$ act on $w$ by

$$\begin{aligned} g[w] &= g\left[ \Big( \mathrm{Re}\, w_1 + i|w'|^2, w' \Big) + \Big( i\rho(w), 0, \ldots, 0 \Big) \right] \\ &\equiv g\left[ \Big( \mathrm{Re}\, w_1 + i|w'|^2, w' \Big) \right] + \Big( i\rho(w), 0, \ldots, 0 \Big)\,. \end{aligned} \tag{3.4.1}$$

In other words, we let $g$ act on level sets of the height function.

It is our job now to calculate this last line and to see that it is a holomorphic action on $\mathcal{U}$. We have, for $g = (z', t)$,

$$\begin{aligned} & g[w] \\ &= g\left[ \Big( \mathrm{Re}\, w_1 + i|w'|^2, w' \Big) \right] + \Big( i\rho(w), 0, \ldots, 0 \Big) \\ &= \Psi^{-1}\left[ g \cdot (w',\ \mathrm{Re}\, w_1) \right] + \Big( i\rho(w), 0, \ldots, 0 \Big) \\ &= \Psi^{-1}\left[ \big( z' + w', t + \ \mathrm{Re}\, w_1 + 2\,\mathrm{Im}\,(z' \cdot \overline{w}') \big) \right] + \Big( i\rho(w), 0, \ldots, 0 \Big) \\ &\equiv \Big( t + \ \mathrm{Re}\, w_1 + 2\,\mathrm{Im}\,(z' \cdot \overline{w}') + i|z' + w'|^2, z' + w' \Big) + \Big( i\rho(w), 0, \ldots, 0 \Big) \\ &= (t + \ \mathrm{Re}\, w_1 + (-i)[z' \cdot \overline{w}' - \overline{z}' \cdot w'] + i|z'|^2 \\ &\qquad + i|w'|^2 + 2i\,\mathrm{Re}\, z' \cdot \overline{w}' + i\,\mathrm{Im}\, w_1 - i|w'|^2, z' + w') \\ &= (t + w_1 + i|z'|^2 + i[2\,\mathrm{Re}\, \overline{z}' \cdot w' + 2i\,\mathrm{Im}\, \overline{z}' \cdot w'], z' + w') \\ &= (t + i|z'|^2 + w_1 + i2\overline{z}' \cdot w', z' + w')\,. \end{aligned}$$

This mapping is plainly holomorphic in $w$ (but *not* in $z$!). Thus we see explicitly that the action of the Heisenberg group on $\mathcal{U}$ is a (bi)holomorphic mapping.

As we have mentioned previously, the Heisenberg group acts simply transitively on the boundary of $\mathcal{U}$. Thus the group may be identified with the boundary in a natural way. Let us now make this identification explicit. First observe that $\mathbf{0} \equiv (0, \ldots, 0) \in \partial\mathcal{U}$. If $g = (z', t) \in \mathbb{H}^{n-1}$ then

$$g[\mathbf{0}] = \Psi^{-1}[(z', t) \cdot (0', 0)] = \Psi^{-1}[(z', t)] = (t + i|z'|^2, z') \in \partial\mathcal{U} .$$

Conversely, if $(\operatorname{Re} w_1 + i|w'|^2, w') \in \partial\mathcal{U}$ then let $g = (w', \operatorname{Re} w_1)$. Hence

$$g[\mathbf{0}] = \Psi^{-1}[g \cdot \Psi(\mathbf{0})] = \Psi^{-1}[(w', \operatorname{Re} w_1)] = (\operatorname{Re} w_1 + i|w'|^2, w') \in \partial\mathcal{U} .$$

Compare this result with the similar, but much simpler result for the classical upper half plane $U$ that we discussed in the last section.

The upshot of the calculations in this section is that analysis on the boundary of the ball $B$ may be reduced to analysis on the boundary of the Siegel upper half space $\mathcal{U}$. And that in turn is equivalent to analysis on the Heisenberg group $\mathbb{H}^{n-1}$. The Heisenberg group is a step two nilpotent Lie group. In fact all the essential tools of analysis may be developed on this group, just as they were in the classical Euclidean setting. That is our goal in the next several sections.

## 3.5 The Nature of $\partial\mathcal{U}$

**Capsule:** The boundary of the classical upper half plane is *flat*. It is geometrically flat and it is complex analytically flat. In fact it is a line. Not so with the Siegel upper half space. It is strongly pseudoconvex (in point of fact, the upper half space is biholomorphic to the unit ball). This means that the Levi form is strictly positive definite at each boundary point. Thus the boundary of $\mathcal{U}$ is naturally "curved" in a complex analytic sense, and *it cannot be flattened*. These are ineluctable facts about the Siegel upper half space that strongly influence the analysis of this space that we are about to learn.

The boundary of the Siegel upper half space $\mathcal{U}$ is strongly pseudoconvex. This fact may be verified directly—by writing out the Levi form and calculating its eigenvalues—or it may be determined by invoking the facts that the ball is strongly pseudoconvex and $\mathcal{U}$ is biholomorphic to the ball. See [KRA1, Ch. 3] for the chapter and verse on these ideas.

As such, we see that the boundary of $\mathcal{U}$ cannot be "flattened". That is to say, it would be convenient if there were a biholomorphic mapping of $\mathcal{U}$ to a Euclidean half space, but in fact this is impossible. Because the boundary of a Euclidean half space is Levi flat. And Bell's theorem [BE1] says in effect that a strongly pseudoconvex domain can only be biholomorphic to another strongly pseudoconvex domain.

There are other ways to understand the geometry of $\partial\mathcal{U}$. In Section 3.6 we discuss the commutators of vector fields—in the context of the Heisenberg group. The main

point of that discussion is that the Heisenberg group is a *step two nilpotent Lie group*. This means that certain second-order commutators in the Heisenberg group are nonzero, but all other (higher order) commutators are zero. This idea also has a complex-analytic formulation which we now treat briefly.

For simplicity let us restrict attention to $\mathbb{C}^2$. And let $\Omega = \{z \in \mathbb{C}^2 : \rho(z) < 0\} \subseteq \mathbb{C}^2$ be a smoothly bounded domain. If $P \in \partial\Omega$ satisfies $\partial\rho/\partial z_1(P) \neq 0$ then the vector field

$$L = \frac{\partial\rho}{\partial z_1}(P)\frac{\partial}{\partial z_2} - \frac{\partial\rho}{\partial z_2}(P)\frac{\partial}{\partial z_1}$$

is *tangent* to $\partial\Omega$ at $P$ just because $L\rho(P) = 0$. Likewise $\overline{L}$ is also a tangent vector field.

If $P$ is a strongly pseudoconvex point then it may be calculated that the commutator $[L, \overline{L}]\rho(P)$ is *not* zero—that is to say, $[L, \overline{L}]$ has a nonzero component in the normal direction. This is analogous to the Lie algebra structure of the Heisenberg group. And in fact Folland and Stein [FOS1] have shown that the analysis of a strongly pseudoconvex point may be accurately modeled by the analysis of the Heisenberg group. It is safe to say that much of what we present in the current chapters of the present book is inspired by [FOS1].

The brief remarks made here will be put into a more general context, and illustrated with examples, in Section 3.6.

## 3.6 The Heisenberg Group as a Lie Group

**Capsule:** The Heisenberg group is a step two nilpotent Lie group. This is a very strong statement about the complexity of the Lie algebra of the group. In particular, it says something about the Lie brackets of invariant vector fields on the group. These will in turn shape the analysis that we do on the group. It will lead to the notion of homogeneous dimension.

It is convenient now to shift gears a bit and to denote the element of $\mathbb{C}^{n-1} \times \mathbb{R}$ as $[\zeta, t]$, $\zeta \in \mathbb{C}^{n-1}$ and $t \in \mathbb{R}$. Then the space $\mathbb{C}^{n-1} \times \mathbb{R}$ (which is now the Heisenberg group $\mathbb{H}^{n-1}$) has the group structure defined as follows:

$$[\zeta, t] \cdot [\zeta^*, t^*] = [\zeta + \zeta^*, t + t^* + 2\,\mathrm{Im}\,\zeta \cdot \overline{\zeta}^*],$$

where $\zeta \cdot \overline{\zeta^*} = \zeta_1\overline{\zeta_1^*} + \cdots + \zeta_{n-1}\overline{\zeta}_{n-1}^*$. The identity element is $[0, 0]$ and $[\zeta, t]^{-1} = [-\zeta, -t]$. Check the associativity:

$$\begin{aligned}[z, t] \cdot ([w, s] \cdot [\eta, u]) &= [z, t] \cdot [w + \eta, s + u + 2\,\mathrm{Im}\,w \cdot \overline{\eta}] \\ &= [z + w + \eta, t + s + u + 2\,\mathrm{Im}\,w \cdot \overline{\eta} + 2\,\mathrm{Im}\,z \cdot (\overline{w + \eta})]\end{aligned}$$

and

$$\begin{aligned}([z,t]\cdot[w,s])\cdot[\eta,u] &= [z+w, t+s+2\,\mathrm{Im}\, z\cdot\overline{w}]\cdot[\eta,u]\\ &= [z+w+\eta, t+s+u+2\,\mathrm{Im}\, z\cdot\overline{w}+2\,\mathrm{Im}\,(z+w)\cdot\overline{\eta}]\,.\end{aligned}$$

The Heisenberg group $\mathbb{H}^{n-1}$ has $2n-1$ real dimensions and we can define the differentiation of a function in each direction consistent with the group structure by considering 1-parameter subgroups in each direction.

Let $g=[z,t]\in\mathbb{H}^{n-1}$, where $z=(z_1,\ldots,z_{n-1})=(x_1+iy_1,\ldots,x_{n-1}+iy_{n-1})$ and $t\in\mathbb{R}$. If we let

$$\begin{aligned}\gamma_{2j-1}(s) &= [(0,\ldots,0,s+i0,0,\ldots,0),0]\\ \gamma_{2j}(s) &= [(0,\ldots,0,0+is,0,\ldots,0),0]\end{aligned}$$

for $1\le j\le n-1$ and the $s$ term in the $j^{\text{th}}$ slot, and if we let

$$\gamma_{2n-1}(s)=\gamma_t(s)=[0,s]$$

[with $(n-1)$ zeros and one $s$], then each forms a one-parameter subgroup of $\mathbb{H}^{n-1}$. Just as an example,

$$[(0,\ldots,s+i0,\ldots,0),0]\cdot[(0,\ldots,s'+i0,\ldots,0),0]=[(0,\ldots,(s+s')+i0,\ldots,0),0]\,.$$

We define the differentiation of $f$ at $g=[z,t]$ in each one-parameter group direction as follows:

$$\begin{aligned}X_jf(g) &\equiv \frac{d}{ds}f(g\cdot\gamma_{2j-1}(s))\bigg|_{s=0}\\ &= \frac{d}{ds}f([(x_1+iy_1,\ldots,x_j+s+iy_j,\ldots,x_{n-1}+iy_{n-1}),t+2y_js])\bigg|_{s=0}\\ &= \left(\frac{\partial f}{\partial x_j}+2y_j\frac{\partial f}{\partial t}\right)[z,t],\quad 1\le j\le n-1\,,\\ Y_jf(g) &\equiv \frac{d}{ds}f(g\cdot\gamma_{2j}(s))\bigg|_{s=0}\\ &= \frac{d}{ds}f([(x_1+iy_1,\ldots,x_j+i(y_j+s),\ldots,x_{n-1}+iy_{n-1}),t-2x_js])\bigg|_{s=0}\\ &= \left(\frac{\partial f}{\partial y_j}-2x_j\frac{\partial f}{\partial t}\right)[z,t],\quad 1\le j\le n-1\,,\\ Tf(g) &\equiv \frac{d}{ds}f(g\cdot\gamma_t(s))\bigg|_{s=0}\end{aligned}$$

$$= \frac{d}{ds} f([x, t+s])\Big|_{s=0}$$
$$= \frac{\partial f}{\partial t}[z, t]\,.$$

We think of $X_j$, $Y_j$, and $T$ as vector fields on the Heisenberg group. These three objects embody the structure of the group in an analytic manner (as we shall see below). Although we shall not belabor this point of view in the present book, we note now that it is appropriate to think of $X_j$, $Y_j$, $T$ as a basis for the *Lie algebra* of the Heisenberg group.

Central to geometric analysis and symplectic geometry is the concept of the commutator of vector fields. We review the idea here in the context of $\mathbb{R}^N$.

A *vector field* on a domain $U \subseteq \mathbb{R}^N$ is a function

$$\lambda : U \to \mathbb{R}^N$$

with $\lambda(x) = \sum_{j=1}^N a_j(x)\partial/\partial x_j$. We think of $\partial/\partial x_1, \ldots, \partial/\partial x_N$ as a basis for the range space $\mathbb{R}^N$. If $\lambda_1, \lambda_2$ are two such vector fields then we define their *commutator* to be

$$[\lambda_1, \lambda_1] = \lambda_1\lambda_2 - \lambda_2\lambda_1\,. \tag{3.6.1}$$

Of course a vector field is a linear partial differential operator. It acts on the space of testing functions. So, if $\varphi \in C_c^\infty$ then it is useful to write (3.6.1) as

$$[\lambda_1, \lambda_2]\varphi = \lambda_1(\lambda_2\varphi) - \lambda_2(\lambda_1\varphi)\,.$$

Let us write this out in coordinates. We set

$$\lambda_1 = \sum_{j=1}^N a_j^1(x)\frac{\partial}{\partial x_j}$$

and

$$\lambda_2 = \sum_{j=1}^N a_j^2(x)\frac{\partial}{\partial x_j}\,.$$

Then

$$\begin{aligned}
[\lambda_1, \lambda_2]\varphi &= \lambda_1(\lambda_2\varphi) - \lambda_2(\lambda_1\varphi) \\
&= \sum_{j=1}^N a_j^1(x)\frac{\partial}{\partial x_j}\left(\sum_{j=1}^N a_j^2(x)\frac{\partial}{\partial x_j}\varphi\right)\varphi - \sum_{j=1}^N a_j^2(x)\frac{\partial}{\partial x_j}\left(\sum_{j=1}^N a_j^1(x)\frac{\partial}{\partial x_j}\varphi\right)\varphi \\
&= \left[\sum_{j,\ell=1}^N a_j^1(x)\left(\frac{\partial}{\partial x_j}a_\ell^2(x)\right)\frac{\partial}{\partial x_\ell} + \sum_{j,\ell=1}^N a_j^1(x)a_\ell^2(x)\frac{\partial^2}{\partial x_j \partial x_\ell}\right]\varphi -
\end{aligned}$$

$$\left[\sum_{j,\ell=1}^{N} a_j^2(x)\left(\frac{\partial}{\partial x_j}a_\ell^1(x)\right)\frac{\partial}{\partial x_\ell}+\sum_{j,\ell=1}^{N} a_j^2(x)a_\ell^1(x)\frac{\partial^2}{\partial x_j\partial x_\ell}\right]\varphi$$

$$=\left(\sum_{j,\ell=1}^{N} a_j^1(x)\left(\frac{\partial}{\partial x_j}a_\ell^2(x)\right)\frac{\partial}{\partial x_\ell}-\sum_{j,\ell=1}^{N} a_j^2(x)\left(\frac{\partial}{\partial x_j}a_\ell^1(x)\right)\frac{\partial}{\partial x_\ell}\right)\varphi\,.$$

The main thing to notice is that $[\lambda_1,\lambda_2]$ is ostensibly—by its very definition—a second order linear partial differential operator. But in fact the top-order terms cancel out. So that, in the end, $[\lambda_1,\lambda_2]$ is a *first-order* linear partial differential operator. In other words—and this point is absolutely essential—*the commutator of two vector fields is another vector field*. This is what will be important for us in our study of the Heisenberg group.

Let

$$X_j=\frac{\partial}{\partial x_j}+2y_j\frac{\partial}{\partial t},\quad 1\le j\le n-1\,,$$

$$Y_j=\frac{\partial}{\partial y_j}-2x_j\frac{\partial}{\partial t},\quad 1\le j\le n-1\,,$$

$$T=\frac{\partial}{\partial t}\,.$$

Note that $[X_j,X_k]=[Y_j,Y_k]=[X_j,T]=[Y_j,T]=0$ for all $1\le j,k\le n$ and $[X_j,Y_k]=0$ if $j\neq k$. As an instance,

$$\big[X_j,X_k\big]=\left(\frac{\partial}{\partial x_j}+2y_j\frac{\partial}{\partial t}\right)\left(\frac{\partial}{\partial x_k}+2y_k\frac{\partial}{\partial t}\right)-\left(\frac{\partial}{\partial x_k}+2y_k\frac{\partial}{\partial t}\right)\left(\frac{\partial}{\partial x_j}+2y_j\frac{\partial}{\partial t}\right)\,.$$

And now a simple calculation shows that everything cancels out and the commutator equals 0.

The only nonzero commutator in the Heisenberg group is $[X_j,Y_j]$, and we calculate that right now:

$$\begin{aligned}[X_j,Y_j]&=\left(\frac{\partial}{\partial x_j}+2y_j\frac{\partial}{\partial t}\right)\left(\frac{\partial}{\partial y_j}-2x_j\frac{\partial}{\partial t}\right)-\left(\frac{\partial}{\partial y_j}-2x_j\frac{\partial}{\partial t}\right)\left(\frac{\partial}{\partial x_j}+2y_j\frac{\partial}{\partial t}\right)\\&=-2\left(\frac{\partial}{\partial x_j}x_j\right)\frac{\partial}{\partial t}-2\left(\frac{\partial}{\partial y_j}y_j\right)\frac{\partial}{\partial t}\\&=-4\frac{\partial}{\partial t}\\&=-4T\,.\end{aligned}$$

Thus we see that

$$[X_j, Y_j] = -4T\,.$$

To summarize: all commutators $[X_j, X_k]$, $[Y_j, Y_k]$ and $[X_j, T]$ equal 0. The only nonzero commutator is $[X_j, Y_j] = -4T$, $j = 1, 2, \ldots, n-1$. One upshot of these simple facts is that any *third-order commutator* $[[A, B], C]$ will be zero—just because $[A, B]$ will be either 0 or $-4T$. Thus the vector fields on the Heisenberg group form a nilpotent Lie algebra of step two.

We have discussed in Section 3.4 how the Heisenberg group acts holomorphically on the Siegel upper half space. Here we collect some facts about the invariant measure for this action.

**Definition 3.6.2** Let $G$ be a topological group that is locally compact and Hausdorff. A (left) Haar measure on $G$ is a Radon measure that is invariant under the group operation. Among other things, this means that if $K$ is a compact set and $g \in G$ then the measure of $K$ and the measure of $g \cdot K \equiv \{g \cdot k : k \in K\}$ are equal.

**Exercise for the Reader**
In $\mathbb{H}^{n-1}$, Haar measure coincides with the Lebesgue measure. [This is an easy calculation using elementary changes of variable.]

Let $g = [z, t] \in \mathbb{H}^{n-1}$. The *dilation* on $\mathbb{H}^{n-1}$ is defined to be

$$\alpha_\delta g = [\delta z, \delta^2 t].$$

We can easily check that $\alpha_\delta$ is a group homomorphism:

$$\alpha_\delta \left([z, t] \cdot [z^*, t^*]\right) = \alpha_\delta [z, t] \cdot \alpha_\delta [z^*, t^*]\,.$$

A *ball* with center $[z, t]$ and radius $r$ is defined as

$$B([z, t], r) = \{[\zeta, s] : |\zeta - z|^4 + |s - t|^2 < r^2\}\,.$$

[Later on, in the Exercises and Chapter 4, we shall examine this idea in the language of the Heisenberg group norm.] For $f, g \in L^1(\mathbb{H}^{n-1})$, we can define the convolution of $f$ and $g$:

$$f * g(x) = \int f(y^{-1} \cdot x) g(y)\, dy.$$

## 3.7 Classical Analysis

**Capsule:** In this section we study an important result of Folland and Stein about fractional integration. It generalizes a classical result of Riesz, and provides a powerful analytic tool.

In preparation for our detailed hard analysis of the Heisenberg group in Sections 3.8 Chapter 4, we use this section to review a some ideas from classical real analysis. This is all a setup so that we can prove the Folland–Stein theorem.

### 3.7.1 The Folland–Stein Theorem

We review an idea from Section 1.1. Let $(X, \mu)$ be a measure space and $f : X \to \mathbb{C}$ a measurable function. We say $f$ is *weak* $r$, $0 < r < \infty$ (or sometimes weak-type $r$) if there exists some constant $C$ such that

$$\mu\{x : |f(x)| > \lambda\} \leq \frac{C}{\lambda^r}, \quad \text{for any} lambda > 0.$$

**Remark 3.7.1** If $f \in L^r$, then $f$ is weak-type $r$. But not vice versa. For suppose that $f \in L^r$; then

$$C \geq \int |f|^r \, d\mu \geq \int_{\{|f|>\lambda\}} |f|^r \, d\mu \geq \lambda^r \cdot \mu\{|f| > \lambda\}$$

hence $f$ is of weak-type $r$. For the other assertion suppose that $X = \mathbb{R}^+$. Then $f(x) = \frac{1}{x^{1/r}}$ is weak-type $r$ but not $r^{\text{th}}$-power integrable. □

**Prelude** In the classical theory of fractional integration—due to Riesz and others—the $L^p$ mapping properties of the fractional integral operators

$$f \longmapsto \int_{\mathbb{R}^N} \frac{f(t)}{|x-t|^{N-\alpha}} \, dt$$

were established using particular, Euclidean properties of the kernels $|x|^{-N+\alpha}$, $0 < \alpha < N$. It was a remarkable insight of Folland and Stein that all that mattered was the distribution of values of the kernel. This fact is captured in the next theorem.

**Theorem 3.7.2 (Folland, Stein [FOS1])** *Let $(X, \mu), (Y, \nu)$ be measurable spaces. Let*

$$k : X \times Y \to \mathbb{C}$$

*satisfy*

$$\mu\{x : |k(x, y)| > \lambda\} \leq \frac{C}{\lambda^r}, \quad (\textit{for each fixed } y)$$

$$\nu\{y : |k(x, y)| > \lambda\} \leq \frac{C'}{\lambda^r}, \quad (\textit{for each fixed } x)$$

*where $C$ and $C'$ are independent of $y$ and $x$ respectively and $r > 1$. Then*

$$f \longmapsto \int_Y f(y)k(x,y)\,d\nu(y)$$

*maps $L^p(Y,\nu)$ to $L^q(X,\mu)$ where $\frac{1}{q} = \frac{1}{p} + \frac{1}{r} - 1$, for $1 < p < \frac{r}{r-1}$.*

**Remark 3.7.3** We shall prove this important theorem below. Certainly you should compare and contrast this result with the classical Riesz fractional integration result that is treated in [STE4, Ch. 5]. The Folland–Stein result is a far reaching generalization which frees the result from the structure of Euclidean space and shows quite plainly how the key idea is measure-theoretic.

It is worth noting here that the norms of the operators *must* blow up as $p \to 1$ or $p \to r/(r-1)$. This is so because, if not, then one could use the semi-continuity of the integral to derive boundedness at the endpoints (which is false). □

**Prelude** Schur's lemma is probably *the* most basic fact about integral kernels and their corresponding integral operators. Many of the more sophisticated ideas—including the Folland–Stein theorem—are based on Schur. We include below the most fundamental formulation of Schur.

A key tool in our proof of the Folland–Stein result is the following idea of Issai Schur (which is in fact a rather basic version of the Folland–Stein theorem):

**Lemma 3.7.4 (Schur)** *Let $1 \leq r \leq \infty$. Let $(X,\mu), (Y,\nu)$ be measurable spaces and let $k : X \times Y \to \mathbb{C}$ satisfy*

$$\left(\int |k(x,y)|^r\,d\mu(x)\right)^{\frac{1}{r}} \leq C,$$

$$\left(\int |k(x,y)|^r\,d\nu(x)\right)^{\frac{1}{r}} \leq C',$$

*where $C$ and $C'$ are independent of $y$ and $x$ respectively. Then*

$$f \longmapsto \int_Y f(y)k(x,y)\,dy$$

*maps $L^p(Y,\nu)$ to $L^q(X,\mu)$, where $\frac{1}{q} = \frac{1}{p} + \frac{1}{r} - 1$, for $1 \leq p \leq \frac{r}{r-1}$.*

Schur's lemma is a standard result, with an easy proof, and the details may be found in [STE2, Ch. 1]. Note that, for Schur's lemma, which of course has a stronger hypothesis than Folland–Stein, the boundedness in the conclusions is also true at the endpoints.

In order to prove the Folland–Stein theorem, we shall use the idea of distribution function that was introduced in Section 3.6.

**Proof of Theorem 3.7.2** By the Marcinkiewicz interpolation theorem (see Exercise 2 in Chapter 1), it is enough to show that $f \longmapsto Tf$ is weak-type $(p,q)$ for

each $p$ and corresponding $q$. Fix $s > 0$. Let $\kappa > 0$ be a constant to be specified later. Let us define

$$k_1(x,y) = \begin{cases} k(x,y) & \text{if } |k(x,y)| \geq \kappa \\ 0 & \text{otherwise,} \end{cases}$$

$$k_2(x,y) = \begin{cases} k(x,y) & \text{if } |k(x,y)| < \kappa \\ 0 & \text{otherwise.} \end{cases}$$

Note that $k(x,y) = k_1(x,y) + k_2(x,y)$ and $k_2$ is bounded. Let

$$T_1 f(x) = \int k_1(x,y) f(y)\, d\nu(y)$$

$$T_2 f(x) = \int k_2(x,y) f(y)\, d\nu(y)$$

Then $Tf = T_1 f + T_2 f$. For $g$ a function and $\lambda > 0$, let $\alpha_g(\lambda) = \mu\{g > \lambda\}$. Then

$$\begin{aligned} \alpha_{Tf}(2s) &\equiv \mu\{|Tf| > 2s\} \\ &= \mu\{|T_1 f + T_2 f| > 2s\} \\ &\leq \mu\{|T_1 f| + |T_2 f| > 2s\} \\ &\leq \mu\{|T_1 f| > s\} + \mu\{|T_2 f| > s\} \\ &= \alpha_{T_1 f}(s) + \alpha_{T_2 f}(s)\,. \end{aligned} \tag{3.7.2.1}$$

Let $f \in L^p(X)$ and assume $\|f\|_{L^p} = 1$. Choose $p'$ such that $\frac{1}{p} + \frac{1}{p'} = 1$. Then we get

$$|T_2 f(x)| = \left| \int k_2(x,y) f(y)\, d\nu(y) \right| \leq \left( \int |k_2(x,y)|^{p'}\, d\nu(y) \right)^{\frac{1}{p'}} \left( \int |f(y)|^p\, d\nu(y) \right)^{\frac{1}{p}}.$$

We then see that

$$\int |k_2(x,y)|^{p'}\, d\nu(y) = \int_0^\kappa p' s^{p'-1} \alpha_{k_2(x,\cdot)}(s)\, ds$$

$$\leq \int_0^\kappa p' s^{p'-1} \frac{C}{s^r}\, ds = Cp' \int_0^\kappa s^{p'-1-r}\, ds = C' \kappa^{p'-r}$$

The last equality holds since

$$p' - 1 - r = \frac{1}{1 - \frac{1}{p}} - 1 - r = \frac{p}{p-1} - 1 - r > -1.$$

Thus we get

$$|T_2 f(x)| \le (C'\kappa^{p'-r})^{\frac{1}{p'}} \|f\|_{L^p} = C''\kappa^{1-\frac{r}{p'}}.$$

Let $\kappa = (s/C'')^{q/r}$. Then

$$|T_2 f(x)| \le C'' \left(\frac{s}{C''}\right)^{\frac{q}{r}\left(1-\frac{r}{p'}\right)} = s.$$

Therefore we get $\alpha_{T_2 f}(s) = 0$. Hence, from (3.7.2.1), we get

$$\alpha_{Tf}(2s) \le \alpha_{T_1 f}(s).$$

Since $|k_1(x,y)| \ge \kappa$ whereever $k_1$ does not vanish, we have $\alpha_{k_1(x,\cdot)}(s) = \alpha_{k_1(x,\cdot)}(\kappa)$ if $s \le \kappa$. Thus

$$\begin{aligned}
\int_Y |k_1(x,y)|\, dv(y) &= \int_0^\infty \alpha_{k_1(x,\cdot)}(s)\, ds \\
&= \int_0^\kappa \alpha_{k_1(x,\cdot)}(s)\, ds + \int_\kappa^\infty \alpha_{k_1(x,\cdot)}(s)\, ds \\
&\le \kappa \alpha_{k_1(x,\cdot)}(\kappa) + \int_\kappa^\infty \frac{C}{s^r}\, ds \\
&\le \kappa \frac{C}{\kappa^r} + \frac{C}{1-r}\kappa^{1-r} \\
&= C'\kappa^{1-r}. \qquad (3.7.2.2)
\end{aligned}$$

Similarly, we get

$$\int_X |k_1(x,y)|\, d\mu(x) \le C\kappa^{1-r}.$$

Recall that if $m(x,y)$ is a kernel and

$$\int |m(x,y)|\, d\mu(x) \le C$$

$$\int |m(x,y)|\, dv(y) \le C$$

then, by Schur's lemma, $f \longmapsto \int_Y f(y) m(x,y)\, dv(y)$ is bounded on $L^p(X)$, $1 \le p \le \infty$. Thus, $T_1$ is bounded on $L^p$.

$$\|T_1 f\|_{L^p(X)} \le C\kappa^{1-r}\|f\|_{L^p} = C\kappa^{1-r}.$$

By Tchebycheff's inequality,

$$\mu\{x : |f(x)| > \lambda\} \leq \frac{\|f\|_{L^p}^p}{\lambda^p}.$$

Therefore

$$\alpha_{T_1 f}(s) \leq \frac{\|T_1 f\|_{L^p}^p}{s^p} \leq \frac{(C\kappa^{1-r})^p}{s^p} = C' \frac{\left(\frac{s}{C''}\right)^{\frac{q}{r}\cdot(1-r)p}}{s^p} = C''' s^{\frac{qp(1-r)}{r} - p} = C''' \frac{1}{s^q}.$$

Hence

$$\alpha_{Tf}(2s) \leq \frac{C'''}{s^q}.$$

This is the weak-type estimate that we seek. □

Now we have a rather universal fractional integral result at our disposal, and it certainly applies to fractional integration on the Heisenberg group $\mathbb{H}^{n-1}$. For example, set $|\cdot|_h$ on $\mathbb{H}^{n-1}$ to be equal to

$$|g|_h = (|z|^4 + t^2)^{\frac{1}{4}}$$

when $g = (z, t) \in \mathbb{H}^{n-1}$. If

$$k_\beta(x) = |x|_h^{-2n-2+\beta}\,,\ 0 < \beta < 2n + 2\,,$$

is a kernel and we wish to consider the operator

$$\mathcal{I}^\beta : f \mapsto f * k_\beta$$

on $\mathbb{H}^{n-1}$, then the natural way to proceed now is to calculate the weak-type of $k_\beta$. Then the Folland–Stein theorem will instantly tell us the mapping properties of $\mathcal{I}^\beta$. Now

$$m\{x : |k_\beta(x)| > \lambda\} = m\left\{x : |x|_h \leq \left(\frac{1}{\lambda}\right)^{1/[2n+2-\beta]}\right\}$$

$$\leq c \cdot \left(\frac{1}{\lambda}\right)^{[2n+2]/[2n+2-\beta]}.$$

We see immediately that $k_\beta$ is of weak-type $[2n + 2]/[2n + 2 - \beta]$. Thus the hypotheses of the Folland–Stein theorem are satisfied with $r = [2n+2]/[2n+2-\beta]$. We conclude that $\mathcal{I}^\beta$ maps $L^p$ to $L^q$ with

$$\frac{1}{q} = \frac{1}{p} - \frac{\beta}{2n+2}\,,\ 1 < p < \frac{2n+2}{\beta}\,.$$

## 3.8 Calderón–Zygmund Theory

**Capsule:** One of the truly profound ideas of the postwar period in analysis has been the Calderón–Zygmund theory. This set of ideas describes integral operators on $\mathbb{R}^N$ that generalize the Hilbert transform in one dimension, and proves $L^p$ boundedness properties of these integral operators. The Calderón–Zygmund theory truly shaped harmonic analysis for the second half of the twentieth century.

We now wish to turn our attention to singular integral operators. One of the key tools for the classical approach to this subject is the Whitney decomposition of an open set. That is an important tool in geometric analysis that first arose in the context of the Whitney extension theorem [WHI]. We begin with a review of that idea.

**Prelude** Extending a smooth function from a smooth submanifold of $\mathbb{R}^N$ to all of $\mathbb{R}^N$ is an intuitively obvious and appealing process. For the implicit function theorem tells us that we may as well assume that the submanifold is a linear subspace, and then the extension process is just the trivial extension in the orthogonal variables. Extending from a more arbitrary subset is a very meaningful idea, but much more subtle. It was Hassler Whitney who cracked the problem and wrote the fundamental paper [WHI].

**Theorem 3.8.1 (Whitney Extension Theorem)** *Suppose $F \subseteq \mathbb{R}^N$ is any closed set. Let $f \in C^k(F)$. Then $f$ can be extended to a $C^k$ function on all $\mathbb{R}^N$.*

The idea of $f \in C^k(F)$ for $F$ an arbitrary closed set requires some explanation. If $F$ is a closed set without interior, or more generally just any old set, then what does $C^k(F)$ mean? The answer is rather technical, but quite natural in view of the theory of Taylor series. The tract [FED, Ch. 3] contains all the details.

The crux of the proof is that one decomposes ${}^cF$ into carefully chosen boxes—each box has diameter comparable to its distance from $F$ itself—and then extends $f$ to each box in the appropriate manner. We shall not provide the details here, but see [FED, Ch. 3]. As already noted, one of the main tools in the proof of Whitney's result is this decomposition theorem:

**Prelude** The Whitney decomposition theorem arose originally as a fundamental tool in the proof of the Whitney extension theorem. The extension of the function was achieved box-by-box. Today Whitney's decomposition is a basic tool in harmonic analysis, used to prove the Calderón–Zygmund theorem and other Martingale-type results.

**Theorem 3.8.2 (Whitney Decomposition Theorem)** *Let $F \subseteq \mathbb{R}^N$ be a closed set and $\Omega = \mathbb{R}^N \setminus F$. Then there exists a collection of closed cubes $\mathcal{F} = \{Q_j\}_{j=1}^{\infty}$ in $\Omega$ such that*

1. *$\cup_j Q_j = \Omega$*
2. *$Q_j$'s have pairwise disjoint interiors*
3. *there exist constants $C_1 < C_2$ such that $C_1 \cdot diam\,(Q_j) \leq dist\,(Q_j, F) \leq C_2 \cdot diam\,(Q_j)$.*

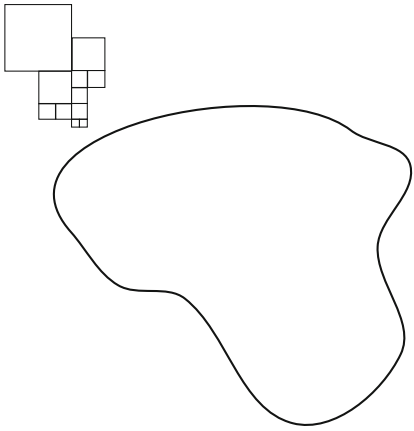


**Figure 3.1** The Whitney decomposition.

**Remark 3.8.3** It may be noted that, in one real dimension, things are deceptively simple. For if $F$ is a closed subset of $\mathbb{R}^1$ then the complement is open and may be written as the disjoint union of open intervals. Nothing like this is true in several real variables. The Whitney decomposition is a substitute for that simple and elegant decomposition. □

**Proof of the Theorem** Examine Figure 3.1 as you read the proof. Consider the collection of closed cubes $\mathcal{C}_0$ in $\mathbb{R}^N$ with vertices having integer coordinates and side length 1. Let $\mathcal{C}_1$ be the set of closed cubes obtained by slicing the cubes in $\mathcal{C}_0$ in half in each coordinate direction. $\mathcal{C}_2$ is gotten from $\mathcal{C}_1$ by slicing in half in each coordinate direction. Continue the process.

Also we may get a collection of closed cubes $\mathcal{C}_{-1}$ such that slicing the cubes in $\mathcal{C}_{-1}$ in half in each direction produces $\mathcal{C}_0$. We obtain $\mathcal{C}_{-2}, \mathcal{C}_{-3}$, and so on in a similar fashion.

Ultimately, we have collections of cubes $\mathcal{C}_j$, $j \in \mathbb{Z}$, where the cubes in $\mathcal{C}_j$ have side length $2^{-j}$ and diameter $\sqrt{N}2^{-j}$.

Define

$$\Omega_j = \{x \in \mathbb{R}^N : C \cdot 2^{-j} < \text{ dist } (x, F) \leq C \cdot 2^{-j+1}\},$$

where the constant $C$ will be specified later. Then obviously $\Omega = \cup_{j=1}^{\infty} \Omega_j$, and the union is pairwise disjoint. Now we select a cube in $\mathcal{C}_j$ if if has nonempty intersection with $\Omega_j$. Then the collection of such cubes, say $\{Q_\alpha\}$, covers $\Omega$:

$$\Omega \subset \cup_\alpha Q_\alpha.$$

If we let $C = 2\sqrt{N}$, then each selected cube $Q$ is disjoint from $F$. Suppose that $Q \in \mathcal{C}_j$ is one of the selected cubes. Then $\exists\, x \in Q \cap \Omega_j$. By the definition of $\Omega_j$, we get

$$2 \operatorname{diam} Q = 2\sqrt{N}2^{-j} \leq \text{dist } (x, F) \leq 2\sqrt{N}2^{-j+1} = 4 \operatorname{diam} Q. \tag{3.8.2.1}$$

Therefore

$$\operatorname{dist}(Q,F) \geq \operatorname{dist}(x,F) - \operatorname{diam} Q \geq 2\operatorname{diam} Q - \operatorname{diam} Q = \operatorname{diam} Q > 0\,.$$

Hence, for each selected $Q_\alpha$, $Q_\alpha \cap F = \emptyset$ and thus $\cup_\alpha Q_\alpha \subset \Omega$. Therefore we have

$$\Omega = \cup_\alpha Q_\alpha\,.$$

The key fact about our cubes is that if two cubes have nontrivially intersecting interiors then one is contained in the other. Thus we can find a disjoint collection of cubes $Q'_\alpha$ such that $\Omega = \cup_\alpha Q'_\alpha$. □

**Prelude** Next is the result, previously mentioned, in which we use the Whitney decomposition (the original proof of Calderón and Zygmund does *not* use the Whitney decomposition). When the Calderón–Zygmund decomposition was first proved it was a revelation: a profound, geometric way to think about singular integrals. It continues today to be influential and significant. Certainly it has affected the way that the subject has developed. The atomic theory, pseudodifferential operators, the David–Journé theorem, and many other essential parts of our subject have been shaped by the Calderón–Zygmund theorem.

**Theorem 3.8.4 (Calderón–Zygmund Decomposition)** *Let $f$ be a nonnegative, integrable function in $\mathbb{R}^N$. Then, for $\alpha > 0$ fixed, there is a decomposition of $\mathbb{R}^N$ such that*

**(1)** $\mathbb{R}^N = F \cup \Omega$, $F \cap \Omega = \emptyset$, *$F$ is closed.*
**(2)** $f(x) \leq \alpha$ *for almost every* $x \in F$.
**(3)** $\Omega = \cup_j Q_j$, *where $Q_j$'s are closed cubes with disjoint interiors and $f$ satisfies*

$$\alpha < \frac{1}{m(Q_j)} \int_{Q_j} f(x)\,dx \leq 2^N \alpha.$$

*($m(Q_j)$ denotes the measure of the cube $Q_j$.)*

**Proof** Decompose $\mathbb{R}^N$ into a mesh of equal closed cubes, whose interiors are disjoint, and whose common diameter is so large that

$$\frac{1}{|Q|} \int_Q f(x)\,dx \leq \alpha \tag{3.8.4.1}$$

for every cube $Q$ in this mesh.

Let $Q'$ be a fixed cube in this mesh. We subdivide it into $2^N$ congruent cubes by bisecting each of the sides of $Q'$. Let $Q''$ be on of the new subcubes.
**Case I:**

$$\frac{1}{|Q''|} \int_{Q''} f(x)\,dx \leq \alpha\,.$$

**Case II:**

$$\frac{1}{|Q''|}\int_{Q''} f(x)\,dx > \alpha\,.$$

In **Case II**, one does not subdivide $Q''$ any further, and $Q''$ is selected as one of the cubes $Q_j$ appearing in the statement of the theorem. One has for such a cube the inequality

$$\alpha < \frac{1}{|Q''|}\int_{Q''} f(x)\,dx \;\leq \frac{1}{2^{-N}Q'}\int_{Q'} f(x)\,dxleq 2^N\alpha\,.$$

In **Case I**, we proceed with the subdivision of $Q''$, and repeat this process until (if ever) we are forced into the second case. We write

$$\Omega = \bigcup_j Q_j\,,$$

the union of cubes obtained from **Case II**. Note that we have started this process with with all possible cubes $Q'$ from the original mesh and satisfying (3.8.4.1).

Now we claim that $f(x) \leq \alpha$ almost everywhere in $F = {}^c\Omega$. In fact, for almost every point $x \in F$ we know by the Lebesgue differentiation theorem (see [STE4]) that

$$f(x) = \lim_{\text{diam } Q\to 0} \frac{1}{|Q|}\int_Q f(t)\,dt\,.$$

But each of the cubes that enters into our decomposition that contains an $x \in F$ is a cube for which the **Case I** holds. That completes the proof. □

An immediate corollary of this theorem is the following:

**Corollary 3.8.5** *Let $f, \alpha, F, \Omega$, and $Q_j$ have the same meaning as in the theorem. Then there exist two constants A and B so that*

**(a)** $\mathbb{R}^N = F \cup \Omega$, $F \cap \Omega = \emptyset$.
**(b)** $f(x) \leq \alpha$ *almost everywhere on* $F$.
**(c)**

$$|\Omega| \leq \frac{A}{\alpha}\|f\|_{L^1}\,.$$

**(d)**

$$\frac{1}{|Q_j|}\int_{Q_j} f(x)\,dx \leq B\alpha\,.$$

**Definition 3.8.6** [Calderón–Zygmund Kernel] A Calderón–Zygmund kernel $K(x)$ in $\mathbb{R}^N$ is one having the form

$$K = \frac{\Omega(x)}{|x|^N}$$

where

1. $\Omega(x)$ is homogeneous of degree 0
2. $\Omega(x) \in C^1(\mathbb{R}^N \setminus \{0\})$
3. $\int_\Sigma \Omega(x) d\sigma(x) = 0$, ($\Sigma$ is the unit sphere in $\mathbb{R}^N$ and $d\sigma$ is area measure on the sphere).)

It should be noted that Theorem 3.8.9 below implies that one can make sense of the convolution $f * K$. This is usually done by means of the so-called *Cauchy principal value*, which is defined by

$$\text{P.V. } f * K \equiv \lim_{\epsilon \to 0} \int_{|t|>\epsilon} f(x-t)K(t)\, dt\,.$$

We shall encounter this point of view in Theorem 4.3.2 below.

The next result is the key to our study of singular integral kernels and operators.

**Prelude** The Calderón–Zygmund theorem is the cornerstone of a whole "real variable" theory of Hardy spaces. It has played a major role in the development of the Fefferman–Stein–Weiss theory of harmonic analysis.

In order to prove our result on singular integrals, we need to recall a certain integral of Marcinkiewicz.

**Lemma 3.8.7** *Let $F$ be a closed set in $\mathbb{R}^N$ whose complement has finite measure. If $x \in \mathbb{R}^N$, then let $\delta(x)$ be the distance of $x$ from $F$. Define*

$$I_*(x) \equiv \int_{\mathbb{R}^N} \frac{\delta(x+t)}{|t|^{N+1}}\, dt\,.$$

*Then $I_*(x) < \infty$ for almost every $x \in F$. Furthermore,*

$$\int_F I_*(t)\, dt \leq c \cdot |{}^cF|\,. \tag{3.8.7.1}$$

**Proof** Since the integrand is positive, it suffices to prove (3.8.7.1). We now see that

$$\begin{aligned}\int_F I_*(x)\, dx &= \int_F \int_{\mathbb{R}^N} \frac{\delta(x+t)}{|t|^{N+1}}\, dtdx \\ &= \int_F \int_{\mathbb{R}^N} \frac{\delta(t)}{|x-t|^{N+1}}\, dtdx\end{aligned}$$

$$= \int_F \int_{{}^cF} \frac{\delta(t)}{|x-t|^{N+1}}\, dtdx$$
$$= \int_{{}^cF} \left( \int_F \frac{dx}{|x-t|^{N+1}} \right) \delta(t)\, dt$$

Now consider

$$\int_F \frac{dx}{|x-t|^{N+1}} \quad \text{for} \quad t \in {}^cF\,.$$

The least value of $|x-t|$, as $x$ varies over $F$, is just $\delta(t)$—which is simply the distance of $t$ from $F$. Hence

$$\int_F \frac{dx}{|x-t|^{N+1}} \leq \int_{|x|\geq\delta(t)} \frac{dx}{|x|^{N+1}} \leq c(\delta(t))^{-1}\,.$$

Thus we have shown that

$$\int_F I_*(x)\, dx \leq \int_{{}^cF} c(\delta(t))^{-1} \cdot \delta(t)\, dt = c|{}^cF|\,.$$

That completes the proof. □

**Theorem 3.8.8** *Let $K \in L^2(\mathbb{R}^N)$. Assume that*

**(a)** $|\widehat{K}| \leq B$
**(b)** $K \in C^1(\mathbb{R}^N \setminus \{0\})$ *and* $|\nabla K(x)| \leq C|x|^{-N-1}$.

*For $1 < p < \infty$, and $f \in L^1 \cap L^p(\mathbb{R}^N)$, set*

$$Tf(x) = K * f(x) = \int_{\mathbb{R}^N} K(x-t)f(t)\, dt\,.$$

*Then there exists a constant $A_p$ such that*

$$\|Tf\|_p \leq A_p \|f\|_p.$$

**Proof** Hypothesis **(a)** tells us that the Fourier transform of the kernel is bounded. Since

$$\widehat{Tf} = \widehat{K} \cdot \widehat{f}\,,$$

we see immediately from the Plancherel theorem that $T$ is bounded on $L^2$.

If we can show that $T$ is weak type $(1,1)$, then it will follow from the Marcinkiewicz interpolation theorem that $T$ is bounded on $L^p$, $1 < p < 3$. Then a simple duality argument shows that $T$ is bounded on $L^p$ for $2 < p < \infty$.

So it all comes down to proving the weak typ $(1,1)$ inequality. It is here that the Calderón–Zygmund decomposition plays a crucial role. Consider the set

$$S_\alpha \equiv \{x : |Tf(x)| > \alpha\}\,.$$

Now apply Corollary 3.8.5 to the function $|f|$ and the constant $\alpha$. So we have

**(i)** $\mathbb{R}^N = F \cup \Omega$,
**(ii)** $F \cap \Omega = \emptyset$,
**(iii)** $|f(x)| \leq \alpha$ for a.e. $x \in F$,
**(iv)** $\Omega = \cup_j Q_j$, with the interiors of the $Q_j$ pairwise disjoint,
**(v)** $|\Omega| \leq (A/\alpha) \int_{\mathbb{R}^N} |f(x)|dx$,
**(vi)** $(1/|Q_j|) \int_{Q_j} |f(x)|\,dx \leq C\alpha$.

We define

$$g(x) = \begin{cases} f(x) & \text{if} \quad x \in F\,, \\ \frac{1}{|Q_j|} \int_{Q_j} f(x)\,dx & \text{if} \quad x \in \overset{\circ}{Q}_j\,. \end{cases}$$

This defines $g$ almost everywhere. Next set $f(x) = g(x) + b(x)$. This equation *defines* the function $b$.

Notice that the "good" function $g$ has the property that it is bounded by $\alpha$. The "bad" function $b$ has no such favorable bound; but it compensates by having mean value 0 on each of the cubes $Q_j$.

Since $Tf = Tg + Tb$, we see that

$$|\{x : |Tf(x)| > \alpha\}| \leq |\{x : |Tg(x)| > \alpha/2\}| + |\{x : |Tb(x)| > \alpha/2\}|\,.$$

Thus we need to estimate the two terms on the right.

To get favorable estimates on $Tg$, we used the $L^2$ theory. Namely,

$$\begin{aligned} \|g\|_{L^2}^2 &= \int_{\mathbb{R}^N} |g(x)|^2\,dx \\ &= \int_F |g(x)|^2\,dx + \int_\Omega |g(x)|^2\,dx \\ &\leq \int_F \alpha \cdot |f(x)|\,dx + C^2\alpha^2|\Omega| \\ &\leq (C^2A + 1)\alpha\|f\|_{L^1}\,. \end{aligned} \tag{3.8.8.1}$$

Now we apply the fact that $T$ is strongly bounded on $L^2$ so it is certainly weak type $(2,2)$. Thus we have

$$|\{x : |Tg(x)| > \alpha\}| \leq \frac{K}{\alpha^2} \int_{\mathbb{R}^N} |g(t)|^2\,dt\,.$$

Combining this with (3.8.8.1) yields

$$|\{x : |Tg(x)| > \alpha\} \leq \frac{C}{\alpha}\|f\|_{L^1} . \tag{3.8.8.2}$$

Now we turn our attention to $Tb$. We write

$$b_j(x) \equiv \begin{cases} b(x) & \text{if} \quad x \in Q_j \\ 0 & \text{if} \ \ x \in Q_j . \end{cases}$$

Then of course $b(x) = \sum_j b_j(x)$ and $Tb(x) = \sum_j Tb_j(x)$. Also

$$Tb_j(x) = \int_{Q_j} K(x-t)f(t)\,dt . \tag{3.8.8.3}$$

Of course $F = {}^c(\cup_j Q_j)$. We shall be able to obtain a favorable estimate of (3.8.8.3) when $x \in F$. Let $t^j$ be the center of $Q_j$. We exploit the fact that each $b_j$ has mean value zero to write

$$Tb_j(x) = \int_{Q_j} [K(x-t) - K(x-t^j)]b_j(t)\,dt .$$

Because $\nabla K(x) \leq C|x|^{-N-1}$, we use the mean value theorem to see that

$$|K(x-t) - K(x-t^j)| \leq C \cdot \frac{\text{diam}\,(Q_j)}{|x-\overline{t}^j|^{N+1}} ,$$

where $\overline{t}^j$ is a point on the line segment connecting $t$ to $t^j$.

Now it is critical that we remember that the diameter of $Q_j$ is comparable to the distance of $Q_j$ from $F$. Therefore, if $x$ is a fixed point of $F$, then the set of distances $\{|x-t|\}$, as $t$ ranges over $Q_j$, are all comparable with each other. Thus

$$|Tb_j(x)| \leq C \cdot \text{diam}\,(Q_j) \int_{Q_j} \frac{|b(t)|}{|x-t|^{N+1}} dt, . \tag{3.8.8.4}$$

Notice, however, that

$$\int_{Q_j} |b(t)|\,dt \leq \int_{Q_j} |f(t)|\,dt + C\alpha \int_{Q_j} 1\,dt .$$

Hence

$$\int_{Q_j} |b(t)|\,dt \leq (1+C)\alpha|Q_j| . \tag{3.8.8.5}$$

Now let $\delta(t)$ denote the distance of $t$ from $F$. Observe that

$$\text{diam}\,(Q_j) \cdot |Q_j| \leq C \int_{Q_j} \delta(t)\, dt$$

so that

$$|Q_j| \leq C \cdot \frac{1}{\text{diam}\,(Q_j)} \int_{Q_j} \delta(t)\, dt\,.$$

Combining this with (3.8.8.4) and (3.8.8.5) yields

$$|Tb_j(x)| \leq C\alpha \int_{Q_j} \frac{\delta(t)}{|x-t|^{N+1}}\, dt \leq C\alpha \int_{\mathbb{R}^N} \frac{\delta(t)}{|x-t|^{N+1}}\, dt \quad \text{if} \quad x \in F\,.$$

Now we bring the Marcinkiewicz integral (Lemma 3.8.7) into play. In fact this lemma now tells us that

$$\int_F |Tb(x)|\, dx \leq C\alpha|\Omega| \leq C\|f\|_{L^1}\,.$$

As a result,

$$|\{x \in F : |Tb(x)| > \alpha/2\} \leq \frac{C}{\alpha} \cdot \|f\|_{L^1}\,.$$

Remembering that

$$|^cF| = |\Omega| \leq \frac{C}{\alpha}\|f\|_{L^1}\,,$$

we have obtained the estimate

$$|\{x : |Tb(x)| > \alpha/2\} \leq \frac{C}{\alpha} \cdot \|f\|_{L^1}\,. \qquad \square$$

Combining our estimates for $Tg$ and $Tb$ yields the desired result. $\square$

**Theorem 3.8.9 (Calderón–Zygmund)** *Let $K(x)$ be a Calderón–Zygmund kernel. Then $T : f \longmapsto K * f$ is bounded on $L^p$, $1 < p < \infty$.*

We shall treat some of these matters in the next chapter, particularly in the context of the Heisenberg group.

## Exercises

1. Refer to Section 3.8. Formulate and prove a version of the Whitney decomposition on the Heisenberg group. [**Hint:** You may find it useful to consider the homogeneous norm $|g|_h = (|z|^4 + t^2)^{1/4}$ which will be covered in detail later on.]
2. Give an example of a nilpotent group of step 3.
3. Find a group of $3 \times 3$ matrices that is isomorphic to $\mathbb{H}^1$.
4. What is the center of the Heisenberg group?
5. What is the context for the Heisenberg group in mathematical physics? The reference [STE2] will give you some hints.
6. Which automorphisms of the unit ball in $\mathbb{C}^2$ preserve the point $(1/2, 0)$?
7. How many fixed points can a biholomorphic map of the unit ball in $\mathbb{C}^n$ have? Finitely many? Countably many? Uncountably many?
8. We know that the Heisenberg group has infinitely many abelian subgroups of dimension 1 (these are 1-parameter subgroups). Are there any abelian subgroups of dimension 2 or higher?
9. What are the normal subgroups of the Heisenberg group? [*Hint:* Think about the center of the group.]
10. What are the Lie quotient groups of the Heisenberg group?
11. Describe all the homomorphisms from the Heisenberg group $\mathbb{H}^n$ to the Heisenberg group $\mathbb{H}^{n-1}$.

# Chapter 4
# Analysis on the Heisenberg Group

**Prologue:** And now we reach a high point of our work. We have spent a long chapter laying the background and motivation for what we are doing. Chapter 3 set the stage for Heisenberg group analysis, and laid out the foundations for the subject. Now we are going to dive in and prove a big theorem.

Our goal in this chapter is to prove the $L^p$ boundedness of the Szegő projection on the Heisenberg group, which is the boundary of the Siegel upper half space (which is biholomorphically equivalent to the unit ball in $\mathbb{C}^n$). We also make some remarks about the Poisson–Szegő integral. It must be emphasized here that this is a singular integral, but not in the traditional, classical sense. The singularity is *not* isotropic, and the mapping properties of the operator cannot simply be analyzed using traditional techniques. The anisotropic Heisenberg analysis that we have developed here is what is needed. The foundational reference for all these ideas is [FOS1] as well as the earlier works [KOV], [COG1, Ch. 3], [KNS].

So our program has several steps. The first is to develop the theory of the Calderón–Zygmund integral. The next is to consider the Szegő kernel on the Siegel upper half space in some detail. We must calculate it explicitly, and render it in a form that is useful to us. In particular, we must see that it is a convolution operator on the Heisenberg group.

Third, we must see that the Szegő projection is in fact a singular integral operator in the sense of this book. Then all of our analysis will come to bear and we can derive a significant theorem about mapping properties of the Szegő projection.

Along the way we shall also learn about the Poisson–Szegő operator. It is an essential feature of the analysis of the Heisenberg group. We shall be able to say something about its mapping properties as well.

Many of the calculations in this chapter are inspired by ideas of E. M. Stein.

## 4.1 A Deeper Look at the Heisenberg Group

**Capsule:** Now we wish to develop a Calderón–Zygmund theory on the Heisenberg group. This will entail a re-thinking of many of the key ideas, and also taking into account the special geometry of the Heisenberg group. This study is a model for how harmonic analysis should be done on Lie groups.

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When we were thinking of the Heisenberg group as the boundary of a domain in $\mathbb{C}^n$, then the appropriate Heisenberg group to consider was $\mathbb{H}^{n-1}$, as that Lie group has dimension $2n-1$ (the correct dimension for a boundary). Now we are about to study the Heisenberg intrinsically, in its own right, so it is appropriate (and it simplifies the notation a bit) to focus our attention on $\mathbb{H}^n$. No confusion should result.

In $\mathbb{H}^n = \mathbb{C}^n \times \mathbb{R}$, the group operation is defined as

$$(z,t)\cdot(z',t') = (z+z', t+t'+2\mathrm{Im}\, z\cdot\overline{z'}), \quad z,z'\in\mathbb{C}^n,\ t,t'\in\mathbb{R}\,.$$

Let $g=(z,t)=(z_1,z_2,\ldots,z_n,t)=(x_1+iy_1,x_2+iy_2,\ldots,x_n+iy_n,t)\in\mathbb{H}^n$. We write

$$dV(g) = dx_1dy_1\cdots dx_ndy_ndt\,,$$

so that

$$dV(\delta g) = d(\delta x_1)\,d(\delta y_1)\cdots d(\delta x_n)d(\delta y_n)d(\delta^2 t) = \delta^{2n+2}\,dV(g)\,.$$

We call $2n+2$ the *homogeneous dimension* of $\mathbb{H}^n$. [Note that the topological dimension of $\mathbb{H}^n$ is $2n+1\neq 2n+2$.]

Define $|(z,t)| = (|z|^4+t^2)^{1/4}$. This is the nonisotropic norm on the Heisenberg group. The critical index $N$ for a singular integral is such that

$$\int_{B(0,1)} \frac{1}{|(z,t)|^\alpha}\,dV(z,t) \quad \begin{cases} = \infty \text{ if } & \alpha \geq N \\ < \infty \text{ if } & 0<\alpha<N\,. \end{cases}$$

and the critical index coincides with the homogeneous dimension. Thus the critical index for a singular integral in $\mathbb{H}^n$ is $2n+2$, which is different from the topological dimension. See the next page for further development of these ideas.

Put in slightly different words, a classical Calderón–Zygmund singular integral kernel on $\mathbb{R}^N$ is isotropically homogeneous of degree $-N$. By contrast, a singular integral kernel on the Heisenber group $\mathbb{H}^n$ will be nonistropically homogeneous of degree $-2n-2$.

As previously noted, it is necessary to reinvent many of the most basic classical notions of analysis in order to conduct our studies on the Heisenberg group. We treat some of these ideas in this and the next few sections.

Set $|\cdot|_h$ on $\mathbb{H}^n$ to be equal to

$$|g|_h = (|z|^4+t^2)^{\frac{1}{4}}$$

when $g=(z,t)\in\mathbb{H}^n$ (we discuss this norm in greater detail below).

For $x, y \in \mathbb{H}^n$, we define the distance $d(x, y)$ as follows:

$$d(x, y) \equiv |x^{-1} \cdot y|_h$$

Then $d(x, y)$ satisfies the following properties:

1. $d(x, y) = 0 \iff x = y$;
2. $d(x, y) = d(y, x)$;
3. $\exists \gamma_0 > 0$ such that $d(x, y) \leq \gamma_0[d(x, w) + d(w, y)]$.

Notice that the third stipulation for the norm is not the usual triangle inequality. But it is a useful substitute, and is sufficient for many applications in analysis. But it should be noted that we are *not* saying that $d$ is a metric. We sometimes call such a $d$ a *semimetric* or a *quasimetric*. But the literature is not consistent in this matter.

Define balls in $\mathbb{H}^n$ by $B(x, r) = \{y \in \mathbb{H}^n : d(x, y) < r\}$. Then, equipped with the Lebesgue measure on $\mathbb{R}^{2n+1}$, $\mathbb{H}^n$ is a space of homogeneous type (Subsection 2.1.1). Because our balls come from a semi-metric, the enveloping property is straightforward—it simply follows from the semi-triangle inequality. The other two properties of a space of homogeneous type are also routine.

## 4.2 $L^2$ Boundedness of Calderón–Zygmund Integrals

Thinking back to the classical situation on $\mathbb{R}^N$, we recall that we first treated the action of singular integral operators on $L^2$. We were able to do so effectively and expeditiously by exploiting Plancherel's theorem. While there is also a Plancherel theorem on the Heisenberg group, it is not nearly as useful as an analytic tool. Therefore we shall develop another approach.

Let us now treat this matter in a bit more detail. In $\mathbb{R}^N$, for a Calderón–Zygmund kernel $K(z)$, we know that $f \longmapsto f * K$ is bounded on $L^2$. This result can be established using Plancherel's theorem. See also Theorem 3.8.5. Since $K$ has mean value zero, it induces a distribution, hence it has a Fourier transform. Now $K$ is homogeneous of degree $-N$, so we know that $\widehat{K}$ is homogeneous of degree $-N - (-N) = 0$. Thus $\widehat{K}$ is bounded and

$$\|f * K\|_2 = \|\widehat{f * K}\|_2 = \|\widehat{f} \cdot \widehat{K}\|_2 \leq C\|\widehat{f}\|_2 = C\|f\|_2.$$

But we cannot use the same technique in $\mathbb{H}^n$ since we do not have the Fourier transform in $\mathbb{H}^n$ as a useful analytic tool. Instead we use the so-called Cotlar–Knapp–Stein lemma.

## 4.3 The Cotlar–Knapp–Stein Lemma

**Capsule:** Let $H$ be a Hilbert space. Suppose we have operators $T_j : H \to H$ that have uniformly bounded norm, $\|T_j\|_{\text{op}} = 1$. Then what can we say about $\|\sum_{j=1}^N T_j\|_{\text{op}}$?

It was Mischa Cotlar who first understood how to conceptualize this idea. Cotlar and Knapp/Stein independently found a much more flexible formulation of the result which has proved to be quite useful in the practice of harmonic analysis. We now formulate a version of their theorem (see [COT] and [KNS]).

**Lemma 4.3.1 (Cotlar–Knapp–Stein)** *Let $H$ be a Hilbert space and $T_j : H \to H$ be bounded operators, $j = 1, \ldots, N$. Let $\{a_j\}_{j=-\infty}^{\infty}$ be a positive, bi-infinite sequence of numbers such that $A = \sum_{-\infty}^{\infty} a_j < \infty$. Also assume that*

$$\|T_j T_k^*\|_{\text{op}} \leq a_{j-k}^2\,, \quad \|T_j^* T_k\|_{\text{op}} \leq a_{j-k}^2. \tag{4.3.1.1}$$

*Then*

$$\left\| \sum_{j=1}^N T_j \right\|_{\text{op}} \leq A\,.$$

**Proof** We will use the fact that $\|TT^*\| = \|T^*T\| = \|T\|^2 = \|T^*\|^2$. Also, since $TT^*$ is self-adjoint, we have $\|(TT^*)^k\| = \|TT^*\|^k$. Let $T = \sum_{j=1}^N T_j$. Then we get

$$(TT^*)^m = \Big[\Big(\sum_{j=1}^N T_j\Big)\Big(\sum_{j=1}^N T_j^*\Big)\Big]^m = \sum_{1\leq j_k\leq N} T_{j_1} T_{j_2}^* T_{j_3} T_{j_4}^* \cdots T_{j_{2m-1}} T_{j_{2m}}^*$$

By (4.3.1.1), we get

$$\|T_{j_1} T_{j_2}^* \cdot T_{j_3} T_{j_4}^* \cdots T_{j_{2m-1}} T_{j_{2m}}^*\| \leq \|T_{j_1} T_{j_2}^*\| \|T_{j_3} T_{j_4}^*\| \cdots \|T_{j_{2m-1}} T_{j_{2m}}^*\| \leq a_{j_1-j_2}^2 \cdots a_{j_{2m-1}-j_{2m}}^2.$$

Also

$$\begin{aligned}\|T_{j_1} T_{j_2}^* \cdot T_{j_3} T_{j_4}^* \cdots T_{j_{2m-1}} T_{j_{2m}}^*\| &\leq \|T_{j_1}\| \|T_{j_2}^* T_{j_3}\| \|T_{j_4}^* T_{j_5}\| \cdots \|T_{j_{2m-2}}^* T_{j_{2m-1}}\| \|T_{j_2 m}\| \\ &\leq A a_{j_2-j_3}^2 \cdots a_{j_{2m-2}-j_{2m-1}}^2 A\end{aligned}$$

Therefore we may conclude that

$$\begin{aligned}\|T_{j_1} T_{j_2}^* \cdots T_{j_{2m-1}} T_{j_{2m}}^*\| &= \|T_{j_1} T_{j_2}^* \cdots T_{j_{2m-1}} T_{j_{2m}}^*\|^{\frac{1}{2}} \|T_{j_1} T_{j_2}^* \cdots T_{j_{2m-1}} T_{j_{2m}}^*\|^{\frac{1}{2}} \\ &\leq A a_{j_1-j_2} a_{j_2-j_3} \cdots a_{j_{2m-2}-j_{2m-1}} a_{j_{2m-1}-j_{2m}}\,.\end{aligned}$$

Hence

$$\|TT^*\|^m \leq \sum_{\substack{1\leq j_k\leq N\\ k=1,2,\ldots,2m}} A a_{j_1-j_2} a_{j_2-j_3}\cdots a_{j_{2m-2}-j_{2m-1}} a_{j_{2m-1}-j_{2m}} .$$

Now we sum over each $j_m$ in succession, beginning with $j_{2m}, j_{2m-1}$, and so on down. There are $2m-1$ such sums, and each gives rise, by estimation, to a factor of $A$. With the last sum in $j_1$, we get a factor of $N$. So the estimate in the end is

$$\|TT^*\|^m \leq \sum_{j=1}^{N} A^{2m-1} = N\cdot A^{2m-1} .$$

Taking the $m$-th root, we get

$$\|TT^*\| \leq N^{1/m}\cdot A^{(2m-1)/m} .$$

Letting $m\to\infty$, we get $\|T\|^2 \leq A^2$ or $\|T\| \leq A$. □

**Prelude** The next result is the cornerstone of basic Heisenberg group analysis. It set the stage for a whole new generation of the analysis of singular integrals.

**Theorem 4.3.2 (Knapp, Stein. 1971)** *Let $K$ be a function on $\mathbb{H}^n$ that is smooth away from 0 and homogeneous of degree $-2n-2$. Assume that*

$$\int_{|z|_h=1} K d\sigma = 0 ,$$

*where $d\sigma$ is area measure (i.e., Hausdorff measure) on the unit sphere $\Sigma$ in the Heisenberg group. Define*

$$Tf(z) = \mathrm{PV}(K * f) = \lim_{\epsilon\to 0}\int_{|t|_h>\epsilon} K(t) f(t^{-1}z) dt \tag{4.3.2.1}$$

*Then the limit exists pointwise and in norm and*

$$\|Tf\|_2 \leq C\|f\|_2 \tag{4.3.2.2}$$

**Remark 4.3.3** In fact, $T : L^p \to L^p$, for $1 < p < \infty$. We shall discuss the details of this assertion a bit later.

Theorem 4.3.2 is proved by breaking the integral up into dyadic pieces to which the Cotlar–Knapp–Stein theorem applies. We cannot provide the details here, but refer the reader instead to [STE2, Ch. 12, 13]. □

## 4.4 $L^p$ Boundedness of Calderón–Zygmund Integrals

In the last subsection we established $L^2$ boundedness of the Calderón–Zygmund operators on the Heisenberg group. Given the logical development that we have seen thus far in the subject, the natural next step for us would be to prove a weak-type (1,1) estimate for these operators. Then one could apply the Marcinkiewicz interpolation theorem to get strong $L^p$ estimates for $1 < p < 2$. Finally, a simple duality argument would yield strong $L^p$ estimates for $2 < p < \infty$. And that would complete the picture for singular integrals on the Heisenberg group.

The fact is that a general paradigm for proving weak-type (1,1) estimates on a space of homogeneous type has been worked out in [COG1, Ch. 3]. There are a number of interesting new twists and turns in this treatment—for instance the geometry connected with the Whitney decomposition is rather challenging—and we encourage readers to consult this original source as interest dictates.

What we shall do here is to sketch out how the theory works particularly for the Szegő kernel $S(z, \zeta)$ on the Siegel upper half space. Refer to Theorem 3.8.8 in our treatment of the classical real variable theory of singular integrals. Instead of the hypothesis that the Fourier transform of the kernel is bounded, we instead use the fact that integration against the Szegő kernel is the canonical projection from $L^2$ of the boundary to $H^2$ of the boundary. So of course it is bounded on $L^2$.

For the weak type $(1, 1)$ result we follow the proof of Theorem 3.8.8 rather closely. Here are the steps:

**(1)** We formulate and prove a result about a Marcinkiewicz-type integral.
**(2)** We formulate and prove a Whitney decomposition. [This is a bit tricky, but see [COG1, Ch. 3].]
**(3)** We use Step **(2)** to formulate and prove a Calderón–Zygmund decomposition.
**(4)** We observe that $|\nabla_\zeta S(z, \zeta)|$ and $\nabla_z S(z, \zeta)|$ satisfy the sort of estimates that are analogous to hypothesis **(b)** in Theorem 3.8.8.
**(5)** Given an $f \in L^1$, we decompose $f = g + b$ as in the proof of Theorem 3.8.8.
**(6)** We obtain the desired estimates on $Sg$ just as we did in the classical setup in the proof of Theorem 3.8.8.
**(7)** We use Step **(4)** to obtain estimates on $Sb$ just as we did in the classical setup in the proof of Theorem 3.8.8.

In the remainder of this treatment, we shall take it for granted that $L^p$ boundedness for Calderón–Zygmund operators has been established, $1 < p < \infty$, and we shall use the result to good effect. A nice treatment of the circle of ideas being discussed here appears in [FOL].

We note that it is straightforward to prove that

$$\int_{r_1 < ||z|^2 + it| < r_2} (|z|^2 + it)^{-n-1} \, d\sigma(z, t) = 0 \, .$$

Just introduce a suitable nonisotropic version of polar coordinates (the ideas is also discussed in the text). Thus we have a version of the mean-value zero property for the Szegő kernel. And that gives another approach to treating the Szegő kernel as a singular integral kernel in the sense of Calderón and Zygmund. See [CHA2, p. 54 ff.] for more on this matter.

## 4.5 Calderón–Zygmund Applications

One fundamental result that can be derived from the preceding material concerns the mapping properties of the Szegő projection. Recall that the Szegő kernel $S(z, \zeta)$ is the canonical reproducing kernel for the space $H^2(\Omega)$. Put in other words,

$$f \longmapsto \int f(\zeta) S(z, \zeta)\, d\sigma(\zeta)$$

is the projection of $L^2(\partial\Omega)$ to $H^2(\Omega)$ (where we identify each $H^2$ function with its boundary function). See [KRA1, Ch. 1] for details. See the discussion later in this section. Here the Hardy space is defined as follows:

**(a)** In case $\Omega$ is the unit disc $D$ in $\mathbb{C}$, and $0 < p < \infty$, we define

$$H^p(D) = \left\{ f \text{ holomorphic on } D : \sup_{0<r<1} \int_0^{2\pi} |f(re^{i\theta})|^p \, d\theta^{1/p} \equiv \|f\|_{H^p(D)} < \infty \right\} .$$

**(b)** In case $\Omega$ is the upper half plane $U = \{\zeta \in \mathbb{C} : \operatorname{Im} \zeta > 0\}$, and $0 < p < \infty$, we define

$$H^p(U) = \left\{ f \text{ holomorphic on } U : \sup_{0<y<\infty} \int_{-\infty}^{\infty} |f(x+iy)|^p \, dx^{1/p} \equiv \|f\|_{H^p(U)} < \infty \right\} .$$

**(c)** In case $\Omega$ is the unit ball $B$ in $\mathbb{C}^n$, and $0 < p < \infty$, we define

$$H^p(B) = \left\{ f \text{ holomorphic on } B : \sup_{0<r<1} \int_{|\zeta|=1} |f(r\zeta)|^p \, d\sigma(\zeta)^{1/p} \equiv \|f\|_{H^p(B)} < \infty \right\} .$$

Here $d\sigma$ is rotationally invariant area measure on $\partial B$.

**(d)** For an arbitrary bounded domain $\Omega \subseteq \mathbb{C}^n$ with $C^2$ boundary, the notion of dilation is not available. So we must work a bit harder. We think of $\Omega$ as the sublevel set of a defining function $\rho$: $\Omega = \{z \in \mathbb{C}^n : \rho(z) < 0\}$, with $\nabla\rho \neq 0$ on $\partial\Omega$. Now, for $\epsilon$ positive and small, $0 < \epsilon < \epsilon_0$, we set

$$H^p(\Omega) = \left\{ f \text{ holomorphic on } \Omega : \sup_{0<\epsilon<\epsilon_0} \int_{\rho(\zeta)=-\epsilon} |f(\zeta)|^p \, d\sigma_\epsilon(\zeta)^{1/p} \equiv \|f\|_{H^p(\Omega)} < \infty \right\} .$$

Here $d\sigma_\epsilon$ is Hausdorff $(2n-1)$-dimensional measure on $\partial\Omega_\epsilon \equiv \{z \in \mathbb{C}^n : \rho(z) = -\epsilon\}$. It is known (see Lemma 2.3.2) that this definition is independent of the choice of defining function $\rho$.

It is known that a Hardy space function in $H^p(\Omega)$ has a boundary function in $L^p(\partial\Omega)$. See, for instance, [KRA1, Ch. 8].

For a fixed, bounded domain $\Omega \subseteq \mathbb{C}^n$ with $C^2$ boundary, and $z \in \Omega$, we consider the linear functional

$$e_z : H^2(\Omega) \ni f \longmapsto f(z) \in \mathbb{C}\,.$$

It is easy to show (see, for instance, [KRA18, Ch. 1]) that this is in fact a bounded linear functional on the Hilbert space $H^2$. Let $k_z$ be a Hilbert space representative for the functional. Define the Szegő kernel to be $S(z,\zeta) = \overline{k_z}(z,\zeta)$.

The mapping

$$L^2(\partial\Omega) \ni f \longmapsto \int_{\partial\Omega} f(\zeta) S(z,\zeta)\, d\sigma(\zeta)$$

is the Szegő projection. Obviously, by construction, the Szegő projection is bounded on $L^2(\partial\Omega)$. But is it bounded on $L^p$?

The Szegő kernel is a Calderón–Zygmund singular integral on the Heisenberg group. [It is tricky to check the mean-value zero property, and we shall not do so. Our approach to Theorem 3.8.8 avoided this issue.] The Szegő projection comes up very naturally in other contexts of the function theory of several complex variables. See [CHSTG] and [KRA9] for examples. The paper [NARSW] contains generalizations to weakly pseudoconvex domains.

We next review a few of the key ideas about the Szegő and Poisson–Szegő kernels.

## 4.6 The Szegő Integral on the Heisenberg Group

**Capsule:** We have met the Szegő kernel in earlier parts of this book. It is the canonical reproducing kernel for $H^2$ of any bounded domain with reasonably nice boundary. It fits the paradigm of Hilbert space with reproducing kernel (see [ARO]). And it is an important operator for complex function theory. But it does not have the invariance properties of the Bergman kernel. It is not quite as useful in geometric contexts. Nonetheless the Szegő kernel is an important artifact of complex function theory and certainly worthy of our studies.

Let $B$ be the unit ball in $\mathbb{C}^n$. Consider the Hardy space $H^2(B)$:

$$H^2(B) = \left\{ f \text{ holomorphic on } B : \sup_{0<r<1} \left[ \int_{\partial B} |f(r\xi)|^2 d\sigma(\xi) \right]^{\frac{1}{2}} = \|f\|_{H^2(B)} < \infty \right\}$$

Note that $H^2(B)$ is a Hilbert space. The Szegő kernel is a canonical reproducing kernel for $H^2$. Now we turn to the particular properties of this kernel, and the allied Poisson–Szegő kernel, on the Siegel upper half space and the Heisenberg group.

## 4.7 The Poisson–Szegő Integral

**Capsule:** The Poisson–Szegő kernel, invented by Hua and Korányi, is interesting because it is a positive kernel that reproduces $H^2$. It is quite useful in function-theoretic contexts. But it is not nearly so well known as perhaps it should be. Like the Bergman and Szegő kernels, it is quite difficult in practice to compute. But various asymptotic expansions make it accessible in some theoretical situations.

We begin with some review.

Let $\Omega \in \mathbb{C}^n$ be a bounded domain. We would like to construct a positive reproducing kernel related to the Szegő kernel. Let $S(z, \zeta)$ be the Szegő kernel on $\Omega$.

**Definition 4.7.1** We define the Poisson–Szegő kernel as follows:

$$\mathcal{P}(z, \zeta) = \frac{|S(z, \zeta)|^2}{S(z, z)}.$$

**Prelude** The Poisson–Szegő kernel is a fairly modern idea that grew perhaps out of representation-theoretic considerations (see [HUA, Ch. 3]). It arises in a variety of contexts but is particularly useful in the harmonic analysis of several complex variables. Its chief virtue is that it is a positive kernel that reproduces $H^2$, and that it has a "structure" (i.e., the shape of its singularity) that is closely tied to the complex structure of the domain in question. It happens that the Poisson–Szegő kernel also solves the Dirichlet problem for the invariant Laplacian on the ball in $\mathbb{C}^n$, but that seems to be a special effect true only for that domain and a very few other special domains. This circle of ideas has been explored in [GRA1, GRA2]. See also [KRA5, Ch. 6].

**Lemma 4.7.2** *Let $\Omega$ be a bounded domain in $\mathbb{C}^n$ with $C^2$ boundary. The Poisson–Szegő kernel on $\Omega$ is well defined and nonnegative.*

**Proof** Let $\{\phi_j\}$ be a complete orthonormal basis for $H^2(\Omega)$. Note that

$$S(z, z) = \sum_{j=1}^{\infty} \phi_j(z)\overline{\phi_j(z)} = \sum_{j=1}^{\infty} |\phi_j(z)|^2 \geq 0.$$

This follows from elementary Hilbert space theory, or see [KRA1, §1.5]. Suppose that there exists a $z_0 \in \Omega$ such that $S(z_0, z_0) = 0$. Then $\phi_j(z_0) = 0$, for all $j \geq 1$. Thus

$$f(z_0) = \sum_{j=1}^{\infty} \alpha_j \phi_j(z_0) = 0, \quad \forall f \in H^2(\Omega).$$

Again this is just Hilbert space theory. But $f \equiv 1 \in H^2(\Omega)$. Contradiction.

As a consequence, the expression in the denominator of our defining expression for the Poisson–Szegő kernel never vanishes. So this new kernel is well defined, and it is obviously nonnegative. □

It is a fact that integration against the Poisson–Szegő kernel reproduces holomorphic functions that extend continuously to the boundary. This matter is treated in detail in Section 6.2 below.

## 4.8 Applications of the Paley–Wiener Theorem

**Capsule:** Now it is time to focus in on the Siegel half space and the canonical kernels on that space. Both the Szegő and the Poisson–Szegő kernels will play a prominent role. Their mapping properties are of particular interest.

The Siegel upper half space $\mathcal{U}$ is biholomorphically equivalent to the unit ball via the generalized Cayley map. Thus $\mathcal{U}$ has a Bergman and a Szegő kernel.

Let $U$ be a domain in $\mathbb{C}^n$. A set $S \subseteq U \subseteq \mathbb{C}^n$ is called a *set of determinacy* if any holomorphic function on $U$ that vanishes on $S$ must be identically zero on $U$.

**EXAMPLE 4.8.1** The set $S = \{(s + it, 0) : s, t \in \mathbb{R}\} \subset \mathbb{C}^2$ is *not* a set of determinacy on $\mathbb{C}^2$. Because $f(z_1, z_2) = z_2$ is holomorphic and vanishes on $S$. □

**EXAMPLE 4.8.2** The set $S = \{(s + i0, t + i0) : s, t \in \mathbb{R}\} \subset \mathbb{C}^2$ *is* a set of determinacy. This assertion follows from elementary one-variable power series considerations. □

**Remark** If $S \subset \mathbb{C}^n$ is a totally real,[1] $n$-dimensional manifold, then $S$ is a set of determinacy. See [WEL, Ch. 1]. □

**Prelude** There is an important idea lurking in the background in the next lemma. Let $S$ be a 2-dimensional real linear subspace of $\mathbb{C}^2$. Let $f$ be a holomorphic function. If $f\big|_S = 0$, does it follow that $f \equiv 0$? The answer depends on how $S$ is positioned in space; put in other words, it depends on the complex structure of $S$.

**Lemma 4.8.3** *Let $\Omega \subseteq \mathbb{C}^n$ be a domain and let $f(z, w)$ be defined on $\Omega \times \Omega$. Assume that $f$ is holomorphic in $z$ and conjugate holomorphic in $w$. If $f(z, z) = 0$ for all $z$, then $f(z, w) \equiv 0$ on $\Omega \times \Omega$ (i.e., the diagonal is a set of determinacy in a certain sense—not quite the same as that defined above).*

[1] A manifold $M \subseteq \mathbb{C}^n$ is said to be *totally real* if, whenever $\alpha$ lies in the tangent space to $M$, then $i\alpha$ does not lie in the tangent space.

**Proof** Consider the mapping

$$\phi : (z, w) \to \left( \frac{z + \overline{w}}{2}, \frac{z - \overline{w}}{2i} \right).$$

Then

$$\phi^{-1} : (\alpha, \beta) \to (\alpha + i\beta, \overline{\alpha - i\beta}).$$

Let $\widetilde{f} = f \circ \phi^{-1}(\alpha, \beta)$. We can easily check that it is holomorphic in $\alpha$ and $\beta$:

$$\frac{\partial \widetilde{f}}{\partial \overline{\alpha}} = \frac{\partial f}{\partial z}\frac{\partial z}{\partial \overline{\alpha}} + \frac{\partial f}{\partial \overline{w}}\frac{\partial \overline{w}}{\partial \overline{\alpha}} = 0$$
$$\frac{\partial \widetilde{f}}{\partial \overline{\beta}} = \frac{\partial f}{\partial z}\frac{\partial z}{\partial \overline{\beta}} + \frac{\partial f}{\partial \overline{w}}\frac{\partial \overline{w}}{\partial \overline{\beta}} = 0\,.$$

Since $f(z, z) = 0$ and

$$\phi : (z, z) \to (\operatorname{Re} z, \operatorname{Im} z),$$

we know that

$$\widetilde{f}(\operatorname{Re} z, \operatorname{Im} z) = f \circ \phi^{-1}(\operatorname{Re} z, \operatorname{Im} z) = f(z, z) = 0.$$

But $\{(\operatorname{Re} z, \operatorname{Im} z)\} \subset \mathbb{C}^{2n}$ is a totally real $2n$-dimensional manifold, thus a set of determinacy. Therefore, $\widetilde{f} = f \circ \phi^{-1} \equiv 0$. Hence $f \equiv 0$. □

Recall the height function $\rho$ on $\mathcal{U}$:

$$\rho(w) = \operatorname{Im} w_1 - |w'|^2\,,$$

where $w' = (w_2, \ldots, w_n)$. We look at the almost analytic extension of $\rho$:

$$\rho(z, w) = \frac{i}{2}(\overline{w}_1 - z_1) - \sum_{k=2}^{n} z_k \overline{w}_k.$$

Note that $\rho$ is holomorphic in $z$ and conjugate holomorphic in $w$ and $\rho(w, w) = \rho(w)$.

**Prelude** The next theorem is key to the principal result of this chapter. The Szegő kernel is a Heisenberg singular integral, hence can be analyzed using the machinery that we have developed. Of course analogous results are true in one dimension as well.

**Theorem 4.8.4** *On the Siegel upper half space $\mathcal{U}$, the Szegő kernel $S(z,\zeta)$ is:*

$$S(z,\zeta) = \frac{n!}{4\pi^{n+1}} \cdot \frac{1}{\rho(z,\zeta)^{n+1}}.$$

For $\rho > 0$ and $F \in H^2(\mathcal{U})$, we let

$$F_\rho(z',t) = F\big(z', t + i(|z'|^2 + \rho)\big).$$

Lurking in the background here is the map $\Psi$ that takes elements of $\partial\mathcal{U}$ to elements of the Heisenberg group. In the current discussion we will find it convenient *not* to mention $\Psi$ explicitly. The role of $\Psi$ will be understood from context.

We know that, for $z \in \mathcal{U}$,

$$F(z) = \int_{\partial\mathcal{U}} F_0(w) S(z,w) d\sigma(w).$$

This is just the standard reproducing property of the Szegő kernel acting on $H^2$ of the Siegel upper half space, with $F_0$ the $L^2$ boundary limit function of $F$.

The classical structure theorem for the Hardy space $H^2(U)$ on the upper half plane is as follows.

**Prelude** The Paley–Wiener theorem is one of the big ideas of modern harmonic analysis. It dates back to the important book [PAW]. Today Paley–Wiener theory shows itself in signal processing, in wavelet theory, in partial differential equations, and many other parts of mathematical analysis.

**Theorem 4.8.5 (Paley–Wiener Theorem)** *The equation*

$$f \mapsto F(z) \equiv \int_0^\infty e^{2\pi i z\cdot\lambda} f(\lambda) d\lambda$$

*yields an isomorphism between $L^2(0,\infty)$ and $H^2(U)$.*

This theorem is of particular value because the elements of $L^2(0,\infty)$ are easy to understand, whereas the elements of $H^2(U)$ are less so. Observe that one direction of the proof of this theorem is easy: given a function $f \in L^2(0,\infty)$, the integral above converges (absolutely) as soon as $\operatorname{Im} z > 0$. Furthermore, for any $y > 0$, we set $F_y(x) \equiv F(x+iy)$ and see that

$$\begin{aligned} \|F_y\|^2_{L^2(\mathbb{R}^1)} &= \int_{\mathbb{R}^1} \left| \int_0^\infty e^{-2\pi y\lambda} \cdot e^{2\pi i x\lambda} f(\lambda) d\lambda \right|^2 dx \\ &\leq \int_{\mathbb{R}^1} \left| \int_0^\infty e^{2\pi i x\lambda} f(\lambda) d\lambda \right|^2 dx \end{aligned}$$

$$= \|\widehat{f}(x)\|^2_{L^2(\mathbb{R}^1)}$$
$$= \|f\|^2_{L^2(0,\infty)} .$$

It is also clear that

$$\|F\|_{H^2(\mathbb{R}^2_+)} = \sup_{y>0} \|F_y\|_{L^2(\mathbb{R}^1)} = \|f\|_{L^2(0,\infty)}.$$

The more difficult direction is the assertion that the map $f(\lambda) \mapsto F(z)$ is actually *onto* $H^2$. We shall not treat it in detail (although our proof below specializes down to the half plane in $\mathbb{C}$), but refer the reader instead to [KAT, Ch. 1].

We would like to develop an analogue for the Paley–Wiener theorem on the Siegel upper half space $\mathcal{U}$. First we must discuss integration on $\mathbb{H}^{n-1}$. Recall that a measure $d\lambda$ on a topological group is the Haar measure (unique up to multiplication by a constant) if it is a Borel measure that is invariant under left translation. Our measure $d\zeta dt$ (ordinary Lebesgue measure) turns out to be both left and right invariant, i.e., it is *unimodular*. The proof is simply a matter of carrying out the integration:

$$\begin{aligned}
\int\int f([\xi, s] \cdot [\zeta, t]) d\zeta\, dt &= \int\int f(\zeta + \xi, t + s + 2\mathrm{Im}\, \xi \cdot \overline{\zeta}) d\zeta\, dt \\
&= \int\int f(\zeta + \xi, t + s + 2\mathrm{Im}\, \xi \cdot \overline{\zeta})\, dt\, d\zeta \\
&= \int\int f(\zeta + \xi, t)\, dt\, d\zeta \\
&= \int\int f(\zeta, t)\, dt\, d\zeta\, .
\end{aligned}$$

Observe now that the map $[\zeta, t] \mapsto [-\zeta, -t]$ preserves the measure but also sends an element of $\mathbb{H}^{n-1}$ onto its inverse. Thus it sends left translation into right translation, and so the left invariance of the measure implies its right invariance.

With that preliminary step out of the way, we can make the following definition.

**Definition 4.8.6** For $f$ holomorphic on the Siegel upper half space $\mathcal{U}$, we define

$$\|f\|_{H^2} = \sup_{\rho>0} \left( \int\int |f(z', t + i|z'|^2 + i\rho)|^2\, d|z'|\, dt \right)^{\frac{1}{2}} .$$

Then we set

$$H^2(\mathcal{U}) = \{f : f \text{ is holomorphic on } \mathcal{U}, \|f\|_{H^2} < \infty\}.$$

Here $\rho$ is the height function that we have introduced for $\mathcal{U}$. Just as in the case of $U \subset \mathbb{C}^1$, where we integrated over parallels to $\mathbb{R}^1 = \partial\mathbb{R}^2_+$, so here we integrate over parallels to $\mathbb{H}^{n-1} = \partial\mathcal{U}$.

We will now see that $H^2(\mathcal{U})$ is a Hilbert space, and we will develop a Fourier analysis for it. The substitute for $L^2(0,\infty)$ in the present context will be $\widehat{H^2}$ which consists of all functions $\widehat{f} = \widehat{f}(z',\lambda)$ with $z' \in \mathbb{C}^{n-1}$, $\lambda \in \mathbb{R}^+$ such that

**(1)** $\widehat{f}$ is jointly measurable in $z'$ and $\lambda$;
**(2)** For almost every $\lambda$, the function $z' \mapsto \widehat{f}(z',\lambda)$ is entire on $\mathbb{C}^n$;
**(3)** $\|\widehat{f}\|^2_{\widehat{H^2}} \equiv \int_{\mathbb{C}^{n-1}} \int_{\mathbb{R}^+} |\widehat{f}(z',\lambda)|^2 e^{-4\pi\lambda|z'|^2}\, d\lambda\, d|z'| < \infty$.

Now we define an equivalence relation on the set of such functions as follows: $f \sim g$ if, for almost every $\lambda$, $f(z,\lambda) = g(z,\lambda)$ for *every* $z$. Then $\widehat{H^2}$ is the set of equivalence classes. Note that the Fourier transform takes each of these equivalence classes to a unique, well-defined function.

We have the following basic structure theorem:

**Prelude** Now we find our long-sought version of the Paley–Wiener theorem for the Siegel upper half space $\mathcal{U}$. It will bear nice consequences for us.

**Theorem 4.8.7** *Consider the equation*

$$F(z) = F(z_1, z') = \int_0^\infty e^{2\pi i\lambda z_1} \widehat{f}(z',\lambda)\, d\lambda\,. \tag{4.8.7.1}$$

**(1)** *Given an $\widehat{f} \in \widehat{H^2}$, the integral in (4.8.7.1) converges absolutely for $z \in \mathcal{U}$ and uniformly for $z \in K \subset\subset \mathcal{U}$. Thus we can interchange the order of differentiation and integration, and we see that the function $F$ given by the integral is holomorphic.*

**(2)** *The function $F$ defined in (4.8.7.1) from an $\widehat{f} \in \widehat{H^2}$ is an element of $H^2$, and the resulting map $\widehat{f} \mapsto F$ is an isometry of $\widehat{H^2}$ onto $H^2$; i.e., it is an isomorphism of Hilbert spaces.*

**(3)** *Let $\widetilde{\imath} = (0,0,\ldots,0,i) \in \mathbb{C}^{n+1}$ and let $F \in H^2$. Set $F_\varepsilon = F(z + \varepsilon\widetilde{\imath})\Big|_{\partial\mathcal{U}}$. Then $F_\varepsilon$ is a function on $\mathbb{H}^{n-1}$, $F_\varepsilon \to F_0$ in $L^2(\mathbb{H}^{n-1})$ as $\varepsilon \to 0$, and*

$$\|F_0\|_{L^2(\mathbb{H}^{n-1})} = \|F\|_{H^2}\,.$$

The idea of the proof is to freeze $z'$ and look at the Paley–Wiener representation of the half space $\operatorname{Im} z_1 > |z'|^2$. There are several nontrivial technical problems with this program, so we shall have to develop the proof in stages. First, we want $\widehat{H^2}$ to be complete so that it is a Hilbert space.

**Lemma 4.8.8** *$\widehat{H^2}$ is a Hilbert space.*

**Proof** Since $\widehat{H^2}$ is defined as $L^2(\mathbb{C}^n \times \mathbb{R}^+, d\mu)$ for a certain measure $\mu$, its inner product is already determined. The troublesome part of the lemma is the completeness.

Since we are dealing with analytic functions, the $L^2$ convergence will lead to a very strong (i.e., uniformly on compact subsets) type of convergence on the interior of $\mathbb{C}^n \times \mathbb{R}^1$. Now suppose we are given a Cauchy sequence $\{\widehat{f}_j\}$ in $\widehat{H^2}$; we must show that some subsequence converges to an element of $\widehat{H^2}$. Since $H^2$ is an $L^2$ space, some subsequence converges in $L^2$ ($L^2$ being complete), and we can extract from that a subsequence converging both in $L^2$ and pointwise almost everywhere. Next take a compact set $K \subset\subset \mathbb{C}^n$ which is the closure of an open set and $L \subset\subset \mathbb{R}^+$, and a subsequence $\{\widehat{f}_{j_\ell}\}$ such that

$$\int_L \int_K |\widehat{f}_{j_\ell} - \widehat{f}_{j_{\ell+1}}|^2 \, d|z'| \, d\lambda \leq \frac{1}{2^{2\ell}}.$$

Thus

$$\sum_\ell \|\widehat{f}_{j_\ell} - \widehat{f}_{j_{\ell+1}}\|^2_{K,L} = \sum_\ell \int_L \int_K |\widehat{f}_{j_\ell}(z', \lambda) - \widehat{f}_{j_{\ell+1}}(z', \lambda)|^2 \, d|z'| \, d\lambda < \infty \, .$$

If we set

$$\triangle_\ell(\lambda) = \int_K |\widehat{f}_{j_\ell}(z', \lambda) - \widehat{f}_{j_{\ell+1}}(z', \lambda)|^2 \, d|z'| \, ,$$

then we have

$$\int_L \sum_\ell \triangle_\ell(\lambda) \, d\lambda < \infty \, .$$

Thus $\sum_\ell \triangle_\ell(\lambda) < \infty$ for almost every $\lambda \in L$. Passing to a set $K' \subset\subset K$ we find a number $r > 0$ such that $B(z', r) \subset\subset K$ for all $z' \in K'$. Since, for a fixed $\lambda$, the functions $\widehat{f}_{j_\ell}$ are holomorphic on $K$, they obey the mean value property. Therefore

$$\begin{aligned}
\left|\widehat{f}_{j_\ell}(z', \lambda) - \widehat{f}_{j_{\ell+1}}(z', \lambda)\right| &\leq \frac{1}{c_n r^{2n}} \cdot \int_{B(z', r)} \left|\widehat{f}_{j_\ell}(w', \lambda) - \widehat{f}_{j_{\ell+1}}(w', \lambda)\right| \, d|w'| \\
&\leq \frac{1}{\widetilde{c_n} r^n} \cdot \left( \int_{B(z', r)} \left|\widehat{f}_{j_\ell}(w', \lambda) - \widehat{f}_{j_{\ell+1}}(w', \lambda)\right|^2 \, d|w'| \right)^{\frac{1}{2}} \\
&\leq C_r \cdot \triangle_\ell^{\frac{1}{2}}(\lambda)
\end{aligned}$$

for all $z' \in K'$ and $\lambda$ fixed. Therefore the sequence $\{\widehat{f}_{j_\ell}(\cdot, \lambda)\}$ converges uniformly on compact subsets of $\mathbb{C}^n$ for almost every $\lambda$. Since, for almost every $\lambda$ the functions

$\widehat{f}_{j_\ell}$ are holomorphic, the limit is then holomorphic. Since the functions $\widehat{f}_{j_\ell}$ already converge in $L^2(\mathbb{C}^n \times \mathbb{R}^+, d\mu)$ and pointwise almost everywhere, our limit is in $\widehat{H^2}$. □

Next we prove

**Lemma 4.8.9** *If $\widehat{f} \in \widehat{H^2}$, then, for $(z', z_1) \in K \subset\subset \mathcal{U}$, we have that*

$$\int_0^\infty e^{2\pi i \lambda z_1} \widehat{f}(z', \lambda)\, d\lambda$$

*converges absolutely. Its absolute value is $\leq C_K \|\widehat{f}\|_{\widehat{H^2}}$.*

**Proof** For $z = (z', z_1) \in K \subset\subset \mathcal{U}$ there is an $\epsilon > 0$ such that $\text{Im}\, z_1 - |z'|^2 \geq \varepsilon > 0$. Also observe that, for almost every $\lambda$ (since $\widehat{f}(z', \lambda)$ is entire in $z'$), we have by the mean value property that

$$|\widehat{f}(z', \lambda)| \leq \frac{1}{\text{Vol}[B(z', \delta)]} \cdot \int_{B(z', \delta)} |\widehat{f}(w', \lambda)| d|w'|. \tag{4.8.9.1}$$

The number $\delta > 0$ will be selected later.

Since $-\text{Im}\, z_1 \leq -\varepsilon - |z'|^2$, we calculate that

$$\begin{aligned} I &= \left| \int_0^\infty e^{2\pi i \lambda z_1} \widehat{f}(z', \lambda)\, d\lambda \right| \\ &\leq \int_0^\infty e^{-2\pi\varepsilon\lambda} \cdot e^{-2\pi\lambda|z'|^2} \cdot |\widehat{f}(z', \lambda)|\, d\lambda \\ &\leq \left( \int_0^\infty e^{-2\pi\varepsilon\lambda}\, d\lambda \right)^{\frac{1}{2}} \cdot \left( \int_0^\infty e^{-2\pi\varepsilon\lambda} \cdot e^{-4\pi\lambda|z'|^2} |\widehat{f}(z', \lambda)|^2\, d\lambda \right)^{\frac{1}{2}} \end{aligned}$$

by Schwarz's inequality. Now set

$$C_0 = \left( \int_0^\infty e^{-2\pi\varepsilon\lambda}\, d\lambda \right)^{\frac{1}{2}}$$

and apply (4.8.9.1) to obtain

$$I \leq \frac{C_0}{\text{Vol}[B(z', \delta)]} \cdot \left( \int_0^\infty e^{-2\pi\varepsilon\lambda} \cdot e^{-4\pi\lambda|z'|^2} \left( \int_{B(z', \delta)} |\widehat{f}(w', \lambda)| d|w'| \right)^2 d\lambda \right)^{\frac{1}{2}}.$$

But an application of the generalized Minkowski inequality to the $w'$ integration yields

$$I^2 \leq C_0^2 \cdot \int_0^\infty e^{-2\pi\varepsilon\lambda} \cdot e^{-4\pi\lambda|z'|^2} \int_{B(z',\delta)} |\widehat{f}(w',\lambda)|^2 \, d|w'| \, d\lambda \, .$$

Now we would like to replace the expression $e^{-4\pi\lambda|z'|^2}$ by $e^{-4\pi\lambda|w'|^2}$ and then apply condition **(3)** of the definition of $\widehat{H^2}$. Since $w' \in B(z', \delta)$, we see that

$$e^{-4\pi\lambda|w'|^2} \geq e^{-4\pi\lambda|z'|^2} \cdot e^{-4\pi\lambda\delta} .$$

We choose $0 < \delta < \frac{\varepsilon}{2}$. It follows that

$$e^{-2\pi\varepsilon\lambda} \cdot e^{-4\pi\lambda|z'|^2} = e^{-2\pi\varepsilon\lambda} \cdot e^{4\pi\delta\lambda} \cdot e^{-4\pi\delta\lambda} \cdot e^{-4\pi\lambda|z'|^2} \leq e^{-4\pi\lambda|w'|^2}$$

and we find that

$$I^2 \leq C_0^2 \cdot \int_0^\infty \int_{B(z',\delta)} |\widehat{f}(w',\lambda)|^2 e^{-4\pi\lambda|w'|^2} \, d|w'| \, d\lambda \, .$$

Hence we have

$$I \leq C_0 \cdot \|\widehat{f}\|_{\widehat{H^2}} \, .$$

This completes the proof of the lemma. □

Now that we have the absolute convergence of our integral (and uniform convergence for $z \in K \subset\subset \mathcal{U}$), we are allowed to differentiate under the integral sign and it is clear that the $F$ which is created from $\widehat{f} \in \widehat{H}^2$ is holomorphic.

Before continuing with the proof of Theorem 4.8.4 we make two observations:

**(i)** There would appear to be an ambiguity in the definition

$$F(z', \lambda) = \int_{-\infty}^{\infty} e^{-2\pi\lambda(x_1+iy_1)} \widehat{f}(z', x_1 + iy_1) \, dx_1 \;\text{ with }\; y_1 > |z'|^2 \, .$$

After all, the righthand side explicitly depends on $y_1$, and yet the lefthand side is independent of $y_1$. The fact is that the righthand side is also independent of $y_1$. After all, $\widehat{f}$ is holomorphic in the variable $x_1 + iy_1$, as it ranges over the half plane $y_1 > |z'|^2$. Then our claim is simply that the integral of $\widehat{f}$ over a line parallel to the $z$-axis is independent of the particular line we choose (as long as $y_1 > |z'|^2$). This statement is a consequence of Cauchy's integral theorem: the difference of the integral of $\widehat{f}$ over two parallel horizontal lines is the limit of the integral of $\widehat{f}$ over long horizontal rectangles—from $-N$ to $N$ say. Now the integral of $\widehat{f}$ over a rectangle is zero, and we will see that $\widehat{f}$ has sufficiently rapid decrease at $\infty$ so that the integrals over the ends of the rectangle tend to 0 as $N \to +\infty$. Thus

$$\int_{-\infty}^{\infty} e^{-2\pi\lambda(x_1+iy_1)}\widehat{f}(z', x_1 + iy_1)\, dx_1$$
$$= \int_{-\infty}^{\infty} e^{-2\pi\lambda(x_1+i\widetilde{y_1})}\widehat{f}(z', x_1 + i\widetilde{y}_1)\, dx_1 \qquad \text{for } 0 < y_1 < \widetilde{y}_1 .$$

**(ii)** Fix a point $(z_1, z') \in \mathcal{U}$. Consider the functional which sends $\widehat{f} \in \widehat{H^2}$ to $F(z_1, z')$, where $F$ is the function created by the Fourier integral of $\widehat{f}$. Then this functional is continuous on $\widehat{H^2}$. By the way that we defined $\widehat{H^2}$, everything is well defined. There are no ambiguities.

We next prove

**Lemma 4.8.10** *Let $F \in H^2(\mathcal{U})$. Then, for a fixed $z'$, $F_\varepsilon(z', \cdot) \in H^2(\{y_1 > |z'|^2\})$ (as a function of one complex variable) where*

$$F_\varepsilon(z', x_1 + iy_1) = \widehat{f}(z', x_1 + iy_1 + i\varepsilon), \ \textit{for } \varepsilon > 0.$$

**Proof** We may assume $z' = 0$. Apply the mean value theorem to $F(0, x_1 + iy_1 + i\varepsilon)$ on $B(0, \delta) \times D(x_1 + iy_1 + i\epsilon, \delta')$, where $B \subset \mathbb{C}^n$ and $D$ is a disc in the plane. We see that

$$|F(0, x_1 + iy_1 + i\varepsilon)|^2$$
$$\leq C_{\delta,\delta'} \cdot \int_{D(x_1+iy_1+i\varepsilon,\delta')} \int_{B(0,\delta)} |F(z', x_1 + iy_1 + w + i\varepsilon)|^2\, d|z'|\, d|w| .$$

Hence

$$\int_{-\infty}^{\infty} |F(0, x_1 + iy_1 + i\varepsilon)|^2\, dx_1$$
$$\leq C_{\delta,\delta'} \cdot \int_{-\infty}^{\infty} \int_{D(x_1+iy_1+i\varepsilon,\delta')} \int_{B(0,\delta)} |F|^2\, d|z'|\, d|w|\, dx_1 .$$

We write $w = u + iv$. Now

$$\int_{-\infty}^{\infty} \int_{D(w,\delta')} |G(z + w)|\, d|w|\, dx \leq \int_{-\infty}^{\infty} \int_{-\delta'}^{\delta'} |G(z + iv)|\, dv\, dx$$

for any $G$ (use the fact that $D(w, \delta')$ lies in a box centered at $w$ of side $2\delta'$ and sides parallel to the axes). Choose $\delta' = \frac{\varepsilon}{3}$ and set $\varepsilon' = \frac{2\varepsilon}{3}$, to obtain

$$\int_{-\delta'}^{\delta'} F_\varepsilon(z + iv)\, dv = \int_0^{2\delta'} F_{\varepsilon'}(z + iv)\, dv .$$

Thus

$$\int_{-\infty}^{\infty} |F_\varepsilon(0, x_1 + iy_1)|^2 \, dx_1$$
$$\leq C \cdot \int_{-\infty}^{\infty} \int_0^{2\delta'} \int_{B(0,\delta)} |F(z', x_1 + iy_1 + iv + i\varepsilon')|^2 d|z'| dv \, dx_1 \, .$$

Now $|z'| < \delta$; we choose $\delta = \sqrt{\frac{\varepsilon'}{2}}$ so that $|z'|^2 < \frac{\varepsilon'}{2}$. Therefore

$$\|F_\varepsilon(0, \cdot)\|^2_{H^2}$$
$$\leq C \cdot \int_{-\infty}^{\infty} \int_{B(0,\delta)} \int_0^{2\delta'} \left| F(z', x_1 + i(|z'|^2 + y_1 + v + \varepsilon'/2 + (\varepsilon'/2 - |z'|^2))) \right|^2 \, dv \, d|z'| \, dx_1 \, .$$

Next set $\widetilde{v} = v + \frac{\varepsilon}{2} - |z'|^2$ and observe that

$$\int_0^{2\delta'} |F(0, v + \frac{\varepsilon'}{2} - |z'|^2)| \, dv \leq \int_0^{2\delta' + \frac{\varepsilon'}{2}} |F(0, \widetilde{v})| \, d\widetilde{v} \, .$$

But we know that $2\delta' + \frac{\varepsilon'}{2} = \varepsilon$ so we have

$$\begin{aligned} \|F_\varepsilon(0, \cdot)\|^2_{H^2} &\leq C \int_0^{\varepsilon} \int_{-\infty}^{\infty} \int_{B(0,\delta)} \left| F(z', x_1 + i(|z'|^2 + y_1 + v + \varepsilon'/2)) \right|^2 d|z'| \, dx_1 \, dv \\ &\leq C \int_0^{\varepsilon} \|F\|^2_{H^2(\mathcal{U})} \, dv \\ &= C \cdot \varepsilon \cdot \|F\|^2_{H^2(\mathcal{U})} \\ &< \infty \, . \end{aligned}$$

This completes the proof. □

Notice that this lemma is not necessarily true for the boundary limit function $F(z', z_1 + i|z'|^2)$. For the constant $C_\delta \sim \delta^{-n}$, hence the righthand side blows up as $\varepsilon \to 0^+$.

We are finally in a position to bring our calculations together and to prove Theorem 4.8.7. We have seen that, from a given $\widehat{f} \in \widehat{H^2}$, we obtain a function $F(z_1, z')$, holomorphic in $\mathcal{U}$. We now show that this function is in $H^2$ and in fact that its $H^2$ norm equals $\|\widehat{f}\|_{\widehat{H^2}}$.

Now

$$\int \int_{-\infty}^{\infty} |F(z', z_1 + i\rho + i|z'|^2)|^2 d|z_1| d|z'| = \int \int_0^{\infty} |\widehat{F}(z', \lambda)|^2 e^{-4\pi\lambda(|z'|^2 + \rho)} d\lambda d|z'|$$

and the integral on the right increases to $\int_{\mathbb{C}^n}\int_0^\infty |\widehat{F}(z',\lambda)|^2 e^{-4\pi\lambda|z'|^2} d\lambda d|z'|$ as $\rho \to 0^+$. Hence

$$\begin{aligned}\|F\|^2_{H^2(\mathcal{U})} &= \sup_{\rho>0} \int\int |F(z', z_1 + i|z'|^2 + i\rho)|^2 d|z_1| d|z'| \\ &= \int\int |\widehat{F}(z',\lambda)|^2 e^{-4\pi\lambda|z'|^2} d\lambda d|z'| \\ &= \|\widehat{f}\|^2_{\widehat{H^2}}.\end{aligned}$$

Furthermore, this equality of norms implies that our map from $\widehat{H}^2$ to $H^2$ is injective. All that remains is to show that an arbitrary $F \in H^2$ has such a representation.

Given $F = F(z', z_1) \in H^2(\mathcal{U})$, Lemma 4.8.10 tells us that, for any fixed $\varepsilon > 0$ and $z' \in \mathbb{C}^n$, the function $F_\varepsilon(z', z_1) \equiv F(z', z_1 + i\varepsilon)$ has a classical Paley–Wiener representation. We leave it as an exercise to check that the resulting function $F_\varepsilon(z', \lambda)$ is holomorphic in $z'$. Since we have the relation

$$F_\varepsilon(z', z_1) = \int_0^\infty \widehat{F}_\varepsilon(z', \lambda) e^{2\pi\lambda z_1} d\lambda$$

and since the functions $\{F_\varepsilon\}$ are uniformly bounded in $\widehat{H}^2$ as $\varepsilon \to 0$, it follows that the functions $\{\widehat{F}_\varepsilon\}$ are uniformly bounded in $\widehat{H^2}$. We can therefore extract a subsequence $F_{\varepsilon_j}$ such that $\widehat{F}_{\varepsilon_j} \to f_0$ weakly as $j \to \infty$. Observe that, since $F_0 \in \widehat{H^2}$, we can recover $\widehat{f}$ from $F_0 \in H^2(\mathcal{U})$.

Lemma 4.8.9 tells us that, for $(z_1, z') \in K \subset\subset \mathcal{U}$, the (continuous) linear functional on $\widehat{H^2}$ given by Fourier inversion and then evaluation at the point $(z_1, z')$ is uniformly bounded:

$$|F(z', z_1)| \leq C_K \cdot \|\widehat{F}\|_{\widehat{H^2}}.$$

Thus $F_{\varepsilon_j}(z', z_1) \to F_0(z', z_1)$ uniformly on compact subsets of $\mathcal{U}$. However,

$$F_{\varepsilon_j}(z', z_1) \equiv F(z', z_1 + i\varepsilon_j) \to F(z', z_1)$$

pointwise, so we know that $F_0 \equiv \widehat{f}$. Thus $F$ has a representation in terms of a function in $\widehat{H^2}$ because $F_\varepsilon$ does.

Finally we must show that, if $F_\varepsilon$ is defined as above, then $F_\varepsilon$ converges to the function $\widehat{f}$ in $L^2(\partial\mathcal{U})$. But we see that

$$F_\varepsilon(z) = F(z', z_1 + i\varepsilon) = \int_0^\infty e^{2\pi\lambda z_1} \cdot e^{-2\pi\lambda\varepsilon} \widehat{F}(z', \lambda) d\lambda$$

so that

$$\int_{\mathbb{H}^{n-1}} |F(z', z_1 + i\varepsilon)|^2 d|z'|d|z_1| = \int\int e^{-4\pi\varepsilon\lambda} |\widehat{F}(z', \lambda)|^2 e^{-4\pi\lambda|z'|^2} d|z'|d\lambda$$

and

$$\int_{\mathbb{H}^{n-1}} |F_{\varepsilon_1}(z) - F_{\varepsilon_2}(z)|^2 d|z'|d|z_1| = \int\int |e^{-2\pi\lambda\varepsilon_1} - e^{-2\pi\lambda\varepsilon_2}|^2 \cdot |\widehat{F}(z', \lambda)|^2 e^{-4\pi\lambda|z'|^2} d|z'|d\lambda.$$

Thus the dominated convergence theorem tells us that $\{F_\varepsilon\}$ is a Cauchy sequence in $L^2(\mathbb{H}^{n-1})$. Therefore $F$ has boundary values in $\mathcal{L}^2(\mathbb{H}^{n-1})$. □

As a direct consequence of Lemma 4.8.10, we have the following corollary:

**Corollary 4.8.11** *The space $H^2(\mathcal{U})$ is a Hilbert space with reproducing kernel.*

The reproducing kernel for $H^2$ is the Szegő kernel; we shall see, by symmetry considerations, that it is uniquely determined up to a constant. We let $S(z, w)$ denote the reproducing kernel for $H^2(\mathcal{U})$.

Although it may not be immediately apparent from the statement of Theorem 4.8.4, it will turn out that the Szegő kernel is a singular integral kernel on the Heisenberg group. This will be important for our applications.

Now we shall examine the proof of Theorem 4.8.4. Before we prove the theorem we will formulate an important corollary. Since all our constructs are canonical, the Szegő representation ought to be modeled on a simple convolution operator on the Heisenberg group. Let us determine how to write the reproducing formula as a convolution.

A function $F$ defined on $\mathcal{U}$ induces, for each value of the "height" $\rho$, a function on the Heisenberg group:

$$F_\rho(\zeta, t) = F\left(\zeta, t + i(|\zeta|^2 + \rho)\right).$$

Since $S(z, w)$ is the reproducing kernel, we know that

$$F(z) = \int_{\mathbb{H}^{n-1}} F(w) S(z, w) d\beta(w) \tag{4.8.12}$$

where $d\beta(w) = dw'du_1$ is the Haar measure on $\mathbb{H}^{n-1}$ with $w$ written as $w = (u_1 + iv_1, w')$. Recall that part **(3)** of Theorem 4.8.7 guarantees the existence of $L^2$ boundary values for $F$, and the boundary of $\mathcal{U}$ is $\mathbb{H}^{n-1}$. Thus the integral (4.8.12) is well defined.

Observe that, since the Heisenberg group is *not* commutative, we must be careful when discussing convolutions. We will deal with *right* convolutions, namely an integral of right translates of the given function $F$:

$$(F * K)(z) = \int_{\mathbb{H}^{n-1}} F(z \cdot y^{-1}) K(y) dy = \int_{\mathbb{H}^{n-1}} F(y) K(y^{-1} \cdot z) dy.$$

The result we seek is

**Corollary 4.8.13** *We have that*

$$F_\rho(\zeta, t) = F_0 * K_\rho(\zeta, t),$$

*where $F_0$ is the $L^2$ boundary limit of $F$, and*

$$K_\rho(\zeta, t) = 2^{n+1} c_n (|\zeta|^2 - it + \rho)^{-n-1}.$$

**Proof of the Corollary (Assuming the Theorem)** We write

$$F_\rho(\zeta, t) = \int_{\mathbb{H}^{n-1}} S\left((\zeta, t + i\rho + i|\zeta|^2), (w', u_1 + i|w'|^2)\right) F(w', u_1 + i|w'|^2) d\beta(w).$$

Therefore

$$\begin{aligned}
&F_\rho(\zeta, t) \\
&= \int_{\mathbb{H}^{n-1}} c_n \left( \frac{i}{2}(u_1 - i|w'|^2 - t - i\rho - i|\zeta|^2) - \sum_{k=2}^{n} \zeta_k \overline{w}_k \right)^{-n-1} F(w', u_1 + i|w'|^2) d\beta(w) \\
&= \int_{\mathbb{H}^{n-1}} 2^{n+1} c_n \frac{F(w', u_1 + i|w'|^2)}{\left[ |\zeta|^2 + |w'|^2 - 2\mathrm{Re}\, \zeta \cdot \overline{w}' + \rho - i(t - u_1 + 2\mathrm{Im}\, \zeta \cdot \overline{w}') \right]^{n+1}} d\beta(w) \\
&= \int_{\mathbb{H}^{n-1}} F_0(w', u_1) K_\rho \left( (\zeta, t)^{-1} \cdot (u_1, w') \right) d\beta(w) .
\end{aligned}$$

That completes the proof. □

**Proof of Theorem 4.8.4** First we need the following elementary uniqueness result from complex analysis:

We know that if $\rho(z, w)$ is holomorphic in $z$ and antiholomorphic in $w$ then it is uniquely determined by $\rho(z, z) = \rho(z)$. Next we demonstrate

**Claim (i):** If $g$ is an element of $\mathbb{H}^{n-1}$ then $S(gz, gw) \equiv S(z, w)$.
After all, if $F \in H^2(\mathcal{U})$ then the map $F \mapsto F_g$ (where $F_g(z) = F(g^{-1}z)$) is a unitary map of $H^2(\mathcal{U})$ to itself. Now

$$F(g^{-1}z) = \int_{\mathbb{H}^{n-1}} S(z, w) F(g^{-1}w) d\beta(w).$$

We make the change of variables $w' = gw$; since $d\beta$ is Haar measure, it follows that $d\beta(w') = d\beta(w)$. Thus

$$F(g^{-1}z) = \int_{\mathbb{H}^{n-1}} S(z, gw') F(w') d\beta(w')$$

so

$$F(z) = \int_{\mathbb{H}^{n-1}} S(gz, gw)F(w)d\beta(w).$$

We conclude that $S(z, w)$ and $S(gz, gw)$ are *both* reproducing kernels for $H^2(\mathcal{U})$ hence they are equal.

**Claim (ii):** If $\delta$ is the natural dilation of $\mathcal{U}$ by $\delta(z_1, z') = (\delta^2 z_1, \delta z')$ then

$$S(\delta z, \delta w) = \delta^{-2n-2} S(z, w).$$

The proof is just as above:

$$\begin{aligned} F(\delta z) &= \int_{\mathbb{H}^{n-1}} S(z, w)F(\delta w)d\beta(w) \\ &= \int_{\mathbb{H}^{n-1}} S(z, \delta^{-1}w')F(w') \cdot \delta^{-2n-2} d\beta(w) \end{aligned}$$

so that

$$F(z) = \delta^{-(2n+2)} \cdot \int_{\mathbb{H}^{n-1}} S(\delta^{-1}z, \delta^{-1}w)F(w)d\beta(w).$$

Then the uniqueness of the reproducing kernel yields

$$S(z, w) = \delta^{-(2n+2)} \cdot S(\delta^{-1}z, \delta^{-1}w) \text{ for } \delta > 0.$$

Now the uniqueness result following Theorem 4.8.4 shows that $S(z, w)$ will be completely determined if we can prove that

$$S(z, z) = c_n \cdot [\rho(z)]^{-n-1}.$$

However, $\rho(z)$ is invariant under translation of $\mathcal{U}$ by elements of the Heisenberg group (i.e., $\rho(gz) = \rho(z)$, for all $g \in \mathbb{H}^{n-1}$) and

$$\rho(\delta z) = \text{Im}\,(\delta^2 z_1) - \delta^2 |z'|^2 = \delta^2 \rho(z).$$

Therefore the function

$$S(z, z) \cdot [\rho(z)]^{n+1}$$

has homogeneity zero and is invariant under the action of the Heisenberg group. Since the Heisenberg group acts simply transitively on "parallels" to $\partial\mathcal{U}$, and since dilations enables us to move from any one parallel to another, any function with these two invariance properties must be constant. Hence we have

$$S(z,z) \equiv c_n[\rho(z)]^{-n-1}.$$

It follows that

$$S(z,w) = c_n[\rho(z,w)]^{-n-1}.$$

At long last we have proved Theorem 4.8.4. We have not taken the trouble to calculate the exact value of the constant in front of the canonical kernel. That value has no practical significance for us here. □

## Exercises

1. Theorem 3.8.8 presents an approach to singular integrals that is a bit different from the classical $\Omega(x)/|x|^N$ idea with $\Omega$ having mean value zero. The latter approach is generally attributed to Calderón and Zygmund, while the approach in Theorem 3.7.6 appears in [STE4] and is discussed in [HOR7]. For many purposes, this point of view is the more flexible and useful one. Formulate and prove a version of singular integrals on the Heisenberg group from the point of view of Theorem 3.8.8.
2. The classical Calderón–Zygmund theory concerns $L^p$ boundedness of singular integral operators. But there is also interest in how the singular integral operators act on other Banach spaces of functions. Consider the Lipschitz spaces. One must be careful with those spaces, because they are defined in terms of the supremum norm, and singular integrals are not bounded in the $L^\infty$ norm. Nonetheless, come up with a formulation of boundedness of singular integrals (on $\mathbb{R}^N$ and also on the Heisenberg group) on Lipschitz spaces. The paper [TAI] will give you some hints.
3. It is a standard result in the classical Calderón–Zygmund theory that an integral kernel that is homogeneous of degree $-N$ *must* have mean value zero on spheres in order to give a viable integral operator that is bounded on $L^p$. Formulate and prove such a result on the Heisenberg group.
4. It is classical to formulate the mean-value-zero condition in terms of integration on spherical shells. Show that that point of view is logically equivalent to the mean-value-zero condition formulated in terms of Hausdorff measure on spheres. Do so both in $\mathbb{R}^N$ and on the Heisenberg group.
5. Refer to Exercise 5, Chapter 1 for the definition of Sobolev space.
   What can you say about the projection from the Sobolev space $W^1$ on the sphere to the holomorphic functions on that Sobolev space? Is it a Hilbert space with reproducing kernel? What would the kernel be?

**6.** The Bergman kernel on the ball $B$ in $\mathbb{C}^n$ is given by

$$K(z,\zeta) = \frac{n!}{\pi^n} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^{n+1}} .$$

The important property of the Bergman kernel is that it is the reproducing kernel for the Hilbert space

$$A^2(B) = \left\{ f \text{ holomorphic on } B : \int_B |f(z)|^2 \, dV(z)^{1/2} \equiv \|f\|_{A^2(B)} < \infty \right\} .$$

The space $A^2$ is the Bergman space.

Stokes's theorem suggests that, at least on the ball in $\mathbb{C}^n$, the Bergman kernel should be a (normal) derivative of the Szegő kernel. Formulate and prove such a statement. The reference [KRA23] has results of this nature.

**7.** What is the Poisson–Szegő kernel on the Siegel upper half space $\mathcal{U}$? What can you say about its mapping properties on $L^p$?

**8.** Do an Internet search and determine what the Berezin transform is. What is the Berezin kernel on the Siegel upper half space?

**9.** Give an example of a singular integral on the boundary of the Siegel upper half space that is distinct from the Szegő integral.

**10.** There are intrinsic reasons why a singular integral needs to have a certain homogeneity as specified in the text. If the homogeneity is *less* than that mandated (i.e., the singularity of the kernel is *less* singular), then the induced integral operator will be a smoothing operator. If the homogeneity is greater than that mandated (i.e., the singularity of the kernel is *more* singular), then the induced integral operator will be unbounded on $L^p$. Explain these remarks.

# Chapter 5
# Reproducing Kernels

**Prologue:** Reproducing kernels are a big part of the world of harmonic analysis. Going back to the Poisson kernel and the Cauchy kernel, and evolving later on to the Bergman kernel and the Szegő kernel, and then to more constructive kernels such as the Henkin-Ramirez kernel, this is a whole world of hard analysis and interesting functional analysis. In the present chapter we explore some of these ideas.

## 5.1 Reproducing Kernels

**Capsule:** Here we lay out the basics of reproducing formulas in complex analysis.

The Cauchy integral formula and the Poisson integral formula are perhaps the two most central and important examples of integral reproducing formulas in the classical context. These are examples of *constructive* reproducing formulas (kernels) because the integral formulas (kernels) can often be written down explicitly or perhaps asymptotically (see [KRA1, Ch. 1], [KRA10], [KRA18, Ch. 1], and especially [KRA8]). What is of interest for our purpose here is that there are other integral reproducing formulas, which are canonical in nature, but for which the formulas (kernels) generally *cannot* be written down explicitly. Often the canonical kernels have many attractive features, but the fact that they are not explicit means that we do not necessarily understand their singularities, and therefore it is difficult to analyze them or to make estimates on them.

But there are techniques for making peace between the canonical and the constructive. The techniques presented here, due to Kerzman and Stein [KES], are useful in other contexts.

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## 5.2 Canonical Integral Formulas

**Capsule:** There are canonical integral formulas—these are usually nonconstructive (i.e., they are shown to exist by abstract means)—and constructive integral formulas. The latter are satisfying because they can be quite explicit. But they can only be created on a limited collection of domains.

There are interesting connections between the two types of integral formulas. We begin to explore these ideas here.

In what follows, in $\mathbb{C}^n$, we use $d\sigma$ to represent area measure on a hypersurface. Let $\Omega$ be a smoothly bounded domain in $\mathbb{C}$ or $\mathbb{C}^n$. Define

$$H^2(\Omega) = \left\{ f \text{ holomorphic on } \Omega : \sup_{0<\epsilon<\epsilon_0} \left( \int_{\partial\Omega} |f(\zeta - \epsilon\nu_\zeta)|^2 \, d\sigma(\zeta) \right)^{1/2} < \infty \right\},$$

where $\nu_\zeta$ is the unit outward normal vector at the boundary point $\zeta \in \partial\Omega$. Clearly $H^2(\Omega)$ is a complex linear space. It is commonly called the *Hardy space*. We equip $H^2(\Omega)$ with the norm coming from its definition.

We have the following important preliminary result.

**Lemma 5.2.1** *There is a constant $C = C(K, \Omega)$, depending only on the domain $\Omega$ and on $K$ compact in $\Omega$, such that, if $f \in H^2(\Omega)$, then*

$$\sup_{z\in K} |f(z)| \leq C \cdot \|f\|_{H^2(\Omega)} . \tag{5.2.1.1}$$

It is convenient to prove the lemma using the Bochner–Martinelli formula (see below), for instance. Details are provided in [KRA18, Ch. 1].

With the indicated norm, the space $H^2(\Omega)$ is a Hilbert space. First note that any element of $H^2$ may be canonically identified with an element of $L^2(\partial\Omega)$ (see [KRA1, Ch. 8] and also our Chapter 2 for the details). For the completeness of $H^2(\Omega)$, note that if $\{f_j\}$ is a Cauchy sequence, then it will converge in the $L^2$ topology to some limit function $g$. But the lemma tells us that, for holomorphic functions, $L^2$ convergence implies uniform convergence on compact sets (sometimes called *normal convergence*). Hence the limit function is holomorphic and $L^2$, thus a member of $H^2$.

As a result of the lemma, the point evaluation

$$H^2(\Omega) \ni f \longmapsto f(z) ,$$

for $z$ fixed in $\Omega$, is a bounded linear functional on $H^2$. Let $k_z$ be the Riesz representative of this functional.

We then define a function

$$S : \Omega \times \Omega \to \mathbb{C}$$

by the formula

$$S(z,\zeta) \equiv \overline{k_z(\zeta)}\,.$$

Of course $S$ is easily extended (by Poisson integration) to $\Omega \times \Omega$. The function $S$ is the *Szegő kernel* for the Hilbert space $H^2(\Omega)$.

We see that $S$ is uniquely determined because, again by the Riesz representation theorem, the element $k_z$ for each $z \in \Omega$ is unique.

It is a classical fact, and we shall not provide the details here (but see [KRA1, Ch. 1] for all the particulars), that the kernel $S$ may (at least in principle) be constructed by way of a complete orthonormal basis for $H^2$. To wit, let $\{\varphi_j\}$ be such a basis. Then

$$S(z,\zeta) = \sum_{j=1}^{\infty} \varphi_j(z)\overline{\varphi_j(\zeta)}\,.$$

Here the convergence is in the Hilbert space topology in each variables. And in fact the lemma shows that the convergence is uniform on compact subsets of $\Omega \times \Omega$.

We now need to present some preliminary results about the constructible Bochner–Martinelli kernel and integral representation formula.

**Definition 5.2.2** On $\mathbb{C}^n$ we let

$$\begin{aligned}\omega(z) &\equiv dz_1 \wedge dz_2 \wedge \cdots \wedge dz_n \\ \eta(z) &\equiv \sum_{j=1}^{n} (-1)^{j+1} z_j dz_1 \wedge \cdots \wedge dz_{j-1} \wedge dz_{j+1} \wedge \cdots \wedge dz_n\,.\end{aligned}$$

The form $\eta$ is sometimes called the *Leray form.* We shall often write $\omega(\overline{z})$ to mean $d\overline{z}_1 \wedge \cdots \wedge d\overline{z}_n$ and likewise $\eta(\overline{z})$ to mean $\sum_{j=1}^{n}(-1)^{j+1}\overline{z}_j d\overline{z}_1 \wedge \cdots \wedge d\overline{z}_{j-1} \wedge d\overline{z}_{j+1} \wedge \cdots \wedge d\overline{z}_n$.

**Prelude** We next introduce our first constructive integral formula. This formula has several interesting features:

**(a)** The formula is the same for every smoothly bounded domain.
**(b)** The formula is completely explicit.
**(c)** In one complex variable the formula reduces to the classical Cauchy integral formula.
**(d)** While the formula reproduces holomorphic functions, it does *not* create holomorphic functions. That is to say, the kernel is not holomorphic in the free variable $z$.

**Theorem 5.2.3 (Bochner–Martinelli)** *Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain with $C^1$ boundary. Let $f \in C^1(\overline{\Omega})$. Then, for any $z \in \Omega$, we have*

$$f(z) = \frac{1}{nW(n)} \int_{\partial\Omega} \frac{f(\zeta)\eta(\overline{\zeta} - \overline{z}) \wedge \omega(\zeta)}{|\zeta - z|^{2n}} - \frac{1}{nW(n)} \int_{\Omega} \frac{\overline{\partial} f(\zeta)}{|\zeta - z|^{2n}} \wedge \eta(\overline{\zeta} - \overline{z}) \wedge \omega(\zeta) \, .$$

**Remark** We see that the Bochner–Martinelli formula is a quintessential example of a constructible integral formula. The kernel is quite explicit, and it is the same for all domains. For the Bergman kernel, and for other canonical kernels that we shall see below, this latter property does not hold. □

**Corollary 5.2.4** *If $\Omega \subseteq \mathbb{C}^n$ is bounded and has $C^1$ boundary and if $f \in C^1(\overline{\Omega})$ and $\overline{\partial} f = 0$ on $\Omega$, then*

$$f(z) = \frac{1}{nW(n)} \int_{\partial\Omega} \frac{f(\zeta)\eta(\overline{\zeta} - \overline{z})}{|\zeta - z|^{2n}} \wedge \omega(\zeta) \, . \tag{5.2.4.1}$$

We note that the classical Cauchy integral formula in one complex variable is an immediate consequence of our new Bochner–Martinelli formula.

**Corollary 5.2.5** *In complex dimension 1, the last corollary says that*

$$f(z) = \frac{1}{2\pi i} \oint_{\partial\Omega} \frac{f(\zeta)}{\zeta - z} \, d\zeta \, .$$

Corollary 5.2.4 is particularly interesting. Like the classical Cauchy formula, it gives a constructible integral reproducing formula that is the same on all domains. Unlike the classical Cauchy formula, its kernel is *not* holomorphic in the free variable $z$. This makes the Bochner–Martinelli formula of limited utility in *constructing* holomorphic functions.

We note that Corollary 5.2.4 holds for broader classes of holomorphic functions—such as the Hardy classes. One sees this by a simple limiting argument. See our discussion of $H^2$ below.

We conclude this section by noting that the integral

$$\mathbf{S}g(z) = \int_{\partial\Omega} S(z, \zeta) g(\zeta) \, d\zeta$$

defines a projection from $L^2(\partial\Omega)$ to $H^2(\Omega)$. This is because the mapping is self-adjoint, idempotent, and fixes $H^2$. We call this mapping the *Szegő projection*. [Note that the Bergman projection is constructed similarly.] However, the Bochner–Martinelli kernel in Corollary 5.2.4 does *not* give a projection. Why not?

## 5.3 Formulas with Holomorphic Kernel

In one complex variable it is easy to construct integral formulas. The Cauchy formula is quite trivial to write down. And it is the same kernel for any domain. One may also write down formulas on the ball and polydisc in $\mathbb{C}^n$. After that things become complicated. Certainly one should mention here the classic work [HUA] in which the Bergman kernel is calculated for each of the Cartan bounded symmetric domains.

It was not until about 1970 that people found ways to explicitly write down integral reproducing formulas with holomorphic kernels in several complex variables. Here we discuss the idea. [It should be noted that both Bungart [BUN] and Gleason [GLE] proved some time ago—by abstract means—that reproducing kernels that are holomorphic in the free variable *always* exist. But the methods of [BUN] and [GLE] are nonconstructive, and the proofs quite abstract.]

Fix a non-negative integer $k$ and a strongly pseudoconvex domain $\Omega \subset\subset \mathbb{C}^n$ with $C^{k+3}$ boundary. Let $\rho : \mathbb{C}^n \to \mathbb{R}$ be a $C^{k+3}$ defining function for $\Omega$ with the property that it is *strictly plurisubharmonic* in a neighborhood of $\partial\Omega$. It is a standard fact (see [KRA1, Ch. 3]), for which we do not provide the details, that the function (known as the *Levi polynomial*)

$$L : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$$

given by

$$L_P(z) = L(z, P) \equiv \rho(P) + \sum_{j=1}^{n} \frac{\partial \rho}{\partial z_j}(P)(z_j - P_j) + \frac{1}{2} \sum_{j,k=1}^{n} \frac{\partial^2 \rho(P)}{\partial z_j \partial z_k}(z_j - P_j)(z_k - P_k)$$

satisfies the following properties:

(5.3.1) For each $P \in \mathbb{C}^n$, the function $z \mapsto L(z, P)$ is holomorphic (indeed, it is a polynomial);
(5.3.2) For each $z \in \mathbb{C}^n$, the function $P \mapsto L(z, P)$ is $C^{k+1}$;
(5.3.3) For each $P \in \partial\Omega$, there is a neighborhood $U_P$ such that if $z \in \overline{\Omega} \cap \{w \in U_P : L_P(w) = 0\}$, then $z = P$.

Our goal is to remove the need to restrict to a small neighborhood of $P \in \partial\Omega$ (property (5.3.3)) while preserving properties (5.3.1)–(5.3.2). We proceed through a sequence of lemmas. We shall not provide here all the detailed proofs of the results. These may be found in [KRA1, Ch. 5]. Here we only outline the key ideas.

Following Henkin, we use the notation

$$\Omega_\delta = \{z \in \mathbb{C}^n : \rho(z) < \delta\};$$
$$U_\delta = \{z \in \mathbb{C}^n : |\rho(z)| < \delta\}, \quad \delta > 0\,.$$

Further, let us fix the following constants:
(5.3.4) Choose $\delta > 0$ and $\gamma > 0$ such that

$$\sum_{j,k=1}^{n} \frac{\partial^2 \rho}{\partial z_j \partial \overline{z}_k}(P) w_j \overline{w}_k \geq \gamma |w|^2, \quad \text{all } P \in U_\delta, \quad \text{all } w \in \mathbb{C}^n .$$

(5.3.5) Shrinking $\delta$ if necessary, we may select $\kappa > 0$ so that

$$|\text{grad}\, \rho(z)| \geq \kappa \quad \text{for all } z \in U_\delta .$$

(5.3.6) With $\delta$ as above, let

$$K = \sum_{|\alpha|+|\beta| \leq 3} \left\| \left(\frac{\partial}{\partial z}\right)^\alpha \left(\frac{\partial}{\partial \overline{z}}\right)^\beta \rho(z) \right\|_{L^\infty(U_\delta)} .$$

**Lemma 5.3.7** *There is a $\lambda > 0$ such that, if $P \in \partial\Omega$ and $|z - P| < \lambda$, then*

$$2\text{Re}\, L_P(z) \leq \rho(z) - \gamma |z - P|^2/2 .$$

**Corollary 5.3.8** *Let $\epsilon = \gamma\lambda^2/20$. If $P \in \partial\Omega, z \in \Omega_\epsilon, \lambda/3 \leq |z - P| \leq 2\lambda/3$, then*

$$\text{Re}\, L_P(z) < 0 .$$

We may assume that $\epsilon < \lambda < \delta < 1$ (where $\delta$ is as in (5.3.4)–(5.3.6)). Let $\eta : \mathbb{R} \to [0, 1]$ be a $C^\infty$ function that satisfies $\eta(x) = 0$ for $x \geq 2\lambda/3$ and $\eta(x) = 1$ for $x \leq \lambda/3$.

**Lemma 5.3.9** *Fix $P \in \partial\Omega$. The $(0, 1)$ form*

$$f_P(z) = \begin{cases} -\overline{\partial}_z \{\eta(|z - P|)\} \cdot \log L_P(z) & \text{if } |z - P| < \lambda, z \in \Omega_\epsilon \\ 0 & \text{if } |z - P| \geq \lambda, z \in \Omega_\epsilon \end{cases}$$

*is well-defined (if we take the principal branch for logarithm) and has $C^\infty$ coefficients for $z \in \Omega_\epsilon$. If $z$ is fixed, then $f_P(z)$ depends $C^k$ on $P$. Finally, $\overline{\partial}_z f_P(z) = 0$ on $\Omega_\epsilon$. [One may note that this construction is valid even for $P$ sufficiently near $\partial\Omega$.]*

**Lemma 5.3.10** *There is a $C^\infty$ function $u_P$ on $\Omega_\epsilon$ such that $\overline{\partial} u_P = f_P$.*

This is just Hörmander's result—see Chapter 4 of [KRA1].

We now define

$$\Phi(z, P) = \begin{cases} [\exp u_P(z)] \cdot L_P(z) & \text{if } |z - P| < \lambda/3 \\ \exp\,[u_P(z) + \eta(|z - P|) \log L_P(z)] & \text{if } \lambda/3 \leq |z - P| < \lambda \\ \exp\,(u_P(z)) & \text{if } \lambda \leq |z - P| . \end{cases}$$

Notice that $\Phi$ is unambiguously defined. To study the properties of $\Phi$, we require two technical results.

**Lemma 5.3.11** *If $U \subseteq \mathbb{C}^n$ is any open set and $K \subset\subset U$, then any $u \in C^1(U)$ satisfies*

$$\sup_K |u| \leq C \left( \|u\|_{L^2(U)} + \|\overline{\partial} u\|_{L^\infty(U)} \right) .$$

*Here the constant $C$ depends on $U$ and $K$ but not on $u$.*

**Corollary 5.3.12** *Let $\Omega \subset\subset \mathbb{C}^n$ be pseudoconvex and $K \subset\subset \Omega$. Let $f$ be a $\overline{\partial}$–closed $(0,1)$ form with $C^1$ coefficients. If $u = Mf$ is the Hörmander solution to $\overline{\partial} u = f$ (see [KRA1, Ch. 4] and Section 10 below), then we have*

$$\|u\|_{L^\infty(K)} \leq C \|f\|_{L^\infty(\Omega)} ,$$

*where $C$ depends only on $K$ and $\Omega$ (and not on $f$ or $u$).*

**Proposition 5.3.13** *Assume once more that $\Omega \subset\subset \mathbb{C}^n$ has $C^{k+3}$ boundary. Then $\Phi(\cdot, P)$ is holomorphic on $\Omega_\epsilon$. Also there is a $C > 0$, independent of $P$, such that for all $z \in \Omega_{\epsilon/2}$ we have*

*(5.3.13.1)* *if $|z - P| < \lambda/3$, then* $|\Phi(z,P)| \geq C|L_P(z)|$;

*(5.3.13.2)* *if $|z - P| \geq \lambda/3$, then* $|\Phi(z,P)| \geq C$.

Now we would like to consider the smooth dependence of $\Phi$ on $P$.

Now fix $z \in \Omega$. Let $\theta \in C_c^\infty(\Omega_\epsilon)$ satisfy $\theta(z) = 1$. Let $s > 2n$. Let $M_s^\theta$ be the right inverse to $\overline{\partial}_{0,0}$ (the Hörmander solution operator) for the pseudoconvex domain $\Omega_\epsilon$. Let notation be as in (5.3.4) through (5.3.6). Let $\mu$ be in the dual space of $W^s(\Omega, \phi_1)$ [naturally this dual space is just $W^s(\Omega, \phi_1)$ itself]. Then

$$\langle \mu, M_s^\theta f_P \rangle = \langle (M_s^\theta)^* \mu, f_P \rangle ,$$

which depends $C^k$ on $P$ because $f_P$ does.

**Proposition 5.3.14** *The function $\Phi(z,P)$ depends in a $C^k$ fashion on $P$ for fixed $z \in \Omega_\epsilon$.*

**Proposition 5.3.15** *Let $\Omega \subseteq \mathbb{C}^n$ be pseudoconvex. Let $\Omega_m = \Omega \cap \{z \in \mathbb{C}^n : z_1, \ldots, z_m = 0\}, m = 1, \ldots, n$. Let $A_m(\Omega) = \{f \text{ holomorphic on } \Omega : f|_{\Omega_m} = 0\}$. Then there are linear operators*

$$Q_i^m : A_m(\Omega) \to \{f \text{ holomorphic on } \Omega\} \ , \quad i = 1, \ldots, m ,$$

*such that*

$$f(z) = \sum_{i=1}^m z_i \cdot (Q_i^m f)(z)$$

*for all $f \in A_m(\Omega)$.*

**Prelude** Key in the result that we are about to present is the following extension result. It is worth noting that Theorem 5.3.16 is essentially logically equivalent to the Levi problem.

**Theorem 5.3.16** *Let $\Omega \subseteq \mathbb{C}^n$ be pseudoconvex (no assumptions about boundary smoothness, or even boundedness, need be made). Let $\omega = \Omega \cap \{(z_1, \ldots, z_n) : z_n = 0\}$. Let $f : \omega \to \mathbb{C}$ satisfy the property that the map*

$$(z_1, \ldots, z_{n-1}) \mapsto f(z_1, \ldots, z_{n-1}, 0)$$

*is holomorphic on $\widetilde{\omega} = \{(z_1, \ldots, z_{n-1}) \in \mathbb{C}^{n-1} : (z_1, \ldots, z_{n-1}, 0) \in \omega\}$. Then there is a holomorphic $F : \Omega \to \mathbb{C}$ such that $F|_\omega = f$. Indeed there is a linear operator*

$$\mathcal{E}_{\omega,\Omega} : \{\text{holomorphic functions on } \omega\} \to \{\text{holomorphic functions on } \Omega\}$$

*such that $(\mathcal{E}_{\omega,\Omega} f)|_\omega = f$. The operator is continuous in the topology of normal convergence.*

**Corollary 5.3.17** *Let $\Omega \subseteq \mathbb{C}^n$ be pseudoconvex. Then there are continuous linear operators*

$$T_i : \{\textit{holomorphic functions on } \Omega\} \to \{\textit{holomorphic functions on } \Omega \times \Omega\}$$

*such that, for any holomorphic $f : \Omega \to \mathbb{C}$, we have*

$$f(z) - f(w) = \sum_{i=1}^{n} (z_i - w_i) T_i f(z, w), \quad \textit{all } z, w \in \Omega\,.$$

**Proposition 5.3.18 (Hefer's Lemma)** *Let $\Omega \subseteq \mathbb{C}^n$ be strongly pseudoconvex with $C^4$ boundary. Let $\Phi : \Omega_{\epsilon/2} \times \partial\Omega \to \mathbb{C}$ be the $C^1$ singular function constructed above. Then we may write*

$$\Phi(z, \zeta) = \sum_{i=1}^{n} (\zeta_i - z_i) \cdot P_i(z, \zeta), \quad z \in \Omega_{\epsilon/2}, \quad \zeta \in \partial\Omega\,,$$

*where each $P_i$ is holomorphic in $z \in \Omega_{\epsilon/2}$ and $C^1$ in $\zeta \in \partial\Omega$.*

We now quickly review the Cauchy-Fantappié formula. See [KRA1, Ch. 5] for the details.

**Theorem 5.3.19** *Let $\Omega \subset\subset \mathbb{C}^n$ be a domain with $C^1$ boundary. Let $w(z, \zeta) = (w_1(z, \zeta), \ldots, w_n(z, \zeta))$ be a $C^1$, vector-valued function on $\overline{\Omega} \times \overline{\Omega} \setminus \{\text{diagonal}\}$ that satisfies*

$$\sum_{j=1}^{n} w_j(z, \zeta)(\zeta_j - z_j) \equiv 1.$$

*Then, using our earlier notation, we have for any*

$$f \in C^1(\overline{\Omega}) \cap \{\text{holomorphic functions on } \Omega\}$$

*and any* $z \in \Omega$ *the formula*

$$f(z) = \frac{1}{nW(n)} \int_{\partial\Omega} f(\zeta)\eta(w) \wedge \omega(\zeta).$$

We see that the Cauchy–Fantappié formula is a direct generalization of the Bochner–Martinelli formula discussed above. Now we can give the punchline of this development.

**Theorem 5.3.20 (Henkin [HEN1])** *Let* $\Omega \subseteq \mathbb{C}^n$ *be a strongly pseudoconvex domain with* $C^4$ *boundary. Let* $\Phi : \Omega_{\epsilon/2} \times \partial\Omega \to \mathbb{C}$ *be the Henkin singular function. Define*

$$w_i(z,\zeta) = \frac{P_i(z,\zeta)}{\Phi(z,\zeta)}\,,\ i = 1,\ldots,n\,.$$

*Here* $P_i(z,\zeta)$ *are as in Proposition 5.3.18. Just as in our earlier discussion, let*

$$\eta(w) = \sum_{i=1}^{n} (-1)^{i+1} w_i dw_1 \wedge \cdots \wedge dw_{i-1} \wedge dw_{i+1} \wedge \cdots \wedge dw_n$$

*and*

$$\omega(\zeta) = d\zeta_1 \wedge \cdots \wedge d\zeta_n\,.$$

*Then, for any* $f \in C^1(\overline{\Omega}) \cap \{\text{holomorphic functions on } \Omega\}$, *we have the integral representation*

$$f(z) = \int_{\partial\Omega} f(\zeta)\eta(w) \wedge \omega(\zeta)\,.$$

**Corollary 5.3.21** *With notation as in the theorem, we have*

$$f(z) = \int_{\partial\Omega} f(\zeta) \frac{K(z,\zeta)}{\Phi^n(z,\zeta)} d\sigma(\zeta)\,, \tag{5.3.21.1}$$

*where* $K : \Omega_{\epsilon/2} \times \partial\Omega$ *is holomorphic in* $z$ *and continuous in* $\zeta$. *In fact,* $K(z,\zeta)d\sigma(\zeta) = \eta(z) \wedge \omega(\zeta)$.

Of course Corollary 5.3.21 gives us a constructive integral reproducing formula with kernel that is holomorphic in the free $z$ variable. This is a very useful device, and important for the function theory of several complex variables.

## 5.4 Asymptotic Expansion for the Kernel

C. Fefferman [FEF1] made an important contribution in 1974 when he produced an asymptotic expansion for the Bergman kernel of a strongly pseudoconvex domain. Basically he was able to write

$$K(z,\zeta) = P(z,\zeta) + \mathcal{E}(z,\zeta)\,,$$

where $P$ (the principal term) is, in suitable local coordinates, the Bergman kernel of the ball and $\mathcal{E}$ (the error term) is a term of strictly lower order (in some measurable sense). This powerful formula gives one a means for calculating mapping properties of the Bergman integral. Fefferman himself used the formula to calculate the boundary asymptotics of Bergman metric geodesics (for the purpose of proving the smooth boundary extension of biholomorphic mappings). Fefferman states in his paper—although the details have never been worked out—that there is a similar asymptotic expansion for the Szegő kernel of a strongly pseudoconvex domain.

At about the same time, Boutet de Monvel and Sjöstrand [BOUTS] used the technique of Fourier integral operators [HOR3, HOR6] to directly derive an asymptotic expansion for the Szegő kernel of a strongly pseudoconvex domain. This expansion is quite similar to Fefferman's: there is a principal term, which in suitable local coordinates is the Szegő kernel of the ball, and there is an error term which is of lower order. Theorem 1.5 of [BOUTS] treats means of applying these Fourier integral operator techniques to the study of the Bergman kernel on strongly pseudoconvex domains.

The main purpose of the present discussion is to consider another method, due to Kerzman and Stein, for deriving asymptotic expansions for the canonical kernels that is more elementary and uses less machinery. At this time there are virtually no results about asymptotic expansions for the canonical kernels on weakly pseudoconvex domains. Some interesting partial results appear in [HAN]. See also [KRA8].

## 5.5 Constructive Kernels vs. Canonical Kernels

In previous sections, we have defined the Szegő projection $\mathbf{S} : L^2(\partial\Omega) \to H^2(\Omega)$. We also have a mapping $\mathbf{H} : L^2(\partial\Omega) \to H^2(\Omega)$ that is determined by the Henkin kernel of Corollary 5.3.21. We note that $\mathbf{H}$ defines a bounded operator from $L^2(\partial\Omega)$ to $H^2(\Omega)$ (the Hardy space—see [KRA1, Chapter 8]) for the following reason.

It is known that $\partial\Omega$, when equipped with balls coming from the complex structure and the usual boundary area measure (see [KRA1, Ch. 8]), is a space of homogeneous type in the sense of Coifman and Weiss [COG1, Ch. 1]. Further, it is straightforward to verify that the Henkin operator $\mathbf{H}$ satisfies the hypotheses of the David–Journé $T1$ theorem for spaces of homogeneous type (see [CHST1, Ch. 4] for

a nice exposition of these ideas). Thus we may conclude that the Henkin operator maps $L^2(\partial\Omega)$ to $L^2(\partial\Omega)$. Since the Henkin kernel also obviously maps $L^2(\partial\Omega)$ to holomorphic functions, we may conclude that the Henkin integral maps $L^2(\partial\Omega)$ to $H^2(\Omega)$. This mapping, however, is *not* a projection. For instance, the kernel is not self adjoint.

Now of course $\mathbf{S}$, being a projection, is self-adjoint. So $\mathbf{S} = \mathbf{S}^*$. It is not at all true that $\mathbf{H} = \mathbf{H}^*$, but one may calculate that $\mathcal{A} \equiv \mathbf{H}^* - \mathbf{H}$ is small in a measurable sense.

We also have

$$\mathbf{HS} = \mathbf{S}\,, \qquad \mathbf{SH}^* = \mathbf{S}\,,$$

$$\mathbf{SH} = \mathbf{H}\,, \qquad \mathbf{H}^*\mathbf{S} = \mathbf{H}^*\,.$$

Let us discuss these four identities for a moment.

For the first, notice that $\mathbf{S}$ is the projection onto $H^2$, and $\mathbf{H}$ preserves holomorphic functions. So certainly $\mathbf{HS} = \mathbf{S}$. For the second, we calculate that

$$\langle \mathbf{SH}^* x, y\rangle = \langle \mathbf{H}^* x, \mathbf{S}y\rangle = \langle x, \mathbf{HS}y\rangle = \langle x, \mathbf{S}y\rangle$$

(because $\mathbf{H}$ preserves holomorphic functions) and thus $= \langle \mathbf{S}x, y\rangle$. Hence $\mathbf{SH}^* = \mathbf{S}$. For the third, notice that $\mathbf{H}$ maps into the holomorphic functions, and $\mathbf{S}$ preserves holomorphic functions. And, for the fourth, we calculate that

$$\langle \mathbf{H}^*\mathbf{S}x, y\rangle = \langle \mathbf{S}x, \mathbf{H}y\rangle = \langle x, \mathbf{SH}y\rangle = \langle x, \mathbf{H}y\rangle = \langle \mathbf{H}^* x, y\rangle\,.$$

In conclusion, $\mathbf{H}^*\mathbf{S} = \mathbf{H}^*$.

Now we see that

$$\mathbf{S}\mathcal{A} = \mathbf{S}(\mathbf{H}^* - \mathbf{H}) = \mathbf{SH}^* - \mathbf{SH} = \mathbf{S} - \mathbf{H}\,.$$

As a result,

$$\mathbf{S} = \mathbf{H} + \mathbf{S}\mathcal{A}$$

so

$$\mathbf{S}(\mathbf{I} - \mathcal{A}) = \mathbf{H}\,.$$

In conclusion,

$$\mathbf{S} = \mathbf{H}(\mathbf{I} - \mathcal{A})^{-1}\,.$$

If indeed we know that $\mathcal{A}$ is norm small in a suitable sense (details omitted), then $(\mathbf{I} - \mathcal{A})^{-1}$ is well defined by a Neumann series. Thus we may write

$$\mathbf{S} = \mathbf{H} + \mathbf{H}\mathcal{A} + \mathbf{H}\mathcal{A}^2 + \cdots + \mathbf{H}\mathcal{A}^j + \mathbf{H}\mathcal{A}^{j+1} + \cdots .$$

Hence we have expressed the Szegő projection $\mathbf{S}$ as an asymptotice expansion in terms of the Henkin projection $\mathbf{H}$. By applying this asymptotic expansion to the Dirac delta mass, this last formula can be translated into saying that the Szegő *kernel* $S$ can be written as an asymptotic expansion in terms of the Henkin *kernel*.

It should be noted that Ewa Ligocka [LIG1] has shown that these same ideas may be applied to expand the Bergman kernel in an asymptotic expansion in terms of the Henkin kernel. We shall not treat the details of her argument here.

## Exercises

**1.** *The automorphism group of the ball.* Let $B \subseteq \mathbb{C}^2$ be the unit ball. Complete the following outline to calculate the set of biholomorphic self-maps of $B$.

**(a)** Let $a \in \mathbb{C}, |a| < 1$. Then

$$\phi_a(z_1, z_2) = \left( \frac{z_1 - a}{1 - \bar{a}z_1}, \frac{\sqrt{1 - |a|^2} z_2}{1 - \bar{a}z_1} \right)$$

is a biholomorphic mapping of $B$ to itself.

**(b)** If $\rho$ is a unitary mapping of $\mathbb{C}^2$ (that is, the inverse of $\rho$ is its conjugate transpose; equivalently, $\rho$ preserves the Hermitian inner product on $\mathbb{C}^2$) then $\rho$ is a biholomorphic mapping of the ball.

**(c)** Prove that if $\psi : B \to B$ is holomorphic and $\psi(0) = 0$ then all eigenvalues of Jac $\psi(0)$ have modulus not exceeding 1 (*Hint:* apply the one-variable Schwarz lemma in a clever way).

**(d)** Apply part **(c)** to any biholomorphic mapping of $B$ that preserves the origin, and to its inverse, to see that such a mapping has Jacobian matrix at the origin with all eigenvalues of modulus 1.

**(e)** Use the result of part **(d)** to see that any biholomorphic mapping of the ball that preserves the origin must be linear, indeed unitary.

**(f)** If now $\alpha$ is any biholomorphic mapping of the ball, choose a unitary mapping $\rho$ and a complex constant $a$ such that $\phi_a \circ \rho \circ \alpha$ is a biholomorphic mapping that preserves the origin. Thus, by part **(e)**, $\phi_a \circ \rho \circ \alpha$ is unitary.

**(g)** Conclude that the mappings $\phi_a$ and the unitary mappings generate the automorphism group of the ball in $\mathbb{C}^2$.

**2.** Let $\mathbf{a}(\partial D)$ be the space of those continuous functions that are the boundary functions of harmonic functions on the disc. Mimic the construction of the Szegö kernel to obtain a reproducing kernel. What reproducing kernel do you obtain? Why is the space $\mathbf{a}(\partial D)$ defined incorrectly (see [EPS, p. 63])?

**3.** Is there a Bergman kernel for the square integrable harmonic functions on the disc? Can you write it down explicitly?

**4.** Derive a formula for the Green's function and the Poisson kernel for the ball $B(x_0, r) \subseteq \mathbb{R}^N$ by using invariance properties of the Laplacian (do not imitate the proof in the text for the unit ball).

**5.** Use the mean value characterization of harmonic functions to prove that the Poisson kernel $P_\Omega(x, t)$ for a domain $\Omega$ must be harmonic in $x$.

**6.** Here is an alternative proof that the Poisson kernel for the ball is harmonic in the $x$-variable (for which I am indebted to C. Berg): Assume without loss of generality that $y = (1, 0, \ldots, 0)$. For $N > 2$ we write

$$\begin{aligned}\omega_{N-1} \cdot P(x, y) &= -\frac{1}{|x-y|^{N-2}} - \frac{2(x-y)\cdot y}{|x-y|^N} \\ &= \frac{-1}{|x-y|^{N-2}} + \frac{1}{N-2}\frac{\partial}{\partial x_1}\left(\frac{2}{|x-y|^{N-2}}\right).\end{aligned}$$

**7.** Suppose that $\Omega \subseteq \mathbb{C}$ is a domain and $f : \Omega \to \mathbb{C}$ is conformal. Show that

$$\int_\Omega |f'(z)|^2 dA(z)$$

is precisely the area of the image of $f$.

**8.** Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with $C^2$ boundary. Let $G(x, y)$ be the Green's function for the Laplacian on $\Omega$. Prove that $G(P, Q) = G(Q, P)$, all $P \neq Q$. In particular, assuming the results of Section 2.2, $G$ extends to a smooth ($C^{2-\epsilon}$) function on $\bar{\Omega} \times \bar{\Omega} \setminus \{(x, x) : x \in \bar{\Omega}\}$ and is harmonic in each variable. [*Hint:* Fix $P, Q \in \Omega, P \neq Q$. Apply Green's theorem on

$$\Omega_\epsilon \equiv \Omega \setminus \left(\bar{B}(P, \epsilon) \cup \bar{B}(Q, \epsilon)\right),$$

$\epsilon$ small, to the functions $G(P, \cdot)$ and $G(Q, \cdot)$.

**9.** Hopf's lemma, originally proved (see R. Courant and D. Hilbert [COU]) for the sake of establishing the maximum principle for solutions of second order elliptic equations, has proved to be a powerful tool in the study of functions of several complex variables. Here we state and outline a proof of this result. While simpler proofs are available (see [KRA5]) this one has the advantage of applying in rather general circumstances.

**Theorem** Let $\Omega \subseteq \mathbb{R}^N$ be a bounded domain with $C^2$ boundary. Let $f : \bar{\Omega} \to \mathbb{R}$ be harmonic and non-constant on $\Omega$, $C^1$ on $\bar{\Omega}$. Suppose that $f$ assumes a (not necessarily strict) maximum at $P \in \partial\Omega$. If $\nu = \nu_P$ is the unit outward normal to $\partial\Omega$ at $P$, then $(\partial f/\partial \nu)(P) > 0$.

**Outline of Proof**

**(a)** Let $B_1$ be a ball internally tangent to $\partial\Omega$ at $P$ with $\partial B_1 \cap \partial\Omega = \{P\}$. Let $r > 0$ be the radius of $B_1$. Assume without loss of generality that the center of $B_1$ is at the origin. Let $B_2$ be a ball centered at $P$ of radius $r_1 < r$. Let $B' = B_1 \cap B_2$. Notice that $\partial B' = S_1' \cup S_2'$.

**(b)** Let $\alpha > 0$ and set $h(x) = e^{-\alpha|x|^2} - e^{-\alpha r^2}$. Then

$$\Delta h = e^{-\alpha|x|^2}\{4\alpha^2|x|^2 - 2\alpha N\}.$$

**(c)** If $\alpha > 0$ is sufficiently large then $\Delta h > 0$ on $B'$.

**(c)** Set $v(x) \equiv f(x) + \epsilon h(x)$. If $\epsilon > 0$ is sufficiently small then $v(x) < f(P)$ for $x \in S_1'$. Also $v(x) = f(x) < f(P)$ for $x \in S_2' \setminus \{P\}$. Use the maximum principle.

**(d)** $\max_{x\in\bar{B'}} v(x) = f(P)$.

**(e)** $\frac{\partial v}{\partial \nu}(P) = \frac{\partial f}{\partial \nu}(P) + \epsilon\frac{\partial h}{\partial \nu}(P) \geq 0$.

**(f)** $\frac{\partial f}{\partial \nu}(P) > 0$.

Now suppose only that $f \in C(\bar{\Omega})$ and harmonic on $\Omega$ and that the point $P \in \partial\Omega$ is a local (not necessarily strict) maximum of $f$. Modify the preceding argument to prove that

$$\liminf_{\epsilon\to 0^+}\left(\frac{f(P) - f(P - \epsilon\nu)}{\epsilon}\right) > 0.$$

**10.** Holomorphic mappings in $\mathbb{C}^2$ are not necessarily conformal (i.e. infinitesimally angle preserving). Show that the example $F(z_1, z_2) = (z_1^2, z_1^2 + z_2)$ confirms this statement.

**11.** Use the classical Hurwitz theorem of one complex variable to give another proof that a holomorphic function of several variables cannot have an isolated zero.

**12.** Are holomorphic functions open? Are holomorphic mappings open?

**13.** Let $z \in \Omega_1 \subseteq \Omega_2 \subseteq \mathbb{C}^n$. Let $K_j$ be the Bergman kernel for $\Omega_j$. Then show that $K_2(z, z) \leq K_1(z, z)$ for all $z \in \Omega_1$. Further show that $\|K_2(z, \cdot)\|_{L^2(\Omega_2)} \leq \|K_1(z, \cdot)\|_{L^2(\Omega_1)}$.

**14.** Let $\Omega \subseteq \mathbb{R}^N$ be a domain and $f$ harmonic on $\Omega$. Let $P \in \Omega$ and assume that the function $\phi$ is compactly supported in $\Omega$ and is radially symmetric about $P$ (that is, $\phi(x) = \phi(x')$ if $|x - P| = |x' - P|$). Further assume that $\int \phi(x)dx = 1$. Then prove that $f(P) = \int f(x)\phi(x)dx$.

**15.** Let $\{a_{ij}\}_{i,j=1}^n$ be complex constants such that $\sum a_{ij}w_i\bar{w}_j \geq C|w|^2$ for some $C > 0$ and all $w \in \mathbb{C}^n$. Let

$$\Omega = \left\{z \in \mathbb{C}^n : -2\mathrm{Re}\, z_1 + \sum_{i,j=1}^n a_{ij}z_i\overline{z}_j < 0\right\}.$$

Prove that the Bergman kernel for $\Omega$ is given on the diagonal by

$$K(z,z) = \frac{n!\det\left(a_{ij}\right)_{i,j=1}^{n}}{\pi^n(2\mathrm{Re}\, z_1 - \sum_{i,j=1}^{n} a_{ij}z_i\bar{z}_j)^{n+1}}.$$

(*Hint:* The domain $\Omega$ is biholomorphic to the ball. See [GRAI] for details.)

**16.** Let $\Omega \subseteq \mathbb{C}^n$ be a domain. Let $\{f_j\}$ be a sequence of holomorphic functions on $\Omega$ that converges pointwise to a function $f$ at every point of $\Omega$. Prove that $f$ is holomorphic on a dense open subset of $\Omega$. [*Hint:* Let $\bar{U} \subseteq \Omega$ be the closure of any open subset $U$ of $\Omega$. Apply the Baire category theorem to the sets $S_M = \{z \in \bar{U} : |f_j(z)| \leq M, \text{all } j\}$.]

**17.** Check that $(\partial/\partial z_j)z_k \equiv \delta_{jk}$, $(\partial/\partial\bar{z}_j)\bar{z}_k \equiv \delta_{jk}$. Also $(\partial/\partial z_j)\bar{z}_k \equiv 0$ and $(\partial/\partial\bar{z}_j)z_k \equiv 0$ for every $j$ and $k$. Finally, $\langle dz_j, \partial/\partial z_k\rangle \equiv \delta_{jk}$, $\langle d\bar{z}_j, \partial/\partial\bar{z}_k\rangle \equiv \delta_{jk}$, $\langle dz_j, \partial/\partial\bar{z}_k\rangle \equiv 0$, $\langle d\bar{z}_j, \partial/\partial z_k\rangle \equiv 0$ for all $j,k$. These equalities make the definitions of $\partial/\partial z_j$, $\partial/\partial\bar{z}_j$, $dz_j$, and $d\bar{z}_j$ seem natural.

Prove the following complex versions of the chain rule (supply the right hypotheses as well)

$$\frac{\partial}{\partial z_j}(f\circ g) = \sum_{k=1}^{n}\frac{\partial f}{\partial z_k}\cdot\frac{\partial g_k}{\partial z_j} + \sum_{\ell=1}^{n}\frac{\partial f}{\partial\overline{z}_\ell}\cdot\frac{\partial\bar{g}_\ell}{\partial z_j},$$

$$\frac{\partial}{\partial\bar{z}_j}(f\circ g) = \sum_{k=1}^{n}\frac{\partial f}{\partial z_k}\cdot\frac{\partial g_k}{\partial\bar{z}_j} + \sum_{\ell=1}^{n}\frac{\partial f}{\partial\overline{z}_\ell}\cdot\frac{\partial\bar{g}_\ell}{\partial\bar{z}_j},$$

How does the first of these equations simplify in the case that either $f$ or $g$ is holomorphic?

**18.** Derive the complex form of Taylor's theorem for a function $f$ (not necessarily holomorphic) on a ball $B(z^0,r) \subseteq \mathbb{C}^n$: if $f \in C^k(B(z^0,r))$, $w \in B(z^0,r)$, then

$$f(w) = \sum_{|\alpha|+|\beta|\leq k}\left(\frac{\partial}{\partial z}\right)^{\alpha}\left(\frac{\partial}{\partial\overline{z}}\right)^{\beta} f(z^0)\cdot\frac{(w-z^0)^\alpha(\bar{w}-\overline{z^0})^\beta}{\alpha!\beta!}$$
$$+o(|z^0-w|^k).$$

**19.** Use the notation of the Bochner–Martinelli formula. Let $\Omega \subset\subset \mathbb{C}^n$ be a domain with $C^1$ boundary. Let $a(\zeta) = (a_1(\zeta),\ldots,a_n(\zeta))$ with $a_j(\zeta) = p_j(\zeta)/q(\zeta), j = 1,\ldots,n$. Suppose that the $a_j, p_j, q$ are $C^1$ functions on $\partial\Omega$. Let $d\sigma$ be area measure on $\partial\Omega$. Then

$$\eta(a)\wedge\omega(\zeta)\Big|_{\partial\Omega} = \frac{h(\zeta)}{q(\zeta)^n}d\sigma(\zeta)$$

for some continuous $h$ on $\partial\Omega$. [*Hint:* This is a purely algebraic fact. Write out the case $n = 2$ explicitly to see it.]

**20.** Here is a useful way to generate an orthonormal basis for the Bergman space. Fix $z_0 \in \Omega$. Let $\phi_0$ be the (unique!) element of $A^2$ with $\phi_0(z_0)$ real, $\|\phi_0\| = 1$, and $\phi_0(z_0)$ maximal. (Why does such a $\phi_0$ exist?) Let $\phi_1$ be the (unique) element of $A^2$ with $\phi_1(z_0) = 0, (\partial\phi_1/\partial z_1)(z_0)$ real, $\|\phi_1\| = 1$, and $(\partial\phi_1/\partial z_1)(z_0)$ maximal. (Why does such a $\phi_1$ exist?) Now $\phi_1$ is orthogonal to $\phi_0$, else $\phi_1$ has non-zero projection on $\phi_0$, leading to a contradiction. Continue this proces to create an orthogonal system on $\Omega$. Use Taylor series to see that it is complete. This circle of ideas comes from the elegant paper [KOB1].

**21.** Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain and let $(g_{ij})$ be its Bergman metric. Prove that the matrix $(g_{ij}(z))$ is positive definite, each $z \in \Omega$. [*Hint:* Refer to the preceding exercise. The crucial fact is that for each $z \in \Omega$ and each $j$ there is an element $f \in A^2(\Omega)$ such that $\partial f/\partial z_j(z) \neq 0$.]

**22.** Let $K$ be the Bergman kernel for the ball $B$. Show that $K(0, 0) = 1/V(B)$. Use the automorphism group of $B$, together with the invariance of the kernel, to calculate $K(z, z)$ for every $z \in B$. The values of $K$ on the diagonal then completely determine $K(z, \zeta)$.

**23.** Let $\Omega$ be the Hartogs triangle $\{(z_1, z_2) : |z_1| < |z_2| < 1\}$. Every analytic function on a neighborhood of $\bar{\Omega}$ continues to the bidisc. Find an orthonormal basis for $A^2(\Omega)$.

# Chapter 6
# More on the Bergman and Szegő Kernels

**Prologue:** In the early days of functional analysis—the early twentieth century—people did not yet know what a Banach space was nor a Hilbert space. They frequently studied a *particular* complete, infinite-dimensional space from a more abstract point of view. The most common space to be studied in this regard was of course $L^2$. It was when Stefan Bergman took a course from Erhard Schmidt on $L^2$ of the unit interval $I$ that he conceived of the idea of the Bergman space of square-integrable holomorphic functions on the unit disc $D$. And the rest is history.

It is important for Bergman's theory that his space of holomorphic functions have an inner product structure, and that it is complete. The first of these properties follows from the fact that it is a subspace of $L^2$; the second follows from a fundamental inequality that we shall consider in the next section.

## 6.1 The Bergman Kernel

**Capsule:** The Bergman kernel is the prototype for all the canonical kernels that exist today. While the idea was anticipated in the thesis work of S. Bochner [BOC], it was Bergman who really developed the idea of the kernel function and established its place in the firmament. It may be said that the Szegő kernel is modeled on the idea of the Bergman kernel (although it actually anticipated the Bergman kernel by a few years), and the same may be said for other kernels that arise in harmonic analysis and the theory of partial differential equations.

It is difficult to create an explicit integral formula, with holomorphic reproducing kernel, for holomorphic functions on an arbitrary domain[1] in $\mathbb{C}^n$. Classical studies which perform such constructions tend to concentrate on domains having a great deal of symmetry (see, for instance, [HUA]). We now examine one of several non-constructive approaches to this problem. This circle of ideas, due to S. Bergman

[1] Here a *domain* is a connected, open set.

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[BERM1, BERM2, Ch. 1, Ch. 2] and to G. Szegő [SZE] (some of the ideas presented here were anticipated by the thesis of S. Bochner [BOC]), will later be seen to have profound applications to the boundary regularity of holomorphic mappings.

Lutz Bungart [BUN] and Andrew Gleason [GLE] have shown that *any* bounded domain in $\mathbb{C}^n$ will have a reproducing kernel for holomorphic functions such that the kernel itself is holomorphic in the free variable. In other words, the formula has the form

$$f(z) = \int_\Omega f(\zeta) K(z, \zeta)\, dV(\zeta)\,,$$

and $K$ is holomorphic in the $z$ variable. Of course Bungart's and Gleason's proofs are highly nonconstructive, and one can say almost nothing about the actual form of the kernel $K$. The venerable Bochner–Martinelli kernel is easily constructed on any bounded domain with reasonable boundary (just as an application of Stokes's theorem) and the kernel is *explicit*—just like the Cauchy kernel in one complex variable. Also the kernel is the same for every domain. But the Bochner–Martinelli kernel is definitely *not* holomorphic in the free variable. On the other hand, G. M. Henkin [HEN1], N. Kerzman [KER1], E. Ramirez [RAM], and Grauert–Lieb [GRAL] have given very explicit constructions of reproducing kernels on strictly pseudoconvex domains (see the definition below). And their kernels *are* holomorphic in the $z$ variable. This matter is treated in [KRA1, Ch. 10].

In fact this last described result was considered to be quite a dramatic advance. For Henkin, Kerzman, Ramirez, and Grauert–Lieb provided us with a fairly explicit kernel, with an explicit and measurable singularity, that can not only reproduce but also create holomorphic functions. Such a kernel is very much like the Cauchy kernel in one complex variable. Thus, at least on strictly pseudoconvex domains, we can perform many of the activities to which we are accustomed from the function theory of one complex variable. We can get formulas for derivatives of holomorphic functions, we can analyze power series, we can consider an analogue of the Cauchy transform, and (perhaps most importantly) we can write down solution operators for the $\overline{\partial}$ problem. People were optimistic that these new integral formulas would give a shot in the arm to the theory of function algebras—that they would now be able to study $H^\infty(\Omega)$ and $A(\Omega)$ on a variety of domains in $\mathbb{C}^n$ (see [GAM, Ch. II, IV] for the role model in $\mathbb{C}^1$). But this turned out to be too difficult.

The Bergman kernel is a canonical kernel that can be defined on any bounded domain. It has wonderful invariance properties, and is a powerful tool for geometry and analysis. But it is difficult to calculate explicitly.

In this section we will see some of the invariance properties of the Bergman kernel. This will lead in later sections to the definition of the Bergman metric (in which all biholomorphic mappings become isometries) and to such other canonical constructions as representative coordinates. The Bergman kernel has certain extremal properties that make it a powerful tool in the theory of partial differential equations (see S. Bergman and M. Schiffer [BERN]). Also the form

of the singularity of the Bergman kernel (calculable for some interesting classes of domains) explains many phenomena of the theory of several complex variables.

Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain (it is possible, but often tricky, to treat unbounded domains as well). If the domain is smoothly bounded, then we may think of it as specified by a *defining function*:

$$\Omega = \{z \in \mathbb{C}^n : \rho(z) < 0\}\,.$$

It is customary to require that $\nabla\rho \neq 0$ on $\partial\Omega$. One can demonstrate the existence of a defining function by using the implicit function theorem. See [KRA1, Ch. 3] for a detailed consideration of defining functions.

Given a domain $\Omega$ as described in the last paragraph, and a point $P \in \partial\Omega$, we say that $w$ is a *complex tangent vector* at $P$, and write $w \in \mathcal{T}_P(\partial\Omega)$, if

$$\sum_{j=1}^{n} \frac{\partial\rho}{\partial z_j}(P) w_j = 0\,.$$

The point $P$ is said to be *(weakly) pseudoconvex* if

$$\sum_{j,k=1}^{n} \frac{\partial^2\rho}{\partial z_j \partial\overline{z}_k}(P) w_j \overline{w}_k \geq 0$$

for $0 \neq w \in \mathcal{T}_P(\partial\Omega)$.

The point $P$ is said to be *strongly pseudoconvex* if

$$\sum_{j,k=1}^{n} \frac{\partial^2\rho}{\partial z_j \partial\overline{z}_k}(P) w_j \overline{w}_k > 0$$

for $0 \neq w \in \mathcal{T}_P(\partial\Omega)$. In fact a little elementary analysis shows that we can write the defining property of strong pseudoconvexity as

$$\sum_{j,k=1}^{n} \frac{\partial^2\rho}{\partial z_j \partial\overline{z}_k}(P) w_j \overline{w}_k \geq C|w|^2 \quad \text{for all } w \in \mathbb{C}^n\,,$$

and make the estimate uniform when $P$ ranges over a compact, strongly pseudoconvex boundary neighborhood of $\Omega$. Again, the book [KRA1, Ch. 3] has extensive discussion of the notion of strong pseudoconvexity.

Now let us return to the Bergman theory. Let $dV$ denote Lebesgue volume measure on $\Omega$. Define the *Bergman space*

$$A^2(\Omega) = \left\{ f \text{ holomorphic on } \Omega : \left( \int_\Omega |f(z)|^2\, dV(z) \right)^{1/2} \equiv \|f\|_{A^2(\Omega)} < \infty \right\}\,.$$

Of course we equip the Bergman space with the inner product

$$\langle f, g\rangle = \int_\Omega f(z)\overline{g}(z)\, dV(z)\,.$$

**Lemma 6.1.1** *Let $K \subseteq \Omega \subseteq \mathbb{C}^n$ be compact. There is a constant $C > 0$, depending on $K$ and on $n$, such that*

$$\sup_{z\in K} |f(z)| \leq C_K \|f\|_{A^2(\Omega)}, \quad \text{all } f \in A^2(\Omega).$$

**Proof** Since $K$ is compact, there is an $r(K) = r > 0$ so that, for any $z \in K$, $B(z, r) \subseteq \Omega$. Here $B(z, r)$ is the usual Euclidean ball with center $z$ and radius $r$.

Therefore, for each $z \in K$ and $f \in A^2(\Omega)$, the mean-value property for holomorphic functions implies that

$$\begin{aligned}
|f(z)| &= \left| \frac{1}{V(B(z,r))} \int_{B(z,r)} f(t)\, dV(t) \right| \\
&= \left| \frac{1}{V(B(z,r))} \int_{\mathbb{C}^n} f(t)\chi_{B(z,r)}(t)\, dV(t) \right| \\
&\leq (V(B(z,r)))^{-1/2} \|f\|_{L^2(B(z,r))} \\
&\leq C(n) r^{-n} \|f\|_{A^2(\Omega)} \\
&\equiv C_K \|f\|_{A^2(\Omega)}.
\end{aligned}$$

□

**Lemma 6.1.2** *The space $A^2(\Omega)$ is a Hilbert space with the inner product $\langle f, g\rangle \equiv \int_\Omega f(z)\overline{g(z)}\, dV(z)$.*

**Proof** Everything is clear except for completeness. Let $\{f_j\} \subseteq A^2$ be a sequence that is Cauchy in norm. Since $L^2$ is complete there is an $L^2$ limit function $f$. We need to see that $f$ is holomorphic. But Lemma 6.1.1 yields that norm convergence implies normal convergence (i.e., uniform convergence on compact sets). Certainly holomorphic functions are closed under normal limits (just use the Cauchy theory of one complex variable). Therefore $f$ is holomorphic and $A^2(\Omega)$ is complete. □

**Lemma 6.1.3** *For each fixed $z \in \Omega$, the functional*

$$\Phi_z : f \mapsto f(z), \quad f \in A^2(\Omega)$$

*is a continuous linear functional on $A^2(\Omega)$.*

**Proof** This is immediate from Lemma 6.1.1 if we take $K$ to be the singleton $\{z\}$. □

We may now apply the Riesz representation theorem to see that there is an element $K_z \in A^2(\Omega)$ such that the linear functional $\Phi_z$ is represented by inner product with $K_z$: if $f \in A^2(\Omega)$ then, for all $z \in \Omega$, we have

$$f(z) = \Phi_z(f) = \langle f, K_z \rangle.$$

**Definition 6.1.4** The *Bergman kernel* is the function $K(z, \zeta) = K_\Omega(z, \zeta) \equiv \overline{K_z(\zeta)}$, $z, \zeta \in \Omega$. It has the reproducing property

$$f(z) = \int_\Omega K(z, \zeta) f(\zeta)\, dV(\zeta), \quad \forall f \in A^2(\Omega).$$

**Proposition 6.1.5** *The Bergman kernel $K(z, \zeta)$ is conjugate symmetric: $K(z, \zeta) = \overline{K(\zeta, z)}$.*

**Proof** By its very definition, $\overline{K(\zeta, \cdot)} \in A^2(\Omega)$ for each fixed $\zeta$. Therefore the reproducing property of the Bergman kernel gives

$$\int_\Omega K(z, t)\overline{K(\zeta, t)}\, dV(t) = \overline{K(\zeta, z)}.$$

On the other hand,

$$\begin{aligned} \int_\Omega K(z, t)\overline{K(\zeta, t)}\, dV(t) &= \overline{\int K(\zeta, t)\overline{K(z, t)}\, dV(t)} \\ &= \overline{\overline{K(z, \zeta)}} = K(z, \zeta). \end{aligned}$$

□

**Proposition 6.1.6** *The Bergman kernel is uniquely determined by the properties that it is an element of $A^2(\Omega)$ in $z$, is conjugate symmetric, and reproduces $A^2(\Omega)$.*

**Proof** Let $K'(z, \zeta)$ be another such kernel. Then

$$\begin{aligned} K(z, \zeta) = \overline{K(\zeta, z)} &= \int K'(z, t)\overline{K(\zeta, t)}\, dV(t) \\ &= \overline{\int K(\zeta, t)\overline{K'(z, t)}\, dV(t)} \\ &= \overline{\overline{K'(z, \zeta)}} = K'(z, \zeta). \end{aligned}$$

□

Since $L^2(\Omega)$ is a separable Hilbert space then so is its subspace $A^2(\Omega)$. Thus there is a countable, complete orthonormal basis $\{\phi_j\}_{j=1}^\infty$ for $A^2(\Omega)$.

**Proposition 6.1.7** *Let $L$ be a compact subset of $\Omega$. Then the series*

$$\sum_{j=1}^\infty \phi_j(z)\overline{\phi_j(\zeta)}$$

*sums uniformly on $L \times L$ to the Bergman kernel $K(z, \zeta)$.*

**Proof** By the Riesz–Fischer and Riesz representation theorems, we obtain

$$\begin{aligned}\sup_{z\in L}\left(\sum_{j=1}^{\infty}|\phi_j(z)|^2\right)^{1/2} &= \sup_{z\in L}\left\|\{\phi_j(z)\}_{j=1}^{\infty}\right\|_{\ell^2}\\ &= \sup_{\substack{\|\{a_j\}\|_{\ell^2}=1\\ z\in L}}\left|\sum_{j=1}^{\infty}a_j\phi_j(z)\right|\\ &= \sup_{\substack{\|f\|_{A^2}=1\\ z\in L}}|f(z)|\\ &\le C_L\,. \end{aligned} \tag{6.1.7.1}$$

In the last inequality we have used Lemma 6.1.1. Therefore

$$\sum_{j=1}^{\infty}\left|\phi_j(z)\overline{\phi_j(\zeta)}\right| \le \left(\sum_{j=1}^{\infty}|\phi_j(z)|^2\right)^{1/2}\left(\sum_{j=1}^{\infty}|\phi_j(\zeta)|^2\right)^{1/2}$$

and the convergence is uniform over $z, \zeta \in L$. For fixed $z \in \Omega$, (6.1.7.1) shows that $\{\phi_j(z)\}_{j=1}^{\infty} \in \ell^2$. Hence we have that $\sum \phi_j(z)\overline{\phi_j(\zeta)} \in \overline{A^2(\Omega)}$ as a function of $\zeta$. Let the sum of the series be denoted by $K'(z,\zeta)$. Notice that $K'$ is conjugate symmetric by its very definition. Also, for $f \in A^2(\Omega)$, we have

$$\int K'(\cdot,\zeta)f(\zeta)\,dV(\zeta) = \sum \widehat{f}(j)\phi_j(\cdot) = f(\cdot),$$

where convergence is in the Hilbert space topology. [Here $\widehat{f}(j)$ is the $j^{\text{th}}$ Fourier coefficient of $f$ with respect to the basis $\{\phi_j\}$.] But Hilbert space convergence dominates pointwise convergence (Lemma 6.1.1) so

$$f(z) = \int K'(z,\zeta)f(\zeta)\,dV(\zeta), \quad \text{all } f \in A^2(\Omega).$$

Therefore $K'$ is the Bergman kernel. □

**Remark 6.1.8** It is worth noting explicitly that the proof of 6.1.7 shows that

$$\sum \phi_j(z)\overline{\phi_j(\zeta)}$$

equals the Bergman kernel $K(z,\zeta)$ *no matter what the choice* of complete orthonormal basis $\{\phi_j\}$ for $A^2(\Omega)$. This can be very useful information in practice. □

**Proposition 6.1.9** *If $\Omega$ is a bounded domain in $\mathbb{C}^n$ then the mapping*

$$P : f \mapsto \int_\Omega K(\cdot, \zeta) f(\zeta)\, dV(\zeta)$$

*is the Hilbert space orthogonal projection of $L^2(\Omega, dV)$ onto $A^2(\Omega)$. We call $P$ the Bergman projection.*

**Proof** Notice that $P$ is idempotent and self-adjoint and that $A^2(\Omega)$ is precisely the set of elements of $L^2$ that are fixed by $P$. □

**Definition 6.1.10** Let $\Omega \subseteq \mathbb{C}^n$ be a domain and let $f : \Omega \to \mathbb{C}^n$ be a *holomorphic mapping*, that is $f(z) = (f_1(z), \ldots, f_n(z))$ with $f_1, \ldots, f_n$ holomorphic on $\Omega$. Let $w_j = f_j(z), j = 1, \ldots, n$. Then the *holomorphic Jacobian matrix* of $f$ is the matrix

$$J_{\mathbb{C}} f = \frac{\partial(w_1, \ldots, w_n)}{\partial(z_1, \ldots, z_n)}.$$

Write $z_j = x_j + iy_j, w_k = \xi_k + i\eta_k, j, k = 1, \ldots, n$. Then the *real Jacobian matrix* of $f$ is the matrix

$$J_{\mathbb{R}} f = \frac{\partial(\xi_1, \eta_1, \ldots, \xi_n, \eta_n)}{\partial(x_1, y_1, \ldots, x_n, y_n)}.$$

**Proposition 6.1.11** *With notation as in the definition, we have*

$$\det J_{\mathbb{R}} f = |\det J_{\mathbb{C}} f|^2$$

*whenever $f$ is a holomorphic mapping.*

**Proof** We exploit the functoriality of the Jacobian. Let $w = (w_1, \ldots, w_n) = f(z) = (f_1(z), \ldots, f_n(z))$. Write $z_j = x_j + iy_j, w_j = \xi_j + i\eta_j, j = 1, \ldots, n$. Then, using the fact that $f$ is holomorphic,

$$\begin{aligned} &d\xi_1 \wedge d\eta_1 \wedge \ldots \wedge d\xi_n \wedge d\eta_n \\ &\qquad = (\det J_{\mathbb{R}} f(x, y)) dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n . \end{aligned} \tag{6.1.11.1}$$

On the other hand,

$$\begin{aligned} &d\xi_1 \wedge d\eta_1 \wedge \cdots \wedge d\xi_n \wedge d\eta_n \\ &= \frac{1}{(2i)^n} d\overline{w}_1 \wedge dw_1 \wedge \cdots \wedge d\overline{w}_n \wedge dw_n \\ &= \frac{1}{(2i)^n} \overline{(\det J_{\mathbb{C}} f(z))} (\det J_{\mathbb{C}} f(z)) d\overline{z}_1 \wedge dz_1 \wedge \cdots \wedge d\overline{z}_n \wedge dz_n \\ &= |\det J_{\mathbb{C}} f(z)|^2 dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n. \end{aligned} \tag{6.1.11.2}$$

Equating (6.1.11.1) and (6.1.11.2) gives the result. □

**Exercise for the Reader** Prove Proposition 6.1.11 using only matrix theory (no differential forms). This will give rise to a great appreciation for the theory of differential forms (see L. Bers [BERS, Ch. 7] for help).

Now we can prove the holomorphic implicit function theorem:

**Prelude** The holomorphic implicit function theorem is one of many possible implicit function theorems (see [KRA1, Ch. 2]). The holomorphic version is particularly useful, and is encountered frequently in the literature.

**Theorem 6.1.12** *Let $f_j(w,z), j = 1,\ldots,m$ be holomorphic functions of $(w,z) = ((w_1,\ldots,w_m),(z_1,\ldots,z_n))$ near a point $(w^0,z^0) \in \mathbb{C}^m \times \mathbb{C}^n$. Assume that*

$$f_j(w^0,z^0) = 0, \quad j = 1,\ldots,m,$$

*and that*

$$\det\left(\frac{\partial f_j}{\partial w_k}\right)_{j,k=1}^{m} \neq 0 \text{ at } (w^0,z^0).$$

*Then the system of equations*

$$f_j(w,z) = 0 \ , \ j = 1,\ldots,m,$$

*has a unique holomorphic solution $w(z)$ in a neighborhood of $z^0$ that satisfies $w(z^0) = w^0$.*

**Proof** We rewrite the system of equations as

$$\mathrm{Re}\, f_j(w,z) = 0, \ \ \mathrm{Im} f_j(w,z) = 0$$

for the $2m$ real variables $\mathrm{Re}\, w_k, \mathrm{Im}\, w_k, k = 1,\ldots,m$. By Proposition 6.1.11, the determinant of the Jacobian over $\mathbb{R}$ of this new system is the modulus squared of the determinant of the Jacobian over $\mathbb{C}$ of the old system. By our hypothesis, this number is nonvanishing at the point $(w^0,z^0)$. Therefore the classical implicit function theorem (see [KRPA3]) implies that there exist $C^1$ functions $w_k(z), k = 1,\ldots,m$, with $w(z^0) = w^0$ and that solve the system. Our job is to show that these functions are in fact holomorphic. When properly viewed, this is purely a problem of geometric algebra:

Applying exterior differentiation to the equations

$$0 = f_j(w(z),z), \ \ j = 1,\ldots,m,$$

yields that

$$0 = df_j = \sum_{k=1}^{m} \frac{\partial f_j}{\partial w_k} dw_k + \sum_{k=1}^{m} \frac{\partial f_j}{\partial z_k} dz_k.$$

There are no $d\overline{z}_j$'s and no $d\overline{w}_k$'s because the $f_j$'s are holomorphic.

The result now follows from linear algebra only: the hypothesis on the determinant of the matrix $(\partial f_j/\partial w_k)$ implies that we can solve for $dw_k$ in terms of $dz_j$. Therefore $w$ is a holomorphic function of $z$. □

A holomorphic mapping $f : \Omega_1 \to \Omega_2$ of domains $\Omega_1 \subseteq \mathbb{C}^n, \Omega_2 \subseteq \mathbb{C}^n$ is said to be *biholomorphic* if it is one-to-one, onto, and $\det J_{\mathbb{C}} f(z) \neq 0$ for every $z \in \Omega_1$.

**Exercise for the Reader** Use Theorem 6.1.12 to prove that a biholomorphic mapping has a holomorphic inverse (hence the name).

**Remark 6.1.13** It is true, but not at all obvious, that the nonvanishing of the Jacobian determinant is a superfluous condition in the definition of "biholomorphic mapping;" that is, the nonvanishing of the Jacobian follows from the univalence of the mapping. A proof of this assertion is sketched in Exercise 37 at the end of [KRA1, Ch. 11]. □

In what follows we shall frequently denote the Bergman kernel for a given domain $\Omega$ by $K_\Omega$.

**Proposition 6.1.14** *Let $\Omega_1, \Omega_2$ be domains in $\mathbb{C}^n$. Let $f : \Omega_1 \to \Omega_2$ be biholomorphic. Then*

$$\det J_{\mathbb{C}} f(z) K_{\Omega_2}(f(z), f(\zeta)) \det \overline{J_{\mathbb{C}} f(\zeta)} = K_{\Omega_1}(z, \zeta).$$

**Proof** Let $\phi \in A^2(\Omega_1)$. Then, by change of variable,

$$\begin{aligned}
&\int_{\Omega_1} \det J_{\mathbb{C}} f(z) K_{\Omega_2}(f(z), f(\zeta)) \det \overline{J_{\mathbb{C}} f(\zeta)} \phi(\zeta)\, dV(\zeta) \\
&= \int_{\Omega_2} \det J_{\mathbb{C}} f(z) K_{\Omega_2}(f(z), \widetilde{\zeta}) \det \overline{J_{\mathbb{C}} f(f^{-1}(\widetilde{\zeta}))} \phi(f^{-1}(\widetilde{\zeta})) \\
&\qquad \times \det J_{\mathbb{R}} f^{-1}(\widetilde{\zeta})\, dV(\widetilde{\zeta}).
\end{aligned}$$

By Proposition 6.1.11 this simplifies to

$$\det J_{\mathbb{C}} f(z) \int_{\Omega_2} K_{\Omega_2}(f(z), \widetilde{\zeta}) \left\{ \left( \det J_{\mathbb{C}} f(f^{-1}(\widetilde{\zeta})) \right)^{-1} \phi\left( f^{-1}(\widetilde{\zeta}) \right) \right\} dV(\widetilde{\zeta}).$$

By change of variables, the expression in braces $\{\ \}$ is an element of $A^2(\Omega_2)$. So the reproducing property of $K_{\Omega_2}$ applies and the last line equals

$$= \det J_{\mathbb{C}} f(z) \left( \det J_{\mathbb{C}} f(z) \right)^{-1} \phi\left( f^{-1}(f(z)) \right) = \phi(z).$$

By the uniqueness of the Bergman kernel, the proposition follows. □

**Proposition 6.1.15** *For $z \in \Omega \subset\subset \mathbb{C}^n$ it holds that $K_\Omega(z, z) > 0$.*

**Proof** Now

$$K_\Omega(z,z) = \sum_{j=1}^{\infty} |\phi_j(z)|^2 \geq 0.$$

If in fact $K(z,z) = 0$ for some $z$ then $\phi_j(z) = 0$ for all $j$ hence $f(z) = 0$ for every $f \in A^2(\Omega)$. This is absurd. □

**Definition 6.1.16** For any bounded domain $\Omega \subseteq \mathbb{C}^n$, we define a Hermitian metric on $\Omega$ by

$$g_{jk}(z) = \frac{\partial^2}{\partial z_j \partial \overline{z}_k} \log K(z,z), \quad z \in \Omega.$$

This means that the square of the length of a tangent vector $\xi = (\xi_1, \ldots, \xi_n)$ at a point $z \in \Omega$ is given by

$$|\xi|^2_{B,z} = \sum_{j,k} g_{jk}(z) \xi_j \overline{\xi}_k.$$

The metric that we have defined is called the *Bergman metric*.

In a Hermitian metric $\{g_{jk}\}$, the length of a $C^1$ curve $\gamma : [0,1] \to \Omega$ is given by

$$\ell(\gamma) = \int_0^1 \left( \sum_{j,k} g_{j,k}(\gamma(t)) \gamma_j'(t) \overline{\gamma_k'(t)} \right)^{1/2} dt.$$

If $P, Q$ are points of $\Omega$, then their distance $d_\Omega(P,Q)$ in the metric is defined to be the infimum of the lengths of all piecewise $C^1$ curves connecting the two points.

**Remark 6.1.17** It is not *a priori* obvious that the Bergman metric for a bounded domain $\Omega$ is given by a positive definite matrix at each point. We now outline a proof of this fact.

First we generate an orthonormal basis for the Bergman space. Fix $z_0 \in \Omega$. Let $\phi_0$ be the (unique!) element of $A^2$ with $\phi_0(z_0)$ real, $\|\phi_0\| = 1$, and $\phi_0(z_0)$ maximal. (Why does such a $\phi_0$ exist?) Let $\phi_1$ be the (unique) element of $A^2$ with $\phi_1(z_0) = 0$, $(\partial\phi_1/\partial z_1)(z_0)$ real, $\|\phi_1\| = 1$, and $(\partial\phi_1/\partial z_1)(z_0)$ maximal. (Why does such a $\phi_1$ exist?) Now $\phi_1$ is orthogonal to $\phi_0$, else $\phi_1$ has nonzero projection on $\phi_0$, leading to a contradiction. Continue this process to create an orthogonal system on $\Omega$. Use Taylor series to see that it is complete. This circle of ideas comes from the elegant paper S. Kobayashi [KOB1].

Now let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain and let $(g_{jk})$ be its Bergman metric. Use the ideas in the last paragraph to prove that the matrix $(g_{jk}(z))$ is positive definite, each $z \in \Omega$. [*Hint:* The crucial fact is that, for each $z \in \Omega$ and each $k$, there is an element $f \in A^2(\Omega)$ such that $\partial f/\partial z_k(z) \neq 0$.] □

**Proposition 6.1.18** *Let $\Omega_1, \Omega_2 \subseteq \mathbb{C}^n$ be domains and let $f : \Omega_1 \to \Omega_2$ be a biholomorphic mapping. Then $f$ induces an isometry of Bergman metrics:*

$$|\xi|_{B,z} = |(J_{\mathbb{C}} f)\xi|_{B,f(z)}$$

*for all $z \in \Omega_1, \xi \in \mathbb{C}^n$. Equivalently, $f$ induces an isometry of Bergman distances in the sense that*

$$d_{\Omega_2}(f(P), f(Q)) = d_{\Omega_1}(P, Q).$$

**Proof** This is a formal exercise but we include it for completeness: From the definitions, it suffices to check that

$$\sum g^{\Omega_2}_{j,k}(f(z))\,(J_{\mathbb{C}} f(z)w)_j \left(\overline{J_{\mathbb{C}} f(z)w}\right)_k = \sum_{j,k} g^{\Omega_1}_{jk}(z) w_j \overline{w}_k \tag{6.1.18.1}$$

for all $z \in \Omega, w = (w_1, \ldots, w_n) \in \mathbb{C}^n$. But, by Proposition 6.1.14,

$$\begin{aligned} g^{\Omega_1}_{jk}(z) &= \frac{\partial^2}{\partial z_j \overline{z}_k} \log K_{\Omega_1}(z, z) \\ &= \frac{\partial^2}{\partial z_j \overline{z}_k} \log \left\{ |\det J_{\mathbb{C}} f(z)|^2 K_{\Omega_2}(f(z), f(z)) \right\} \\ &= \frac{\partial^2}{\partial z_j \overline{z}_k} \log K_{\Omega_2}(f(z), f(z)) \end{aligned} \tag{6.1.18.2}$$

since $\log |\det J_{\mathbb{C}} f(z)|^2$ is locally

$$\log (\det J_{\mathbb{C}} f) + \log \left(\overline{\det J_{\mathbb{C}} f}\right) + C$$

hence is annihilated by the mixed second derivative. But line (6.1.18.2) is nothing other than

$$\sum_{\ell,m} g^{\Omega_2}_{\ell,m}(f(z)) \frac{\partial f_\ell(z)}{\partial z_j} \frac{\partial \overline{f_m(z)}}{\partial \overline{z}_k}$$

and (6.1.18.1) follows. □

**Proposition 6.1.19** *Let $\Omega \subset\subset \mathbb{C}^n$ be a domain. Let $z \in \Omega$. Then*

$$K(z, z) = \sup_{f \in A^2(\Omega)} \frac{|f(z)|^2}{\|f\|^2_{A^2}} = \sup_{\|f\|_{A^2(\Omega)} = 1} |f(z)|^2.$$

**Proof** Now

$$\begin{aligned} K(z,z) &= \sum |\phi_j(z)|^2 \\ &= \left( \sup_{\|\{a_j\}\|_{\ell^2}=1} \left| \sum \phi_j(z) a_j \right| \right)^2 \\ &= \sup_{\|f\|_{A^2}=1} |f(z)|^2, \end{aligned}$$

by the Riesz–Fischer theorem,

$$= \sup_{f \in A^2} \frac{|f(z)|^2}{\|f\|_{A^2}^2}.$$

□

We shall use this proposition in a moment. Meanwhile, we should like to briefly mention some open problems connected with the Bergman kernel:

**The Lu Qi-Keng Conjecture** We have already noticed that $K_\Omega(z,z) > 0$, all $z \in \Omega$, any bounded $\Omega$. It is reasonable to ask whether $K_\Omega(z,\zeta)$ is ever equal to zero. In fact various geometric constructions connected with the Bergman metric and associated biholomorphic invariants (which involve division by $K$) make it particularly desirable that $K$ be nonvanishing.

If $\Omega = D$, the unit disc, then explicit calculation (which we perform below) shows that

$$K(z,\zeta) = \frac{1}{\pi} \frac{1}{(1 - z\bar{\zeta})^2},$$

hence $K(z,\zeta)$ is nonvanishing on $D \times D$. Proposition 6.1.14 and the Riemann mapping theorem then show that the Bergman kernel for any proper, simply connected subdomain of $\mathbb{C}$ is nonvanishing.

The Bergman kernel for the annulus was studied in M. Skwarczynski [SKW] and was seen to vanish at some points. It is shown in N. Suita and A. Yamada [SUY] that, if $\Omega \subseteq \mathbb{C}$ is a multiply connected domain with smooth boundary, then $K_\Omega$ must vanish—this is proved by an analysis of differentials on the Riemann surface consisting of the double of $\Omega$. By using the easy fact that the Bergman kernel for a product domain is the product of the Bergman kernels (exercise), we may conclude that any domain in $\mathbb{C}^2$ of the form $A \times \Omega$, where $A$ is multiply connected, has a Bergman kernel with zeroes. The Lu Qi-Keng conjecture can be formulated as

**Conjecture:** A topologically trivial domain in $\mathbb{C}^n$ has nonvanishing Bergman kernel.

It is known (R. E. Greene and S. G. Krantz [GRK1, GRK2]) that a domain that is $C^\infty$ sufficiently close to the ball in $\mathbb{C}^n$ has nonvanishing Bergman kernel. Also, if a domain $\Omega$ has Bergman kernel that is bounded from zero (and satisfies a

modest geometric condition) then all "nearby" domains have Bergman kernel that is bounded from zero. Thus it came as a bit of a surprise when in H. Boas [BOA1] it was shown that there exist topologically trivial domains—even ones with real analytic boundary and satisfying all reasonable additional geometric conditions—for which the Bergman kernel has zeroes. See also J. Wiegerinck [WIE], where interesting ideas contributing to the solution of this problem first arose. A survey of results related to the Lu Qi-Keng conjecture appears in [BOA3]. See also [BOA4].

**Exercise for the Reader** The set of smoothly bounded domains for which the Lu Qi-Keng conjecture is true is closed in the Hausdorff topology on domains.

### *6.1.1 Smoothness to the Boundary of $K_\Omega$*

It is of interest to know whether $K_\Omega$ is smooth on $\overline{\Omega} \times \overline{\Omega}$. We can see from the formula for the Bergman kernel of the disc that $K_D(z, z)$ blows up as $z \to 1^-$. In fact this property of blowing up prevails at any boundary point of a domain at which there is a peaking function. The reference T. Gamelin [GAM, p. 52 ff.] contains background information on peaking functions.

Certainly the assertion is true on any smoothly bounded domain in $\mathbb{C}^1$. For simply connected domains, this follows from the result on the disc by conformal mapping. For a finitely connected domain, it follows from the Ahlfors map (see [KRA17, Ch. 4]) or from the uniformization theorem. However, there is strong evidence that—as long as $\Omega$ is smoothly bounded—on compact subsets of

$$\overline{\Omega} \times \overline{\Omega} \setminus ((\partial\Omega \times \partial\Omega) \cap \{z = \zeta\})$$

the Bergman kernel will be smooth. For strictly pseudoconvex domains, this statement is true; its proof (see [KER2]) uses deep and powerful methods of partial differential equations. Unfortunately, on the Diederich–Fornæss worm domain [DIRF2] (which is smoothly bounded, pseudoconvex but has many pathological properties), the Bergman kernel is *not* smooth as just indicated. See also [KRPE1] and [KRPE2] as well as [LIG1]. We discuss the worm in Chapter 9.

In what follows, a multi-index in $\mathbb{C}^n$ is an $n$-tuple $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$ of nonnegative integers. We write

$$z^\alpha \equiv z_1^{\alpha_1} \cdot z_2^{\alpha_2} \cdot \cdots \cdot z_n^{\alpha_n} .$$

Also

$$\frac{\partial^\alpha}{\partial z^\alpha} \equiv \frac{\partial^{\alpha_1}}{\partial z_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial z_2^{\alpha_2}} \cdots \frac{\partial^{\alpha_n}}{\partial z_n^{\alpha_n}} .$$

Perhaps the most central open problem in the function theory of several complex variables is to prove that a biholomorphic mapping of two smoothly bounded, pseudoconvex domains extends to a diffeomorphism of the closures. It is known (see S. Bell and H. Boas [BEB]) that a sufficient condition for this problem to have an affirmative answer on a smoothly bounded domain $\Omega \subseteq \mathbb{C}^n$ is that, for any multi-index $\alpha$, there are constants $C = C_\alpha$ and $m = m_\alpha$ such that the Bergman kernel $K_\Omega$ satisfies

$$\sup_{z\in\Omega} \left| \frac{\partial^\alpha}{\partial z^\alpha} K_\Omega(z,\zeta) \right| \leq C \cdot \delta_\Omega(\zeta)^{-m}$$

for all $\zeta \in \Omega$. Here $\delta_\Omega(w)$ denotes the distance of the point $w \in \Omega$ to the boundary of the domain.

### 6.1.2 Calculating the Bergman Kernel

The Bergman kernel can almost never be calculated explicitly; unless the domain $\Omega$ has a great deal of symmetry—so that a useful orthonormal basis for $A^2(\Omega)$ can be determined—there are few techniques for determining $K_\Omega$. Sometimes one can exploit the automorphism group of the domain (see [HUA] for an exemplary instance of this technique). We shall explore some of these ideas below.

In 1974 C. Fefferman [FEF1, Part I] introduced a new technique for obtaining an asymptotic expansion for the Bergman kernel on a large class of domains. (For an alternative approach see L. Boutet de Monvel and J. Sjöstrand [BOUTS].) This work enabled rather explicit estimations of the Bergman metric and opened up an entire branch of analysis on domains in $\mathbb{C}^n$ (see C. Fefferman [FEF2], S. S. Chern and J. Moser [CHSM], P. Klembeck [KLE] and R. E. Greene and S. G. Krantz [GRK1, GRK2, GRK3, GRK4] for example).

The Bergman theory that we have presented here would be a bit hollow if we did not at least calculate the kernel in a few instances. We complete the section by addressing that task.

Restrict attention to the ball $B \subseteq \mathbb{C}^n$. The functions $z^\alpha$, $\alpha$ a multi-index, are each in $A^2(B)$ and are pairwise orthogonal by the symmetry of the ball. By the uniqueness of the power series expansion for an element of $A^2(B)$, the elements $z^\alpha$ form a complete orthogonal system on $B$ (their closed linear span is $A^2(B)$). Setting

$$\gamma_\alpha = \int_B |z^\alpha|^2 \, dV(z),$$

we see that $\{z^\alpha / \sqrt{\gamma_\alpha}\}$ is a complete orthonormal system in $A^2(B)$. Thus, by Proposition 6.1.7,

$$K(z,\zeta) = \sum_{\alpha} \frac{z^{\alpha}\overline{\zeta}^{\alpha}}{\gamma_{\alpha}}. \tag{6.1.20}$$

If we want to calculate the Bergman kernel for the ball in closed form, we need to calculate the $\gamma_\alpha$s. This requires some lemmas from real analysis. These lemmas will be formulated and proved on $\mathbb{R}^N$ and $B_N = \{x \in \mathbb{R}^N : |x| < 1\}$.

**Lemma 6.1.21** *We have that*

$$\int_{\mathbb{R}^N} e^{-\pi|x|^2} dx = 1.$$

**Proof** The case $N = 1$ is familiar from calculus. For the $N$-dimensional case, write

$$\int_{\mathbb{R}^N} e^{-\pi|x|^2} dx = \int_{\mathbb{R}} e^{-\pi x_1^2} dx_1 \cdots \int_{\mathbb{R}} e^{-\pi x_N^2} dx_N$$

and apply the one-dimensional result. □

Let $\sigma$ be the unique rotationally invariant area measure on $S_{N-1} = \partial B_N$ and let $\omega_{N-1} = \sigma(\partial B)$.

**Lemma 6.1.22** *We have*

$$\omega_{N-1} = \frac{2\pi^{N/2}}{\Gamma(N/2)},$$

*where*

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

*is Euler's Gamma function.*

**Proof** Introducing polar coordinates we have

$$1 = \int_{\mathbb{R}^N} e^{-\pi|x|^2} dx = \int_{S^{N-1}} d\sigma \int_0^{\infty} e^{-\pi r^2} r^{N-1} dr$$

or

$$\frac{1}{\omega_{N-1}} = \int_0^{\infty} e^{-\pi r^2} r^N \frac{dr}{r}.$$

Letting $s = r^2$ in this last integral and doing some obvious manipulations yields the result. □

Now we return to $B \subseteq \mathbb{C}^n$. We set

$$\eta(k) = \int_{\partial B} |z_1|^{2k} d\sigma \ , \ N(k) = \int_B |z_1|^{2k} \, dV(z) \ , \ k = 0, 1, \ldots .$$

**Lemma 6.1.23** *We have*

$$\eta(k) = \pi^n \frac{2(k!)}{(k+n-1)!} \ , \ N(k) = \pi^n \frac{k!}{(k+n)!}.$$

**Proof** Polar coordinates show easily that $\eta(k) = 2(k+n)N(k)$. So it is enough to calculate $N(k)$. Let $z = (z_1, z_2, \ldots, z_n) = (z_1, z')$. We write

$$\begin{aligned}
N(k) &= \int_{|z|<1} |z_1|^{2k} \, dV(z) \\
&= \int_{|z'|<1} \left( \int_{|z_1| \leq \sqrt{1-|z'|^2}} |z_1|^{2k} \, dV(z_1) \right) dV(z') \\
&= 2\pi \int_{|z'|<1} \int_0^{\sqrt{1-|z'|^2}} r^{2k} r dr \, dV(z') \\
&= 2\pi \int_{|z'|<1} \frac{(1-|z'|^2)^{k+1}}{2k+2} \, dV(z') \\
&= \frac{\pi}{k+1} \omega_{2n-3} \int_0^1 (1-r^2)^{k+1} r^{2n-3} dr \\
&= \frac{\pi}{k+1} \omega_{2n-3} \int_0^1 (1-s)^{k+1} s^{n-1} \frac{ds}{2s} \\
&= \frac{\pi}{2(k+1)} \omega_{2n-3} \beta(n-1, k+2),
\end{aligned}$$

where $\beta$ is the classical beta function of special function theory (see G. Carrier, M. Crook, and C. Pearson [CARR, p. 191] or E. Whittaker and G. Watson [WHW, pp. 235 ff.]). By a standard identity for the beta function we then have

$$\begin{aligned}
N(k) &= \frac{\pi}{2(k+1)} \omega_{2n-3} \frac{\Gamma(n-1)\Gamma(k+2)}{\Gamma(n+k+1)} \\
&= \frac{\pi}{2(k+1)} \frac{2\pi^{n-1}}{\Gamma(n-1)} \frac{\Gamma(n-1)\Gamma(k+2)}{\Gamma(n+k+1)} \\
&= \frac{\pi^n k!}{(k+n)!}.
\end{aligned}$$

This is the desired result. □

**Lemma 6.1.24** *Let $z \in B \subseteq \mathbb{C}^n$ and $0 < r < 1$. The symbol $\mathbf{1}$ denotes the point $(1, 0, \ldots, 0)$. Then*

$$K_B(z, r\mathbf{1}) = \frac{n!}{\pi^n} \frac{1}{(1 - rz_1)^{n+1}}.$$

**Proof** We see that

$$\begin{aligned} K_B(z, r\mathbf{1}) &= \sum_\alpha \frac{z^\alpha (r\mathbf{1})^\alpha}{\gamma_\alpha} = \sum_{k=0}^\infty \frac{z_1^k r^k}{N(k)} \\ &= \frac{1}{\pi^n} \sum_{k=0}^\infty (rz_1)^k \cdot \frac{(k+n)!}{k!} \\ &= \frac{n!}{\pi^n} \sum_{k=0}^\infty (rz_1)^k \binom{k+n}{n} \\ &= \frac{n!}{\pi^n} \cdot \frac{1}{(1 - rz_1)^{n+1}}. \end{aligned}$$

This is the desired result. □

**Prelude** The next theorem gives an explicit formula for the Bergman kernel on the unit ball in $\mathbb{C}^n$. There are *very few* domains for which this can be done. The book [HUA] treats such domains exhaustively. What we can hope for, in a more general setting, is to get an asymptotic expansion for the Bergman kernel. The paper [FEF1] gives such an expansion on a strongly pseudoconvex domain.

**Theorem 6.1.25** *If $z, \zeta \in B$, then the Bergman kernel for the unit ball in $\mathbb{C}^n$ is*

$$K_B(z, \zeta) = \frac{n!}{\pi^n} \frac{1}{(1 - z \cdot \overline{\zeta})^{n+1}},$$

*where $z \cdot \overline{\zeta} = z_1\overline{\zeta}_1 + z_2\overline{\zeta}_2 + \cdots + z_n\overline{\zeta}_n$.*

**Proof** Let $z = r\widetilde{z} \in B$, where $r = |z|$ and $|\widetilde{z}| = 1$. Also fix $\zeta \in B$. Choose a unitary rotation $\rho$ such that $\rho\widetilde{z} = \mathbf{1}$. We have

$$\begin{aligned} K_B(z, \zeta) &= K_B(r\widetilde{z}, \zeta) = K(r\rho^{-1}\mathbf{1}, \zeta) \\ &= K(r\mathbf{1}, \rho\zeta) = \overline{K(\rho\zeta, r\mathbf{1})} \\ &= \frac{n!}{\pi^n} \cdot \frac{1}{\left(1 - r(\overline{\rho\zeta})_1\right)^{n+1}} \end{aligned}$$

$$= \frac{n!}{\pi^n} \cdot \frac{1}{\left(1 - (r\mathbf{1}) \cdot (\overline{\rho\zeta})\right)^{n+1}}$$

$$= \frac{n!}{\pi^n} \cdot \frac{1}{\left(1 - (r\rho^{-1}\mathbf{1}) \cdot \overline{\zeta}\right)^{n+1}}$$

$$= \frac{n!}{\pi^n} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^{n+1}}.$$

□

**Corollary 6.1.26** *The Bergman kernel for the unit disc in the complex plane is*

$$K_D(z, \zeta) = \frac{1}{\pi} \frac{1}{(1 - z\overline{\zeta})^2}.$$

**Proposition 6.1.27** *The Bergman metric for the ball $B = B(0, 1) \subseteq \mathbb{C}^n$ is given by*

$$g_{ij}(z) = \frac{n+1}{(1 - |z|^2)^2} \left[(1 - |z|^2)\delta_{ij} + \overline{z}_i z_j\right].$$

**Proof** Since $K(z, z) = n!/(\pi^n(1 - |z|^2)^{n+1})$, this is a routine computation that we leave to the reader. □

**Corollary 6.1.28** *The Bergman metric for the disc (i.e., the ball in dimension one) is*

$$g_{ij}(\zeta) = \frac{2}{(1 - |\zeta|^2)^2}, \ i = j = 1,$$

*This is (up to a constant multiple) the well-known Poincaré, or Poincaré–Bergman, metric.*

**Proposition 6.1.29** *The Bergman kernel for the polydisc $D^n(0, 1) \subseteq \mathbb{C}^n$ is the product*

$$K(z, \zeta) = \frac{1}{\pi^n} \prod_{j=1}^{n} \frac{1}{(1 - z_j\overline{\zeta}_j)^2}.$$

**Proof** Exercise for the reader. Use the uniqueness property of the Bergman kernel. □

**Exercise for the Reader** Calculate the Bergman metric for the polydisc.

### 6.1.3 The Poincaré–Bergman Distance on the Disc

If $D \subseteq \mathbb{C}$ is the unit disc, $\zeta \in D$, then we know that that

$$|w|_{B,\zeta} = \left\{ \frac{2|w|^2}{(1-|\zeta|^2)^2} \right\}^{1/2} = \frac{\sqrt{2}|w|}{1-|\zeta|^2},$$

where the subscript $B$ indicates that we are working in the Bergman metric. We now use this formula to derive an explicit expression for the Poincaré–Bergman *distance* from $0 \in D$ to $r + i0 \in D, 0 < r < 1$. Call this distance $d(0, r)$. Then

$$d(0,r) = \inf \left\{ \int_0^1 |\gamma'(t)|_{B,\gamma(t)} dt : \right.$$
$$\left. \gamma \text{ is a piecewise smooth curve in } D, \gamma(0) = 0, \gamma(1) = r + i0 \right\}.$$

Elementary comparisons show that, among curves of the form $\psi(t) = t + iw(t), 0 \leq t \leq 1$, the curve $\gamma(t) = tr + i0$ is the shortest in the Poincaré metric. Further elementary arguments show that a general curve of the form $\psi(t) = v(t) + iw(t)$ is always longer than some corresponding curve of the form $t + i\widetilde{w}(t)$. We leave the details of these assertions to the reader. Thus

$$\begin{aligned} d(0,r) &= \int_0^1 \frac{\sqrt{2}r}{(1-(rt)^2)} dt \\ &= \sqrt{2} \int_0^r \frac{1}{1-t^2} dt \\ &= \frac{1}{\sqrt{2}} \log \left( \frac{1+r}{1-r} \right). \end{aligned}$$

Since rotations are conformal maps of the disc we may next conclude that

$$d(0, re^{i\theta}) = \frac{1}{\sqrt{2}} \log \left( \frac{1+r}{1-r} \right).$$

Finally, if $w_1, w_2$ are arbitrary, then the Möbius transformation

$$\phi : z \mapsto \frac{z - w_1}{1 - \overline{w}_1 z}$$

satisfies $\phi(w_1) = 0, \phi(w_2) = (w_2 - w_1)/(1 - \overline{w}_1 w_2)$. Then Proposition 6.1.18 yields that

$$\begin{aligned} d(w_1, w_2) &= d\left(0, \frac{w_2 - w_1}{1 - \overline{w}_1 w_2}\right) \\ &= \frac{1}{\sqrt{2}} \left( \frac{1 + \left| \frac{w_2 - w_1}{1 - \overline{w}_1 w_2} \right|}{1 - \left| \frac{w_2 - w_1}{1 - \overline{w}_1 w_2} \right|} \right). \end{aligned}$$

We note in passing that the expression $\rho(w_1, w_2) \equiv |(w_2 - w_1)/(1 - \overline{w}_1 w_2)|$ is called the *pseudohyperbolic distance*. It is also a conformally invariant metric, but it does *not* arise from integrating an infinitesimal metric (i.e., lengths of tangent vectors at a point). A fuller discussion of both the Poincaré metric and the pseudohyperbolic metric on the disc may be found in [KRA16, Ch. 1, 3] and [GARN, Ch. 1].

### 6.1.4 Construction of the Bergman Kernel by Way of Differential Equations

It is actually possible to obtain the Bergman kernel of a domain in the plane from the Green's function for that domain (see [KRA5, Section 1.3.3]). Let us now summarize the key ideas. Unlike the first Bergman kernel construction, the present one will work for *any* domain with $C^2$ boundary. Thanks to work of Garabedian [GARA], one can say rather precisely what the Green's function of any planar domain is (see also [JAK]).

First, the fundamental solution for the Laplacian in the plane is the function

$$\Gamma(\zeta, z) = \frac{1}{2\pi} \log |\zeta - z| \, .$$

This means that $\triangle_\zeta \Gamma(\zeta, z) = \delta_z$ in the sense of distributions. [Observe that $\delta_z$ denotes the Dirac "delta mass" at $z$ and $\triangle_\zeta$ is the Laplacian in the $\zeta$ variable.] In more prosaic terms, the condition is that

$$\int \Gamma(\zeta, z) \cdot \triangle \varphi(\zeta) \, d\xi d\eta = \varphi(z)$$

for any $C^\infty$ function $\varphi$ with compact support. We write, as usual, $\zeta = \xi + i\eta$. [This topic is treated in detail in [KRA1, Ch. 0, 1].]

Given a domain $\Omega \subseteq \mathbb{C}$, the *Green's function* is posited to be a function $G(\zeta, z)$ on $\Omega \times \Omega$ that satisfies

$$G(\zeta, z) = \Gamma(\zeta, z) + F_z(\zeta) \, ,$$

where $F_z(\zeta) = F(\zeta, z)$ is a particular harmonic function in the $\zeta$ variable (to be specified momentarily). Moreover, it is mandated that $G(\,\cdot\,, z)$ vanish on the boundary of $\Omega$. One constructs the function $F(\,\cdot\,, z)$, for each fixed $z$, by solving a Dirichlet problem with boundary data $-\Gamma(\,\cdot\,, z)$. Again, the reference [KRA1, p. 40] has all the particulars. It is worth noting, and this point is not completely obvious but is discussed in [KRA1, Ch. 1], that the Green's function is a symmetric function of its arguments.

The next proposition establishes a striking connection between the Bergman kernel and the classical Green's function.

**Prelude** The next result is of considerable philosophical interest because it relates the Bergman kernel to the more explicit and accessible Green's function. This only works in the setting of one complex variable, but see [GARA] for some ideas in several complex variables and [HARV] for an attempt at higher-dimensional generalization of Proposition 6.1.30.

**Proposition 6.1.30** *Let $\Omega \subseteq \mathbb{C}$ be a bounded domain with $C^2$ boundary. Let $G(\zeta, z)$ be the Green's function for $\Omega$ and let $K(z, \zeta)$ be the Bergman kernel for $\Omega$. Then*

$$K(z,\zeta) = 4 \cdot \overline{\frac{\partial^2}{\partial\zeta\partial\overline{z}} G(\zeta, z)}\,. \tag{6.1.30.1}$$

**Proof** Our proof will use a version of Stokes's theorem written in the notation of complex variables. It says that, if $u \in C^1(\overline{\Omega})$, then

$$\oint_{\partial U} u(\zeta)\, d\zeta = 2i \cdot \iint_U \frac{\partial u}{\partial \overline{\zeta}}\, d\xi\, d\eta\,, \tag{6.1.30.2}$$

where again $\zeta = \xi + i\eta$. The reader is invited to convert this formula to an expression in $\xi$ and $\eta$ and to confirm that the result coincides with the standard real-variable version of Stokes's theorem that can be found in any calculus book (see, e.g., [THO], [BLK]).

Now we already know that

$$G(\zeta, z) = \frac{1}{4\pi} \log(\zeta - z) + \frac{1}{4\pi} \log \overline{(\zeta - z)} + F(\zeta, z)\,. \tag{6.1.30.3}$$

Here we think of the logarithm as a multivalued holomorphic function; after we take a derivative, the ambiguity (which comes from an additive multiple of $2\pi i$) goes away.

Differentiating with respect to $z$ (and using subscripts to denote derivatives), we find that

$$G_z(\zeta, z) = \frac{1}{4\pi} \frac{-1}{\zeta - z} + F_z(\zeta, z)\,.$$

We may rearrange this formula to read

$$\frac{1}{\zeta - z} = -4\pi \cdot G_z(\zeta, z) + 4\pi F_z(\zeta, z)\,.$$

We know that $G$, as a function of $\zeta$, vanishes on $\partial\Omega$. Hence so does $G_z$. Let $f \in C^2(\overline{\Omega})$ be holomorphic on $\Omega$. It follows that the Cauchy formula

$$f(z) = \frac{1}{2\pi i} \oint_{\partial\Omega} \frac{f(\zeta)}{\zeta - z}\, d\zeta$$

can be rewritten as

$$f(z) = -2i \oint_{\partial\Omega} f(\zeta) F_z(\zeta, z)\, d\zeta\,.$$

Now we apply Stokes's theorem (in the complex form) to rewrite this last as

$$f(z) = 4 \cdot \iint_{\Omega} (f(\zeta) F_z)_{\overline{\zeta}}(\zeta, z)\, d\xi\, d\eta\,,$$

where $\zeta = \xi + i\eta$. Since $f$ is holomorphic and $F$ is real-valued, we may conveniently write this last formula as

$$f(z) = 4 \cdot \iint_{\Omega} f(\zeta) \overline{F_{\zeta\overline{z}}}(\zeta, z)\, d\xi\, d\eta\,.$$

Now formula (6.1.30.3) tells us that $F_{\zeta\overline{z}} = G_{\zeta\overline{z}}$. Therefore we have

$$f(z) = \iint_{\Omega} f(\zeta) 4\overline{G_{\zeta\overline{z}}}(\zeta, z)\, d\xi\, d\eta\,. \tag{6.1.30.4}$$

With a suitable limiting argument, we may extend this formula from functions $f$ that are holomorphic and in $C^2(\overline{\Omega})$ to functions in $A^2(\Omega)$.

It is straightforward now to verify that $4\overline{G_{\zeta\overline{z}}}$ satisfies the first three characterizing properties of the Bergman kernel, just by examining our construction. The crucial reproducing property is of course formula (6.1.30.4). Then it follows that

$$K(z, \zeta) = 4 \cdot \overline{\frac{\partial^2}{\partial\zeta\partial\overline{z}} G(\zeta, z)}\,.$$

That is the desired result. □

It is worth noting that the proposition we have just established gives a practical method for confirming the existence of the Bergman kernel—by relating it to the Green's function, whose existence is elementary. See [HARV] for a version of these techniques in the several complex variables context.

Now let us calculate. Of course the Green's function of the unit disc $D$ is

$$G(\zeta, z) = \frac{1}{2\pi} \log|\zeta - z| - \frac{1}{2\pi} \log|1 - \zeta\overline{z}|\,,$$

as a glance at any classical complex analysis text will tell us (see, for example, [AHL1] or [HIL]). Verify the defining properties of the Green's function for yourself.

With formula (6.1.30.1) in mind, we can make life a bit easier by writing

$$\begin{aligned} G(\zeta, z) = &\frac{1}{4\pi}\log(\zeta - z) + \frac{1}{4\pi}\log(\overline{\zeta - z}) \\ &-\frac{1}{4\pi}\log(1 - \zeta\overline{z}) - \frac{1}{4\pi}\log\left(\overline{1 - \zeta\overline{z}}\right) . \end{aligned}$$

Here we think of the expression on the right as the concatenation of four multivalued functions, in view of the ambiguity of the logarithm function. This ambiguity is irrelevant for us because the derivative of the Green's function is still well defined (i.e., the derivative annihilates additive constants).

Now we readily calculate that

$$\frac{\partial G}{\partial \overline{z}} = \frac{1}{4\pi} \cdot \frac{-1}{\overline{\zeta} - \overline{z}} + \frac{1}{4\pi} \cdot \frac{\zeta}{1 - \zeta\overline{z}}$$

and

$$\frac{\partial^2 G}{\partial \zeta \partial \overline{z}} = \frac{1}{4\pi} \cdot \frac{1}{(1 - \zeta\overline{z})^2} .$$

In conclusion, we may apply Proposition 6.1.30 to see that

$$K(z, \zeta) = \frac{1}{\pi} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^2} .$$

This result is consistent with that obtained in the other calculation of the Bergman kernel of the disc. The Bergman metric, as before, is obtained by differentiation.

### 6.1.5 *Construction of the Bergman Kernel by Way of Conformal Invariance*

Let $D \subseteq \mathbb{C}$ be the unit disc. First we notice that, if either $f \in A^2(D)$ or $\overline{f} \in A^2(D)$, then

$$f(0) = \frac{1}{\pi} \iint_D f(\zeta)\, dA(\zeta) . \tag{6.1.5.1}$$

This is the standard, two-dimensional area form of the mean-value property for holomorphic or harmonic functions.

Of course the constant function $u(z) \equiv 1$ is in $A^2(D)$, so it is reproduced by integration against the Bergman kernel. Hence, for any $w \in D$,

$$1 = u(w) = \iint_D K(w,\zeta)u(\zeta)\, dA(\zeta) = \iint_D K(w,\zeta)\, dA(\zeta)\,,$$

or

$$\frac{1}{\pi} = \frac{1}{\pi}\iint_D K(w,\zeta)\, dA(\zeta)\,.$$

By (6.1.5.1), we may conclude that

$$\frac{1}{\pi} = K(w,0)$$

for any $w \in D$.

Now, for $a \in D$ fixed, consider the Möbius transformation

$$h(z) = \frac{z-a}{1-\overline{a}z}\,.$$

We know that

$$h'(z) = \frac{1-|a|^2}{(1-\overline{a}z)^2}\,.$$

We may thus apply Proposition 6.1.14 with $\phi = h$ to find that

$$\begin{aligned} K(w,a) &= h'(w)\cdot K(h(w),h(a))\cdot \overline{h'(a)} \\ &= \frac{1-|a|^2}{(1-\overline{a}w)^2}\cdot K(h(w),0)\cdot \frac{1}{1-|a|^2} \\ &= \frac{1}{(1-\overline{a}w)^2}\cdot\frac{1}{\pi} \\ &= \frac{1}{\pi}\cdot\frac{1}{(1-w\overline{a})^2}\,. \end{aligned}$$

This is our formula for the Bergman kernel.

## 6.2 The Szegő and Poisson–Szegő Kernels

**Capsule:** The Szegő kernel and integral formula are very natural generalizations of the Cauchy kernel and formula in one complex variable. The Poisson–Szegő kernel is an interesting generalization of the Poisson kernel to the several complex variable setting. Both are useful tools, and well worth mastering.

The basic theory of the Szegő kernel is similar to that for the Bergman kernel—they are both special cases of a general theory of "Hilbert spaces with reproducing kernel" (see [ARO]). Thus we only outline the basic steps here, leaving details to the reader. See also the book [KRA11, Ch. 8].

Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain with $C^2$ boundary. Let $A(\Omega)$ be those functions continuous on $\overline{\Omega}$ that are holomorphic on $\Omega$. Let $H^2(\partial\Omega)$ be the space consisting of the closure in the $L^2(\partial\Omega, d\sigma)$ topology of the restrictions to $\partial\Omega$ of elements of $A(\Omega)$. Then $H^2(\partial\Omega)$ is a proper Hilbert subspace of $L^2(\partial\Omega)$. Each element $f \in H^2(\partial\Omega)$ has a natural holomorphic extension to $\Omega$ given by its Poisson integral $Pf$. It is a standard fact—see [KRA1, Ch. 8] or our Chapter 2—that

$$\lim_{\epsilon \to 0^+} f(\zeta - \epsilon \nu_\zeta) = f(\zeta)$$

for almost every $\zeta$ in the boundary of $\Omega$. Here, as usual, $\nu_\zeta$ is the unit outward normal to $\partial\Omega$ at the point $\zeta$.

For each fixed $z \in \Omega$, the functional

$$\psi_z : H^2(\Omega) \ni f \mapsto Pf(z)$$

is continuous. (Why?—You may find the Bochner–Martinelli formula, Theorem 5.2.3, useful here.) Let $\widetilde{k}_z(\zeta)$ be the Hilbert space representative (coming from the Riesz representation theorem) for the functional $\psi_z$. Define the Szegő kernel $S(z, \zeta)$ by the formula

$$S(z, \zeta) = \overline{\widetilde{k}_z(\zeta)} \ , z \in \Omega, \ \ \zeta \in \partial\Omega.$$

If $f \in H^2(\partial\Omega)$ then

$$f(z) = \int_{\partial\Omega} S(z, \zeta) f(\zeta) d\sigma(\zeta)$$

for all $z \in \Omega$. Here $d\sigma$ is $(2n-1)$-dimensional Hausdorff measure on $\partial\Omega$. We shall not explicitly formulate and verify the various uniqueness and extremal properties for the Szegő kernel. The statements and proofs are exactly like those for the Bergman kernel.

We note that the integral

$$\mathbf{S}g(z) = \int_{\partial\Omega} S(z,\zeta)g(\zeta)\,d\zeta$$

defines a projection from $L^2(\partial\Omega)$ to $H^2(\Omega)$. This is because the mapping is self-adjoint, idempotent, and fixes $H^2$. We call this mapping the *Szegő projection*. [Note that the Bergman projection is constructed similarly.]

Let $\{\phi_j\}_{j=1}^{\infty}$ be any complete, orthonormal basis for $H^2(\partial\Omega)$. Define

$$S'(z,\zeta) = \sum_{j=1}^{\infty} \phi_j(z)\overline{\phi_j(\zeta)}, \quad z,\zeta \in \Omega\,.$$

For convenience we tacitly identify here each function with its Poisson extension to the interior of the domain. Then, for $K \subseteq \Omega$ compact, the series defining $S'$ converges uniformly on $K \times K$. By a Riesz–Fischer argument, $S'(\cdot,\zeta)$ is the Poisson integral of an element of $H^2(\partial\Omega)$ and $S'(z,\cdot)$ is the conjugate of the Poisson integral of an element of $H^2(\partial\Omega)$. So $S'$ extends to $(\overline{\Omega}\times\Omega)\cup(\Omega\times\overline{\Omega})$, where it is understood that all functions on the boundary are defined only almost everywhere. The kernel $S'$ is conjugate symmetric. Also, by Riesz–Fischer theory, $S'$ reproduces $H^2(\partial\Omega)$. Since the Szegő kernel is unique, it follows that $S = S'$.

We can actually calculate the Szegő kernel on the disc. We first note that $\{z^j\}_{j=0}^{\infty}$ forms a basis for the Hilbert space $H^2(D)$. This follows from the standard theory of power series for holomorphic functions on $D$. It is orthogonal by parity. It is complete by the uniqueness of the power series expansion. With a simple calculation, we can normalize the basis to the complete orthonormal basis $\{(1/\sqrt{2\pi})z^j\}_{j=0}^{\infty}$. Thus we see that

$$S(z,\zeta) = \sum_{j=0}^{\infty} \frac{1}{2\pi} z^j \overline{\zeta}^j = \frac{1}{2\pi} \cdot \frac{1}{1 - z\cdot\overline{\zeta}}\,.$$

Now it is instructive to write out the Szegő integral for a function in $A(D)$:

$$\begin{aligned} f(z) &= \int_0^{2\pi} f(e^{i\theta}) \cdot \frac{1}{2\pi} \cdot \frac{1}{1 - z\cdot e^{-i\theta}}\,d\theta \\ &= \frac{1}{2\pi i}\int_0^{2\pi} \frac{f(e^{i\theta})}{e^{i\theta} - z} \cdot ie^{i\theta}\,d\theta \\ &= \frac{1}{2\pi i}\oint_{\partial D} \frac{f(\zeta)}{\zeta - z}\,d\zeta\,. \end{aligned}$$

Thus we see that the canonical Szegő integral formula is in fact nothing other than the constructive Cauchy integral formula. But only on the disc!

The Poisson–Szegő kernel is obtained by a formal procedure from the Szegő kernel: this procedure manufactures a *positive* reproducing kernel from one that is not necessarily positive. The origin of this kernel may be found in [HUA, Ch. 3]. Note in passing that, just as we argued for the Bergman kernel in the last section, $S(z, z)$ is never 0 when $z \in \Omega$.

**Proposition 6.2.1** *Define*

$$\mathcal{P}(z, \zeta) = \frac{|S(z, \zeta)|^2}{S(z, z)}, \quad z \in \Omega, \ \zeta \in \partial\Omega.$$

*Then, for any $f \in A(\Omega)$ and $z \in \Omega$, it holds that*

$$f(z) = \int_{\partial\Omega} f(\zeta)\mathcal{P}(z, \zeta)d\sigma(\zeta).$$

**Remark 6.2.2** In passing to the Poisson–Szegő kernel we gain the advantage of positivity of the kernel (for more on this circle of ideas, see [KRA1, Ch. 8] and also [KAT, Ch. 1]). However we lose something in that $\mathcal{P}(z, \zeta)$ is no longer holomorphic in the $z$ variable nor conjugate holomorphic in the $\zeta$ variable. The literature on this kernel is rather sparse and there are many unresolved questions. The paper [KRA20] discusses some of the mapping properties of the Poisson–Szegő kernel. See also the more recent book [KRA11, Ch. 8]. It is an interesting historical fact that the Poisson–Szegő kernel was invented by L.-K. Hua in [HUA], though he did *not* give the kernel this name. □

As an exercise, use the paradigm of Proposition 6.2.1 to construct a positive kernel from the Cauchy kernel on the disc (be sure to first change notation in the usual Cauchy formula so that it is written in terms of arc length measure on the boundary). What familiar kernel results?

Like the Bergman kernel, the Szegő and Poisson–Szegő kernels can almost never be explicitly computed. They can be calculated asymptotically in a number of important instances, however (see C. Fefferman [FEF1, Part I], L. Boutet de Monvel and J. Sjöstrand [BOUTS]). We will give explicit formulas for these kernels on the ball.

**Lemma 6.2.3** *The functions $\{z^\alpha\}$, where $\alpha$ ranges over multi-indices, are pairwise orthogonal and span $H^2(\partial B)$.*

**Proof** The orthogonality follows from symmetry considerations. For the completeness, notice that it suffices to see that the span of $\{z^\alpha\}$ is dense in $A(B)$ in the uniform topology on the boundary. By the Stone–Weierstrass theorem, the closed algebra generated by $\{z^\alpha\}$ and $\{\overline{z}^\alpha\}$ is all of $C(\partial B)$. But the monomials $\overline{z}^\alpha, \alpha \neq 0$, are orthogonal to $A(B)$ (use the power series expansion about the origin to see this). The claimed density follows. □

**Lemma 6.2.4** *Let* $\mathbf{1} = (1, 0, \ldots, 0)$. *Then*

$$S(z, \mathbf{1}) = \frac{(n-1)!}{2\pi^n} \frac{1}{(1 - z_1)^n}.$$

**Proof** We have that

$$\begin{aligned} S(z, \mathbf{1}) &= \sum_\alpha \frac{z^\alpha \cdot \mathbf{1}^\alpha}{\|z_1^\alpha\|^2_{L^2(\partial B)}} \\ &= \sum_{k=0}^\infty \frac{z_1^k}{\eta(k)} \\ &= \frac{1}{2\pi^n} \sum_{k=0}^\infty \frac{z_1^k (k+n-1)!}{k!} \\ &= \frac{(n-1)!}{2\pi^n} \sum_{k=0}^\infty \binom{k+n-1}{n-1} z_1^k \\ &= \frac{(n-1)!}{2\pi^n} \frac{1}{(1-z_1)^n}. \end{aligned}$$

□

**Lemma 6.2.5** *Let* $\rho$ *be a unitary rotation on* $\mathbb{C}^n$. *For any* $z \in \overline{B}, \zeta \in \partial B$, *we have that* $S(z, \zeta) = S(\rho z, \rho\zeta)$.

**Proof** This is a standard change of variables argument and we omit it. □

**Prelude** The next theorem presents an explicit formula for the Szegő kernel on the unit ball in $\mathbb{C}^n$. There are not many domains on which the Szegő kernel is known explicitly.

**Theorem 6.2.6** *The Szegő kernel for the ball is*

$$S(z, \zeta) = \frac{(n-1)!}{2\pi^n} \frac{1}{(1 - z \cdot \overline{\zeta})^n}.$$

**Proof** Let $z \in B$ be arbitrary. Let $\rho$ be the unique unitary rotation such that $\rho z$ is a multiple of $\mathbf{1}$. Then

$$\begin{aligned} S(z, \zeta) &= S(\rho^{-1}\mathbf{1}, \zeta) \\ &= S(\mathbf{1}, \rho\zeta) = \overline{S(\rho\zeta, \mathbf{1})} \\ &= \frac{(n-1)!}{2\pi^n} \frac{1}{\left(1 - (\overline{\rho\zeta}) \cdot \mathbf{1}\right)^n} \end{aligned}$$

$$= \frac{(n-1)!}{2\pi^n} \frac{1}{\left(1 - \overline{\zeta} \cdot (\rho^{-1}\mathbf{1})\right)^n}$$

$$= \frac{(n-1)!}{2\pi^n} \frac{1}{(1 - z \cdot \overline{\zeta})^n} .$$

□

**Corollary 6.2.7** *The Poisson–Szegő kernel for the ball is*

$$\mathcal{P}(z,\zeta) = \frac{(n-1)!}{2\pi^n} \frac{(1-|z|^2)^n}{|1 - z \cdot \overline{\zeta}|^{2n}}.$$

**Exercise for the Reader** Calculate the Szegő and Poisson–Szegő kernel for the polydisc.

Let us now review some of our key ideas. We let $\Omega$ be a bounded domain in $\mathbb{C}$ or $\mathbb{C}^n$. Let $L^2(\Omega)$ be the square-integrable functions on $\Omega$ (with respect to ordinary Lebesgue measure) and let $A^2(\Omega) \subseteq L^2(\Omega)$ be the subspace consisting of the holomorphic functions. This last space is known as the *Bergman space*. Then $A^2$ is a closed, Hilbert subspace of $L^2$. Thus we may consider the projection

$$P : L^2(\Omega) \to A^2(\Omega) .$$

It is well known that this projection mapping is given by an integration kernel $K(z,\zeta)$ which is called the *Bergman kernel* (see [KRA1, Ch. 1] for details in this matter). As an instance, the Bergman kernel for $\Omega$ the unit disc $D$ in the complex plane is given by

$$K_D(z,\zeta) = \frac{1}{\pi} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^2} . \tag{6.2.8}$$

Of particular interest for us is the obvious fact that this $K$ is smooth on $\overline{D} \times \overline{D} \setminus (\Delta \cap [\partial D \times \partial D])$, where $\Delta$ is the diagonal.

We might also note that the Bergman kernel for the unit ball $B$ in $\mathbb{C}^n$ is given by the formula

$$K_B(z,\zeta) = \frac{n!}{\pi^n} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^{n+1}} .$$

Again, one may see by inspection that $K$ is smooth on $\overline{B} \times \overline{B} \setminus \mathcal{D}$.

Norberto Kerzman [KER2] has shown that a similar result is true on any smoothly bounded, strictly pseudoconvex domain: the Bergman kernel is smooth on the product of the closures less the boundary diagonal. In fact we now know (see the remark at the end of Kerzman's paper) that the result holds on any domain

for which the $\overline{\partial}$-Neumann operator $N$ is known to be pseudolocal (see [KER2] for an explication of these ideas). This just means that $N\varphi$ is smooth wheerever $\varphi$ is smooth.

To see this, note that if $f \in A^2(\Omega)$ and $z \in \Omega$, then

$$\begin{aligned} f(z) &= \langle f, \delta_z \rangle \\ &= \langle Pf, \delta_z \rangle \\ &= \langle f, P\delta_z \rangle \, . \end{aligned}$$

Here, of course, $\delta_z$ is the Dirac delta mass at $z$. It follows that

$$K(z, \zeta) = \overline{P(\delta_z)} \, . \tag{6.2.9}$$

Here $P$ is the Bergman projection. Of course $P(\delta_z)$ is an element of $A^2(\Omega)$, so it is certainly smooth on $\Omega$. We are interested in the behavior of this function (of the variables $z$ and $\zeta$) as $z$ and $\zeta$ tend to the boundary.

Now the well-known formula of Kohn is given by

$$P = I - \overline{\partial}^* N \overline{\partial} \, . \tag{6.2.10}$$

Here $N$ is the $\overline{\partial}$-Neumann operator. See [KRA5, Ch. 7] for the details.

To be more specific, we can combine (6.2.9) and (6.2.10) to see that

$$K(z, \zeta) = \delta_\zeta - \overline{\partial}^* N \overline{\partial} \delta_\zeta \, . \tag{6.2.11}$$

Of course $\overline{\partial}$ and $\overline{\partial}^*$ are classical partial differential operators. So it is certainly the case that, whereever a function $\varphi$ is smooth, then also $\overline{\partial}\varphi$ and $\overline{\partial}^* \varphi$ are smooth. The pseudolocality of $N$—a noted result of J. J. Kohn [KOH2]—means precisely that $N\psi$ is smooth whereever $\psi$ is smooth. All in all then, line (6.2.11) tells us that, if $z$ is fixed inside $\Omega$, then $K(z, \zeta)$ is smooth up to the boundary in the $\zeta$ variable (just because $\delta_z$ is). The estimates that come with the pseudolocality of $N$ (see [KRA5, Ch. 7] for the strictly pseudoconvex case) tell us further that, as $\zeta \to \partial\Omega$, and if $z$ stays in a compact subset of the closure that is bounded from $\zeta$, then $K$ is still smooth.

With all this information in hand, it is a matter of interest to know for which domains the Bergman kernel will have a singularity when the two variables approach the same boundary point. In those circumstances the question of boundedness of the Bergman projection on various $L^p$ and other spaces has a hope of being tractable, and the study of Condition $R$ is accessible. But in fact Ligocka [LIG2] has shown that this contention fails on the Diederich–Fornaess worm domain (see [DIRF2] as well as [CHS, Ch. 6] and [KRPE1]). Of course it is known (see [KRPE2] or [CHS, Ch. 6]) that the worm is bounded, pseudoconvex, and has smooth boundary. We treat the worm in Chapter 9.

We note in passing that the Bergman kernel for the bidisc does *not* satisfy the condition described in the last paragraph (of course the bidisc does not have smooth boundary). One might speculate that a similar failure occurs on a smoothly bounded, convex domain in $\mathbb{C}^2$ which has an analytic disc in the boundary.

Our purpose in the next sections is to show that the assertions being discussed here fail dramatically on the disc $D \subseteq \mathbb{C}$ when one considers the reproducing kernels for certain closed subspaces of $A^2(D)$.

## 6.3 Aronszajn Theory

**Capsule:** Aronszajn's ideas date to the early 1950s, and they were quite prescient. He generalized the idea of the Bergman and Szegő kernels and created an abstract framework that one can study in their stead. And it has become quite fashionable today to do so. We introduce this set of ideas in the present section.

One of the first canonical integral formulas ever created was that of Stefan Bergman [BERM1]. We shall present Bergman's idea in the context of a more general construction due to Nachman Aronszajn [ARO]. This is the idea of a Hilbert space with reproducing kernel. Fortunately Aronszajn's idea also entails the Szegő kernel and several other important reproducing kernels.

**Definition 6.3.1** Let $X$ be any set and let $\mathcal{H}$ be a Hilbert space of complex-valued functions on $X$. We say that $\mathcal{H}$ is a *Hilbert space with reproducing kernel* if, for each $x \in X$, the linear (point evaluation) map of the form

$$\begin{aligned} L_x : \mathcal{H} &\longrightarrow \mathbb{C} \\ f &\longmapsto f(x)\,, \end{aligned}$$

is continuous. We write this as

$$|f(x)| \leq C \cdot \|f\|_{\mathcal{H}}\,. \tag{6.3.1.1}$$

In this circumstance, the classical Riesz representation theorem (see [KRA14, Ch. 3]) tells us that, for each $x \in X$, there is a unique element $k_x \in \mathcal{H}$ such that

$$f(x) = \langle f, k_x \rangle \qquad \forall f \in \mathcal{H}\,. \tag{6.3.1.2}$$

We then define a function

$$K : X \times X \to \mathbb{C}$$

by the formula

$$K(x, y) \equiv \overline{k_x(y)}\,.$$

The function $K$ is the *reproducing kernel* for the Hilbert space $\mathcal{H}$.

We see that $K$ is uniquely determined by $\mathcal{H}$ because, again by the Riesz representation theorem, the element $k_x$ for each $x \in X$ is unique.

We know from our earlier discussion that, if $\{\varphi_j\}$ is a complete orthonormal basis for $\mathcal{H}$, then

$$K(x, y) = \sum_{j=1}^{\infty} \varphi_j(x)\overline{\varphi_j(y)}\,.$$

Here the convergence is in the Hilbert space topology in each variables.

## 6.4 A New Basis

**Capsule:** In this section we explore the idea, on the disc, of examining a nonstandard basis for a Bergman space. The goal is to determine what sort of Bergman kernel it generates. We in fact look at several different examples.

We do our work in this section on the unit disc $D$ in the complex plane $\mathbb{C}$.

Of course an orthogonal basis for the Bergman space of the disc $D$ is given by $\{\zeta^j\}_{j=0}^{\infty}$. Calculating the $L^2$ norm of each of these elements, we find that an orthonormal basis for $A^2(D)$ is

$$\varphi_j(\zeta) = \frac{\sqrt{j+1}}{\sqrt{\pi}} \cdot \zeta^j\,, \quad j = 0, 1, 2, \ldots.$$

The basis is seen to be complete just by the theory of power series. Now it is a standard fact that the Bergman kernel can be constructed from such a basis by the formula

$$K(z, \zeta) = \sum_{j=0}^{\infty} \varphi_j(z) \cdot \overline{\varphi_j(\zeta)}\,.$$

This series converges, uniformly on compact subsets of $D \times D$, to the kernel

$$K(z, \zeta) = \frac{1}{\pi} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^2}\,.$$

But now let us consider[2] the subspace $\mathcal{X}$ of $A^2(D)$ generated by the basis elements $\varphi_{2j}(\zeta)$, $j = 0, 1, 2, \ldots$. This is certainly a closed subspace of $A^2(D)$, and *its* Bergman kernel is given by

[2]The discussion here is based on unpublished work [BOAKP] of Boas, Krantz, and Peloso.

$$\sum_{j=0}^{\infty} \varphi_{2j}(z) \cdot \overline{\varphi_{2j}(\zeta)} \,. \tag{6.4.1}$$

Let us calculate explicitly the sum in (6.4.1). It is given by

$$\sum_{j=0}^{\infty} \frac{2j+1}{\pi} z^{2j} \cdot \overline{\zeta^{2j}} \,.$$

We may sum this series by examining the auxiliary expression

$$\begin{aligned}
\sum_{j=0}^{\infty} \frac{2j+1}{\pi} \alpha^{2j} &= \frac{d}{d\alpha} \left[ \frac{1}{\pi} \cdot \sum_{j=0}^{\infty} \alpha^{2j+1} \right] \\
&= \frac{d}{d\alpha} \left[ \frac{1}{\pi} \cdot \alpha \cdot \frac{1}{1-\alpha^2} \right] \\
&= \frac{1}{\pi} \cdot \frac{\alpha^2+1}{(1-\alpha^2)^2} \,.
\end{aligned}$$

We conclude that the Bergman kernel $K_{\mathcal{X}}$ for $\mathcal{X}$ is given by

$$K_{\mathcal{X}}(z,\zeta) = \frac{1}{\pi} \cdot \frac{(z\overline{\zeta})^2+1}{(1-(z\overline{\zeta})^2)^2} \,. \tag{6.4.2}$$

The notable fact is that $K_{\mathcal{X}}$ blows up when either $z \cdot \overline{\zeta}$ tends to 1, or when $z \cdot \overline{\zeta}$ tends to $-1$. In other words, $K_{\mathcal{X}}$ blows up when either $z$ and $\zeta$ tend to the *same* boundary point or when $z$ and $\zeta$ tend to antipodal boundary points.

A companion result is obtained when one considers instead the space $\mathcal{Y}$ generated by the basis $\{\varphi_{2j+1}\}, j = 0, 1, 2, \ldots$. Its Bergman kernel is given by

$$\sum_{j=0}^{\infty} \varphi_{2j+1}(z) \cdot \overline{\varphi_{2j+1}(\zeta)} \,. \tag{6.4.3}$$

Let us calculate explicitly the sum in (6.4.3). It is given by

$$\sum_{j=0}^{\infty} \frac{2j+2}{\pi} z^{2j+1} \cdot \overline{\zeta^{2j+1}} \,.$$

We may sum this series by examining the auxiliary expression

$$\begin{aligned}\sum_{j=0}^{\infty} \frac{2j+2}{\pi}\alpha^{2j+1} &= \frac{d}{d\alpha}\left[\frac{1}{\pi}\cdot\sum_{j=0}^{\infty}\alpha^{2j+2}\right] \\ &= \frac{d}{d\alpha}\left[\frac{1}{\pi}\cdot\alpha^2\cdot\frac{1}{1-\alpha^2}\right] \\ &= \frac{1}{\pi}\cdot\frac{2\alpha}{(1-\alpha^2)^2}\,.\end{aligned}$$

We conclude that the Bergman kernel $K_{\mathcal{Y}}$ for $\mathcal{Y}$ is given by

$$K_{\mathcal{Y}}(z,\zeta) = \frac{1}{\pi}\cdot\frac{2z\overline{\zeta}}{(1-(z\overline{\zeta})^2)^2}\,. \tag{6.4.4}$$

Again, this new Bergman kernel has boundary singularities when either $z$ and $\zeta$ tend to the *same* boundary point or when $z$ and $\zeta$ tend to antipodal boundary points.

It is notable that

$$K_{\mathcal{X}}(z,\zeta) + K_{\mathcal{Y}}(z,\zeta) = \frac{1}{\pi}\cdot\frac{1}{(1-z\cdot\overline{\zeta})^2} = K_D(z,\zeta)\,.$$

It may be noted that the even part (in the $z$ variable) of the classical Bergman kernel for the disc $D$ is

$$\frac{1}{\pi}\cdot\frac{(z\overline{\zeta})^2+1}{(1-(z\overline{\zeta})^2)^2}\,.$$

This is precisely the kernel that we found for the Bergman space $\mathcal{X}$ generated by the basis of monomials with even index. Likewise the odd part (in the $z$ variable) of the classical Bergman kernel for the disc $D$ is

$$\frac{1}{\pi}\cdot\frac{2z\overline{\zeta}}{(1-(z\overline{\zeta})^2)^2}\,.$$

This is precisely the kernel that we found for the Bergman space $\mathcal{Y}$ generated by the basis of monomials with odd index.

## 6.5 Additional Examples

**Capsule:** This section continues the theme of the last section.

Of course one is by no means restricted to doing analysis modulo 2. One could examine the Bergman space generated by $\varphi_0$, $\varphi_3$, $\varphi_6$, etc. and calculate the corresponding Bergman kernel. We shall not do so here, but simply note that the resulting kernel has boundary singularities at the *third* roots of unity. In other words, there are three boundary singularities.

A similar result obtains if one considers the Bergman kernel for the space generated by $\varphi_{jm}$, where $m$ is any fixed positive integer. In that case, there are boundary singularities at the $m^{\text{th}}$ roots of unity.

A rather more dramatic example is obtained when we consider the space $\mathcal{Z}$ generated by the basis $\{\varphi_{2^j}\}, j = 0, 1, 2, \ldots$. The corresponding Bergman kernel is

$$K_{\mathcal{Z}}(z, \zeta) = \sum_{j=0}^{\infty} \frac{2^j + 1}{\pi} z^{2^j} \cdot \overline{\zeta^{2^j}} = \sum_{j=0}^{\infty} \frac{2^j + 1}{\pi} (z \cdot \overline{\zeta})^{2^j} \,.\,.$$

We shall not sum this series explicitly. But we instead analyze the auxiliary holomorphic function

$$\Phi(\alpha) = \sum_{j=0}^{\infty} \frac{2^j + 1}{\pi} \alpha^{2^j} \,. \tag{6.5.1}$$

The sequence of exponents is lacunary, and the radius of convergence of the series is 1. Thus the Hadamard gap theorem applies and we see that the series in (6.5.1) defines a holomorphic function with a singularity at every boundary point of the disc. In other words, the holomorphic functions $\Phi$ does not analytically continue past any boundary point. From this we conclude that the kernel $K_{\mathcal{Z}}$ has boundary singularities at *every* boundary point of the unit disc $D$.

If we consider the complementary basis $\varphi_1, \varphi_3, \varphi_5, \varphi_6, \varphi_7$, etc. and the Bergman kernel $K'_{\mathcal{Z}}$ associated to the space it generates, then we must conclude that it, too, has boundary singularities at every boundary point. This is true because

$$K_{\mathcal{Z}} + K'_{\mathcal{Z}} = K_D \,.$$

What is particularly interesting is that the first basis in this example (the one with a lacunary sequence of indices) is quite sparse, while the second is not.

## 6.6 The Behavior of the Singularity

**Capsule:** This section continues some of the ideas of the preceding sections and in particular studies the nature of the singularity of the Bergman kernel.

In this section we build (at least philosophically) on the earlier material and prove the next result. We note that, in this theorem, "boundary singularity" for the Bergman kernel means a boundary point of the unit disc $D \subseteq \mathbb{C}$ which has no neighborhood to which the Bergman kernel directly analytically continues as $z, \zeta$ both approach the point. In particular, "boundary singularity" does not necessarily mean that the kernel is blowing up at the indicated point.

**Theorem 6.6.1** *Let $\mathbb{Z}^+$ denote the collection of nonnegative integers. Write*

$$\mathbb{Z}^+ = \mathcal{I}_1 \cup \mathcal{I}_2 \,,$$

*where $\mathcal{I}_1 \cap \mathcal{I}_2 = \emptyset$ and each of $\mathcal{I}_1$, $\mathcal{I}_2$ is an infinite set.*

*Let $B_1$ be the Bergman space on the unit disc $D$ generated by the basis with elements in $\mathcal{I}_1$ and $B_2$ the Bergman space on $D$ generated by the basis with elements in $\mathcal{I}_2$. Correspondingly, let $K_1$ be the Bergman kernel for $B_1$ and $K_2$ the Bergman kernel for $B_2$. Then $K_1$, $K_2$ each have more than one boundary singularity.*

**Proof** Let $\{\varphi_j\}_{j=1}^{\infty}$ be a basis for $B_1$. Then the corresponding Bergman kernel is

$$K_1(z, \zeta) = \sum_{j=1}^{\infty} \varphi_j(z) \cdot \overline{\varphi}_j(\zeta) \,.$$

In particular, we see immediately that the kernel depends on $z \cdot \overline{\zeta}$. Since we are considering $\{\varphi_j\}$ as a subset of the particular basis $\{\sqrt{(j+1)/\pi}\,\zeta^j\}$, we know that the Bergman kernel for $B_1$ is a sum of *some* of the terms $(j+1)(z\overline{\zeta})^j$ (for convenience we omit the factor of $1/\pi$). Thus it suffices for us to study the single-variable power series

$$\sum_j (j+1)t^j \,.$$

Here the sum is taken over *some*, but not *all*, of the $j$. Also $t$ is a complex variable. Certainly a series of this form has radius of convergence 1 and has a singularity at the boundary point 1. This corresponds to the singularity of the Bergman kernel on the boundary diagonal. The question that we must study is whether this one-variable power series has any other singular points on the boundary of $D$.

Note that the derivative or the integral of the series has the very same singularities. So we may as well study the series

$$\sum_j t^j \,,$$

where the sum is taken over *some* of the $j$, but not *all* of the $j$.

Now a classical result of Szegő (see [SZE]) comes into play. This result says that a power series with finitely many distinct coefficients (in our case the coefficients are all either 0 or 1) either has the boundary circle as its natural boundary (meaning that *every* boundary point is singular) or else the series sums to a rational function (whose Maclaurin series coefficients form a sequence that is eventually periodic). We know from our earlier calculations that, in the second case, there are polynomials $p$ and $q$ such that the series represents a rational function of the form

$$p(\zeta) + \frac{q(\zeta)}{1 - \zeta^m} .$$

Here $m$ is the period of the coefficient sequence.

Translating these ideas back into the language of $z$ and $\zeta$, we find that we have the following two cases for our Bergman kernel:

**(1)** The subspace $B_1$ of the Bergman space has a basis of monomials with an eventually periodic structure. In this case, the kernel has, for each fixed $z \in \partial D$, singularities at finitely many values (more than one) of $\zeta \in \partial D$.

**(2)** Our subspace of the Bergman space has a basis of monomials that lacks an eventually periodic structure. In this case, the kernel function is singular when $z$, $\zeta$ approach two arbitrary (and, in general, distinct) boundary points. □

The second case here holds in particular for either $\mathcal{I}_1$ or $\mathcal{I}_2$ when the set $\mathcal{I}_1$ is lacunary as in Section 6.5. We concluding by noting that the results of this section and the last are unpublished work of Boas, Krantz, and Peloso.

## 6.7 A Real Bergman Space

**Capsule:** In the present section we consider a Bergman space of harmonic functions and examine the corresponding kernel.

The space $\mathbf{h}^2(D)$ of square-integrable *harmonic* functions on the disc $D$ is a Hilbert space with reproducing kernel in the sense of Aronszajn (see [ARO]). In particular, if $K \subseteq D$ is compact then there is a constant $C = C(K)$ so that if $f \in \mathbf{h}^2(D)$ then

$$\sup_{z \in K} |f(z)| \leq C \cdot \|f\|_{\mathbf{h}^2} .$$

As a result, there is a "Bergman kernel" for this space.

It may be noted that a basis for $\mathbf{h}^2(D)$ consists of

$$\{\zeta^j\}_{j=0}^{\infty} \bigcup \{\overline{\zeta}^j\}_{j=1}^{\infty} .$$

The Bergman kernel for $\mathbf{h}^2$ is easily calculated; it is essentially a derivative of the usual Poisson kernel.

If we instead calculate the Bergman kernel for the space generated just by $\{\zeta^j\}$ or for the space generated just by $\{\overline{\zeta}^j\}$ then we obtain the usual Bergman kernel or (up to an additive constant) the conjugate of the usual Bergman kernel.

On the other hand, one might consider the space $\mathtt{h}^2(D)$ of harmonic functions $u$ on the disc that satisfy

$$\sup_{0<r<1} \int_0^{2\pi} |u(re^{i\theta})|^2 \, d\theta < \infty \, .$$

This is analogous to the Hardy space, but now we are focusing on *harmonic* functions. It is easy to verify that this is a Hilbert space with reproducing kernel. Calculating the kernel, one finds that it is the classical Poisson kernel. Given Stokes's theorem, it is no surprise that the kernel for $\mathbf{h}^2$ is a derivative of the kernel for $\mathtt{h}^2$.

One should note that, in this example, the Bergman kernel for a subspace does *not* have extra singularities. All the kernels being discussed here have just the classical singularity at 1.

## 6.8 Relation Between Bergman and Szegő

**Capsule:** This section explores an explicit connection between the Bergman and the Szegő kernels on the disc, the ball, and other domains. Intuition and Stokes's theorem suggest that the former kernel should be a (normal) derivative of the latter. The results of the present section bear out that insight.

### *6.8.1 Introduction*

Two of the most classical and well established reproducing formulas in complex analysis are those of S. Bergman and G. Szegő. The first of these is a formula for the Bergman space, and the associated integral lives on the interior of the domain in question. The latter of these is a formula for the Hardy space, and the associated integral lives on the boundary of the domain. For formal reasons, the Bergman integral gives rise to a projection from $L^2(\Omega)$ to $A^2(\Omega)$ (the Bergman space); likewise, the Szegő integral gives rise to a projection from $L^2(\partial\Omega)$ to $H^2(\partial\Omega)$ (the Hardy space).

Since both of the artifacts in question here are canonical, it is natural to suspect that there is some relationship between the two integral formulas. After all, they both reproduce functions that are continuous on the closure of the domain

and holomorphic on the interior. In the present discussion we establish such a connection—very explicitly—on a variety of domains in $\mathbb{C}^1$ and $\mathbb{C}^n$. This is done by way of a moderately subtle calculation using Stokes's theorem. The calculation itself has some intrinsic interest, but the main point is the relationship between the canonical integrals and the associated projections. See [KRA23] for the details of these ideas.

In separate calculations we treat the situation on the disc $D$, the unit ball $B$, and on a strongly pseudoconvex domain $\Omega$. While the first two calculations are very explicit, in some sense it is the third of these calculations which is most natural and most satisfying.

It is a pleasure to thank Jürgen Leiterer, who contributed a number of useful ideas to these calculations.

### 6.8.2 *The Case of the Disc*

Let $D$ be the unit disc in $\mathbb{C}$. In this context, the Szegő kernel is

$$S(z,\zeta) = \frac{1}{2\pi} \cdot \frac{1}{1 - z \cdot \overline{\zeta}}$$

and the Bergman kernel is

$$K(z,\zeta) = \frac{1}{\pi} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^2} \, .$$

Take $f$ to be real analytic on a neighborhood of $\overline{D}$. Now we can calculate

$$\begin{aligned}
&\frac{1}{2\pi} \int_{\partial D} f(\zeta) S(z,\zeta) d\sigma(\zeta) \\
&= \frac{1}{2\pi} \int_{\partial D} f(\zeta) \cdot \frac{1}{1 - z\overline{\zeta}} \frac{\left[\overline{\zeta} d\zeta - \zeta d\overline{\zeta}\right]}{2i} \\
&= \frac{1}{4\pi i} \int_{\partial D} \frac{f(\zeta)\overline{\zeta}}{1 - z \cdot \overline{\zeta}} d\zeta - \frac{1}{4\pi i} \int_{\partial D} \frac{f(\zeta)\zeta}{1 - z \cdot \overline{\zeta}} d\overline{\zeta} \\
&\overset{\text{(Stokes)}}{=} \frac{1}{4\pi i} \iint_D \frac{f(\zeta)}{1 - z \cdot \overline{\zeta}} d\overline{\zeta} \wedge d\zeta + \frac{1}{4\pi i} \iint_D \frac{f(\zeta)\overline{\zeta} z}{(1 - z \cdot \overline{\zeta})^2} d\overline{\zeta} \wedge d\zeta \\
&\qquad - \frac{1}{4\pi i} \iint_D \frac{\partial (f \cdot \zeta)/\partial \zeta}{1 - z \cdot \overline{\zeta}} d\zeta \wedge d\overline{\zeta} + \frac{1}{4\pi i} \iint_D \frac{\partial f / \partial \overline{\zeta} \cdot \overline{\zeta}}{1 - z \cdot \overline{\zeta}} d\overline{\zeta} \wedge d\zeta
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}\wedge d\zeta - \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)z\overline{\zeta}}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}\wedge d\zeta \\
&\qquad + \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)\overline{\zeta}z}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}\wedge d\zeta - \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial(f\cdot\zeta)/\partial\zeta}{1-z\cdot\overline{\zeta}}\,d\zeta\wedge d\overline{\zeta} \\
&\qquad + \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\overline{\zeta}\cdot\overline{\zeta}}{1-z\cdot\overline{\zeta}}\,d\overline{\zeta}\wedge d\zeta \\
&= \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}\wedge d\zeta - \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial(f\cdot\zeta)/\partial\zeta}{1-z\cdot\overline{\zeta}}\,d\zeta\wedge d\overline{\zeta} \\
&\qquad + \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\overline{\zeta}\cdot\overline{\zeta}}{1-z\cdot\overline{\zeta}}\,d\overline{\zeta}\wedge d\zeta \\
&= \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}\wedge d\zeta - \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\zeta\cdot\zeta}{1-z\cdot\overline{\zeta}}\,d\zeta\wedge d\overline{\zeta} \\
&\qquad - \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)}{1-z\cdot\overline{\zeta}}\,d\zeta\wedge d\overline{\zeta} + \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\overline{\zeta}\cdot\overline{\zeta}}{1-z\cdot\overline{\zeta}}\,d\overline{\zeta}\wedge d\zeta \\
&= \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}\wedge d\zeta - \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\zeta\cdot\zeta}{1-z\cdot\overline{\zeta}}\,d\zeta\wedge d\overline{\zeta} \\
&\qquad - \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)}{(1-z\cdot\overline{\zeta})^2}\,d\zeta\wedge d\overline{\zeta} + \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)z\overline{\zeta}}{(1-z\cdot\overline{\zeta})^2}\,d\zeta\wedge d\overline{\zeta} \\
&\qquad + \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\overline{\zeta}\cdot\overline{\zeta}}{1-z\cdot\overline{\zeta}}\,d\overline{\zeta}\wedge d\zeta \\
&= \frac{1}{2\pi i}\int\!\!\int_D \frac{f(\zeta)}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}\wedge d\zeta - \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\zeta\cdot\zeta}{1-z\cdot\overline{\zeta}}\,d\zeta\wedge d\overline{\zeta} \\
&\qquad + \frac{1}{4\pi i}\int\!\!\int_D \frac{f(\zeta)z\overline{\zeta}}{(1-z\cdot\overline{\zeta})^2}\,d\zeta\wedge d\overline{\zeta} + \frac{1}{4\pi i}\int\!\!\int_D \frac{\partial f/\partial\overline{\zeta}\cdot\overline{\zeta}}{1-z\cdot\overline{\zeta}}\,d\overline{\zeta}\wedge d\zeta \\
&= A - B + C + D\,.
\end{aligned}
$$

Certainly $A = \int_D f(\zeta)K(z,\zeta)\,dA(\zeta)$, where $K$ is the Bergman kernel of the disc. So this is the Bergman projection. Now we shall analyze the terms $-B + C + D$ to

see how the Szegő kernel differs from the Bergman kernel. We shall determine that, for certain monomials in a basis for the Bergman (Szegő) space, the sum of these three terms is zero. While for others it is not. In the latter case the difference will be controllable.

First we consider a monomial $f(\zeta) = \zeta^k \overline{\zeta}^m$ with $k = 0, 1, \ldots, m = 1, 2, \ldots$, and $k < m$. In this case,

$$\begin{aligned} Sf(z) &= \int_{\partial D} f(\zeta) S(z, \zeta)\, d\sigma(\zeta) \\ &= \int_{\partial D} \zeta^k \overline{\zeta}^m \frac{1}{1 - z\overline{\zeta}}\, d\sigma(\zeta)\,. \end{aligned}$$

Now we expand the Szegő kernel in a Neumann series and, counting powers of $\zeta$ and powers of $\overline{\zeta}$, we see immediately by parity that the integral is zero.

A similar calculation shows that, for these values of $m$ and $k$, $Bf(z) \equiv 0$.

But matters are different when $k \geq m$. In this case, for the Szegő integral, we may write $\zeta^k \overline{\zeta}^m = \zeta^{k-m} \equiv h(z)$ for $|\zeta| = 1$. Hence

$$\begin{aligned} Sf(z) &= Sh(z) \\ &= h(z) \\ &= z^{k-m}\,. \end{aligned}$$

By contrast,

$$\begin{aligned} Bf(z) &= \frac{1}{2\pi i} \int_D \frac{\zeta^k \overline{\zeta}^m}{(1 - z\overline{\zeta})^2}\, d\overline{\zeta} \wedge d\zeta \\ &= \frac{1}{2\pi i} \int_D \zeta^k \overline{\zeta}^m \left( \sum_{j=0}^{\infty} z^j \overline{\zeta}^j \right)^2 d\overline{\zeta} \wedge d\zeta \\ &= \frac{1}{2\pi i} \sum_{j,\ell=0}^{\infty} z^{j+\ell} \int_D \zeta^k \overline{\zeta}^{m+j+\ell}\, d\overline{\zeta} \wedge d\zeta \\ &= \frac{1}{2\pi i} \sum_{\substack{0 \leq j,\ell \leq k-m \\ j+\ell = k-m}} z^{k-m} \int_D |\zeta|^{2k}\, d\overline{\zeta} \wedge d\zeta \\ &= \frac{1}{2\pi i} \sum_{\substack{0 \leq j,\ell, \leq k-m \\ j+\ell = k-m}} z^{k-m} \frac{2\pi i}{k+1} \\ &= \frac{k-m+1}{k+1} z^{k-m} \\ &= \left(1 - \frac{m}{k+1}\right) z^{k-m}\,. \end{aligned}$$

Thus we see that

$$Sf(z) - Bf(z) = \frac{m}{k+1} z^{k-m} .$$

As a consequence, if now $f$ is a function that is real analytic in a neighborhood of $\overline{D}$, then we may write

$$f(\zeta) = \sum_{k \geq 0, m > 0} \alpha_{k,m} \zeta^k \overline{\zeta}^m .$$

As a consequence of our calculations above, we then find that

$$Sf(z) - Bf(z) = \sum_{k \geq 0, k \geq m} \frac{m}{k+1} z^{k-m} .$$

This series of course converges uniformly on compact subsets of the open unit disc $D$.

We treat the case of the Bergman and Szegő projections on the ball below.

Given Fefferman's asymptotic expansion for the Bergman kernel [FEF1], and Boutet de Monvel–Sjöstrand's asymptotic expansion for the Szegő kernel [BOUTS], one would expect a like calculation (up to a controllable error term) on a smoothly bounded, strongly pseudoconvex domain. Unfortunately we do not know enough about the canonical kernels on domains of finite type to be able to predict what will happen there. We explore the strongly pseudoconvex case below.

In a more recent work, Chen and Fu [CHEF] have explored some new comparisons of the Bergman and Szegő kernels. A sample theorem is this:

**Prelude** The next result of Chen and Fu gives a new and novel set of results about the relationship between the Bergman and Szegő kernels.

**Theorem 6.8.1** *Let $\Omega \subseteq \mathbb{C}^n$ be a pseudoconvex domain with $C^2$ boundary. Then*

**(1)** *For any $0 < a < 1$, there exists a constant $C > 0$ such that*

$$\frac{S(z,z)}{K(z,z)} \leq C\delta(z) |\log \delta(z)|^{n/a} .$$

**(2)** *If there is a neighborhood $U$ of $\partial\Omega$, a bounded, continuous plurisubharmonic function $\varphi$ on $U \cap \Omega$, and a defining function $\rho$ of $\Omega$ satisfying $i\partial\overline{\partial}\varphi \geq i\rho^{-1}\partial\overline{\partial}\rho$ on $U \cap \Omega$ as currents, then there exists constants $0 < a < 1$ and $C > 0$ such that*

$$\frac{S(z,z)}{K(z,z)} \geq C\delta(z) |\log \delta(z)|^{-1/a} .$$

These authors further show that, for a $C^2$-bounded *convex* domain the quotient $S/K$ is comparable to $\delta$ without any logarithmic factor.

The techniques used in the work of Chen and Fu are weighted estimates for the $\overline{\partial}$ operator (in the spirit of Hörmander's work [HOR1]) and also an innovative use of the Diederich–Fornæss index (see [DIRF1]). We can say no more about the details here.

We turn next to an examination of the situation on the unit ball $B$ in $\mathbb{C}^n$.

### 6.8.3 The Unit Ball in $\mathbb{C}^n$

For simplicity we shall in fact restrict attention to complex dimension 2. In that situation, the area measure $d\sigma$ on the boundary is given by

$$d\sigma = \frac{1}{16}\left[\zeta_1 d\zeta_2 \wedge d\overline{\zeta}_1 \wedge d\overline{\zeta}_2 - \zeta_2 d\zeta_1 \wedge d\overline{\zeta}_1 \wedge d\overline{\zeta}_2 + \overline{\zeta}_1 d\overline{\zeta}_2 \wedge d\zeta_1 \wedge d\zeta_2 - \overline{\zeta}_2 d\overline{\zeta}_1 \wedge d\zeta_1 \wedge d\zeta_2\right].$$

As a result, we have

$$\begin{aligned}
&\int_{\partial B} f(\zeta) S(z,\zeta)\, d\sigma(\zeta) \\
&= \frac{1}{2\pi^2}\cdot\frac{1}{16}\iiint_{\partial B} f(\zeta)\frac{1}{(1-z\cdot\overline{\zeta})^2}\left[\zeta_1 d\zeta_2\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2 - \zeta_2 d\zeta_1\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2\right. \\
&\qquad \left. + \overline{\zeta}_1 d\overline{\zeta}_2\wedge d\zeta_1\wedge d\zeta_2 - \overline{\zeta}_2 d\overline{\zeta}_1\wedge d\zeta_1\wedge d\zeta_2\right] \\
&= \frac{1}{32\pi^2}\iiiint_B \frac{\partial f/\partial\zeta_1}{(1-z\cdot\overline{\zeta})^2}\cdot\zeta_1\, d\zeta_1\wedge d\zeta_2\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2 \\
&\qquad + \frac{1}{32\pi^2}\iiiint_B \frac{f}{(1-z\cdot\overline{\zeta})^2}\, d\zeta_1\wedge d\zeta_2\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2 \\
&\qquad - \frac{1}{32\pi^2}\iiiint_B \frac{\partial f/\partial\zeta_2}{(1-z\cdot\overline{\zeta})^2}\cdot\zeta_2\, d\zeta_2\wedge d\zeta_1\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2 \\
&\qquad - \frac{1}{32\pi^2}\iiiint_B \frac{f}{(1-z\cdot\overline{\zeta})^2}\, d\zeta_2\wedge d\zeta_1\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2 \\
&\qquad + \frac{1}{32\pi^2}\iiiint_B \frac{\partial f/\partial\overline{\zeta}_1}{(1-z\cdot\overline{\zeta})^2}\cdot\overline{\zeta}_1\, d\overline{\zeta}_1\wedge d\overline{\zeta}_2\wedge d\zeta_1\wedge d\zeta_2
\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{f}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}_1\wedge d\overline{\zeta}_2\wedge d\zeta_1\wedge d\zeta_2\\
&+\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{2f\cdot\overline{\zeta}_1 z_1}{(1-z\cdot\overline{\zeta})^3}\,d\overline{\zeta}_1\wedge d\overline{\zeta}_2\wedge d\zeta_1\wedge d\zeta_2\\
&-\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{\partial f/\partial\overline{\zeta}_2}{(1-z\cdot\overline{\zeta})^2}\cdot\overline{\zeta}_2\,d\overline{\zeta}_2\wedge d\overline{\zeta}_1\wedge d\zeta_1\wedge d\zeta_2\\
&-\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{f}{(1-z\cdot\overline{\zeta})^2}\,d\overline{\zeta}_2\wedge d\overline{\zeta}_1\wedge d\zeta_1\wedge d\zeta_2\\
&-\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{2f\cdot\overline{\zeta}_2 z_2}{(1-z\cdot\overline{\zeta})^3}\,d\overline{\zeta}_2\wedge d\overline{\zeta}_1\wedge d\zeta_1\wedge d\zeta_2\,.
\end{aligned}
$$

Now we may group together like terms to obtain

$$
\begin{aligned}
&= -\frac{1}{8\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{f(\zeta)}{(1-z\cdot\overline{\zeta})^3}\,d\overline{\zeta}_1\wedge d\zeta_1\wedge d\overline{\zeta}_2\wedge d\zeta_2\\
&\qquad+\frac{3}{16\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{f(\zeta)\cdot(z\cdot\overline{\zeta})}{(1-z\cdot\overline{\zeta})^3}\,d\overline{\zeta}_1\wedge d\zeta_1\wedge d\overline{\zeta}_2\wedge d\zeta_2\\
&\qquad+\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{\partial f/\partial\zeta_1}{(1-z\cdot\overline{\zeta})^2}\cdot\zeta_1\,d\zeta_1\wedge d\zeta_2\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2\\
&\qquad-\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{\partial f/\partial\zeta_2}{(1-z\cdot\overline{\zeta})^2}\cdot\zeta_2\,d\zeta_2\wedge d\zeta_1\wedge d\overline{\zeta}_1\wedge d\overline{\zeta}_2\\
&\qquad+\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{\partial f/\partial\overline{\zeta}_1}{(1-z\cdot\overline{\zeta})^2}\cdot\overline{\zeta}_1\,d\overline{\zeta}_1\wedge d\overline{\zeta}_2\wedge d\zeta_1\wedge d\zeta_2\\
&\qquad-\frac{1}{32\pi^2}\int\!\!\int_B\!\!\int\!\!\int \frac{\partial f/\partial\overline{\zeta}_2}{(1-z\cdot\overline{\zeta})^2}\cdot\overline{\zeta}_2\,d\overline{\zeta}_2\wedge d\overline{\zeta}_1\wedge d\zeta_1\wedge d\zeta_2\\
&= -A+B+C-D+E-F\,.
\end{aligned}
$$

Now $-A$ is just the usual Bergman integral on the ball $B$ in $\mathbb{C}^2$. We can analyze the other terms just as we did for the disc. As an instance, let us examine $B$. We

take $f(\zeta) = \zeta^\alpha \overline{\zeta}^\beta$, where $\alpha$ and $\beta$ are multi-indices. Then we may write (ignoring dimensionality constants which are of no interest)

$$B = \iiiint_B \frac{f(\zeta) \cdot (z \cdot \overline{\zeta})}{(1 - z \cdot \overline{\zeta})^3} \, dV(\zeta)$$

$$= \iiiint_B (\zeta^\alpha \overline{\zeta}^\beta)(z \cdot \overline{\zeta}) \left( \sum_{j=0}^{\infty} (z \cdot \overline{\zeta})^j \right)^3 \, dV(\zeta) \, .$$

And now we see by inspection that, if $\beta \geq \alpha$, then the integral equals 0.

However, if $\beta < \alpha$, then the integral does not vanish. Take, for instance, the case when $f(\zeta) = \zeta_1^2 \overline{\zeta}_1$. Then our integral becomes

$$\iiiint_B (\zeta_1^2 \overline{\zeta}_1)(z \cdot \overline{\zeta}) \left( \sum_{j=0}^{\infty} (z \cdot \overline{\zeta})^j \right)^3 \, dV = \iiiint_B (\zeta_1^2 \overline{\zeta}_1)(z_1 \overline{\zeta}_1) \cdot 1 \, dV \, ,$$

where we have used parity to see that most of the terms vanish. Now this last is

$$z_1 \iiiint_B |\zeta_1|^4 \, dV(\zeta) = \frac{\pi^2}{12} z_1 \, .$$

The terms $C, D, E, F$ are calculated similarly, and we find that the sum of these error terms is a constant times $z_1$. A similar calculation reveals that the Szegő projection of this same $f$ is a different constant times $z_1$. So the situation is completely analogous to the disc case.

### *6.8.4 Strongly Pseudoconvex Domains*

We again, for simplicity, restrict attention to $\mathbb{C}^2$. In the seminal paper [FEF1], Fefferman shows that, near a strongly pseudoconvex boundary point, the Bergman kernel may be written (in suitable local coordinates) as

$$\frac{2}{\pi^2} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^3} + \mathcal{E}(z, \zeta) \, ,$$

where $\mathcal{E}$ is an error term of strictly lower order (in the sense of pseudodifferential operators) than the Bergman kernel.

In the important paper [BOUTS], Boutet de Monvel and Sjöstrand show that, near a strongly pseudoconvex boundary point, the Szegő kernel may be written (in suitable local coordinates) as

$$\frac{1}{2\pi^2} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^2} + \mathcal{F}(z, \zeta) \,,$$

where $\mathcal{F}$ is an error term of strictly lower order (in the sense of pseudodifferential or Fourier integral operators) than the Szegő kernel.

We now take advantage of these two asymptotic expansions to say something about the relationship between the Bergman and Szegő projections on a smoothly bounded strongly pseudoconvex domain.

Now fix a smoothly bounded, strongly pseudoconvex domain $\Omega$ with defining function $\rho$ (see [KRA1, Ch. 3] for this notion). Let $U$ be a tubular neighborhood of $\partial\Omega$ and let $V$ be a relatively compact subdomain of $U$ that is also a tubular neighborhood of $\partial\Omega$. Let $\varphi_j$ be a partition of unity that is supported in $U$ and sums to be identically 1 on $V$. We assume that each $\varphi_j$ has support so small that both the Fefferman and Boutet de Monvel–Sjöstrand expansions are valid on the support of $\varphi_j$. Then we write

$$\begin{aligned} \int_{\partial\Omega} f(\zeta) S(z, \zeta)\, d\sigma(\zeta) &= \iiint_{\partial\Omega} f(\zeta) S(z, \zeta) \omega(\zeta) \\ &= \sum_j \iiint_{\partial\Omega} \varphi_j(\zeta) f(\zeta) S(z, \zeta) \omega(\zeta) \,, \end{aligned}$$

where $\omega$ is the differential form that is equivalent to area measure on the boundary. And now, using Boutet de Monvel–Sjöstrand, and using the notable lemma of Fefferman [FEF1] that says that a strongly pseudoconvex boundary point is the ball up to fourth order, one can write each term of this last sum as

$$\begin{aligned} \frac{1}{2\pi} \iiint_{\partial B} \widetilde{\varphi_j}(\zeta) f(\zeta) \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^2} \Big[ &\zeta_1 d\zeta_2 \wedge d\overline{\zeta}_1 \wedge d\overline{\zeta}_2 \\ &- \zeta_2 d\zeta_1 \wedge d\overline{\zeta}_1 \wedge d\overline{\zeta}_2 + \overline{\zeta}_1 d\overline{\zeta}_2 \wedge d\zeta_1 \wedge d\zeta_2 - \overline{\zeta}_2 d\overline{\zeta}_1 \wedge d\zeta_1 \wedge d\zeta_2 \Big] + \mathcal{G} \,, \end{aligned}$$

where the error term $\mathcal{G}$ arises from approximating $\partial\Omega$ by $\partial B$, from approximating the Szegő kernel $S$ by the kernel for the ball, by applying a change of variable to $\varphi_j$, and also by approximating $\omega$ by the differential form that we used on the ball.

Now we may carry out the calculations using Stokes's theorem just as in the last subsection to finally arrive at the assertion that the last integral equals

$$\iiint_B \widetilde{\widetilde{\varphi}}_j(\zeta) \frac{f(\zeta)}{(1 - z \cdot \overline{\zeta})^3}\, dV + \mathcal{H}\,.$$

We cannot make the error term $\mathcal{H}$ disappear, but it is smoothly bounded hence negligeble. Finally, we can use the Fefferman asymptotic expansion to relate this last integral to the Bergman projection integral on the strongly pseudoconvex domain $\Omega$.

In summary, we have used Stokes's theorem to relate the Szegő projection integral on a smoothly bounded, strongly pseudoconvex domain to the Bergman projection integral on that domain. In this context, we do not get a literal equality. Instead we get an equality up to a controllable error term.

## 6.9 The Annulus

**Capsule:** Bergman notes in his book [BERM2] that an explicit calculation of the Bergman kernel for the annulus would entail elliptic functions. But, using some analytic tricks, we can come up with a useful asymptotic expansion for the Bergman kernel. That is the subject of the present section.

Let us first consider the Bergman kernel on the domain

$$D' = \{\zeta \in \mathbb{C} : |\zeta| > 1\}\,.$$

One may either utilize the transformation formula

$$K_{\Omega_1}(z, \zeta) = \Phi'(z) \cdot K_{\Omega_2}(\Phi(z), \Phi(\zeta)) \cdot \overline{\Phi'(\zeta)}$$

for a conformal mapping $\Phi : \Omega_1 \to \Omega_2$, or else exploit the fact that if $\{\psi_j\}_{j=1}^{\infty}$ is a complete orthonormal basis for the Bergman space on $\Omega_2$ then the set $\{(\psi \circ \Phi) \cdot \Phi'\}_{j=1}^{\infty}$ is a complete orthonormal basis for the Bergman space on $\Omega_1$. By either means, with the mapping $\Phi(\zeta) = 1/\zeta$ from ${}^c\overline{D}$ to $D$, we find that the Bergman kernel for the complement of the closed unit disc $D'$ is

$$K_{D'}(z, \zeta) = \frac{1}{\pi} \frac{1}{(1 - z\overline{\zeta})^2}\,.$$

We note that the indicated calculations show that a complete orthonormal basis for the Bergman space on $D'$ is $\{1/\zeta^j\}_{j=2}^{\infty}$. Now calculations just like those in Section 6.4 show that, if we let $\mathcal{I}_1 = \{0, 2, 4, 6, \ldots\}$ and $\mathcal{I}_2$ the complementary set, then the Bergman kernels corresponding to these two bases each have two singularities in the boundary (at 1 and $-1$).

Now we may consider the situation on the annulus

$$A = \{\zeta \in \mathbb{C} : 1/2 < |\zeta| < 2\}\,.$$

Bergman [BERM2] has shown that an explicit formula for the Bergman kernel on $A$ would entail elliptic functions. But we can derive an approximate formula that is good enough for our purposes as follows:

Note that $\{\zeta^j\}_{j=-\infty}^{\infty}$ is an orthogonal basis for $A^2(A)$. Moreover, a straightforward calculation shows that

$$\|\zeta^j\|_{A^2(A)} = \sqrt{\frac{\pi}{j+1}} \cdot \sqrt{\frac{2^{4j+4}-1}{2^{2j+2}}}, \quad j \neq -1$$

and

$$\|\zeta^{-1}\|_{A^2(A)} = \sqrt{2\pi} \cdot \sqrt{\log 4}\,.$$

Thus the Bergman kernel for the annulus $A$ is

$$K_A(z,\zeta) = \frac{1}{2\pi \cdot \log 4} z^{-1}\overline{\zeta}^{-1} + \sum_{j\neq -1} \frac{j+1}{\pi} \frac{2^{2j+2}}{2^{4j+4}-1} z^j \overline{\zeta}^j\,.$$

The usual error analysis shows then that

$$\sum_{j=0}^{\infty} \frac{j+1}{\pi} \frac{2^{2j+2}}{2^{4j+4}-1} z^j \overline{\zeta}^j = \sum_{j=0}^{\infty} \frac{j+1}{\pi} \frac{2^{2j+2}}{2^{4j+4}} z^j \overline{\zeta}^j + \mathcal{E}(z,\zeta)\,,$$

where $\mathcal{E}$ is a bounded error term with bounded derivatives of all orders.

An analysis of the terms with index less than or equal to $-2$ gives that

$$\sum_{j=-\infty}^{-2} \frac{j+1}{\pi} \cdot \frac{2^{2j+2}}{2^{4j+4}-1} z^j \overline{\zeta}^j = \sum_{j=2}^{\infty} \frac{j-1}{\pi} \cdot \frac{2^{2j-2}}{2^{4j-4}-1} z^{-j} \overline{\zeta}^{-j} + \mathcal{F}\,,$$

where $\mathcal{F}$ is an error term as usual.

These sums are straightforward to calculate and we find that the Bergman kernel for the annulus $A$ is given by

$$K_A(z,\zeta) = \frac{1}{2\pi \log 4} z^{-1}\overline{\zeta}^{-1} + \frac{4}{\pi} \cdot \frac{1}{(4 - z\cdot\overline{\zeta})^2} + \frac{4}{\pi} \cdot \frac{1}{(1 - 4z\cdot\overline{\zeta})^2} + \mathcal{G}(z,\zeta)\,,$$

where $\mathcal{G}$ is an error term. Notice that the second term reflects the outer boundary of the annulus and the third term reflects the inner boundary.

It is easy to see from these calculations that, if we were to consider the Bergman space on the annulus corresponding to just the basis elements with even index, then the resulting kernel would have two singularities on the outer boundary of the annulus and two singularities on the inner boundary. Refer to [KRA18, Ch. 1], [KRA23] for more on these phenomena.

## 6.10 Multiply Connected Domains

**Capsule:** It is easy to imagine, given what we have learned about analysis on the annulus, that the study of multiply connected domains will have its own complications. We treat some of those here.

Now what about multiply connected domains? A useful result in [KRA23] shows the following. Let $\Omega$ be smoothly bounded with connectivity $k$ and let $S_1, S_2, \ldots, S_k$ be the boundary curves of $\Omega$. Suppose that $S_1$ bounds the unbounded component of the complement. Let $\Omega_1$ be the bounded region bounded by $S_1$ and let $\Omega_j$, $j = 2, \ldots, k$, be the unbounded region bounded by $S_j$. Then, for $z, \zeta \in \Omega$,

$$K_\Omega(z,\zeta) = K_{\Omega_1}(z,\zeta) + K_{\Omega_2}(z,\zeta) + \cdots + K_{\Omega_k}(z,\zeta) + \mathcal{E}(z,\zeta)\,,$$

where $\mathcal{E}$ is a bounded function with bounded derivatives. Thus $K$ can be written as the sum of a Bergman kernel for a domain (namely $\Omega_1$) that is conformally equivalent to the disc plus Bergman kernels for domains which are conformally equivalent to the complement of the closure of the disc (namely $\Omega_2, \ldots, \Omega_k$). We already understand the Bergman kernels for those domains. This matter is treated in some detail in Section 6.14 below.

Meanwhile, we shall explore a slightly different direction. Let $\Omega \subseteq \mathbb{C}$ be a domain—multiply connected or not. Let $\lambda$ be a nontrivial automorphism of that domain. This means that $\lambda$ is a one-to-one, onto, holomorphic mapping of the domain to itself. Let $\{\varphi_j\}$ be a complete orthonormal basis for the Bergman space on $\Omega$. Let $\mathcal{I}_0$ denote the collection of those basis elements $\varphi_j$ such that $\varphi_j \circ \lambda = \varphi_j$ (as an example, think of the even basis elements on the disc, with the automorphism being $\zeta \mapsto -\zeta$). Assume that $\mathcal{I}_0$ is a proper subset of the full basis. Let $X = X_{\mathcal{I}_0}$ be the subspace of the full Bergman space generated by $\mathcal{I}_0$.

Now let us consider the Bergman kernel $K_{\mathcal{I}_0}$ for $X$. Certainly this kernel will have the boundary diagonal $\mathcal{D} = \{(z,\zeta) \in \partial\Omega \times \partial\Omega : z = \zeta\}$ as usual as a singular set. But it will also have the image $\mathcal{S} = \{(z,\zeta) : \zeta \in \partial\Omega, z = \lambda(\zeta)\}$ of $\mathcal{D}$ under $\lambda$ as a singular set. Thus we now have a fairly general criterion for recognizing multiple singular sets.

See Section 6.8 for more on multiply connected domains. The ideas here are developed further in Section 6.14.

## 6.11 The Sobolev Bergman Kernel

**Capsule:** In modern times it was Harold Boas [BOA2] who conceived the idea of studying the reproducing kernel of Sobolev spaces of holomorphic functions (this is much in the spirit of Aronszajn's idea of Hilbert space with reproducing kernel). Such a study reveals new phenomena—particularly kernels with a new sort of singularity.

Refer to Exercise 5, Chapter 1 for the definition of Sobolev space.

We may consider the Bergman kernel on the disc for the Sobolev space $W^1$ of holomorphic functions and it turns out to be, up to a bounded error term (see the calculations below),

$$-\frac{1}{\pi}\log(1-z\overline{\zeta})\,.$$

Specifically, set $\varphi_j(\zeta) = \zeta^j$. We calculate that

$$\iint\limits_D |\varphi_j(\zeta)|^2\,dA = \iint\limits_D |\zeta^j|^2\,dA = \frac{\pi}{j+1}$$

and

$$\iint\limits_D |\varphi_j'(\zeta)|^2\,dA = \iint\limits_D |j\zeta^{j-1}|^2\,dA = j\pi\,.$$

Thus

$$\|\varphi_j\|_{W^1} = \sqrt{\pi}\cdot\sqrt{\frac{j^2+j+1}{j+1}}\,.$$

Thus the full Bergman kernel for $W^1$ is given by

$$\sum_{j=0}^{\infty}\frac{1}{\pi}\cdot\frac{j+1}{j^2+j+1}\cdot z^j\overline{\zeta}^j = \frac{1}{\pi}+\sum_{j=1}^{\infty}\frac{1}{\pi}\cdot\frac{j+1}{j^2+j+1}\cdot z^j\overline{\zeta}^j = \frac{1}{\pi}+\sum_{j=1}^{\infty}\frac{1}{\pi}\cdot\frac{1}{j}\cdot z^j\overline{\zeta}^j+\mathcal{E}\,,$$

where $\mathcal{E}$ is an error term which is bounded and has one bounded derivative. So $\mathcal{E}$ is negligible from the point of view of determining where the kernel has singularities (i.e., where it blows up).

Let $\alpha = z\cdot\overline{\zeta}$. We look at

$$\begin{aligned}\frac{1}{\pi}+\frac{1}{\pi}\sum_{j=1}^{\infty}\frac{1}{j}\alpha^j &= \frac{1}{\pi}+\frac{1}{\pi}\sum_{j=1}^{\infty}\int\alpha^{j-1}\\ &= \frac{1}{\pi}+\frac{1}{\pi}\int\sum_{j=1}^{\infty}\alpha^{j-1}\\ &= \frac{1}{\pi}+\frac{1}{\pi}\int\frac{1}{\alpha}\sum_{j=1}^{\infty}\alpha^j\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{\pi} + \frac{1}{\pi} \int \frac{1}{\alpha} \left[ \sum_{j=0}^{\infty} \alpha^j - 1 \right] \\
&= \frac{1}{\pi} + \frac{1}{\pi} \int \frac{1}{\alpha} \left[ \frac{1}{1-\alpha} - 1 \right] \\
&= \frac{1}{\pi} + \frac{1}{\pi} \int \frac{1}{1-\alpha} \\
&= \frac{1}{\pi} - \frac{1}{\pi} \log(1-\alpha) \, .
\end{aligned}
$$

Thus the Bergman kernel for the order-1 Sobolev space is given by

$$
K(z, \zeta) = \frac{1}{\pi} - \frac{1}{\pi} \log(1 - z\overline{\zeta}) \, .
$$

Also the kernel for the space generated just by the monomials with even index seems to be given by (up to a bounded error term)

$$
\frac{1}{\pi} - \frac{1}{2} \log(1 - z\overline{\zeta}) - \frac{1}{2} \log(1 + z\overline{\zeta}) \, .
$$

To see this, we look at

$$
\sum_{j=0}^{\infty} \frac{1}{\pi} \cdot \frac{2j+1}{(2j)^2 + 2j + 1} z^{2j} \overline{\zeta}^{2j} = \frac{1}{\pi} + \frac{1}{\pi} \sum_{j=1}^{\infty} \frac{1}{2j} z^{2j} \overline{\zeta}^{2j} + \mathcal{F} \, .
$$

Here, as in the first calculation, $\mathcal{F}$ is a bounded term with one bounded derivative. So it is negligible from the point of view of our calculation.

Thus we wish to calculate

$$
\begin{aligned}
\frac{1}{\pi} + \frac{1}{\pi} \sum_{j=1}^{\infty} \frac{1}{2j} \alpha^{2j} &= \frac{1}{\pi} + \frac{1}{\pi} \sum_{j=1}^{\infty} \int \alpha^{2j-1} \\
&= \frac{1}{\pi} + \frac{1}{\pi} \int \frac{1}{\alpha} \sum_{j=1}^{\infty} \alpha^{2j} \\
&= \frac{1}{\pi} + \frac{1}{\pi} \int \frac{1}{\alpha} \left[ \sum_{j=0}^{\infty} \alpha^{2j} - 1 \right] \\
&= \frac{1}{\pi} + \frac{1}{\pi} \int \frac{1}{\alpha} \left[ \frac{1}{1-\alpha^2} - 1 \right]
\end{aligned}
$$

$$= \frac{1}{\pi} + \frac{1}{\pi} \int \frac{\alpha}{1-\alpha^2}$$
$$= \frac{1}{\pi} - \frac{1}{2\pi} \log(1-\alpha^2) \, .$$

In conclusion, with $\alpha = z\overline{\zeta}$, the Bergman kernel for the order 1 Sobolev space using only the basis elements with even index is

$$K'(z,\zeta) = \frac{1}{\pi} - \frac{1}{2\pi} \log(1 - z \cdot \overline{\zeta}) - \frac{1}{2\pi} \log(1 + z \cdot \overline{\zeta}) \, .$$

In short, there are singularities as $z$ and $\zeta$ tend to the *same* disc boundary point, and also as $z$ and $\zeta$ tend to antipodal disc boundary points.

## 6.12 The Theorem of Ramadanov

**Capsule:** In view of the fact that any domain of holomorphy can be exhausted by strongly pseudoconvex domains, it is natural to study the limiting behavior of the Bergman kernel under convergence of a sequence of domains. The pioneer in such a study was Ramadanov [RAM1]. Further developments appear in [KRA21].

In the noted paper [RAM1], I. Ramadanov proved a very useful result about the limit of the Bergman kernels in an increasing sequence of domains. A version of Ramadanov's classical result is this:

**Prelude** The next result is Ramadanov's famous theorem about convergence of the Bergman kernel for an increasing sequence of domains. This theorem has been quite influential in the subject. It is generalized in [KRA21].

**Theorem 6.12.1** *Let $\Omega_1 \subseteq \Omega_2 \subseteq \cdots \subseteq$ be an increasing sequence of bounded domains in $\mathbb{C}^n$ and let $\Omega = \cup_{j=1}^{\infty} \Omega_j$. Assume also that $\Omega$ is bounded. Then*

$$K_\Omega(z,\zeta) = \lim_{j \to \infty} K_{\Omega_j}(z,\zeta) \, ,$$

*with the limit being uniform on compact subsets of $\Omega \times \Omega$.*

In the paper [KRA21], Krantz generalizes Ramadanov's result to a sequence of domains that is not necessarily increasing. Here we present the statement of his theorem and the proof.

**Prelude** Here we present Krantz's [KRA21] generalization of Ramadanov's theorem.

**Theorem 6.12.2** *Let $\Omega_j$ be a sequence of domains that converges to a limit domain $\Omega$ in the Hausdorff metric of domains (see [KRPA2, App. 1]). Then $K_{\Omega_j} \to K_\Omega$ uniformly on compact subsets of $\Omega \times \Omega$.*

**Proof** Let $\Omega_1, \Omega_2, \ldots$ be domains in $\mathbb{C}^n$ and let $\Omega_j \to \Omega$ in the topology of the Hausdorff metric. For convenience, we let $\Phi_j : \Omega \to \Omega_j$ be diffeomorphisms such that the $\Phi_j$ converge to the identity in a suitable topology.

Now fix a point $z$ that lies in all the $\Omega_j$ and in $\Omega$ as well. Then $K_\Omega(z, \cdot\,)$ is the Hilbert space representative (according to Riesz's theorem) of the point evaluation linear functional

$$A^2(\Omega) \ni f \mapsto f(z) .$$

Of course it is also the case that $K_{\Omega_j}(z, \cdot\,)$ is the Hilbert space representative (according to Riesz's theorem) of the point evaluation linear functional

$$A^2(\Omega_j) \ni f \mapsto f(z) .$$

Of course the standard lemma for the Bergman theory (using the mean-value property) tells us that the point evaluation functional at $z$ is bounded with a bound that is independent of $j$ (in fact it only depends on the $(-n)^{\text{th}}$ power of the distance of $z$ to the boundary, and that may be taken to be uniform in $j$). Thus if we set $\psi_j(z) = K_{\Omega_j}(z, \Phi_j(\cdot\,))$ then $\|\psi_j\|_{L^2(\Omega)}$ is bounded, independent of $j$.

By the Banach-Alaoglou theorem, we may conclude that there is a weak-$*$ convergent subsequence $\psi_{j_k}$. Call the weak-$*$ limit $\psi_0$. But then, for $g \in A^2(\Omega_{j_k})$, we see that

$$\begin{aligned} g(z) &= \int_{\Omega_{j_k}} K_{\Omega_{j_k}}(z, \zeta) g(\zeta)\, dV(\zeta) \\ &= \int_\Omega \psi_{j_k}(\xi) g(\Phi_{j_k}(\xi)) \Phi'_{j_k}(\xi) \overline{\Phi'_{j_k}(\xi)}\, dV(\xi) \\ &\equiv \int_\Omega \psi_{j_k}(\xi) h_{j_k}(\xi)\, dV(\xi) \end{aligned}$$

Notice that the $h_{j_k}$ are all defined on $\Omega$ and they converge in the strong topology of $L^2(\Omega)$ to some limit function $\widetilde{h}$. In fact, by applying the $\overline{\partial}$ operator, one can see that $\widetilde{h}$ is a *holomorphic* function on $\Omega$.

Thus our expression converges to

$$\int_\Omega \psi_0(\xi) \widetilde{h}(\xi)\, dV(\xi) . \tag{6.12.2.1}$$

Now in fact we may apply this preceding argument to see that every subsequence of the index $j$ has a subsequence so that we get the indicated convergence. The conclusion is that the $\psi_j$ converge weak-$*$ to $\psi_0$ and the $h_j$ converge strongly to $\widetilde{h}$ so that

$$g(z) = \int_\Omega \psi_0(\xi)\widetilde{h}(\xi)\, dV(\xi)\,.$$

Next we examine the definition of the $h_j$ and the $\Phi_j$ to see that in fact $\widetilde{h}(z) = g(z)$. Thus we may write the last line as

$$g(z) = \int_\Omega \psi_0(\xi)\widetilde{h}(\xi)\, dV(\xi) = \widetilde{h}(z)\,.$$

Finally, one can reason backwards to see that any $L^2$ holomorphic function $\widetilde{h}$ on $\Omega$ can arise in this way. The only possible conclusion is that $\psi_0$ is the representing function for point evaluation at $z$. So $\psi_0$ is the Bergman kernel for $\Omega$. In conclusion $K_{\Omega_j}(z,\,\cdot\,)$ converges weak-$*$ to $K_\Omega(z,\,\cdot\,)$.

Certainly we may now apply the weak-$*$ convergence to a testing function consisting of a $C_c^\infty$ radial function to conclude that $\psi_{j_k}$ in fact converges uniformly on compact sets to $\psi_0$. So we see that $\psi_0$ is conjugate holomorphic. We may therefore conclude that $\psi_0$ is the Bergman kernel of $\Omega$.

Thus we see that the $K_{\Omega_j}(z,\,\cdot\,)$ converge uniformly on compact sets to $K_\Omega(z,\,\cdot\,)$. This is the desired conclusion. □

## 6.13 More on the Szegő Kernel

It is not difficult to see that, suitably formulated, there is a version of Theorem 6.12.2 for the Szegő kernel (see [KRA18, Ch. 1]). To wit,

**Prelude** It is natural to ask whether there is a version of Ramadanov's theorem for the Szegő kernel. Here we present such a result.

**Theorem 6.13.1** *Let $\Omega_j$ be a sequence of domains with $C^2$ boundary that converges to a limit domain $\Omega$ in the $C^2$ topology of domains. This means that each $\Omega_j = \{z \in \mathbb{C}^n : \rho_j(z) < 0\}$, $\nabla\rho_j \neq 0$ on $\partial\Omega_j$, and the defining functions $\rho_j$ converge in the $C^2$ topology. Then the Szegő kernels $S_{\Omega_j} \to S_\Omega$ uniformly on compact subsets of $\Omega \times \Omega$.*

Without much effort, it can also be seen that there is a version of our theorem in the rather general setting of Hilbert space with reproducing kernel. See [ARO] for a thorough treatment of this abstract concept.

## 6.14 Boundary Localization

**Capsule:** In the present section we explore an idea of Krantz [KRA22] about relating the Bergman kernel of a multiply connected domain to the Bergman kernels of component domains (for the complement).

We begin by examining a slightly different avenue for getting one's hands on the Bergman kernel of a domain. The general approach is perhaps best illustrated with an example. Let

$$\Omega = \{\zeta \in \mathbb{C} : 1 < |\zeta| < 2\}.$$

This is the annulus, and any explicit representation of its Bergman kernel will involve elliptic functions (see [BERM2]). One might hope, however, to relate the Bergman kernel $K_\Omega$ of $\Omega$ to the Bergman kernels $K_{\Omega_1}$ and $K_{\Omega_2}$ of

$$\Omega_1 = \{\zeta \in \mathbb{C} : |\zeta| < 2\}$$

and

$$\Omega_2 = \{\zeta \in \mathbb{C} : 1 < |\zeta|\}.$$

The first of these has an explicitly known Bergman kernel (see [KRA1, Ch. 1]) and the second domain is the inversion of a disc, so its kernel is known explicitly as well.

One could pose a similar question for domains of higher connectivity. The question also makes sense, with a suitable formulation, in several complex variables. Our purpose here is to come up with precise formulations of results such as these and to prove them. In one complex variable, we can make decisive use of classical results relating the Bergman kernel to the Green's function (see [KRA18, Ch. 1]). In several complex variables there are analogous results of Garabedian (see [GARA]) that will serve in good stead.

In Section 6.14.1 we introduce appropriate definitions and notation. In Sections 6.14.2, 6.14.3 we prove a basic, representative result in the plane. Subsection 6.14.4 treats the multi-dimensional result.

We thank Richard Rochberg for bringing these questions to our attention.

### *6.14.1 Definitions and Notation*

If $\Omega \subseteq \mathbb{C}^n$ is a bounded domain, then we let $K_\Omega(z, \zeta)$ denote its Bergman kernel. This is the reproducing kernel for

$$A^2(\Omega) \equiv \{f \in L^2(\Omega) : f \text{ is holomorphic on } \Omega\}.$$

It is known, for planar domains, that $K_\Omega(z, \zeta)$ is related to the Green's function $G_\Omega(z, \zeta)$ for $\Omega$ by this formula:

$$K_\Omega(z, \zeta) = 4 \cdot \overline{\frac{\partial^2}{\partial\zeta\partial\overline{z}} G_\Omega(\zeta, z)}.$$

Of course it is essential for our analysis to realize that the Green's function is known quite explicitly on any given domain. If

$$\Gamma(\zeta, z) = \frac{1}{2\pi} \log |\zeta - z|$$

is the fundamental solution for the Laplacian (on all of $\mathbb{C}$) then we construct the Green's function as follows:

Given a domain $\Omega \subseteq \mathbb{C}$ with smooth boundary, the *Green's function* is posited to be a function $G_\Omega(\zeta, z)$ that satisfies

$$G_\Omega(\zeta, z) = \Gamma(\zeta, z) + F_z^\Omega(\zeta)\,,$$

where $F_z^\Omega(\zeta) = F^\Omega(\zeta, z)$ is a particular harmonic function in the $\zeta$ variable. It is mandated that $F^\Omega$ be chosen (and is in fact uniquely determined by the condition) so that $G(\cdot\,, z)$ vanishes on the boundary of $\Omega$. One constructs the function $F^\Omega(\cdot\,, z)$, for each fixed $z$, by solving a suitable Dirichlet problem. Again, the reference [KRA1, p. 40] has all the particulars. It is worth noting that the Green's function is a symmetric function of its arguments.

In our proof, we shall be able to exploit known properties of the Poisson kernel (see especially [KRA13]) and of the solution to the Dirichlet problem (see [KRA5, Ch. 1]) to get the estimates that we need.

We shall first formulate and solve our problem for domains in the plane. Afterward we shall treat matters in higher-dimensional complex space.

### 6.14.2 A Representative Result

We first prove our main result for the domain

$$\Omega = \{\zeta \in \mathbb{C} : 1 < |\zeta| < 2\}\,.$$

This argument will exhibit all the key ideas—at least in one complex variable. The later exposition will be clearer because we took the time to treat this case carefully.

Let

$$\Omega_1 = \{\zeta \in \mathbb{C} : |\zeta| < 2\}$$

and

$$\Omega_2 = \{\zeta \in \mathbb{C} : 1 < |\zeta|\}\,.$$

Certainly $\Omega = \Omega_1 \cap \Omega_2$. For convenience in what follows, we let $S_1$ be the boundary curve of $\Omega_1$ and $S_2$ be the boundary curve of $\Omega_2$. Of course it then follows that $\partial\Omega = S_1 \cup S_2$.

We claim that

$$K_\Omega(z,\zeta) = \frac{1}{2}\left[K_{\Omega_1}(z,\zeta) + K_{\Omega_2}(z,\zeta)\right] + \mathcal{E}(z,\zeta)\,,$$

where $\mathcal{E}$ is an error term that is smooth on $\overline{\Omega \times \Omega}$. In particular, $\mathcal{E}$ is bounded with all derivatives bounded on that domain.

For the proof, we write

$$\begin{aligned}\frac{1}{8}\left[\overline{K_{\Omega_1}(z,\zeta) + K_{\Omega_2}(z,\zeta)}\right] &= \frac{1}{2}\frac{\partial^2}{\partial\zeta\partial\overline{z}}\Big[\big(\Gamma(\zeta,z) + F^{\Omega_1}(\zeta,z)\big) + \big(\Gamma(\zeta,z) + F^{\Omega_2}(\zeta,z)\big)\Big] \\ &= \frac{\partial^2}{\partial\zeta\partial\overline{z}}\left(\Gamma(\zeta,z) + \frac{1}{2}\Big[F^{\Omega_1}(\zeta,z) + F^{\Omega_2}(\zeta,z)\Big]\right).\end{aligned}$$

Now we claim that

$$F^{\Omega_1}(\zeta,z) + F^{\Omega_2}(\zeta,z) = 2F^{\Omega}(\zeta,z) + \mathcal{E}(z,\zeta)$$

for a suitable error term $\mathcal{E}$. We must analyze

$$G(\zeta,z) \equiv \left[F^{\Omega_1}(\zeta,z) + F^{\Omega_2}(\zeta,z)\right] - 2F^{\Omega}(\zeta,z)\,.$$

We think of $G$ as the solution of a Dirichlet problem on $\Omega$, and we must analyze the boundary data. What we see is this:

- For $z$ near $S_1$, $F^\Omega$ and $F^{\Omega_1}$ agree on $S_1$ (in the variable $\zeta$) and equal 0. And $F^{\Omega_2}$ is smooth and bounded by $C \cdot |\log(1/2)|$, just by the form of the Green's function. All three functions are plainly smooth and bounded on $S_2$ (for $z$ still near $S_1$) by similar reasoning. In conclusion, $G$ is smooth and bounded on $\overline{\Omega}$ for $z$ near $S_2$.
- For $z$ near $S_2$, $F^\Omega$ and $F^{\Omega_2}$ agree on $S_2$ (in the variable $\zeta$) and equal 0. And $F^{\Omega_1}$ is smooth and bounded by $C \cdot |\log(1/2)|$, just by the form of the Green's function. All three functions are plainly smooth and bounded on $S_1$ (for $z$ still near $S_2$) by similar reasoning. In conclusion, $G$ is smooth and bounded on $\overline{\Omega}$ for $z$ near $S_1$.
- For $z$ away from both $S_1$ and $S_2$—in the interior of $\Omega$—it is clear that all the terms are bounded and smooth on $\partial\Omega$. So the solution $G$ of the Dirichlet problem will also be smooth as desired.

As a result of these considerations, $G$ is smooth on $\overline{\Omega}$.

That completes our argument and gives, altogether, the error term $\mathcal{E}$. Thus

$$F^{\Omega_1} + F^{\Omega_2} - 2F^{\Omega} = \mathcal{E}\,.$$

It follows that

$$\frac{1}{2}[K_{\Omega_1}(z,\zeta) + K_{\Omega_2}(z,\zeta)] = 4\overline{\frac{\partial^2}{\partial\zeta\partial\overline{z}}\left(\Gamma(\zeta,z) + F^{\Omega}(\zeta,z)\right)} + \mathcal{E}'$$
$$= K_{\Omega}(z,\zeta) + \mathcal{E}''$$

### 6.14.3 The More General Result in the Plane

Now consider a smoothly bounded domain $\Omega \subseteq \mathbb{C}$ with $k$ connected components in its boundary, $k \geq 2$. We denote the boundary components by $S_1, \ldots, S_k$; for specificity, we let $S_1$ be the component of the boundary that bounds the *unbounded* component of the complement of $\Omega$. Let $\Omega_1$ be the *bounded* region in the plane bounded by the single Jordan curve $S_1$. Let $\Omega_2, \ldots, \Omega_k$ be the *unbounded* regions bounded by $S_2, S_3, \ldots, S_k$ respectively.

Then we may analyze, just as in the last subsection, the expression

$$K_{\Omega} - \frac{1}{k}\left[K_{\Omega_1} + K_{\Omega_2} + \cdots + K_{\Omega_k}\right]$$

to obtain a smooth error term

$$\mathcal{E} = \mathcal{E}_1 + \mathcal{E}_2 + \cdots + \mathcal{E}_k\,.$$

That completes our analysis of a smooth, finitely connected domain in the plane.

### 6.14.4 Domains in Higher-Dimensional Complex Space

The elegant paper [GARA] contains the necesarry information about the relationship of the Bergman kernel and a certain Green's function in several complex variables so that we may carry out our program in that more general context.

Fix a smoothly bounded domain $\Omega$ in $\mathbb{C}^k$. Let $t = (t_1, \ldots, t_k)$ be a fixed point in $\Omega$. Following Garabedian's notation, we set

$$r = \sqrt{\sum_{j=1}^{k} |z_j - t_j|^2}\,.$$

Let $\sigma_k$ be constants chosen so that

$$\lim_{\epsilon\to 0} \sigma_k \int_{\Gamma_\epsilon} B \cdot \sum_{j=1}^{k} \frac{\partial r^{-2k+2}}{\partial z_j}\alpha_j\, d\sigma + B(t) = 0\,,$$

where $\Gamma_\epsilon$ is the sphere of radius $\epsilon$ about $t$, $B$ is some continuous function, and $(\alpha_1, \ldots, \alpha_k)$ is a collection of complex-valued direction cosines.

Now set $\theta(z, t)$ to be that function

$$\theta = \sigma_k r^{-2k+2} + \text{regular terms} \tag{6.14.1}$$

on $\Omega$ so that

$$\sum_{j=1}^{k} \frac{\partial \theta}{\partial \overline{z}_j} \cdot \overline{\alpha}_j = 0$$

on $\partial\Omega$,

$$\frac{\partial}{\partial \overline{z}_j} \triangle \theta = 0$$

on $\Omega$ (for $j = 1, \ldots, k$), and such that

$$\int_\Omega \theta \overline{f}\, dV = 0\,,$$

for all functions $f$ analytic in $\Omega$. It follows from standard elliptic theory that such a $\theta$ exists.

In fact, this function $\theta$ that we have constructed is a Green's function for the boundary value problem

$$\begin{aligned} \frac{\partial}{\partial \overline{z}_j} \triangle \beta &= 0 \qquad \text{on } \Omega, \quad j = 1, \ldots, k \\ \sum_{j=1}^{k} \frac{\partial \beta}{\partial \overline{z}_j} \cdot \overline{\alpha}_j &= 0 \qquad \text{on } \partial\Omega\,. \end{aligned}$$

Garabedian goes on to prove that the Bergman kernel for $\Omega$ is related to the Green's function $\theta$ in this way:

$$K_\Omega(z, t) = \triangle_z \theta(z, t)\,.$$

This is just the information that we need to apply the machinery that has been developed here.

In order to flesh out the argument in the context of several complex variables, our primary task is to argue that our new Green's function has a form similar to the classical Green's function from one complex variable. But in fact this is immediate from equation (6.14.1). It follows from this that the maximum principle will go

through as before, and we may establish a version of the result in Subsections 6.14.2 and 6.14.3 in the context of several complex variables. The theorem is this:

**Prelude** The next result is of a new type, but it is intuitively appealing. It was first considered in the paper [KRA22].

**Theorem 6.14.1** *Let $\Omega$ be a smoothly bounded domain in $\mathbb{C}^n$ with boundary having connected components $S_1, S_2, \ldots, S_k$. For specificity, say that $S_1$ is the boundary component that bounds the unbounded portion of the complement of $\overline{\Omega}$. Let $K_\Omega$ be the Bergman kernel for $\Omega$, let $K_1$ be the Bergman kernel for the bounded domain having $S_1$ as its single boundary element, and let $K_j$, for $j \geq 2$, be the Bergman kernel for the unbounded domain having $S_j$ as its single boundary component. Then*

$$K_\Omega = K_1 + K_2 + \cdots + K_k + \mathcal{E} ,$$

*where $\mathcal{E}$ is an error term that is bounded with bounded derivatives.*

The reader can see that this new theorem is completely analogous to the results presented earlier in the one variable setting. But it must be confessed that this theorem is something of a *canard*. For, when $j \geq 2$, any function holomorphic on the unbounded domain with boundary $S_j$ will (by the Hartogs extension phenomenon) extend analytically to all of $\mathbb{C}^n$. And of course there are no $L^2$ holomorphic functions on all of $\mathbb{C}^n$. So it follows that $K_j \equiv 0$. So the theorem really says that

$$K_\Omega = K_1 + \mathcal{E} .$$

This is an interesting fact, but not nearly as important or provocative as the one-variable result. The one other point worth noting is that the statement of the result is now a bit different from that in one complex variable, just because we are dealing with a different Green's function for a different boundary value problem. Basically what we are seeing is that $K_2, \ldots, K_k$ do not count at all, and $K_1$ is the principal and only term.

## Exercises

1. Refer to Exercise 1 in Chapter 5. Derive the automorphism group for the ball in $n$ dimensions.
2. Let $\Omega \subseteq \mathbb{R}^N$ be a domain. Suppose that $\partial\Omega$ is a regularly imbedded $C^j$ manifold, $j = 1, 2, \ldots$. This means that for each $P \in \partial\Omega$ there is a neighborhood $U_P \subseteq \mathbb{R}^N$ and a $C^j$ function $f_P : U_P \to \mathbb{R}$ with $\nabla f_P \neq 0$ and $\{x \in U_P : f_P(x) = 0\} = U_P \cap \partial\Omega$. Prove that there is a function $\rho : \mathbb{R}^N \to \mathbb{R}$ satisfying

   **(a)** $\nabla\rho \neq 0$ on $\partial\Omega$;
   **(b)** $\{x \in \mathbb{R}^N : \rho(x) < 0\} = \Omega$;
   **(c)** $\rho$ is $C^j$.

We call $\rho$ a *defining function* for $\Omega$.

Prove that if $\Omega$ has a $C^j$ defining function, then $\partial\Omega$ is a regularly imbedded $C^j$ submanifold of $\mathbb{R}^N$.

Prove that both of the preceding concepts are equivalent to the following: For each $P \in \partial\Omega$ there is a neighborhood $U_P$, a coordinate system $t_1, \ldots, t_N$ on $U_P$, and a $C^j$ function $\phi(t_1, t_2, \ldots, t_{N-1})$ such that $\{(t_1, \ldots, t_N) \in U_P : t_N = \phi(t_1, \ldots, t_{N-1})\} = \partial\Omega \cap U_P$. This means that $\partial\Omega$ is locally the graph of a $C^j$ function.

**3.** Complete the following outline to prove the Cauchy–Fantappiè formula:

**Theorem** Let $\Omega \subset\subset \mathbb{C}^n$ be a domain with $C^1$ boundary. Let $w(z, \zeta) = (w_1(z, \zeta), \ldots, w_n(z, \zeta))$ be a $C^1$, vector-valued function on $\overline{\Omega} \times \overline{\Omega} \setminus \{\text{diagonal}\}$ that satisfies

$$\sum_{j=1}^{n} w_j(z, \zeta)(\zeta_j - z_j) \equiv 1.$$

Then we have for any $f \in C^1(\overline{\Omega}) \cap \{\text{holomorphic functions on } \Omega\}$ and any $z \in \Omega$ the formula

$$f(z) = \frac{1}{nW(n)} \int_{\partial\Omega} f(\zeta)\eta(w) \wedge \omega(\zeta)\,.$$

Here $\omega(\zeta) \equiv d\zeta_1 \wedge d\zeta_2 \wedge \cdots \wedge d\zeta_n$ and $\eta(w) \equiv \sum_{j=1}^{n} (-1)^{j+1} w_j dw_1 \wedge \cdots \wedge dw_{j-1} \wedge dw_{j+1} \wedge \cdots \wedge dw_n$.

**Proof** We may assume that $z = 0 \in \Omega$.

**(a)** If $\alpha^1 = (a_1^1, \ldots, a_n^1), \ldots, \alpha^n = (a_1^n, \ldots, a_n^n)$ are $n-$tuples of $C^1$ functions on $\overline{\Omega}$ that satisfy $\sum_{j=1}^{n} a_i^j(\zeta) \cdot (\zeta_j - z_j) = 1$, let

$$B(\alpha^1, \ldots, \alpha^n) = \sum_{\sigma \in S_n} \epsilon(\sigma) a^1_{\sigma(1)} \wedge \overline{\partial}(a^2_{\sigma(2)}) \wedge \cdots \wedge \overline{\partial}(a^n_{\sigma(n)}),$$

where $S_n$ is the symmetric group on $n$ letters and $\epsilon(\sigma)$ is the signature of the permutation $\sigma$. Prove that $B$ is independent of $\alpha^1$.

**(b)** It follows that $\overline{\partial}B = 0$ on $\overline{\Omega} \setminus \{0\}$ (indeed $\overline{\partial}B$ is an expression like $B$ with the expression $a^1_{\sigma(1)}$ replaced by $\overline{\partial}a^1_{\sigma(1)}$).

**(c)** Use **(b)**, especially the paranthetical remark, to prove inductively that if $\beta^1 = (b_2^1, \ldots, b_n^1), \ldots, \beta^n = (b_2^n, \ldots, b_n^n)$, then there is a form $\gamma$ on $\Omega \setminus \{0\}$ such that

$$\left[B(\alpha^1, \ldots, \alpha^n) - B(\beta^1, \ldots, \beta^n)\right] \wedge \omega(\zeta) = \overline{\partial}\gamma = d\gamma.$$

**(d)** Prove that if $\alpha^1 = \cdots = \alpha^n = (w_1, \ldots, w_n)$ then $B(\alpha^1, \ldots, \alpha^n)$ simplifies

$$B(\alpha^1, \ldots, \alpha^n) \wedge \omega(\zeta) = (n-1)!\eta(w) \wedge \omega(\zeta).$$

**(e)** Let $\mathcal{S}$ be a small sphere of radius $\epsilon > 0$ centered at 0 such that $\mathcal{S} \subseteq \Omega$. Use part (c) to see that

$$\int_{\partial\Omega} f(\zeta)\eta(w) \wedge \omega(\zeta) = \int_{\mathcal{S}} f(\zeta)\eta(w) \wedge \omega(\zeta).$$

**(f)** Now use **(c)** and **(d)** to see that

$$\int_{\mathcal{S}} f(\zeta)\eta(w) \wedge \omega(\zeta) = \int_{\mathcal{S}} f(\zeta)\eta(v) \wedge \omega(\zeta),$$

where

$$v(z,\zeta) = \frac{\overline{\zeta}_j - z_j}{|\zeta - z|^2}.$$

[Warning: Be careful if you decide to apply Stokes's theorem.] We know that the last line is $n \cdot W(n) \cdot f(0)$.

**4.** Use a limiting argument to show that the hypotheses of the Cauchy–Fantappiè formula (the preceding exercise) may be weakened to $f \in C(\overline{\Omega})$, $w \in C(\overline{\Omega} \times \partial\Omega)$.

Prove, using only linear algebra, that if $w$ is as in the statement of the Cauchy–Fantappiè formula, then there are functions $\psi_1, \ldots, \psi_n, \Psi$ such that $w_j = \psi_j / \Psi, j = 1, \ldots, n$.

**5.** Let $K$ be the Bergman kernel for the ball $B$. Show that $K(0,0) = 1/V(B)$. Use the automorphism group of $B$, together with the invariance of the kernel, to calculate $K(z,z)$ for every $z \in B$. The values of $K$ on the diagonal then completely determine $K(z,\zeta)$. Why?

**6.** What can you say about the Bergman kernel for the domain

$$\Omega = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^4 < 1\}\,?$$

**7.** Generalize the question in the last exercise to the domains

$$\Omega = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^{2j} + |z_2|^{2k} < 1\}$$

for $j, k$ integers at least 2.

**8.** Now we consider Bergman representative coordinates.

Let $\Omega$ be a bounded domain in $\mathbb{C}^n$ and let $q$ be a point of $\Omega$. The "diagonal" Bergman kernel $K_\Omega(q,q)$ is of course real and positive so that there is a neighborhood of $q$ such that, for all $z, w$ in the neighborhood, $K_\Omega(z,w) \neq 0$. Then for all $z, w$ in that neighborhood, we define

$$b_j(z) = b_{j,q}(z) = \frac{\partial}{\partial \overline{w}_j} \log \frac{K(z,w)}{K(w,w)}\bigg|_{w=q}.$$

Note that these coordinates are well-defined, independent of the choice of logarithmic "branches". Each $b_j(z)$ is clearly a holomorphic function of $z$.

Notice that some restriction on $z$ to be in a neighborhood of $q$ may be actually necessary, since it may be that $K_\Omega(z, w)$ vanishes for some pairs $(z, w) \in \Omega \times \Omega$. In any event, the mapping

$$z \longmapsto \big(b_1(z), \ldots, b_n(z)\big) \in \mathbb{C}^n$$

is defined and holomorphic in a neighborhood of the point $q$. Note also that $(b_1(q), \ldots, b_n(q)) = (0, \ldots, 0)$.

By the holomorphic inverse function theorem, these functions give local coordinates if the holomorphic Jacobian

$$\det \left( \frac{\partial b_j}{\partial z_k} \right)_{j,k=1,\ldots,n}$$

is nonzero at $q$. But in fact the non-vanishing of this determinant at $q$ is an immediate consequence of a fact that we have established already, namely, that the Bergman metric is positive definite. Prove this assertion.

The utility of the new coordinates in studying biholomorphic mappings comes from the following lemma:

**Lemma** Let $\Omega_1$ and $\Omega_2$ be two bounded domains in $\mathbb{C}^n$ with $q_1 \in \Omega_1$ and $q_2 \in \Omega_2$. Denote by $b_1^1, \ldots, b_n^1$ the Bergman coordinates as defined near $q_1$ in $\Omega_1$ (using the Bergman kernel for $\Omega_1$) and $b_1^2, \ldots, b_n^2$ the Bergman coordinates defined in the same way near $q_2$ in $\Omega_2$ (using the Bergman kernel for $\Omega_2$). Suppose that there is a biholomorphic mapping $F : \Omega_1 \to \Omega_2$ with $F(q_1) = q_2$. Then the function defined near $0 \in \mathbb{C}^n$ by

$$\begin{aligned}(\alpha_1, \ldots, \alpha_n) \longmapsto &\text{ the } b^2\text{-coordinates of the } F\text{-image of the point} \\ &\text{of } \Omega_1 \text{ with } b^1 \text{ coordinates } (\alpha_1, \ldots, \alpha_n)\end{aligned}$$

is a $\mathbb{C}$-linear transformation.

Prove this lemma.

**9.** Give two examples of unbounded domains with nontrivial Bergman kernel. Give one example of an unbounded domain with trivial Bergman kernel.

**10.** In fact there is a domain in $\mathbb{C}^2$ with finite-dimensional Bergman space. Can you give an example?

# Chapter 7
# The Bergman Metric

**Prologue:** The Bergman metric was the first Kähler metric ever created. It has proved to be a powerful tool in function theory, in differential geometry, and in the theory of partial differential equations. Much of the theory of automorphism groups of domains (at least the part developed by Greene and Krantz) is based on studies of the Bergman metric—see for instance [GRK3].

## 7.1 Smoothness of Biholomorphic Mappings

**Capsule:** It was C. Fefferman [FEF1] who taught us that the Bergman metric, particularly its geodesics, could be used to study the boundary regularity of biholomorphic mappings (thanks to the work of O. Kellogg [KEL], such studies in one complex variable had depended largely on potential theory). Now, thanks to S. Bell [BE1] and others, we have many different techniques (including reflection principles, analytic continuation, and many others) for studying the $C^\infty$ boundary continuation of such mappings.

Poincaré's theorem (see [KRA1, Ch. 0], [KRA16, Ch. 5] for discussion) that the ball and polydisc are biholomorphically inequivalent shows that there is no Riemann mapping theorem (at least in the traditional sense) in several complex variables. More recent results of D. Burns, S. Shnider, and R. Wells [BURS] and of R. E. Greene and S. G. Krantz [GRK1, GRK2] confirm how truly dismal the situation is. First, we need a definition.

**Definition 7.1.1** Let $\rho_0$ be a $C^k$ defining function for a bounded domain $\Omega_0 \subseteq \mathbb{R}^N, k \geq 2$. We define a neighborhood basis for $\Omega_0$ in the $C^k$ topology as follows: Let $\epsilon > 0$ be so small that if $\|\rho - \rho_0\|_{C^k} < \epsilon$ then $\rho$ has nonvanishing gradient on $\{x : \rho(x) = 0\}$. For any such $\rho$, let $\Omega_\rho = \{x \in \mathbb{R}^N : \rho(x) < 0\}$. Define

$$\mathcal{U}_\epsilon^k(\Omega_0) = \{\Omega_\rho \subseteq \mathbb{R}^N : \|\rho - \rho_0\|_{C^k} < \epsilon\}.$$

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Observe that $\mathcal{U}_\epsilon^k(\Omega_0)$ is a *collection of domains*. Then the sets $\mathcal{U}_\epsilon \equiv \mathcal{U}_\epsilon^k(\Omega_0)$ are called *neighborhoods of* $\Omega_0$ *in the* $C^k$ *topology*. Of course, neighborhoods in the $C^\infty$ or $C^\omega$ topology are defined similarly.

If $\Omega_1, \Omega_2$ are domains in $\mathbb{C}^n$, then we say that $\Omega_1 \sim \Omega_2$ if $\Omega_1$ is biholomorphic to $\Omega_2$.

**Prelude** The next result came as something of a surprise to the complex analysis community. Capitalizing on the differential geometric invariants developed by Chern and Moser (see [CHSM]), which were in turn made possible by the boundary smoothness result of Fefferman [FEF1], Burns and Shnider and Wells [BURS] proved some remarkable genericity results for poorly behaved domains. Greene and Krantz (see [GRK1, GRK2, GRK3]) later capitalized on these results and pushed them further.

**Theorem 7.1.2 (Burns–Shnider–Wells)** *Let* $k \in \mathbb{N}$. *Let* $\epsilon > 0$ *be small. Let* $\mathcal{U}_\epsilon^k \equiv \mathcal{U}_\epsilon^k(B)$ *be any neighborhood of the ball* $B \subseteq \mathbb{C}^n$ *in the* $C^k$ *topology as defined above. If* $n \geq 2$, *then* $\mathcal{U}_\epsilon^k / \sim$ *is uncountable, no matter how small* $\epsilon > 0$ *is or how large* $k$ *is (even* $k = \infty$ *or* $k = \omega$*). By contrast, if* $n = 1$ *and* $\epsilon < 1/5$ *then* $\mathcal{U}_\epsilon^k / \sim$ *has just one element.*

The last statement of the theorem perhaps merits some explanation. If $n = 1$ and $\epsilon < 1/5$, then perforce any equivalence class in $\mathcal{U}_\epsilon^k / \sim$ will contain only bounded domains that are simply connected. Thus any such domain will, by the Riemann mapping theorem, be conformally equivalent to the disc.

Greene and Krantz [GRK1, GRK2] have refined the theorem to show that, when $n \geq 2$, in fact each of the equivalence classes is closed and nowhere dense.

We now give a brief accounting of some of the differences between $n = 1$ and $n > 1$. A more detailed discussion appears in R. E. Greene and S. G. Krantz [GRK2, GRK3], in [GKK], and also in the original work of Poincaré. This subject begins with the following breakthrough of C. Fefferman [FEF1, Part I]:

**Prelude** The next (see [FEF1]) is the theorem that gave birth to much of the modern work on biholomorphic mappings (and also automorphism groups) in several complex variables. This result really came as a bolt from the blue. There had never before been a general result about boundary smoothness of biholomorphic mappings.

**Theorem 7.1.3 (Fefferman)** *Let* $\Omega_1, \Omega_2 \subseteq \mathbb{C}^n$ *be strictly pseudoconvex domains with* $C^\infty$ *boundary. If* $\phi : \Omega_1 \to \Omega_2$ *is biholomorphic, then* $\phi$ *extends to a* $C^\infty$ *diffeomorphism of* $\overline{\Omega}_1$ *onto* $\overline{\Omega}_2$.

Fefferman's theorem enables one to see that, if $\Omega_1$ and $\Omega_2$ are biholomorphic under $\phi$, then there are certain differential invariants of $\partial\Omega_1, \partial\Omega_2$ that must be preserved under $\phi$. More precisely, if $k$ is large, then the $k^{\text{th}}$ order Taylor expansion of the defining function $\rho_1$ for $\Omega_1$ (resp. of the defining function $\rho_2$ for $\Omega_2$) has more coefficients than the $k^{\text{th}}$ order Taylor expansion for $\phi$ (the disparity in the number grows rapidly with $k$). Since $\rho_1$ is mapped to $\rho_2$ under composition with $\phi^{-1}$, it follows that some of these coefficients, or combinations thereof, must be invariant

under biholomorphic mappings. N. Tanaka [TAN] and Chern–Moser [CHSM] have made these remarks precise and have shown how to calculate these invariants. A more leisurely discussion of these matters appears in R. E. Greene and S. G. Krantz [GRK3].

Now it is easy to see intuitively that two domains $\Omega$ and $\Omega_0$ can be close in the $C^k$ topology, any $k$, and have entirely different Chern–Moser–Tanaka invariants. This notion is made precise, for instance in D. Burns, S. Shnider, and R. Wells [BURS], by using a transversality argument. [Note that everything we are saying is vacuous in $\mathbb{C}^1$ because the invariants must live in the complex tangent space to the boundary—which is empty in dimension one. See [KRA1].] It is essentially a foregone conclusion that things will go badly in higher dimensions.

If one seeks positive results in the spirit of the Riemann mapping theorem in dimension $n \geq 2$, then one must find statements of a different nature. B. Fridman [FRI] has constructed a "universal domain" $\Omega^*$ which can be used to exhaust any other. He has obtained a number of variants of this idea, using elementary but clever arguments. S. Semmes [SEM] has yet another approach to the Riemann mapping theorem that is more in the spirit of the work of Lempert [LEM1]. We next present, mainly for background, a substitute for the Riemann mapping theorem whose statement and proof is more in the spirit of Fefferman's theorem. We continue to use the notation $\Omega_1 \sim \Omega_2$ to mean that $\Omega_1$ is biholomorphic to $\Omega_2$.

In what follows, we let $\mathrm{Aut}\,(\Omega)$ denote the group (under composition of mappings) of biholomorphic self-maps of the domain $\Omega$.

**Prelude** This next result also was of quite a new character for the complex analysis community. Based on stability results of the Bergman kernel and metric (under smooth deformations of a strongly pseudoconvex domain), Greene and Krantz had new tools available and were able to obtain a number of striking results.

**Theorem 7.1.4 (Greene–Krantz [GRK2])** *Let $B \subseteq \mathbb{C}^n$ be the unit ball. Let $\rho_0(z) = |z|^2 - 1$ be the usual defining function for $B$. If $\epsilon > 0$ is sufficiently small, $k = k(n)$ is sufficiently large, and $\Omega \in \mathcal{U}_\epsilon^k(B)$ then either*

**(7.1.4.1)** $\Omega \sim B$

*or*

**(7.1.4.2)** $\Omega$ *is not biholomorphic to the ball and*

- **(a)** *Aut $(\Omega)$ is compact;*
- **(b)** *Aut $(\Omega)$ has a fixed point. Moreover,*

> *If $K \subset\subset B, \epsilon > 0$ is sufficiently small (depending on $K$), and $\Omega \in \mathcal{U}_\epsilon^k(B)$ has the property that its fixed point set lies in $K$, then there is a biholomorphic mapping $\Phi : \Omega \to \Phi(\Omega) \equiv \Omega' \subseteq \mathbb{C}^n$ such that Aut $(\Omega')$ is the restriction to $\Omega'$ of a subgroup of the group of unitary matrices.*

*The collection of domains to which* **(7.1.4.2)** *applies is both dense and open.*

Theorem 7.1.4 shows, in a weak sense, that domains near the ball that have any automorphisms other than the identity are (biholomorphic to) domains with only Euclidean automorphisms. It should be noted that **(7.1.4.2a)** is already contained in the theorem of Bun Wong and Rosay [WON, ROS1] and that the denseness of

the domains to which **(7.1.4.2)** applies is contained in the work of Burns–Shnider–Wells. The proof of Theorem 7.1.4 involves a detailed analysis of Fefferman's asymptotic expansion for the Bergman kernel and of the $\overline{\partial}$-Neumann problem and would double the length of this book if we were to treat it in any detail.

The purpose of this lengthy introduction has been to establish the importance of Theorem 7.1.3 and to set the stage for what follows. It may be noted that the proof of the result analogous to Fefferman's in $\mathbb{C}^1$, that a biholomorphic mappings of smooth domains extends smoothly to the boundary, was proved in the nineteenth century by P. Painlevé [PAI]. The result in one complex dimension has been highly refined, beginning with work of O. Kellogg [KEL] and more recently by S. Warschawski [WAR1, WAR2, WAR3, WAR4], Rodin and Warschawski [ROR], and others. This classical work uses harmonic estimation, potential theory, and the Jordan curve theorem, devices which have no direct analogue in higher dimensions. A short, self-contained, proof of the one variable result—using ideas closely related to those presented here—appears in [BEK].

We conclude this section by presenting a short and elegant proof of Fefferman's Theorem 7.1.3. The techniques are due to S. Bell [BE1] and S. Bell and E. Ligocka [BELL]. The proof uses an important and non-trivial fact (known as "Condition $R$" of Bell and Ligocka) about the $\overline{\partial}$-Neumann problem. We will actually prove Condition $R$ for a strictly pseudoconvex domain in Chapter 10. (Condition $R$, and more generally the solution of the $\overline{\partial}$-Neumann problem, *is* considered in detail in the book S. G. Krantz [KRA5, Ch. 7]).

Let $\Omega \subset\subset \mathbb{C}^n$ be a domain with $C^\infty$ boundary. We define

**Prelude** Bell's Condition $R$ was an extremely original piece of work. Using it, he was able to hugely simplify Fefferman's [FEF1] terribly difficult proof and to put it on a much more intuitively appealing footing. Condition $R$ has been an important artifact of our studies for thirty years.

> **Condition** $R$ (S. Bell [BE1]) Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded domain. Define an operator on $L^2(\Omega)$ by
>
> $$Pf(z) = \int_\Omega K(z,\zeta) f(\zeta) dV(\zeta)\,,$$
>
> where $K(z,\zeta)$ is the Bergman kernel for $\Omega$. This is the Bergman projection. Then, for each $j > 0$, there is an $m = m(j) > 0$ such that $P$ satisfies the estimates
>
> $$\| Pf \|_{W^j(\Omega)} \leq C_j \| f \|_{W^m(\Omega)}$$
>
> for all testing functions $f$.

Using a little Sobolev theory (see [KRA5, Ch. 1] and also Exercise 5 in our Chapter 1), one can easily see that this formulation of Condition $R$ is equivalent to the condition that the Bergman kernel map $C^\infty(\overline{\Omega})$ to $C^\infty(\overline{\Omega})$.

The remainder of this section is devoted to proving Theorem 7.1.3. We first consider why any strongly pseudoconvex domain satisifies Condition $R$. Let $P : L^2(\Omega) \to L^2(\Omega)$. The operator

$$\overline{\partial} : \bigwedge^{0,j} \to \bigwedge^{0,j+1} \tag{7.1.5}$$

is the usual exterior differential operator of complex analysis. One may show that the second-order, elliptic partial differential operator $\Box = \overline{\partial}\,\overline{\partial}^* + \overline{\partial}^*\,\overline{\partial}$ has a canonical right inverse called $N$. This is the $\overline{\partial}$-Neumann operator. These operators are treated in detail in [FOK] and [KRA5, Ch. 7]. See also our Chapter 10. Then it is a straightforward exercise in Hilbert space theory to verify that

$$P = I - \overline{\partial}^* N \overline{\partial}\,,$$

where $P$ is the Bergman projection. Now the references [FOK] and [KRA5, Ch. 7] prove in detail that $N$ maps $W^s$ (the Sobolev space of order $s$) to $W^{s+1}$ for every $s$. See also Chapter 10 below. It follows from this and formula (7.1.5) that $P$ maps $W^s$ to $W^{s-1}$. That is enough to verify Condition $R$.

We remark in passing that, in general, it does not matter whether $m(j)$ is much larger than $j$ or whether the $m(j)$ in Condition $R$ depends polynomially on $j$ or exponentially on $j$. It so happens that, for a strictly pseudoconvex domain, we may take $m(j) = j$. This assertion is proved in [KRA5, Ch. 7] in detail. On the other hand, Barrett [BARR1] has shown that, on the Diederich–Fornaess worm domain [DIRF2], we must take $m(j) > j$. Later on, Christ [CHST2] showed that Condition $R$ fails altogether on the worm. See Chapter 9 for more on the worm.

Now we build a sequence of lemmas leading to Fefferman's Theorem. First we record some notation.

We let $W^j(\Omega)$ be the usual Sobolev space. See [KRA5, Ch. 1] or Exercise 5 in our Chapter 1 for this idea.

If $\Omega \subset\subset \mathbb{C}^n$ is any smoothly bounded domain and if $j \in \mathbb{N}$, we let

$$WH^j(\Omega) = W^j(\Omega) \cap \{\text{holomorphic functions on } \Omega\},$$

$$WH^\infty(\Omega) = \bigcap_{j=1}^{\infty} WH^j(\Omega) = C^\infty(\overline{\Omega}) \cap \{\text{holomorphic functions on } \Omega\}.$$

Here $W^j$ is the standard Sobolev space on a domain (for which see [KRA5, Ch. 1], [ADA]). Let $W_0^j(\Omega)$ be the $W^j$ closure of $C_c^\infty(\Omega)$. [Exercise: if $j$ is sufficiently large,

then the Sobolev embedding theorem implies trivially that $W_0^j(\Omega)$ is a proper subset of $W^j(\Omega)$.[1]

Let us say that $u, v \in C^\infty(\overline{\Omega})$ agree up to order $k$ on $\partial\Omega$ if

$$\left(\frac{\partial}{\partial z}\right)^\alpha \left(\frac{\partial}{\partial \overline{z}}\right)^\beta (u-v)\Big|_{\partial\Omega} = 0 \quad \forall \alpha, \beta \quad \text{with} \quad |\alpha| + |\beta| \leq k.$$

**Lemma 7.1.6** *Let $\Omega \subset\subset \mathbb{C}^n$ be smoothly bounded and strictly pseudoconvex. Let $w \in \Omega$ be fixed. Let $K$ denote the Bergman kernel for $\Omega$. There is a constant $C_w > 0$ such that*

$$\|K(w, \cdot)\|_{\sup} \leq C_w.$$

**Proof** The function $K(z, \cdot)$ is harmonic. Let $\phi : \Omega \to \mathbb{R}$ be a radial, $C_c^\infty$ function centered at $w$. Assume that $\phi \geq 0$ and $\int \phi(\zeta) dV(\zeta) = 1$. Then the mean value property implies that

$$K(z, w) = \int_\Omega K(z, \zeta)\phi(\zeta) dV(\zeta).$$

But the last expression equals $P\phi(z)$. Therefore

$$\begin{aligned} \|K(w, \cdot)\|_{\sup} &= \sup_{z \in \Omega} |K(w, z)| \\ &= \sup_{z \in \Omega} |K(z, w)| \\ &= \sup_{z \in \Omega} |P\phi(z)|. \end{aligned}$$

By Sobolev's Theorem, this is

$$\leq C(\Omega) \cdot \|P\phi\|_{WH^{2n+1}}.$$

By Condition $R$, this is

$$\leq C(\Omega) \cdot \|\phi\|_{W^{m(2n+1)}} \equiv C_w.$$

□

**Lemma 7.1.7** *Let $u \in C^\infty(\overline{\Omega})$ be arbitrary. Let $s \in \{0, 1, 2, \ldots\}$. Then there is a $v \in C^\infty(\overline{\Omega})$ such that $Pv = 0$ and the functions $u$ and $v$ agree to order $s$ on $\partial\Omega$.*

[1] For the readers's convenience, we recall here that the Sobolev embedding theorem says that, if a function on $\mathbb{R}^N$ has more than $N/2$ derivatives in $L^2$, then in fact it is continuous. See [KRA5, Ch. 1] or [ADA], for instance, for the details.

**Proof** After a partition of unity, it suffices to prove the assertion in a small neighborhood $U$ of $z_0 \in \partial\Omega$. After a rotation, we may suppose that $\partial\rho/\partial z_1 \neq 0$ on $U \cap \overline{\Omega}$, where $\rho$ is a defining function for $\Omega$. Define the differential operator

$$v = \frac{\text{Re}\left\{\sum_{j=1}^n \frac{\partial\rho}{\partial z_j}\frac{\partial}{\partial \overline{z}_j}\right\}}{\sum_{j=1}^n \left|\frac{\partial\rho}{\partial z_j}\right|^2}.$$

Notice that $v\rho = 1$. Now we define $v$ by induction on $s$.

For the case $s = 0$, let

$$w_1 = \frac{\rho u}{\partial\rho/\partial\zeta_1}.$$

Define

$$\begin{aligned} v_1 &= \frac{\partial}{\partial\zeta_1} w_1 \\ &= u + O(\rho). \end{aligned}$$

Then $u$ and $v_1$ agree to order 0 on $\partial\Omega$. Also

$$Pv_1(z) = \int K(z,\zeta)\frac{\partial}{\partial\zeta_1} w_1(\zeta)\, dV(\zeta).$$

This equals, by integration by parts,

$$-\int \frac{\partial}{\partial\zeta_1} K(z,\zeta) w_1(\zeta)\, dV(\zeta).$$

Notice that the integration by parts is valid by Lemma 7.1.6 and because $w_1|_{\partial\Omega} = 0$. Also the integrand in this last line is zero because $K(z,\cdot)$ is conjugate holomorphic.

Suppose inductively that $w_{s-1} = w_{s-2} + \theta_{s-1}\rho^s$ and $v_{s-1} = (\partial/\partial z_1)(w_{s-1})$ have been constructed. We show that there is a $w_s$ of the form

$$w_s = w_{s-1} + \theta_s \cdot \rho^{s+1}$$

such that $v_s = (\partial/\partial z_1)(w_s)$ agrees to order $s$ with $u$ on $\partial\Omega$. By the inductive hypothesis,

$$\begin{aligned} v_s &= \frac{\partial}{\partial z_1} w_s \\ &= \frac{\partial w_{s-1}}{\partial z_1} + \frac{\partial}{\partial z_1}\left[\theta_s \cdot \rho^{s+1}\right] \\ &= v_{s-1} + \rho^s \left[(s+1)\theta_s \frac{\partial\rho}{\partial z_1} + \rho \cdot \frac{\partial\theta_s}{\partial z_1}\right] \end{aligned}$$

agrees to order $s-1$ with $u$ on $\partial\Omega$ so long as $\theta_s$ is smooth. So we need to examine $D(u-v_s)$, where $D$ is an $s$-order differential operator. But if $D$ involves a tangential derivative $D_0$, then write $D = D_0 \cdot D_1$. It follows that $D(u-v_s) = D_0(\alpha)$, where $\alpha$ vanishes on $\partial\Omega$ so that $D_0\alpha = 0$ on $\partial\Omega$. So we need only check $D = \nu^s$.

We have seen that $\theta_s$ must be chosen so that

$$\nu^s(u-v_s) = 0 \qquad \text{on} \quad \partial\Omega.$$

Equivalently,

$$\nu^s(u-v_{s-1}) - \nu^s\left(\frac{\partial}{\partial z_1}\right)(\theta_s\rho^{s+1}) = 0 \quad \text{on} \qquad \partial\Omega$$

or

$$\nu^s(u-v_{s-1}) - \theta_s\left(\nu^s\frac{\partial}{\partial z_1}\rho^{s+1}\right) = 0 \quad \text{on} \qquad \partial\Omega$$

or

$$\nu^s(u-v_{s-1}) - \theta_s\cdot(s+1)!\frac{\partial\rho}{\partial z_1} = 0 \quad \text{on} \qquad \partial\Omega.$$

It follows that we must choose

$$\theta_s = \frac{\nu^s(u-v_{s-1})}{(s+1)!\frac{\partial\rho}{\partial z_1}},$$

which is indeed smooth on $U$. As in the case $s=0$, it holds that $Pv_s = 0$. This completes the induction and the proof. □

**Remark 7.1.8** A retrospection of the proof reveals that we have constructed $v$ by subtracting from $u$ a Taylor-type expansion in powers of $\rho$. □

**Lemma 7.1.9** *For each $s \in \mathbb{N}$ we have $WH^\infty(\Omega) \subseteq P(W_0^s(\Omega))$.*

**Proof** Let $u \in C^\infty(\overline{\Omega})$. Choose $v$ according to Lemma 7.1.7. Then $u - v \in W_0^s$ and $Pu = P(u-v)$. Therefore

$$P(W_0^s) \supseteq P(C^\infty(\overline{\Omega})) \supseteq P(WH^\infty(\Omega)) = WH^\infty(\Omega).$$

□

Henceforth, let $\Omega_1, \Omega_2$ be fixed $C^\infty$ strictly pseudoconvex domains in $\mathbb{C}^n$, with $K_1, K_2$ their Bergman kernels and $P_1, P_2$ the corresponding Bergman projections. Let $\phi : \Omega_1 \to \Omega_2$ be a biholomorphic mapping, and let $u = \det \mathrm{Jac}_{\mathbb{C}}\phi$. For $j = 1, 2$, let $\delta_j(z) = \delta_{\Omega_j}(z) = \mathrm{dist}\,(z, {}^c\Omega_j)$.

**Lemma 7.1.10** *For any $g \in L^2(\Omega_2)$ we have*

$$P_1(u \cdot (g \circ \phi)) = u \cdot ((P_2(g)) \circ \phi).$$

**Proof** Notice that $u \cdot (g \circ \phi) \in L^2(\Omega_1)$ by change of variables. Therefore

$$\begin{aligned} P_1(u \cdot (g \circ \phi))(z) &= \int_{\Omega_1} K_1(z, \zeta) u(\zeta) g(\phi(\zeta)) dV(\zeta) \\ &= \int_{\Omega_1} u(z) K_2(\phi(z), \phi(\zeta)) \overline{u(\zeta)} u(\zeta) g(\phi(\zeta)) dV(\zeta) \,. \end{aligned}$$

Change of variable now yields

$$\begin{aligned} P_1(u \cdot (g \circ \phi))(z) &= u(z) \int_{\Omega_2} K_2(\phi(z), \xi) g(\xi) dV(\xi) \\ &= u(z) \cdot [(P_2(g)) \circ \phi](z). \end{aligned}$$

□

**Lemma 7.1.11** *Let $\psi : \Omega_1 \to \Omega_2$ be a $C^j$ diffeomorphism that satisfies*

$$\left| \frac{\partial^\alpha \psi}{\partial z^\alpha}(z) \right| \leq C \cdot (\delta_1(z))^{-|\alpha|}, \tag{7.1.11.1}$$

*for all multi-indices $\alpha$ with $|\alpha| \leq j \in \mathbb{N}$ and*

$$|\nabla \psi^{-1}(w)| \leq C(\delta_2(w))^{-1}. \tag{7.1.11.2}$$

*Suppose also that*

$$\delta_2(\psi(z)) \leq C\delta_1(z). \tag{7.1.11.3}$$

*Then there is a number $J = J(j)$ such that, whenever $g \in W_0^{j+J}(\Omega_2)$ then $g \circ \psi \in W_0^j(\Omega_1)$.*

**Proof** The subscript 0 causes no trouble by the definition of $W_0^j$. Therefore it suffices to prove an estimate of the form

$$\|g \circ \psi\|_{W_0^j} \leq C \|g\|_{W_0^{j+J}}, \quad \text{all } g \in C_c^\infty(\Omega).$$

By the chain rule and Leibniz's rule, if $\alpha$ is a multi-index of modulus not exceeding $j$, then

$$\left( \frac{\partial}{\partial z} \right)^\alpha (g \circ \psi) = \sum \left[ (D^\beta g) \circ \psi \right] \cdot D^{\gamma_1} \psi \cdots D^{\gamma_\ell} \psi,$$

where $|\beta| \leq |\alpha|, \sum |\gamma_i| \leq |\alpha|$, and the number of terms in the sum depends only on $\alpha$ (a classical formula of Faà de Bruno—see [KRPA1, Ch. 1]—actually gives this sum quite explicitly, but we do not require such detail). Note here that $D^{\gamma_i}\psi$ is used to denote a derivative of *some component* of $\psi$. By hypothesis, it follows that

$$\left|\left(\frac{\partial}{\partial z}\right)^{\alpha}(g\circ\psi)\right| \leq C\sum |(D^{\beta}g)\circ\psi|\cdot(\delta_1(z))^{-j}.$$

Therefore

$$\begin{aligned}\int_{\Omega_1}\left|\left(\frac{\partial}{\partial z}\right)^{\alpha}(g\circ\psi)\right|^2 dV \leq& \\ &\leq C\sum\int_{\Omega_1}|(D^{\beta}g)\circ\psi|^2(\delta_1(z))^{-2j}dV(z)\\ &= C\sum\int_{\Omega_2}|D^{\beta}g(w)|^2\delta_1\left(\psi^{-1}(w)\right)^{-2j}\\ &\qquad\times|\det J_{\mathbb{C}}\psi^{-1}|^2 dV(w).\end{aligned}$$

But (7.1.11.2) and (7.1.11.3) imply that the last line is majorized by

$$C\sum\int_{\Omega_2}|D^{\beta}g(w)|^2\delta_2(w)^{-2j}\delta_2(w)^{-2n}dV(w). \tag{7.1.11.4}$$

Now if $J$ is large enough, depending on the Sobolev embedding Theorem, then

$$|D^{\beta}g(w)| \leq C\|g\|_{W_0^{j+J}}\cdot\delta_2(w)^{2n+2j}.$$

(Remember that $g$ is compactly supported in $\Omega_2$.) Hence (7.1.11.4) is majorized by $C\|g\|_{W_0^{j+J}}$. □

**Lemma 7.1.12** *For each $j \in \mathbb{N}$, there is an integer $J$ so large that if $g \in W_0^{j+J}(\Omega_2)$, then $g\circ\phi \in W_0^j(\Omega_1)$.*

**Proof** The Cauchy estimates give (since $\phi$ is bounded) that

$$\left|\frac{\partial^{\alpha}\phi_{\ell}}{\partial z^{\alpha}}(z)\right| \leq C\cdot(\delta_1(z))^{-|\alpha|}, \qquad \ell = 1,\ldots,n \tag{7.1.12.1}$$

and

$$|\nabla\phi^{-1}(w)| \leq C(\delta_2(w))^{-1}, \tag{7.1.12.2}$$

where $\phi = (\phi_1, \ldots, \phi_n)$. We will prove that

$$C \cdot \delta_1(z) \geq \delta_2(\phi(z)). \tag{7.1.12.3}$$

Then Lemma 7.1.11 gives the result.

To prove (7.1.12.3), let $\rho$ be a smooth strictly plurisubharmonic defining function for $\Omega_1$. Then $\rho \circ \phi^{-1}$ is a smooth plurisubharmonic function on $\Omega_2$. Since $\rho$ vanishes on $\partial\Omega_1$ and since $\phi^{-1}$ is proper, we conclude that $\rho \circ \phi^{-1}$ extends continuously to $\overline{\Omega}_2$. If $P \in \partial\Omega_2$ and $\nu_P$ is the unit outward normal to $\partial\Omega_2$ at $P$, then Hopf's lemma implies that the (lower) one-sided derivative $(\partial/\partial\nu_P)(\rho \circ \phi^{-1})$ satisfies

$$\frac{\partial}{\partial \nu_P}(\rho \circ \phi^{-1}(P)) \geq C.$$

So, for $w = P - \epsilon\nu_P, \epsilon$ small, it holds that

$$-\rho \circ \phi^{-1}(w) \geq C \cdot \delta_2(w).$$

These estimates are uniform in $P \in \partial\Omega_2$. Using the comparability of $|\rho|$ and $\delta_1$ yields

$$C\delta_1(\phi^{-1}(w)) \geq \delta_2(w).$$

Setting $z = \phi^{-1}(w)$ now gives

$$C'\delta_1(z) \geq \delta_2(\phi(z)),$$

which is (7.1.12.3). □

**Exercise for the Reader** Let $\Omega \subset\subset \mathbb{C}^n$ be a smoothly bounded domain. Let $j \in \mathbb{N}$. There is an $N = N(j)$ so large that $g \in W_0^N$ implies that $g$ vanishes to order $j$ on $\partial\Omega$.

**Lemma 7.1.13** *The function $u$ is in $C^\infty(\overline{\Omega}_1)$.*

**Proof** It suffices to show that $u \in W^j(\Omega_1)$, every $j$. So fix $j$. Let $m = m(j)$ as in Condition $R$. According to (7.1.12.1), $|u(z)| \leq C\delta_1(z)^{-2n}$. Then, by Lemma 7.1.12 and the Exercise for the Reader following it, there is a $J$ so large that $g \in W_0^{m+J}(\Omega_2)$ implies $u \cdot (g \circ \phi) \in W^m(\Omega_1)$. Choose, by Lemma 7.1.7, a $g \in W_0^{m+J}(\Omega_2)$ such that $P_2 g \equiv 1$. Then Lemma 7.1.10 yields

$$P_1(u \cdot (g \circ \phi)) = u.$$

By Condition $R$, it follows that $u \in W^j(\Omega_1)$. □

**Lemma 7.1.14** *The function $u$ is bounded from 0 on $\overline{\Omega}_1$.*

**Proof** By symmetry, we may apply Lemma 7.1.13 to $\phi^{-1}$ and $\det J_{\mathbb{C}}(\phi^{-1}) = 1/u$. We conclude that $1/u \in C^\infty(\overline{\Omega}_2)$. Thus $u$ is nonvanishing on $\overline{\Omega}$. □

**Proof of Fefferman's Theorem (Theorem 7.1.3)** Use the notation of the proof of Lemma 7.1.12 Choose $g_1, \ldots, g_n \in W_0^{m+J}(\Omega_2)$ such that $P_2 g_i(w) = w_i$ (here $w_i$ is the $i^{\text{th}}$ coordinate function). Then Lemma 7.1.8 yields that $u \cdot \phi_i \in W^j(\Omega_1), i = 1, \ldots, n$. By Lemma 7.1.12, $\phi_i \in W^j(\Omega_1), i = 1, \ldots, n$. By symmetry, $\phi^{-1} \in W^j(\Omega_2)$. Since $j$ is arbitrary, the Sobolev embedding Theorem finishes the proof. □

It is important to understand the central role of Condition $R$ in this proof. With some emendations, the proof we have presented shows that if $\Omega_1, \Omega_2 \subset\subset \mathbb{C}^n$ are smoothly bounded, pseudoconvex, and both satisfy Condition $R$, then a biholomorphic mapping from $\Omega_1$ to $\Omega_2$ extends smoothly to the boundary (in fact S. Bell [BE1] has shown that it suffices for just one of the domains to satisfy Condition $R$). Condition $R$ is known to hold on domains that have real analytic boundaries (see K. Diederich and J. E. Fornæss [DIRF3]), and more generally on domains of finite type (see [CAT1, CAT2]). There are a number of interesting examples of non-pseudoconvex domains on which Condition $R$ fails (see D. Barrett [BARR2] and C. Kiselman [KIS]). It had been conjectured that Condition $R$ holds on all smoothly bounded pseudoconvex domains. But Christ [CHST2] showed that in fact Condition $R$ fails on the Diederich–Fornæss worm domain (which is smoothly bounded and pseudoconvex). See Chapter 9.

L. Lempert [LEM3] has derived a sharp boundary regularity result for biholomorphic mappings of strictly pseudoconvex domains with $C^k$ boundary. S. Pinchuk and S. I. Tsyganov [PTS] have an analogous result. The correct conclusion turns out to be that there is a loss of smoothness in some directions. So the sharp regularity result is formulated in terms of nonisotropic spaces. It is too technical to describe here.

## 7.2 The Bergman Metric at the Boundary

**Capsule:** It was Fefferman [FEF1] who used the asymptotic expansion for the Bergman kernel to calculate the boundary behavior of Bergman metric geodesics. He smoothly continued a biholomorphic mapping to the boundary by tracing the mapping along these "pseudotransversal" geodesics. It was a remarkably powerful argument, but one that has now been trumped by Bell's Condition $R$.

The Bergman metric on the disc $D$ is given by

$$g_{jk} = \frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}} \log K(z,z) = \frac{\partial}{\partial z}\frac{\partial}{\partial \overline{z}} \left[ -\log \pi - 2\log(1-|z|^2) \right] = \frac{2}{(1-|z|^2)^2} \,.$$

Thus the length of a curve $\gamma : [0, 1] \to D$ is given by

$$\ell_B(\gamma) = \int_0^1 \frac{\sqrt{2}\|\gamma'(t)\|}{(1 - |\gamma(t)|^2)}\, dt\,.$$

It is natural to wonder what one can say about the Bergman metric on a more general class of domains. Let $\Omega \subseteq \mathbb{C}$ be a bounded domain with $C^4$ boundary, and suppose for the moment that $\Omega$ is simply connected. Then the Riemann mapping theorem tells us that there is a conformal mapping $\Phi : \Omega \to D$. Of course then we know that

$$K_\Omega(z, z) = |\Phi'(z)|^2 K_D(\Phi(z), \Phi(z)) = \frac{|\Phi'(z)|^2}{\pi \cdot (1 - |\Phi(z)|^2)^2}\,.$$

Then the Bergman metric on $\Omega$ is given by

$$g_{z\bar{z}} = \frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}} \log\left[\frac{|\Phi'(z)|^2}{\pi \cdot (1 - |\Phi(z)|^2)^2}\right] = \frac{\partial}{\partial z}\frac{\partial}{\partial \bar{z}}\left[\log|\Phi'(z)|^2 - \log\pi - 2\log(1 - |\Phi(z)|^2)\right].$$

Of course $|\Phi'(z)|$ is bounded and bounded from 0 (see [BEK]). Also the derivatives of $\Phi$, up to order 3, are bounded. So the second derivative of $\log|\Phi'(z)|^2$ is a bounded term. The second derivative of $\log\pi$ is of course 0.

The second derivative of the remaining (and most interesting) term may be calculated to be

$$\begin{aligned} g_{z\bar{z}} &= \frac{\partial^2}{\partial\bar{z}\partial z}\left[-2\log(1 - |\Phi(z)|^2)\right] \\ &= \frac{\partial}{\partial\bar{z}}\left(\frac{-2}{1 - |\Phi(z)|^2} \cdot \left[-\Phi'(z)\overline{\Phi}(z)\right]\right) \\ &= \frac{2}{(1 - |\Phi(z)|^2)^2} \cdot \left[-\overline{\Phi}'(z)\Phi(z)\right] \cdot \left[-\Phi'(z)\overline{\Phi}(z)\right] \\ &\qquad -\frac{2}{1 - |\Phi(z)|^2} \cdot \left[-\Phi'(z)\overline{\Phi}'(z)\right]. \end{aligned}$$

This in turn, after some simplification, equals

$$\frac{2|\Phi'(z)|^2}{(1 - |\Phi(z)|^2)^2}\,. \tag{7.2.1}$$

As previously noted, the numerator is bounded and bounded from 0. Hopf's lemma (see [KRA1, Ch. 11]) tells us that $(1 - |\Phi(z)|^2) \approx \text{dist}_{\partial\Omega}(z)$. So that the Bergman metric on $\Omega$ blows up like the reciprocal of the square of the distance to the boundary—just as on the disc.

In the case that $\Omega$ has $C^2$ boundary and is finitely connected—not necessarily simply connected—then one may use the Ahlfors map (see [KRA17, Ch. 4]) instead of the Riemann mapping and obtain a result similar to that in (7.2.1). We omit the details.

H. Bremermann [BRE] showed that any domain in $\mathbb{C}^n$ with complete Bergman metric is a domain of holomorphy. This is in fact not difficult, as one can use the hypothesis of the completeness of the metric to confirm the Kontinuitätssatz (see [KRA1, Ch. 3]), hence derive pseudoconvexity. Bremerman also gave an example to demonstrate that the converse is not true. However, T. Ohsawa [OHS] has shown that any pseudoconvex domain with $C^1$ boundary has complete Bergman metric. This result is important both conceptually and practically. There is no analogous result for either the Carathéodory or Kobayashi metrics.

In the paper [KOB1, Theorem 9.2], Kobayashi shows that any bounded analytic polyhedron has complete Bergman metric.

## 7.3 Inequivalence of the Ball and the Polydisc

**Capsule:** Many mathematicians feel that the subject of the function theory of several complex variables began with Poincaré's 1906 theorem that the ball and the polydisc are biholomorphically inequivalent. His proof of that result, using invariance properties of the automorphism group, was quite original and powerful for its time. And the result itself has been enormously influential. It is as important today as when it was first proved.

In this section we give a Bergman-geometric proof of the following classical result of Poincaré (Poincaré's original proof was more group-theoretic).

**Prelude** Now we have an enunciation and proof of Poincaré's seminal theorem.

**Theorem 7.3.1** *There is no biholomorphic map of the bidisc $D^2(0,1)$ to the ball $B(0,1) \subseteq \mathbb{C}^2$.*

**Proof** Suppose, seeking a contradiction, that there is such a map $\phi$. Since Möbius transformations act transitively on the disc, pairs of them act transitively on the bidisc. Therefore we may compose $\phi$ with a self-map of the bidisc and assume that $\phi$ maps 0 to 0.

If $Y \in \partial B$ then the disc $d_Y = \{z \in B : z = \zeta Y, \zeta \in \mathbb{C}, |\zeta| < 1\}$ is a totally geodesic submanifold of $B$ (informally, this means that if $P, Q$ are points of $d_Y$ then the geodesic connecting them in the Riemannian manifold $d_Y$ is the same as the geodesic connecting them in the Riemannian manifold $B$—see S. Kobayashi and K. Nomizu [KOBN, Ch. 4]).

By our discussion in the calculation of the Poincaré metric, we may conclude that the geodesics, or paths of least length, emanating from the origin in the ball are the rays $\tau_Y : t \mapsto tY$. (This assertion may also be derived from symmetry considerations.)

Likewise, if $\alpha, \beta \in \mathbb{C}, |\alpha| = 1, |\beta| = 1$, then the disc $e_0 = \{(\zeta\alpha, \zeta\beta) : \zeta \in D\} \subseteq D^2(0,1)$ is a totally geodesic submanifold of $D^2(0,1)$. Again we may apply our discussion of the Poincaré metric on the disc to conclude that the geodesic curve emanating from the origin in the bidisc in the direction $X = (\alpha, \beta)$ is $\psi_{\alpha\beta} : t \mapsto tX$. A similar argument shows that the curve $t \mapsto (t, 0)$ is a geodesic in the bidisc.

Now if $t \mapsto tX$ is one of the above mentioned geodesics on the bidisc then it will be mapped under $\phi$ to a geodesic $t \mapsto tY$ in the ball. If $0 < t_1 < t_2 < 1$ then the points $t_1X, t_2X \in D^2$ will be mapped to points $t_1'Y, t_2'Y \in B$ and it must be that $0 < t_1' < t_2' < 1$ since $\phi$ is an isometry hence must map the point $t_2X$ to a point further from the origin than it maps $t_1X$ (because $t_2X$ is further from the origin than $t_1X$). It follows that the limit

$$\lim_{t \to 1^-} \phi(tX)$$

exists for every choice of $X$ and the limit lies in $\partial B$. After composing $\phi$ with a rotation we may suppose that $\{\phi(t(1,0))\}$ terminates at $(1,0)$.

Now consider the function $f(z_1, z_2) = (z_1 + 1)/2$ on $B$. This function has the property that $f(1,0) = 1, f$ is holomorphic on a neighborhood of $\overline{B}$, and $|f(z)| < 1$ for $z \in \overline{B} \setminus \{(1,0\}$. For $0 < r < 1$ we invoke the mean-value property for a harmonic function to write

$$\frac{1}{2\pi} \int_0^{2\pi} f \circ \phi(r, re^{i\theta}) d\theta = f \circ \phi(r, 0). \tag{7.3.1.1}$$

As $r \to 1^-$ the right hand side tends to $\lim_{t \to 1^-} f(t,0) = 1$. However each of the paths $r \to (r, re^{i\theta})$ is a geodesic in the bidisc, as discussed above, and for different $\theta \in [0, 2\pi)$ they are distinct. Thus the curves $r \to \phi(r, re^{i\theta})$ have distinct limits in $\partial B$, and these limits will be different from the point $(1,0) \in \partial B$. In particular, $\lim_{r \to 1^-} f \circ \phi(r, re^{i\theta})$ exists for each $\theta \in [0, 2\pi)$ and assumes a value of modulus strictly less than 1.

By the Lebesgue dominated convergence theorem we may pass to the limit as $r \to 1^-$ in the left side of (7.3.1.1) to obtain a limit that must be strictly less than one in absolute value. That is the required contradiction. □

## Exercises

1. Let us construct an invariant metric on the disc $D$ in the complex plane by hand. Take it that the length of the vector $\langle 1, 0\rangle$ at the base point 0 is 1. Now use the conformal invariance of the metric to calculate the length of any other vector at any other base point. Of course the metric that you obtain in this way should be (a constant multiple of) the Poincaré metric.
2. Imitate Exercise 1 for the unit ball in $\mathbb{C}^n$.

**3.** Consider the domain $\Omega = \{\zeta \in \mathbb{C} : \operatorname{Re}\zeta < 0 \text{ or } \operatorname{Im}\zeta < 0\}$. This domain is conformally equivalent to the upper half plane, which is in turn conformally equivalent to the unit disc. Therefore one can calculate the Bergman kernel for $\Omega$ using its mapping invariance property. What can you say about the boundary behavior of this kernel at the origin? What can you say about the boundary behavior of the Bergman metric at the origin?

**4.** Calculate the Bergman kernel and metric for the punctured disc $D' \equiv \{\zeta \in \mathbb{C} : 0 < |\zeta| < 1\}$. How do they differ from the Bergman kernel and metric for the usual disc $D$?

Calculate the Bergman kernel and metric for the punctured ball $B' \equiv \{z \in \mathbb{C}^n : 0 < |z| < 1\}$. How do they differ from the Bergman kernel and metric for the usual ball $B$?

**5.** Use the results from the text about the Bergman kernel for the annulus to calculate the Bergman metric on the annulus.

**6.** Let $\Omega = B(0,1) \setminus B(\mathbf{1}, 1/2)$, where $B(P,r)$ denotes the open ball with center $P$ and radius $r$ and $\mathbf{1} = (1, 0, \ldots, 0)$. Estimate the distance from the origin to the point $(1/2, 0)$ in the Bergman metric of $\Omega$.

**7.** The infinitesimal Kobayashi–Royden metric on a domain $\Omega \subseteq \mathbb{C}^n$ is defined by $F_K : \Omega \times \mathbb{C}^n \to \mathbb{R}$, where

$$\begin{aligned} F_K(z,\xi) &\equiv \inf\{\alpha : \alpha > 0 \text{ and } \exists f \in \Omega(B) \text{ with } f(0) = z, \left(f'(0)\right)(e_1) = \xi/\alpha\} \\ &= \inf\left\{ \frac{|\xi|}{|(f'(0))(e_1)|} : f \in \Omega(B), (f'(0))(e_1) \text{is a} \right. \\ &\qquad \left. \text{constant multiple of } \xi \right\} \\ &= \frac{|\xi|}{\sup\{|(f'(0))(e_1)| : f \in \Omega(B), (f'(0))(e_1) \text{ is a constant multiple of } \xi\}} \,. \end{aligned}$$

Use the Schwarz lemma to calculate the Kobayashi–Royden metric on the disc. How does it compare with the Poincaré metric? Now calculate the Kobayashi–Royden metric on the ball.

**8.** The infinitesimal Carathéodory metric on a domain $\Omega \subseteq \mathbb{C}^n$ is defined by $F_C : \Omega \times \mathbb{C}^n \to \mathbb{R}$, where

$$F_C(z,\xi) = \sup_{\substack{f \in B(\Omega) \\ f(z)=0}} |f_*(z)\xi| \equiv \sup_{\substack{f \in B(\Omega) \\ f(z)=0}} \left| \sum_{j=1}^{n} \frac{\partial f}{\partial z_j}(z) \cdot \xi_j \right| .$$

Use the Schwarz lemma to calculate the Carathéodory metric on the disc. How does it compare with the Poincaré metric? Now calculate the Carathéodory metric on the ball.

**9.** Refer to Exercises 7 and 8 for terminology. Prove that the Carathéodory metric is always less than or equal to the Kobayashi metric.

**10.** Refer to Exercises 7 and 8 for terminology. Prove that if $\Phi : \Omega_1 \to \Omega_2$ is a holomorphic mapping of complex domains then it is distance non-increasing in the Carathéodory metric. Prove an analogous statement for the Kobayashi–Royden metric.

# Chapter 8
# Further Geometric and Analytic Theory

**Prologue:** Certainly one of the beautiful aspects of Bergman's theory is the geometric properties of the kernel and metric. There are many dimensions to this part of the world: Bergman representative coordinates, the Berezin transform, the invariant Laplacian, Bergman geodesics, and the list could go on at some length. We explore many of these topics here.

## 8.1 Bergman Representative Coordinates

**Capsule:** A little known artifact of the Bergman theory, these are a means by which a biholomorphic mapping may be thought of as (locally) linear. Such a feature is strongly analogous to geodesic normal coordinates in classical Riemannian geometry. We present the basics of Bergman representative coordinates in the present section.

The theory of the Bergman kernel gives rise to many important geometric invariants. Among these are the not-very-well-known Bergman representative coordinates. This is a local coordinate system in which a biholomorphic mapping is realized as a linear mapping. Such a result, while initially quite startling, is in fact completely analogous to the result in Riemannian geometry regarding geodesic normal coordinates. But geodesic normal coordinates are almost never holomorphic—unless the Kähler metric is flat. By contrast, Bergman representative coordinates are always holomorphic.

Bergman representative coordinates are of considerable intrinsic and theoretical interest. But they are also useful in understanding the boundary behavior of biholomorphic mappings. And Greene–Krantz made good use of them in proving their noted semicontinuity theorem (see [KRA18, Ch. 3], [GRK5]).

Bergman's representative coordinates are also essential in the proof of Lu Qi-Keng's Theorem on bounded domains with Bergman metrics of constant holomorphic sectional curvature. This result will be stated and proved in the present section.

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Now let $\Omega$ be a bounded domain in $\mathbb{C}^n$ and let $q$ be a point of $\Omega$. The Bergman kernel $K_\Omega(q,q)$ on the diagonal is of course real and positive so that there is a neighborhood $U$ of $q$ such that, for all $z, w$ in $U$, $K_\Omega(z,w) \neq 0$. Then for all $z, w$ in $U$, we define

$$b_j(z) = b_{j,q}(z) = \frac{\partial}{\partial \overline{w}_j} \log \frac{K(z,w)}{K(w,w)}\bigg|_{w=q}.$$

Note that these coordinates are well-defined, independent of the choice of logarithmic branch. Each representative coordinate $b_j(z)$ is clearly a holomorphic function of $z$.

The mapping

$$z \longmapsto \big(b_1(z), \ldots, b_n(z)\big) \in \mathbb{C}^n$$

is defined and holomorphic in a neighborhood of the point $q$ (a neighborhood on which the kernel does not vanish). Note also that $(b_1(q), \ldots, b_n(q)) = (0, \ldots, 0)$.

We are hoping to use these functions as holomorphic local coordinates in a neighborhood of $q$. By the holomorphic inverse function theorem, these functions give local coordinates if the holomorphic Jacobian

$$\det \left( \frac{\partial b_j}{\partial z_k} \right)_{j,k=1,\ldots,n}$$

is nonzero at $q$.

But in fact the nonvanishing of this determinant at $q$ is an immediate consequence of the fact that the Bergman metric is positive definite. To see this relationship, observe that

$$\begin{aligned} \frac{\partial b_j}{\partial z_k}\bigg|_{z=q} &= \frac{\partial}{\partial z_k} \left( \frac{\partial}{\partial \overline{w}_j} \log K(z,w) \right) \bigg|_{z=w=q} \\ &= \frac{\partial^2}{\partial z_k \partial \overline{z}_j} \log K(z,z)\bigg|_{z=q}. \end{aligned}$$

This last term is of course the Hermitian inner product $\left\langle \frac{\partial}{\partial z_k}, \frac{\partial}{\partial z_j} \right\rangle\Big|_q$ with respect to the Bergman metric. Thus the expression

$$\det \left( \frac{\partial b_j}{\partial z_k} \right) \bigg|_q$$

is the determinant of the inner product matrix of a positive definite Hermitian inner product. Hence this determinant is positive.

The utility of the new coordinates in studying biholomorphic mappings comes from the following:

**Lemma 8.1.1** *Let $\Omega_1$ and $\Omega_2$ be two bounded domains in $\mathbb{C}^n$ with $q^1 \in \Omega_1$ and $q^2 \in \Omega_2$ fixed points. Denote by $b_1^1, \ldots, b_n^1$ the Bergman coordinates as defined near $q^1$ in $\Omega_1$ and $b_1^2, \ldots, b_n^2$ the Bergman coordinates defined in the same way near $q^2$ in $\Omega_2$. Suppose that there is a biholomorphic mapping $F : \Omega_1 \to \Omega_2$ with $F(q^1) = q^2$. Let $\alpha$ denote a variable point near $q^1$. Then the function defined near $0 \in \mathbb{C}^n$ by*

$$b^1 \text{ coordinate at } \alpha \longmapsto b^2 \text{ coordinate of the image of the } \alpha\text{-point under } F$$

*is a $\mathbb{C}$-linear transformation.*

In short, we say that the biholomorphic mapping $F$ is linear when expressed in the Bergman representative coordinates $b^j$. In point of fact, the linear mapping induced by the introduction of Bergman representative coordinates is nothing other than the complex Jacobian of the mapping $F$ at the point $q^1$.

**Proof of the Lemma** To avoid confusion, we write $(z_1, \ldots, z_n)$ and $(w_1, \ldots, w_n)$ for the $\mathbb{C}^n$-coordinates in $\Omega_1$ and $(Z_1, \ldots, Z_n)$ and $(W_1, \ldots, W_n)$ for the $\mathbb{C}^n$-coordinates in $\Omega_2$. Now observe that, for each $j = 1, \ldots, n$,

$$\frac{\partial}{\partial \overline{w}_j} \log \frac{K_{\Omega_2}(F(z), F(w))}{K_{\Omega_2}(F(w), F(w))} = \frac{\partial}{\partial \overline{w}_j} \log \frac{K_{\Omega_1}(z, w)}{K_{\Omega_1}(w, w)} .$$

The reason for this identity is

$$\begin{aligned} \frac{K_{\Omega_2}(F(z), F(w))}{K_{\Omega_2}(F(w), F(w))} = \frac{K_{\Omega_1}(z, w)}{K_{\Omega_1}(w, w)} &\times (\text{a holomorphic function of } z) \\ &\times (\text{a holomorphic function of } w) . \end{aligned}$$

This last follows from the transformation law. Thus we obtain (from the complex chain rule) that

$$\begin{aligned} b_j^1(z) &\stackrel{\text{def}}{=} \frac{\partial}{\partial \overline{w}_j} \log \frac{K_{\Omega_1}(z, w)}{K_{\Omega_1}(w, w)} \bigg|_{w=q_1} \\ &= \frac{\partial}{\partial \overline{w}_j} \Big[ \log K_{\Omega_2}(F(z), F(w)) - \log K_{\Omega_2}(F(w), F(w)) \Big] \bigg|_{w=q_1} \\ &= \sum_k \left[ \frac{\partial \overline{F}^k}{\partial \overline{w}_j} \cdot \frac{\partial}{\partial \overline{W}_k} \log \frac{K_{\Omega_2}(F(z), W)}{K_{\Omega_2}(W, W)} \right] \bigg|_{W=F(q_1)} , \end{aligned}$$

where $F^k$ is the $k^{\text{th}}$ coordinate of $F(w_1, \ldots, w_k)$. But this last expression is exactly

$$\sum_k \frac{\partial \overline{F}^k}{\partial \overline{w}_j}\bigg|_{w=q_1} \cdot b_k^2(F(z)) .$$

Hence

$$b_j^1(z) = \sum_k \frac{\partial \overline{F}^k}{\partial \overline{w}_j}\bigg|_{w=q_1} \cdot b_k^2(F(z)) .$$

Since the Jacobian matrix $(\partial F^k/\partial w_j)$ of $F$ is invertible at $q$, it follows that the $b_k^2(F)$ are linear functions of the $b_j^1$ coordinates. □

The reader may wish to try a hand at calculating Bergman representative coordinates on the unit ball. You may then use these to re-discover the Möbius transformations on the ball.

Note that the whole concept of representative coordinates extends essentially automatically to complex manifolds for which the Bergman metric construction for $(n, 0)$ forms yields a positive definite metric. See [KRA18, Ch. 1]. The construction can still be done locally, using general local holomorphic coordinates, and it remains true that the Bergman coordinates linearize holomorphic mappings.

## 8.2 The Berezin Transform

**Capsule:** In other contexts (see [KRA20] the Berezin transform has been called the Poisson–Bergman projection. It uses the same device that we exploited to study the Poisson–Szegő kernel to create a positive kernel from the Bergman kernel. Such an object has a priori interest and merits our detailed study.

### *8.2.1 Preliminary Remarks*

Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain (i.e., a connected open set) with $C^2$ boundary. Following the general rubric of "Hilbert space with reproducing kernel" laid down by Nachman Aronszajn [ARO], both the Bergman space $A^2(\Omega)$ and the Hardy space $H^2(\Omega)$ have reproducing kernels.

The Bergman kernel (for $A^2$) and the Szegő kernel (for $H^2$) both have the advantage of being canonical. But neither is positive, and this makes them tricky to handle. The Bergman kernel can be treated with the theory of the Hilbert integral (see [PHS]) and the Szegő kernel can often be handled with a suitable theory of singular integrals (see [KRA11, Ch. 9–10]).

It is a classical construction of Hua (see [HUA, Ch. 3]) that one can use the Szegő kernel to produce another reproducing kernel $\mathcal{P}(z, \zeta)$ which also reproduces $H^2$ but which is positive. In this sense it is more like the Poisson kernel of harmonic function theory. In point of fact, this so-called Poisson–Szegő kernel coincides with the Poisson kernel when the domain is the disc $D$ in the complex plane $\mathbb{C}$. Furthermore, the Poisson–Szegő kernel solves the Dirichlet problem for the invariant Laplacian (i.e., the Laplace–Beltrami operator for the Bergman metric) on the ball in $\mathbb{C}^n$. Unfortunately a similar statement about the Poisson–Szegő kernel cannot be made on any other domain. See [GRA1, GRA2] for the full story of these matters.

We want to develop these ideas with the Szegő kernel replaced by the Bergman kernel. This notion was developed independently by Berezin [BERE] in the context of quantization of Kähler manifolds. Indeed, one assigns to a bounded function on the manifold the corresponding Toeplitz operator. This process of assigning a linear operator to a function is called *quantization*. A nice exposition of the ideas appears in [PEE]. Further basic properties may be found in [ZHU, Ch. 2].

Approaches to the Berezin transform are often operator-theoretic (see [ENG1, ENG2]), or sometimes geometric [PEE]. Our point of view here will be more function-theoretic. We shall repeat (in perhaps new language) some results that are known in other contexts. And we shall also enunciate and prove new results. We hope that the mix serves to be both informative and useful.

### 8.2.2 *Introduction to the Poisson–Bergman Kernel*

In the seminal work [HUA, Ch. 3], L. Hua proposed a program for producing a positive kernel from a canonical kernel. He defined

$$\mathcal{P}(z, \zeta) = \frac{|S(z, \zeta)|^2}{S(z, z)},$$

where $S$ is the standard Szegő kernel on a given bounded domain $\Omega$. Now Hua did not consider his construction for the Bergman kernel, but in fact it is just as valid in that context. We may define

$$\mathcal{B}(z, \zeta) = \frac{|K(z, \zeta)|^2}{K(z, z)}.$$

We call this the *Poisson–Bergman kernel*. It is also sometimes called the *Berezin kernel*. Then we have

**Proposition 8.2.1** *Let $\Omega$ be a bounded domain and $K$ its Bergman kernel. With $\mathcal{B}(z,\zeta)$ as defined above, and with $f \in C(\overline{\Omega})$ holomorphic on $\Omega$, we have*

$$f(z) = \int_{\partial\Omega} \mathcal{B}(z,\zeta) f(\zeta)\, dV(\zeta)$$

*for all $z \in \Omega$.*

The proof is just the same as that for the Poisson–Szegő integral, and we omit the details. One of the purposes of the present discussion is to study properties of the Poisson–Bergman kernel $\mathcal{B}$.

Of course the Poisson–Bergman kernel is real, so it will also reproduce the real parts of holomorphic functions. Thus, in one complex variable, the integral reproduces harmonic functions. In several complex variables, it reproduces pluriharmonic functions.

Again, it is natural to ask under what circumstances Proposition 8.2.1 holds for all functions in the Bergman space $A^2(\Omega)$. The question is virtually equivalent to asking when the elements that are continuous on $\overline{\Omega}$ are dense in $A^2$. Catlin [CAT3] has given an affirmative answer to this query on any smoothly bounded pseudoconvex domain.

One of the features that makes the Bergman kernel both important and useful is its invariance under biholomorphic mappings. This fact is useful in conformal mapping theory, and it also gives rise to the Bergman metric. The fundamental result is this:

**Proposition 8.2.2** *Let $\Omega_1, \Omega_2$ be domains in $\mathbb{C}^n$. Let $f : \Omega_1 \to \Omega_2$ be biholomorphic. Then*

$$\det J_{\mathbb{C}} f(z) K_{\Omega_2}(f(z), f(\zeta)) \det \overline{J_{\mathbb{C}} f(\zeta)} = K_{\Omega_1}(z,\zeta).$$

Here $J_{\mathbb{C}} f$ is the complex Jacobian matrix of the mapping $f$. Refer to [KRA1, Ch. 1] for more on this topic.

It is useful to know that the Poisson–Bergman kernel satisfies a similar transformation law:

**Proposition 8.2.3** *Let $\Omega_1, \Omega_2$ be domains in $\mathbb{C}^n$. Let $f : \Omega_1 \to \Omega_2$ be biholomorphic. Then*

$$\mathcal{B}_{\Omega_2}(f(z), f(\zeta)) |\det J_{\mathbb{C}} f(\zeta)|^2 = \mathcal{B}_{\Omega_1}(z,\zeta)\,.$$

**Proof** Of course we use the result of Proposition 8.2.2. Now

$$\begin{aligned} \mathcal{B}_{\Omega_1}(z,\zeta) &= \frac{|K_{\Omega_1}(z,\zeta)|^2}{K_{\Omega_1}(z,z)} \\ &= \frac{|\det J_{\mathbb{C}} f(z) \cdot K_{\Omega_2}(f(z), f(\zeta)) \cdot \overline{\det J_{\mathbb{C}} f(\zeta)}|^2}{\det J_{\mathbb{C}} f(z) \cdot K_{\Omega_2}(f(z), f(z)) \cdot \overline{\det J_{\mathbb{C}} f(z)}} \end{aligned}$$

$$= \frac{|\det J_{\mathbb{C}} f(\zeta)|^2 \cdot |K_{\Omega_2}(f(z), f(\zeta))|^2}{K_{\Omega_2}(f(z), f(z))}$$
$$= |\det J_{\mathbb{C}} f(\zeta)|^2 \cdot \mathcal{B}_{\Omega_2}(f(z), f(\zeta)) .$$

□

We conclude this section with an interesting observation about the Berezin transform—see [ZHU, Ch. 2].

**Proposition 8.2.4** *The operator*

$$\mathcal{B}f(z) = \int_B \mathcal{B}(z, \zeta) f(\zeta)\, dV(\zeta) ,$$

*acting on $L^1(B)$, is univalent.*

**Proof** In fact it is useful to take advantage of the symmetry of the ball. We can rewrite the Poisson–Bergman integral as

$$\int_B f \circ \Phi_z(\zeta)\, dV(\zeta) ,$$

where $\Phi_z$ is a suitable automorphism of the ball. Then it is clear that this integral can be identically zero in $z$ only if $f \equiv 0$. That completes the proof. □

Another, slightly more abstract, way to look at this matter is as follows (we thank Richard Rochberg for this idea, and see also [ENG1]). Let $f$ be any $L^1$ function on $B$. For $w \in B$ define

$$g_w(\zeta) = \frac{1}{(1 - \overline{w} \cdot \zeta)^{n+1}} .$$

If $f$ is bounded on the ball, let

$$T_f : g \mapsto P_B(fg) .$$

We may write the Berezin transform now as

$$\Lambda f(w, z) = \frac{\langle T_f g_z, g_w \rangle}{\langle g_w, g_w \rangle} .$$

This function is holomorphic in $z$ and conjugate holomorphic in $w$. The statement that the Berezin transform $\mathcal{B}f(z) \equiv 0$ is the same as $\Lambda f(z, z) = 0$. But it is a standard fact (see [KRA1, Ch. 7]) that we may then conclude that $\Lambda f(w, z) \equiv 0$. Then $T_f g_z \equiv 0$ and so $f \equiv 0$. So the Berezin transform is univalent. □

### 8.2.3 Boundary Behavior

It is natural to want information about the boundary limits of potentials of the form $\mathcal{B}f$ for $f \in L^2(\Omega)$. We begin with a simple lemma:

**Lemma 8.2.5** *Let $\Omega$ be a bounded domain and $\mathcal{B}$ its Poisson–Bergman kernel. If $z \in \Omega$ is fixed, then*

$$\int_\Omega \mathcal{B}(z,\zeta)\, dV(\zeta) = 1\,.$$

**Proof** Certainly the function $f(\zeta) \equiv 1$ is an element of the Bergman space on $\Omega$. As a result,

$$1 = f(z) = \int_\Omega \mathcal{B}(z,\zeta) f(\zeta)\, dV(\zeta) = \int_\Omega \mathcal{B}(z,\zeta)\, dV(\zeta)$$

for any $z \in \Omega$. □

Our first result is as follows:

**Proposition 8.2.6** *Let $\Omega$ be the ball $B$ in $\mathbb{C}^n$. Then the mapping*

$$f \mapsto \int_\Omega \mathcal{B}(z,\zeta) f(\zeta)\, dV(\zeta)$$

*sends $L^p(\Omega)$ to $L^p(\Omega)$, $1 \le p \le \infty$.*

**Proof** We know from the lemma that

$$\|\mathcal{B}(z,\,\cdot\,)\|_{L^1(\Omega)} = 1$$

for each fixed $z$. An even easier estimate shows that

$$\|\mathcal{B}(\,\cdot\,,\zeta)\|_{L^1(\Omega)} \le 1$$

for each fixed $\zeta$. Now Schur's lemma, or the generalized Minkowski inequality, give the desired conclusion. [Note here that we made decisive use of the fact that $\mathcal{B}(z,\zeta) \ge 0$.] □

**Proposition 8.2.7** *Let $\Omega \subseteq \mathbb{C}^n$ be the unit ball $B$. Let $f \in C(\overline{\Omega})$. Let $F = \mathcal{B}f$. Then $F$ extends to a function that is continuous on $\overline{\Omega}$. Moreover, if $P \in \partial\Omega$, then*

$$\lim_{\Omega \ni z \to P} F(z) = f(P)\,.$$

**Proof** Let $\epsilon > 0$. Choose $\delta > 0$ such that if $z, w \in \overline{\Omega}$ and $|z - w| < \delta$ then $|f(z) - f(w)| < \epsilon$. Let $M = \sup_{\zeta \in \overline{\Omega}} |f(\zeta)|$. Now, for $z \in \Omega$, $P \in \partial\Omega$, and $|z - P| < \epsilon$, we have that

$$\begin{aligned}
|F(z) - f(P)| &= \left| \int_\Omega \mathcal{B}(z,\zeta) f(\zeta)\, dV(\zeta) - f(P) \right| \\
&= \left| \int_\Omega \mathcal{B}(z,\zeta) f(\zeta)\, dV(\zeta) - \int_\Omega \mathcal{B}(z,\zeta) f(P)\, dV(\zeta) \right| \\
&\leq \int_{\substack{\zeta \in \Omega \\ |\zeta - P| < \delta}} \mathcal{B}(z,\zeta) |f(\zeta) - f(P)|\, dV(\zeta) \\
&\qquad + \int_{\substack{\zeta \in \Omega \\ |\zeta - P| \geq \delta}} \mathcal{B}(z,\zeta) |f(\zeta) - f(P)|\, dV(\zeta) \\
&\leq \int_{\substack{\zeta \in \Omega \\ |\zeta - P| < \delta}} \mathcal{B}(z,\zeta) \cdot \epsilon\, dV(\zeta) + \int_{\substack{\zeta \in \Omega \\ |\zeta - P| \geq \delta}} \mathcal{B}(z,\zeta) \cdot 2M\, dV(\zeta) \\
&\equiv I + II\,.
\end{aligned}$$

Now the lemma tells us that $I = \epsilon$. Also we know that the Poisson–Bergman kernel for the ball is

$$\mathcal{B}(z,\zeta) = c_n \frac{(1 - |z|^2)^{n+1}}{|1 - z \cdot \overline{\zeta}|^{2n+2}}\,.$$

Thus, by inspection, $\mathcal{B}(z,\zeta) \to 0$ as $z \to P$ for $|\zeta - P| \geq \delta$. Thus $II$ is smaller than $\epsilon$ as soon as $z$ is close enough to $P$.

In summary, for $z$ sufficiently close to $P$, $|F(z) - f(P)| < 2\epsilon$. That is what we wished to prove. □

Arazy and Engliš [ARE] have in fact shown that the last result is true on any pseudoconvex domain for which each boundary point is a peak point (for the algebra $A(\Omega)$ of functions continuous on the closure and holomorphic inside). Thus the result is true in particular on strictly pseudoconvex domains (see [KRA1, Ch. 3]) and finite type domains in $\mathbb{C}^2$ (see [BEF1]).

Here is another way to look at the matter on strictly pseudoconvex domains. In fact our observation, at the end of the proof of the last proposition, about the vanishing of $\mathcal{B}(z,\zeta)$ for $z \to P$ and $|\zeta - P| \geq \delta$ is a tricky point and not generally known. On a strictly pseudoconvex domain $\Omega$ we have Fefferman's asymptotic expansion [FEF1, Part I]. This says that, in suitable local holomorphic coordinates near a boundary point $P$, we have

$$K_\Omega(z,\zeta) = \frac{c_n}{(1 - z \cdot \overline{\zeta})^{n+1}} + k(z,\zeta) \cdot \log|1 - z \cdot \overline{\zeta}| + \mathcal{E}(z,\zeta)\,. \tag{8.2.8}$$

Thus, using an argument similar to the one that we used earlier for the Poisson–Szegő kernel, one can obtain an asymptotic expansion for the Poisson–Bergman kernel. One sees that, in local coordinates near the boundary.

$$\mathcal{B}_\Omega(z,\zeta) = c_n \cdot \frac{(1-|z|^2)^{n+1}}{|1-z\cdot\overline{\zeta}|^{2n+2}} + \mathcal{E}(z,\zeta)\,,$$

where $\mathcal{E}$ is a kernel that induces a smoothing operator. In particular, the singularity of $\mathcal{E}$ will be measurably less than the singularity of the lead term. So it will still be the case that $\mathcal{B}(z,\zeta) \to 0$ as $z \to P \in \partial\Omega$ and $|\zeta - P| \geq \delta$. Thus we have:

**Proposition 8.2.9** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded, strictly pseudoconvex domain in $\mathbb{C}^n$. Let $f \in C(\overline{\Omega})$. Then the function $\mathcal{B}f$ extends to be continuous on $\overline{\Omega}$. Moreover, if $P \in \partial\Omega$, then*

$$\lim_{\Omega \ni z \to P} \mathcal{B}f(z) = f(P)\,.$$

It is natural, from the point of view of measure theory and harmonic analysis, to want to extend the result of Proposition 8.2.9 to a broader class of functions. To this end we introduce a maximal function to use as a tool.

**Definition 8.2.10** Let $\Omega$ be a smoothly bounded, strictly pseudoconvex domain in $\mathbb{C}^n$. If $z, \zeta \in \overline{\Omega}$ then we set

$$\rho(z,\zeta) = |1 - z\cdot\overline{\zeta}|^{1/2}\,.$$

**Proposition 8.2.11** *When $\Omega = B$, the unit ball, then the function $\rho$ is a metric on $\partial B$. For a more general smoothly bounded, strictly pseudoconvex domain, the function $\rho$ is a pseudometric. That is to say, there is constant $C \geq 1$ such that*

$$\rho(z,\zeta) \leq C\big(\rho(z,\xi) + \rho(\xi,\zeta)\big)\,.$$

**Proof** The first assertion is Proposition 5.3.1 in [KRA4]. The second assertion is proved in [KRA1, pp. 357–358]. We shall provide the details of this argument below. □

**Proposition 8.2.12** *The balls*

$$\beta_2(z,r) = \{\zeta \in \Omega : \rho(z,\zeta) < r\}\,,$$

*together with ordinary Euclidean volume measure $dV$, form a space of homogeneous type in the sense of Coifman and Weiss [COG1].*

**Proof** This is almost immediate from the preceding proposition, but details may be found in [KRA1, §8.6]. □

**Definition 8.2.13** For $z \in \Omega$ and $f \in L^1_{\text{loc}}(\Omega)$ we define

$$\mathcal{M}f(z) = \sup_{r>0} \frac{1}{V(\beta_2(z,r))} \int_{\beta_2(z,r)} |f(\zeta)|\, dV(\zeta)\,.$$

Here $dV$ is volume measure on $\Omega$.

**Theorem 8.2.14** *The operator $\mathcal{M}$ is of weak-type $(1,1)$ and of strong type $(p,p)$, $1 < p \leq \infty$.*

**Proof** Again this is a standard consequence of the previous proposition in the context of spaces of homogeneous type. See [COG1, Ch. 3]. □

**Theorem 8.2.15** *Let $\Omega$ be the unit ball $B$ in $\mathbb{C}^n$. Let $f$ be a locally integrable function on $\Omega$. Then there is a constant $C > 0$ such that, for $z \in \Omega$,*

$$|\mathcal{B}f(z)| \leq C \cdot \mathcal{M}f(z)\,.$$

**Proof** It is easy to see that $|1 - z \cdot \overline{\zeta}| \geq (1/2)(1 - |z|^2)$. Therefore we may perform these standard estimates:

$$\begin{aligned}
|\mathcal{B}f(z)| &= \left| \int_\Omega \mathcal{B}(z,\zeta) f(\zeta)\, dV(\zeta) \right| \\
&\leq \sum_{j=-1}^{\infty} \int_{2^j(1-|z|^2) \leq |1-z\cdot\overline{\zeta}| \leq 2^{j+1}(1-|z|^2)} \mathcal{B}(z,\zeta) |f(\zeta)|\, dV(\zeta) \\
&\leq \sum_{j=-1}^{\infty} \int_{|1-z\cdot\overline{\zeta}| \leq 2^{j+1}(1-|z|^2)} \frac{(1-|z|^2)^{n+1}}{[2^j(1-|z|^2)]^{2n+2}}\, dV(\zeta) \\
&\leq C \cdot \sum_{j=-1}^{\infty} 2^{-j(n+1)} \cdot \left[ \frac{1}{(1-|z|^2)^{n+1} 2^{(j+1)(n+1)}} \right] \int_{|1-z\cdot\overline{\zeta}| \leq 2^{j+1}(1-|z|^2)} |f(\zeta)| dV(\zeta) \\
&\leq C \cdot \sum_{j=-1}^{\infty} 2^{-j(n+1)} \cdot \left[ \frac{1}{V(\beta_2(z, \sqrt{2^{j+1}(1-|z|^2)}))} \right] \int_{\beta_2(z, \sqrt{2^{j+1}(1-|z|^2)})} |f(\zeta)| dV(\zeta)
\end{aligned} \tag{8.2.16.1}$$

The last line is majorized by

$$\begin{aligned}
&\leq C' \cdot \sum_{j=-1}^{\infty} 2^{-j(n+1)} \mathcal{M}f(z) \\
&\leq C \cdot \mathcal{M}f(z)\,.
\end{aligned}$$

□

**Theorem 8.2.16** *Let $\Omega$ be the unit ball $B$ in $\mathbb{C}^n$. Let $f$ be an $L^p(\Omega, dV)$ function, $1 \leq p \leq \infty$. Then $\mathcal{B}f$ has radial boundary limits almost everywhere on $\partial\Omega$.*

**Proof** The proof follows standard lines, using Theorems 8.2.14 and 8.2.15. See the detailed argument in [KRA1, Theorem 8.6.11]. □

In fact a slight emendation of the arguments just presented allows a more refined result.

**Definition 8.2.17** Let $P \in \partial B$ and $\alpha > 1$. Define the *admissible approach region of aperture* $\alpha$ by

$$\mathcal{A}_\alpha(P) = \{z \in B : |1 - z \cdot \overline{\zeta}| < \alpha(1 - |z|^2)\}.$$

Admissible approach regions are a new type of region for Fatou-type theorems. These were first introduced in [KOR1, KOR2] and generalized and developed in [STE1, Ch. 3] and later in [KRA7]. See also our Chapter 2. Now we have

**Theorem 8.2.18** *Let $f$ be an $L^p(B)$ function, $1 \leq p \leq \infty$. Then, for almost every $P \in \partial B$,*

$$\lim_{\mathcal{A}_\alpha(P) \ni z \to P} \mathcal{B}f(z)$$

*exists.*

In fact, using the Fefferman asymptotic expansion (as discussed in detail in the next section), we may imitate the development of Theorems 8.2.14 and 8.2.15 and prove a result analogous to Theorem 8.2.16 on any smoothly bounded, strictly pseudoconvex domain. We omit the details, as they would repeat ideas that we present elsewhere in the present book for slightly different purposes.

## 8.3 Ideas of Fefferman

**Capsule:** Fefferman's 1974 paper [FEF1] has been a great influence over complex geometric analysis for the ensuing forty years. First, the statement of his result was quite dramatic. Second, his techniques were powerful and flexible and could be used in many different contexts. In particular, Fefferman brought new focus onto the Bergman kernel. Since the early 1970s, the Bergman theory has assumed a quite central role in several complex variables.

In the seminal paper [FEF1, Part I], Charles Fefferman produced an asymptotic expansion for the Bergman kernel of a strictly pseudoconvex domain. He used this expansion to get detailed information about the boundary behavior of certain"pseudotransversal" geodesics in the metric; this data in turn was used to show that biholomorphic mappings of strictly pseudoconvex domains continue smoothly to the boundaries.

Fefferman's ideas have been quite influential. For instance, Paul Klembeck [KLE] used the Fefferman expansion to calculate the boundary asymptotics of Bergman metric curvature on a strictly pseudoconvex domain.. This in turn led him to a new and very natural proof of the Bun Wong–Rosay Theorem:

**Theorem 8.3.1** *Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain with $C^2$ boundary. Let $P \in \partial\Omega$ be a point of strong pseudoconvexity. Suppose that $\varphi_j$ are biholomorphic self-maps of $\Omega$ with the property that there is a point $X \in \Omega$ such that $\lim_{j\to\infty} \varphi_j(X) = P$. Then $\Omega$ is biholomorphic to the unit ball $B_n$ in $\mathbb{C}^n$.*

Greene and Krantz ([GRK1, GRK2, GRK3]) showed that the Fefferman asymptotic expansion deforms stably under smooth deformation of the boundary of a strictly pseudoconvex domain $\Omega \subseteq \mathbb{C}^n$. They used that information to prove a variety of results about Bergman geometry, and also about automorphism groups of domains. We state just two of their results here:

**Theorem 8.3.2** *Let $\Omega_0 \subseteq \mathbb{C}^n$ be a fixed strictly pseudoconvex domain with smooth boundary. If $\Omega$ is another smoothly bounded strictly pseudoconvex domain with boundary sufficiently close to $\partial\Omega$ in the smooth domain topology, then the automorphism group (group of biholomorphic self-mappings of $\Omega$) of $\Omega$ is a subgroup of the automorphism group of $\Omega_0$. Indeed, there is a smooth mapping $\Phi : \Omega \to \Omega_0$ such that*

$$Aut(\Omega) \ni \varphi \longmapsto \Phi \circ \varphi \circ \Phi^{-1} \in Aut(\Omega_0)$$

*is an injective homomorphism of the automorphism group of $\Omega$ into the automorphism group of $\Omega_0$.*

**Theorem 8.3.3** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded domain which is sufficiently close to the unit ball $B_n \subseteq \mathbb{C}^n$ in the smooth domain topology and is biholomorphically inequivalent to the ball (such domains are generic—see [GRK1, GRK2]). Then there is a holomorphic embedding $\Psi : \Omega \to \mathbb{C}^n$ so that the automorphism group of $\Omega' \equiv \Psi(\Omega)$ is the restriction to $\Omega'$ of a subgroup of the unitary group on $n$ letters.*

We cannot provide all the details of Fefferman's construction here. It is a long and tedious argument. But we can discuss and describe the asymptotic expansion and indicate some of its uses.

So fix a strictly pseudoconvex domain $\Omega$ with smooth boundary and fix a point $P \in \partial\Omega$. For $z$ and $\zeta$ in $\Omega$ and sufficiently near $P$, Fefferman tells us that (in suitable local coordinates)

$$K_\Omega(z,\zeta) = \frac{c_n}{(1 - z \cdot \overline{\zeta})^{n+1}} + k(z,\zeta) \cdot \log|1 - z \cdot \overline{\zeta}| + \mathcal{E}(z,\zeta)\,. \tag{8.3.4}$$

Here $\mathcal{E}$ is an error term that is smaller, in a measurable sense, than the lead terms.

One can see from formula (8.3.4) that calculations with the Bergman kernel of a strictly pseudoconvex domain are tantamount to calculations with the Bergman kernel for the ball (up to a calculable and estimable error).[1]

Thus one can prove, in the vein of Klembeck, that the curvature tensor of the Bergman metric is asymptotically, as the base point $p$ approaches the boundary of the domain, equal to the curvature tensor for the Bergman metric of the ball. Because the automorphism group of the ball acts transitively on the unit sphere bundle in the tangent bundle to the ball, it follows that the latter curvature tensor must be constant.

## 8.4 The Invariant Laplacian

**Capsule:** Of course every Riemannian metric has associated to it a metrically invariant second-order partial differential operator called the *Laplace-Beltrami operator*. In fact the operator is simply defined to be the divergence of the gradient. For the standard Euclidean metric in $\mathbb{R}^N$ this is the usual Laplacian. This operator is sometimes also called the *invariant Laplacian*.

Given this setup, it is natural to ask what is the invariant Laplacian for the Bergman metric on the unit ball (we confine attention for the moment to the ball just because the ball has a very large automorphism group, and it is the *only* smoothly bounded domain in space with such a large automorphism group). The result of this calculation has a priori interest, and the differential operator has some remarkable properties.

If $g = (g_{jk})$ is a Riemannian metric on a domain $\Omega$ in complex Euclidean space, then there is a second-order partial differential operator, known as the *Laplace–Beltrami operator*, that is invariant under isometries of the metric. In fact, if $g$ denotes the determinant of the metric matrix $g$, and if $(g^{jk})$ denotes the inverse matrix, then this partial differential operator is defined to be

$$\mathcal{L} = \frac{2}{g} \sum_{j,k} \left\{ \frac{\partial}{\partial \overline{z}_j} \left( g g^{jk} \frac{\partial}{\partial z_k} \right) + \frac{\partial}{\partial z_k} \left( g g^{jk} \frac{\partial}{\partial \overline{z}_k} \right) \right\} .$$

[1] The logarithmic term was one of the big surprises of Fefferman's work. It was quite unexpected. And it does not conform to the paradigm that "the Bergman kernel of a strictly pseudoconvex domain is just like that for the ball." But the logarithmic term has only a weak singularity, and is easily estimated. Fefferman provided, in his paper [FEF1, Part I], a concrete example of a domain in which the logarithmic term actually arises. And Burns later proved that the logarithmic term is generic. That is to say, if $\Omega$ is a strictly pseudoconvex domain with smooth boundary and none of the Fefferman asymptotic expansions near any of the boundary points have logarithmic terms, then $\Omega$ is biholomorphic to the unit ball $B$. Burns never published this result. L. Boutet de Monvel [BOUT] in dimension 2 and C. Robin Graham [GRA3] in general gave rigorous proofs of the result. See also the work of Hirachi [HIR1]. There are unbounded domains and also roughly bounded domains on which the analogue of this result for the Szegő kernel is known to fail—see [HIR2].

Now of course we are interested in artifacts of the Bergman theory. If $\Omega \subseteq \mathbb{C}^n$ is a bounded domain and $K = K_\Omega$ its Bergman kernel, then it is well known (see [KRA1, Ch. 1]) that $K(z,z) > 0$ for all $z \in \Omega$. Then it makes sense to define

$$g_{jk}(z) = \frac{\partial^2}{\partial z_j \partial \overline{z}_k} \log K(z,z)$$

for $j,k = 1,\ldots,n$. Then the standard invariance of the Bergman kernel can be used to show that this metric—which is in fact a Kähler metric on $\Omega$—is invariant under biholomorphic mappings of $\Omega$. In other words, any biholomorphic $\Phi : \Omega \to \Omega$ is an isometry in the metric $g$. This is the celebrated *Bergman metric*.

If $\Omega \subseteq \mathbb{C}^n$ is the unit ball $B$, then the Bergman kernel is given by

$$K_B(z,\zeta) = \frac{1}{V(B)} \cdot \frac{1}{(1 - z \cdot \overline{\zeta})^{n+1}},$$

where $V(B)$ denotes the Euclidean volume of the domain $B$. Then

$$\log K(z,z) = -\log V(B) - (n+1)\log(1 - |z|^2).$$

Further,

$$\frac{\partial}{\partial z_j}\big(-(n+1)\log(1-|z|^2)\big) = (n+1)\frac{\overline{z}_j}{1-|z|^2}$$

and

$$\begin{aligned}\frac{\partial^2}{\partial z_j \partial \overline{z}_k}\big(-(n+1)\log(1-|z|^2\big) &= (n+1)\left[\frac{\delta_{jk}}{1-|z|^2} + \frac{\overline{z}_j z_k}{(1-|z|^2)^2}\right] \\ &= \frac{(n+1)}{(1-|z|^2)^2}\big[\delta_{jk}(1-|z|^2) + \overline{z}_j z_k\big] \\ &\equiv g_{jk}(z).\end{aligned}$$

When $n = 2$ we have

$$g_{jk}(z) = \frac{3}{(1-|z|^2)^2}\big[\delta_{jk}(1-|z|^2) + \overline{z}_j z_k\big].$$

Thus

$$\big(g_{jk}(z)\big) = \frac{3}{(1-|z|^2)^2}\begin{pmatrix} 1-|z_2|^2 & \overline{z}_1 z_2 \\ \overline{z}_2 z_1 & 1-|z_1|^2 \end{pmatrix}.$$

Let

$$\left(g^{jk}(z)\right)_{j,k=1}^{2}$$

represents the inverse of the matrix

$$\left(g_{jk}(z)\right)_{j,k=1}^{2}.$$

Then an elementary computation shows that

$$\left(g^{jk}(z)\right)_{j,k=1}^{2} = \frac{1-|z|^2}{3}\begin{pmatrix} 1-|z_1|^2 & -z_2\overline{z}_1 \\ -z_1\overline{z}_2 & 1-|z_2|^2 \end{pmatrix} = \frac{1-|z|^2}{3}\left(\delta_{jk} - \overline{z}_j z_k\right)_{j,k}.$$

Let

$$g \equiv \det\left(g_{jk}(z)\right).$$

Then

$$g = \frac{9}{(1-|z|^2)^3}.$$

Now let us calculate. If $\left(g_{jk}\right)_{j,k=1}^{2}$ is the Bergman metric on the ball in $\mathbb{C}^2$ then we have

$$\sum_{j,k} \frac{\partial}{\partial \overline{z}_j}\left(g g^{jk}\right) = 0$$

and

$$\sum_{j,k} \frac{\partial}{\partial z_j}\left(g g^{jk}\right) = 0.$$

We verify these assertions in detail in dimension 2: Now

$$\begin{aligned} g g^{jk} &= \frac{9}{(1-|z|^2)^3} \cdot \frac{1-|z|^2}{3}(\delta_{jk} - \overline{z}_j z_k) \\ &= \frac{3}{(1-|z|^2)^2}(\delta_{jk} - \overline{z}_j z_k). \end{aligned}$$

It follows that

$$\frac{\partial}{\partial \overline{z}_j}\left[g g^{jk}\right] = \frac{6z_j}{(1-|z|^2)^3}\left(\delta_{jk} - \overline{z}_j z_k\right) - \frac{3z_k}{(1-|z|^2)^2}.$$

Therefore

$$\begin{aligned}\sum_{j,k=1}^{2} \frac{\partial}{\partial \overline{z}_j}\left[g g^{jk}\right] &= \sum_{j,k=1}^{2}\left[\frac{6z_j(\delta_{jk}-\overline{z}_j z_k)}{(1-|z|^2)^3} - \frac{3z_j}{(1-|z|^2)^2}\right] \\ &= 6\sum_k \frac{z_k}{(1-|z|^2)^3} - 6\sum_{j,k}\frac{|z_j|^2 z_k}{(1-|z|^2)^3} - 6\sum_k \frac{z_k}{(1-|z|^2)^2} \\ &= 6\sum_j \frac{z_j}{(1-|z|^2)^2} - 6\sum_k \frac{z_k}{(1-|z|^2)^2} \\ &= 0.\end{aligned}$$

The other derivative is calculated similarly.

Our calculations show that, on the ball in $\mathbb{C}^2$,

$$\begin{aligned}\mathcal{L} &\equiv \frac{2}{g}\sum_{j,k}\left\{\frac{\partial}{\partial \overline{z}_j}\left(g g^{jk}\frac{\partial}{\partial z_k}\right) + \frac{\partial}{\partial z_k}\left(g g^{jk}\frac{\partial}{\partial \overline{z}_j}\right)\right\} \\ &= 4\sum_{j,k} g^{jk}\frac{\partial}{\partial \overline{z}_j}\frac{\partial}{\partial z_k} \\ &= 4\sum_{j,k}\frac{1-|z|^2}{3}\left(\delta_{jk}-\overline{z}_j z_k\right)\frac{\partial^2}{\partial z_k \partial \overline{z}_j}.\end{aligned}$$

This is the *Laplace-Beltrami operator* for the Bergman metric on the ball, sometimes known as the invariant Laplacian. It is distinct from the more classical Laplacian given by

$$\triangle = \sum_{j=1}^{n}\frac{\partial^2}{\partial z_j \partial \overline{z}_j}\,.$$

Now the interesting fact for us is encapsulated in the following proposition:

**Proposition 8.4.1** *The Poisson–Szegő kernel on the ball $B$ solves the Dirichlet problem for the invariant Laplacian $\mathcal{L}$. That is to say, if $f$ is a continuous function on $\partial B$ then the function*

$$u(z) = \begin{cases} \int_{\partial B} \mathcal{P}(z,\zeta)\cdot f(\zeta)\, d\sigma(\zeta) & \text{if} \quad z \in B \\ f(z) & \text{if } z \in \partial B \end{cases}$$

*is continuous on $\overline{B}$ and is annihilated by $\mathcal{L}$ on $B$.*

This fact is of more than passing interest. In one complex variable, the study of holomorphic functions on the disc and the study of harmonic functions on the disc are inextricably linked because the real part of a holomorphic function is harmonic and conversely. Such is not the case in several complex variables. Certainly the real part of a holomorphic function is harmonic. But in fact it is more: such a function is *pluriharmonic*. For the converse direction, any real-valued pluriharmonic function is locally the real part of a holomorphic function. This assertion is false if "pluriharmonic" is replaced by "harmonic."

And the result of Proposition 8.4.1 should not really be considered to be surprising. For the invariant Laplacian is invariant under isometries of the Bergman metric, hence invariant under automorphisms of the ball. And the Poisson–Szegő kernels behaves nicely under automorphisms. E. M. Stein was able to take advantage of these invariance properties to give a proof of Proposition 8.4.1 using Godement's Theorem—that any function that satisfies a suitable mean-value property must be harmonic (i.e., annihilated by the relevant Laplace operator). See [STE1, Ch. 3] for the details. Here we give a more elementary proof of this fact.

**Sketch of the Proof of Proposition 8.4.1** Now

$$\mathcal{L}u = \mathcal{L}\int_{\partial B} \mathcal{P}(z,\zeta)\cdot f(\zeta)\,d\sigma(\zeta) = \int_{\partial B}\left[\mathcal{L}_z\mathcal{P}(z,\zeta)\right]\cdot f(\zeta)\,d\sigma(\zeta)\,.$$

Thus it behooves us to calculate $\mathcal{L}_z\mathcal{P}(z,\zeta)$. Now we shall calculate this quantity for each fixed $\zeta$. Thus, without loss of generality, we may compose with a unitary rotation and suppose that $\zeta = (1+i0, 0+i0)$ so that (in complex dimension 2)

$$\mathcal{P} = c_2 \cdot \frac{(1-|z|^2)^2}{|1-z_1|^4}\,.$$

This will make our calculations considerably easier.

By brute force, we find that

$$\begin{aligned}
\frac{\partial \mathcal{P}}{\partial \overline{z}_1} &= -2(1-z_1)(1-|z|^2)\cdot\left[\frac{-1+z_1+|z_2|^2}{|1-z_1|^6}\right] \\
\frac{\partial^2 \mathcal{P}}{\partial \overline{z}_1 \partial z_1} &= \frac{-2}{|1-z_1|^6}\cdot\left[-|z_1|^2 - |z_1|^2|z_2|^2 + 3|z_2|^2 - z_1|z_2|^2\right. \\
&\qquad \left. -2|z_2|^4 - 1 + z_1 + \overline{z}_1 - \overline{z}_1|z_2|^2\right] \\
\frac{\partial^2 \mathcal{P}}{\partial \overline{z}_1 \partial z_2} &= \frac{-2(1-z_1)}{|1-z_1|^6}\cdot\left[2\overline{z}_2 - \overline{z}_2 z_1 - 2\overline{z}_2|z_2|^2 - \overline{z}_2|z_1|^2\right] \\
\frac{\partial^2 \mathcal{P}}{\partial z_1 \partial \overline{z}_2} &= \frac{-2(1-\overline{z}_1)}{|1-z_1|^6}\cdot\left[2z_2 - z_2\overline{z}_1 - 2z_2|z_2|^2 - z_2|z_1|^2\right]
\end{aligned}$$

$$\frac{\partial \mathcal{P}}{\partial z_2} = \frac{-2z_2 + 2|z_1|^2 z_2 + 2|z_2|^2 z_2}{|1 - z_1|^4}$$
$$\frac{\partial^2 \mathcal{P}}{\partial z_2 \partial \overline{z}_2} = \frac{-2 + 2|z_1|^2 + 4|z_2|^2}{|1 - z_1|^4} \tag{8.4.1.1}$$

Now we know that, in complex dimension two,

$$\mathcal{L}_z \mathcal{P}(z,\zeta) = \frac{4}{3}(1-|z|^2)\cdot(1-|z_1|^2)\cdot\frac{\partial^2 \mathcal{P}_z}{\partial z_1 \partial \overline{z}_1} + \frac{4}{3}(1-|z|^2)\cdot(-\overline{z}_1 z_2)\cdot\frac{\partial^2 \mathcal{P}_z}{\partial z_2 \partial \overline{z}_1}$$
$$+\frac{4}{3}(1-|z|^2)\cdot(-\overline{z}_2 z_1)\cdot\frac{\partial^2 \mathcal{P}_z}{\partial z_1 \partial \overline{z}_2} + \frac{4}{3}(1-|z|^2)\cdot(1-|z_2|^2)\cdot\frac{\partial^2 \mathcal{P}_z}{\partial z_2 \partial \overline{z}_2}.$$

Plugging the values from (8.4.1.1) into this last equation gives

$$\mathcal{L}_z \mathcal{P}(z,\zeta) = \frac{4}{3}(1-|z|^2)\cdot(1-|z_1|^2)\cdot\frac{-2}{|1-z_1|^6}\cdot\Big[-|z_1|^2 - |z_1|^2|z_2|^2$$
$$+3|z_2|^2 - z_1|z_2|^2 - 2|z_2|^4 - 1 + z_1 + \overline{z}_1 - \overline{z}_1|z_2|^2\Big]$$
$$+\frac{4}{3}(1-|z|^2)\cdot(-\overline{z}_1 z_2)$$
$$\times\frac{-2(1-z_1)}{|1-z_1|^6}\cdot\Big[2\overline{z}_2 - \overline{z}_2 z_1 - 2\overline{z}_2|z_2|^2 - \overline{z}_2|z_1|^2\Big]$$
$$+\frac{4}{3}(1-|z|^2)\cdot(-\overline{z}_2 z_1)$$
$$\times\frac{-2(1-\overline{z}_1)}{|1-z_1|^6}\cdot\Big[2z_2 - z_2\overline{z}_1 - 2z_2|z_2|^2 - z_2|z_1|^2\Big]$$
$$+\frac{4}{3}(1-|z|^2)\cdot(1-|z_2|^2)\cdot|1-z_1|^2\cdot\frac{-2+2|z_1|^2+4|z_2|^2}{|1-z_1|^6}.$$

Multiplying out the terms, we find that

$$\mathcal{L}_z \mathcal{P}(z,\zeta) = \frac{-2}{|1-z_1|^6}\cdot\Big[-|z_1|^2 - 4|z_1|^2|z_2|^2 + 3|z_2|^2 - z_1|z_2|^2 - 2|z_2|^4 - 1$$
$$+z_1 + \overline{z}_1 - \overline{z}_1|z_2|^2 + |z_1|^4 + |z_1|^4|z_2|^2 + z_1|z_1|^2|z_2|^2$$
$$+2|z_1|^2|z_2|^4 + |z_1|^2 - z_1|z_1|^2 - \overline{z}_1|z_1|^2 + \overline{z}_1|z_1|^2|z_2|^2\Big]$$
$$-\frac{2}{|1-z_1|^6}\cdot\Big[-2\overline{z}_1|z_2|^2 + 3|z_1|^2|z_2|^2 + 2|z_2|^4\overline{z}_1 + \overline{z}_1|z_2|^2|z_1|^2$$

$$
\begin{aligned}
&-z_1|z_1|^2|z_2|^2 - 2|z_1|^2|z_2|^4 - |z_2|^2|z_1|^4\Big] \\
&-\frac{2}{|1-z_1|^6}\cdot\Big[-2z_1|z_2|^2 + 3|z_1|^2|z_2|^2 + 2|z_2|^4 z_1 + z_1|z_2|^2|z_1|^2 \\
&-\overline{z}_1|z_1|^2|z_2|^2 - 2|z_1|^2|z_2|^4 - |z_2|^2|z_1|^4\Big] \\
-\frac{2}{|1-z_1|^6}\cdot\Big[&1 - |z_1|^2 - 3|z_2|^2 + |z_1|^2|z_2|^2 + 2|z_2|^4 - z_1 + z_1|z_1|^2 \\
&+3z_1|z_2|^2 - z_1|z_1|^2|z_2|^2 - 2z_1|z_2|^4 - \overline{z}_1 + \overline{z}_1|z_1|^2 + 3\overline{z}_1|z_2|^2 - \overline{z}_1|z_1|^2|z_2|^2 \\
&-2\overline{z}_1|z_2|^4 + |z_1|^2 - |z_1|^4 - 3|z_1|^2|z_2|^2 + |z_1|^4|z_2|^2 + 2|z_1|^2|z_2|^4\Big] .
\end{aligned}
$$

And now if we combine all the terms in brackets a small miracle happens: everything cancels. The result is

$$\mathcal{L}_z \mathcal{P}(z,\zeta) \equiv 0 .$$

□

Thus, in some respects, it is inappropriate to study holomorphic functions on the ball in $\mathbb{C}^n$ using the Poisson kernel. The classical Poisson integral does *not* create pluriharmonic functions, and it does not create functions that are annihilated by the invariant Laplacian. In view of Proposition 8.4.1, the Poisson–Szegő kernel is much more apposite. As an instance, Adam Korányi [KOR1, KOR2] made decisive use of this observation in his study (proving boundary limits of $H^2$ functions through admissible approach regions $\mathcal{A}_\alpha$) of the boundary behavior of $H^2(B)$ functions.

It is known that the property described in Proposition 8.4.1 is special to the ball—it is simply untrue on any other domain (see [GRA1, GRA2] for more detail on this matter). Now one of the points that we want to make in this section is that the result of the proposition can nevertheless be extended—in an approximate sense—to a broader class of domains.

**Proposition 8.4.2** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded, strictly pseudoconvex domain and $\mathcal{P}$ its Poisson–Szegő kernel. Then, if $f \in C(\partial\Omega)$, we may write*

$$\mathcal{P}f(z) = \mathcal{P}_1 f(z) + \mathcal{E}f(z) ,$$

*where*

**(i)** *The term $\mathcal{P}_1 f$ is "approximately annihilated" by the invariant Laplacian on $\Omega$;*
**(ii)** *The operator $\mathcal{E}$ is smoothing in the sense of pseudodifferential operators.*

We shall explain the meaning of **(i)** and **(ii)** in the course of the proofs of these statements.

**Proof of Proposition 8.4.2** We utilize of course the asymptotic expansion for the Szegő kernel on a smoothly bounded, strictly pseudoconvex domain (see [FEF1, Part I], [BOUTS]). It says that, for $z, \zeta$ near a boundary point $P$, we have (in suitable biholomorphic local coordinates)

$$S_\Omega(z,\zeta) = \frac{c_n}{(1 - z\cdot\overline{\zeta})^n} + h(z,\zeta)\cdot\log|1 - z\cdot\overline{\zeta}| + \mathcal{E}(z,\zeta)\,. \tag{8.4.2.1}$$

Here $h$ is a smooth function on $\overline{\Omega}\times\overline{\Omega}$.

Now we calculate $\mathcal{P}_\Omega(z,\zeta)$ in the usual fashion:

$$\mathcal{P}_\Omega(z,\zeta) = \frac{|S(z,\zeta)|^2}{S(z,z)} = \frac{\left|\dfrac{c_n}{(1 - z\cdot\overline{\zeta})^n} + h(z,\zeta)\cdot\log|1 - z\cdot\overline{\zeta}|\right|^2}{\dfrac{c_n}{(1-|z|^2)^n} + h(z,z)\cdot\log(1-|z|^2)} + \mathcal{F}(z,\zeta)\,. \tag{8.4.2.2}$$

One can use just elementary algebra to simplify this last expression and obtain that, in suitable local coordinates near the boundary,

$$\begin{aligned}\mathcal{P}_\Omega(z,\zeta) &= c_n\cdot\frac{(1-|z|^2)^n}{|1 - z\cdot\overline{\zeta}|^{2n}} \\ &\quad + \frac{2(1-|z|^2)^n}{|1 - z\cdot\overline{\zeta}|^n}\log|1 - z\cdot\overline{\zeta}| + \mathcal{O}\left[(1-|z|^2)^n\cdot\log|1 - z\cdot\overline{\zeta}|\right] \\ &\equiv c_n\cdot\frac{(1-|z|^2)^n}{|1 - z\cdot\overline{\zeta}|^{2n}} \;+\; \mathcal{G}(z,\zeta)\,.\end{aligned} \tag{8.4.2.3}$$

Now the first expression on the righthand side of (8.4.2.3) is (in the local coordinates in which we are working) the usual Poisson–Szegő kernel for the unit ball in $\mathbb{C}^n$. The second is an error term which we now analyze.

In fact we claim that the error term is integrable in $\zeta$, uniformly in $z$, and the same can be said for the gradient (in the $z$ variable) of the error term. The first of these statements is obvious, as both parts of the error term are clearly majorized by the Poisson–Szegő kernel itself. As for the second part, we note that the gradient of the error gives rise to three types of terms

$$\begin{aligned}\nabla\mathcal{E} &\approx \frac{(1-|z|^2)^{n-1}}{|1 - z\cdot\overline{\zeta}|^n}\cdot\log|1 - z\cdot\overline{\zeta}| \\ &\quad + \frac{(1-|z|^2)^n}{|1 - z\cdot\overline{\zeta}|^{n+1}}\cdot\log|1 - z\cdot\overline{\zeta}| \\ &\quad + \frac{(1-|z|^2)^n}{|1 - z\cdot\overline{\zeta}|^{n+1}} \\ &\equiv I + II + III\,.\end{aligned} \tag{8.4.2.4}$$

Now it is clear by inspection that *I* and *II* are majorized by the ordinary Poisson–Szegő kernel, so they are both integrable in $\zeta$ as claimed. As for *III*, we must calculate:

$$\begin{aligned}
&\int_{\zeta\in\partial\Omega} \frac{(1-|z|^2)^{n-1}}{|1-z\cdot\overline{\zeta}|^{n+1}}\, d\sigma(\zeta) \\
&\leq \sum_{j=-1}^{\infty} \int_{2^j(1-|z|^2)\leq|1-z\cdot\overline{\zeta}|\leq 2^{j+1}(1-|z|^2)} \frac{(1-|z|^2)^{n-1}}{[2^j(1-|z|^2)]^{n+1}}\, d\sigma(\zeta) \\
&\leq \sum_{j=-1}^{\infty} \frac{1}{(1-|z|^2)^2} \int_{|1-z\cdot\overline{\zeta}|\leq 2^{j+1}(1-|z|^2)} 2^{-j(n+1)}\, d\sigma(\zeta) \\
&\leq \sum_{j=-1}^{\infty} C\cdot \frac{2^{-j(n+1)}}{(1-|z|^2)^2} \cdot \left[\sqrt{2^{j+1}(1-|z|^2)}\right]^{2n-2} \\
&\qquad \times\left[2^{j+1}\cdot(1-|z|^2)\right] \\
&\leq \sum_{j=-1}^{\infty} \frac{1}{(1-|z|^2)^2}\cdot(1-|z|^2)^{n-1}\cdot(1-|z|^2) \\
&\qquad \times 2^{-j(n+1)}\cdot 2^{(j+1)(n-1)}\cdot 2^{j+1} \\
&\leq C\cdot 2^n(1-|z|^2)^{n-2}\cdot\sum_{j=-1}^{\infty} 2^{-j} \\
&< \infty\,.
\end{aligned}$$

Thus we see that the Poisson–Szegő kernel for our strictly pseudoconvex domain $\Omega$ can be expressed, in suitable local coordinates, as the Poisson–Szegő kernel for the ball plus an error term whose gradient induces a bounded operator on $L^p$. This means that the error term itself maps $L^p$ to a Sobolev space (see Exercise 5, Chapter 1). In other words, it is a smoothing operator (hence negligible from our point of view).

Before completing the proof of Proposition 8.4.2, we digress a bit. In fact there are several fairly well known results about the interaction of the Poisson–Bergman kernel and the invariant Laplacian. We summarize some of the basic ones here.

**Proposition 8.4.3** *Let $f$ be a $C^2$ function on the unit ball that is annihilated by the invariant Laplacian $\mathcal{L}$. Then, for any $0 < r < 1$ and $\Sigma$ the unit sphere,*

$$\int_{\Sigma} f(r\zeta)\, d\sigma(\zeta) = c(r)\cdot f(0)\,.$$

*Here $d\sigma$ is rotationally invariant measure on the sphere $S$.*

**Proof** Replacing $f$ with the average of $f$ over the orthogonal group, this just becomes a calculation to determine the exact value of the constant $c(r)$—see [RUW3, p. 51]. □

**Proposition 8.4.4** *Suppose that $f$ is a $C^2$ function on the unit ball $B$ that is annihilated by the invariant Laplacian $\mathcal{L}$. Then $f$ satisfies the identity $\mathcal{B}f = f$. In other words, for any $z \in B$,*

$$f(z) = \int_B \mathcal{B}(z, \zeta) f(\zeta)\, dV(\zeta)\,.$$

*[Here $\mathcal{B}$ is the Poisson–Bergman kernel that we studied earlier.]*

**Proof** We have checked the result when $z = 0$ in the last proposition. For a general $z$, compose with a Möbius transformation and use the biholomorphic invariance of the kernel and the differential operator $\mathcal{L}$. □

**Remark 8.4.5** It is a curious fact (see [AFR]) that the converse of this last proposition is only true (as stated) in complex dimensions $1, 2, \ldots, 11$. It is false in dimensions 12 and higher. □

Finally we need to address the question of whether the invariant Laplacian *for the domain* $\Omega$ annihilates the principal term of the righthand side of the formula (8.4.2.3). The point is this. The biholomorphic change of variable that makes (8.4.2.3) valid is *local*. It is valid on a small, smoothly bounded subdomain $\Omega' \subseteq \Omega$ which shares a piece of boundary with $\partial\Omega$. According to Fefferman [FEF1, Part I] (see also the work in [GRK1, GRK2]), there is a smaller subdomain $\Omega'' \subseteq \Omega'$ (which also shares a piece of boundary with $\partial\Omega$ and $\partial\Omega'$) so that the Bergman metric of $\Omega'$ is close—in the $C^2$ topology—to the Bergman metric of $\Omega$ *on the smaller domain* $\Omega''$. It follows then that the Laplace–Beltrami operator $\mathcal{L}_{\Omega'}$ for the Bergman metric of $\Omega'$ will be close to the Laplace–Beltrami operator $\mathcal{L}_\Omega$ of $\Omega$ on the smaller subdomain $\Omega''$. Now, on $\Omega'$, the operator $\mathcal{L}_{\Omega'}$ certainly annihilates the principal term of (8.4.2.3). It follows then that, on $\Omega''$, the operator $\mathcal{L}_\Omega$ *nearly* annihilates the principal term of (8.4.2.3). We shall not calculate the exact sense in which this last statement is true, but leave details for the interested reader.

This discussion completes our consideration of (8.4.2.3). □

It is natural to wonder whether the Poisson–Bergman kernel $\mathcal{B}$ has any favorable properties with respect to important partial differential operators. We have the following positive result:

**Proposition 8.4.6** *Let $\Omega = B$, the unit ball in $\mathbb{C}^n$, and $\mathcal{B} = \mathcal{B}_B(z, \zeta)$ its Poisson–Bergman kernel. Then $\mathcal{B}$ is plurisubharmonic in the $\zeta$ variable.*

**Proof** Fix a point $\zeta \in B$ and let $\Phi$ be an automorphism of $B$ such that $\Phi(\zeta) = 0$. We then have

$$\mathcal{B}_B(z, \zeta) = \mathcal{B}_B(\Phi(z), \Phi(\zeta)) \cdot |\det J_{\mathbb{C}} \Phi(\zeta)|^2 = \mathcal{B}_B(\Phi(z), 0) \cdot |\det J_{\mathbb{C}} \Phi(\zeta)|^2\,. \tag{8.4.6.1}$$

We see that the righthand side is an expression that is independent of $\zeta$ multiplied times a plurisubharmonic function. A formula similar to (8.4.6.1) appears in [HUA].

The same argument shows that $\mathcal{B}(\zeta, \zeta)$ is plurisubharmonic. □

## 8.5 The Dirichlet Problem for the Invariant Laplacian

**Capsule:** In the late 1970s it was independently discovered by C. R. Graham [GRA1] and Garnett/Krantz [KRA5] that the Dirichlet problem for the invariant Laplacian on the ball does not satisfy the expected boundary regularity properties. In particular, it is not the case that smooth boundary data (measured in some typical norm) gives rise to a solution that is smooth on the closure. In this section we consider this phenomenon.

We will study the following Dirichlet problem on $B \subseteq \mathbb{C}^2$:

$$\begin{cases} \triangle_B u = 0 \ \text{ on } B \\ u\big|_{\partial B} = \phi, \end{cases} \tag{8.5.1}$$

where $\phi$ is a given continuous function on $\partial B$. Here $\triangle_B$ is the invariant Laplacian (i.e., the Laplace–Beltrami operator for the Bergman metric) on the unit ball in $\mathbb{C}^n$.

**Exercise** Is this a well-posed boundary value problem (in the sense of Lopatinski)? Consult [KRA5, Ch. 5] for more on this topic.

The remarkable fact about this relatively innocent-looking boundary value problem is that there exist data functions $\phi \in C^\infty(\partial B)$ with the property that the (unique) solution to the boundary value problem is not even $C^2$ on $\overline{B}$. This result appears in [GRA1, GRA2] and was also discovered independently by Garnett–Krantz (see [KRA5, Ch. 6]). It is in striking contrast to the situation that obtains for the Dirichlet problem for a uniformly elliptic operator.

Observe that, for $n = 1$, our Dirichlet problem becomes

$$\begin{cases} (1 - |z|^2)^2 \triangle u = 0 \text{ on } D \subseteq \mathbb{C} \\ u\big|_{\partial D} = \phi, \end{cases}$$

which is just the same as

$$\begin{cases} \triangle u = 0 \quad \text{on } D \subseteq \mathbb{C} \\ u\big|_{\partial D} = \phi. \end{cases}$$

This is the standard Dirichlet problem for the Laplacian—a uniformly strongly elliptic operator. Thus there is a complete existence and regularity theory: the solution $u$ will be as smooth on the closure as is the data $\phi$ (provided that we measure this smoothness in the correct norms). Our problem in dimensions $n > 1$ yields some surprises. We begin by developing some elementary geometric ideas.

Let $\zeta, \xi \in \partial B$. Define

$$\rho(\zeta, \xi) = |1 - \zeta \cdot \overline{\xi}|^{1/2},$$

where $\zeta \cdot \overline{\xi} \equiv \zeta_1 \overline{\xi}_1 + \zeta_2 \overline{\xi}_2$. Then we have

**Proposition 8.5.2** *The binary operator $\rho$ is a metric on $\partial B$.*

**Proof** Let $z, w, \zeta \in \partial B$. We shall show that

$$\rho(z, \zeta) \leq \rho(z, w) + \rho(w, \zeta).$$

Assume for simplicity that the dimension $n = 2$. After applying a unitary rotation, we may suppose that $w = \mathbf{1} = (1, 0)$. Now $|1 - z \cdot \overline{\zeta}| = \frac{1}{2}|z - \zeta|^2$. Therefore it suffices for us to prove that

$$\frac{1}{\sqrt{2}}|z - \zeta| \leq \sqrt{|1 - z_1|} + \sqrt{|1 - \zeta_1|}.$$

But

$$\frac{1}{\sqrt{2}}|z - \zeta| \leq \frac{1}{\sqrt{2}}|z - \mathbf{1}| + \frac{1}{\sqrt{2}}|\mathbf{1} - \zeta|.$$

(Notice that for $z, \zeta$ symmetrically situated about $\mathbf{1}$ and very near to $\mathbf{1}$ this is nearly an equality.) Thus it is enough to prove that

$$|z - \mathbf{1}| \leq \sqrt{2}\sqrt{|1 - z_1|}.$$

Finally, we calculate that

$$\begin{aligned} |z - \mathbf{1}| &= \sqrt{|1 - z_1|^2 + |z_2|^2} \\ &= \sqrt{|1 - z_1|^2 + 1 - |z_1|^2} \\ &= \sqrt{2 - 2\mathrm{Re}\, z_1} \\ &= \sqrt{2}\sqrt{1 - \mathrm{Re}\, z_1} \\ &\leq \sqrt{2}|1 - z_1|\,. \end{aligned}$$

□

Now we define balls using $\rho$: for $P \in \partial B$ and $r > 0$ we define $\beta(P, r) = \{\zeta \in \partial B : \rho(P, \zeta) < r\}$. [These skew balls (see the earlier discussion) play a decisive role in the complex geometry of several variables. We shall get just a glimpse of their use here.] Let $0 \neq z \in B$ be fixed and let $\tilde{z}$ be its orthogonal projection on the boundary: $\tilde{z} = z/|z|$. If we fix $r > 0$ then we may verify directly that

$$\mathcal{P}(z,\zeta) \to 0 \quad \text{uniformly in} \quad \zeta \in \partial B \setminus \beta(\tilde{z}, r) \ \text{ as } \ z \to \tilde{z}.$$

**Proposition 8.5.3** *Let $B \subseteq \mathbb{C}^n$ be the unit ball and $g \in C(\partial B)$. Then the function*

$$G(z) = \begin{cases} \int_{\partial B} \mathcal{P}(z,\zeta) g(\zeta)\, d\sigma(\zeta) & \text{if } z \in B \\ g(z) & \text{if } z \in B \end{cases}$$

*solves the Dirichlet problem (8.5.1) for the Laplace–Beltrami operator $\triangle_B$. Here $\mathcal{P}$ is the Poisson–Szegő kernel.*

**Proof** It is straightforward to calculate that

$$\begin{aligned} \triangle_B G(z) &= \int_{\partial B} [\triangle_B \mathcal{P}(z,\zeta)] g(\zeta)\, d\sigma(\zeta) \\ &= 0 \end{aligned}$$

because $\triangle_B \mathcal{P}(\cdot, \zeta) = 0$.

For simplicity let us now restrict attention once again to dimension $n = 2$. We wish to show that $G$ is continuous on $\overline{B}$. First recall that

$$\mathcal{P}(z,\zeta) = \frac{1}{\sigma(\partial B)} \frac{(1-|z|^2)^2}{|1 - z \cdot \overline{\zeta}|^4}.$$

Notice that

$$\int_{\partial B} |\mathcal{P}(z,\zeta)|\, d\sigma(\zeta) = \int_{\partial B} \mathcal{P}(z,\zeta)\, d\sigma(\zeta) = \int_{\partial B} \mathcal{P}(z,\zeta) \cdot 1\, d\sigma(\zeta) = 1$$

since the identically 1 function is holomorphic on $\Omega$ and is therefore reproduced by integration against $\mathcal{P}$. We have used also the fact that $\mathcal{P} \geq 0$.

Now we enter the proof proper of the proposition. Fix $\epsilon > 0$. By the uniform continuity of $g$ we may select a $\delta > 0$ such that if $P \in \partial B$ and $\zeta \in \beta(P, \delta)$ then $|g(P) - g(\zeta)| < \epsilon$. Then, for any $0 \neq z \in B$ and $P$ its projection to the boundary we have

$$\begin{aligned} |G(z) - g(P)| &= \left| \int_{\partial B} \mathcal{P}(z,\zeta) g(\zeta)\, d\sigma(\zeta) - g(P) \right| \\ &= \left| \int_{\partial B} \mathcal{P}(z,\zeta) g(\zeta)\, d\sigma(\zeta) - \int_{\partial B} \mathcal{P}(z,\zeta) g(P)\, d\sigma(\zeta) \right| \\ &\leq \int_{\partial B} \mathcal{P}(z,\zeta) |g(\zeta) - g(P)|\, d\sigma \\ &= \int_{\beta(P,\delta)} \mathcal{P}(z,\zeta) |g(\zeta) - g(P)|\, d\sigma(\zeta) \\ &\quad + \int_{\partial B \setminus \beta(P,\delta)} \mathcal{P}(z,\zeta) |g(\zeta) - g(P)|\, d\sigma(\zeta) \end{aligned}$$

$$\leq \epsilon + 2\|g\|_{L^\infty} \int_{\partial B \setminus \beta(P,\delta)} \mathcal{P}(z,\zeta)\, d\sigma(\zeta).$$

By the remarks preceding this argument, we may choose $r$ sufficiently close to 1 such that $\mathcal{P}(z,\zeta) < \epsilon$ for $|z| > r$ and $\zeta \in \partial B \setminus \beta(P,\delta)$. Thus, with these choices, the last line does not exceed $C \cdot \epsilon$.

We conclude the proof with an application of the triangle inequality: Fix $P \in \partial B$ and suppose that $0 \neq z \in B$ satisfies both $|P - z| < \delta$ and $|z| > r$. If $\tilde{z} = z/|z|$ is the projection of $z$ to $\partial B$ then we have

$$|G(z) - g(P)| \leq |G(z) - g(\tilde{z})| + |g(\tilde{z}) - g(P)|.$$

The first term is majorized by $\epsilon$ by the argument that we just concluded. The second is less than $\epsilon$ by the uniform continuity of $g$ on $\partial\Omega$.

That concludes the proof. □

Now we know how to solve the Dirichlet problem for $\triangle_B$ and we want next to consider regularity for this operator. The striking fact, in contrast with the uniformly elliptic case, is that for $g$ even in $C^\infty(\partial B)$ we may not conclude that the solution $G$ of the Dirichlet problem is $C^\infty$ on $\overline{B}$. In fact, in dimension $n$, the function $G$ is not generally in $C^n(\overline{B})$. Consider the following example:

**EXAMPLE 8.5.4** Let $n = 2$. Define

$$g(z_1, z_2) = |z_1|^2.$$

Of course $g \in C^\infty(\partial B)$. We now calculate $\mathcal{P}g(z)$ rather explicitly. We have

$$\mathcal{P}g(z) = \frac{1}{\sigma(\partial B)} \int_{\partial B} \frac{(1-|z|^2)^2}{|1 - z \cdot \overline{\zeta}|^4} |\zeta_1|^2\, d\sigma(\zeta).$$

Let us restrict our attention to points $z$ in the ball of the form $z = (r + i0, 0)$. We set

$$\mathcal{P}g(r + i0) \equiv \phi(r).$$

We shall show that $\phi$ fails to be $C^2$ on the interval $[0, 1]$ at the point 1. We have

$$\begin{aligned}
\phi(r) &= \frac{1}{\sigma(\partial B)} \int_{\partial B} \frac{(1-r^2)^2}{|1 - r\zeta_1|^4} |\zeta_1|^2\, d\sigma(\zeta) \\
&= \frac{(1-r^2)^2}{\sigma(\partial B)} \int_{|\zeta_1|<1} \int_{|\zeta_2| = \sqrt{1-|\zeta_1|^2}} \frac{|\zeta_1|^2}{|1 - r\zeta_1|^4} \cdot \frac{1}{\sqrt{1-|\zeta_1|^2}}\, ds(\zeta_2)\, dA(\zeta_1) \\
&= \frac{(1-r^2)^2}{\sigma(\partial B)} \int_{|\zeta_1|<1} \frac{|\zeta_1|^2}{|1 - r\zeta_1|^4} \frac{2\pi\sqrt{1-|\zeta_1|^2}}{\sqrt{1-|\zeta_1|^2}}\, dA(\zeta_1)
\end{aligned}$$

$$= \frac{2\pi}{\sigma(\partial B)}(1-r^2)^2 \int_{|\zeta_1|<1} \frac{|\zeta_1|^2}{|1-r\zeta_1|^4}\, dA(\zeta_1).$$

Now we set $\zeta_1 = \tau e^{i\psi}, 0 \le \tau < 1, 0 \le \psi \le 2\pi$. The integral is then

$$\frac{2\pi}{\sigma(\partial B)}(1-r^2)^2 \int_0^{2\pi} \int_0^1 \frac{\tau^2}{|1-r\tau e^{i\psi}|^4}\, \tau\, d\tau\, d\psi.$$

We perform the change of variables $r\tau = s$ and set $C = 2\pi/\sigma(\partial B)$. The integral becomes

$$C\frac{(1-r^2)^2}{r^4} \int_0^{2\pi} \int_0^r \frac{s^3}{|1-se^{i\psi}|^4}\, ds d\psi$$
$$C\frac{(1-r^2)^2}{r^4} \int_0^r s^3 \int_0^{2\pi} \frac{1}{|1-se^{i\psi}|^4}\, d\psi ds.$$

Now let us examine the inner integral. It equals

$$\int_0^{2\pi} \frac{1}{(1-se^{i\psi})^2(1-se^{-i\psi})^2}\, d\psi$$
$$= \int_0^{2\pi} \frac{e^{2i\psi}}{(1-se^{i\psi})^2(e^{i\psi}-s)^2}\, d\psi$$
$$= \int_0^{2\pi} \frac{\frac{e^{i\psi}}{(1-se^{i\psi})^2}}{(e^{i\psi}-s)^2}\, e^{i\psi}\, d\psi$$
$$= 2\pi \cdot \frac{1}{2\pi i} \oint_{|\eta|=1} \frac{\frac{\eta}{(1-s\eta)^2}}{(\eta-s)^2}\, d\eta.$$

Applying the theory of residues to this Cauchy integral we find that the last line equals

$$2\pi \frac{d}{d\eta}\left(\frac{\eta}{(1-s\eta)^2}\right)\Big|_{\eta=s} = 2\pi \frac{1+s^2}{(1-s^2)^3}.$$

Thus

$$\phi(r) = 2\pi C\frac{(1-r^2)^2}{r^4} \int_0^r \frac{s^3(1+s^2)}{(1-s^2)^3}\, ds$$
$$\overset{(s\mapsto\sqrt{s})}{=} \pi C\frac{(1-r^2)^2}{r^4} \int_0^{r^2} \frac{s(1+s)}{(1-s)^3}\, ds$$

$$
\begin{aligned}
&= \pi C \frac{(1-r^2)^2}{r^4} \int_0^{r^2} \frac{2}{(1-s)^3} - \frac{3}{(1-s)^2} + \frac{1}{(1-s)}\, ds \\
&= \pi C \frac{(1-r^2)^2}{r^4} \left\{ \left[ \frac{1}{(1-s)^2} - \frac{3}{(1-s)} - \log(1-s) \right]_0^{r^2} \right\} \\
&= \pi C \frac{(1-r^2)^2}{r^4} \left\{ \frac{1}{(1-r^2)^2} - \frac{3}{1-r^2} - \log(1-r^2) + 2 \right\} \\
&= \frac{\pi C}{r^4} \left\{ 1 - 3(1-r^2) - (1-r^2)^2 \log(1-r^2) + 2(1-r^2)^2 \right\} \\
&= \pi C \left\{ \frac{1 - 3(1-r^2) + 2(1-r^2)^2}{r^4} - \frac{(1-r^2)^2 \log(1-r^2)}{r^4} \right\}.
\end{aligned}
$$

Thus we see that $\phi(r)$ is the sum of two terms. The first of these is manifestly smooth at the point 1. However the second is not $C^2$ (from the left) at 1. Therefore the function $\phi$ is not $C^2$ at 1 and we conclude that $\mathcal{P}g$ is not smooth at the point $(1, 0) \in \partial B$, even though $g$ itself is. □

The phenomenon described in this example was discovered by Garnett and Krantz in 1977 (unpublished) and independently by C. R. Graham—Graham [GRA1, GRA2]. Graham subsequently developed a regularity theory for $\triangle_B$ using weighted function spaces. He also used Fourier analysis to explain the failure of boundary regularity in the usual function space topologies.

It turns out that these matters were anticipated by G. B. Folland in 1975 (see [FOD]). Using spherical harmonics, one can see clearly that the Poisson–Szegő integral of a function $g \in C^\infty(\partial B)$ will be smooth on $\overline{B}$ if and only if $g$ is the boundary function of a pluriharmonic function (these arise naturally as the real parts of holomorphic functions—see [KRA1, Ch. 3]). We shall explicate these matters later on with a discussion of spherical harmonics.

## 8.6 Concluding Remarks

The idea of reproducing kernels in harmonic analysis is an old one. The Poisson and Cauchy kernels date back to the mid-nineteenth century.

The Cauchy integral formula is special in that its kernel, which is

$$\frac{1}{2\pi i} \cdot \frac{1}{\zeta - z},$$

is the same on any domain. A similar statement is *not* true for the Poisson kernel, although see [KRA13] for a study of the asymptotics of this kernel.

The complex reproducing kernels that are indigenous to several complex variables are much more subtle. It was only in 1974 that C. Fefferman [FEF1] was able to calculate Bergman kernel asymptotics on strictly pseudoconvex domains. Prior to that, the very specific calculations of L. Hua [HUA] on concrete domains with a great deal of symmetry was the standard in the subject. A variant of Fefferman's construction also applies to the Szegő kernel (see also [BOUTS]). Carrying out an analogous program on a more general class of domains has proved to be challenging.

This portion of the book is an invitation to study yet another kernel—the Poisson–Bergman kernel. Inspired by the ideas of [HUA, Ch. 3], this is a positive reproducing kernel for the Bergman space. There are many questions about the role of this new kernel that remain unanswered.

## Exercises

1. Use results from the text to give an approximate formula for the Berezin kernel of the annulus.
2. Up to an error term, calculate Bergman representative coordinates on the annulus.
3. Write down a biholomorphic mapping from the unit ball $B$ in $\mathbb{C}^n$ to the domain

$$\mathcal{U} = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 > |z_2|^2 + \cdots + |z_n|^2\}\,.$$

   [**Hint:** Imitate the Cayley transform from classical function theory.] Use the biholomorphic invariance of the Bergman kernel to determine the Bergman kernel for $\mathcal{U}$.
4. Calculate the Berezin kernel for the domain $\mathcal{U}$ in Exercise 3.
5. What can you say about the Berezin kernel for the domain

$$U = \{\zeta \in \mathbb{C} : 0 < \operatorname{Im} \zeta < 1\}\,?$$

   What is the asymptotic behavior of the kernel as the argument(s) tend to infinity?
6. Use the Fefferman asymptotic expansion for the Bergman kernel of a strictly pseudoconvex domain to give a new asymptotic expansion for the Berezin kernel of a strictly pseudoconvex domain. What sort of logarithmic term do you get?
7. It is certainly the case that the Bergman kernel of the unit disc is a derivative of the Szegő kernel for the disc. Something of this nature is also true on the ball. Explain.

**8.** Use the philosophy of Exercise 7 together with the Fefferman asymptotic expansion to guess what the asymptotic expansion of the Szegő kernel on a strictly pseudoconvex domain should look like. Refer to the paper [BOUTS] to verify your answer.

**9.** Let $\Omega \subseteq \mathbb{C}$ be a bounded domain with $C^2$ boundary. Let $S \subseteq \partial\Omega$ be an arc of the boundary. Then certainly there is a conformal map $\varphi$ of the disc into $\Omega$ so that an arc of the boundary of the image $U$ coincides with $S$. Thus the Bergman kernel of the disc can be transferred to $U$. And the Bergman kernel of $U$ can be compared to the Bergman kernel of $\Omega$ (this is all a poor man's version of Fefferman's procedure for strictly pseudoconvex domains). Fill in the details of this argument to obtain an asymptotic expansion for the Bergman kernel of $\Omega$.

**10.** Let $\Omega \subseteq \mathbb{C}$ be a bounded domain with boundary point $P$ at which the boundary has a corner with angle $\theta$. What can you say about the asymptotic behavior of the Bergman kernel at $P$? How does your answer vary with $\theta$? What happens as $\theta \to \pi$?

# Chapter 9
# Additional Analytic Topics

**Prologue:** In this chapter we treat a variety of geometric/analytic topics, some of which have additional geometric manifestations. These include

- the worm
- irregularity properties of the Bergman projection
- irregularity properties of the Bergman kernel
- the Dirichlet problem
- strongly plurisubharmonic defining functions
- the Monge–Ampère equation
- an example of a smoothly bounded domain that fails Condition *R*
- the Bergman kernel as a Hilbert integral

The reader will find here much grist for his/her mill.

## 9.1 The Worm Domain

**Capsule:** We have mentioned the worm domain several times in the past, but now we will study its particular properties and prove definitive results about the worm. The worm is a counterexample to many geometric analytic desiderata.

The concept of "domain of holomorphy" is central to the function theory of several complex variables. The celebrated solution of the Levi problem tells us that a connected open set (a *domain*) is a domain of holomorphy if and only if it is pseudoconvex (see [KRA1, Ch. 1–5] for the details and proof of this assertion). For us, in the present book, pseudoconvexity is *Levi* pseudoconvexity; this is defined in terms of the positive semi-definiteness of the Levi form. This notion requires the boundary of the domain to be at least $C^2$. When the boundary is not $C^2$ we can still define a notion of pseudoconvexity (called *Hartogs pseudoconvexity*) that coincides with Levi pseudoconvexity in the $C^2$-case. See [KRA1, Ch. 3] for the details. When the Levi form is positive *definite* then we say that the domain is *strictly*

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or *strongly* pseudoconvex. The geometry of pseudoconvex domains has become an integral part of the study of several complex variables. (See [KRA1] for basic ideas about analysis in several complex variables.)

For completeness, we now provide a bit of background. Let $\Omega = \{z \in \mathbb{C}^n : \rho(z) < 0\}$ be a given domain. Here $\rho$ is a defining function for the domain. If $P \in \partial\Omega$, then we say that $w \in \mathbb{C}^n$ is a *complex tangent vector* at $P$ if

$$\sum_{j=1}^{n} \frac{\partial \rho}{\partial z_j}(P) w_j = 0 \,.$$

This condition just says that $w$ is Hermitian orthogonal to the normal vector $\langle \partial\rho/\partial\overline{w}_1, \ldots, \partial\rho/\partial\overline{w}_n \rangle$. We let $\mathcal{T}_P(\partial\Omega)$ denote the complex linear space of complex tangent vectors at $P$. Now we say that $P$ is a *point of Levi pseudoconvexity* if

$$\sum_{j,k=1}^{n} \frac{\partial^2 \rho}{\partial z_j \partial \overline{z}_k}(P) w_j \overline{w}_k \geq 0$$

for every $w \in \mathcal{T}_P(\partial\Omega)$. We say that $P$ is a point of *strong pseudoconvexity* if the inequality in the displayed equation is strict for nonzero $w$.

Consider a pseudoconvex domain $\Omega \subseteq \mathbb{C}^n$. Any such domain has an exhaustion $U_1 \subset\subset U_2 \subset\subset U_3 \subset\subset \cdots \Omega$ with $\cup_j U_j = \Omega$ by smoothly bounded, strictly pseudoconvex domains. This information was fundamental to the solution of the Levi problem (see [BERS] for this classical approach), and is an important part of the geometric foundations of the theory of pseudoconvex domains.

It is natural to ask whether there is a dual result for the exterior of $\Omega$. Specifically, given a pseudoconvex domain $\Omega$, are there smoothly bounded, pseudoconvex domains $W_1 \supset\supset W_2 \supset\supset W_3 \supset\supset \cdots \supset\supset \cdots \overline{\Omega}$ such that $\cap_j W_j = \overline{\Omega}$? A domain having this property is said to have a *Stein neighborhood basis*. A domain failing this property is said to have *nontrivial Nebenhülle*.

Early on, in 1906, F. Hartogs produced the following counterexample (which has come to be known as the *Hartogs triangle*): Let $\mathcal{H} = \{(z_1, z_2) \in \mathbb{C}^2 : 0 < |z_1| < |z_2| < 1\}$.

**Prelude** The Hartogs triangle is a classic example, dating to the early part of the twentieth century, from several-variable complex analysis. It is a domain without a Stein neighborhood basis. It has non-smooth boundary (the worm is an example of this phenomenon with smooth boundary), but is still a striking instance that we must know about.

**Theorem 9.1.1** *Any function holomorphic on a neighborhood of $\overline{\mathcal{H}}$ actually continues analytically to $D^2(0,1) \equiv D \times D$. Thus, $\overline{\mathcal{H}}$ cannot have a neighborhood basis of pseudoconvex domains. Instead, it has a nontrivial Nebenhülle.*

**Proof** Let $U$ be a neighborhood of $\overline{\mathcal{H}}$. For $|z_1| < 1$, the analytic discs

$$\zeta \mapsto (z_1, \zeta \cdot |z_1|)$$

have boundary lying in $U$. But, for $|z_1|$ sufficiently small, the entire disc lies in $U$. Thus a standard argument (as in the proof of the Hartogs extension phenomenon—see [KRA1, Ch. 3]), sliding the discs for increasing $|z_1|$, shows that a holomorphic function on $U$ will analytically continue to $D(0,1) \times D(0,1)$. That proves the result. □

It was, however, believed for many years that the Hartogs example worked only because the boundary of $\mathcal{H}$ is not smooth. Thus, for over 70 years, mathematicians sought a proof that a smoothly bounded pseudoconvex domain *will* have a Stein neighborhood basis. In 1977, it came as quite a surprise when Diederich and Fornæss [DIRF2] produced a smoothly bounded domain—now known as the *worm*—which is pseudoconvex and which does *not* have a Stein neighborhood basis. In fact the Diederich–Fornæss example is the following:

**Definition 9.1.2** Let $\mathcal{W}$ denote the domain

$$\mathcal{W} = \left\{(z_1, z_2) \in \mathbb{C}^2 : \left|z_1 - e^{i \log |z_2|^2}\right|^2 < 1 - \eta(\log |z_2|^2)\right\},$$

where

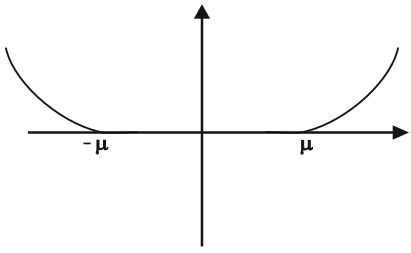
**(i)** $\eta : \mathbb{R} \to \mathbb{R}$, $\eta \geq 0$, $\eta$ is even, $\eta$ is convex;
**(ii)** $\eta^{-1}(0) = I_\mu \equiv [-\mu, \mu]$;
**(iii)** there exists a number $a > 0$ such that $\eta(x) > 1$ if $|x| > a$;
**(iv)** $\eta'(x) \neq 0$ if $\eta(x) = 1$.

We illustrate the function $\eta$ in Figure 9.1. The source [CHS, Ch. 6] has a nice exposition on the worm domain.

Notice that the slices of $\mathcal{W}$ for $z_2$ fixed are discs centered on the unit circle; the centers of these circles wind $\mu/\pi$ times about that circle as $|z_2|$ traverses the range of values for which $\eta(\log |z_2|^2) < 1$.

It is worth commenting here on the parameter $\mu$ in the definition of $\mathcal{W}$. The number $\mu$ in some contexts is selected to be greater than $\pi/2$. The number $\nu = \pi/2\mu$ is half the reciprocal of the number of times that the centers of the circles that make up the worm traverse their circular path.

**Figure 9.1** The function $\eta$.



Many authors use the original choice of parameter $\beta$, where $\mu = \beta - \pi/2$ (see, e.g., [BARR1]). Here, we have preferred to use the notation $\mu$, in accord with the source [CHST2]. See also [[CHS], Ch. 6].

**Proposition 9.1.3** *The domain $\mathcal{W}$ is smoothly bounded and pseudoconvex. Moreover, its boundary is strictly pseudoconvex except at the boundary points $(0, z_2)$ for $\left|\log |z_2|^2\right| \leq \mu$. These points constitute an annulus in $\partial\mathcal{W}$.*

**Proposition 9.1.4** *The smooth worm domain $\mathcal{W}$ has nontrivial Nebenhülle.*

The proofs of these propositions are deferred to Section 9.2.

As Diederich and Fornæss [DIRF2] showed, the worm provides a counterexample to a number of interesting questions in the geometric function theory of several complex variables. As an instance, the worm gives an example of a smoothly bounded, pseudoconvex domain which lacks a global plurisubharmonic defining function. It also provides counterexamples in holomorphic approximation theory. Clearly, the worm showed considerable potential for a central role in the function theory of several complex variables. But in point of fact the subject of the worm lay dormant for nearly 15 years after the appearance of [DIRF2]. It was the remarkable article of Kiselman [KIS] that re-established the importance and centrality of the worm.

In order to put Kiselman's work into context, we must provide a digression on the subject of biholomorphic mappings of pseudoconvex domains. In the present discussion, all domains $\Omega$ are smoothly bounded. We are interested in one-to-one, onto, invertible mappings (i.e., biholomorphic mappings or biholomorphisms) of domains

$$\Phi : \Omega_1 \longrightarrow \Omega_2 \,.$$

Thanks to a classical theorem of Liouville (see [KRPA2, Ch. 5]), there are no conformal mappings, other than trivial ones, in higher-dimensional (dimension greater than 2) complex Euclidean space. Thus biholomorphic mappings are studied instead. It is well known that the Riemann mapping theorem fails in several complex variables (see [KRA1, Ch. 11], [GRK2, GRK3, ISK]). It is thus a matter of considerable interest to find means to classify domains up to biholomorphic equivalence.

Poincaré's program for such a classification consisted of two steps: **(1)** to prove that a biholomorphic mapping of smoothly bounded pseudoconvex domains extends smoothly to a diffeomorphism of the closures of the domains and **(2)** to then calculate biholomorphic differential invariants on the boundaries of the domains. His program was stymied for more than 60 years because the machinery did not exist to tackle step **(1)** The breakthrough came in 1974 with Fefferman's seminal article [FEF1]. In it he used remarkable techniques of differential geometry, singular integrals, and partial differential equations to prove that a biholomorphic mapping of smoothly bounded, *strictly* pseudoconvex domains will extend to a diffeomorphism of the closures.

Fefferman's proof was quite long and difficult, and left open the question of **(a)** whether there was a more accessible and more natural approach to the question and **(b)** whether there were techniques that could be applied to a more general class of domains. Steven Bell [BE1] as well as Bell and Ewa Ligocka [BELL] provided a compelling answer.

Let $\Omega$ be a fixed, bounded domain in $\mathbb{C}^n$. Let $A^2(\Omega)$ be the square-integrable holomorphic functions on $\Omega$. Then $A^2(\Omega)$ is a closed subspace of $L^2(\Omega)$. The Hilbert space projection

$$P : L^2(\Omega) \longrightarrow A^2(\Omega)$$

can be represented by an integration formula

$$Pf(z) = \int_\Omega K(z, \zeta) f(\zeta)\, dV(\zeta)\,.$$

The kernel $K(z, \zeta) = K_\Omega(z, \zeta)$ is called the *Bergman kernel*. It is an important biholomorphic invariant. See [ZHU, Ch. 2], [KRA1, Ch. 1], [KRA18] for all the basic ideas concerning the Bergman kernel. Also refer to our Chapters 5 and 6.

Clearly, the Bergman projection $P$ is bounded on $L^2(\Omega)$. Notice that, if $\Omega$ is smoothly bounded, then $C^\infty(\overline{\Omega})$ is dense in $L^2(\Omega)$. In fact, more is true: If $\Omega$ is Levi pseudoconvex and smoothly bounded, then $C^\infty(\overline{\Omega}) \cap \{\text{holomorphic functions}\}$ is dense in $A^2(\Omega)$ (see [CAT3]).

Bell [BE1] has formulated the notion of Condition $R$ for the domain $\Omega$. We say that $\Omega$ satisfies *Condition R* if $P : C^\infty(\overline{\Omega}) \longrightarrow C^\infty(\overline{\Omega})$. It is known, thanks to the theory of the $\overline{\partial}$-Neumann problem, that strictly pseudoconvex domains satisfy Condition $R$. Deep work of Diederich–Fornæss [DIRF3] and Catlin [CAT1, CAT2] shows that domains with real analytic boundary, and also finite type domains, satisfy Condition $R$. An important formula of Kohn relates the $\overline{\partial}$-Neumann operator to the Bergman projection in a useful way (see also [KRA5, Ch. 8]). The fundamental result of Bell and Bell–Ligocka is as follows.

**Prelude** This next is the result that really put Condition $R$ on the map. It is a striking improvement over Fefferman's theorem, and applies to a much broader class of domains.

**Theorem 9.1.5** *Let $\Omega_j \subset \mathbb{C}^n$ be smoothly bounded, Levi pseudoconvex domains. Suppose that one of the two domains satisfies Condition R. If $\Phi : \Omega_1 \longrightarrow \Omega_2$ is a biholomorphic mapping then $\Phi$ extends to be a $C^\infty$ diffeomorphism of $\overline{\Omega}_1$ to $\overline{\Omega}_2$.*

This result established the centrality of Condition $R$. The techniques of proof are so natural and accessible that it seems that Condition $R$ is certainly the "right" approach to questions of boundary regularity of biholomorphic mappings. Work of

Boas–Straube in [BOAS1] shows that Condition $R$ is virtually equivalent to natural regularity conditions on the $\overline{\partial}$-Neumann operator.[1]

For later reference, and for its importance in its own right, we mention here that the above Theorem 9.1.5 can be "localized". To be precise, we say that a given smoothly bounded domain $\Omega$ satisfies the *Local Condition R* at a point $p_0 \in \partial\Omega$ if there exists a neighborhood $U$ of $p_0$ such that $P : C^\infty(\overline{\Omega}) \longrightarrow L^2(\Omega) \cap C^\infty(\overline{\Omega} \cap U)$. Then Bell's local result is as follows; see [BE3].

**Theorem 9.1.6** *Let $\Omega_j \subset \mathbb{C}^n$ be smoothly bounded, pseudoconvex domains, $j = 1, 2$. Suppose that $\Omega_1$ satisfies Local Condition R at $p_0 \in \partial\Omega_1$. If $\Phi : \Omega_1 \longrightarrow \Omega_2$ is a biholomorphic mapping then there exists a neighborhood $U$ of $p_0$ such that $\Phi$ extends to be a $C^\infty$ diffeomorphism of $\overline{\Omega}_1 \cap U$ onto its image.*

We might mention, as important background information, a result of David Barrett [BARR2] from 1984. This considerably predates the work on which we now concentrate. It does not concern the worm, but it does concern the regularity of the Bergman projection.

**Theorem 9.1.7** *There exists a smoothly bounded, non-pseudoconvex domain $\Omega \subseteq \mathbb{C}^2$ on which Condition R fails.*

Although Barrett's result is *not* on a pseudoconvex domain, it provides some insight into the trouble that can be caused by rapidly varying normals to the boundary. See [BARR3] for some pioneering work on this idea.

As indicated above, it was Kiselman [KIS] who established an important connection between the worm domain and Condition $R$. He proved that, for a certain non-smooth version of the worm (see below), a form of Condition $R$ fails.

For $s > 0$, let $W^s(\Omega)$ denote the usual Sobolev space on the domain $\Omega$ (see, for instance, [HOR2, Ch. 1], [KRA5, Ch. 1] or Exercise 5, Chapter 1). Building on Kiselman's idea, Barrett [BARR1] used an exhaustion argument to show that the Bergman projection fails to preserve the Sobolev spaces of sufficiently high order on the *smooth* worm.

**Prelude** Christ's theorem, which we now present, was a major breakthrough. It showed that there is a smoothly bounded, pseudoconvex domain which fails Condition $R$. This put a real kink in our program to prove boundary smoothness for all biholomorphic mappings.

**Theorem 9.1.8 (Barrett)** *For $\mu > 0$, let $\mathcal{W}$ be the smooth worm, defined as in Definition 9.1.2, and let $\nu = \pi/2\mu$. Then the Bergman projection $P$ on $\mathcal{W}$ does not map $W^s(\mathcal{W})$ to $W^s(\mathcal{W})$ when $s \geq \nu$.*

The capstone of results, up until 1996, concerning analysis on the worm domain is the seminal article of M. Christ. Christ finally showed that Condition $R$ fails on the smooth worm. Precisely, his result is the following (see [CHST2]).

[1] Here the $\overline{\partial}$-Neumann operator $N$ is the natural right inverse to the $\overline{\partial}$-Laplacian $\Box = \overline{\partial}^* \overline{\partial} + \overline{\partial}\overline{\partial}^*$.

**Theorem 9.1.9 (Christ)** *Let $\mathcal{W}$ be the smooth worm. Then there is a function $f \in C^\infty(\overline{\mathcal{W}})$ such that its Bergman projection $Pf$ is not in $C^\infty(\overline{\mathcal{W}})$.*

We note explicitly that the result of this theorem is closely tied to, indeed is virtually equivalent with, the assertion that the $\overline{\partial}$-Neumann problem is not hypoelliptic on the smooth worm, [BOAS1].

In the work of Kiselman et al., the geometry of the boundary of the worm plays a fundamental role in the analysis. In particular, the fact that for large $\mu$ the normal does considerable winding is fundamental to all of the negative results. It is of interest to develop a deeper understanding of the geometric analysis of the worm domain, because it will clearly play a seminal role in future work in the analysis of several complex variables.

We conclude this discussion of biholomorphic mappings with a consideration of biholomorphic mappings of the worm. It is at this time unknown whether a biholomorphic mapping of the smooth worm $\mathcal{W}$ to another smoothly bounded, pseudoconvex domain will extend to a diffeomorphism of the closures. [The paper [CHE1] offers a study of the biholomorphic *selfmaps* of the worm, but the arguments there seem to be incomplete.] Of course the worm does *not* satisfy Condition *R*, so the obvious tools for addressing this question are not available.

Our discussion is organized as follows:

Section 9.2 gives particulars of the Diederich–Fornæss worm. Specifically, we prove that the worm is smoothly bounded and Levi pseudoconvex, and we establish that there is no global plurisubharmonic defining function. We also examine the Diederich–Fornæss bounded plurisubharmonic exhaustion function on the smooth worm.

Section 9.3 considers non-smooth versions of the worm (these originated with Kiselman). We outline some of Kiselman's results.

Section 9.4 discusses the irregularity of the Bergman projection on the worm. In particular, we reproduce some of Kiselman's and Barrett's analysis.

Sections 9.4 and 9.5 discuss the failure of Condition *R* on the worm domains.

## 9.2 Additional Worm Ideas

**Capsule:** We devote this section to proving the basic geometric properties of the worm domain.

We now present the details of the first basic properties of the Diederich–Fornæss worm domain $\mathcal{W}$. Recall that $\mathcal{W}$ is defined in Definition 9.1.2. Some material of this section can also be found in the excellent monograph [CHS, Ch. 6].

We begin by proving Proposition 9.1.3.

**Proof of Proposition 9.1.3** Property **(iii)** of the worm shows immediately that the worm domain is bounded. Let

$$\rho(z_1, z_2) = \left| z_1 + e^{i \log |z_2|^2} \right|^2 - 1 + \eta(\log |z_2|^2) \, .$$

Then $\rho$ is (potentially) a defining function for $\mathcal{W}$. If we can show that $\nabla\rho \neq 0$ at each point of $\partial\mathcal{W}$ then the implicit function theorem guarantees that $\partial\mathcal{W}$ is smooth.

If it happens that $\partial\rho/\partial z_1(p) = 0$ at some boundary point $p = (p_1, p_2)$, then we find that

$$\frac{\partial\rho}{\partial z_1}(p) = \overline{p}_1 + e^{-i\log|p_2|^2} = 0$$

Now let us look at $\partial\rho/\partial z_2$ at the point $p$. The first term in $\rho$ differentiates to 0, and we find that

$$\frac{\partial\rho}{\partial z_2}(p) = \eta'(\log|p_2|^2)\cdot\frac{\overline{p}_2}{|p_2|^2}\,.$$

Since $\rho(p) = 0$, we have that $\eta(\log|p_2|^2) = 1$. Hence, by property **(iv)**, $\eta'(\log|p_2|^2) \neq 0$. It follows that $\partial\rho/\partial z_2(p) \neq 0$. We conclude that $\nabla\rho(z) \neq 0$ for every boundary point $z$.

For the pseudoconvexity, we write

$$\rho(z) = |z_1|^2 + 2\mathrm{Re}\left(z_1 e^{-i\log|z_2|^2}\right) + \eta(\log|z_2|^2)\,.$$

Multiplying through by $e^{\arg z_2^2}$, we have that locally $\mathcal{W}$ is given by

$$|z_1|^2 e^{\arg z_2^2} + 2\mathrm{Re}\left(z_1 e^{-i\log z_2^2}\right) + \eta(\log|z_2|^2)e^{\arg z_2^2} < 0\,.$$

The function $e^{-i\log z_2^2}$ is locally well defined and holomorphic, and its modulus is $e^{\arg z_2^2}$. Thus the first two terms are plurisubharmonic. Therefore, we must check that the last term is plurisubharmonic. Since it only depends on $z_2$, we merely have to calculate its Laplacian. We have, arguing as before, that:

$$\Delta\Big(\eta(\log|z_2|^2)e^{\arg z_2^2}\Big) = \Big(\Delta\eta(\log|z_2|^2)\Big)e^{\arg z_2^2} + \eta(\log|z_2|^2)\Delta e^{\arg z_2^2} \geq 0\,.$$

Because $\eta$ is convex and nonnegative (property **(i)**), the nonnegativity of this last expression follows. This shows that $\mathcal{W}$ is smoothly bounded and pseudoconvex.

It may be worth noting explicitly that we have proved that the (locally defined) defining function for the worm is plurisubharmonic. This does not contradict the fact (proved by Diederich and Fornæss) that the worm has no *globally defined* plurisubharmonic defining function.

In order to describe the locus of weakly pseudoconvex points, we consider again the local defining function

$$\rho(z_1, z_2) = |z_1|^2 e^{\arg z_2^2} + 2\mathrm{Re}\left(z_1 e^{-i\log z_2^2}\right) + \eta(\log|z_2|^2)e^{\arg z_2^2}\,.$$

This function is strictly plurisubharmonic at all points $(z_1, z_2)$ with $z_1 \neq 0$ because of the first two terms, or where $\left|\log |z_2|^2\right| > \mu$, because of the last term. Thus consider the annulus $\mathcal{A} \subset \partial\mathcal{W}$ given by

$$\mathcal{A} = \left\{(z_1, z_2) \in \partial\mathcal{W} : z_1 = 0 \text{ and } \left|\log |z_2|^2\right| \leq \mu\right\}.$$

A direct calculation shows that the complex Hessian for $\rho$ at a point $z \in \mathcal{A}$ acting on $v = (v_1, v_2) \in \mathbb{C}^2$ is given by

$$|v_1|^2 + 2\text{Re}\left(v_1\overline{v}_2 \frac{e^{i\log|z_2|^2}}{z_2}\right).$$

By pseudoconvexity, such an expression must be nonnegative for all complex tangential vectors $v$ at $z$. But such vectors are of the form $v = (0, v_2)$, so that the Levi form $\mathcal{L}_\rho \equiv 0$ on $A$. This proves the result. □

It is appropriate now to give the proof of Diederich and Fornæss that the worm has nontrivial Nebenhülle. What is of interest here, and what distinguishes the worm from the older example of the Hartogs triangle, is that the worm is a bounded, pseudoconvex domain with *smooth boundary*.

We now show (Proposition 9.1.4) that $\overline{\mathcal{W}}$ does not have a Stein neighborhood basis.

**Proof of Proposition 9.1.4** What we actually show is that if $U$ is *any* neighborhood of $\overline{\mathcal{W}}$ then $U$ will contain

$$K = \{(0, z_2) : -\pi \leq \log |z_2|^2 \leq \pi\} \cup \{(z_1, z_2) : \log |z_2|^2 = \pi \text{ or } -\pi \text{ and } |z_1 - 1| < 1\}.$$

In fact, this assertion is immediate by inspection.

By the usual Hartogs extension phenomenon argument, it then follows immediately that if $U$ is pseudoconvex then $U$ must contain

$$\widehat{K} = \left\{(0, z_2) : -\pi \leq \log |z_2|^2 \leq \pi \text{ and } |z_1 - 1| < 1\right\}.$$

Thus, there can be no Stein neighborhood basis. □

We turn next to a few properties of the smooth worm $\mathcal{W}$ connected with potential theory. The significance of the next result stems from the article [BOAS2]. In that article, Boas and Straube established the following:

**Theorem 9.2.1** *Let $\Omega$ be a smoothly bounded pseudoconvex domain that admits a defining function that is plurisubharmonic on the boundary. Then, for every $s > 0$,*

$$P : W^s(\Omega) \to W^s(\Omega)$$

*is bounded. In particular, $\Omega$ satisfies Condition R.*

For sake of completeness, we mention here that, if the Bergman projection $P$ on a domain $\Omega$ is such that $P : C^\infty(\overline{\Omega}) \longrightarrow C^\infty(\overline{\Omega})$ is bounded (i.e., $\Omega$ satisfies

Condition *R*), then $P$ is said to be *regular*, while if $P : W^s(\Omega) \to W^s(\Omega)$ for every $s > 0$ (and hence, $\Omega$ satisfies Condition *R a fortiori*) $P$ is said to be *exactly regular*.

Thanks to the result of Christ [CHST2], we now know that $\mathcal{W}$ does not satisfy Condition *R*, hence *a fortiori* it cannot admit a defining function which is plurisubharmonic on the boundary. However, it is simpler to give a direct proof of this fact.

**Proposition 9.2.2** *There exists no defining function $\widetilde{\rho}$ for $\mathcal{W}$ that is plurisubharmonic on the entire boundary.*

**Proof** Suppose that such a defining function $\widetilde{\rho}$ exists. Then, there exists a smooth positive function $h$ such that $\widetilde{\rho} = h\rho$. A direct calculation shows that the complex Hessian for $\widetilde{\rho}$ at a point $z \in \mathcal{A}$ acting on $v = (v_1, v_2) \in \mathbb{C}^2$ is given by

$$\mathcal{L}_{\widetilde{\rho}}(z; (v_1, v_2)) = 2\mathrm{Re}\left[\overline{v}_1 v_2 \left(\frac{ih}{z_2} + \partial_{z_2} h\right) e^{i\log|z_2|^2}\right] + \left[h + 2\mathrm{Re}\left(\partial_{z_1} h \cdot e^{i\log|z_2|^2}\right)\right]|v_1|^2 .$$

Since this expression is assumed to be always nonnegative, we must have

$$\left(\frac{ih}{z_2} + \partial_{z_2} h\right) e^{i\log|z_2|^2} = \overline{\partial_{\overline{z}_2}\left(he^{-i\log|z_2|^2}\right)} \equiv 0 ,$$

on $\mathcal{A}$. Therefore, the function $g(z_2) = h(0, z_2)e^{-i\log|z_2|^2}$ is a holomorphic function on $\mathcal{A}$. Hence, $g(z_2)e^{i\log|z_2|^2} = h(0, z_2)e^{2\arg z_2}$ is locally a holomorphic function. Thus it must be locally a constant, hence a constant $c$ on all of $\mathcal{A}$.

Therefore, on $\mathcal{A}$,

$$h(0, z_2) = ce^{-2\arg z_2}$$

which is impossible. This proves the result. □

We conclude this section with another important result about the Diederich–Fornæss worm domain $\mathcal{W}$. This result is part of potential theory, and is related to the negative result Proposition 9.2.2. In what follows, we say that $\lambda$ is a *bounded plurisubharmonic exhaustion function* for a domain $\Omega$ if

**(a)** $\lambda$ is continuous on $\overline{\Omega}$;
**(b)** $\lambda$ is strictly plurisubharmonic on $\Omega$;
**(c)** $\lambda = 0$ on $\partial\Omega$;
**(d)** $\lambda < 0$ on $\Omega$;
**(e)** For any $c < 0$, the set $\Omega_c = \{z \in \Omega : \lambda(z) < c\}$ is relatively compact in $\Omega$.

A bounded plurisubharmonic exhaustion function carries important geometric information about the domain $\Omega$. It has been used in several papers to measure the geometric nature of the domain $\Omega$—in particular the deviation of the domain from being strongly pseudoconvex. There are many important features of plurisubharmonic exhaustion functions that are still unknown.

Now Diederich–Fornæss have proved the following [DIRF1] (see also [RAN1] for a simpler proof when the boundary is smoother).

**Prelude** The idea in the next theorem is quite striking for its elegance and simplicity. One thing that it teaches us is the value of calculating.

**Theorem 9.2.3** *Let $\Omega$ be any smoothly bounded pseudoconvex domain, $\Omega = \{z \in \mathbb{C} : \varrho(z) < 0\}$. Then there exists $\delta$, $0 < \delta \leq 1$, and a defining function $\widetilde{\varrho}$ for $\Omega$ such that $-(-\widetilde{\varrho})^{\delta}$ is a bounded strictly plurisubharmonic exhaustion function for $\Omega$.*

The importance of this result in the setting of the regularity of the Bergman projection appears in the following related result, proved by Berndtsson–Charpentier [BERNC] and [KOH1], respectively.

**Prelude** Certainly the next theorem is the outgrowth of ideas that Kohn has been developing for many years. It is still a phenomenon that we do not understand well.

**Theorem 9.2.4** *Let $\Omega$ be a smoothly bounded pseudoconvex domain and let $P$ denotes its Bergman projection. Let $\widetilde{\rho}$ be a smooth defining function for $\Omega$ such that $-(-\widetilde{\rho})^{\delta}$ is strictly plurisubharmonic. Then there exists $s_0 = s_0(\Omega, \delta)$ such that*

$$P : W^s(\Omega) \to W^s(\Omega)$$

*is continuous for all $0 \leq s < s_0$.*

**Remark 9.2.5** The sharp value of $s_0$ is not known, and most likely the exact determination of such a value might prove a very difficult task. The two sources [BERNC] and [KOH1] present completely different approaches and descriptions of $s_0$, that is of the range $[0, s_0)$ for which $P$ is bounded on $W^s$, with $s \in [0, s_0)$. In [BERNC] it is proved that such a range is at least $[0, \delta/2)$, i.e., they show that $s_0 \geq \delta/2$, while in [KOH1] the parameter $s_0$ is not so explicit, but it tends to infinity as $\delta \to 1$. The value found in [BERNC] has the advantage of providing an explicit lower bound for the regularity of the Bergman projection on a given domain, while the value given in [KOH1] is sharp in the sense given by Boas and Straube's Theorem 10.7.1. □

**Remark 9.2.6** In fact the exponent $\delta$ in Theorem 9.2.3 is a matter of some interest. Known today as "the Diederich–Fornæss exponent", it has been used as an interesting geometric hypothesis in a number of results. Further exploration of this idea is well justified. □

The domain $\mathcal{W}$ serves as an example that the exponent $\delta$ may be arbitrarily small. To illustrate this point, the following result is essentially proved in [DIRF1]. Here we add the precise estimate that such an exponent $\delta$ is less than the value $\nu$.

**Prelude** This theorem tells us something about the Diederich–Fornæss exponent for the worm.

**Theorem 9.2.7** *Let $\delta_0 > 0$ be fixed. Then there exists $\mu_0 > 0$ such that, for all $\mu \geq \mu_0$, the following holds. If $\widetilde{\varrho}$ is a defining function for $\mathcal{W} = \mathcal{W}_\mu$, with $\mu \geq \mu_0$ and $\delta > 0$ is such that $-(-\widetilde{\varrho})^{\delta}$ is a bounded plurisubharmonic exhaustion function for $\mathcal{W}$, then $\delta < \delta_0$.*

*More precisely, we show that, in the notation above, $\delta < \nu = \pi/2\mu$.*

**Proof** We may assume that $\widetilde{\rho} = h\rho$, where $\rho = \rho_\mu$ is defined in 9.2.2 and $h$ is a smooth positive function on $\overline{\mathcal{W}}$. Then, by hypothesis $-h^\delta(-\rho)^\delta$ is strictly plurisubharmonic on $\mathcal{W}$.

Let

$$\begin{aligned}\sigma(z_1,z_2) &= -\frac{1}{2\pi}\int_0^{2\pi} h^\delta(z_1,e^{i\theta}z_2)\big(-\rho(z_1,e^{i\theta}z_2)\big)^\delta\,d\theta\\ &= -\frac{1}{2\pi}\int_0^{2\pi} h^\delta(z_1,e^{i\theta}z_2)\,d\theta\big(-\rho(z_1,z_2)\big)^\delta\\ &= -\widetilde{h}(z_1,z_2)\big(-\rho(z_1,z_2)\big)^\delta.\end{aligned}$$

Obviously, $\sigma$ is also strictly plurisubharmonic on $\mathcal{W}$, and $\widetilde{h}$ is strictly positive and smooth on $\mathcal{W}$. We can also write $\widetilde{h}(z_1,z_2) = h^\#(z_1,|z_2|^2)$, where $h^\#$ is defined for $(z_1,t)\in\mathbb{C}\times\mathbb{R}^+$ such that if $|z_2|^2 = t$ then $(z_1,z_2)\in\mathcal{W}$. For simplicity of notation, we rename such a function $h$ again.

Thus we have that

$$\sigma(z_1,z_2) = -h(z_1,|z_2|^2)\big(-\rho(z_1,z_2)\big)^\delta$$

is strictly plurisubharmonic on $\mathcal{W}$.

Now consider the points in $\mathcal{W}$ of the form $p = (z_1,z_2) = (\varepsilon e^{i\log|z_2|^2},z_2)$ with $e^{-\mu/2}\le|z_2|\le e^{\mu/2}$. For these points one has that

$$\partial\rho(p) = \big((1-\varepsilon)e^{i\log|z_2|^2},0\big).$$

A straightforward computation shows that, at such points $p\equiv(\varepsilon e^{i\log|z_2|^2},z_2)$ the Levi form $\mathcal{L}_\sigma$ of $\sigma$ calculated at vectors $v=(v_1,v_2)\in\mathbb{C}^2$ equals (all the functions are evaluated at the points $p$ and we write $\zeta$ in place of $e^{i\log|z_2|^2}$)

$$\begin{aligned}\mathcal{L}_\sigma\big(p;(v_1,v_2)\big) = \varepsilon^{\delta-2}(2-\varepsilon)^{\delta-2}\Big\{&(2-\varepsilon)\Big(-\varepsilon^2(2-\varepsilon)\partial^2_{z_1\overline{z}_1}h\\ &+2\delta\varepsilon(1-\varepsilon)\mathrm{Re}\,\big(\zeta\partial_{z_1}h\big)+\delta\varepsilon h\\ &+\delta(1-\delta)\frac{(1-\varepsilon)^2}{2-\varepsilon}h\Big)|v_1|^2\\ &+2w\varepsilon(2-\varepsilon)\mathrm{Re}\Big[\big(-\varepsilon(2-\varepsilon)\partial^2_{z_1\overline{z}_2}h\\ &+\delta(1-\varepsilon)\partial_{z_2}h+\delta\frac{i\zeta}{z_2}h\big)v_1\overline{v}_2\Big]\\ &+\delta^2(2-\delta)\Big(-(2-\delta)\partial^2_{z_2\overline{z}_2}h+\frac{2\delta}{|z_2|^2}h\Big)|v_2|^2\Big\}.\end{aligned}$$

Next, we evaluate the above Levi form at vectors of the form $(v_1, v_2) = (u_1, \varepsilon u_2)$. Making the obvious simplification, we see that the necessary condition in order for $\sigma$ to be strictly plurisubharmonic is that and $0 < \varepsilon < 1$ and, for all $(u_1, u_2) \in \mathbb{C}^2$,

$$\begin{aligned}
&\Big( -\varepsilon^2(2-\varepsilon)\partial^2_{z_1\overline{z}_1} h + 2\delta\varepsilon(1-\varepsilon)\mathrm{Re}\left(\zeta\partial_{z_2} h\right) + \delta\varepsilon h \\
&\quad + \delta(1-\delta)\frac{(1-\varepsilon)^2}{2-\varepsilon} h\Big)|u_1|^2 \\
&\qquad + 2\mathrm{Re}\left[\Big( -\varepsilon(2-\varepsilon)\partial^2_{z_1\overline{z}_2} h + \delta(1-\varepsilon)\partial_{z_2} h + \delta\frac{i\zeta}{z_2} h\Big) u_1\overline{u}_2\right] \\
&\qquad + \Big( -(2-\varepsilon)\partial^2_{z_2\overline{z}_2} h + \frac{2\delta}{|z_2|^2} h\Big)|u_2|^2 \geq 0\,. \qquad (9.2.7.1)
\end{aligned}$$

Since $h \in C^\infty(\overline{\mathcal{W}})$ this inequality must hold also for $\varepsilon = 0$ and $(0, z_2) \in \mathcal{A}$. Then we have

$$\Big(\frac{1}{2}\delta(1-\delta)h\Big)|u_1|^2 + 2\mathrm{Re}\left[\Big(\delta\zeta\partial_{z_2} h + \delta\frac{i\zeta}{z_2} h\Big) u_1\overline{u}_2\right] + \Big(-(2-\varepsilon)\partial^2_{z_2\overline{z}_2} h + \frac{2\delta}{|z_2|^2}\Big)|u_2|^2 \geq 0\,. \qquad (9.2.7.2)$$

Next, we substitute for $h$ the function $\widetilde{h}$ defined on $\mathbb{C} \times \mathbb{R}^+$ such that $h(z_1, z_2) = \widetilde{h}(z_1, |z_2|^2)$. Then

$$\partial_{z_2} h(0, z_2) = \overline{z}_2 \partial_t \widetilde{h}(0, |z_2|^2) \qquad \text{and} \qquad \partial^2_{z_2\overline{z}_2} h(0, z_2) = |z_2|^2 \partial_t^2 \widetilde{h}(0, |z_2|^2) + \partial_t \widetilde{h}(0, |z_2|^2)\,.$$

Plugging these into (9.2.7.1) we then obtain the differential inequality for the function $\widetilde{h}$:

$$\frac{1}{2}\delta(1-\delta)\widetilde{h}|u_1|^2 + 2\mathrm{Re}\left[\delta\zeta\Big(\partial_t\widetilde{h} + \frac{i}{|z_2|^2}\widetilde{h}\Big) u_1\overline{u}_2\right] + \Big( -2|z_2|^2\partial_t^2\widetilde{h} - 2\partial_t\widetilde{h} + \frac{2\delta}{|z_2|^2}\widetilde{h}\Big)|u_2|^2 \geq 0$$

for all $(u_1, u_2) \in \mathbb{C}^2$, $e^{-\mu/2} \leq |z_2| \leq e^{\mu/2}$ (and the function $\widetilde{h}$ being evaluated at the points $(0, |z_2|^2)$). Now if we choose $(u_1, u_2)$ of the form $(2e^{i\theta}/|z_2|, 1)$ in such a way that the second term in the above display becomes non-positive, we obtain that the function $\sigma$ is plurisubharmonic only if

$$\frac{\delta(1-\delta)}{|z_2|^2}\widetilde{h} - 2\delta\Big((\partial_t\widetilde{h})^2 + \frac{\widetilde{h}^2}{|z_2|^4}\Big)^{1/2} - |z_2|^2\partial_t^2\widetilde{h} - \partial_t\widetilde{h} + \frac{\delta}{|z_2|^2}\widetilde{h} \geq 0$$

which in turns gives

$$-\delta^2\widetilde{h} - |z_2|^4\partial_t^2\widetilde{h} - |z_2|^2\partial_t\widetilde{h} \geq 0 \qquad (9.2.7.3)$$

for all points $(0, |z_2|^2)$ with $e^{-\mu/2} \leq |z_2| \leq e^{\mu/2}$.

We now set $g(s) = \widetilde{h}(0, e^s)$ for $s \in [-\mu, \mu]$. Notice that $|z_2|^2 = e^s$ and that

$$g' = e^s \partial_t \widetilde{h} \qquad \text{and} \qquad g'' = e^s \partial_t \widetilde{h} + e^{2s} \partial_t^2 \widetilde{h} \,. \tag{9.2.7.4}$$

From (9.2.7.2) we obtain the differential inequality

$$g'' + \delta^2 g \leq 0 \,,$$

for $s \in [-\mu, \mu]$, where $g$ is a smooth strictly positive function. From the strict positivity of $g$ it follows that, for all $0 < \delta' < \delta$, it must be that

$$g'' + \delta'^2 g < 0 \,,$$

again for all $s \in [-\mu, \mu]$. Setting $\widetilde{g}(s) = g(s/\delta')$ the differential inequality above can be re-written as

$$\widetilde{g}'' + \widetilde{g} < 0$$

for all $s \in [-\mu\delta', \mu\delta']$. Finally, by translation (calling the new function $g$ again), i.e., setting $g(s) = \widetilde{g}(s + \mu\delta')$, we obtain that

$$g'' + g < 0 \tag{9.2.7.5}$$

for a smooth strictly positive function $g$, for all $s \in [0, 2\mu\delta']$.

We now claim that there exists a smooth strictly positive function $\varphi$ such that

$$\varphi'' + \varphi < 0 \qquad \text{and} \qquad \varphi' < 0 \tag{9.2.7.6}$$

$s \in [0, \mu\delta']$. For notice that if $g$ as above is such that $g'(a) < 0$, then $g'(s) < 0$ for $s \in [a, 2\mu\delta']$, while, if instead $g'(a) \geq 0$, then $g'(s) > 0$ for $s \in [0, a)$, since $g'' < 0$ on $[0, 2\mu\delta']$. In this latter case, making the substitution $s \mapsto 2\mu\delta' - s$ that preserves (9.2.7.6), we obtain a function with negative derivative on $[a, 2\mu\delta')$. By the arbitrariness of $\delta' < \delta$ we establish the claim.

Now, the argument at the end of the proof of Theorem 6 in [DIRF1] shows that the differential inequalities (9.2.7.3) above are possible only if $\mu\delta' < \pi/2$, i.e.,

$$\delta' < \frac{\pi}{2\mu} = \nu \,.$$

This proves the result. □

## 9.3 Alternative Versions of the Worm Domain

**Capsule:** Christer Kiselman [KIS] deserves credit for coming up with the idea of studying a non-smooth version of the worm. Such a domain has more symmetries than the smooth worm, and can more easily be analyzed. David Barrett took advantage of this idea in his paper [BARR1]. Krantz and Peloso [KRPE1, KRPE2] have conducted extensive studies of the non-smooth worm. See also [KRPS1, KRPS2].

In order to perform certain analyses on $\mathcal{W}$ some simplifications of the domain turn out to be particularly useful.

In the first instance, one can simplify the expression of the defining function $\rho$ for $\mathcal{W}$ by taking $\eta$ to be 1 minus the characteristic function of the interval $[-\mu, \mu]$. This has the effect of truncating the two caps and destroying in part the smoothness of the boundary. Precisely, we can define

$$\mathcal{W}' = \left\{(z_1, z_2) \in \mathbb{C}^2 : \left|z_1 - e^{i \log |z_2|^2}\right|^2 < 1, \ \left|\log |z_2|^2\right| < \mu\right\}.$$

We remark that $\mathcal{W}'$ is a bounded, pseudoconvex domain with boundary that is $C^\infty$ except at points that satisfy

**(i)** $|z_2| = e^{\mu/2}$ and $|z_1 - e^{-i \log |z_2|^2}| = 1$;
**(ii)** $|z_2| = e^{-\mu/2}$ and $|z - e^{-i \log |z_2|^2}| = 1$.

Of interest are also two non-smooth, *unbounded* worms. Here, in order to be consistent with the results obtained in [KRPE1], we change the notation a bit. (In practice, we set $\mu = \beta - \pi/2$.)

For $\beta > \pi/2$ we define

$$D_\beta = \left\{\zeta \in \mathbb{C}^2 : \mathrm{Re}\left(\zeta_1 e^{-i \log |\zeta_2|^2}\right) > 0, \ \left|\log |\zeta_2|^2\right| < \beta - \frac{\pi}{2}\right\}$$

and

$$D'_\beta = \left\{z \in \mathbb{C}^2 : \left|\mathrm{Im}\, z_1 - \log |z_2|^2\right| < \frac{\pi}{2}, \ |\log |z_2|^2| < \beta - \frac{\pi}{2}\right\}.$$

It should be noted that these latter two domains are biholomorphically equivalent via the mapping

$$(z_1, z_2) \ni D'_\beta \mapsto (e^{z_1}, z_2) \ni D_\beta \, .$$

Neither of these domains is bounded. Moreover, these domains are *not* smoothly bounded. Each boundary is only Lipschitz, and, in particular, their boundaries are Levi flat.

We notice in passing that the slices of $D_\beta$, for each fixed $\zeta_2$, are half planes in the variable $\zeta_1$. Likewise the slices of $D'_\beta$, for each fixed $\zeta_2$, are strips in the variable $\zeta_1$.

The geometries of these domains are rather different from that of the smooth worm $\mathcal{W}$, which has smooth boundary and all boundary points, except those on a singular annulus $(0, e^{i\log|z_2|^2})$ in the boundary, are strictly pseudoconvex. However our worm domain $D_\beta$ is actually a model for the smoothly bounded $\mathcal{W}$ (see, for instance, [BARR1]), and it can be expected that phenomena that are true on $D_\beta$ or $D'_\beta$ will in fact hold on $\mathcal{W}$ as well. We will say more about this symbiotic relationship below.

## 9.4 Pathologies of the Bergman Projection

**Capsule:** In this section we detail some of the ideas of David Barrett about the irregularity of the Bergman projection on the worm. These calculations have proven to be of lasting value.

We begin this section by discussing the proof of Barrett's result Theorem 9.1.8 in [BARR1]. Now let us describe these ideas in some detail. We begin with some of Kiselman's main ideas.

Let the Bergman space $\mathcal{H} = A^2$ be the collection of holomorphic functions that are square integrable with respect to Lebesgue volume measure $dV$ on a fixed domain. Following Kiselman [KIS] and Barrett [BARR1], using the rotational invariance in the $z_2$-variable, we decompose the Bergman space for the domains $D_\beta$ and $D'_\beta$ as follows. Using the rotational invariance in $z_2$ and elementary Fourier series, each $f \in \mathcal{H}$ can be written as

$$f = \sum_{j=-\infty}^{\infty} f_j \,,$$

where each $f_j$ is holomorphic and satisfies $f_j(z_1, e^{i\theta} z_2) = e^{ij\theta} f_j(z_1, z_2)$ for $\theta$ real. In fact, such an $f_j$ must have the form

$$f_j(z_1, z_2) = g_j(z_1, |z_2|) z_2^j \,,$$

where $g_j$ is holomorphic in $z_1$ and locally constant in $z_2$.

Therefore

$$\mathcal{H} = \bigoplus_{j\in\mathbb{Z}} \mathcal{H}^j \,,$$

where

$$\mathcal{H}^j = \left\{ f \in L^2 : f \text{ is holomorphic and } f(w_1, e^{i\theta} w_2) = e^{ij\theta} f(w_1, w_2) \right\} .$$

If $K$ is the Bergman kernel for $\mathcal{H}$ and $K_j$ the Bergman kernel for $\mathcal{H}^j$, then we may write

$$K = \sum_{j=-\infty}^{\infty} K_j \, .$$

Notice that, by the invariance property of $\mathcal{H}^j$, with $z = (z_1, z_2)$ and $w = (w_1, w_2)$, we have that

$$K_j(z, w) = H_j(z_1, w_1) z_2^j \overline{w}_2^j \, .$$

Our job, then, is to calculate each $H_j$, and thereby each $K_j$. The first step of this calculation is already done in [BARR1]. We outline the calculation here for the sake of completeness.

**Proposition 9.4.1** *Let $\beta > \pi/2$. Then*

$$H_j(z_1, w_1) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{e^{i(z_1 - \overline{w}_1)\xi} \xi \left(\xi - \frac{j+1}{2}\right)}{\sinh(\pi\xi) \sinh\left[(2\beta - \pi)\left(\xi - \frac{j+1}{2}\right)\right]} \, d\xi \, . \tag{9.4.1.1}$$

The articles [KIS] and [BARR1] calculate and analyze only the Bergman kernel for $\mathcal{H}^{-1}$ (i.e., the Hilbert subspace with index $j = -1$). This is attractive to do because certain "resonances" cause cancellations that make the calculations tractable when $j = -1$. One of the main thrusts of the work [KRPE1] is to perform the more difficult calculations for all $j$ and then to sum them over $j$.

We begin by following the calculations in [KIS] and [BARR1] in order to get our hands on the Bergman kernels of the $\mathcal{H}^j$. Let $f_j \in \mathcal{H}^j$ and fix $w_2$. Then $f_j(w_1, w_2) = h_j(w_1) w_2^j$ (where we of course take into account the local independence of $h_j$ from $w_2$). Now, writing $w_1 = x + iy$, $w_2 = re^{i\theta}$, and then making the change of variables $\log r^2 = s$, we have

$$\begin{aligned}
\|f_j\|_{\mathcal{H}}^2 &= \int_{D'_\beta} |h_j(w_1)|^2 |w_2|^{2j} \, dV(w) \\
&= \int_{-\infty}^{\infty} \int_{|y - \log r^2| < \frac{\pi}{2}} 2\pi |h_j(x + iy)|^2 \int_{|\log r^2| < \beta - \frac{\pi}{2}} r^{2j+1} \, dr dy dx \\
&= \pi \int_{\mathbb{R}} \int_{|y - s| < \frac{\pi}{2}} |h_j(x + iy)|^2 \int_{|s| < \beta - \frac{\pi}{2}} e^{s(j+1)} \, ds dy dx \\
&= \pi \int_{|y| < \beta, x \in \mathbb{R}} |h_j(x + iy)|^2 \int_{-\infty}^{\infty} e^{(j+1)s} \chi_{\pi/2}(y - s) \chi_{\beta - \pi/2}(s) \, ds dx dy \\
&= \int_{S_\beta} |h_j(w_1)|^2 \Big( \chi_{\pi/2} * \left[ e^{(j+1)(\cdot)} \chi_{\beta - \pi/2}(\cdot) \right] \Big)(y) \, dx dy \, ;
\end{aligned} \tag{9.4.2}$$

here we have set

$$S_\beta = \{x + iy \in \mathbb{C} : |y| < \beta\}$$

and used the notation

$$\chi_\alpha(y) = \begin{cases} 1 \text{ if } |y| < \alpha\,, \\ 0 \text{ if } |y| \geq \alpha\,. \end{cases}$$

For $\beta > \pi/2$, we now set

$$\lambda_j(y) = \left(\chi_{\pi/2} * \left[e^{(j+1)(\cdot)}\chi_{\beta-\pi/2}(\cdot)\right]\right)(y)\,.$$

So line (9.4.2) equals

$$\int_{S_\beta} |h_j(w_1)|^2 \lambda_j(y)\, dxdy\,.$$

Thus we have shown that, if $f_j \in \mathcal{H}^j, f_j = h_j(w_1)w_2^j$, then

$$\|f_j\|^2_{\mathcal{H}} = \int_{S_\beta} |h_j(w_1)|^2 \lambda_j(y)\, dxdy\,.$$

Now let $\varphi \in A^2(S_\beta, \lambda_j\, dA)$. That is, $\varphi$ is square-integrable on $S_\beta$ with respect to the measure $\lambda_j\, dA$ (here $dA = dxdy$ is two-dimensional area measure). Note that $\lambda_j$ depends only on the single variable $y$. Let $\widetilde{\varphi}$ denote the partial Fourier transform of $\varphi(x + iy)$ in the $x$-variable. Then (by standard Littlewood-Paley theory)

$$\widetilde{\varphi}(\xi, y) = \int \varphi(x + iy)e^{-ix\xi}\, dx = e^{-y\xi}\widetilde{\varphi_0}(\xi)\,,$$

where $\varphi_0(x) = \varphi(x + i0)$. Therefore, denoting by $B_\beta = B_\beta^{(j)}$ the Bergman kernel for the strip $S_\beta$ with respect to the weight $\lambda_j$ and writing $\omega = s + it$ and denoting by $\xi$ the variable dual to $s$, we have

$$\begin{aligned}\int_{\mathbb{R}} \widetilde{\varphi_0}(\xi)e^{i\zeta\xi}\, d\xi &= 2\pi\varphi(\zeta) = 2\pi \int_{S_\beta} \varphi(\omega)B_\beta(\zeta, \omega)\lambda_j(\operatorname{Im}\omega)\, dA(\omega) \\ &\equiv \int_{-\beta}^{\beta}\int_{\mathbb{R}} \widetilde{\varphi}(\xi, t)\widetilde{B}_\beta(\zeta, (\xi, t))\lambda_j(t)\, d\xi dt \\ &= \int_{\mathbb{R}} \widetilde{B}_\beta(\zeta, (\xi, 0)) \int_{-\beta}^{\beta} \widetilde{\varphi_0}(\xi)e^{-2\xi t}\lambda_j(t)\, dt\, d\xi\,.\end{aligned}$$

Notice that there is a factor of $e^{-\xi t}$ from each of the Fourier transform functions in the integrand.

This gives a formula for $\widetilde{B}_\beta$:

$$\widetilde{B}_\beta(\zeta,(\xi,0)) = \frac{e^{i\zeta\xi}}{\int_{-\beta}^{\beta} e^{-2t\xi}\lambda_j(t)\,dt} = \frac{e^{i\zeta\xi}}{\widehat{\lambda}_j(-2i\xi)}\,.$$

Amalgamating all our notation, and using the fact that the (Hermitian) diagonal in $\mathbb{C}^2$ is a set of determinacy, we find that

$$B_\beta(z,w) = \frac{1}{2\pi}\int_{\mathbb{R}} \frac{e^{i(z-\overline{w})\xi}}{\widehat{\lambda}_j(-2i\xi)}\,d\xi\,.$$

But of course $\widehat{(\chi_{\pi/2})}(\xi) = (e^{i\xi\pi/2} - e^{-i\xi\pi/2})/\xi$, so that

$$\widehat{(\chi_{\pi/2})}(-2\xi i) = \frac{1}{\xi}\sinh(\pi\xi)\,.$$

Furthermore,

$$\left(e^{(j+1)s}\chi_{\beta-\frac{\pi}{2}}(s)\right)\widehat{\ } = \frac{\sinh\left((2\beta-\pi)\left(\xi-\frac{j+1}{2}\right)\right)}{\xi-\frac{j+1}{2}}\,.$$

Thus

$$\widehat{\lambda}_j(-2i\xi) = \frac{\sinh(\pi\xi)\sinh((2\beta-\pi)(\xi-\frac{j+1}{2}))}{\xi(\xi-\frac{j+1}{2})}\,,$$

and

$$\frac{1}{\widehat{\lambda}_j(-2i\xi)} = \frac{\xi\left(\xi-\frac{j+1}{2}\right)}{\sinh(\pi\xi)\sinh((2\beta-\pi)(\xi-\frac{j+1}{2}))}\,.$$

In conclusion,

$$H_j(z_1,w_1) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\left(e^{i(w_1-\overline{z}_1)\xi}\right)\xi\left(\xi-\frac{j+1}{2}\right)}{\sinh(\pi\xi)\sinh\left((2\beta-\pi)\left(\xi-\frac{j+1}{2}\right)\right)}\,d\xi\,,$$

thus proving 9.4.1.1. □

At this point we sketch the proof of the main result of Barrett in [BARR1]. Namely, we show that the Bergman projection on the worm does not act continuously on the Sobolev space $W^s$ (see Exercise 5, Chapter 1 for the definition).

**Proof** The proof starts from the observation that the Bergman projection $\mathcal{P}$ on $\mathcal{W}$ preserves each $\mathcal{H}^j$. Therefore, in order to show that $\mathcal{P}$ is not continuous on $W^s$, for some $s$, it suffices to show that $\mathcal{P}$ fails to be continuous in this topology when restricted to some $\mathcal{H}^j$.

The first step is to calculate the asymptotic expression for the kernel when $j = -1$. Recall that we are working on the non-smooth domain $D'_\beta$. Using the method of contour integrals it is not difficult to obtain that

$$K'_{-1}(z,w) = \left(e^{-\nu_\beta|z_1-\overline{w}_1|} + \mathcal{O}(e^{-\nu|\mathrm{Re}\, z_1-\mathrm{Re}\, w_1|})\right)\cdot(z_2\overline{w}_2)^{-1}$$

as $|\mathrm{Re}\, z_1 - \mathrm{Re}\, w_1| \longrightarrow +\infty$, uniformly in all closed strips $\{|\mathrm{Im}\, z_1|,\ |\mathrm{Im}\, w_1| \leq \lambda\}$, with $\nu > \nu_b$.

By applying the biholomorphic transformation between $D_\beta$ and $D'_\beta$, one obtains an asymptotic expression for the kernel $K_{-1}$ relative to the domain $D_\beta$:

$$K_{-1}(\zeta,\omega) = (|\zeta_1||\omega_1|)^{-1}\cdot\left(\frac{|\omega_1|^{\nu_\beta}}{|\zeta_1|^{\nu_\beta}} + \mathcal{O}(|\omega_1|^{\nu_\beta}/|\zeta_1|^{\nu_\beta})^{-\nu}\right)\cdot(\zeta_2\overline{\omega}_2)^{-1}\ ,$$

with $\nu > \nu_\beta$, as $|\zeta_1| - |\omega_1| \longrightarrow 0^+$, The proof of these two assertions can be found in [BARR1] (or see [CHS]).

The next step is a direct calculation to show that $K_{-1}(\cdot,w) \notin W^s(D_\beta)$ for $s \geq \nu_\beta$. This assertion is proved by using the characterization of Sobolev norms for holomorphic functions on a domain $\Omega$: For $-1/2 < t < 1/2$, $m$ a nonnegative integer, the norm

$$\sum_{|\alpha|\leq m} \left\| |\rho|^t \partial_z^\alpha h \right\|_{L^2(\Omega)}$$

is equivalent to the $H^{m-t}$-norm of the holomorphic function $h$. The proof of such a characterization can be found in [LIG1].

Next, one notices that the reproducing kernel $K_{-1}(\cdot,w)$ can be written as the projection of a radially symmetric smooth cut-off function $\chi$, translated at $w$. That is, if we denote by $P_{-1}$ the projection relative to the subspace $\mathcal{H}^{-1}$, then

$$K_{-1}(\cdot,w) = P_{-1}\big(\chi(\cdot - w)\big)\,.$$

Therefore, since $K_{-1}(\cdot,w) \notin W^s(D_\beta)$ for $s \geq \nu_\beta$, then $P_{-1}$, and therefore, $P_{D_\beta}$ is not continuous on $W^s(D_\beta)$.

The final step of the proof is to transfer this negative result from $D_\beta$ to $\mathcal{W}$. This is achieved by an exhaustion argument—see [BARR1]. We adapt this kind of argument to obtain a negative result in the $L^p$-norm for the Bergman projection on the smooth worm of Diederich and Fornaess. We refer the reader to the literature for all the details. □

## 9.5 Pathologies of the Bergman Kernel

**Capsule:** Of course we know, just by inspection, that the Bergman kernel of the disc or the ball is smooth on the product of the closures of the domain minus the boundary diagonal. One might hope that something similar would be true on a broad class of domains. And it is true on strongly pseudoconvex domains—see [KER2]. It is true on some other classes of domains as well. But it is *not* true on the worm. It is this matter that we consider in the present section.

We now examine the boundary asymptotics for the Bergman kernel on the domains $D_\beta$ and $D'_\beta$ and determine various irregularity properties of the corresponding Bergman kernels.

Begin with the asymptotic formula in the discussion above of Barrett's result. We point out particularly that there are two kinds of behavior: one kind at the "finite portion of the boundary" and the other one as $|\operatorname{Re} z_1 - \operatorname{Re} w_1| \longrightarrow +\infty$.

These two different behaviors are expressed by the first and second terms in the expansion. For the former type, we notice that the lead terms have expressions in the denominator of products of two terms like

$$(i(z_1 \pm \overline{w}_1) + 2\beta)^2\,, \quad (z_2\overline{w}_2 - e^{\pm(\beta-\pi/2)})^2\,, \quad \text{and} \quad (z_2\overline{w}_2 - e^{-[i(z_1-\overline{w}_1)\pm\pi]/2})^2\,.$$

These singularities are similar to the ones of a Bergman kernel of a domain in $\mathbb{C}^2$ which is essentially a product domain. It is important to observe that the kernel does not become singular only when $z, w$ tend to the same point on the boundary. For instance, it becomes singular as $(i(z_1 \pm \overline{w}_1) + 2\beta) \longrightarrow 0$, while there is no restriction on the behavior of $z_2$ and $w_2$. We will be more detailed below in the case of the domain $D_\beta$. For the case of this domain, we finally notice that the main term at infinity, that is when $|\operatorname{Re} z_1 - \operatorname{Re}_1| \longrightarrow +\infty$, behaves like $e^{-\nu_\beta|z_1-\overline{w}_1|}\cdot(z_2\overline{w}_2)^{-1}$.

Next we consider the case of $D_\beta$. The mapping $(z_1, z_2) \in D'_\beta \mapsto (\zeta_1, \zeta_2) = (e^{z_1}, z_2) \in D_\beta$ sends the point at infinity (in $z_1$) into the origin (in $\zeta_1$). Keeping into account the Jacobian factor, when $|\zeta_1| - |\omega_1| \longrightarrow 0^+$, the kernel on $D_\beta$ is asymptotic to

$$\frac{|\omega_1|^{\nu_\beta-1}}{|\zeta_1|^{\nu_\beta+1}}\cdot(\zeta_2\overline{\omega}_2)^{-1}\,.$$

Recall the inequalities that define $D_\beta$:

$$D_\beta = \left\{\zeta \in \mathbb{C}^2 : \operatorname{Re}\left(\zeta_1 e^{-i\log|\zeta_2|^2}\right) > 0,\ \left|\log|\zeta_2|^2\right| < \beta - \frac{\pi}{2}\right\}.$$

If we take $\zeta,\ \omega \in D_\beta$ and let $\omega_1$ tend to 0, then clearly $\omega \longrightarrow \partial D_\beta$ and $\zeta_1,\ \zeta_2,\ \omega_2$ are unrestricted. Therefore $K_{D_\beta}(\cdot, \omega) \notin C^\infty(D_\beta)$ for $\omega \in \{(0, \omega_2)\}$, with $|\log|\omega_2|^2| < \beta - \pi/2$.

Notice that this is in contrast, for instance, to the situation on the ball or, more generally, on a strictly pseudoconvex domain. On either of those types of domains $\Omega$, the kernel is known to be smooth on $\overline{\Omega} \times \overline{\Omega} \setminus (\Delta \cap [\partial\Omega \times \partial\Omega])$. See [KER2] and [KRA1, Ch. 1].

By the same token (by almost the same calculation), it is easy to conclude that the Bergman projection on $D_\beta$ *cannot* map functions in $C^\infty(\overline{D}_\beta)$ to functions in $C^\infty(\overline{D}_\beta)$. This, of course, is the failure of Condition $R$ on these domains.

In Section 9.1 we have seen that $P_{\mathcal{W}} : C^\infty(\overline{\mathcal{W}}) \not\longrightarrow C^\infty(\overline{\mathcal{W}})$, that is, that $\mathcal{W}$ does not satisfy Condition $R$. A philosophically related fact, due to Chen [CHE2] and Ligocka [LIG1] independently, is that the Bergman kernel of $\mathcal{W}$ cannot lie in $C^\infty(\overline{\mathcal{W}} \times \overline{\mathcal{W}} \setminus (\Delta \cap [\partial\mathcal{W} \times \partial\mathcal{W}])$ (where $\Delta$ is the diagonal). In fact, in [CHE2] it is shown that this phenomenon is a consequence of the presence of a complex variety in the boundary of $\mathcal{W}$.

The proof of the general result of So-Chin Chen follows a classical paradigm for establishing propagation of singularities for the $\overline{\partial}$-Neumann problem and similar phenomena.

**Prelude** It is a nice insight that a complex variety in the boundary implies ill behavior of the Bergman kernel. This theorem formalizes that fact.

**Theorem 9.5.1** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded, pseudoconvex domain with $n \geq 2$. Assume that there is a complex variety $V$, of complex dimension at least 1, in $\partial\Omega$. Then*

$$K_\Omega(z, w) \notin C^\infty(\overline{\Omega} \times \overline{\Omega} \setminus \triangle(\partial\Omega)) \,,$$

*where $\triangle(\partial\Omega) = \{(z, z) : z \in \partial\Omega\}$.*

**Proof** Let $p \in V$ be a regular point. Let $\nu_p$ be the unit outward normal vector at $p$. Then there are small numbers $\delta, \varepsilon_0 > 0$ such that $w - \varepsilon\nu_p \in \Omega$ for all $w \in \partial\Omega \cap B(p, \delta)$ and all $0 < \varepsilon < \varepsilon_0$. Let $\mathbf{d}$ be an analytic disc in $\partial\Omega \cap B(p, \delta) \cap V$. We may assume that this disc is centered at $p$. In other words, $\mathbf{d}$ is the image of the unit disc in the plane mapped into $\mathbb{C}^n$ with the origin going to $p$.

Seeking a contradiction, we assume that $K_\Omega(z, w) \in C^\infty(\overline{\Omega} \times \overline{\Omega} \setminus \triangle(\partial\Omega))$. Then we certainly have

$$\sup_{w \in \partial\mathbf{d}} |K_\Omega(p, w)| \leq M < +\infty \tag{9.5.1.1}$$

for some positive, finite number $M$. On the other hand, we know (see [BLP]) that

$$\lim_{\varepsilon \longrightarrow 0} K_\Omega(p - \varepsilon\nu_p, p - \varepsilon\nu_p) = +\infty \,. \tag{9.5.1.2}$$

By the maximum modulus principle we then obtain

$$\sup_{w \in \partial\mathbf{d}_\varepsilon} |K_\Omega(p - \varepsilon\nu_p, w)| \geq K_\Omega(p - \varepsilon\nu_p, p - \varepsilon\nu_p) \,,$$

where $\mathbf{d}_\varepsilon = \mathbf{d} - \varepsilon v_p \subseteq \Omega$. We conclude that

$$\sup_{w\in\partial\mathbf{d}} |K_\Omega(p,w)| = \lim_{\varepsilon\longrightarrow 0} \sup_{w\in\partial\mathbf{d}_\varepsilon} |K_\Omega(p-\varepsilon v_p, w)| = +\infty\,.$$

This gives a contradiction, and the result is established. □

## 9.6 Kohn's Projection Formula

**Capsule:** This celebrated formula of J. J. Kohn (see [KRA5, Ch. 7–8]) is quite old, and rather elementary. But it is extremely important. It helps to explain why Condition *R* holds automatically on any domain on which the $\overline{\partial}$ problem is well behaved. And it has many other important implications.

In the 1960s, J. J. Kohn produced an elegant formula that relates the Bergman projection to the $\overline{\partial}$-Neumann problem. We refer the reader to [FOK, Ch. 3–4] or [KRA5, Ch. 7–8] for details of this important topic. Here we only briefly review the key concepts.

In studying the $\overline{\partial}$ operator, it is convenient to treat the second order, self-adjoint operator given by

$$\Box = \overline{\partial}\overline{\partial}^* + \overline{\partial}^*\overline{\partial}\,.$$

It is shown that this partial differential operator has a canonical right inverse $N$, which is known as the $\overline{\partial}$-Neumann operator.

Let $\Omega \subset\subset \mathbb{C}^n$ be a fixed domain on which the equation $\overline{\partial}u = \alpha$ is always solvable when $\alpha$ is a $\overline{\partial}$-closed $(0,1)$ form (i.e., a domain of holomorphy—in other words, a pseudoconvex domain). Let $P : L^2(\Omega) \to A^2(\Omega)$ be the Bergman projection. If $u$ is any solution to $\overline{\partial}u = \alpha$, then $w = w_\alpha = u - Pu$ is the unique solution that is orthogonal to holomorphic functions. Thus $w$ is well-defined, independent of the choice of $u$. Define the mapping

$$T : \alpha \mapsto w_\alpha.$$

Then, for $f \in L^2(\Omega)$ it holds that

$$Pf = f - T(\overline{\partial}f). \tag{9.6.1}$$

To see this, first notice that $\overline{\partial}[f - T(\overline{\partial}f)] = \overline{\partial}f - \overline{\partial}f = 0$, where all derivatives are interpreted in the weak sense. Thus $f - T(\overline{\partial}f)$ is holomorphic. Also $f - [f - T(\overline{\partial}f)]$ is orthogonal to holomorphic functions by design. This establishes the identity (9.6.1). But we have a more useful way of expressing $T$: namely $T = \overline{\partial}^* N$. Thus we have derived the following important result:

$$P = I - \overline{\partial}^* N \overline{\partial}. \tag{9.6.2}$$

This is Kohn's formula.

Formula (9.6.2) is particularly useful for studying Condition $R$. For if $\Omega$ is a strictly pseudoconvex domain, or a finite type domain in the sense of Kohn–D'Angelo–Catlin, then it is known that $N$ satisfies a regularity estimate. In the strictly pseudoconvex case the estimate is

$$\|N\alpha\|_{W^{s+1}} \leq C\|\alpha\|_{W^s} .$$

Here $W^s$ is the standard Sobolev space of order $s$ (see [KRA5, Ch. 1] or Exercise 5, Chapter 1). It immediately follows from (9.6.2) that the Bergman projection $P$ maps $W^s$ to $W^{s-2}$. And that is a form of Condition $R$. We conclude, then, that a biholomorphic mapping of smoothly bounded, strongly pseudoconvex domains extends to a diffeomorphism of the closures. A similar result holds for finite type domains.

We can also learn something from applying formula (9.6.2) to the Dirac $\delta$ mass at a point $w$ of the domain. For the Bergman kernel $K(z, w) = P(\delta_w)$. Thus, according to (9.6.2),

$$K(z, w) = P(\delta_w) = \delta_w - \overline{\partial}^* N \overline{\partial} \delta_w .$$

By the pseudolocality of $N$ on a finite type domain, we may conclude that the kernel is smooth—up to the boundary—away from $w$. [Here an operator $T$ is said to be *pseudolocal* if $T\varphi$ is smooth whereever $\varphi$ is smooth. Thus partial differential operators are trivially pseudolocal. What is interesting, and nontrivial, is when an integral operator is pseudolocal.]

## 9.7 Boundary Behavior of the Kernel

**Capsule:** Here we have a look at various approaches to the question of boundary behavior of the Bergman kernel. As noted previously, the behavior on the disc and the ball is our paradigm. We explore which other domains have similar behavior, and which do not.

The earliest work on the boundary behavior of the Bergman kernel was done by S. Bergman himself [BERM2]. A more modern approach, based on estimates for the $\overline{\partial}$ problem, appears in [HOR1]. Hörmander's results later were given some technical refinements in [DIR1, DIR2]. Here we present the statements and proofs of Hörmander's results.

We will see that the main tool used in the proof is a comparison technique. We determine information about the Bergman kernel for the domain $\Omega$ under study by comparing that kernel with the kernel of another (nearby) domain for which the kernel is more accessible. This is a very powerful idea, and is used pervasively in this field. Even Fefferman's decisive result about the boundary behavior of the Bergman kernel uses a (rather sophisticated) comparison technique. The elegant

paper [GRAI] makes use of comparison techniques that are similar to the ones presented here.

### 9.7.1 Hörmander's Result on Boundary Behavior

**Theorem 9.7.1** *Let $\Omega \subseteq \mathbb{C}^n$ be a bounded domain of holomorphy with defining function given by*

$$\rho(z) = \begin{cases} -\text{dist}_{\partial\Omega}(z) & \text{if} \quad z \in \Omega \\ \text{dist}_{\partial\Omega}(z) & \text{if} \quad z \notin \Omega \,. \end{cases}$$

*Assume that the operator $\overline{\partial} : L^2_{(0,0)}(\Omega) \to L^2_{0,1}(\Omega)$ has closed range. Let $P$ be a point of $\partial\Omega$ such that $\partial\Omega$ is $C^2$ in a neighborhood of $P$. Also suppose that $\partial\Omega$ is strictly pseudoconvex at $P$. Let $k(P)$ be the product of the $n-1$ eigenvalues of the Levi form at $P$. Then*

$$\lim_{\Omega \ni z \to P} |\rho(z)|^{n+1} |K_\Omega(z,z)| \to k(P) \cdot \frac{n!}{4\pi^n} \,.$$

The proof will be broken up into a sequence of lemmas, and will occupy most of the rest of this section. At the end we shall comment on Fefferman's asymptotic expansion, and how it trumps the work of Bergman, Hörmander, and Diederich. We note that, by the main theorem of [HOR1], the hypothesis of closedness of the $\overline{\partial}$ operator is automatically fulfilled on a bounded domain of holomorphy.

**Lemma 9.7.2** *If $\Omega$ is as in the theorem and $\Omega' \subseteq \Omega$ is another domain, then*

$$|K_{\Omega'}(z,z)| \geq |K_\Omega(z,z)| \,.$$

**Proof** This is obvious from the characterization

$$|K_\Omega(z,z)| = \sup_{\substack{u \in A^2(\Omega) \\ \|u\|_{A^2} = 1}} |u(z)|^2 \,. \tag{9.7.2.1}$$

□

**Lemma 9.7.3** *Let $\Omega$ be a bounded, pseudoconvex domain. Let $P \in \partial\Omega$ and suppose that, for some neighborhood $U$ of $P$, there is a holomorphic function $u_0$ on $\Omega' \equiv \Omega \cap U$ such that $|u_0| \leq 1$ in $\Omega'$ and $|u_0(z)| \to 1$ when $z \to P$. We also suppose that $|u_0(z)|$ has an upper bound less than 1 in $\Omega' \cap {}^c U_0$ for some neighborhood $U_0$ of $P$ with compact closure contained in $U$. Then we have*

$$\lim_{z \to P} \frac{|K_\Omega(z,z)|}{|K_{\Omega'}(z,z)|} = 1 \,.$$

**Remark 9.7.4** Certainly a holomorphic peaking function will suffice for the function $u_0$ in this lemma. It is known (see [KRA1, Ch. 3]), for example on strictly pseudoconvex domains, that holomorphic peaking functions always exist. □

**Proof of Lemma 9.7.3** Let $\chi$ be a $C^\infty$ function with compact support in $U$ which is identically equal to 1 on $U_0$. Assume that $0 \leq \chi \leq 1$ everywhere. If $u' \in L^2_{(0,0)}(\Omega')$ and is holomorphic there, then (for an integer $\nu$ to be specified later) we set

$$u = \chi \cdot u' \cdot u_0^\nu - v \,.$$

Our goal is to choose $v$ so that $\overline{\partial} u = 0$ on $\Omega'$. Thus we must solve the equation

$$\overline{\partial} v = (\overline{\partial}\chi) u' u_0^\nu \,.$$

Of course the righthand side is $\overline{\partial}$-closed, and supported in $U \cap \Omega$. We solve this $\overline{\partial}$ problem on the domain $\Omega$. Hörmander's theorem in [HOR1] tells us that there will be a solution $v$ satisfying

$$\int_\Omega |v|^2 \, dV \leq C \int_\Omega |u' u_0^\nu|^2 \, dV = C \int_{\Omega' \cap {}^c U_0} |u' u_0^\nu|^2 \, dV \tag{9.7.3.1}$$

because $\overline{\partial}\chi = 0$ on $U_0$.

If $\epsilon > 0$, then we have

$$\int_{\Omega'} |u - u' u_0^\nu|^2 \, dV \leq 2(C+1) \int_{\Omega' \cap {}^c U_0} |u' u_0^\nu|^2 \, dV \leq \epsilon^2 \int_{\Omega'} |u'|^2 \, dV \,,$$

provided that $\nu$ is chosen so large that $|u_0|^{2\nu} \leq \epsilon^2/[2(C+1)]$ in $\Omega' \cap {}^c U_0$. From the definition of the kernel function in $\Omega'$ and (9.7.3.1), we see now that for $z \in \Omega'$ we have

$$|u(z) - u'(z) u_0(z)^\nu|^2 \leq \epsilon^2 |K_{\Omega'}(z,z)| \int_{\Omega'} |u'|^2 \, dV \,.$$

Thus

$$|u(z)| \geq |u'(z)||u_0(z)|^\nu - \epsilon \left( |K_{\Omega'}(z,z)| \int_{\Omega'} |u'|^2 \, dV \right)^{1/2} .$$

Since the supremum in (9.7.2.1) is clearly attained, we can, for every $z \in \Omega'$, choose $u' \not\equiv 0$ so that

$$|u'(z)|^2 = |K_{\Omega'}(z,z)| \int_{\Omega'} |u'|^2 \, dV \,.$$

For the corresponding function $u$ we then obtain the estimate

$$|u(z)|^2 \geq |K_{\Omega'}(z,z)| \, (|u_0(z)|^\nu - \epsilon)^2 \int_{\Omega'} |u'|^2 \, dV \,, \tag{9.7.3.2}$$

as long as $z \in \Omega'$ and $|u_0(z)|^\nu > \epsilon$. Now the triangle inequality and (9.7.3.1) tell us that

$$\int_\Omega |u|^2 \, dV \leq (1+\epsilon)^2 \int_{\Omega'} |u'|^2 \, dV \,.$$

Together with (9.7.3.2), this last estimate implies that

$$|K_\Omega(z,z)| \geq |K_{\Omega'}(z,z)| \cdot (|u_0(z)|^\nu - \epsilon)^2 \, (1+\epsilon)^{-2} \quad \text{if } z \in \Omega' \text{ and } |u_0(z)|^\nu > \epsilon \,.$$

Thus

$$\liminf_{z \to z_0} \frac{|K_\Omega(z,z)|}{|K_{\Omega'}(z,z)|} \geq (1-\epsilon)^2 (1+\epsilon)^2$$

and, since $\epsilon > 0$ is arbitrary, this (together with the preceding lemma) proves the current lemma. □

This last lemma will enable us to reduce the proof of our main result to the study of a special domain on which the kernel function is relatively easy to compute.

**Lemma 9.7.5** *Let $E_0$ be the ellipsoid in $\mathbb{C}^n$ defined by*

$$E_0 = \{z \in \mathbb{C}^n : a_1|z_1|^2 + z_2|z_2|^2 + \cdot + a_n|z_n|^2 < a_0\}$$

*where $a_0, a_1, a_2, \ldots$ are positive numbers. Then*

$$|K_{E_0}(z,z)| = n! \pi^{-n} a_0 \cdot a_1 \cdot \cdots \cdot a_n \cdot (a_0 - a_1|z_1|^2 - a_2|z_2|^2 - \cdots - a_n|z_n|^2)^{-n-1} \,.$$

**Proof** We may assume that $a_0 = 1$. After a linear change of variables, we may also suppose that $a_1 = a_2 = \cdots a_n = 1$. Because of the unitary invariance of the kernel $K$, we may let $z = (0, 0, \ldots, 0, \zeta)$. If $u$ is an element of $A^2(E_0)$, then a unitary transformation $\lambda$ of the variables $z_1, z_2, \ldots, z_{n-1}$ leaves $E_0$, $u(0, 0, \ldots, 0, \zeta)$, $\int |u|^2 \, dV$, and $\int |u|^2 \, dV$ invariant. Let $z' = (z_1, z_2, \ldots, z_{n-1})$. If we form

$$u_1(z) = \int u(\lambda z', z_n) \, d\mu \,,$$

where $d\mu$ is normalized Haar measure on the unitary group, we then obtain a function $u_1 \in A^2(E_0)$ so that

$$u_1(0, 0, \ldots, 0, \zeta) = u(0, 0, \ldots, 0, \zeta) \,.$$

By Minkowski's inequality, we now see that

$$\int_{E_0} |u_1|^2 \, dV \leq \int_{E_0} |u|^2 \, dV \, .$$

But $u_1$ is invariant for unitary transformations of $z_1, z_2, \ldots, z_{n-1}$, and therefore must be a function of $z_n$ only. In determining the supremum in the proof of the first lemma, we may therefore assume that $u$ is a holomorphic function of $z_n$ for $|z_n| < 1$.

Now put

$$u(z) = \sum_0^\infty c_j z_n^j \, .$$

Since the volume of the unit ball in $\mathbb{R}^{n2-2}$ is $\pi^{n-1}/(n-1)!$, we find that

$$\begin{aligned} \int_{E_0} |u|^2 \, dV &= \frac{\pi^{n-1}}{(n-1)!} \int_0^{2\pi} \int_0^1 |u(re^{i\theta}|^2 r(1-r^2)^{n-1} \, dr d\theta \\ &= \frac{\pi^{n-1}}{(n-1)!} \sum_0^\infty |c_j|^2 2\pi \int_0^1 r^{2j+1}(1-r^2)^{n-1} \, dr \\ &= \pi^n \sum_0^\infty \frac{|c_j|^2 j!}{(j+n)!} \, . \end{aligned}$$

By the Cauchy–Schwarz inequality, it follows now that

$$|u(0, 0, \ldots, 0, \zeta)|^2 \leq \pi^{-n} \sum_0^\infty |\zeta|^{2j} \frac{(j+n)!}{j!} \int_{E_0} |u|^2 \, dV \, ,$$

where equality is attained for some $u$. Since the sum of the series is $n!(1-|\zeta|^2)^{-n-1}$, the lemma is now proved. □

The following variant of the last lemma will be particularly useful in the proof of our main result:

**Lemma 9.7.6** *Let $a_{jk}$ $(j, k = 1, \ldots, n)$ be a positive definite hermitian symmetric matrix. Set*

$$F_0 = \left\{ z \in \mathbb{C}^n : \operatorname{Im} z_n > \sum_{j,k=1}^n a_{jk} z_j \overline{z}_k \right\} \, .$$

*Then*

$$|K_{F_0}(z, z)| = \sigma n! 4^{-1} \pi^{-n} \left( \operatorname{Im} z_n - \sum_{j,k=1}^n a_{jk} z_j \overline{z}_k \right)^{-n-1} \, ,$$

*where $\sigma = det(a_{jk})_{j,k=1}^{n-1}$.*

**Proof** By a unitary transformation of the variables $z_1, z_2, \ldots, z_{n-1}$, we may reduce the matrix $(a_{jk})_{j,k=1}^{n-1}$ to diagonal form; and the statement of the theorem remains invariant. Assuming this reduction to have been made, we can introduce new variables

$$w_j = a_j + \frac{z_n a_{nj}}{a_{jj}}\,, \quad j = 1, 2 \ldots, n-1$$

and $w_n = z_n$. The determinant of this transformation of variables is equal to 1, so again the statement of the theorem is invariant. So we may assume that the entire matrix $(a_{jk})$ has diagonal form. If we write

$$\operatorname{Im} z_n - a_{nn}|z_n|^2 = \frac{1}{4a_{nn}} - a_{nn}\left|z_n - \frac{i}{2a_{nn}}\right|^2\,,$$

then the lemma now follows from Lemma 9.7.3 □

**Proof of Theorem 9.7.1** It is a standard fact (see Lemma 3.3.3 of [HOR1]) that there is a real-valued function $\psi \in C^2$ which is strictly plurisubharmonic in a neighborhood of $P$ and so that $\Omega$ is defined by the equation $\psi < 0$ and grad$\psi$ is the exterior unit normal to $\partial\Omega$ at $P$. We choose local coordinates at $P$ so that $P$ is the origin and the differentials $dz_j$ are orthonormal at $P$. Thus we have that the Riemannian element of integration has density 1 with respect to the Lebesgue measure in the coordinate space.

We further choose our coordinates so that $\psi(z) + \operatorname{Im} z_n = O(|z|^2)$ at $P$. By Taylor's formula, $\Omega$ is thus defined in a neighborhood of $P$ by an inequality of the form

$$\operatorname{Im} z_n > \sum_{j,k=1}^{n} \frac{\partial^2\psi(0)}{\partial z_j \partial \overline{z}_k} z_j \overline{z}_k + \operatorname{Re} A(z) + o(|z|^2)\,,$$

where $A$ is an analytic, homogeneous, second degree polynomial. If we replace the coordinate $z_n$ by $z_n - iA(z)$, the differential at $P$ is not affected. So we may assume without loss of generality that $A = 0$ from the outset. Put $a_{jk} = \partial^2\psi(0)/\partial z_j \partial \overline{z}_k$, which is a hermitian symmetric, positive definite matrix at $P$.

For any $\epsilon > 0$, we set

$$\Omega_\epsilon = \left\{ z : \operatorname{Im} z_n > \sum_{j,k=1}^{n} a_{jk} z_j \overline{z}_k + \epsilon |a|^2 \right\}\,.$$

Then

$$\Omega_\epsilon^\delta \equiv \Omega_\epsilon \cap \{z : |z| < \delta\}$$

is contained in $\Omega$ if $\delta$ is sufficiently small. We now see, using Lemma 9.5.2, that

$$|K_\Omega(z,z)| \le e^\epsilon |K_{\Omega_\epsilon^\delta}(z,z)| \,.$$

If we let $z \to 0$ so that $\operatorname{Im} z_n/|a|$ has a positive lower bound, then it follows from Lemmas 9.7.3 and 9.7.6 applied to $\Omega_\epsilon^\delta$ and $\Omega_\epsilon$ that

$$\begin{aligned}\limsup_{z\to 0} (\operatorname{Im} z_n)^{n+1} |K_\Omega(z,z)| &\le e^\epsilon \limsup_{z\to 0} (\operatorname{Im} z_n)^{n+1} |K_{\Omega_\epsilon^\delta}(z,z)| \\ &= e^\epsilon \limsup_{z\to 0} (\operatorname{Im} z_n)^{n+1} |K_{\Omega_\epsilon}(z,z)| \\ &= n! 4^{-1} \pi^{-n} e^\epsilon \det(a_{jk} + \epsilon\delta_{jk})_{j,k=1}^{n-1} \,.\end{aligned}$$

Since $\epsilon$ is arbitrary, this last proves (with the notation from our main theorem) that

$$\limsup_{z\to P} |\rho(z)|^{n+1} |K_\Omega(z,z)| \le \frac{k(P) n!}{4\pi^n} \,, \tag{9.7.1.1}$$

assuming that $z$ approaches $P$ inside a small cone in the coordinate space around the normal to $\partial\Omega$ at $P$. But we can see that in fact the result is valid uniformly in $P$, so it remains true for arbitrary approach to $P$.

Let $\epsilon$ be positive but smaller than the least eigenvalue of the matrix $(a_{jk})$. For sufficiently small $\delta > 0$, we have

$$\Omega^\delta = \{z : z \in \Omega \,, |z| < \delta\} \subseteq \Omega_{-\epsilon} \,.$$

Hence Lemma 9.7.3 can be applied with $U = \{z : |z| < \delta\}$ and $u_0(z) = e^{izn}$. From Lemma 9.7.3 and the monotonocity in Lemma 9.5.2, we then find, for $\delta$ sufficiently small, that

$$\begin{aligned}\limsup_{z\to P} (\operatorname{Im} z_n)^{n+1} |K_\Omega(z,z)| &= \limsup_{z\to P} (\operatorname{Im} z_n)^{n+1} |K_{\Omega^\delta}(z,z)| \\ &\ge e^{-\epsilon} \limsup_{z\to P} (\operatorname{Im} z_n)^{n+1} |K_{\Omega_{-\epsilon}}(z,z)|\end{aligned}$$

when $z \to P$ and remains in a small cone about the normal to $\partial\Omega$. Arguing exactly as in the proof of (9.7.1.1), we conclude now that

$$\liminf_{z\to P} |\rho(z)|^{n+1} |K_\Omega(z,z)| \ge \frac{k(P) n!}{4\pi^n} \,.$$

Details are left to the reader. And that completes the proof. □

### 9.7.2 *Fefferman's Asymptotic Expansion*

The 1974 result of Fefferman [FEF1] subsumes all the earlier work. For Fefferman shows that, near a boundary point of a smooth, strictly pseudoconvex domain[2] the Bergman kernel for $\Omega$ can be written (in suitable local coordinates) as

$$K_\Omega(z,w) = K_B(z,w) + \mathcal{E}(z,w)\,.$$

Here $K_B$ is the Bergman kernel for the unit ball, and $\mathcal{E}$ is an error term which is of measurably lower order than $K_B$. So we see that this theorem is much more explicit than 9.7.1 which we proved above, and it is also valid *off the diagonal*.

Fefferman's argument is quite lengthy and complicated, and we cannot present it here. What we *can* do, however, is to explicate the approximation part of his reasoning. This is a nice piece of logic, and shows how to approximate a strictly pseudoconvex point by (the image of) the unit ball.

We shall exploit the following fact

> **FACT:** Given $p \in \partial\Omega$ strictly pseudoconvex, we can find a region $\widetilde{\Omega}$ internally tangent to $\partial\Omega$ to fourth order at $p$, and an explicit biholomorphic change of coordinates $F$ mapping a neighborhood of $p$ in $\partial\widetilde{\Omega}$ to a neighborhood of $F(p)$ in the boundary of the unit ball. Of course we can then pull back the Bergman kernel from the unit ball to obtain an explicit formula for the kernel $K_{\widetilde{\Omega}}$ of $\widetilde{\Omega}$.

Now we will give a more formal enunciation of this fact and provide a proof.

**Proposition 9.7.7** *Let $p \in \partial\Omega$ be a strictly pseudoconvex point. Then there is a neighborhood $V$ of $p$ with the following property: For any point $Q \in V$ there is a biholomorphic mapping $\zeta_Q$ sending $V$ to a neighborhood of the origin, sending $Q$ to the point $(\psi(Q), 0, \ldots, 0)$, and sending $\Omega \cap V$ to a region of the form*

$$\operatorname{Re}\zeta_1 = |\zeta'|^2 - p_4(\zeta', \operatorname{Im}\zeta_1) + (\textit{5th and higher-order terms in } \zeta', \overline{\zeta}', \operatorname{Im}\zeta_1)$$

*with $(\zeta_1, \zeta') \in \zeta_w(V)$. Here $p_4(\zeta', \operatorname{Im}\zeta_1)$ is a real-valued, 4th-order polynomial in $\zeta', \overline{\zeta}', \operatorname{Im}\zeta_1$ satisfying $p_4(\zeta', \operatorname{Im}\zeta_1) \geq C(|\zeta'|^4 + |\operatorname{Im}\zeta_1|^4)$. Furthermore, we can make $\zeta_Q(\cdot)$ depend smoothly on $Q \in V$.*

*Here we may think (refer to the* **FACT** *above) of $\{\operatorname{Re}\zeta_1 = |\zeta'|^2\}$ as the boundary of $\widetilde{\Omega}$.*

**Remark 9.7.8** Earlier work of Hörmander [HOR1] had an approximation result of this type, but it was not as sharp. The order of tangency in that work was strictly less

[2] At the time that Fefferman wrote his paper, it really was necessary to assume that the entire domain was strictly pseudoconvex in order to get certain global estimates for the $\overline{\partial}$-Neumann problem. However, more recent results of Catlin [CAT1, CAT2] and others show that one need only assume that the boundary is strictly pseudoconvex near the boundary point in question. The more global hypotheses can be something considerably weaker.

than the fourth-order result obtained here. People were actually somewhat startled by Fefferman's approximation result.

**Proof of the Proposition** Let $q$ be the point of $\partial\Omega$ that is closest to $Q$. Then $Q - q$ is normal to $\partial\Omega$. After a suitable translation and rotation of $\mathbb{C}^n$, we may suppose that $q = 0$ and that the tangent plane to $\partial\Omega$ at $q$ is $\{z : \operatorname{Re} z_1 = 0\}$. In particular, in the new coordinate system we have $Q = (\tau, 0, \ldots, 0)$ with $\tau$ real and positive. In the complex part of the tangent space, specified by $\{z : z_1 = 0\}$, make a linear change of coordinates so that the Levi form at $Q$ becomes the identity matrix. As a result, we have a new coordinate system, $(z_1, z')$, in which $Q = (\tau, 0)$ and $\partial\Omega$ takes the form

$$\operatorname{Re} z_1 = \operatorname{Re}\left\{a(i\operatorname{Im} z_1)^2 + \sum_{j=2}^{n} b_j \cdot (i\operatorname{Im} z_1)\cdot z_j + \sum_{j,k=2}^{n} c_{jk} z_j z_k\right\}$$
$$+|z'|^2 + (\text{3rd and higher order terms})\,.$$

We can make the coefficent $a$ real. Here $|z'|^2$ appears as the Levi form $(z', \overline{z}')$.

If we set

$$\zeta_1 = z_1 - a z_1^2 - \sum_{j=2}^{n} b_j z_1 z_j - \sum_{j,k=2}^{n} c_{jk} z_j z_k\,,$$
$$\zeta' = z'\,,$$

then, in the new $(\zeta_1, \zeta')$ coordinate system, $\partial\Omega$ takes the form

$$\operatorname{Re}\zeta_1 = |\zeta'|^2 + (\text{3rd and higher terms})\,, \tag{9.7.7.1}$$

and $Q$ still has the form $(\tau, 0)$ for $\tau$ real. If we write out the most general real-valued homogeneous third degree polynoimial in $\operatorname{Im}\zeta_1$, $\zeta'$, $\overline{\zeta}'$, then we find that (9.7.7.1) is equivalent to

$$\begin{aligned}\operatorname{Re}\zeta_1 = \quad & |\zeta'|^2 + \operatorname{Re}\left\{\sum_{j,k,\ell=2}^{n} a_{jk\ell}\zeta_j\zeta_k\zeta_\ell + \sum_{j,k,\ell=2}^{n} b_{jk\ell}\zeta_j\zeta_k\overline{\zeta}_\ell\right.\\ & \sum_{j,k=2}^{n} c_{jk}(i\operatorname{Im}\zeta_1)\zeta_j\zeta_k + \sum_{j,ik=2}^{n} d_{jk}(i\operatorname{Im}\zeta_1)\zeta_j\overline{\zeta}_k\\ & \left.\sum_{j=2}^{n} e_j(i\operatorname{Im}\zeta_1)^2\zeta_j + f\cdot(i\operatorname{Im}\zeta_1)^3\right\}\,,\end{aligned} \tag{9.7.7.2}$$

where the $a$s, $b$s, $c$s, $d$s, $e$s, and $f$s are complex numbers. We may take $f$ to be purely imaginary. In the new coordinates then

$$z_1 = \zeta_1 - f\zeta_1^3 - \sum_{j=2}^{n} e_j \zeta_1^2 \zeta_j - \sum_{j,k=2}^{n} c_{jk}\zeta_1\zeta_j\zeta_k - \sum_{j,k,\ell=2}^{n} a_{jk\ell}\zeta_j\zeta_k\zeta_\ell \,,$$

$$z_r = \zeta_r + \frac{1}{2}\sum_{j,k=2}^{n} b_{jkr}\zeta_j\zeta_k + \frac{1}{2}\sum_{j=2}^{n} d_{jr}\zeta_1\zeta_j \,,$$

the surface (9.7.7.2) is transformed to $\operatorname{Re} z_1 = |z'|^2 +$(4th and higher $r$-order terms), while $Q$ is mapped to $(\tau + if\tau^3, 0)$.

The rest of the proof depends on the following claim:

**Claim** *There is a linear fractional transformation of the Siegel upper half-space* $\{(z_1, z') : \operatorname{Im} z_1 > |z'|^2\}$ *taking 0 to 0 and taking* $(\tau + if\tau^3, 0)$ *to* $(\tau, 0)$. *Moreover, the transformation is biholomorphic in a fixed neighborhood of* $(0, 0)$.

To see this, it is enough to check the case of the upper half plane in

$$\mathbb{C}^1 = \{z = (z_1, z') : z' = 0\} \,.$$

For, because of the explicit formulas for linear fractional transformations of the Siegel upper half space, we see that every linear fractional transformation of the upper half plane to itself extends to a linear fractional transformation of the full Siegel domain to itself. Moreover, the property of being biholomorphic in a fixed neighborhood of 0 is preserved. The easy 1-dimensional case of the claim is left for the interested reader.

Therefore we can make a holomorphic change of coordinates transforming $\partial\Omega$ to the surface

$$\operatorname{Re}\zeta_1 = |\zeta'|^2 - (\text{4th-order terms}) + (\text{5th- and higher-order terms}) \,. \tag{9.7.7.3}$$

It remains to show that the 4th-order terms can be made positive. However, the reader may easily check that the form (9.7.7.3) of the surface is preserved, while the 4th-order terms are forced to be nonnegative if we simply make the change of coordinates

$$z' = \zeta' + C\zeta_1\zeta'$$
$$z_1 = \zeta_1 - C\zeta_1^4 \,,$$

for $C > 0$ large enough. The reader may check that all the above changes of coordinates depend smoothly on $Q$. □

So we see quite explicitly with this proposition that a strictly pseudoconvex point has defining function which (after suitable modifications of the local coordinates) agrees with the ball up to fourth order. This result is tricky but elementary. Now we proceed to the guts of the approximation argument.

Fix a domain $\Omega$ and a point $p$ in the boundary which is strictly pseudoconvex. This means that the boundary near $p$ is $C^2$ and that the Levi form at $p$ is positive definite. Of course an obvious implication is that boundary points near to $p$ are also strictly pseudoconvex. Now we consider $A^2(\Omega)$ and the orthogonal space $A^2(\Omega)^\perp$. For a fixed point $w \in \Omega$, the Dirac mass $\delta_w$ breaks up uniquely into $A^2$ and $(A^2)^\perp$ components by the equation

$$\delta_w = K_\Omega(\cdot, w) + \big(\delta_w - K_\Omega(\cdot, w)\big). \tag{9.7.9}$$

We approximate the Bergman kernel $K_\Omega$ by producing an explicit $K^0 \in A^2$ and $K^+ \in (A^2)^\perp$ which add up to a small perturbation of $\delta_w$. Let $q$ be the point of $\partial\Omega$ which is nearest to $w$. We apply Proposition 9.7.7, setting $K^0(\cdot) = K_{\widetilde{\Omega}}(\cdot, w)$, $\widetilde{K}^+(\cdot) = \delta_w - K_{\widetilde{\Omega}}(\cdot, w)$ on $\widetilde{\Omega}$. We have $K^0 \in A^2(\widetilde{\Omega})$, $\widetilde{K}^+ \in (A^2_{\widetilde{\Omega}})^\perp$, and $\delta_w = K^0 + \widetilde{K}^+$ on $\widetilde{\Omega}$, so that $\delta_w = K^0\chi_{\widetilde{\Omega}} + \widetilde{K}^+\chi_{\widetilde{\Omega}}$ on $\Omega$. Clearly, since $\widetilde{K}^+ \in (A^2_{\widetilde{\Omega}})^\perp$ and $\widetilde{\Omega} \subseteq \Omega$, we know that $K^+ = \widetilde{K}^+\chi_{\widetilde{\Omega}} \in (A^2_\Omega)^\perp$. Furthermore, from the explicit formula for $K_{\widetilde{\Omega}}$, it follows immediately that $K^0$ continues analytically from $\widetilde{\Omega}$ to all of $\Omega$. [For fixed $w$ in the unit ball, $(1 - z \cdot \overline{w})^{-(n+1)}$ continues analytically beyond the unit sphere.] Thus we may write

$$\delta_w + K^0\chi_{\Omega\setminus\widetilde{\Omega}} = K^0 + K^+ \tag{9.7.10}$$

on $\Omega$, with $K^0 \in A^2_\Omega$ and $K^+ \in (A^2_\Omega)^\perp$ given by explicit formulas in terms of $(z, w)$.

In a sense that can be made precise, the term $K^0\chi_{\Omega\setminus\widetilde{\Omega}}$ on the lefthand side is small, since $\Omega \setminus \widetilde{\Omega}$ is such a thin subset of $\Omega$ (recall that $\partial\widetilde{\Omega}$ is highly tangent to $\partial\Omega$). To make this idea precise and quantitative, we associate to each kernel $A(z, w)$ on $\Omega \times \Omega$ the operator $\mathbf{A}f(z) = \int_\Omega A(z, w)f(w)\, dV(w)$ on $L^2(\Omega)A$. Thus (9.7.10) becomes an operator equation

$$\mathbf{I} + \mathcal{E} = \mathbf{K^0} + \mathbf{K}^+,$$

where $K^0$, $K^+$, $\mathcal{E}$ are given by kernels with explicit formulas and $\mathbf{K^0}f \in A^2(\Omega)$, $\mathbf{K}^+f \in (A^2_\Omega)^\perp$ for all $f \in L^2(\Omega)$. The thinness of $\Omega \setminus \widetilde{\Omega}$ shows that $\mathcal{E}$ has small norm as an operator on $L^2(\Omega)$, so that for $f \in L^2(\Omega)$,

$$\begin{aligned} f &= (\mathbf{I} + \mathcal{E})(\mathbf{I} + \mathcal{E})^{-1}f \\ &= (\mathbf{K^0} - \mathbf{K^0}\mathcal{E} + \mathbf{K^0}\mathcal{E^2} - \cdots)f + (\mathbf{K}^+ - \mathbf{K}^+\mathcal{E} + \cdots)f \\ &\equiv F + G \end{aligned}$$

with $F \in A^2_\Omega$ and $G \in (A^2_\Omega)^\perp$. Comparing this result with (9.7.8), we see that the Bergman kernel for $\Omega$ must be given by the operator equation

$$\mathbf{P}_\Omega = \mathbf{K^0} - \mathbf{K^0}\mathcal{E} + \mathbf{K^0}\mathcal{E^2} - \cdots, \tag{9.7.11}$$

the series converging in the norm topology of operators on $L^2(\Omega)$A. Note that $\mathbf{P}_\Omega$ is the Bergman projection on $\Omega$. We can obtain an asymptotic expansion for the Bergman kernel itself by apply both sides of this last operator equation to the Dirac delta mass. It is actually rather difficult to evaluate the righthand side of (9.7.11). Fefferman needs to develop an entire calculus of integral operators in order to do it. We cannot provide the details here, but instead refer the reader to [FEF1, Part I]. We note that a consequence of these calculations (and this should be compared to the result of Hörmander discussed in the earlier part of this section) is that

$$K_\Omega(z,z) = \Phi(z)\delta_\Omega^{-(n+1)}(z) + \widetilde{\Phi}(z)\log\delta_\Omega(z)$$

for $z$ near $\partial\Omega$. Here $\Phi, \widetilde{\Phi} \in C^\infty(\Omega)$, $\delta_\Omega(z)$ is the distance of $z$ to $\partial\Omega$, and $\Phi(z) \neq 0$ for $z \in \partial\Omega$. This is in fact the Corollary on page 45 of [FEF1].

It is worth noting here that Boutet de Monvel and Sjöstrand [BOUTS] show (Corollary 1.7) that, for the Szegő kernel,

$$K_\Omega(z,w) = \Phi(z,w)\delta_\Omega^{-(n+1)}(z,w) + \widetilde{\Phi}(z,w)\log\delta_\Omega(z,w)$$

for some $\Phi, \widetilde{\Phi} \in C^\infty(\overline{\Omega} \times \overline{\Omega})$. They are then able to relate their formula to Fefferman's.

A word of explanation is needed for our last statement. Where did the logarithmic term come from? This was one of the dramatic results of Fefferman's work—that the Bergman kernel of a strictly pseudoconvex domain can (at least in principle) contain a logarithmic term. And in fact Fefferman provides in his paper an explicit example of domain for which the logarithmic term actually occurs (see Example 9.7.12 below).

It is a noted conjecture of I. Ramadanov (see [RAM2]) that a strongly pseudoconvex domain with no logarithmic term for the Bergman kernel asymptotic expansion at any boundary point must be spherical. Dan Burns (unpublished) in fact proved such a result. L. Boutet de Monvel [BOUT] in dimension 2 and C. Robin Graham [GRA3] in general gave rigorous proofs of the result. See also the work of Hirachi [HIR1]. There are unbounded domains and also roughly bounded domains on which the analogue of this result for the Szegő kernel is known to fail—see [HIR2].

The logarithmic term arises in Fefferman's calculations because he is analyzing certain integral expressions—which are in effect negative powers of a nonisotropic distance function—using integration by parts. When the power of the nonisotropic distance is $-1$, then the next integration gives rise to a log term.

One of the nice features of the work of Boutet de Monvel and Sjőstrand is that they derive their asymptotic expansion for the Szegő kernel in a rather natural fashion from the explicit formula on the ball using the theory of Fourier integral operators (see [HOR3]). In particular, they produce a nice map from the given strongly pseudoconvex domain to the ball and then pull back the Bergman kernel via that map.

From this study they obtained the formula (see Theorem (1.5) in [BOUTS])

$$K(z,\zeta) = \int_0^\infty e^{i\delta_\Omega(z,w)} b(z,w,t)\, dt\,,$$

where $b \in C^\infty(\overline{\Omega}, \overline{\Omega}, \mathbb{R}^+)$.

We conclude this section with a presentation of Fefferman's example of a domain whose Bergman kernel has a logarithmic term. We will exploit this form of the asymptotic expansion:

$$K_\Omega(z,w) = C|\text{grad}\delta_\Omega(w)|^2 \det\mathcal{L}(w) X^{-(n+1)}(z,w) + \widetilde{K}(z,w)\,,$$

where $\widetilde{K}$ is an admissible kernel of weight $\geq n-1$. Here $\mathcal{L}$ is the Levi form. Also "weight" is a concept that Fefferman introduces as part of his integral calculus. We can safely take it for granted.

**EXAMPLE 9.7.12** Let $\Omega$ be the connected component of

$$\{(z_1,z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^2 - c|z_2|^8 < 1\}$$

containing the origin, where $c$ is small and positive. Note that this $\Omega$ is highly tangential to the unit ball $B$ at the point $(1,0)$. Also $B \subseteq \Omega$. Now set

$$\psi(w) = 1 - |w_1|^2 - |w_2|^2 + c|w_3|^8$$

and $w^0 = (\tau, 0)$ with $0 < \tau < 1$. Now the above formulation of the asymptotic expansion for the Bergman kernel tells us that

$$K_\Omega(z,w^0)) = \varphi_0(w^0)(1-\tau z_1)^{-3} + \sum_{j=1}^{M} \varphi_j(z,w^0)(1-\tau z_1)^{-m_j}$$
$$+\widetilde{\varphi}(z,w^0)\log(1-\tau z_1) + O(1)\,,$$

with $\varphi_0(w^0) \neq 0$ and $\text{weight}(\varphi_j) - m_j \geq -5/2$. We shall show that

$$\lim_{t\to 1^-} \widetilde{\varphi}(w^0,w^0) \neq 0\,.$$

To do so, we apply the reproducing property of $K_\Omega$ to the anti-holomorphic function $F(w) = K_B(w^0,w) = c_1(1-\tau\overline{w}_1)^{-3}$ to obtain

$$c_1(1-\tau^2)^{-3} = F(w^0)$$
$$= \int_\Omega F(z) K_\Omega(z,w^0)\, dV(z)$$

$$
\begin{aligned}
&= \int_\Omega c_1(1-\tau\overline{z}_1)^{-3} K_\Omega(z,w^0)\,dV(z) \\
&= \int_B c_1(1-\tau\overline{z}_1)^{-3} K_\Omega(z,w^0)\,dV(z) \\
&\qquad + \int_{\Omega\setminus B} c_1(1-\tau\overline{z}_1)^{-3} K_\Omega(z,w^0)\,dV(z) \\
&= K_\Omega(w^0,w^0) + \int_{\Omega\setminus B} c_1(1-\tau\overline{z}_1)^{-3} K_\Omega(z,w^0)\,dV(z) \qquad (9.7.12.1)
\end{aligned}
$$

(since $c_1(1-\tau\overline{z}_1)^{-3} = K_B(w^0,z)$ and $K_\Omega(\cdot\,,w^0)$ is holomorphic on $B \subseteq \Omega$). We find that

$$
\begin{aligned}
&c_1(1-\tau^2)^{-3} - \varphi_0(w^0)(1-\tau^2)^{-3} - \sum_{j=1}^{M} \varphi_j(w^0,w^0)(1-\tau^2)^{-m_j} \\
&\qquad -\widetilde{\varphi}(w^0,w^0)\log(1-\tau^2) + O(1) \\
&= c_1\varphi_0(w^0)\int_{\Omega\setminus B} |1-\tau z_1|^{-6}\,dV(z) \\
&\qquad + c_1\Bigg\{\int_{\Omega\setminus B}(1-\tau\overline{z}_1)^{-3}\Bigg[\sum_{j=1}^{M}\varphi_j(z,w^0)\cdot(1-\tau z_1)^{-m_j} \\
&\qquad\qquad + \widetilde{\varphi}(z,w^0)\log(1-\tau z_1) + O(1)\Bigg]dV(z)\Bigg\} \\
&= c_1\varphi_0(w^0)\int_{\Omega\setminus B} |1-\tau z_1|^{-6}\,dV(z) + O(1) \\
&\sim \log(1-\tau^2) \qquad \text{as } \tau \to 1^-\,.
\end{aligned}
$$

We see immediately that $\lim_{\tau\to 1^-}\widetilde{\varphi}(w^0,w^0)$ could not be zero. Therefore the term $\widetilde{\varphi}(z)$ is really present in $K_\Omega(z,z)$. □

We conclude this section by noting that Fefferman developed his ideas further by formulating a program for studying the geometry and analysis of strictly pseudoconvex domains. See [FEF2]. The main idea is to consider the Bergman and Szegő kernels as analogues of the heat kernel of a Riemannian manifold. In Riemannian geometry, the coefficients of the asymptotic expansion of the heat kernel can be expressed in terms of the curvature of the metric. Integrating the coefficients, one may obtain index theorems in various settings. See [BEA] and [HIR3] for some of the details.

We also mention that, in his famous problem list, S.-T. Yau [YAU] raises the question of classifying pseudoconvex domains whose Bergman metrics are Kähler–Einstein. S.-Y. Cheng [CHSG] conjectured that, if the Bergman metric of a strictly

pseudoconvex domain is Kähler–Einstein, then the domain is biholomorphic to the ball. This conjecture was proved by Fu–Wong [FUW] in the case of a simply connected, strictly pseudoconvex domain with smooth boundary.

## 9.8 Regularity for the Dirichlet Problem

**Capsule:** In this section we explore some properties of the classical Dirichlet problem for the Laplacian. We are particularly interested in boundary regularity. And there are applications to boundary smoothness of conformal mappings.

We begin by giving a precise definition of a domain "with smooth boundary:"

**Definition 9.8.1** Let $U \subseteq \mathbb{C}$ be a bounded domain. We say that $U$ has *smooth boundary* if the boundary consists of finitely many curves and each of these is locally the graph of a $C^\infty$ function.

In practice it is more convenient to have a different definition of domain with smooth boundary. A function $\rho$ is called a *defining function* for $U$ if $\rho$ is defined in a neighborhood $W$ of $\partial U$, $\nabla\rho \neq 0$ on $\partial U$, and $W \cap U = \{z \in W : \rho(z) < 0\}$. Now we say that $U$ has smooth (or $C^k$) boundary if $U$ has a defining function $\rho$ that is smooth (or $C^k$). Yet a third definition of smooth boundary is that it consists of finitely many curves $\gamma_j$, each of which is the trace of a smooth curve $\mathbf{r}(t)$ with nonvanishing gradient. We invite the reader to verify that these three definitions are equivalent. See also the Appendix I in [KRA1].

Our motivating question for the present section is as follows:

**MAIN QUESTION:** Let $\Omega \subseteq \mathbb{C}$ be a bounded domain with smooth boundary. Let $0 < \alpha < 1$. Assume that $f \in \Lambda_\alpha(\partial\Omega)$. If $u \in C(\overline{\Omega})$ satisfies (i) $u$ is harmonic on $\Omega$ and (ii) $u\big|_{\partial\Omega} = f$, then does it follow that $u \in \Lambda_\alpha(\overline{\Omega})$?

Here $\Lambda_\alpha$ is the usual Lipschitz space (see [KRA6, KRA12]). Here is a scheme for answering this question:

**Step 1:** Suppose at first that $U$ is bounded and simply connected.
**Step 2:** By the Riemann mapping theorem, there is a conformal mapping $\phi : U \to D$. Here $D$ is the unit disc. We would like to reduce our problem to the Dirichlet problem on $D$ for the data $f \circ \phi^{-1}$.

In order to carry out this program, we need to know that $\phi$ extends smoothly to the boundary. It is a classical result of Caratheodory [CAR] that, if a simply connected domain $U$ has boundary consisting of a Jordan curve, then any conformal map of the domain to the disc extends univalently and bicontinuously to the boundary. It is less well known that Painlevé, in his thesis [PAI], proved that when $U$ has smooth boundary then the conformal mapping extends smoothly to the boundary. In fact Painlevé's result long precedes that of Carathéodory.

We shall present here a modern approach to smoothness to the boundary for conformal mappings. These ideas come from [KER2]. See also [BEK] for a self-contained approach to these matters, as well as our presentation in Section 7.1. Our

purpose here is to tie the smoothness-to-the-boundary issue for mappings directly to the regularity theory of the Dirichlet problem for the Laplacian.

Let $W$ be a collared neighborhood of $\partial U$. Set $\partial U' = \partial W \cap U$ and let $\partial D' = \phi(\partial U')$. Define $B$ to be the region bounded by $\partial D$ and $\partial D'$. We solve the Dirichlet problem on $B$ with boundary data

$$f(\zeta) = \begin{cases} 1 \text{ if } \zeta \in \partial D \\ 0 \text{ if } \zeta \in \partial D' \end{cases}$$

Call the solution $u$.

Consider $v \equiv u \circ \phi : U \to \mathbb{R}$. Then of course $v$ is still harmonic. By Carathéodory's theorem, $v$ extends to $\partial\Omega, \partial\Omega'$, and

$$v = \begin{cases} 1 \text{ if } \zeta \in \partial U \\ 0 \text{ if } \zeta \in \partial U' \end{cases}$$

Suppose that we knew that solutions of the Dirichlet problem on a smoothly bounded domain with $C^\infty$ data are in fact $C^\infty$ on the closure of the domain. Then, if we consider a first-order derivative $\mathcal{D}$ of $v$, we obtain

$$|\mathcal{D}v| = |\mathcal{D}(u \circ \phi)| = |\nabla u|\,|\nabla \phi| \leq C.$$

It follows that

$$|\nabla \phi| \leq \frac{C}{|\nabla u|}. \tag{9.8.2}$$

This will prove to be a useful estimate once we take advantage of the following:

**Lemma 9.8.3 (Hopf)** *Let $\Omega \subset\subset \mathbb{R}^N$ have $C^2$ boundary. Let $u \in C(\overline{\Omega})$ with $u$ harmonic and non-constant on $\Omega$. Let $P \in \overline{\Omega}$ and assume that $u$ takes a local minimum at $P$. Then*

$$\frac{\partial u}{\partial \nu}(P) < 0.$$

**Proof** Suppose without loss of generality that $u > 0$ on $\Omega$ near $P$ and that $u(P) = 0$. Let $B_R$ be a ball that is internally tangent to $\overline{\Omega}$ at $P$. We may assume that the center of this ball is at the origin and that $P$ has coordinates $(R, 0, \ldots, 0)$. Then, by Harnack's inequality (see [GRK6]), we have for $0 < r < R$ that

$$u(r, 0, \ldots, 0) \geq c \cdot \frac{R^2 - r^2}{R^2 + r^2}$$

hence

$$\frac{u(r, 0, \ldots, 0) - u(R, 0, \ldots, 0)}{r - R} \leq -c' < 0.$$

Therefore

$$\frac{\partial u}{\partial v}(P) \leq -c' < 0.$$

This is the desired result. □

Now let us return to the $u$ from the Dirichlet problem that we considered prior to line (9.8.2). Hopf's lemma tells us that $|\nabla u| \geq c' > 0$ near $\partial D$. Thus, from (9.8.2), we conclude that

$$|\nabla \phi| \leq C. \tag{9.8.4}$$

Hence we have bounds on the first derivatives of $\phi$.

To control the second derivatives, we calculate that

$$\begin{aligned} C \geq |\nabla^2 v| &= |\nabla(\nabla v)| = |\nabla(\nabla(u \circ \phi))| \\ &= |\nabla(\nabla u(\phi) \cdot \nabla \phi)| = |\left(\nabla^2 u \cdot [\nabla \phi]^2\right) + \left(\nabla u \cdot \nabla^2 \phi\right)|. \end{aligned}$$

Here the reader should think of $\nabla$ as representing a generic first derivative and $\nabla^2$ a generic second derivative. We conclude that

$$|\nabla u|\,|\nabla^2 \phi| \leq C + |\nabla^2 u|\,|(\nabla \phi)^2| \leq C'.$$

Hence (again using Hopf's lemma)

$$|\nabla^2 \phi| \leq \frac{C}{|\nabla u|} \leq C''.$$

In the same fashion, we may prove that $|\nabla^k \phi| \leq C_k$, any $k \in \{1, 2, \ldots\}$. This means (use the fundamental theorem of calculus) that $\phi \in C^\infty(\overline{\Omega})$.

We have arrived at the following situation: Smoothness to the boundary of conformal maps implies regularity of the Dirichlet problem on a smoothly bounded domain. Conversely, regularity of the Dirichlet problem can be used, together with Hopf's lemma, to prove the smoothness to the boundary of conformal mappings. We must find a way out of this impasse.

A possible solution to the problem posed in the last paragraph is to study the Dirichlet problem for a more general class of operators that is invariant under *smooth* changes of coordinates. We study these operators by (i) localizing the problem and (ii) mapping the smooth domain under a diffeomorphism to an upper half space. It will turn out that *elliptic operators* are invariant under these operations. We can then use the calculus of pseudo-differential operators to prove local boundary regularity for elliptic operators.

We do not actually carry out here the program described in the last paragraph. The reference [KRA5, Ch. 4–5] contains all the details.

There is an important point implicit in our discussion that deserves to be brought into the foreground. The Laplacian is invariant under conformal transformations (exercise). This observation was useful in setting up the discussion in the present section. But it turned out to be a point of view that is too narrow: we found ourselves in a situation of circular reasoning. We thus expanded our point of view to a wider universe in which our operators are invariant under *diffeomorphisms*. This type of invariance will give us more flexibility and more power. See [KRA5, Ch. 3] for background to this discussion.

Let us conclude this section by exploring how the Laplacian behaves under a diffeomorphic change of coordinates. For simplicity we restrict attention to $\mathbb{R}^2$ with coordinates $(x, y)$. Let

$$\phi(x, y) = (\phi_1(x, y), \phi_2(x, y)) \equiv (x', y')$$

be a diffeomorphism of $\mathbb{R}^2$. Let

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

In $(x', y')$ coordinates, the operator $\Delta$ becomes

$$\phi_*(\Delta) = |\nabla\phi_1|^2 \frac{\partial^2}{\partial x'^2} + |\nabla\phi_2|^2 \frac{\partial^2}{\partial y'^2}$$

$$+2\left[\frac{\partial x'}{\partial x}\frac{\partial y'}{\partial x} + \frac{\partial x'}{\partial y}\frac{\partial y'}{\partial y}\right]\frac{\partial^2}{\partial x \partial y} + (\text{first-order terms}).$$

In an effort to see what the new operator has in common with the old one, we introduce the notation

$$D = \sum a_\alpha \frac{\partial}{\partial x^\alpha},$$

where

$$\frac{\partial}{\partial x^\alpha} = \frac{\partial}{\partial x_1^{\alpha_1}} \frac{\partial}{\partial x_2^{\alpha_2}} \cdots \frac{\partial}{\partial x_n^{\alpha_n}}$$

is a differential monomial. Its "symbol" is defined to be

$$\sigma(D) = \sum a_\alpha(x)\xi^\alpha, \qquad \xi^\alpha = \xi_1^{\alpha_1}\xi_2^{\alpha_2}\cdots\xi_n^{\alpha_n}.$$

The symbol of the Laplacian $\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ is:

$$\sigma(\Delta) = \xi_1^2 + \xi_2^2.$$

Now associate to $\sigma(\Delta)$ a matrix $A_\Delta = (a_{ij})_{1\le i,j\le 2}$, where $a_{ij} = a_{ij}(x)$ is the coefficient of $\xi_i\xi_j$ in the symbol. Thus

$$A_\Delta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The symbol of the transformed Laplacian (in the new coordinates) is

$$\begin{aligned}\sigma(\phi_*(\Delta)) = {} & |\nabla\phi_1|^2\xi_1^2 + |\nabla\phi_2|^2\xi_2^2 \\ & + 2\left[\frac{\partial x'}{\partial x}\frac{\partial y'}{\partial y} + \frac{\partial x'}{\partial y}\frac{\partial y'}{\partial y}\right]\xi_1\xi_2 \\ & + (\text{lower order terms}).\end{aligned}$$

Then

$$A_{\sigma(\phi^*(\Delta))} = \begin{pmatrix} |\nabla\phi_1|^2 & \left[\frac{\partial x'}{\partial x}\frac{\partial y'}{\partial x} + \frac{\partial x'}{\partial y}\frac{\partial y'}{\partial y}\right] \\ \left[\frac{\partial x'}{\partial x}\frac{\partial y'}{\partial x} + \frac{\partial x'}{\partial y}\frac{\partial y'}{\partial y}\right] & |\nabla\phi_2|^2 \end{pmatrix}$$

The matrix $A_{\sigma(\phi^*(\Delta))}$ is positive definite provided that the change of coordinates $\phi$ is a diffeomorphism (i.e., has non-degenerate Jacobian). It is this positive definiteness property of the symbol that is crucial to the success of the theory of pseudodifferential operators (see [KRA5, Ch. 3]). For our study of the boundary regularity of conformal mappings, the transformation properties of the Laplacian under holomorphic mappings was sufficient.

## 9.9 Plurisubharmonic Defining Functions

**Capsule:** Just as the boundary regularity for the classical Dirichlet problem can be used to study the boundary regularity of conformal mappings, so the boundary behavior of the Monge–Ampère equation can be used to study the boundary regularity of biholomorphic mappings of strongly pseudoconvex domains.

In fact this is an old idea of Kerzman, Kohn, and Nirenberg [KER3]. But an example of Bedford and Fornaess [BEF2] suggested that the needed regularity of the Monge–Ampère equation is false. It was only later that Krantz and Li [KRL4] found a way around this difficulty and did carry out the program proposed in [KER3]. We discuss these matters in the present section.

In classical analysis, an important theorem of Painlevé [PAI] and Kellogg [KEL] states that any conformal mapping between two smoothly bounded domains in the complex plane $\mathbb{C}$ can be extended to be a diffeomorphism on the closures of the domains. This theorem was generalized by C. Fefferman [FEF1, Part I] in 1974 to strictly pseudoconvex domains in $\mathbb{C}^n$. Fefferman's original proof of this theorem

is very technical, relying as it does on deep work on the boundary asymptotics of the Bergman kernel and on the regularity of $\overline{\partial}$-Neumann operator that is due to J. J. Kohn [FOK]. Bell–Ligocka [BELL], and later Bell [BE1], gave a simpler proof which deals with more general domains, including pseudoconvex domains of finite type, by using regularity of the Bergman projection and the $\overline{\partial}$-Neumann operator as studied by Kohn [FOK], Catlin [CAT2], Boas–Straube [BOAS1], and others.

We know from [KER3] that Painlevé and Kellogg's theorem can be proved by using the regularity of the Dirichlet problem for the Laplacian in a smoothly bounded planar domain, where the property of the Laplacian being conformally invariant plays an important role in the proof. [See [BEK] for another point of view.] The natural generalization of the Laplacian in one complex variable to several complex variables, with these considerations in mind, is the complex Monge–Ampère equation. In [KER3], Kerzman observed that the proof of the Fefferman mapping theorem would follow from the $C^\infty$ global regularity of the Dirichlet problem of a degenerate complex Monge–Ampère equation. However, counterexamples in Bedford–Fornæss [BEF2] as well as in Gamelin–Sibony [GAN] show that, in general, the degenerate Dirichlet problem for the complex Monge–Ampère equation does not have $C^2$ boundary regularity. Thus Kerzman's idea does not work in the sense of its original formulation.

The main purpose of the present section is to construct a plurisubharmonic defining function $\rho$ for a smoothly bounded strictly pseudoconvex domain in $\mathbb{C}^n$ with $\det H(\rho)$ vanishing to higher order near the boundary, where $H(\rho)$ denotes the complex hessian of $\rho$. In other words, we shall prove the following theorem.

**Prelude** This next is the main result of the present section. It is our technical tool for obtaining regularity for the Monge–Ampère equation and then for studying the boundary regularity of biholomorphic mappings.

**Theorem 9.9.1** *Let $\Omega$ be a bounded strictly pseudoconvex domain in $\mathbb{C}^n$ with $C^\infty$ boundary $\partial\Omega$. For any $0 < \epsilon << 1$ and any positive integer $q$ there is a plurisubharmonic defining function $\rho_q \in C^\infty(\overline{\Omega})$ for $\Omega$ so that*

$$\det H(\rho_q)(z) \leq C\text{dist}\,(z, \partial\Omega)^q, \quad z \in \Omega\,. \tag{9.9.1.1}$$

Combining Theorem 9.9.1 with a result of Caffarelli–Kohn–Nirenberg–Spruck [CAF] on the Dirichlet problem for the complex Monge–Ampère equation, we provide a new proof for the following result, which was proved in [BERNC] and [DIRF4].

**Corollary 9.9.2** *Let $\Omega_j, j = 1, 2$ be two bounded strictly pseudoconvex domains in $\mathbb{C}^n$ with $C^\infty$ boundary. Let $\varphi : \Omega_1 \to \Omega_2$ be a proper holomorphic mapping. Then for any $\epsilon > 0$ we have*

*(i)* *$\varphi$ can be extended as a $Lip_1(\overline{\Omega}_1)$ mapping;*
*(ii)* *$\det(\varphi') \in Lip_{1/2}(\overline{\Omega}_1)$;*
*(iii)* *There is a $C^2$ defining function $\rho$ for $\Omega_2$ so that, for any $k = 1, \ldots, n$, we have $\sum_{j=1}^n \left[\frac{\partial\rho}{\partial w_j} \circ \varphi\right] \cdot \frac{\partial\varphi_j}{\partial z_k} \in Lip_1(\overline{\Omega}_1)$.*

As a corollary of (ii) and of a theorem in [FOST, NWY, PIV] and [WEB2], we may give a new proof of the Fefferman mapping theorem.

Our presentation is organized as follows. In the next section, Theorem 9.9.1 is proved. In the following section, some applications are given.

## 9.10 Proof of Theorem 9.9.1

Now we prove the main result of the last section.

We begin with the proof of the first theorem.

**Proof of Theorem 9.9.1** Let $\delta(z)$ denote the (signed) distance function from $z$ to $\partial\Omega$ (positive inside and negative outside). Since $\Omega$ is a bounded domain in $\mathbb{C}^n$ with $C^\infty$ boundary $\partial\Omega$, then $\delta(z) \in C^\infty(\overline{\Omega})$ (after modification of the distance function on a compact set in the interior). By a rotation we see that, for any fixed point $z^0 \in \Omega$ near the boundary, we may assume that the $z_n$ direction is the normal direction $(\frac{\partial\delta(z)}{\partial\overline{z}_1}, \cdots, \frac{\partial\delta}{\partial\overline{z}_n})$ at the point $z^0$. Let

$$H(\delta)_{n-1}(z) = \left[\frac{\partial^2\delta}{\partial z_\ell \partial\overline{z}_q}(z)\right]_{1\le\ell,q\le n-1}$$

Since $\Omega$ is strictly pseudoconvex, there is an $\epsilon > 0$ so that

$$-H(\delta)_{n-1}(z^0) \ge \epsilon I_{n-1}$$

for all $z \in \overline{\Omega}$ with $\delta(z) \le \epsilon$. We may assume that $H(\delta)_{n-1}$ is diagonalized at $z^0$. Note that

$$H(-\delta)(z) = \begin{bmatrix} -H(\delta)_{n-1}(z) & 0 \\ 0 & \frac{-\partial^2\delta}{\partial z_n\partial\overline{z}_n} \end{bmatrix}$$

If it happens that $\sum_{ij}\frac{\partial\delta}{\partial z_i}\frac{\partial\delta}{\partial\overline{z}_j}\frac{\partial^2\delta}{\partial z_i\partial\overline{z}_j} = 0$, then $\rho_q = -\delta(z) + \delta(z)^{2+q}$ is the desired defining function. [In fact it is these terms that distinguish the study of the real Hessian from the more subtle study of the complex Hessian. In particular, we know that $\frac{\partial^2\delta}{\partial x_n^2}$ equals 0, but the term $\partial^2\delta/\partial z_n\partial\overline{z}_n$ may not be zero. Therefore estimate (9.9.1.1) is much easier to check for the determinant of the real Hessian of $\delta$; matters are much trickier for the complex Hessian.]

Now we let

$$\rho(z) = -\delta(z)$$

and

$$r^{[m]}(z) = \rho(z) + \sum_{k=2}^{m} a_k(z)\delta(z)^k \quad z \in \Omega\,.$$

We will prove inductively that

$$\det(H(r^{[m]})(z)) = b_{m-1}(z) \cdot \rho^{m-1},$$

where $b_{m-1}$ is some smooth function. In particular, $\det(H(r^{[m]}))$ vanishes to order $m-1$ at the boundary.

Now

$$\partial_{i\bar{j}} r^{[m]}(z) = \partial_{i\bar{j}}\rho(z) + \sum_{k=2}^{m}\left[\partial_{i\bar{j}}a_k\,\rho(z)^2 + k(k-1)a_k\partial_i\rho\partial_{\bar{j}}\rho)\right]\rho^{k-2}$$
$$+\sum_{k=2}^{m} k\left[\partial_i a_k\partial_{\bar{j}}\rho + \partial_i\rho\partial_{\bar{j}}a_k + a_k\rho_{i\bar{j}}\right]\rho^{k-1}$$

Let $H(r^{[m]})_{n-1}(z) = [\frac{\partial^2 r^{[m]}}{\partial z_i \partial\bar{z}_j}]_{(n-1)\times(n-1)}$, and let $B(z)^* = [r_{n\bar{1}}, \cdots, r_{n\overline{n-1}}]$ be a row-vector. Then

$$H(r^{[m]})(z) = \begin{bmatrix} H(r^{[m]})_{n-1}(z) & B(z) \\ B(z)^* & r^{[m]}_{n\bar{n}}(z) \end{bmatrix}$$

Then

$$\det\left(H(r^{[m]})(z)\right) = \det\left(H(r^{[m]})_{n-1}(z)\right)[r^{[m]}_{n\bar{n}} - \{B^*H(r^{[m]})_{n-1}(z)\}^{-1}B(z)]$$

We know that

$$H(r^{[m]})_{n-1}(z) \geq \epsilon I_{n-1}$$

for all $z \in \Omega_\epsilon$, where $\epsilon$ is a positive number depending only on $\Omega$ and $\|a_j\|_{C^2(\overline{\Omega})}$. At first, we let

$$(d\rho)(z) = \sum_{i,j}\rho_i\rho_{\bar{j}}\rho_{i\bar{j}}\,;$$

then we define

$$a_2(z) = -\frac{(d\rho)(z)}{2((d\rho)(z) + |\partial\rho|^4)}.$$

Thus we have

$$\begin{aligned} r_{n\overline{n}}(z) &= \rho_{n\overline{n}} + \sum_{k=2}^{m}[\partial_{n\overline{n}}a_k\rho^k + k(k-1)a_k|\partial\rho|^2\rho^{k-2}] \\ &\quad + \sum_{k=2}^{m} k[\partial_n a_k \partial_{\overline{n}}\rho + \partial_n\rho\partial_{\overline{n}}a_k + a_k\rho_{n\overline{n}}]\rho^{k-1} \\ &= \sum_{k=2}^{m}[\partial_{n\overline{n}}a_k\rho^k + k(k+1)a_{k+1}|\partial\rho|^2\rho^{k-1}] \\ &\quad + \sum_{k=2}^{m} k[\partial_n a_k \partial_{\overline{n}}\rho + \partial_n\rho\partial_{\overline{n}}a_k + a_k\rho_{n\overline{n}}]\rho^{k-1} \\ &= O(\rho) \end{aligned}$$

Therefore

$$\det\left(H(r^{[2]})(z)\right) = b_1(z)\rho(z)$$

Assume that we have constructed $r^{[m]} = \rho(z) + \sum_{k=2}^{m} a_k\rho(z)^k$ such that

$$\det\left(H(r^{[m]})(z)\right) = b_{m-1}(z)\rho^{m-1}, \quad z \in \Omega.$$

We consider

$$r^{[m+1]} = r^{[m]} + a_{m+1}\rho(z)^{m+1}.$$

Since

$$\begin{aligned} &H(r^{[m+1]})(z) \\ &= H(r^{[m]}) + \rho(z)^{m+1}H(a_{m+1})(z) + (m+1)a_{m+1}\rho(z)^m H(\rho) \\ &\quad +(m+1)ma_{m+1}\rho^{m-1}\partial\rho(z)\otimes\overline{\partial\rho} \\ &\quad +(m+1)\rho(z)^m[\partial\rho\otimes\overline{\partial a_{m+1}} + \partial a_{m+1}\otimes\overline{\partial\rho}], \end{aligned}$$

it is easy to see that

$$\det\left(H(r^{[m+1]})(z)\right) = b_m(z)\rho(z)^m) + \det\left[H(r^{[m]}) + (m+1)ma_{m+1}\rho^{m-1}\right].$$

By a rotation, we may let $z_n$ be the complex normal direction of $\partial\Omega_{\delta(z)}$ at $z$. Thus

$$\det\left(H(r^{[m]})(z) + (m+1)m\rho(z)^{m-1}\partial\rho(z)\otimes\overline{\partial\rho}\right) = \det M,$$

where

$$M = \begin{bmatrix} H(r^{[m]})_{n-1}(z) & B(m)(z) \\ B(m)(z)^* & r^{[m]}_{n\overline{n}}(z) + (m+1)m\rho^{m-1}\rho_n\rho_{\overline{n}} \end{bmatrix}.$$

But this

$$\begin{aligned} &= \det\left(H(r^{[m]})_{n-1}(z)[r^{[m]}_{n\overline{n}} + B(m)^* H(r^{[m]})_{n-1}(z)^{-1} B(m)(z)\right. \\ &\qquad \left. +(m+1)m\rho^{m-1}\rho_n\rho_{\overline{n}}\right] \\ &= \det\left(H(r^{[m]})(z)\right) + \det\left(H(r^{[m]})_{n-1}(z)\right)(m+1)m\rho_n\rho_{\overline{n}}\rho(z)^{m-1} \\ &= b_{m-1}(z)\rho(z)^{m-1} + \det\left(H(r^{[m]})_{n-1}(z)\right)(m+1)m\rho_n\rho_{\overline{n}}\rho(z)^{m-1}. \end{aligned}$$

We need to choose $a_{m+1}$ such that

$$b_{m-1}(z) + (m+1)m\det\left(H(r^{[m]})_{n-1}(z)\right)\rho_n\rho_{\overline{n}}(z) = 0$$

From this it will follow that

$$\det\left(H(r^{[m+1]})(z)\right) = b_m(z)\rho(z)^m.$$

From the construction, we know that $H(r^{[m]})_{n-1}(z)$ is positive definite with least positive eigenvalue $\epsilon_m$ for all $z \in \Omega \setminus \Omega_{\epsilon_m}$. Thus if $c_m$ is large enough so that $|\nabla\delta|^2 c_m \geq 2|b_m(z)|$ then, by choosing $\epsilon_m$ small enough, we will easily see that the function

$$\rho_m(z) \equiv r^{[m+1]}(z) + c_m\delta(z)^{m+2} \tag{9.9.1.2}$$

is positive definite in $\Omega \setminus \Omega_{\epsilon_m}$ and (9.9.1.1) holds on $\Omega \setminus \Omega_{\epsilon_m}$. Here $\Omega_t = \{z \in \Omega : \delta(z) < t\}$. Then we use arguments in [CAF] and [LI1] to extend $\rho_m$ to be defined on $\Omega$ and strictly pseudoconvex on $\Omega$. The proof of the theorem is complete. □

## 9.11 Uses of the Monge–Ampère Equation

**Capsule:** It is in this section that we apply our earlier results of this chapter to study the boundary regularity of biholomorphic mappings.

In this section, we shall prove Corollary 9.9.2. Let us recall a theorem of Caffarelli, Kohn, Nirenberg, and Spruck [CAF].

**Prelude** This next theorem is the inspiration for our technique. It is a profound idea of Caffarelli, Kohn, Nirenberg, and Spruck.

**Theorem 9.11.1** *Let $\Omega$ be a bounded, strictly pseudoconvex domain in $\mathbb{C}^n$ with $C^4$ boundary $\partial\Omega$. Let $f$ be a nonnegative function on $\overline{\Omega}$ such that $f(z)^{1/n} \in C^{1,1}(\overline{\Omega})$. Let $H(u)$ denote the complex Hessian matrix of the function $u$. Then there is a unique plurisubharmonic function $v \in C^{1,1}(\overline{\Omega})$ satisfying*

$$\begin{aligned} \det H(v)(z) &= f(z) \quad \textit{for } z \in \Omega; \\ v &= 0 \qquad \textit{for } z \in \partial\Omega\,. \end{aligned} \tag{9.11.1.1}$$

**Proof of Corollary 9.9.2** Let $\varphi : \Omega_1 \to \Omega_2$ be a proper holomorphic mapping, and let $\rho_{4n}$ be the plurisubharmonic defining function for $\Omega_2$, with $m = 4n$, that we constructed in Theorem 9.9.1. We let

$$r(z) = \rho_{4n}(\varphi(z)), \qquad z \in \Omega_1.$$

Then we have

$$\det H(r)(z) = \det\left(H(\rho_{4n})(\varphi(z))\right)|\det \varphi'(z)|^2 = f(z), \quad z \in \Omega_1$$

Since $\det\left(H(\rho_{4n}(\varphi(z)) \le O(\rho(\varphi(z))^{4n} \le C\text{dist}\,(z, \partial\Omega_1)^4 n$ and $|\det \varphi'(z)|^2 \le C\text{dist}\,(z, \partial\Omega)^{-2n}$. We have $f(z)^{1/n} \in C^{1,1}(\overline{\Omega}_1)$ and $f \ge 0$. We know that $r \in C^{1,1}(\overline{\Omega}_1) \cap C^2(\Omega_1)$. So

$$\|r\|_{C^{1,1}(\overline{\Omega}_1)} \le C.$$

It is obvious that $\exp(\rho_{4n})$ is strictly plurisubharmonic in $\Omega_2$ and that there is a constant $\epsilon > 0$ so that

$$H(\exp(\rho_{4n})(w) \ge \epsilon I_n, \quad \text{for all} \quad w \in \Omega_2.$$

Thus

$$\begin{aligned} \frac{\partial^2 \exp(r(z))}{\partial z_\ell \partial \overline{z}_\ell} &= \sum_{p,\ell=1}^{n} \frac{\partial^2 \exp(\rho_{4n})}{\partial w_p \partial \overline{w}_\ell}(\varphi(z)) \frac{\partial \varphi_p}{\partial z_\ell} \overline{\frac{\partial \varphi_\ell}{\partial z_\ell}}(z) \\ &\ge \epsilon \sum_{p=1}^{n} \left| \frac{\partial \varphi_p}{\partial z_\ell}(z) \right|^2. \end{aligned}$$

This shows that

$$\|\varphi\|^2_{\text{Lip}(\overline{\Omega}_1)} \le C\lambda_0^{-1} \|\exp(r)\|_{C^{1,1}(\overline{\Omega}_1)}.$$

Thus we have $\det(\varphi'(z)) \in L^\infty(\Omega_1)$.

If we apply $\frac{\partial^2}{\partial z_\ell \partial z_m}$ to $r(z)$ and use the above result, then we have

$$\left| \sum_{p=1}^{n} \frac{\partial \rho_{4n}}{\partial w_p}(\varphi(z)) \frac{\partial^2 \varphi_p}{\partial z_\ell \partial z_m} \right| \leq C\,. \tag{9.9.2.1}$$

Let $z^0 \in \Omega_1$ be sufficiently near to $\partial\Omega_1$. Without loss of generality, by applying a rotation, we may assume that $z_1^0, \cdots, z_{n-1}^0$ are complex tangential at $z^0$ and also that at the point $\varphi(z_0)$ the directions $w_1, \cdots, w_{n-1}$ are complex tangential. Thus

$$\begin{aligned}
&\frac{\partial}{\partial z_p} \log(\det(\varphi'(z_0)))(z) \\
&= \sum_{\ell,m=1}^{n} \varphi^{\ell m} \frac{\partial^2 \varphi_m}{\partial z_\ell \partial z_p}(z) \\
&= \sum_{\ell<n} \varphi^{\ell m} \frac{\partial^2 \varphi_m}{\partial z_\ell \partial z_p}(z) + \sum_{m<n} \varphi^{nm} \frac{\partial^2 \varphi_m}{\partial z_n \partial z_p}(z) + \varphi^{nn} \frac{\partial^2 \varphi_n}{\partial z_n \partial z_p}(z),
\end{aligned}$$

where $(\varphi^{\ell m})$ is the inverse matrix of $\varphi'(z)$. Since $\varphi \in \mathrm{Lip}_1(\overline{\Omega}_1)$, we have

$$\left| \frac{\partial^2 \varphi_m}{\partial z_\ell \partial z_p}(z_0) \right| \leq C\delta_1(z)^{-1/2}$$

for all $1 \leq \ell \leq n-1$ and $1 \leq m, p \leq n$. By (9.9.2.1), we have

$$\left| \frac{\partial^2 \varphi_n}{\partial z_n \partial z_p}(z_0) \right| \leq C.$$

Now we consider the terms with $1 \leq m \leq n-1$. Since $(0, \cdots, 0, \varphi_n)$ is normal at $\varphi(z_0)$ and $z_1, \cdots, z_{n-1}$ are complex tangential to $\partial\Omega_1$ at $z_0$, we have

$$\left| \frac{\partial \varphi_n}{\partial z_\ell}(z_0) \right| \leq C\delta_2(\varphi(z_0))^{1/2} \leq C\delta_1(z_0)^{1/2}.$$

Since $\det(\varphi'(z))$ is bounded and

$$|\det(\varphi(z))\varphi^{\ell m}(z)| \leq C,$$

we see that

$$\left| \det \varphi'(z) \varphi^{nm}(z_0) \frac{\partial^2 \varphi_m}{\partial z_n \delta z_n}(z_0) \right|$$

$$\leq C\delta_1(z_0)^{-1}\left|\det\left(\frac{\partial\varphi_p}{\partial z_\ell}\right)_{1\leq\ell\leq n-1,p\neq m}\right|$$
$$\leq C\delta_1(z)^{-1/2}.$$

Combining all the estimates, we have proved that

$$|\nabla\det(\varphi'(z))| = |\det(\varphi'(z))\nabla\det(\varphi'(z)\log\det(\varphi'(z))| \leq C\delta(z)^{-1/2},$$

for all $z \in \Omega_1$. Hence $\det(\varphi'(z)) \in \mathrm{Lip}_{1/2}(\overline{\Omega}_1)$. The proof of Corollary 9.9.2 is complete. □

**Note:** Combining Corollary 9.9.2 and results in [FOST, NWY] [PIV] and [WEB1], we obtain a new proof of Fefferman's mapping theorem in [FEF1].

## 9.12 An Example of Barrett

**Capsule:** David Barrett's example of a smoothly bounded (non-pseudoconvex) domain that fails Condition *R* is very pleasing. It is quite elementary, but is nonetheless important.

We have seen that Bell's Condition *R* is a matter of some interest. While there was originally hope that Condition *R* would hold on any smoothly bounded, pseudoconvex domain, we now know (see [CHST2]) that Condition *R* fails on the worm domain of Diederich and Fornæss. Before Christ's result, David Barrett [BARR2] exhibited a smoothly bounded *non-pseudoconvex* domain on which Condition *R* fails. We treat that example now.

We begin by defining three real-valued functions, $r_1$, $r_2$, and $c$, on the interval $[1, 6]$. Each of these is smooth on the interval $(1, 6)$. They are given by

$$r_1(x) = \begin{cases} 3-\sqrt{x-1} & \text{for } x \geq 1 \text{ near } 1 \\ \text{decreasing} & \text{for } 1 \leq x \leq 2 \\ 1 & \text{for } 2 \leq x \leq 5 \\ \text{increasing} & \text{for } 5 \leq x \leq 6 \\ 3-\sqrt{6-x} & \text{for } x \leq 6 \text{ near } 6\,. \end{cases}$$

The function is exhibited in Figure 9.2.

Second,

$$r_2(x) = \begin{cases} 3+\sqrt{x-1} & \text{for } x \geq 1 \text{ near } 1 \\ \text{increasing} & \text{for } 1 \leq x \leq 2 \\ 4 & \text{for } 2 \leq x \leq 5 \\ \text{decreasing} & \text{for } 5 \leq x \leq 6 \\ 3+\sqrt{6-x} & \text{for } x \leq 6 \text{ near } 6\,. \end{cases}$$

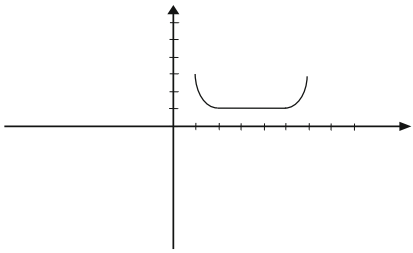


**Figure 9.2** The function $y = r_1(x)$.

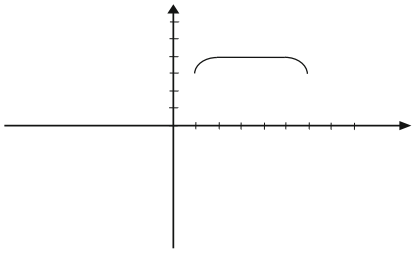


**Figure 9.3** The function $y = r_2(x)$.

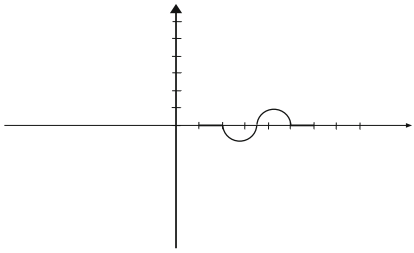


**Figure 9.4** The function $y = c(x)$.

The function is exhibited in Figure 9.3.

Finally, for a positive integer $k$,

$$c(x) = \begin{cases} 0 & \text{for} \quad 1 \le x \le 2 \\ \text{decreasing} & \text{for} \quad 2 \le x \le 3 \\ (x-3)^{2k} - 1 & \text{for } 3 - \epsilon < x < 3 + \epsilon, \epsilon > 0 \text{ small} \\ \text{increasing} & \text{for} \quad 3 \le x \le 4 \\ -(x-4)^{2k} + 1 & \text{for } 4 - \epsilon < x < 4 + \epsilon, \epsilon > 0 \text{ small} \\ \text{decreasing} & \text{for} \quad 4 \le x \le 5 \\ 0 & \text{for} \quad 5 \le x \le 6\,. \end{cases}$$

The function is exhibited in Figure 9.4.

Now define

$$\Omega\{(z,w) \in \mathbb{C}^2 : 1 < |w| < 6, |z| < r_2(|w|), |z - c(|w|)| > r_1(|w|)\}\,.$$

Let

$$\Omega_w = \{z \in \mathbb{C} : (z, w) \in \Omega\}$$

be the cross-section of $\Omega$ at $w$.

We note the following:

**(a)** If $2 \leq |w| \leq 5$, then $\Omega_w$ consists of the disc $|z| < 4$ minus a unit disc of varying center.

**(b)** If $1 < |w| \leq 2$ or $5 \leq |w| < 6$, then the slices $\Omega_w$ are annuli collapsing towards the circle $|z| = 3$ at the limiting values $|w| = 1$ and $|w| = 6$.

**(c)** The domain $\Omega$ is smooth near the limiting values $|w| = 1$ and $|w| = 6$ since it is defined there by the inequalities $(|z| - 3)^2 < |w| - 1$ and $(|z| - 3)^2 < 6 - |w|$ respectively.

**(d)** The union of the $\Omega_w$ for $1 \leq |w| \leq 6$ is the punctured disc $0 < |z| < 4$.

We now have three key lemmas. In what follows, we let $P$ denote the Bergman projection on $\Omega$ and $\mathcal{O}(\Omega)$ the space of all holomorphic functions on $\Omega$. Further we let $C_c^\infty(\Omega)$ be the $C^\infty$ functions with compact support in $\Omega$.

**Lemma 9.12.1** *The space $P(C_c^\infty(\Omega))$ is dense in $L^2(\Omega) \cap \mathcal{O}(\Omega)$.*

**Lemma 9.12.2** *The function $1/z$ lies in $L^2(\Omega) \cap \mathcal{O}(\Omega)$.*

**Lemma 9.12.3** *Let $p \geq 2 + 1/k$. Any function $g \in L^p(\Omega) \cap \mathcal{O}(\Omega)$ which is independent of the variable $w$ extends to a holomorphic function of $z$ on the disc $|z| < 4$.*

These three lemmas will imply the main result. We now prove the lemmas.

**Proof of Lemma 9.12.1** Let $h \in L^2(\Omega) \cap \mathcal{O}(\Omega)$ be orthogonal to $P(C_c^\infty(\Omega))$. Then, for every $\eta \in C_c^\infty(\Omega)$,

$$\int_\Omega \eta \cdot |h|^2 \, dV(z, w) = \int_\Omega (\eta h) \cdot \overline{h} \, dV(z, w) = \int_\Omega P(\eta h) \cdot \overline{h} \, dV(z, w) = 0 \, .$$

If $\eta$ is taken to be nonnegative, then we must conclude that $h = 0$ on the support of $\eta$. Since the choice of $\eta$ is otherwise arbitrary, we see that $h \equiv 0$. That concludes the proof. □

**Proof of Lemma 9.12.2** We calculate that

$$\begin{aligned} \|1/z\|^2_{L^2(\Omega)} &= \int_\Omega 1/|z|^2 \, dV(z, w) \\ &= 2\pi \int_1^6 |w| \, d|w| \int_{\Omega_{|w|}} 1/|z|^2 \, dA(z) \, . \end{aligned} \tag{9.12.2.1}$$

Now define

$$I(t) = t \int_{D_t} 1/|z|^2 \, dA(z)$$

for $t$ real. Clearly, in view of (9.12.2.1), our job is to show that $\int_1^6 I(t)\, dt < \infty$.

It is plain that, for $t$ away from 3 and 4, $I(t)$ is well behaved (indeed bounded). So we must examine the behavior for $t$ near 3 and for $t$ near 4. For $t$ near 3,

$$\Omega_t \subseteq \{z \in \mathbb{C} : (t-3)^{2k} < |z| < 4\}\,.$$

Hence

$$\begin{aligned} I(t) &\leq t \int_{(t-3)^{2k}<|z|<4} 1/|z|^2 \, dA(z) \\ &= 2\pi t \log[4/(t-3)^{2k}]\,. \end{aligned}$$

Certainly the logarithm function is integrable. So $I(t)$ is integrable near 3. A similar argument can be applied to analyze the situation for $t$ near 4. Thus $1/|z|$ is in $L^2$ and the result is proved. □

**Proof of Lemma 9.12.3** This is the trickiest of the three lemmas.

First observe that if $p = \infty$ then the result follows from the Riemann removable singularities theorem. So assume that $p < \infty$, and let $h \in L^p(\Omega) \cap \mathcal{O}(\Omega)$ be independent of $w$. Write

$$h(z, w) = h(z) = \sum_{j=-\infty}^{\infty} a_j z^j\,.$$

Certainly this series converges for $0 < |z| < 4$. We calculate that

$$\begin{aligned} \|h\|^p_{L^p(\Omega)} &= 2\pi \int_1^6 |w| \, d|w| \int_{\Omega_{|w|}} |h(z)|^p \, dA(z) \\ &\geq 2\pi \int_{-\epsilon}^{\epsilon} dt \int_{\Omega_{3+t} \cup \Omega_{4+t}} |h(z)|^p \, dA(z)\,. \end{aligned}$$

For small $t$, we see that

$$\begin{aligned} \Omega_{3+t} \cup \Omega_{4+t} &= \{z \in \mathbb{C} : |z| < 4\} \setminus \{z \in \mathbb{C} : |z - 1 + t^{2k}| < 1, |z + 1 - t^{2k}| < 1\} \\ &\supseteq \{z \in \mathbb{C} : 2|t|^{2k} < |z| < 4\}\,. \end{aligned}$$

As a result,

$$\|h\|_{L^p(\Omega)}^p \geq 4\pi \int_0^\epsilon dt \int_{2t^k<|z|<4} |h(z)|^p \, dA(z) \,. \tag{9.12.3.1}$$

Since $h \in L^2(\Omega) \cap \mathcal{O}(\Omega)$, we may use line (9.12.3.1) to calculate that

$$\begin{aligned}
\infty &> 4\pi \int_0^\epsilon dt \int_{2t^k<|z|<4} \left| \sum_j a_j z^j \right|^2 dA(z) \\
&= 4\pi \sum_j |a_j|^2 \int_0^\epsilon dt \int_{2t^k<|z|<4} |z|^{2j} \, dA(z) \\
&= 8\pi^2 \left\{ |a_{-1}|^2 \int_0^\epsilon \log(2t^{-k}) \, dt + \sum_{j \neq -1} |a_j|^2 \frac{4^{j+1}}{2j+2} \int_0^\epsilon (4^{j+1} - t^{2(j+1)k}) \, dt \right\} .
\end{aligned}$$

We see then that $a_j = 0$ when $2(j+1)k \leq -1$, in other words when $j \leq -2$. Thus $h$ may be written as $h(z) = b/z + g(z)$, where $g$ is holomorphic in the disc $|z| < 4$. Certainly $g \in L^p(\Omega)$ (because $1/z \notin L^2$). But line (9.12.3.1) now tells us that

$$\begin{aligned}
\|1/z\|_{L^p(\Omega)} &\geq 4\pi \int_0^\epsilon dt \int_{2t^k<|z|<4} |z|^{-p} \, dA(z) \\
&= 8\pi^2 \cdot \frac{2^{2-p}}{2-p} \int_0^\epsilon \left(2^{2-p} - t^{(2-p)k}\right) dt \\
&= +\infty
\end{aligned}$$

if $(2-p)k \leq -1$, that is if $p \geq 2 + 1/k$. Thus the only way that $h$ can be in $L^p(\Omega)$ is if $b = 0$. That proves the lemma. □

**Theorem 9.12.4** *Let $p \geq 2 + 1/k$. Then the space $P(C_c^\infty(\Omega))$ is not a subset of $L^p(\Omega)$.*

**Proof** Lemma 9.12.1 tells us that it suffices to show that $L^p(\Omega) \cap \mathcal{O}(\Omega)$ is not dense in $L^2(\Omega) \cap \mathcal{O}(\Omega)$ when $p \geq 2 + 1/k$. By Lemma 9.12.2., it thus suffices to show that the function $1/z$ cannot be approximated in the $L^2$ topology by functions in $L^p(\Omega) \cap \mathcal{O}(\Omega)$.

Seeking a contradiction, let us then suppose that we have functions $\{h_j\}_{j=1}^\infty$ in $L^p(\Omega) \cap \mathcal{O}(\Omega)$ with $h_j \to 1/z$ in $L^2(\Omega)$. Set

$$g_j(z, w) = \frac{1}{2\pi} \int_0^{2\pi} h_j(z, e^{i\theta} w) \, d\theta \,.$$

Then $g_j \in L^p(\Omega) \cap \mathcal{O}(\Omega)$ and $g_j \to 1/z$ in $L^2(\Omega)$. Moreover, each $g_j$ is constant on circles of the form $\{(z, e^{i\theta}w) : 0 \leq \theta \leq 2\pi\}$; thus each $g_j$ is locally independent of $w$.

We claim that actually each $g_j$ is *globally* independent of $w$. We already know that $g_j$ is independent of $w$ in a neighborhood of the product set

$$\{(z, w) \in \mathbb{C}^2 : 1 < |w| < 6, |z| = 3\} \subseteq \Omega \, .$$

Thus, for any two points $w_1, w_2$ in the annulus $1 < |w| < 6$, the functions $g_j(\cdot, w_1)$ and $g_j(\cdot, w_2)$ have as common domain of definition the connected open set $\Omega_{w_1} \cap \Omega_{w_2}$. Since the two functions agree near the circle $|z| = 3$, we have that $g_j(z, w_1) = g_j(z, w_2)$ whenever $(z, w_1)$ and $(z, w_2)$ are in $\Omega$, as we wished to show.

As a consequence, Lemma 9.12.3 tells us that the $g_j$ extend to holomorphic functions of $z$ on the disc $|z| < 4$, so that the residue

$$\frac{1}{2\pi i} \int_{|z|=3} g_j(z) \, dz$$

must vanish. But $g_j \to 1/z$ uniformly on compact subsets of $\Omega$, so that

$$\frac{1}{2\pi i} \int_{|z|=3} g_j(z) \, dz \to \frac{1}{2\pi i} \int_{|z|=3} \frac{1}{z} \, dz = 1 \, .$$

This contradiction proves the theorem. □

This example of Barrett has the advantage of being elementary and accessible. But it must be stressed that the domain he produces is *not* pseudoconvex. As previously noted, M. Christ has given us the more dramatic and important example in that he shows that the Diederich–Fornaess worm domain does not satisfy Condition $R$ (see [CHS] for information about the worm). Y.-T. Siu [SIU] has offered another proof of Christ's result. Neither proof is easy.

Here we shall offer a few words about why Christ's proof works. We begin with a quick review of the properties of the worm. Refer to Sections 9.1, 9.2 for basic properties of worm domains.

Now we turn to Christ's theorem. In order to discuss the failure of Condition $R$ on the Diederich–Fornæss worm domain, we recall the basic facts about the $\overline{\partial}$-Neumann problem.

Let $\Omega \subset \mathbb{C}^n$ be a bounded domain with smooth boundary and let $\rho$ be a smooth defining function for $\Omega$. The $\overline{\partial}$-Neumann problem on $\Omega$ is a boundary value problem for the elliptic partial differential operator

$$\Box = \overline{\partial}\overline{\partial}^* + \overline{\partial}^*\overline{\partial} \, .$$

Here $\overline{\partial}^*$ denotes the $L^2$-Hilbert space adjoint of the (unbounded) operators $\overline{\partial}$. Of course $\Box$ acts componentwise, just as a multiple of the Laplacian.

In order to apply $\Box$ to a form or current $u$, one needs to require that $u, \overline{\partial} u \in \text{dom}\,(\overline{\partial}^*)$. These conditions translate into two differential equations on the boundary for $u$ and the are called the $\overline{\partial}$-*Neumann boundary conditions* (see [FOK] or [TRE]). These equations are

$$u \lrcorner \overline{\partial}\rho = 0\,, \qquad \text{and} \quad \overline{\partial} u \lrcorner \overline{\partial}\rho = 0\,, \qquad \text{on } \partial\Omega\,. \tag{9.12.5}$$

Thus the equation $\Box\, u = f$ becomes a boundary value problem.

$$\begin{aligned} \Box\, u &= f \quad \text{on } \Omega \\ u \lrcorner \overline{\partial}\rho = 0\,, \overline{\partial} u \lrcorner \overline{\partial}\rho &= 0 \quad \text{on } \partial\Omega\,. \end{aligned} \tag{9.12.6}$$

This is an equation defined on *forms*. The significant problem is for $(0,1)$-forms, and we restrict to this case in the present discussion.

It follows from Hörmander's original article on the solution of the $\overline{\partial}$-equation [HOR1] that the $\overline{\partial}$-Neumann problem is always solvable on a bounded pseudoconvex domain $\Omega$ in $\mathbb{C}^n$ with smooth boundary for any data $f \in L^2(\Omega)$. We denote by $N$—the Neumann operator—such a solution operator. Moreover, $N$ turns out to be continuous in the $L^2$-topology:

$$\|Nu\|_{L^2} \leq c\|u\|_{L^2}\,.$$

An important formula of Kohn says that

$$P = I - \overline{\partial}^* N \overline{\partial}\,.$$

The proof of this is a formal calculation—see our Section 9.6. Important work by Boas and Straube [BOAS2] essentially established that the Neumann operator $N$ has a certain regularity (that is, it maps some Sobolev space $W^s$ to itself, for instance) if and only if $P$ will have the same regularity property. In particular, if $N$ is continuous on a Sobolev space $W^s$ for some $s > 0$ (of $(0,1)$-forms), then the Bergman projection $P$ is continuous on the same Sobolev space $W^s$ (of functions). See Exercise 5, Chapter 1.

Such regularity is well known to hold on strictly pseudoconvex domains [FOK, KRA5, Ch. 7–8]). In addition, Catlin proved a similar regularity result on finite type domains (see [CAT1, CAT2, KRA1, Ch. 11]).

Michael Christ's milestone result [CHST2] has proved to be of central importance for the field. It demonstrates concretely the seminal role of the worm, and points to future directions for research. Certainly the research program being described here, including the calculations in [KRPE1], is inspired by Christ's work.

Christ's work is primarily concerned with global regularity, or global hypoellipticity. A partial differential operator $L$ is said to be *globally hypoelliptic* if, whenever $Lu = f$ and $f$ is globally $C^\infty$, then $u$ is globally $C^\infty$. We measure regularity, here

and in what follows, using the standard Sobolev spaces $W^s$, $0 < s < \infty$ (see [KRA5, Ch. 1], [HOR2, Ch. 1], or Exercise 5 of Chapter 1).

Christ's proof of the failure of global hypoellipticity is a highly complex and recondite calculation with pseudodifferential operators. We cannot replicate it here. But the ideas are so important that we feel it worthwhile to outline his argument. We owe a debt to the elegant and informative article [CHST2] for these ideas.

As a boundary value problem for an elliptic operator, the $\overline{\partial}$-Neumann problem may be treated by Calderón's method of reduction to a pseudodifferential equation on $\partial\Omega$. The sources [HOR3] and [TRE, Ch. 3] and [KRA5, Ch. 5] give full explanations of the classical ideas about this reduction. In the more modern reference [CNS], Chang et al elaborate the specific application of these ideas to the $\overline{\partial}$-Neumann problem in $\mathbb{C}^2$. (Thus, in the remaining part of this discussion, $\Omega$ will denote a smoothly bounded pseudoconvex domain in $\mathbb{C}^2$.) The upshot is that one reduces the solution of the equation $\Box\, u = f$ to the solution of an equation $\Box^+ v = g$ on the boundary. Here, $u$ and $f$ are $(0,1)$-forms, while $v$ and $g$ are sections of a certain complex line bundle on $\partial\Omega$. (The fact that this bundle is 1-dimensional is a consequence of the inclusion $\Omega \subset \mathbb{C}^2$.)

To be more explicit, the solution $u$ of (9.12.6) can be written as $u = \mathcal{G}f + \mathcal{R}v$, where $\mathcal{G}$ is the *Green's operator* and $\mathcal{R}$ is the *Poisson operator*[3] for the operator $\Box$ and $v$ is chosen in such a way to satisfy the boundary conditions. In fact,

$$\begin{aligned}
&\Box\,(\mathcal{G}f + \mathcal{R}v) = f + 0 = f \qquad \text{on } \Omega \\
&(\mathcal{G}f + \mathcal{R}v) \lrcorner \overline{\partial}\rho = v \lrcorner \overline{\partial}\rho = 0 \qquad \text{on } \partial\Omega \\
&\overline{\partial}(\mathcal{G}f + \mathcal{R}v) \lrcorner \overline{\partial}\rho = \overline{\partial}\mathcal{G} \llcorner \overline{\partial}\rho + \overline{\partial} v \lrcorner \overline{\partial}\rho = 0 \qquad \text{on } \partial\Omega\,.
\end{aligned}$$

The section $v$ has two components, but one of these vanishes because of the first $\overline{\partial}$-Neumann boundary condition. The second $\overline{\partial}$-Neumann boundary condition may be written as an equation $\Box^+ v = g$ on $\partial\Omega$, where $\Box^+$ is a pseudodifferential operator of order 1. Also we note that $g = (\overline{\partial}\mathcal{G}f \lrcorner \overline{\partial}\rho)$ restricted to $\partial\Omega$.

Christ's argument begins with a real-variable model for the $\overline{\partial}$-Neumann problem that meshes well with the geometry of the boundary of the worm domain $\mathcal{W}$.

Let $M$ be the 2-torus $\mathbb{T}^2$ and let $X, Y$ two smooth real vector fields on $M$. Fix a coordinate patch $V_0$ in $M$ and suppose that $V_0$ has been identified with $\{(x,t) \in (-2,2) \times (-2\delta, -2\delta)\} \subset \mathbb{R}^2$. Let $J = [-1,1] \times \{0\} \subset V_0$.

Call a piecewise smooth path $\gamma$ on $M$ *admissible* if every tangent to $\gamma$ is in the span of $X, Y$. Assume that

**(i)** The vector fields $X, Y, [X,Y]$ span the tangent space to $M$ at every point of $M \setminus J$.

[3]Thus $\mathcal{G}$ is the solution operator for the elliptic boundary value problem $\Box\,(\mathcal{G}f) = f$ on $\Omega$ and $\mathcal{G}f = 0$ on $\partial\Omega$, while $\mathcal{R}$ is the solution operator for the elliptic boundary value problem $\Box\,(\mathcal{R}v) = 0$ on $\Omega$ and $\mathcal{R}v = v$ on $\partial\Omega$.

**(ii)** In $V_0$, $X \equiv \partial_x$ and $Y \equiv b(x,t)\partial_t$.
**(iii)** For all $|x| \leq 1$ and $|t| \leq \delta$, we have that $b(x,t) = \alpha(x)t + \mathcal{O}(t^2)$, where $\alpha(x)$ is nowhere vanishing.

It follows then that every pair $x, y \in M$ is connected by an admissible path.

**Theorem 9.12.7** *With $X, Y, M$ as above, let $L$ be any partial differential operator on $M$ of the form $L = -x^2 - Y^2 + a$, where $a \in C^\infty(M)$ and*

$$\|u\|^2 \leq C\langle Lu, u\rangle \tag{9.12.7.1}$$

*for all $u \in C^2(M)$. Then $L$ is not globally regular in $C^\infty$.*

We note that our hypotheses, particularly inequality (9.12.7.1), imply that $L$ has a well-defined inverse $L^{-1}$ which is a bounded linear operator on $L^2(M)$.

The following theorem gives a more complete, and quantitative, version of this result:

**Theorem 9.12.8** *Let $X, Y, M, L$ be as above. Then $L$ has the following global properties:*

**(a)** *There is a positive number $s_0$ such that, for every $0 < s < s_0$, $L^{-1}$ preserves $W^s(M)$;*
**(b)** *For each $s > s_0$, $L^{-1}$ fails to map $C^\infty(M)$ to $W^s(M)$;*
**(c)** *There is a sequence of values $s < r$ tending to infinity such that if $u \in W^s(M)$ satisfies $Lu \in H^r(M)$ then $u \in H^r$;*
**(d)** *There are arbitrarily large values of $s$ with a constant $C = C_s$ such that if $u \in W^s(M)$ is such that $Lu \in W^s(M)$ then*

$$\|u\|_{W^s} \leq C\|Lu\|_{W^s}\,. \tag{9.12.8.1}$$

**(e)** *For each value of $s$ as in part* **(d)**, *$\{f \in W^s(M) : L^{-1}f \in W^s(M)\}$ is a closed subspace of $W^s$ with finite codimension.*

The proof of Theorem 9.12.8 breaks into two parts. The first part consists of proving the *a priori* inequality (9.12.8.1). The second part, following ideas of Barrett in [BARR2], shows that, for any $s \geq s_0$, the operator $L$ cannot be exactly regular on $W^s(M)$. We refer the reader to [CHST2] for the details. Section 8 of [CHST2] also provides a nice outline of the analysis.

The next step is to reduce the analysis of the worm domain to the study of the manifold $M$ as above. With this idea in mind we set $\overline{L} = \overline{\partial}_b$ and $L$ its complex conjugate. The characteristic variety[4] of $\overline{L}$ is a real line bundle $\Sigma$ that splits smoothly as two rays: $\Sigma = \Sigma^+ \cup \Sigma^-$.

The principal symbol of $\Box^+$ vanishes only on $\Sigma^+$ that is half the characteristic variety. We may compose $\Box^+$ with an elliptic pseudodifferential operator of order $+1$ to change $\Box^+$ to the form

[4]The characteristic variety of a pseudodifferential operator is the conic subset of the cotangent bundle on which its principal symbol vanishes.

$$\mathcal{L} = \overline{L}L + B_1\overline{L} + B_2 L + B_3 \tag{9.12.9}$$

microlocally in a conical neighborhood of $\Sigma^+$, where each $B_j$ is a pseudodifferential operator with order not exceeding 0. Since $\Box^+$ is elliptic on the complement of $\Sigma^+$, our analysis may thus be microlocalized to a small conical neighborhood of $\Sigma^+$.

For a worm domain $\mathcal{W}$, there is circular symmetry in the second variable. This induces a natural action on functions and on forms. As indicated earlier, the Hilbert space of square-integrable $(0,k)$-forms has the orthogonal decomposition $\oplus_j \mathcal{H}^j_k$. The Bergman projection and the Neumann operator preserve $\mathcal{H}^j_0$ and $\mathcal{H}^j_1$. We now have the following key result.

**Proposition 9.12.10** *Let $\mathcal{W}$ be the worm. Then there is a discrete subset $S \subset \mathbb{R}^+$ such that, for each $s \notin S$ and each $j \in \mathbb{Z}$, there is a constant $C = C(s,j) < \infty$ such that, for each $(0,1)$ form $u \in \mathcal{H}^j_1 \cap C^\infty(\overline{\mathcal{W}})$ such that $Nu \in C^\infty$, it holds that*

$$\|Nu\|_{W^s(\mathcal{W})} \leq C \cdot \|u\|_{W^s(\mathcal{W})} .$$

The operators $\mathcal{L}$, $\overline{L}$, $L$, $B_j$ in (9.12.9) may be constructed so as to commute with the circle action in the second variable, hence they will preserve each $\mathcal{H}^j$. In summary, for each $j$, the action of $\mathcal{L}$ on $\mathcal{H}^j(\partial\mathcal{W})$ may be identified with the action of an operator $\mathcal{L}_j$ on $L^2(\partial\mathcal{W}/S^1)$.

Of course $\partial\mathcal{W}$ is 3-dimensional, hence $\partial\mathcal{W}/S^1$ is a real 2-dimensional manifold. It is convenient to take coordinates $(x,\theta,t)$ on $\partial\mathcal{W}$ so that

$$z_2 = \exp(x + i\theta) \quad \text{and} \quad z_1 = \exp(i2x)(e^{it} - 1) ;$$

here $|\log|z_2|^2| \leq r$ and $\mathcal{L}_j$ takes the form $\overline{L}L + B_1\overline{L} + B_2 L + B_3$ (just as in (9.12.9)!). In this last formula, $\overline{L}$ is a complex vector field which has the form $\overline{L} = \partial_x + it\alpha(t)\partial_t$, where $|x| \leq r/2$, $\alpha(0) \neq 0$, and each $B_j$ is a classical pseudodifferential operator of order not exceeding 0—depending on $j$ in a non-uniform manner.

We set $J = \{(x,t) : |x| \leq r/2, t = 0\}$ and write $\overline{L} = X + iY$; then the vector fields $X$, $Y$, $[X,Y]$ span the tangent space to $\partial\mathcal{W}/S^1$ at each point of the complement of $J$, and are tangent to $J$ at every point of $J$. We conclude that the operator $\mathcal{L}_j$ on $\partial\mathcal{W}/S^1$ is quite similar to the two-dimensional model that we discussed above.

There are two complications which we must note (and which are not entirely trivial): **(1)** There are pseudodifferential factors, and the reduction of the $\overline{\partial}$-Neumann problem to $\mathcal{L}$, and thereafter to $\mathcal{L}_j$, requires only a microlocal *a priori* estimate for $\mathcal{L}_j$ in a conic subset of phase space; **(2)** The lower-order terms $B_1\overline{L}$, $B_2 L$, $B_3$ are *not* negligible, indeed they determine the values of the exceptional Sobolev exponents, but the analysis can be carried out for these terms as well.

It should be noted that a special feature of the worm is that the rotational symmetry in $z_2$ makes possible (as we have noted) a reduction to a 2-dimensional analysis, and this in turn produces a certain convenient ellipticity. There is no uniformity of estimates with respect to $j$, but the analysis can be performed for each fixed $j$.

## 9.13 A Hilbert Integral

**Capsule:** Of course the Hilbert transform is well known to every harmonic analyst. The Hilbert integral is somewhat less well known. It is a one-sided version of the Hilbert transform, one in which the kernel does *not* have any cancellation property. Nonetheless, the Hilbert integral does have favorable mapping properties.

Because the Bergman kernel operates on a domain with boundary (so that there is no translation structure), Phong and Stein [PHS] were able to show that the Hilbert integral models the Bergman integral. We describe some of these ideas in the present section.

Perhaps the most important and ubiquitous integral operator in all of analysis is the *Hilbert transform*:

$$f \longmapsto \int_{\mathbb{R}} \frac{f(t)}{x-t}\, dt\,. \tag{9.13.1}$$

This integral is interesting because it does not converge in the sense of the Lebesgue integral. That is to say, the singularity of the function $1/(x-t)$ is non-integrable. So the integral (9.13.1) must be interpreted in the sense of the *Cauchy principal value*. It is the most classical example of a *singular integral*. Essential to the understanding of this integral is that the function $1/(x-t)$ has certain cancellation built into it. In simplest terms, $1/t$ is odd, so it integrates to 0 on "spheres" (in dimension 1 a sphere is just a pair of points centered at the origin) centered at 0. Because of this cancellation property, the convolution with the kernel $1/t$ induces a distribution. Thus the integral turns out to be tractable.

The concept of singular integral, at least in its classical form, really only makes sense on a space with a homogeneous structure. Such a structure arises naturally from a (Euclidean) group acting on the space. For $\mathbb{R}$ or $\mathbb{R}^N$, the natural group to consider is the group of translations (although rotations and dilations also play a distinctive role). The translations give rise to the notion of convolution, and the Hilbert transform (9.13.1) is a convolution operator.

The Hilbert transform is important in classical analysis because it governs the norm convergence (and, in a more subtle form, also the pointwise convergence) of Fourier series. It also arises in the study of the regularity of important partial differential operators. It is a fundamental result of M. Riesz that the Hilbert transform is bounded on $L^p$ for $1 < p < \infty$. It is unbounded on $L^1$ and $L^\infty$.

The Bergman kernel does *not* fit into the context just described. This kernel is defined on a *domain* $\Omega$. There certainly is no translation structure, and no concept of convolution. A better model for the Bergman kernel is the classical *Hilbert integral*. In its most basic form, the Hilbert integral is the operator given by

$$f \longmapsto Hf(x) \equiv \int_0^\infty \frac{f(t)}{x+t}\, dt\,.$$

Here we think of $f$ as a function with domain $[0, \infty)$ and the Hilbert integral $Hf$ also has domain $[0, \infty)$.

Certainly, on the halfline, the kernel $1/(x+t)$ has no cancellation built into it. Nonetheless, $H$ is bounded on $L^p$ for $1 < p < \infty$. Our treatment of the Hilbert integral derives from [PHS].

**Proposition 9.13.2** *Let $1 < p < \infty$. We have the estimate*

$$\int_0^\infty |Hf(x)|^p \, dx \leq C_p \int_0^\infty |f(x)|^p \, dx \, .$$

**Proof** Let $\epsilon > 0$ and consider the integral

$$\int_0^\infty \frac{\tau^{-\epsilon}}{x+\tau} \, d\tau \, . \tag{9.13.2.1}$$

The substitution $\tau = x\mu$ gives that

$$(9.13.2.1) = x^{-\epsilon} \cdot \int_0^\infty \frac{\mu^{-\epsilon}}{1+\mu} \, d\mu \, .$$

If $0 < \epsilon < 1$ then the new integrand is clearly integrable at the origin. Also it is of size $\mu^{-1-\epsilon}$ at infinity, so it is integrable there. Thus

$$(9.13.2.1) = \gamma_\epsilon \cdot x^{-\epsilon} \, .$$

Here $\gamma_\epsilon$ is a positive constant whose value can be determined, but is of no intrinsic interest.

Now we write the Hilbert integral as

$$Hf(x) = \int_0^\infty \left( \frac{t^{\epsilon/q} f(t)}{(x+t)^{1/p}} \right) \cdot \left( \frac{t^{-\epsilon/q}}{(x+t)^{1/q}} \right) dt \, .$$

Here $p$ and $q$ are conjugate Hölder exponents: $1/p + 1/q = 1$ with $1 < p, q < \infty$. Now apply Hölder's inequality with exponent $p$ on the first integral and exponent $q$ on the second integral. The result is

$$|Hf(x)| \leq \left( \int_0^\infty \frac{t^{\epsilon p/q} |f(t)|^p}{x+t} \, dt \right)^{1/q} \cdot \left( \int_0^\infty \frac{t^{-\epsilon}}{x+t} \, dt \right)^{1/q} \, .$$

So we see that

$$|Hf(x)|^p \leq \gamma_\epsilon^{p/q} \cdot x^{-\epsilon p/q} \cdot \left( \int_0^\infty \frac{t^{\epsilon p/q} |f(t)|^p}{x+t} \, dt \right) \, .$$

We integrate both sides with respect to $x$ on the interval $[0, \infty)$ and apply Fubini. Of course we must use (9.13.2.1) again (with $\epsilon$ replaced by $\epsilon p/q$) to see that

$$\int_0^\infty |Hf(x)|^p \, dx \le C_p \int_0^\infty |f(x)|^p \, dx \, .$$

Note that, by the second application of (9.13.2.1), the term $t^{\epsilon p/q}$ cancels out. This all works provide $0 < \epsilon < 1$ and $0 < \epsilon p/q < 1$. That completes the proof. □

Next we turn to a higher-dimensional version of the Hilbert integral:

**Proposition 9.13.3** *Let $L(x,t)$ be a nonnegative kernel for $(x,t) \in \mathbb{R}^N \times \mathbb{R}_+$ which satisfies*

**1.** $L(\lambda x, \lambda t) = \lambda^{-(N+1)} \cdot L(x,t)$ *for all* $\lambda > 0$, *all* $(x,t) \in \mathbb{R}^N \times \mathbb{R}_+$;
**2.** $\int_{\mathbb{R}^N} L(x,1)\, dx = C < \infty$.

*Assume that the kernel $K(x,y,t,u)$ for $(x,y,t,u) \in \mathbb{R}^N \times \mathbb{R}^N \times \mathbb{R}_+ \times \mathbb{R}_+$ satisfies*

$$|K(x,y,t,u)| \le L(x-y, t+u) \, .$$

*Define*

$$Hf(x,t) = \int_{\mathbb{R}^N} \int_0^\infty K(x,y,t,u) f(y,u) \, dy du \, , \quad (x,t) \in \mathbb{R}^N \times \mathbb{R}_+ \, .$$

*Then*

$$\int_{\mathbb{R}^N} \int_0^\infty |H(f)(x,t)|^p \, dx dt \le \widetilde{C}_p \int_{\mathbb{R}^N} \int_0^\infty |f(y,u)|^p \, dy du \tag{9.13.3.1}$$

*whenever $1 < p < \infty$ and $f \in L^p(\mathbb{R}^{N+1}_+)$.*

**Proof** It is enough to prove (9.13.3.1) with $K$ replaced by $L(x-y, t+u)$ and we do so. By homogeneity and change of variables in the integral,

$$\begin{aligned} \int_{\mathbb{R}^N} L(x, t+u) \, dx &= (t+u)^{-(N+1)} \int_{\mathbb{R}^N} L\left(\frac{x}{t+u}, 1\right) dx \\ &= (t+u)^{-1} \int_{\mathbb{R}^N} L(x,1) \, dx \\ &= C \cdot (t+u)^{-1} \, . \end{aligned}$$

Thus our estimate of line (9.13.3.1) gives

$$\int_{\mathbb{R}^N} \int_0^\infty L(x, t+u) u^{-\epsilon} \, du = C \cdot \gamma_\epsilon \cdot t^{-\epsilon} \, . \tag{9.13.3.2}$$

Now, just as in the proof of Proposition 3.1.2, we see that

$$\begin{aligned}
|H(f)(x,t)| &= \left| \int_{\mathbb{R}^N} \int_0^\infty L(x-y,t+u) f(y,u)\, dydu \right| \\
&= \left| \int_{\mathbb{R}^N} \int_0^\infty \left[ u^{\epsilon/q} \cdot f(y,u) \cdot (L(x-y,t+u))^{1/p} \right] \right. \\
&\qquad \left. \times \left[ u^{-\epsilon/q} \cdot (L(x-y,t+u))^{1/q} \right] dydu \right| \\
&\leq \int_{\mathbb{R}^N} \int_0^\infty u^{\epsilon p/q} |f(y,u)|^p L(x-y,t+u)\, dydu^{1/p} \\
&\qquad \times \int_{\mathbb{R}^N} \int_0^\infty u^{-\epsilon} L(x-y,t+u)\, dydu^{1/q} \\
&\leq (C\gamma_\epsilon t^{-\epsilon})^{1/q} \cdot \int_{\mathbb{R}^N} \int_0^\infty u^{\epsilon p/q} |f(y,u)|^p L(x-y,t+u)\, dydu^{1/p} .
\end{aligned}$$

As a result,

$$|Hf(x,t)|^p \leq (C\gamma_\epsilon t^{-\epsilon})^{p/q} \cdot \int_{\mathbb{R}^N} \int_0^\infty u^{\epsilon p/q} |f(y,u)|^p \cdot L(x-y,t+u)\, dydu .$$

Now we integrate in $x$ and $t$, apply Fubini, and utilize (9.13.3.2). Then, just as in the proof of Proposition 9.13.2, we get the desired estimate. □

Now let us split $\mathbb{R}^N = \mathbb{R}^p \times \mathbb{R}^q$, where $x = (x^p, x^q)$, $x \in \mathbb{R}^N$, $x^p \in \mathbb{R}^p$, $x^q \in \mathbb{R}^q$. Of course $N = p + q$. For $s > 0$ we consider the nonisotropic dilations

$$(x^p, x^q) \longmapsto (\lambda x^p, \lambda^s x^q) , \quad \lambda > 0 .$$

We assume that $\mathbb{R}^N$ is equipped with a group structure so that Lebesgue measure $dx = dx^p dx^q$ is the bi-invariant Haar measure. [An example of such a group structure is ordinary translation. Another is the Heisenberg group structure, which we treated in Chapters 3 and 4.] We denote the group operation by $\cdot$ and the group inverse of $g$ by $g^{-1}$. Now a similar proof to that of the last proposition gives the following generalization:

**Proposition 9.13.4** *Let $L(x,t)$ be a nonnegative kernel satisfying*

1. $L(\lambda x^p, \lambda^s x^q, \lambda^s t) = \lambda^{-(p+(q+1)s)} L(x^p, x^q, t)$;
2. $\int_{\mathbb{R}^N} L(x,1)\, dx = C < \infty$;
3. $|K(x,y,t,u)| \leq L(y^{-1} \cdot x, t+u)$.

*Then the operator*

$$H(f)(x,t) = \int_{\mathbb{R}^N} \int_0^\infty K(x,y,t,u) f(y,u)\, dydu$$

*is bounded on $L^p$.*

The paper [PHS] applies these results on the Hilbert integral to derive boundedness theorems for the Bergman integral. The methods are rather detailed and technical, and we cannot treat them here.

## Exercises

1. Show that the smoothly bounded worm domain may be exhausted by biholomorphic copies of the non-smooth worm. [**Hint:** See [BARR2] for more on this idea.]
2. Let $\mathbf{h}^2(\Omega)$ be the real-valued harmonic functions on a domain $\Omega$ in $\mathbb{R}^N$ which satisfy a suitable square integrability condition (as in the definition of $H^2$). Show that $\mathbf{h}^2$ is a Hilbert space with reproducing kernel in the sense of Aronszajn. How is the corresponding kernel related to the Poisson kernel?
3. Use the Fefferman asymptotic expansion for the Bergman kernel to actually calculate an asymptotic formula for the holomorphic sectional curvature of the Bergman metric on a strictly pseudoconvex domain. See [KLE] for details.
4. Show that the complex Monge–Ampère operator is invariant under biholomorphic mappings.
5. Refer to Exercise 5 of Chapter 1 for terminology. Calculate explicitly the Bergman kernel for the Sobolev space $W^2$ on the unit disc in the plane.
6. Refer to Exercise 5 of Chapter 1 or Exercise 2 of Chapter 10 for terminology. Calculate explicitly the Bergman kernel for the Sobolev space $W^1$ on the unit ball in complex space.
7. The Kohn projection formula easily shows that Condition $R$ holds on any smoothly bounded domain in the plane. Explain.
8. Consider the domains

$$\Omega_m = \{z \in \mathbb{C} : \operatorname{Re} z > 0, |\operatorname{Im} z| < (\operatorname{Re} z)^m\}$$

   for $m$ a positive, even integer. What can you say about Condition $R$ on such a domain?
9. The original Diederich–Fornæss worm domain is strictly pseudoconvex except on a boundary set of $(2n-1)$-dimensional measure zero. Explain.
10. Consider the linear space $C^\infty(\overline{\Omega})$ for a smoothly bounded domain $\Omega$ in $\mathbb{R}^N$. Explain why this space is *not* a Hilbert space with reproducing kernel.
11. Is there any smoothly bounded domain in the complex plane on which Condition $R$ fails?
12. Show that the Diederich–Fornæss worm domain is topologically trivial.
13. Show that the Diederich–Fornæss worm domain is not biholomorphic to the unit ball.

# Chapter 10
# The Solution of the Inhomogeneous Cauchy–Riemann Equations

## 10.1 The Inhomogeneous Cauchy–Riemann Equations

All readers will know the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \qquad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

in the complex plane. A $C^1$ function $h = u + iv$ satisfies these equations if and only if $h$ is holomorphic. The Cauchy–Riemann equations are a powerful tool in elementary function theory.

In $\mathbb{C}$, we may define

$$\frac{\partial}{\partial z} \equiv \frac{1}{2}\left(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}\right)$$

and

$$\frac{\partial}{\partial \overline{z}} \equiv \frac{1}{2}\left(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y}\right) .$$

Thus $h = u + iv$ is holomorphic if and only if $\partial h/\partial \overline{z} \equiv 0$. This is just another way, in complex notation, of writing the Cauchy–Riemann equations from the preceding paragraph.

In $\mathbb{C}^n$ we introduce the partial differential operators

$$\frac{\partial}{\partial z_j} \equiv \frac{1}{2}\left(\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right) , \quad j = 1, 2, \ldots, n$$

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and

$$\frac{\partial}{\partial \overline{z}_j} \equiv \frac{1}{2}\left(\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right), \quad j = 1, 2, \ldots, n.$$

We say that a $C^1$ function $f$ of $n$ complex variables is *holomorphic* if and only if $\partial f/\partial \overline{z}_j \equiv 0$ for $j = 1, \ldots, n$. This is just a manner of saying that $f$ satisfies the classical Cauchy–Riemann equations in each complex variable separately; thus it is holomorphic in each variable separately.

In $\mathbb{C}^n$ we also study the *inhomogeneous* Cauchy–Riemann equations. These are a useful tool for *constructing* holomorphic functions. To appreciate this device, we define for a $C^1$ function $u$ the operator

$$\overline{\partial} u \equiv \sum_{j=1}^{n} \frac{\partial u}{\partial \overline{z}_j} d\overline{z}_j .$$

Thus $\overline{\partial}$ is an operator from functions to $(0,1)$ forms—here a $(0,1)$ form is a linear combination of the $d\overline{z}_j$. We also define

$$\overline{\partial}\left(\sum_{j=1}^{n} \alpha_j(z) d\overline{z}_j\right) = \sum_{j=1}^{n}\sum_{k=1}^{n} \frac{\partial \alpha_j}{\partial \overline{z}_k} d\overline{z}_k \wedge d\overline{z}_j .$$

Clearly $\overline{\partial} \circ \overline{\partial} = 0$.

Thus it makes sense to study the partial differential equation

$$\overline{\partial} u = f , \tag{10.1.1}$$

where $f$ is a given $(0,1)$ form and $u$ is an unknown function.[1] If we apply the operator $\overline{\partial}$ to both sides of (10.1.1), then we see that a necessary condition for solvability is that $\overline{\partial} f = 0$. We call this the condition of being $\overline{\partial}$*-closed*. Equation (10.1.1) is commonly called the $\overline{\partial}$*-problem*.

Here is a standard and useful result on the $\overline{\partial}$-problem. The reference is [HOR1].

**Theorem 10.1.2** *Let $\Omega \subseteq \mathbb{C}^n$ be a bounded, pseudoconvex domain. Let $f$ be a $\overline{\partial}$-closed $(0,1)$ form on $\Omega$ with $L^2$ coefficients. Then there is an $L^2$ function $u$ on $\Omega$ such that $\overline{\partial} u = f$ and*

$$\|u\|_{L^2} \leq C \|f\|_{L^2} .$$

[1] One can also study this PDE in one complex variable. But, because of issues connected with the support of the solution, this is a much less useful tool in that context—see [KRA1, Ch. 0, 1] for the details.

*The constant C depends only on the dimension and the diameter of the domain. If f has $C^\infty$ coefficients then so does u.*

Hörmander's theorem is in fact true in much greater generality. In that broader context, the relevant function spaces are formulated in terms of a plursubharmonic weight function.

This theorem has been profoundly influential in many fields, including algebraic and complex geometry. A nice exposition of the latter ideas appears in [DEM].

A typical application of the partial differential equation (10.1.1) is as follows.

**Theorem 10.1.3** *Let $\Omega \subseteq \mathbb{C}^n$ be pseudoconvex with $C^2$ boundary. Let $\omega = \Omega \cap \{(z_1, \ldots, z_n) : z_n = 0\}$. Assume that $\omega$ is a nontrivial, $(n-1)$-dimensional domain.*

*Let $f : \omega \to \mathbb{C}$ satisfy the property that the map*

$$(z_1, \ldots, z_{n-1}) \mapsto f(z_1, \ldots, z_{n-1}, 0)$$

*is holomorphic on $\widetilde{\omega} = \{(z_1, \ldots, z_{n-1}) \in \mathbb{C}^{n-1} : (z_1, \ldots, z_{n-1}, 0) \in \omega\}$. Then there is a holomorphic $F : \Omega \to \mathbb{C}$ such that $F|_\omega = f$. In fact there is a linear operator*

$$\mathcal{E}_{\omega,\Omega} : \{\text{holomorphic functions on } \omega\} \to \{\text{holomorphic functions on } \Omega\}$$

*such that $(\mathcal{E}_{\omega,\Omega} f)|_\omega = f$. The operator is continuous in the topology of normal convergence.*

**Proof** Let $\pi : \mathbb{C}^n \to \mathbb{C}^n$ be the Euclidean projection $(z_1, \ldots, z_n) \mapsto (z_1, \ldots, z_{n-1}, 0)$. Let $\mathcal{B} = \{z \in \Omega : \pi z \notin \omega\}$. Then $\mathcal{B}$ and $\omega$ are *relatively closed* disjoint subsets of $\Omega$. Hence there is a function $\Psi : \Omega \to [0, 1]$, $\Psi \in C^\infty(\Omega)$, such that $\Psi \equiv 1$ on a relative neighborhood of $\omega$ and $\Psi \equiv 0$ on a relative neighborhood of $\mathcal{B}$.[2] Set

$$F(z) = \Psi(z) \cdot f(\pi(z)) + z_n \cdot v(z),$$

where $v$ is an unknown smooth function to be determined.

Notice that $f(\pi(z))$ is well defined on $\operatorname{supp} \Psi$. We wish to select $v \in C^\infty(\Omega)$ so that $\overline{\partial} F = 0$. Then the function $F$ defined by the displayed equation will be the holomorphic function that we seek.

Thus we ask that

$$\overline{\partial} v(z) = \frac{\left(-\overline{\partial}\Psi(z)\right) \cdot f\big(\pi(z)\big)}{z_n}. \tag{10.1.3.1}$$

Now the right side of this equation is $C^\infty$ since $\overline{\partial}\Psi \equiv 0$ on a neighborhood of $\omega$—in particular, on a neighborhood of the set where $z_n = 0$. Also, by inspection, the right

[2] This last assertion is intuitively non-obvious. It is a version of the $C^\infty$ Urysohn lemma, for which see [HIR, Ch. 2]. It is also a good exercise for the reader to construct $\Psi$ by hand.

side is annihilated by the $\overline{\partial}$ operator (remember that $\overline{\partial}^2 = 0$). By Theorem 10.1.2, there exists a $v \in C^\infty(\Omega)$ that satisfies (10.1.3.1). Therefore the extension $F$ exists and is holomorphic.

Note finally that, since the $\overline{\partial}$ solution operator is linear (see [KRA1, Ch. 4–5] for details), it follows that $F$ depends linearly on $f$.

Following the proof of Theorem 10.1.2, we may check that the operator $\mathcal{E}_{\omega,\Omega}$ is bounded in $L^2$ of any compact set. But then the Cauchy estimates show that the operator is bounded in the topology of normal convergence. □

**Remark 10.1.4** An obverse of Theorem 10.1.3 is the noted Ohsawa–Takegoshi extension theorem (see [OHT]). This is a widely used result, with notable applications in algebraic geometry. □

It is a widely studied matter to determine estimates for the equation (10.1.1). That is, if the coefficients of $f$ lie in a particular Banach space, then what can we say about the attributes of $u$? In what Banach space will $u$ lie?

Studies of strongly elliptic partial differential equations (see [KRA5, Ch. 5]) lead one to think that, if $f$ has coefficients in $W^r$ (the standard Sobolev space—see Exercise 5 in Chapter 1), then $u$ should show an improvement in smoothness corresponding to the degree of the partial differential operator. This is what happens with the Laplacian. But the $\overline{\partial}$ operator is *not* elliptic up to the boundary. Instead it is *subelliptic*. So it has a weaker regularity theory.

The first estimates for the $\overline{\partial}$-problem were proved by J. J. Kohn (see [FOK] and the references therein). He showed that, when $\Omega$ is a smoothly bounded, strongly pseudoconvex domain, then

$$\|u\|_{W^{s+1/2}} \leq C\|f\|_{W^s} .$$

so long as $u$ is the so-called "canonical solution" to the $\overline{\partial}$ problem. Here the canonical solution is that which is orthogonal to holomorphic functions; it is given by $u_c = u - Pu$, where $u$ is any solution to (10.1.1) and $P$ is the Bergman projection.

A few years later, Lars Hörmander [HOR1] proved that the equation $\overline{\partial}u = f$ has a solution $u$ that satisfies

$$\|u\|_{L^2} \leq C\|f\|_{L^2}$$

on *any* bounded pseudoconvex domain.

It was considered a breakthrough when, around 1970, Grauert/Lieb [GRAL] and Kerzman [KER1] and Ramirez [RAM] and Henkin [HEN1] independently proved uniform estimates for the $\overline{\partial}$-problem. More precisely, they showed that, on a strongly pseudoconvex domain $\Omega$, a particular solution[3] (which can be explicitly constructed) of the equation $\overline{\partial}u = f$ satisfies

[3] But *not* the canonical solution!

$$\|u\|_{L^\infty} \leq C\|f\|_{L^\infty} .$$

Kerzman also proved $L^p$ estimates. Key to their attack was to construct an explicit integral formula that gives the solution to (10.1.1). Thus the estimates in the uniform norm for the solution reduce to proving suitable estimates on the integral kernel. The integral formula solution considered here is *not* the canonical solution. But it is closely associated with the canonical solution, and shares many of its important properties.

Later, Siu [SIU] showed that the Henkin solution $u$ satisfies

$$\|u\|_{C^k} \leq C_k\|f\|_{C^k}$$

for every positive integer $k$ on a smoothly bounded, strongly pseudoconvex domain.

In his 1974 Ph.D. thesis, Krantz [KRA10] obtained precise estimates for the $\overline{\partial}$-problem in strongly pseudoconvex domains. In particular, he showed that, for an explicitly constructible solution (due to Henkin) $u$ of the $\overline{\partial}$-problem,

$$\|u\|_{L^q} \leq C \cdot \|f\|_{L^p}$$

for $1 < p < 2n + 2$ and $\frac{1}{q} = \frac{1}{p} - \frac{1}{2n+2}$. Also

$$\|u\|_{\mathrm{Lip}_c} \leq C \cdot \|f\|_{L^p}$$

for $2n + 2 < p \leq \infty$ and $c = 1/2 - (n + 1)/p$.

Krantz also proved estimates in various nonisotropic Lipschitz norms, but we shall not provide the details here. See [KRA10] for the particulars.

It is certainly possible to combine the techniques of Siu [SIU] and Krantz [KRA10] to obtain sharp estimates on the derivatives of the Henkin solution $u$ of the equation $\overline{\partial}u = f$. These details have never been worked out.

There are further developments in the realm of estimates for the $\overline{\partial}$-problem. As an example, Greiner and Stein [GRS] proved sharp estimates in many different norms on strongly pseudoconvex domains for $\overline{\partial}$ for the *canonical solution* (the one that is orthogonal to holomorphic functions). This is profound work, and ties together many of the earlier ideas that we have described above. Key to the work of Greiner and Stein is that Greiner constructed a parametrix for the $\overline{\partial}$-Neumann problem using pseudodifferential operators.

## 10.2 Some Notation

The $\overline{\partial}$-Neumann problem is a very particular generalization to several complex variables of the Cauchy–Riemann equations of one complex variable. While the groundwork for this problem was laid by D. C. Spencer, Charles Morrey, P. Conner,

and others, it was J. J. Kohn [KOH2] who tamed the problem. It is the key to many of the important hard analysis techniques of the function theory of several complex variables.

The $\overline{\partial}$-Neumann problem was also one of the first non-elliptic problems for which a regularity theory was established. In fact, for many years, people were quite skeptical of there being any sort of regularity theory that was less than optimal (i.e., at the elliptic level). Whereas in an elliptic problem of order $m$ we have learned that the solution is $m$ degrees smoother than the data, the $\overline{\partial}$-Neumann problem does not exhibit the maximal degree of smoothing. Indeed the $\overline{\partial}$-Neumann problem is *subelliptic* rather than elliptic; roughly speaking, this means that the solution gains a predictable number of derivatives, but that number is less than the degree of the partial differential operator in question.

Subelliptic regularity was quite unexpected in the early 1960's when it was first discovered (see [KOH2]). It cannot be treated by a naive application of the theory of pseudodifferential operators or by other classical techniques. The monograph [GRS] describes a special calculus of pseudodifferential operators designed for the study of the $\overline{\partial}$-Neumann problem on an important special class of domains; indeed, it is the same special class that we shall study here.

The method that we present here is not Kohn's original, which is rather complicated, but is a simpler method, called *elliptic regularization* that he developed later in collaboration with Nirenberg (see [KOHN1, FOK] and references therein). The monograph [FOK] is the canonical reference for matters related to the $\overline{\partial}$-Neumann problem. However the presentation there is perhaps too complicated for our purposes, since it is in the setting of an arbitarary metric structure on an almost complex manifold. In an effort to maintain simplicity, we shall formulate and solve the $\overline{\partial}$-Neumann problem only on a smoothly bounded strongly pseudoconvex domain in $\mathbb{C}^n$. We do follow the basic steps in [FOK], but by specializing we can provide considerably more detail and context. See also [KRA5, Ch. 7].

Those interested in the most general setting for the $\overline{\partial}$-Neumann problem will, after reading the material here, be well-prepared to consult [FOK] and other more recent sources (see, for instance, [RAN2] and [CAT1, CAT2, CAT3]) on the $\overline{\partial}$-Neumann problem.

As previously mentioned, we shall be working strictly on domains in $\mathbb{C}^n$. We may thus bypass a certain amount of formalism by relying on the standard coordinates and Hermitian geometry in space.

If $Q \in \mathbb{C}^n \cong \mathbb{R}^{2n}$ then the (real) tangent space to $\mathbb{R}^{2n}$ at $Q$ is spanned by

$$\frac{\partial}{\partial x_1}, \frac{\partial}{\partial y_1}, \ldots, \frac{\partial}{\partial x_n}, \frac{\partial}{\partial y_n}.$$

It clearly has (real) dimension $2n$. We set

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\left[\frac{\partial}{\partial x_j} - i\frac{\partial}{\partial y_j}\right],$$

$$\frac{\partial}{\partial \overline{z}_j} = \frac{1}{2}\left[\frac{\partial}{\partial x_j} + i\frac{\partial}{\partial y_j}\right].$$

Then we let $T^{1,0}$ denote the complex linear space spanned by

$$\frac{\partial}{\partial z_1}, \ldots, \frac{\partial}{\partial z_n}$$

and $T^{0,1} \equiv \overline{T^{1,0}}$ denote the complex linear space spanned by

$$\frac{\partial}{\partial \overline{z}_1}, \ldots, \frac{\partial}{\partial \overline{z}_n}.$$

The *complexified tangent space* is then

$$\mathbb{C}T_Q(\mathbb{C}^n) \equiv T^{1,0} \oplus_{\mathbb{C}} T^{0,1}.$$

It obviously has complex dimension $2n$. The complexified tangent space is, in a natural sense, the tensor product of the real tangent space with $\mathbb{C}$. However we shall have no use for this formalism (see [WEL, Ch. 1, 3] for more on this point of view).

There is associated to the complexified tangent space a complexified cotangent space. We set

$$dz_j = dx_j + i dy_j$$
$$d\overline{z}_j = dx_j - i dy_j.$$

A trivial calculation shows that

$$\left\langle dz_j, \frac{\partial}{\partial z_j}\right\rangle = 1 \quad , \quad \left\langle dz_j, \frac{\partial}{\partial \overline{z}_j}\right\rangle = 0,$$

$$\left\langle d\overline{z}_j, \frac{\partial}{\partial z_j}\right\rangle = 0 \quad , \quad \left\langle d\overline{z}_j, \frac{\partial}{\partial \overline{z}_j}\right\rangle = 1.$$

If $\alpha = (a_1, \ldots, a_s)$ is a multi-index then we set

$$dz^\alpha \equiv dz_{a_1} \wedge dz_{a_2} \wedge \cdots \wedge dz_{a_s},$$

and

$$d\overline{z}^\alpha \equiv d\overline{z}_{a_1} \wedge d\overline{z}_{a_2} \wedge \cdots \wedge d\overline{z}_{a_s}.$$

In this context the *magnitude* of the multi-index is $|\alpha| \equiv s$. Then $\bigwedge^{p,q}$ denotes the space of forms of the type

$$\omega = \sum_{|\alpha|=p,|\beta|=q} a_{\alpha\beta} dz^\alpha \wedge d\overline{z}^\beta .$$

It is an exercise in linear algebra to see, for $r$ a positive integer, that if $\bigwedge^r$ denotes the space of all classical (real variable) $r$-forms on $\mathbb{C}^n \cong \mathbb{R}^{2n}$, then

$$\bigwedge^r = \bigoplus_{p+q=r} \bigwedge^{p,q} .$$

Next we turn to the exterior derivative. Let

$$\omega = \sum_{\alpha,\beta} a_{\alpha,\beta} dz^\alpha \wedge d\overline{z}^\beta$$

be a differential form. The exterior derivative $d$ may be written as $d = \partial + \overline{\partial}$, where

$$\partial\omega = \sum_{\alpha,\beta} \sum_{j=1}^{n} \frac{\partial a_{\alpha,\beta}}{\partial z_j} dz_j \wedge dz^\alpha \wedge d\overline{z}^\beta$$

and

$$\overline{\partial}\omega = \sum_{\alpha,\beta} \sum_{j=1}^{n} \frac{\partial a_{\alpha,\beta}}{\partial \overline{z}_j} d\overline{z}_j \wedge dz^\alpha \wedge d\overline{z}^\beta .$$

Clearly

$$\partial : \bigwedge^{p,q} \to \bigwedge^{p+1,q} \qquad \text{and} \qquad \overline{\partial} : \bigwedge^{p,q} \to \bigwedge^{p,q+1} .$$

Since

$$0 = d^2 = (\partial + \overline{\partial})(\partial + \overline{\partial}) ,$$

we see by counting degrees that

$$\partial^2 = 0 , \quad \overline{\partial}^2 = 0 , \quad \partial\overline{\partial} = -\overline{\partial}\partial .$$

We define a Hermitian inner product on $\mathbb{C}T_Q(\mathbb{C}^n)$ as follows:

$$\left\langle \frac{\partial}{\partial z_j}, \frac{\partial}{\partial z_k} \right\rangle_Q = \left\langle \frac{\partial}{\partial \overline{z}_j}, \frac{\partial}{\partial \overline{z}_k} \right\rangle_Q = \frac{1}{2}\delta_{jk}$$

and

$$\left\langle \frac{\partial}{\partial z_j}, \frac{\partial}{\partial \overline{z}_k} \right\rangle_Q = 0 \qquad \text{for all } j, k.$$

Thus we see that

$$\mathbb{C}T_Q(\mathbb{C}^n) = T_Q^{1,0} \oplus T_Q^{0,1}$$

is an orthogonal decomposition.

There is thus an inner product on covectors. We have

$$\left\langle dz_j, dz_k \right\rangle_Q = \left\langle d\overline{z}_j, d\overline{z}_k \right\rangle_Q = 2\delta_{jk}$$

and

$$\langle dz_j, d\overline{z}_k \rangle_Q = 0 \qquad \text{for all } j, k.$$

By functoriality, if $\phi, \psi$ are both $(p, q)$ forms then

$$\phi = \sum_{\substack{|I|=p \\ |J|=q}} \phi_{IJ} dz^I \wedge d\overline{z}^J \qquad \text{and} \qquad \psi = \sum_{\substack{|I|=p \\ |J|=q}} \psi_{IJ} dz^I \wedge d\overline{z}^J,$$

hence

$$\langle \phi, \psi \rangle_Q = 2^{p+q} \sum_{I,J} \phi_{IJ}(Q) \cdot \overline{\psi}_{IJ}(Q). \tag{10.2.1}$$

It follows in particular that, if $\phi$ is a form of type $(p, q)$ and $\psi$ is a form of type $(p', q')$ with $(p, q) \neq (p', q')$, then $\phi$ is orthogonal to $\psi$.

When $\mathbb{C}^n$ is identified with $\mathbb{R}^{2n}$ then the volume form, in real coordinates, is

$$dV = dx_1 \wedge dy_1 \wedge \cdots \wedge dx_n \wedge dy_n.$$

It is more convenient for us to use complex coordinates: since $dx_j \wedge dy_j = (i/2)(dz_j \wedge d\overline{z}_j)$, we find that

$$dV = \left(\frac{i}{2}\right)^n dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge dz_n \wedge d\overline{z}_n.$$

We now use the volume form to produce an inner product on *forms* that is consistent with the inner product that we have defined on covectors. Let $\Omega \subseteq \mathbb{C}^n$ be a domain. Then, with $\phi, \psi$ as above,

$$\langle \phi, \psi \rangle_\Omega = \int_\Omega \langle \phi, \psi \rangle_Q \, dV \, .$$

More explicitly,

$$\langle \phi, \psi \rangle_\Omega = 2^{p+q} \int_\Omega \sum_{I,J} \phi_{IJ} \overline{\psi}_{IJ} \, dV \, .$$

We next define certain linear spaces of forms with coefficients that satisfy regularity properties. Set

$$\bigwedge^{p,q} = \bigwedge^{p,q}(\Omega) \equiv \left\{ \phi = \sum_{\substack{|I|=p \\ |J|=q}} \phi_{IJ} \, dz^I \wedge d\overline{z}^J : \phi_{IJ} \in C^\infty(\overline{\Omega}) \text{ for all } I, J \right\} ;$$

$$\bigwedge_c^{p,q} \equiv \left\{ \phi = \sum_{\substack{|I|=p \\ |J|=q}} \phi_{IJ} \, dz^I \wedge d\overline{z}^J : \phi_{IJ} \in C_c^\infty(\Omega) \text{ for all } I, J \right\} ;$$

$$W_s^{p,q} \equiv \left\{ \phi = \sum_{\substack{|I|=p \\ |J|=q}} \phi_{IJ} \, dz^I d\overline{z}^J : \phi_{IJ} \in W_s(\Omega) \text{ for all } I, J \right\} .$$

Here $W_s = W^s$ is the standard Sobolev space.[4] In what follows, when we write $\bigwedge_c^{p,q}(\overline{\Omega})$ or $\bigwedge_c^{p,q}(\overline{U} \cap \Omega)$, then we interpret the closure bar on the domain to mean that the support of the form is allowed to intersect the boundary. Thus the support is compact *in the closure*.

The final piece of notation that we need is the Hodge star operator. First consider the context of real analysis on $\mathbb{R}^N$. Let

$$\gamma = dx_1 \wedge \cdots \wedge dx_N$$

be the volume form, $\bigwedge^k$ the space of $k$-alternating forms, and $\bigwedge_Q^k$ the $k$-covectors at $Q \in \mathbb{R}^N$. Define

$$* : \bigwedge_Q^k \to \bigwedge_Q^{N-k}$$

by the equality

$$\langle \psi, \phi \rangle_Q \gamma = \psi_Q \wedge (*\phi_Q) \tag{10.2.2}$$

[4] It is also common in the literature to denote these spaces by $H_s$ or $H^s$.

for all $\psi \in \bigwedge_Q^k$. Now that we have defined the $*$ operation (pointwise) on covectors; we extend it to forms by setting

$$\langle \psi, \phi \rangle \gamma = \psi \wedge (*\phi)$$

for all $\psi \in \bigwedge^k$.

**EXAMPLE 10.2.3** Here is an example of the utility of the $*$ operator in a classical setting. Let

$$\Omega = \{x \in \mathbb{R}^N : r(x) < 0\}$$

be a smoothly bounded domain. We assume that $dr \neq 0$ on $\partial\Omega$. We may replace $r$ by $r/|dr|$ so that $|dr| = 1$ on $\partial\Omega$. Then the area form on $\partial\Omega$ is $*dr$. (Exercise: use Green's theorem.) □

Now let us turn our attention to $\mathbb{C}^n$. Define

$$* : \bigwedge_Q^{p,q} \to \bigwedge_Q^{n-q,n-p}$$

by the identity

$$\langle \psi, \phi \rangle_Q \gamma = \psi_Q \wedge \overline{(*\phi_Q)}$$

for $\psi \in \bigwedge_Q^{p,q}$. Likewise

$$* : \bigwedge^{p,q} \to \bigwedge^{n-p,n-q}$$

is defined by

$$\langle \psi, \phi \rangle \gamma = \psi \wedge \overline{(*\phi)}\,,$$

where $\langle \psi, \phi \rangle$ is the function of $z$ given by

$$\langle \phi, \psi \rangle(z) = 2^{p+q} \sum_{|I|=p, |J|=q} \phi_{IJ}(z) \overline{\psi_{IJ}}(z)\,.$$

We want to study existence and regularity for the partial differential equation

$$\overline{\partial} w = f\,. \tag{10.2.4}$$

We shall restrict attention to the case that $w$ is a function and $f$ a $(0,1)$ form. If $w$ exists then we may apply $\overline{\partial}$ to both sides of the equation to obtain

$$0 = \overline{\partial}^2 w = \overline{\partial} f.$$

Thus *a necessary condition for the equation* (10.2.4) *to have a solution is that* $\overline{\partial} f = 0$. This point bears a moment's discussion.

In the one-dimensional space $\mathbb{C}^1$, the Cauchy–Riemann equations for $C^1$ functions $w = \xi + i\eta$ and $f = u + iv$ can be written either as

$$\frac{1}{2}\left(\frac{\partial \xi}{\partial x} - \frac{\partial \eta}{\partial y}\right) = u \qquad \text{and} \qquad \frac{1}{2}\left(\frac{\partial \eta}{\partial x} + \frac{\partial \xi}{\partial y}\right) = v$$

or as

$$\frac{\partial w}{\partial \overline{z}} = f.$$

In either notation, the number of unknown functions is in balance with the number of equations. However in dimensions two and greater the situation is different. Write $f = f_1 dz_1 + \cdots + f_n dz_n$. Then our system (10.2.4) is

$$\frac{\partial v}{\partial \overline{z}_j} = f_j, \quad j = 1, \ldots, n,$$

and the number of equations $n$ exceeds the number of unknown functions 1. Passing to real notation results in no improvement. We call such a system *over-determined*. Algebraic considerations (or dimension) suggest that some compatibility conditions will be needed on the data in order for the system to be solvable. And indeed that is what we have discovered. (A formal theory of over-determined systems of partial differential equations has been developed by D. C. Spencer; see, for instance [SPE].)

There is also a problem with uniqueness for solutions of (10.2.4). For if $u$ is one solution to this equation and $h$ is any holomorphic function then $u + h$ is also a solution. In order to obtain a workable theory for this partial differential equation, we shall need an effective method for choosing a "canonical" solution. For this we shall exploit the metric structure introduced above. First let us see that the $\overline{\partial}$ operator is elliptic in a natural sense (we shall see that it is the boundary conditions associated with the $\overline{\partial}$ operator that are not elliptic). Since we are dealing with an operator on *forms*, a certain amount of formalism is ultimately necessary. However the $\overline{\partial}$ operator acting on functions may be thought of as an $n$- tuple composed of the operators $\partial/\partial \overline{z}_1, \ldots, \partial/\partial \overline{z}_n$. Each of these is plainly elliptic on a suitable copy of $\mathbb{C}$, and they span all directions. Hence so is the operator $\overline{\partial}$ itself elliptic.

Now we turn to the question of ellipticity on forms. Let $\Omega \subseteq \mathbb{C}^n$ be a domain. Think of $E = E^{p,q}$ as the vector bundle of $(p, q)$ covectors over $\Omega$. Then a differential form $\omega$ of type $(p, q)$ is a section of this vector bundle: we write $\omega \in \Gamma(E)$. Letting $F = E^{p,q+1}$, we then have

$$\overline{\partial} : \Gamma(E) \to \Gamma(F).$$

Fix $z \in \Omega$. The *symbol* of $\overline{\partial}$ assigns to each covector $\eta$ at $z$ a linear mapping

$$\sigma(\overline{\partial}, \eta) : E_z \to F_z.$$

This is done as follows: It is elementary to construct a scalar-valued function $\rho$ such that $\rho(z) = 0$ and $d\rho(z) = \eta$ (just use Tayor series—or even the fundamental theorem of calculus). If $\mu \in E_z$, then we let $\widetilde{\mu}$ be a local section of $E$ such that $\widetilde{\mu}(z) = \mu$ (just work in a neighborhood of $z$ over which $E$ is trivial). We define

$$\sigma(\overline{\partial}, \eta)\mu = \overline{\partial}(\rho \cdot \widetilde{\mu})\big|_z.$$

Of course in the case of functions the vector bundles $E$ and $F$ are just $\Omega \times \mathbb{C}$.

With this definition of symbol, we now explicitly calculate the symbol of $\overline{\partial}$. With $\eta, \rho, \mu$ as in the definition we have

$$\sigma(\overline{\partial}, \eta)\mu_z = \overline{\partial}(\rho\widetilde{\mu})_z = \left[\overline{\partial}\rho \wedge \widetilde{\mu} + \rho\overline{\partial}\widetilde{\mu}\right]_z = \{\Pi_{0,1}\eta\} \wedge \mu_z.$$

Here $\Pi_{0,1}$ is the projection of a 1-form into $\bigwedge^{0,1}$. We see that

$$\sigma(\overline{\partial}, \eta)(\,\cdot\,) = \Pi_{0,1}\eta \wedge (\,\cdot\,). \tag{10.2.5}$$

Now in the present general setting an operator is said to be elliptic if the complex

$$0 \longrightarrow E_z \overset{\sigma(\overline{\partial},\eta)}{\longrightarrow} F_z$$

is exact, that is if $\sigma(\overline{\partial}, \eta)$ is injective. In the case that $p = 0, q = 1$, that is when $\overline{\partial}$ is acting on functions, it is then clear that $\overline{\partial}$ is elliptic. When $\overline{\partial}$ is acting on forms of higher order, we must speak of *ellipticity of a complex*. Namely, a sequence

$$\Gamma(E) \overset{D_1}{\longrightarrow} \Gamma(F) \overset{D_2}{\longrightarrow} \Gamma(G)$$

is called elliptic if the induced sequence

$$E_z \overset{\sigma(D_1,\eta)}{\longrightarrow} F_z \overset{\sigma(D_2,\eta)}{\longrightarrow} G_z$$

is exact at $F_z$—that is, if the kernel of the second mapping equals the image of the first. We invite the reader to check that the $\overline{\partial}$ complex

$$\bigwedge^{p,q} \overset{\overline{\partial}}{\longrightarrow} \bigwedge^{p,q+1} \overset{\overline{\partial}}{\longrightarrow} \bigwedge^{p,q+2}$$

is elliptic (or see [FOK, pp. 10–11] for further details).

Let us now discuss the formal adjoint of the $\overline{\partial}$ operator. We work on a domain $\Omega \subseteq \mathbb{C}^n$. If we think of $\overline{\partial} : \bigwedge^{p,q} \to \bigwedge^{p,q+1}$ then the formal adjoint $\vartheta : \bigwedge_c^{p,q+1} \to \bigwedge_c^{p,q}$ is defined by the relation

$$\langle \vartheta\phi, \psi \rangle = \langle \phi, \overline{\partial}\psi \rangle$$

for all $\psi \in \bigwedge_c^{p,q}$. Here the inner product is the Hermitian inner product on forms that we introduced above. It is easy to see that the operator $\vartheta$ is well defined.

Let us calculate $\vartheta$ when $p = q = 0$. Thus we are looking at $\overline{\partial}$ acting on functions and sending them to (0,1) forms. If $\psi$ is a function then of course

$$\overline{\partial}\psi = \sum_{j=1}^{n} \frac{\partial \psi}{\partial \overline{z}_j} \, d\overline{z}_j \, .$$

Then, for $\phi$ a $(0, 1)$ form, we have

$$\begin{aligned} \langle \vartheta\phi, \psi \rangle &= \langle \phi, \overline{\partial}\psi \rangle \\ &= \left\langle \sum_j \phi_j d\overline{z}_j, \sum_j \frac{\partial \psi}{\partial \overline{z}_j} \, d\overline{z}_j \right\rangle \\ &= \sum_j \left\langle \phi_j d\overline{z}_j, \frac{\partial \psi}{\partial \overline{z}_j} d\overline{z}_j \right\rangle \\ &= \sum_j 2 \int_\Omega \phi_j \, \overline{\frac{\partial \psi}{\partial \overline{z}_j}} \, dV \, . \end{aligned}$$

Now we can perform integration by parts in each of these integrals; *because the function $\psi$ is compactly supported, the boundary terms in the integration by parts vanish.* The last line therefore equals

$$-\sum_j 2 \int \frac{\partial \phi_j}{\partial z_j} \, \overline{\psi} \, dV = \int \left( -2 \sum_j \frac{\partial \phi_j}{\partial z_j} \right) \overline{\psi} \, dV \, .$$

Comparing the far left and far right sides of our calculations, and using the fact that $C_c^\infty$ functions are dense in $L^2(\Omega)$, we find that

$$\vartheta\phi = -2 \sum_j \frac{\partial \phi_j}{\partial z_j} \, .$$

**Exercise** Check that, for $\overline{\partial}$ operating on general $(p, q)$ forms, the adjoint $\vartheta$ is

$$\vartheta\phi = 2(-1)^{p+1} \sum_{I,H,J,k} \epsilon_{kH}^{J} \frac{\partial \phi_{IJ}}{\partial z_k} \, dz^I \wedge d\overline{z}^H \, ,$$

where $\epsilon_{kH}^J$ is the sign of the permutation changing $(k, h_1, \ldots, h_q)$ into $J = (j_1, \ldots, j_{q+1})$.

One checks that $\overline{\partial}\vartheta + \vartheta\overline{\partial}$ is elliptic as an operator just because the $\overline{\partial}$ complex is elliptic.

In setting up the $\overline{\partial}$-Neumann problem, a principal task for us is to compare the formal adjoint $\vartheta$ of $\overline{\partial}$ with the the Hilbert space adjoint $\overline{\partial}^*$. Recall that, if $\mathcal{H}^1, \mathcal{H}^2$ are Hilbert spaces and $L : \mathcal{H}^1 \to \mathcal{H}^2$ is a (not necessarily bounded) linear operator defined on a dense subset of $\mathcal{H}^1$, then the Hilbert space adjoint $L^*$ is defined on the set

$$\operatorname{dom} L^* \equiv \left\{\phi \in \mathcal{H}^2 : \exists c > 0 \text{ with } |\langle \phi, L\psi\rangle_{\mathcal{H}^2}| \leq c\|\psi\|_{\mathcal{H}^1} \ \forall\ \psi \in \operatorname{dom} L\right\}.$$

Then the map

$$\mathcal{H}^1 \ni \psi \mapsto \langle \phi, L\psi\rangle_{\mathcal{H}^2}$$

is linear and satisfies $|\langle \phi, L\psi\rangle_{\mathcal{H}^2}| \leq c\|\psi\|_{\mathcal{H}^1}$. Since the domain of $L$ is dense, the functional extends to a bounded linear functional on all of $\mathcal{H}^1$ with the same bound. Therefore the Riesz representation theorem guarantees that there is an element $\alpha_\phi \in \mathcal{H}^1$ such that

$$\langle \phi, L\psi\rangle_{\mathcal{H}^2} = \langle \alpha_\phi, \psi\rangle_{\mathcal{H}^1}$$

for all $\psi \in \operatorname{dom} L$. We set $L^*\phi \equiv \alpha_\phi$.

If our Hilbert spaces are $L^2$ spaces then the operator $\overline{\partial}$ is densely defined on $\bigwedge_c^{p,q}$. We wish to determine the domain of $\overline{\partial}^*$, and to relate $\overline{\partial}^*$ to $\vartheta$. As a first step we prove the following lemma:

**Lemma 10.2.6** *Let $\phi, \psi \in C_c^\infty(\overline{\mathbb{R}^{N+1}_+})$ (that is, the supports of these functions may intersect the boundary). Let the partial differential operator $L$ be given by*

$$L = \sum_{j=1}^N a_j \frac{\partial}{\partial t_j} + b\frac{\partial}{\partial r},$$

*where $r$ is the downward (negative) normal coordinate to $\mathbb{R}^N = \partial\mathbb{R}^{N+1}_+$ and $a_j, b$ are functions. Then*

$$\langle L\phi, \psi\rangle = \langle \phi, L'\psi\rangle + \int_{\mathbb{R}^N} \left[\phi b\overline{\psi}\right]_{r=0} dt,$$

*where $L'$ is the formal adjoint for $L$ which is determined, as usual, by inner product with functions that are compactly supported in $\mathbb{R}^{N+1}_+$.*

**Proof** Now

$$\begin{aligned}\langle L\phi, \psi\rangle &= \int_{\mathbb{R}^{N+1}_+} \left(\sum_{j=1}^N a_j \frac{\partial \phi}{\partial t_j} + b\frac{\partial \phi}{\partial r}\right)\overline{\psi}\, dt\, dr \\ &= \sum_{j=1}^N \int_{\mathbb{R}^{N+1}_+} a_j \frac{\partial \phi}{\partial t_j}\overline{\psi}\, dt\, dr \;+\; \int_{\mathbb{R}^{N+1}_+} b\frac{\partial \phi}{\partial r}\overline{\psi}\, dt\, dr \\ &= -\sum_{j=1}^N \int_{\mathbb{R}^{N+1}_+} \phi \frac{\partial}{\partial t_j}\left(a_j\overline{\psi}\right) dt\, dr + \int_{\mathbb{R}^N} \phi b\overline{\psi}\Big|_{r=-\infty}^{0} dt \\ &\quad - \int_{\mathbb{R}^{N+1}_+} \phi \frac{\partial}{\partial r}\left(b\overline{\psi}\right) dt\, dr\,.\end{aligned}$$

Notice that, in the first group of integrals, we have used the fact that the $t_j$ directions are *tangential*, together with the compact support of $\psi$, to see that no boundary terms result from the integrations by parts. Now the last line is

$$\begin{aligned}&\int_{\mathbb{R}^{N+1}_+} \phi \left[-\sum_{j=1}^N \frac{\partial}{\partial t_j}\left(\overline{a_j}\psi\right) - \frac{\partial}{\partial r}\left(\overline{b}\psi\right)\right] dt\, dr \\ &\qquad + \int_{\mathbb{R}^N} \left[\phi b\overline{\psi}\right]_{r=0} dt \\ &\equiv \langle \phi, L'\psi\rangle + \int_{\mathbb{R}^N} \left[\phi b\overline{\psi}\right]_{r=0} dt\,.\end{aligned}$$

That completes the proof. □

A simple computation shows that

$$\sigma\left(L, \sum_{j=1}^N \eta_j dt_j + \eta_r dr\right) = \sum_{j=1}^N \eta_j a_j(t,r) + \eta_r b(t,r)\,.$$

In particular, $\sigma(L, dr) = b$ and the integral from our lemma satisfies

$$\int_{\mathbb{R}^N} \left[\phi b\overline{\psi}\right]_{r=0} dt = \int_{\partial\mathbb{R}^{N+1}_+} \left\langle \sigma(L, dr)\phi, \psi\right\rangle dt\,.$$

The arguments that we have presented are easily adapted from the half space to a smoothly bounded domain (just use a local boundary coordinate patch). The result is that

$$\langle Lf, g\rangle = \langle f, L'g\rangle + \int_{\partial\Omega} \langle \sigma(L, dr)f, g\rangle\,.$$

The arguments also are easily applied to partial differential operators on a vector bundle (we leave details to the interested reader). Thus, in particular,

$$\left\langle \overline{\partial}\phi, \psi \right\rangle = \left\langle \phi, \vartheta\psi \right\rangle + \int_{\partial\Omega} \left\langle \sigma(\overline{\partial}, dr)\phi, \psi \right\rangle \tag{10.2.7}$$

and

$$\left\langle \vartheta\phi, \psi \right\rangle = \left\langle \phi, \overline{\partial}\psi \right\rangle + \int_{\partial\Omega} \left\langle \sigma(\vartheta, dr)\phi, \psi \right\rangle. \tag{10.2.8}$$

Set

$$\mathcal{D}^{p,q} \equiv \bigwedge^{p,q}(\overline{\Omega}) \cap \operatorname{dom} \overline{\partial}^* .$$

Then we have

**Proposition 10.2.9** *The linear space $\mathcal{D}^{p,q}$ is equal to the space of those $\phi \in \bigwedge^{p,q}(\overline{\Omega})$ such that $\sigma(\vartheta, dr)\phi = 0$ on $\partial\Omega$. Moreover, $\overline{\partial}^* = \vartheta$ on $\mathcal{D}^{p,q}$.*

**Proof** Let $\phi \in \mathcal{D}^{p,q}$ and $\psi \in \bigwedge_c^{p,q-1}(\Omega)$. According to formula (10.2.8) above, we have

$$\begin{aligned} \left\langle \overline{\partial}^*\phi, \psi \right\rangle &= \left\langle \phi, \overline{\partial}\psi \right\rangle \\ &= \left\langle \vartheta\phi, \psi \right\rangle - \int_{\partial\Omega} \left\langle \sigma(\vartheta, dr)\phi, \psi \right\rangle \\ &= \left\langle \vartheta\phi, \psi \right\rangle . \end{aligned} \tag{10.2.9.1}$$

In the last equality we have used the fact that $\psi$ vanishes on $\partial\Omega$ to make the integral vanish.

But $\bigwedge_c^{p,q-1}$ is dense in $H_0^{p,q-1}$. Therefore

$$\left\langle \overline{\partial}^*\phi, \psi \right\rangle = \left\langle \vartheta\phi, \psi \right\rangle$$

for all $\psi \in H_0^{p,q-1}$; that is,

$$\overline{\partial}^*\phi = \vartheta\phi$$

for all $\phi \in \mathcal{D}^{p,q}$. But then (10.2.8) implies that

$$\int_{\partial\Omega} \left\langle \sigma(\vartheta, dr)\phi, \psi \right\rangle = 0$$

for $\phi \in \mathcal{D}^{p,q}$, $\psi \in \bigwedge^{p,q-1}$. Therefore $\sigma(\vartheta, dr)\phi = 0$ on $\partial\Omega$. We have proved that

$$\mathcal{D}^{p,q} \subseteq \left\{\phi \in \bigwedge^{p,q}(\overline{\Omega}) : \sigma(\vartheta, dr)\phi = 0 \text{ on } \partial\Omega\right\} .$$

Conversely, let $\phi \in \bigwedge^{p,q}(\overline{\Omega})$ satisfy $\sigma(\vartheta, dr)\phi = 0$ on $\partial\Omega$. From the equation

$$\langle \phi, \overline{\partial}\psi\rangle = \langle \vartheta\phi, \psi\rangle - \int_{\partial\Omega} \langle \sigma(\vartheta, dr)\phi, \psi\rangle$$

for $\psi \in \bigwedge_c^{p,q-1}(\overline{\Omega})$, we learn that

$$\langle \phi, \overline{\partial}\psi\rangle = \langle \vartheta\phi, \psi\rangle$$

and therefore

$$|\langle \phi, \overline{\partial}\psi\rangle| \leq \|\vartheta\phi\|_{L^2}\|\psi\|_{L^2} .$$

Therefore $\phi \in \operatorname{dom} \overline{\partial}^*$ and we see that $\overline{\partial}^*\phi = \vartheta\phi$. That proves the proposition. □

## 10.3 Statement of the $\overline{\partial}$-Neumann Problem

**Capsule:** The $\overline{\partial}$-problem has proved to be one of the most central, and one of the most powerful, tools of complex analysis. We see over and over again that the tools of partial differential equations are more effective than the tools of topology and geometry. This is another instance of that phenomenon.

Let $L$ be a uniformly elliptic partial differential operator of order $m$. Classically, the heart of the study of an elliptic boundary value problem for $L$ is a *coercive estimate* of the form

$$\sum_{|\alpha|\leq m} \|D^\alpha u\|_{W^0} \leq C\big(\|Lu\|_{W^0} + \|u\|_{W^0}\big).$$

However the condition that we discovered earlier, namely that $\sigma(\vartheta, dr)\phi = 0$ on $\partial\Omega$, is not a coercive boundary condition. We shall need to develop a substitute for the classical approach, and that will require building up some machinery. We begin with some elementary functional analysis:

**Lemma 10.3.1 (Friedrichs)** *Let $H$ be a Hilbert space equipped with the inner product $\langle \cdot\,, \cdot \rangle$ and corresponding norm $\|\ \|$. Let $Q(\cdot\,, \cdot)$ be a densely defined hermitian form on $H$. Let $\mathcal{D}$ be the domain of $Q$ and assume that*

$$Q(\phi, \phi) \geq c \cdot \|\phi\|^2$$

*for all $\phi \in \mathcal{D}$. Assume also that $\mathcal{D}$ itself is a Hilbert space when it is equipped with the inner product $Q$.*

*Then there exists a canonical self-adjoint map $F$ on $H$ associated with $Q$ such that*

**(1)** *$\mathrm{dom}\, F \subseteq \mathcal{D}$;*
**(2)** *$Q(\phi, \psi) = (F\phi, \psi)$ for all $\phi \in \mathrm{dom}\, F, \psi \in \mathcal{D}$.*

**Proof** After rescaling, we may assume that $c = 1$. Let $\alpha \in H$ and consider the linear functional on $\mathcal{D}$ given by

$$\mu_\alpha : \mathcal{D} \ni \psi \mapsto \langle \psi, \alpha \rangle.$$

We have

$$\begin{aligned} |\mu_\alpha(\psi)| &\leq \|\alpha\| \cdot \|\psi\| \\ &\leq \|\alpha\| \cdot Q(\psi, \psi)^{1/2}. \end{aligned}$$

Therefore the Riesz representation theorem implies the existence of an element $\phi_\alpha \in \mathcal{D}$ such that

$$\begin{aligned} \langle \psi, \alpha \rangle &= \mu_\alpha(\psi) \\ &= Q(\psi, \phi_\alpha), \end{aligned}$$

that is,

$$\langle \alpha, \psi \rangle = Q(\phi_\alpha, \psi).$$

Define

$$T : H \ni \alpha \mapsto \phi_\alpha \in \mathcal{D}.$$

Then

$$\begin{aligned} \|T\alpha\|^2 &\leq Q(T\alpha, T\alpha) \\ &= \langle \alpha, T\alpha \rangle \\ &\leq \|\alpha\| \cdot \|T\alpha\|. \end{aligned}$$

It follows that $\|T\alpha\| \leq \|\alpha\|$ and $T$ is bounded as an operator from $H$ to $H$.

The operator $T$ is injective since $T\alpha = 0$ implies that $\langle \alpha, \psi \rangle = Q(T\alpha, \psi) = 0$ for every $\psi \in \mathcal{D}$ and $\mathcal{D}$ is dense in $H$. Furthermore $T$ is self-adjoint because

$$\begin{aligned} \langle T\alpha, \beta \rangle &= \overline{\langle \beta, T\alpha \rangle} \\ &= \overline{Q(T\beta, T\alpha)} \\ &= Q(T\alpha, T\beta) \\ &= \langle \alpha, T\beta \rangle. \end{aligned}$$

Next, set $U = \operatorname{range} T \subseteq \mathcal{D}$ and define

$$F = T^{-1} : U \to H.$$

By the equality $Q(T\alpha, \psi) = \langle \alpha, \psi \rangle$ we obtain

$$Q(\beta, \psi) = \langle F\beta, \psi \rangle$$

for $\psi \in \mathcal{D}, \beta \in U$. That completes the proof. □

**Exercise** Show that $U$ is dense both in $\mathcal{D}$ and in $H$. (Hint: Use the fact that $T$ is self-adjoint.)

We intend to apply the Friedrichs lemma to the $\overline{\partial}$-Neumann problem. To this end, we introduce the following notation:

**(1)** Let $H = H_0^{p,q} = \big\{(p,q)-$ forms with $L^2$ coefficients on $\Omega\big\}$;
**(2)** Let $Q(\phi, \psi) = (\overline{\partial}\phi, \overline{\partial}\psi) + (\vartheta\phi, \vartheta\psi) + (\phi, \psi)$;
**(3)** Let $\mathcal{D} = \widetilde{\mathcal{D}}^{p,q} \equiv$ the closure of $\mathcal{D}^{p,q}$ in the $Q$-topology.

Now we have to do some formal checking: There is a natural continuous inclusion of $\mathcal{D}^{p,q}$ in $H_0^{p,q}$. We will see that this induces an inclusion of $\widetilde{\mathcal{D}}^{p,q}$ in $H_0^{p,q}$.

Let $\{\phi_n\}$ be a $Q$-Cauchy sequence in $\mathcal{D}^{p,q}$. By definition of $Q$, we see that $\{\phi_n\}, \{\vartheta\phi_n\}$, and $\{\overline{\partial}\phi_n\}$ are Cauchy sequences in $H_0^{p,q}$ (i.e., in the $L^2$ topology). Let $\phi$ be the $L^2$ limit of $\{\phi_n\}$. When interpreted in the weak or distribution sense, $\overline{\partial}$ and $\vartheta$ are closed operators. Hence we have

$$\overline{\partial}\phi_n \to \overline{\partial}\phi$$

and

$$\vartheta\phi_n \to \vartheta\phi.$$

If $\phi = 0$ in $L^2$ then

$$Q(\phi, \phi) = \lim_n Q(\phi_n, \phi_n) = \lim_{n\to\infty} \Big[ \langle \phi_n, \phi_n \rangle + \langle \overline{\partial}\phi_n, \overline{\partial}\phi_n \rangle + \langle \vartheta\phi_n, \vartheta\phi_n \rangle \Big]$$

$$= \langle \phi, \phi \rangle + \langle \overline{\partial}\phi, \overline{\partial}\phi \rangle + \langle \vartheta\phi, \vartheta\phi \rangle = 0.$$

Therefore $\phi_n \to 0$ in the $Q$-topology. It follows that the inclusion $\mathcal{D}^{p,q} \hookrightarrow H_0^{p,q}$ extends to the inclusion $\widetilde{\mathcal{D}}^{p,q} \hookrightarrow H_0^{p,q}$.

Now we may apply the Friedrichs theorem to obtain a (canonical) self-adjoint map

$$F : \operatorname{dom} F\big( \underset{\text{dense}}{\subseteq} \widetilde{\mathcal{D}}^{p,q} \big) \to H_0^{p,q}$$

such that

$$Q(\phi, \psi) = \langle F\phi, \psi \rangle$$

for all $\phi \in \operatorname{dom} F$, $\psi \in \mathcal{D}$. Now we need to identify $F$ and determine its domain.

Set

$$\mathcal{D}_c^{p,q} \equiv \bigwedge_c^{p,q} \bigcap \operatorname{dom} \overline{\partial}^* = \{\rho \in \mathcal{D}^{p,q} : \operatorname{supp} \rho \text{ is compact}\}.$$

If $\phi, \psi \in \mathcal{D}_c^{p,q}$ then

$$\begin{aligned} Q(\phi, \psi) &= \langle \overline{\partial}\phi, \overline{\partial}\psi \rangle + \langle \vartheta\phi, \vartheta\psi \rangle + \langle \phi, \psi \rangle \\ &= \langle \vartheta\overline{\partial}\phi, \psi \rangle + \langle \overline{\partial}\vartheta\phi, \psi \rangle + \langle \phi, \psi \rangle \\ &= \left\langle \left( (\vartheta\overline{\partial} + \overline{\partial}\vartheta) + I \right)\phi, \psi \right\rangle \\ &\equiv \big\langle (\Box + I)\phi, \psi \big\rangle. \end{aligned}$$

It must be noted here explicitly that $\Box = \vartheta\overline{\partial} + \overline{\partial}\vartheta$. This is the $\overline{\partial}$ Laplacian. As we shall see below, the $\overline{\partial}$-Neumann operator $N$ will be a canonical right inverse for $\Box$.

Notice also that passing $\vartheta$ and $\overline{\partial}$ from side to side in the inner products is justified here because $\phi, \psi$ are compactly supported in $\Omega$. The reader may also check that, because we are working in $\mathbb{C}^n$ with the standard Hermitian inner product, it turns out that $\Box = -\triangle$. We use the notation $\Box$ by convention.

**Proposition 10.3.2** *Let $\phi \in \mathcal{D}^{p,q}$. Then $\phi \in \operatorname{dom} F$ if and only if $\overline{\partial}\phi \in \mathcal{D}^{p,q+1}$. In this case it holds that*

$$F\phi = (\Box + I)\phi.$$

**Proof** Let $\phi \in \mathcal{D}^{p,q}$ lie in $\operatorname{dom} F$. We know that, for $\psi \in \bigwedge_c^{p,q}$,

$$\langle F\phi, \psi \rangle = \big\langle (\Box + I)\phi, \psi \big\rangle. \tag{10.3.2.1}$$

Since $\bigwedge_c^{p,q}$ is dense in $H_0^{p,q}$, it follows that (10.3.2.1) holds for $\psi \in \mathcal{D}^{p,q}$. We now see that

$$\begin{aligned} \langle \vartheta\phi, \vartheta\psi \rangle &= \langle \overline{\partial}\vartheta\phi, \psi \rangle - \int_{\partial\Omega} \langle \sigma(\overline{\partial}, dr)\vartheta\phi, \psi \rangle \\ &= \langle \overline{\partial}\vartheta\phi, \psi \rangle - \int_{\partial\Omega} \langle \vartheta\phi, \sigma(\vartheta, dr)\psi \rangle \\ &= \langle \overline{\partial}\vartheta\phi, \psi \rangle \end{aligned} \tag{10.3.2.2}$$

since $\psi \in \mathcal{D}^{p,q}$ (hence $\sigma(\vartheta, dr)\psi = 0$ on $\partial\Omega$). Also,

$$\langle \overline{\partial}\phi, \overline{\partial}\psi\rangle = \langle \vartheta\overline{\partial}\phi, \psi\rangle - \int_{\partial\Omega} \langle \sigma(\vartheta, dr)\overline{\partial}\phi, \psi\rangle.$$

We cannot argue as before since we have no control over the support of $\phi$. Instead we proceed as follows:

$$\begin{aligned} Q(\phi, \psi) &= \langle \overline{\partial}\phi, \overline{\partial}\psi\rangle + \langle \vartheta\phi, \vartheta\psi\rangle + \langle \phi, \psi\rangle \\ &= \langle \vartheta\overline{\partial}\phi, \psi\rangle - \int_{\partial\Omega} \langle \sigma(\vartheta, dr)\overline{\partial}\phi, \psi\rangle + \langle \vartheta\phi, \vartheta\psi\rangle + \langle \phi, \psi\rangle \\ &= \big\langle (\Box + I)\phi, \psi\big\rangle - \int_{\partial\Omega} \langle \sigma(\vartheta, dr)\overline{\partial}\phi, \psi\rangle. \end{aligned}$$

Here we have used (10.3.2.2). Since $\phi \in \operatorname{dom} F$, we know that

$$Q(\phi, \psi) = \langle F\phi, \psi\rangle = \big\langle (\Box + I)\phi, \psi\big\rangle.$$

Therefore

$$\int_{\partial\Omega} \langle \sigma(\vartheta, dr)\overline{\partial}\phi, \psi\rangle = 0 \tag{10.3.2.3}$$

for all $\psi \in \bigwedge_c^{p,q}$; by density, (10.3.2.3) persists for $\psi \in \widetilde{\mathcal{D}}^{p,q}$. We wish to conclude that $\sigma(\vartheta, dr)\overline{\partial}\phi = 0$ on $\partial\Omega$. If we choose $\psi$ to be $\sigma(\vartheta, dr)\overline{\partial}\phi$ then

$$\begin{aligned} \sigma(\vartheta, dr)\psi &= \sigma(\vartheta, dr)\big(\sigma(\vartheta, dr)\overline{\partial}\phi\big) \\ &= \sigma(\vartheta^2, dr)\overline{\partial}\phi \\ &= 0. \end{aligned}$$

Hence $\psi = \sigma(\vartheta, dr)\overline{\partial}\phi \in \mathcal{D}^{p,q}$ and putting this choice of $\psi$ into (10.3.2.3) yields

$$\int_{\partial\Omega} \langle \sigma(\vartheta, dr)\overline{\partial}\phi, \sigma(\vartheta, dr)\overline{\partial}\phi\rangle = 0.$$

Therefore

$$\sigma(\vartheta, dr)\overline{\partial}\phi = 0 \qquad \text{on } \partial\Omega.$$

We conclude that $\overline{\partial}\phi \in \widetilde{\mathcal{D}}^{p,q+1}$. That $F\phi = (\Box + I)\phi$ is then automatic from what has gone before.

Conversely, if $\phi \in \mathcal{D}^{p,q}$ and $\overline{\partial}\phi \in \mathcal{D}^{p,q+1}$ we let $\psi \in \mathcal{D}^{p,q}$. Then

$$Q(\phi, \psi) = \langle (\Box + I)\phi, \psi\rangle$$

(just calculate) hence $\phi \in \operatorname{dom} F$ and $F\phi = (\Box + I)\phi$. □

We have made the following important discovery:

**The $\overline{\partial}$-Neumann Boundary Conditions**

Let $\phi \in \bigwedge^{p,q}(\overline{\Omega})$. Then $\phi \in \operatorname{dom} F$ if and only if

**(1)** $\phi \in \mathcal{D}^{p,q}$ (that is, $\sigma(\vartheta, dr)\phi = 0$ on $\partial\Omega$);
**(2)** $\overline{\partial}\phi \in \mathcal{D}^{p,q+1}$ (that is, $\sigma(\vartheta, dr)\overline{\partial}\phi = 0$ on $\partial\Omega$).

In studying the $\overline{\partial}$-Neumann problem, we will follow the paradigm laid out in the study of elliptic boundary value problems in Chapter 5 of [KRA5]. Namely, we will prove an *a priori* estimate which will in turn lead to a sharp existence and regularity result.

We remark in passing that the $\overline{\partial}$-Neumann problem as formulated and solved here is in the Hilbert space $L^2 = H_0$. It would be of some interest to solve the problem in other Hilbert spaces, such as the Sobolev space $W^s$. The groundwork for studying those Hilbert space versions of the $\overline{\partial}$-Neumann problem has been laid in [FOSP].

## 10.4 The Main Estimate

**Capsule:** The main estimate is the key to understanding the $\overline{\partial}$-Neumann problem. All of the subsequent results—compactness, subelliptic estimates, and so forth—depend on the main estimate. It is worth taking some time to understand where the main estimate comes from and how it is used.

If $\alpha \in H_0^{p,q}$ then the construction of the operator $F$ guarantees the existence of a unique $\phi \in \widetilde{\mathcal{D}}^{p,q}$ such that $F\phi = \alpha$. We are interested in the regularity up to the boundary of this solution $\phi$. If we had a classical coercive inequality

$$Q(\phi, \phi) \geq c\|\phi\|^2_{W^1} ,$$

then it would be straightforward to prove the desired estimates. We do not have such an inequality, but will find a substitute that will do the job.

**Lemma 10.4.1** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded domain with defining function $r$: $\Omega = \{z \in \mathbb{C}^n : r(z) < 0\}$ and $|\nabla r| = 1$ on $\partial\Omega$. Then*

**(1)** *If $\phi \in \bigwedge^{0,1}$, then $\phi \in \mathcal{D}^{0,1}$ if and only if $\sum_j (\partial r/\partial z_j)\phi_j = 0$ on $\partial\Omega$;*
**(2)**

$$\frac{\partial r}{\partial z_j}(*dr) = \left(-\frac{1}{2i}\right)^n dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge \widehat{dz_j} \wedge \cdots \wedge dz_n \wedge d\overline{z}_n$$

*on $\partial\Omega$;*

**(3)** *If $f, g \in C^\infty(\overline{\Omega})$ then we have*

$$d\left[f\overline{g}\left(-\frac{1}{2i}\right)^n dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge \widehat{dz_j} \wedge \cdots \wedge dz_n \wedge \overline{z}_n\right] = \left(\frac{\partial f}{\partial z_j}\overline{g} + f\frac{\partial \overline{g}}{\partial z_j}\right)\gamma,$$

*where $\gamma$ is the volume element;*

**(4)**

$$\left(-\frac{1}{2i}\right)^n \sum_j \int_{\partial\Omega} f\overline{g}\frac{\partial r}{\partial z_j}\, d\sigma = \sum_j \int_\Omega \left(\frac{\partial f}{\partial z_j}\overline{g} + f\frac{\partial \overline{g}}{\partial z_j}\right)\gamma.$$

**(5)** *The analogue of* **(4)** *with $\partial/\partial z_j$ replaced by $\partial/\partial \overline{z}_j$.*

**Proof (1)** This is just definition checking: Let $p \in \partial\Omega$ and let $r$ be a defining function with $|dr| = 1$ on $\partial\Omega$. Of course $r(p) = 0$. Let $\phi = \sum_{i=1}^n \phi_i d\overline{z}_i$. Then $\phi \in \mathcal{D}^{0,1}$ if and only if

$$\begin{aligned} 0 &= \sigma(\vartheta, dr)\phi\big|_p \\ &= \vartheta(r\phi)\big|_p \\ &= (\vartheta r)_p(\phi_p) \\ &= -\sum_{j=1}^n \frac{\partial r}{\partial z_j}(p)\phi_j(p). \end{aligned}$$

If we apply $\wedge dr$ to both sides of the equation in the statement of **(2)**, then we obtain $\partial r/\partial z_j \cdot \gamma = \partial r/\partial z_j \cdot \gamma$. Result **(2)** follows.

To prove **(3)**, we calculate that

$$\begin{aligned} &d\left[f\overline{g}\left(-\frac{1}{2i}\right)^n dz_1 \wedge d\overline{z}_1 \wedge \cdots \widehat{dz_j} \wedge \cdots \wedge dz_n \wedge d\overline{z}_n\right] \\ &= \left[\frac{\partial}{\partial z_j}(f\overline{g})\left(-\frac{1}{2i}\right)^n dz_j \wedge dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge \widehat{dz_j} \wedge \cdots \wedge dz_n \wedge d\overline{z}_n\right] \\ &= \left(-\frac{1}{2i}\right)^n \left(\frac{\partial f}{\partial z_j}\overline{g} + f\frac{\partial \overline{g}}{\partial z_j}\right)\gamma. \end{aligned}$$

To prove **(4)**, we apply Stokes's theorem to **(3)**. Thus

$$\sum_{j=1}^n \int_{\partial\Omega} f\overline{g}\left(-\frac{1}{2i}\right)^n dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge \widehat{dz_j} \wedge \cdots \wedge dz_n \wedge d\overline{z}_n$$

$$= \sum_{j=1}^{n} \int_{\Omega} d\left[ f\overline{g} \left( -\frac{1}{2i} \right)^n dz_1 \wedge d\overline{z}_1 \wedge \cdots \wedge \widehat{d\overline{z}_j} \wedge \cdots \wedge dz_n \wedge d\overline{z}_n \right]$$

$$= \sum_{j=1}^{n} \int_{\Omega} \left( \frac{\partial f}{\partial z_j} \overline{g} + f \frac{\partial \overline{g}}{\partial z_j} \right) \gamma.$$

Now we use **(2)** and the fact that $*dr = d\sigma$ to see that the left side is equal to

$$\left( -\frac{1}{2i} \right)^n \sum_{j=1}^{n} \int_{\partial\Omega} f\overline{g} \frac{\partial r}{\partial z_j} \, d\sigma. \qquad \square$$

**Proposition 10.4.2** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded domain in $\mathbb{C}^n$ with defining function $r$. Let $\phi \in \mathcal{D}^{0,1}$. Then*

$$Q(\phi, \phi) = \sum_{j,k} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2 + \sum_{j,k} \int_{\partial\Omega} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} \phi_j \overline{\phi}_k + \|\phi\|^2.$$

*(Here $\| \cdot \|$ denotes the $L^2$ norm.)*

**Proof** If $\phi \in \bigwedge^{0,1}(\overline{\Omega})$ then

$$\begin{aligned} \overline{\partial}\phi &= \overline{\partial} \left( \sum_{j=1}^{n} \phi_j \, d\overline{z}_j \right) \\ &= \sum_j \sum_k \frac{\partial \phi_j}{\partial \overline{z}_k} \, d\overline{z}_k \wedge d\overline{z}_j \\ &= \sum_{j>k} \left( \frac{\partial \phi_j}{\partial \overline{z}_k} - \frac{\partial \phi_k}{\partial \overline{z}_j} \right) d\overline{z}_k \wedge d\overline{z}_j. \end{aligned}$$

Now

$$\begin{aligned} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} - \frac{\partial \phi_k}{\partial \overline{z}_j} \right\|^2 &= \left\langle \frac{\partial \phi_j}{\partial \overline{z}_k} - \frac{\partial \phi_k}{\partial \overline{z}_j}, \frac{\partial \phi_j}{\partial \overline{z}_k} - \frac{\partial \phi_k}{\partial \overline{z}_j} \right\rangle \\ &= \left\| \frac{\partial \phi_k}{\partial \overline{z}_j} \right\|^2 + \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2 - \left\langle \frac{\partial \phi_k}{\partial \overline{z}_j}, \frac{\partial \phi_j}{\partial \overline{z}_k} \right\rangle - \left\langle \frac{\partial \phi_j}{\partial \overline{z}_k}, \frac{\partial \phi_k}{\partial \overline{z}_j} \right\rangle. \end{aligned}$$

Therefore

$$\|\overline{\partial}\phi\|^2 = \sum_{j>k} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} - \frac{\partial \phi_k}{\partial \overline{z}_j} \right\|_{W^0}^2$$

$$= \frac{1}{2}\sum_{j,k=1}^{n}\left\|\frac{\partial\phi_j}{\partial\overline{z}_k}-\frac{\partial\phi_k}{\partial\overline{z}_j}\right\|_{W^0}^2$$

$$= \sum_{j,k=1}^{n}\left\|\frac{\partial\phi_j}{\partial\overline{z}_k}\right\|_{W^0}^2-\sum_{j,k=1}^{n}\left\langle\frac{\partial\phi_j}{\partial\overline{z}_k},\frac{\partial\phi_k}{\partial\overline{z}_j}\right\rangle. \tag{10.4.2.1}$$

Now

$$\sum_{j,k=1}^{n}\left\langle\frac{\partial\phi_j}{\partial\overline{z}_k},\frac{\partial\phi_k}{\partial\overline{z}_j}\right\rangle = \sum_{j,k=1}^{n}\int_\Omega\frac{\partial\phi_j}{\partial\overline{z}_k}\frac{\partial\overline{\phi_k}}{\partial z_j}\,\gamma$$

$$= -\sum_{j,k=1}^{n}\int_\Omega\frac{\partial^2\phi_j}{\partial\overline{z}_k\partial z_j}\overline{\phi}_k\,\gamma+\sum_{j,k=1}^{n}\int_{\partial\Omega}\frac{\partial\phi_j}{\partial\overline{z}_k}\overline{\phi}_k\frac{\partial r}{\partial z_j}\,d\sigma$$

$$= \sum_{j,k=1}^{n}\int_\Omega\frac{\partial\phi_j}{\partial z_j}\frac{\partial\overline{\phi_k}}{\partial\overline{z}_k}\,\gamma-\sum_{j,k=1}^{n}\int_{\partial\Omega}\frac{\partial\phi_j}{\partial z_j}\overline{\phi}_k\frac{\partial r}{\partial\overline{z}_k}\,d\sigma$$

$$+\sum_{j,k=1}^{n}\int_{\partial\Omega}\frac{\partial\phi_j}{\partial\overline{z}_k}\overline{\phi}_k\frac{\partial r}{\partial z_j}\,d\sigma, \tag{10.4.2.2}$$

where we have used part **(4)** of the last lemma twice. By part **(1)** of that lemma,

$$\sum_k\frac{\partial r}{\partial\overline{z}_k}\overline{\phi}_k=\overline{\sum_k\left(\frac{\partial r}{\partial z_k}\phi_k\right)}=0$$

on $\partial\Omega$. Hence the second term on the right hand side of (10.4.2.2) vanishes.

Notice that the condition

$$\sum_j\frac{\partial r}{\partial z_j}\phi_j=0$$

on $\partial\Omega$ (since $\phi\in\mathcal{D}^{0,1}$) implies that any tangential derivative of this expression vanishes on $\partial\Omega$. Further observe that

$$\overline{\sum_k\phi_k\frac{\partial}{\partial z_k}}$$

is a tangential derivative. Therefore, on $\partial\Omega$,

$$\sum_k\overline{\phi}_k\frac{\partial}{\partial\overline{z}_k}\left(\sum_j\phi_j\frac{\partial r}{\partial z_j}\right)=0\,.$$

This means that on $\partial\Omega$ we have

$$\sum_{j,k} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} \phi_j \overline{\phi_k} = -\sum_{j,k} \frac{\partial r}{\partial z_j} \frac{\partial \phi_j}{\partial \overline{z}_k} \overline{\phi_k}. \tag{10.4.2.3}$$

Now equations (10.4.2.2) and (10.4.2.3) yield that

$$\sum_{j,k} \left\langle \frac{\partial \phi_j}{\partial \overline{z}_k}, \frac{\partial \phi_k}{\partial \overline{z}_j} \right\rangle = \sum_{j,k} \left\langle \frac{\partial \phi_j}{\partial z_j}, \frac{\partial \phi_k}{\partial z_k} \right\rangle - \sum_{j,k} \int_{\partial\Omega} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} \phi_j \overline{\phi_k}\, d\sigma. \tag{10.4.2.4}$$

Substituting (10.4.2.4) into (10.4.2.1) yields that

$$\begin{aligned} \|\overline{\partial}\phi\|^2 &= \sum_{j,k} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2 - \sum_{j,k} \left\langle \frac{\partial \phi_j}{\partial z_j}, \frac{\partial \phi_k}{\partial z_k} \right\rangle + \int_{\partial\Omega} \sum_{j,k} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} \phi_j \overline{\phi_k}\, d\sigma \\ &= \sum_{j,k} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2 - \|\vartheta\phi\|^2 + \int_{\partial\Omega} \sum_{j,k} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} \phi_j \overline{\phi_k}\, d\sigma. \end{aligned}$$

As a result,

$$\begin{aligned} Q(\phi,\phi) &= \|\overline{\partial}\phi\|^2 + \|\vartheta\phi\|^2 + \|\phi\|^2 \\ &= \sum_{j,k} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2 + \int_{\partial\Omega} \sum_{j,k} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} \phi_j \overline{\phi_k}\, d\sigma + \|\phi\|^2. \end{aligned}$$

That completes the proof. □

**Definition 10.4.3** Let $\Omega \equiv \{z \in \mathbb{C}^n : r(z) < 0\}$ be a smoothly bounded domain. If $p \in \partial\Omega$ then let $a$ be a complex tangent vector to $\partial\Omega$ at $p$, i.e.

$$\sum_{j=1}^{n} \frac{\partial r}{\partial z_j}(p) a_j = 0.$$

The *Levi form* at $p$ is defined to be the quadratic form

$$L_p(a) = \sum_{j,k=1}^{n} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k}(p) a_j \overline{a}_k.$$

We say that $\Omega$ is *pseudoconvex* at $p$ if $L_p$ is positive semi-definite on the space of complex tangent vectors. We say that $\Omega$ is *strongly pseudoconvex* at $p$ if $L_p$ is positive definite on the space of complex tangent vectors. We say that $\Omega$ is pseudoconvex (resp. strongly pseudoconvex) if every boundary point is pseudoconvex (resp. strongly pseudoconvex).

For emphasis, a pseudoconvex domain is sometimes called *weakly pseudoconvex*.

The geometric meaning of strong pseudoconvexity does not lie near the surface. It is a biholomorphically invariant version of strong convexity (which concept is *not* biholomorphically invariant). A detailed discussion of pseudoconvexity and strong pseudoconvexity appears in [KRA1, Ch. 3]. In particular, it is proved in that reference that if $p$ is a point of strong pseudoconvexity then there is a biholomorphic change of coordinates in a neighborhood of $p$ so that $p$ becomes strongly convex.

**Definition 10.4.4** We define a norm $E(\cdot)$ on $\bigwedge^{0,1}(\overline{\Omega})$ by setting

$$E(\phi)^2 \equiv \sum_{j,k=1}^{n} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2 + \int_{\partial\Omega} |\phi|^2 d\sigma + \|\phi\|^2.$$

**Proposition 10.4.5** *Let $\Omega$ be a smoothly bounded domain. Then there exists a constant $c > 0$ such that, for all $\phi \in \mathcal{D}^{0,1}$,*

$$Q(\phi, \phi) \leq cE(\phi)^2. \tag{10.4.5.1}$$

*Moreover, if $\Omega$ is strongly pseudoconvex then there is a constant $c' > 0$ such that for $\phi \in \mathcal{D}^{0,1}$ we have*

$$Q(\phi, \phi) \geq c'E(\phi)^2. \tag{10.4.5.2}$$

**Proof** Given our identity

$$Q(\phi, \phi) = \sum_{j,k} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2 + \int_{\partial\Omega} \sum_{j,k} \frac{\partial^2 r}{\partial z_j \partial \overline{z}_k} \phi_j \overline{\phi_k}\, d\sigma + \|\phi\|^2,$$

part (10.4.5.1) is obvious. Part (10.4.5.2) follows also from this identity and the definitions of $E(\phi)$, the Levi form, and of strong pseudoconvexity. □

**Remark 10.4.6** Inequality (10.4.5.2) is our substitute for the coercive estimate (compare them at this time). Notice that it was necessary for us to give up some regularity. Indeed, $E(\phi)$ contains no information about derivatives of the form $\partial\phi_j/\partial z_k$.

Notice, moreover, that (10.4.5.2) has the *form* of the hypothesis of the Friedrichs lemma. However it is not the same since $E(\phi)$ is not compatible with $\|\phi\|$. □

**Definition 10.4.7 (The Main Estimate)** Let $\Omega$ be a smoothly bounded domain in $\mathbb{C}^n$ and let $E(\phi)$ be defined as above. We say that the *basic estimate* or *main estimate* holds for elements of $\mathcal{D}^{p,q}(\Omega)$ provided that there is a constant $c > 0$ such that

$$Q(\phi, \phi) \geq cE(\phi)^2$$

for all $\phi \in \mathcal{D}^{p,q}(\Omega)$.

Putting our definitions together, we see that the basic estimate holds for elements of $\mathcal{D}^{p,q}(\Omega)$ when $\Omega$ is strongly pseudoconvex.

**Exercise** Show that, for $\psi \in \bigwedge_c^{p,q}(\Omega)$, we have $Q(\psi,\psi) \geq c\|\psi\|^2_{W^1}$ (Hint: integrate by parts). Then on $\bigwedge_c^{p,q}(\Omega)$ we have a classical coercive estimate. The lack of full regularity in some directions for the $\overline{\partial}$-Neumann problem is due to the complex geometry of the boundary.

**Exercise** Show that, on any smoothly bounded domain $\Omega$, the expression $E(\,\cdot\,)$ satisfies

$$E(\phi) \leq C \cdot \|\phi\|^2_{W^1}$$

for all $\phi \in \bigwedge^{p,q}(\overline{\Omega})$ but that in general there is no constant $C' > 0$ such that

$$E(\phi) \geq C'\|\phi\|^2_{W^1}$$

for all $\phi \in \bigwedge^{p,q}(\overline{\Omega})$.

Now we are ready to formulate the main theorem of this chapter. Recall that, by construction, the equation $F\phi = \alpha$ admits a unique solution $\phi \in \mathrm{dom}\,(F)$ for every $\alpha \in H_0^{p,q}(\Omega)$.

**Theorem 10.4.8 (The Main Theorem)** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded domain. Assume that the main estimate holds for elements of $\mathcal{D}^{p,q}(\Omega)$. For $\alpha \in H_0^{p,q}$, we let $\phi$ denote the unique solution to the equation $F\phi = \alpha$. Then we have:*

**(1)** *If $W$ is a relatively open subset of $\overline{\Omega}$ and if $\alpha\big|_W \in \bigwedge^{p,q}(W)$ then $\phi\big|_W \in \bigwedge^{p,q}(W)$;*

**(2)** *Let $\rho, \rho_1$ be smooth functions with $\mathrm{supp}\,\rho \subseteq \mathrm{supp}\,\rho_1 \subseteq W$ and $\rho_1 \equiv 1$ on $\mathrm{supp}\,\rho$. Then:*

**(a)** *If $W \cap \partial\Omega = \emptyset$ then $\forall s \geq 0$ there is a constant $c_s > 0$ (depending on $\rho, \rho_1$ but independent of $\alpha$) such that*

$$\|\rho\phi\|^2_{W^{s+2}} \leq c_s\big(\|\rho_1\alpha\|^2_{W^s} + \|\alpha\|^2_{W^0}\big).$$

**(b)** *If $W \cap \partial\Omega \neq \emptyset$ then $\forall s \geq 0$ there exists a constant $c_s \geq 0$ such that*

$$\|\rho\phi\|^2_{W^{s+1}} \leq c_s\big(\|\rho_1\alpha\|^2_{W^s} + \|\alpha\|^2_{W^0}\big).$$

*Here of course we are using Sobolev norms.*

**Remark 10.4.9** Observe that **(1)** states that $F$ is hypoelliptic. Statement **(2a)** asserts that, in the interior or $\Omega$, the operator $F$ enjoys the regularity of a strongly elliptic operator. That is, $F$ is of order 2 and the solution of $F\phi = \alpha$ exhibits a gain of two derivatives.

On the other hand, **(2b)** states that at the boundary $F$ enjoys only subelliptic regularity—the solution enjoys a gain of only one derivative. Examples ([GRN], [KRA19]) show that this estimate is sharp. □

The proof of the Main Theorem is quite elaborate and will take up most of the remainder of the chapter. We will begin by building up some technical machinery. Next, and what is of most interest, we study the boundary estimate. Of course **(1)** follows from **(2a)**, **(2b)**. All the hard work goes into proving this last statement. The tradeoff between existence and regularity is rather delicate in this context. To address this issue we shall use the technique, developed by Kohn and Nirenberg, of elliptic regularization (see [KOHN1]).

Before we end the section, we wish to stress that part **(2a)** is the least interesting of all the parts of the Main Theorem. For notice that, if $\phi, \psi \in \bigwedge_0^{p,q}$, then

$$\langle F\phi, \psi\rangle = \big((\Box + I)\phi, \psi\big).$$

Now $\Box + I$ is elliptic, so the regularity statement follows from standard elliptic theory.

## 10.5 Special Boundary Charts and Technical Matters

**Capsule:** The study of the $\overline{\partial}$-Neumann problem entails many technical devices—see [FOK]. These include some sophisticated versions of integration by parts, and also some novel uses of the Schwarz inequality. This section presents the background on those ideas.

The proof of the Main Theorem has at its heart a number of sophisticated applications of the method of integration by parts. As a preliminary exercise we record here some elementary but useful facts that will be used along the way.

In learning about the calculus of finite differences we shall use $\mathbb{R}^N$ as our setting. If $u$ is a function on $\mathbb{R}^N$, $j \in \{1, \ldots, N\}$, and $h \in \mathbb{R}$, then we define

$$\triangle_h^j u(x_1, \ldots, x_N) = \frac{1}{2ih}\big[u(x_1, \ldots, x_{j-1}, x_j + h, x_{j+1}, \ldots, x_N) - u(x_1, \ldots, x_{j-1}, x_j - h, x_{j+1}, \ldots, x_N)\big].$$

Also if $\beta = (\beta_1, \ldots, \beta_N)$ is a multi-index and

$$H = \big(h_{jk}\big)_{\substack{j=1,\ldots,N \\ k=1,\ldots,\beta_j}}$$

is an array of real numbers then we define

$$\triangle_H^\beta = \prod_{j=1}^{N} \prod_{k=1}^{\beta_j} \triangle_{h_{jk}}^j.$$

Here the symbol $\prod$ is to be interpreted as a *composition* of the $\triangle$ operators.

**EXAMPLE 10.5.1** Let $N = 1$, $\beta = (2)$, and $H = (h)$ with $h > 0$. If $u$ is a function then

$$\begin{aligned} \triangle_H^\beta u(x) &= \triangle_h^1\big(\triangle_h^1 u(x)\big) \\ &= \triangle_h^1\big(u(x+h) - u(x-h)\big) \\ &= u(x+2h) + u(x-2h) - 2u(x). \end{aligned}$$

This is a standard second difference operator such as one encounters in [KRA6]. □

A comprehensive consideration of finite difference operators may be found in [KRA6]. We shall limit ourselves here to a few special facts.

**Lemma 10.5.2** *If $u \in L^2(\mathbb{R}^N$ then*

$$\big(\triangle_h^j u\big)\widehat{\ }(\xi) = -\frac{\sin(h\xi_j)}{h}\widehat{u}(\xi).$$

*Here $\widehat{\ }$ denotes the standard Fourier transform.*

**Proof** We calculate that

$$\begin{aligned} \big(\triangle_h^j u\big)\widehat{\ }(\xi) &= \frac{1}{2ih}\int e^{ix\xi}\big[u(x_1,\ldots,x_j+h,\ldots,x_N) - u(x_1,\ldots,x_j-h,\ldots,x_N)\big]dx \\ &= \frac{1}{2ih}\big[e^{-ih\xi_j} - e^{ih\xi_j}\big]\widehat{u}(\xi) \\ &= -\frac{\sin(h\xi_j)}{h}\widehat{u}(\xi)\,. \end{aligned}$$

□

Notice that

$$\left|\frac{\sin(h\xi_j)}{h}\right| \approx |\xi_j|$$

as $h \to 0$.

**Lemma 10.5.3** *Let $u \in W^{s'}(\mathbb{R}^N)$, $\beta$ a multi-index, $|\beta| = s$. Assume that $s \le s'$. If $r \le s' - s$ then for any $H$ we have*

$$\|\triangle_H^\beta u\|_{W^r} \le \|D^\beta u\|_{W^r}.$$

*[Here the norms are Sobolev space norms.]*

**Proof** We calculate that

$$\begin{aligned}
\| \triangle_H^{\beta} u\|_{W^r}^2 &= \int (1+|\xi|^2)^r \left|(\triangle_H^{\beta} u)\widehat{\ }(\xi)\right|^2 d\xi \\
&= \int (1+|\xi|^2)^r \prod_{j=1}^{N}\prod_{k=1}^{\beta_j} \left|\frac{\sin h_{j_k}\xi_j}{h_{j_k}}\right|^2 |\widehat{u}(\xi)|^2\, d\xi \\
&\lesssim \int (1+|\xi|^2)^r \prod_{j=1}^{N} |\xi_j|^{2\beta_j} |\widehat{u}(\xi)|^2\, d\xi \\
&= \int (1+|\xi|^2)^r \left|(D^{\beta} u)\widehat{\ }(\xi)\right|^2 d\xi \\
&= \|D^{\beta} u\|_{W^r}^2 .
\end{aligned}$$

□

**Lemma 10.5.4** *If $D_1, D_2$ are partial differential operators of degrees $k_1, k_2$ respectively then*

$$[D_1, D_2] \equiv D_1 D_2 - D_2 D_1$$

*has degree not exceeding $k_1 + k_2 - 1$.*

**Proof** Exercise: write it out. □

**Definition 10.5.5** If $A$ and $B$ are numerical quantities then we shall use Landau's notation

$$A = \mathcal{O}(B)$$

to indicate that

$$A \leq C \cdot B$$

for some constant $C$. We will sometimes write $A \lesssim B$, which has the same meaning.

We write $A \approx B$ to mean both $A \lesssim B$ and $B \lesssim A$.

**Lemma 10.5.6** *For every $\epsilon > 0$ there is a $K > 0$ such that for any $a, b \in \mathbb{R}$ we have*

$$ab \leq \epsilon a^2 + K b^2.$$

**Proof** Recall that $2\alpha\beta \leq \alpha^2 + \beta^2$ for all $\alpha, \beta \in \mathbb{R}$. Hence

$$\begin{aligned}
2ab &= 2\left(\sqrt{2\epsilon}a\right)\left(\frac{1}{\sqrt{2\epsilon}} b\right) \\
&\leq 2\epsilon a^2 + \frac{1}{2\epsilon} b^2.
\end{aligned}$$

Thus $K = 1/4\epsilon$ does the job. □

For the moment now let us identify $\mathbb{C}^n$ with $\mathbb{R}^{2n}$. We consider a domain $\Omega = \{x \in \mathbb{R}^{2n} : r(x) < 0\}$ with $\nabla r \neq 0$ on $\partial\Omega$. Let $U \subseteq \overline{\Omega}$ be a relatively open set that has non-trivial intersection with $\partial\Omega$. Coordinates $(t_1, \ldots, t_{2n-1}, r)$ constitute a *special boundary chart* if $r$ is the defining function for $\Omega$ and $(t_1, \ldots, t_{2n-1})$ form coordinates for $\partial\Omega \cap U$. (This construct is quite standard in differential geometry and is called "giving $U$ a product structure.") We associate to a special boundary chart an orthonormal basis $\omega_1, \ldots, \omega_n$ for $\bigwedge^{1,0}$ such that $\omega_n = \sqrt{2}\partial r$. It is frequently convenient to take the function $r$ to be (signed) Euclidean distance to the boundary. Obviously $\partial r/\partial z_n \neq 0$ on $U$.

If $U$ is a special boundary chart then we set

$$L_j = \frac{\partial r}{\partial z_j}\frac{\partial}{\partial z_n} - \frac{\partial r}{\partial z_n}\frac{\partial}{\partial z_j} \quad , \quad j = 1, \ldots, n-1,$$

$$L_n = \frac{\partial}{\partial z_n}.$$

Then $L_1, \ldots, L_n$ generate $T^{1,0}_P(\Omega)$ for $P \in \Omega$ and $L_1, \ldots, L_{n-1}$ are tangential; this last statement means that $L_j r = 0$ on $U$, $j = 1, \ldots, n-1$.

**Exercise** Assume that $|\nabla r| \equiv 1$ on $\partial\Omega$ (this can always be arranged by replacing $r$ by $r/|\nabla r|$). Set

$$\nu = \sum_{j=1}^{n} \frac{\partial r}{\partial \overline{z}_j}\frac{\partial}{\partial z_j}.$$

Then $\nu$ is normal to $U \cap \partial\Omega$. Complete $\nu$ to an orthonormal basis $\widetilde{L}_1, \ldots, \widetilde{L}_{n-1}, \nu$ for $T^{1,0}_P(\Omega), P \in U$. The canonical dual basis $\widetilde{\omega}_1, \ldots, \widetilde{\omega}_n \in \bigwedge^{1,0}(\Omega)$ satisfies $\omega_n = \sqrt{2}\partial r$.

Check that $\widetilde{L}_1, \ldots \widetilde{L}_{n-1}, \overline{\widetilde{L}}_1, \ldots, \overline{\widetilde{L}}_{n-1}, \operatorname{Im} \nu$ form a real basis for $T_q(\partial\Omega), q \in \partial\Omega$.

On a special boundary chart, the $\overline{\partial}$-Neumann boundary conditions can be easily expressed in terms of coordinates:

**Lemma 10.5.7** *If $U$ is a special boundary chart and*

$$\phi = \sum_{I,J} \phi_{IJ} \omega^I \wedge \overline{\omega}^J$$

*on $U$ then $\phi \in \mathcal{D}^{p,q}$ if and only if $\phi \in \bigwedge^{p,q}(\overline{\Omega})$ and $\phi_{IJ} = 0$ on $\partial\Omega$ whenever $n \in J$.*

**Proof** Exercise. This is just definition chasing. (Or see [FOK, p. 33].) □

**Lemma 10.5.8 (Schur)** *Let $(X, \mu), (Y, \nu)$ be measure spaces. Let $K : X \times Y \to \mathbb{C}$ be a jointly measurable function such that*

$$\int |K(x,y)|\, d\mu(x) \le C,$$

*uniformly in* $y \in Y$, *and*

$$\int |K(x,y)|\, d\nu(y) \le C,$$

*uniformly in* $x \in X$. *Then the operator*

$$f \mapsto \int K(x,y) f(y)\, d\nu(y)$$

*maps* $L^p(Y,\nu)$ *to* $L^p(X,\mu)$, $1 \le p \le \infty$.

**Proof** This is a special case of Lemma 3.7.4. □

We now introduce an important analytic tool that is useful in studying smoothness of functions.

**Definition 10.5.9** The *Bessel potential* of order $r$ is defined by the Fourier analytic expression

$$\Lambda^r : \phi \mapsto \left((1+|\xi|^2)^{r/2}\widehat{\phi}\right)^{\vee},$$

where $\phi \in C_c^\infty$. This operation extends to $W^r$ in a natural way.

Observe that $\phi \in W^r$ if and only if $\Lambda^r \phi \in L^2 (\equiv W^0)$. Notice also that $\Lambda^r : W^t \to W^{t-r}$ for any $t, r \in \mathbb{R}$.

**Lemma 10.5.10** *Let* $a \in C_c^\infty(\mathbb{R}^N)$. *Suppose that* $0 \le s$ *and* $1 \le s'$ *with* $s \le s'$ *are integers. If* $u \in W^{s'-1}$ *then, for all* $r \le s' - s$ *and multi-indices* $\beta$ *with* $|\beta| = s$, *we have*

$$\left\| [a, \triangle_H^\beta] u \right\|_{W^r} \lesssim \left\| u \right\|_{W^{r+s-1}},$$

*that is,* $[a, \triangle_H^\beta]$ *behaves like an operator of order* $s-1$.

**Proof** The case $s = 0$ is of no interest so we begin by considering $s = 1$. Thus $|\beta| = 1$. We want to show that

$$\|[a, \triangle_h^j] u\|_{W^r} \lesssim \|u\|_{W^r}$$

for any $r$ and any $u \in W^{s'}$, $s' \ge r$. Using Bessel potentials, this inequality is seen to be equivalent to

$$\|\Lambda^r [a, \triangle_h^j] \Lambda^{-r} v\| \lesssim \|v\|_{W^0}.$$

Now, if $\mathcal{F}$ denotes the Fourier transform then

$$\mathcal{F}\big(\Lambda^r[a,\triangle_h^j]\Lambda^{-r}v\big)(\eta)=\int K(\xi,\eta)\widehat{v}(\xi)\,d\xi, \tag{10.5.10.1}$$

where

$$K(\xi,\eta)=\left(\frac{1+|\eta|^2}{1+|\xi|^2}\right)^{r/2}\cdot\left[\frac{\sin(h\eta_j)}{h}-\frac{\sin(h\xi_j)}{h}\right]\widehat{a}(\eta-\xi).$$

In fact let us do the calculation that justifies this assertion: According to the definition of the Bessel potential we have

$$\mathcal{F}\big(\Lambda^r[a,\triangle_h^j]\Lambda^{-r}v\big)(\eta)=(1+|\eta|^2)^{r/2}\mathcal{F}\big([a,\triangle_h^j]\Lambda^{-r}v\big)(\eta). \tag{10.5.10.2}$$

Next,

$$\begin{aligned}
\mathcal{F}([a,\triangle_h^j]\Lambda^{-r}v)(\eta)&=\mathcal{F}\left(a\,\triangle_h^j\,\Lambda^{-r}(v)-\triangle_h^j a\Lambda^{-r}(v)\right)(\eta)\\
&=\widehat{a}*(\triangle_h^j\Lambda^{-r}v)\widehat{\ }(\eta)+\frac{\sin(h\eta_j)}{h}\Big(\widehat{a}*(\Lambda^{-r}v)\widehat{\ }\Big)(\eta)\\
&=-\left(\widehat{a}*\left[\frac{\sin(h\xi_j)}{h}(1+|\xi|^2)^{-r/2}\widehat{v}(\xi)\right]\right)(\eta)\\
&\quad+\frac{\sin(h\eta_j)}{h}\left(\widehat{a}*(1+|\xi|^2)^{-r/2}\widehat{v}(\xi)\right)(\eta)\\
&=\int\widehat{a}(\eta-\xi)\left[\frac{\sin(h\xi_j)}{h}-\frac{\sin(h\eta_j)}{h}\right](1+|\xi|^2)^{-r/2}\widehat{v}(\xi)\,d\xi.
\end{aligned}$$

The last line, combined with (10.5.10.2), gives (10.5.10.1).

Now we will verify that the kernel $K(\xi,\eta)$ satisfies the hypotheses of Schur's lemma. Note that

$$\begin{aligned}
\left(\frac{1+|\eta|^2}{1+|\xi|^2}\right)^{r/2}&\le C\cdot\left(\frac{1+|\eta|}{1+|\xi|}\right)^r\\
&\le C\cdot\big(1+|\eta-\xi|\big)^{|r|}\\
&\le C(1+|\eta-\xi|^2)^{|r|/2}.
\end{aligned}$$

Then

$$|K(\xi,\eta)|\le C\big(1+|\eta-\xi|^2\big)^{|r|/2}|\xi-\eta|\cdot|\widehat{a}(\eta-\xi)|$$

so that $K(\xi, \eta)$ is uniformly integrable in $\xi$ and $\eta$. By Schur's lemma we have

$$\begin{aligned}\|\Lambda^r[a, \triangle_h^j]\Lambda^{-r}v\|_{W^0} &= \left\|\mathcal{F}\big(\Lambda^r[a, \triangle_h^j]\Lambda^{-r}v\big)\right\|_{W^0} \\ &= \Big\| \int K(\xi, \eta)\widehat{v}(\xi)\, d\xi \Big\|_{W^0} \\ &\leq C\|v\|_{W^0}.\end{aligned}$$

This proves the desired inequality, and we have handled the case $|\beta| = 1$.

If now $|\beta| = s > 1$, we claim that $[a, \triangle_H^\beta]$ is a sum of terms of the form

$$\triangle_{H'}^{\beta'}[a, \triangle_h^j]\triangle_{H''}^{\beta''}$$

with $|\beta'| + |\beta''| = s - 1$. To see this assertion in case $s = 2$, we calculate that

$$\begin{aligned}[a, \triangle_H^\beta] &= [a, \triangle_h^j \triangle_k^\ell] \\ &= a\, \triangle_h^j\, \triangle_k^\ell - \triangle_h^j\, \triangle_k^\ell\, a \\ &= \triangle_h^j a\, \triangle_k^\ell + [a, \triangle_h^j]\, \triangle_k^\ell - \triangle_h^j\, \triangle_k^\ell a \\ &= \triangle_h^j[a, \triangle_k^\ell] + [a, \triangle_h^j]\, \triangle_k^\ell\,.\end{aligned}$$

The claim now follows easily by induction.

Finally, using the preceding lemma, (10.5.10.2), and the claim we have

$$\begin{aligned}\big\|[a, \triangle_H^\beta]u\big\|_{W^r} &\leq \sum \left\| \big(\, \triangle_{H'}^{\beta'}\, [a, \triangle_h^j]\, \triangle_{H'}^{\beta''}\, \big)u \right\|_{W^r} \\ &\leq \sum \big\|[a, \triangle_h^j]\, \triangle_{H'}^{\beta''}\, u\big\|_{W^{r+|\beta'|}} \\ &\leq \sum \big\|\, \triangle_{H''}^{\beta''}\, u\big\|_{W^{r+|\beta'|}} \\ &\leq C \cdot \|u\|_{W^{r+|\beta''|+|\beta'|}} \\ &= C \cdot \|u\|_{W^{r+s-1}}\,.\end{aligned}$$

□

**Lemma 10.5.11** *Let $u, v \in W^s$, $0 \leq r \leq s$. Then*

$$\langle \triangle_h^j u, v\rangle = \langle u, \triangle_h^j v\rangle.$$

**Proof** This is just an elementary change of variable. □

Next we have

**Lemma 10.5.12 (The Generalized Schwarz Inequality)** *Let $f, g$ be $L^2$ functions and $s \in \mathbb{R}$. Then*

$$\langle f, g\rangle \leq \|f\|_{W^s}\|g\|_{W^{-s}}.$$

**Proof** Look at the Fourier transform side and use the standard Schwarz inequality. □

**Proposition 10.5.13** *Let $K \subseteq \mathbb{R}^N$ be compact and let $u, v \in L^2(K)$. Let $D$ be a first order differential operator and let $\beta$ be a multi-index with $|\beta| = s$. Then*

**(1)** *If $u \in W^s$ then $\| \triangle_H^\beta u\|_{W^0} \leq \|u\|_{W^s}$ uniformly in $H$ as $|H| \to 0$;*
**(2)** *If $u \in W^s$ then $\|[D, \triangle_H^\beta]u\|_{W^0} \lesssim \|u\|_{W^s}$ uniformly in $H$ as $|H| \to 0$;*
**(3)** *If $u \in W^{s-1}$ then*

$$\langle \triangle_H^\beta u, v\rangle = \langle u, \triangle_H^\beta v\rangle;$$

**(4)** *If $u \in W^{s-1}, v \in H^1$ then*

$$\left|\left\langle u, [\triangle_H^\beta, D]v\right\rangle\right| \leq \|u\|_{W^{s-1}}\|v\|_{W^1},$$

*uniformly as $|H| \to 0$;*
**(5)** *If $u \in W^s$ and $\| \triangle_H^\beta u\|_{W^s}$ is bounded as $|H| \to 0$ then $D^\beta u \in W^s$.*

**Proof**

**(1)** By Lemma 10.5.7 we have

$$\begin{aligned}\| \triangle_H^\beta u\|_{W^0} &\leq \|D^\beta u\|_{W^0}\\ &\leq \|u\|_{W^s}\end{aligned}$$

uniformly in $H$ as $|H| \to 0$.

**(2)** Write $D = \sum_{j=1}^N a_j(x)D^j$. Assume that $\operatorname{supp} a_j \subseteq K$. Notice that the finite difference operators commute with $D^j$. Then $[D, \triangle_H^\beta] = \sum_{j=1}^N [a_j, \triangle_H^\beta]D^j$. Thus

$$\begin{aligned}\left\|[D, \triangle_H^\beta]u\right\|_{W^0} &\leq \sum_{j=1}^N \left\|[a_j, \triangle_H^\beta]D^j u\right\|_{W^0}\\ &\leq \sum_{=1}^N \|D^j u\|_{W^{s-1}}\\ &\leq C \cdot \|u\|_{W^s}.\end{aligned}$$

Note that we have used Lemma 10.5.10 in the penultimate inequality.

**(3)** This is an elementary change of variable.

**(4)** Write $D = \sum_{j=1}^{N} a_j(x) D^j$. Then $[D, \triangle_H^\beta] = \sum_{j=1}^{B} [a_j, \triangle_H^\beta] D^j$. Therefore

$$\begin{aligned}
\left| \left\langle u, [\triangle_H^\beta, D] v \right\rangle \right| &\leq \sum_{j=1}^{N} \left| \left\langle u, [a_j, \triangle_H^\beta] D^j v \right\rangle \right| \\
&\lesssim \sum_{j=1}^{N} \|u\|_{W^{s-1}} \left\| [a_j, \triangle_H^\beta] D^j v \right\|_{W^{1-s}} \\
&\lesssim \sum_{j=1}^{N} \|u\|_{W^{s-1}} \|D^j v\|_{W^0} \\
&\lesssim \|u\|_{W^{s-1}} \|v\|_{W^1}.
\end{aligned}$$

Here we have used the generalized Schwarz inequality.

**(5)** If $u \in W^s$ and $\| \triangle_H^\beta u\|_{W^s}$ is bounded as $|H| \to 0$ then we want to show that $D^\beta u \in W^s$. It suffices to prove the result for $|\beta| = 1$, i.e. $\triangle_H^\beta = \triangle_h^j$. Saying that $\| \triangle_h^j u\|_{W^s}$ is bounded as $h \to 0$ means that

$$\int \left| \frac{\sin(\xi_j h)}{h} (1 + |\xi|^2)^{s/2} \widehat{u}(\xi) \right|^2 d\xi$$

is bounded as $h \to 0$. Thus the dominated convergence theorem gives the result. □

One of the significant features of the analysis of the $\overline{\partial}$-Neumann problem is that it is *nonisotropic*. This means that in different directions the analysis is different. In particular, a normal derivative behaves like two tangential derivatives. It turns out that the reason for this is that, when the boundary is strongly pseudoconvex, the (complex) normal derivative is a commutator of tangential derivatives (exercise: calculate $[L_j, \overline{L}_j]$ in the boundary of the ball to see this). This assertion will be brought to the surface in the course of our calculations.

Our nonisotropic analysis will be facilitated by the introduction of some specialized function spaces known as the tangential Sobolev spaces. Consider $\mathbb{R}_+^{N+1}$ with coordinates $(t_1, \ldots, t_N, r)$, $r < 0$. Define the *partial Fourier transform* (PFT) by

$$\widetilde{u}(\tau, r) = \int_{\mathbb{R}^N} e^{-it\tau} u(t, r)\, dt.$$

Then the *tangential Bessel potential* is defined to be

$$\left(\Lambda_t^s u\right)\widetilde{\ }(\tau, r) = (1 + |\tau|^2)^{s/2} \widetilde{u}(\tau, r), \quad (\tau, r) \in \mathbb{R}_+^{N+1}.$$

We then define *tangential Sobolev norms* by

$$|||u|||_{W^s}^2 \equiv \|\Lambda_t^s u\|_{W^0}^2 = \int_{\mathbb{R}^N} \int_{-\infty}^0 (1+|\tau|^2)^s |\widetilde{u}(\tau, r)|^2 \, dr d\tau.$$

For the remainder of the section, we will write $\widetilde{\Delta}_H^\beta$ to denote a finite difference operator acting on the first $N$ variables only. We have

**(1)** If $u \in W^{s'}$, $|\beta| = s$, and $r \leq s' - s$ then

$$\|\widetilde{\Delta}_H^\beta u\|_{W^r} \lesssim |||D^\beta u|||_{W^r}.$$

This is proved by imitating the proof of the isotropic result 10.5.13, part **(1)**, using the tangential Fourier transform and integrating out in the $r$ variable.

**(2)** If $\rho \in C_c^\infty$, $u \in W^{s'-1}$, $|\beta| = s$, and $r \leq s' - s$ then

$$\left\| [\rho, \widetilde{\Delta}_H^\beta] u \right\|_{W^r} \lesssim |||u|||_{W^{r+s-1}}.$$

We prove this by first reducing to the case of a single difference (as we have done before). Then we express the lefthand side as an integral with kernel (as in the isotropic case) and use the Schwarz inequality.

**(3)** If $u \in W^r$, $r \geq 0$, then

$$\left\langle \widetilde{\Delta}_h^j u, v \right\rangle = \left\langle u, \widetilde{\Delta}_h^j v \right\rangle.$$

From **(1)**, **(2)**, and **(3)** we can obtain that, for $u \in W^s$ and $|\beta| = s$,

$$\|\widetilde{\Delta}_H^\beta u\|_{W^0} \lesssim |||u|||_{W^s}.$$

Also, for $D$ a first order partial differential operator,

$$\|[D, \widetilde{\Delta}_H^\beta] u\|_{W^0} \lesssim \sum_{j=1}^N |||D^j u|||_{W^{s-1}}$$

and

$$\left\langle \widetilde{\Delta}_H^\beta u, v \right\rangle = \left\langle u, \widetilde{\Delta}_H^\beta v \right\rangle$$

Finally,

$$|\langle u, v \rangle| \leq |||u|||_{W^{s-1}} |||v|||_{W^{1-s}}.$$

We invite the reader to fill in the details in the proofs of these assertions about the tangential Sobolev norms.

## 10.6 Beginning of the Proof of the Main Theorem

**Capsule:** These are J. J. Kohn's historic arguments from the early 1960s. They have been profoundly influential in the subject for a good many years.

The method of proof presented here is not the one in Kohn's original work [KOH2]. In fact the method presented here was developed in the later work [KOHN1]. In the latter paper, a method of elliptic regularizaion is developed that allows us to exploit the usual elliptic regularity theory to avoid the nasty question of existence for the $\overline{\partial}$-Neumann problem. The idea is to add to the (degenerate) quadratic form $Q$ an expression consisting of $\delta$ times the quadratic form for the classical Laplacian. Certain estimates are proved, uniformly in $\delta$, and then $\delta$ is allowed to tend to 0. We begin with a definition:

**Definition 10.6.1** Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded domain. (*We do not assume that the basic estimate holds in* $\mathcal{D}^{p,q}$.) For convenience we identify $(z_1, z_2, \ldots, z_n)$ with $(x_1 + ix_2, x_3 + ix_4, \ldots, x_{2n-1} + ix_{2n})$. For $\delta > 0$ and $D^j = i\partial/\partial x_j$, we define

$$Q^\delta(\phi, \psi) = Q(\phi, \psi) + \delta \sum_{j=1}^{2n} \langle D^j \phi, D^j \psi \rangle.$$

Observe that

$$Q^\delta(\phi, \phi) \geq \delta \|\phi\|^2_{W^1}. \tag{10.6.2}$$

The point of creating $Q^\delta$ is that it has better properties at the boundary than does $Q$. Indeed, in the interior we have $Q(\cdot\,, \cdot\,) = \langle (\Box + I)\cdot\,, \cdot \rangle$ and this is perfectly suited to our purposes. But $Q$ does not satisfy a coercive estimate at the boundary. Since $Q^\delta$ *does* satisfy such an estimate, it is much more useful.

Recall that we defined $\widetilde{\mathcal{D}}^{p,q}$ to be the closure of $\mathcal{D}^{p,q}$ (which we know equals $\operatorname{dom} \overline{\partial}^* \cap \bigwedge^{p,q}$) *in the Q-topology*. We let $\widetilde{\mathcal{D}}^{p,q}_\delta$ be the closure of $\mathcal{D}^{p,q}$ *in the* $Q^\delta$ *topology*. This setup would be intractible if $\widetilde{\mathcal{D}}^{p,q}_\delta$ varied with different values of $\delta$. Fortunately, that is not the case:

**Lemma 10.6.3** *For all* $\delta, \delta' > 0$ *it holds that* $\widetilde{\mathcal{D}}^{p,q}_\delta = \widetilde{\mathcal{D}}^{p,q}_{\delta'}$. *All of the spaces* $\widetilde{\mathcal{D}}^{p,q}_\delta$ *are contained in* $\widetilde{\mathcal{D}}^{p,q} \cap W^{p,q}_1$.

**Proof** A sequence $\{\phi_k\}$ is Cauchy in the $Q^\delta$-topology if and only if $\{\phi_k\}$ and $\{D^j \phi_k\}$ are Cauchy in $L^2, j = 1, \ldots, 2n$. And this statement does not depend on $\delta$. Thus the notion of closure is independent of $\delta$.

The second statement is now obvious. □

Notice that we are *not* saying that $\widetilde{\mathcal{D}}^{p,q} = \widetilde{\mathcal{D}}^{p,q}_\delta$. In fact this equality is not true. To see this, suppose that the two spaces *were* equal. Then the open mapping principal implies that the $Q$-norm and the $Q^\delta$-norm are equivalent. This would imply that $Q$

contains information (as does $Q^\delta$) about the $L^2$ norm of $\partial\phi_j/\partial z_k$ when $\phi \in \bigwedge^{0,1}$. But this is clearly not the case. That gives a contradiction.

Next we wish to apply the Friedrichs theory to the $Q^\delta$s in $\widetilde{\mathcal{D}}^{p,q}$. Note that $Q^\delta \geq Q \geq \|\cdot\|^2_{W^0}$. Also, by construction, $\widetilde{\mathcal{D}}^{p,q}_\delta$ is complete in the $Q^\delta$-topology. Thus, by the Friedrichs theory, there is a self-adjoint $F^\delta$ with $\operatorname{dom} F^\delta \subseteq W^{p,q}_0$ and

$$F^\delta : \operatorname{dom} F^\delta \to W^{p,q}_0$$

univalently and surjectively. Evidently $F^\delta$ will correspond to $\Box + I + \delta \sum_j D^{j'} D^j$.

We see that, given a form $\alpha \in W^{p,q}_0$, there exists a unique $\phi^\delta \in \widetilde{\mathcal{D}}^{p,q}_\delta$ such that

$$F^\delta \phi^\delta = \alpha.$$

Moreover, $\phi^\delta$ satisfies interior estimates uniformly in $\delta$ since

$$\sigma_P(F^\delta) = \begin{pmatrix} (1+\delta)|\xi|^2 & & 0 \\ & \ddots & \\ 0 & & (1+\delta)|\xi|^2 \end{pmatrix}$$

and we see that the eigenvalues of the principal symbol are bounded from below, uniformly in $\delta$.

**Theorem 10.6.4** *Let $U$ be a special boundary chart and $V \subseteq \overline{V} \subseteq U$. Let $\rho_1$ be a smooth, real-valued, function with support in $U$ and $\rho_1 \equiv 1$ on $\overline{V}$. If $\phi \in \operatorname{dom} F^\delta$ and $\rho_1 F^\delta \phi \in \bigwedge^{p,q}(\overline{\Omega})$ then, for every smooth $\rho$ which is supported in $\overline{V}$, we have $\rho\phi \subseteq \bigwedge^{p,q}(\overline{\Omega})$.*

**Remark 10.6.5** Recall that $F^\delta = I + \Box + \delta \sum_j D^{j'} D^j$. The theorem says that $F^\delta$ is hypoelliptic. From the proof (below) we will see that

$$\|\rho\phi\|^2_{W^{s+2}} \lesssim \|\rho_1 F^\delta \phi\|_{W^s} + \|\phi\|_{W^0}.$$

*However the constant in the inequality will depend heavily on $\delta$.* It will in fact be of size $\delta^{-s-2}$; so it blows up as $\delta \to 0$. This means in particular that the uniformity that we need will not come cheaply. □

**Proof of the Theorem** We know that $\phi \in \operatorname{dom} F^\delta$ implies that $\phi \in W^{p,q}_1$. We will prove that if $\rho\phi \in W^{p,q}_s$ for all cutoff functions $\rho$ supported in $V$ then $\rho\phi \in W^{p,q}_{s+1}$. The theorem will then follow from the Sobolev lemma.

We begin by proving the following claim:

**Claim 1:** If $\rho\phi \in W^{p,q}_s$ for all $\rho$ then $D^\beta_t \rho\phi \in W^{p,q}_1$ whenever $|\beta| = s$.

Notice that this claim is not the full statement that we are proving: it gives us information about the derivatives of order $s+1$ only when all but one of the derivatives is in the tangential directions. We deal with tangential derivatives first

since tangential derivatives and tangential differences preserve $\mathcal{D}^{p,q}$ and tangential differences preserve $\widetilde{\mathcal{D}}^{p,q}$.

For the claim, it suffices to show that

$$\|\widetilde{\triangle}_H^\beta \rho\phi\|_{W^1}^2$$

is bounded as $|H| \to 0$ when $|\beta| = s$.

Now $\|\psi\|_{W^1}^2 \lesssim Q^\delta(\psi,\psi)$ (where the constant depends on $\delta$); therefore it suffices to estimate $Q^\delta(\triangle_H^\beta \rho\phi, \triangle_H^\beta \rho\phi)$. In order to perform this estimate, we need to analyze:

**(1)** $\langle \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle$
**(2)** $\langle \vartheta\widetilde{\triangle}_H^\beta \rho\phi, \vartheta\widetilde{\triangle}_H^\beta \rho\phi\rangle$
**(3)** $\langle \widetilde{\triangle}_H^\beta \rho\phi, \widetilde{\triangle}_H^\beta \rho\phi\rangle$
**(4)** $\langle D^j\widetilde{\triangle}_H^\beta \rho\phi, D^j\widetilde{\triangle}_H^\beta \rho\phi\rangle$.

**Analysis of (1):** Now

$$\langle \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle = \langle \widetilde{\triangle}_H^\beta \overline{\partial}\rho\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle + \langle [\overline{\partial}, \widetilde{\triangle}_H^\beta]\rho\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle.$$

Recall that $[\overline{\partial}, \widetilde{\triangle}_H^\beta]$ acts like an operator of order $s$. Therefore

$$\begin{aligned}\left|\langle [\overline{\partial}, \widetilde{\triangle}_H^\beta]\rho\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle\right| &\lesssim \left\|[\overline{\partial}, \widetilde{\triangle}_H^\beta]\rho\phi\right\|_{W^0} \cdot \left\|\overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\right\|_{W^0} \\ &\lesssim \left\|\rho\phi\right\|_{W^s} \left\|\widetilde{\triangle}_H^\beta \rho\phi\right\|_{W^1}.\end{aligned}$$

Next,

$$\langle \widetilde{\triangle}_H^\beta \overline{\partial}\rho\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle = \langle \widetilde{\triangle}_H^\beta \rho\overline{\partial}\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle + \langle \widetilde{\triangle}_H^\beta [\overline{\partial}, \rho]\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle.$$

Obviously $[\overline{\partial}, \rho]$ is an operator of order 0, i.e. it consists of multiplication by a smooth function with support in $V$: call it $\rho'$. In fact the support of $\rho'$ lies in the support of $\rho$. Then

$$\begin{aligned}\left|\langle \widetilde{\triangle}_H^\beta [\overline{\partial}, \rho]\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle\right| &\lesssim \left\|\widetilde{\triangle}_H^\beta \rho'\phi\right\|_{W^0} \left\|\overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\right\|_{W^0} \\ &\lesssim \|\rho'\phi\|_{W^s} \|\widetilde{\triangle}_H^\beta \rho\phi\|_{W^1}.\end{aligned}$$

It follows that

$$\langle \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle = \langle \widetilde{\triangle}_H^\beta \rho\overline{\partial}\phi, \overline{\partial}\widetilde{\triangle}_H^\beta \rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s} \cdot \|\widetilde{\triangle}_H^\beta \rho\phi\|_{W^1}\big).$$

Here and in what follows we use $\rho'$ to denote a smooth function with support a subset of the support of $\rho$—in particular, the support of $\rho'$ lies in $V$.

Now we see that see that

$$\begin{aligned}
\langle \overline{\partial}\widetilde{\Delta}_H^{\beta}\rho\phi, \overline{\partial}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle &= \langle \rho\overline{\partial}\phi, \widetilde{\Delta}_H^{\beta}\overline{\partial}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\cdot\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\\
&= \big(\rho\overline{\partial}\phi, \overline{\partial}\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \langle\rho\overline{\partial}\phi, [\widetilde{\Delta}_H^{\beta}, \overline{\partial}]\widetilde{\Delta}_H^{\beta}\rho\phi\rangle\\
&\qquad +\mathcal{O}\big(\|\rho'\phi\|_{W^s}\cdot\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\\
&= \big(\rho\overline{\partial}\phi, \overline{\partial}\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle\\
&\qquad +\mathcal{O}\big(\|\rho\overline{\partial}\phi\|_{W^{s-1}}\|[\widetilde{\Delta}_H^{\beta}, \overline{\partial}]\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^{1-s}}\big)\\
&\qquad +\mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\\
&= \big(\rho\overline{\partial}\phi, \overline{\partial}\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\,.
\end{aligned}$$

Write $[\overline{\partial}, \rho] = \rho'$ and $\widetilde{\Delta}_H^{\beta} = \widetilde{\Delta}_{H'}^{\beta'}\widetilde{\Delta}_h^{j}$. Then the last line equals

$$\begin{aligned}
&= \langle\overline{\partial}\phi, \overline{\partial}\rho\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \langle\overline{\partial}\phi, \rho'\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle\\
&\qquad +\mathcal{O}\big(\|\rho'\phi\|_{W^s}\cdot\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\\
&= \langle\overline{\partial}\phi, \overline{\partial}\rho\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \langle\rho'\overline{\partial}\phi, \widetilde{\Delta}_{H'}^{\beta'}\widetilde{\Delta}_h^{j}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle\\
&\quad +\mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\\
&= \langle\overline{\partial}\phi, \overline{\partial}\rho\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \langle\widetilde{\Delta}_{H'}^{\beta'}\rho'\overline{\partial}\phi, \widetilde{\Delta}_h^{j}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle\\
&\quad +\mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\\
&= \langle\overline{\partial}\phi, \overline{\partial}\rho\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \mathcal{O}\big(\|\widetilde{\Delta}_{H'}^{\beta'}\rho'\overline{\partial}\phi\|_{W^0}\|\widetilde{\Delta}_h^{j}\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^0}\big)\\
&\quad +\mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big)\\
&= \langle\overline{\partial}\phi, \overline{\partial}\rho\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big).
\end{aligned}$$

We have thus proved that

$$\langle\overline{\partial}\widetilde{\Delta}_H^{\beta}\rho\phi, \overline{\partial}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle = \langle\overline{\partial}\phi, \overline{\partial}\rho\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big).$$

The very same argument may be used to prove (statement **(2)**) that

$$\langle\vartheta\widetilde{\Delta}_H^{\beta}\rho\phi, \vartheta\widetilde{\Delta}_H^{\beta}\rho\phi\rangle = \langle\vartheta\phi, \vartheta\rho\widetilde{\Delta}_H^{\beta}\widetilde{\Delta}_H^{\beta}\rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^{\beta}\rho\phi\|_{W^1}\big).$$

As an exercise, the reader may apply similar arguments (reference statements **(3)** and **(4)**) to

$$\langle\widetilde{\Delta}_H^{\beta}\rho\phi, \widetilde{\Delta}_H^{\beta}\rho\phi\rangle \qquad \text{and} \qquad \delta\sum_j\langle D^j\widetilde{\Delta}_H^{\beta}\rho\phi, D^j\widetilde{\Delta}_H^{\beta}\rho\phi\rangle$$

to obtain

$$Q^\delta\big(\widetilde{\Delta}_H^\beta \rho\phi, \widetilde{\Delta}_H^\beta \rho\phi\big) = Q^\delta\big(\phi, \rho\widetilde{\Delta}_H^\beta \widetilde{\Delta}_H^\beta \rho\phi\big) + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}\big). \tag{10.6.4.1}$$

Let $\rho_1 \in C_c^\infty$ with $\rho_1 \equiv 1$ on $\overline{V}$. Since $\phi \in \operatorname{dom} F^\delta$, we may rewrite the righthand side of (10.6.4.1) as

$$\begin{aligned}
&\langle F^\delta\phi, \rho\widetilde{\Delta}_H^\beta \widetilde{\Delta}_H^\beta \rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}\big)\\
&= \langle \rho\rho_1 F^\delta\phi, \widetilde{\Delta}_H^\beta \widetilde{\Delta}_H^\beta \rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}\big)\\
&= \langle \widetilde{\Delta}_{H'}^{\beta'} \rho\rho_1 F^\delta\phi, \widetilde{\Delta}_h^j \widetilde{\Delta}_H^\beta \rho\phi\rangle + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}\big).
\end{aligned}$$

Now we apply the Schwarz inequality to see that the righthand side in modulus does not exceed

$$\begin{aligned}
&\lesssim \|\widetilde{\Delta}_{H'}^{\beta'} \rho\rho_1 F^\delta\phi\|_{W^0}\|\widetilde{\Delta}_h^j \widetilde{\Delta}_H^\beta \rho\phi\|_{W^0} + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}\big)\\
&\lesssim \|\rho_1 F^\delta\phi\|_{W^{s-1}}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1} + \mathcal{O}\big(\|\rho'\phi\|_{W^s}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}\big)\\
&\lesssim \frac{2}{\epsilon}\|\rho_1 F^\delta\phi\|_{W^{s-1}}^2 + \frac{\epsilon}{2}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}^2 + \frac{2}{\epsilon}\|\rho'\phi\|_{W^s}^2 + \frac{\epsilon}{2}\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}^2.
\end{aligned}$$

Here $\epsilon > 0$ is to be specified. Thus, using (10.6.4.1), we have

$$\begin{aligned}
\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1} &\lesssim Q^\delta\big(\widetilde{\Delta}_H^\beta \rho\phi, \widetilde{\Delta}_H^\beta \rho\phi\big)\\
&\lesssim \frac{2}{\epsilon}\|\rho_1 F^\delta\phi\|_{W^{s-1}}^2 + \frac{2}{\epsilon}\|\rho'\phi\|_{W^s}^2 + \epsilon\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}^2.
\end{aligned}$$

If we select $\epsilon$ to be positive and smaller than $1/4$, then we find that

$$\|\widetilde{\Delta}_H^\beta \rho\phi\|_{W^1}^2 \lesssim \|\rho_1 F^\delta\phi\|_{W^{s-1}}^2 + \|\rho'\phi\|_{W^s}.$$

Applying part **(5)** of 10.5.13 now yields that

$$\|D_t^\beta \rho\phi\|_{W^1} < \infty.$$

We have proved the claim.

Now we will prove:

**Claim 2:** If $m \geq 2$ and $|\beta| + m = s + 1$ then

$$D_t^\beta D_r^m(\rho\phi) \in W_0^{p,q}.$$

[Equivalently, $\rho D_t^\beta D_r^m \phi \in W_0^{p,q}$.]

The case $m = 1$ of the claim has already been covered in Claim 1. Now let $m = 2$. In local coordinates the operator $F^\delta$ looks like

$$F^\delta = A_{2n} D_r^2 + \sum_{j=1}^{2n-1} A_j D_t^j D_r + \sum_{j,k=1}^{2n-1} A_{j,k} D_t^j D_t^k,$$

where the $A_1, \ldots, A_{2n}, A_{j,k}, B_j, C$ are matrices of smooth functions. The ellipticity of $F^\delta$ is expressed by the invertibility of the matrix of its symbols $A_1, \ldots, A_{2n}$. In particular, $A_{2n}$ is invertible. Therefore, recalling that $\rho_1 \equiv 1$ on $\operatorname{supp} \rho$, we see that

$$\rho D_r^2 \phi = \rho A_{2n}^{-1} \left[ \rho_1 F^\delta \phi - \sum_{j=1}^{2n-1} A_j D_t^j D_r \phi - \sum_{j,k=1}^{2n-1} A_{j,k} D_t^j D_t^k \phi \right].$$

Applying $\rho D_t^\beta$ with $|\beta| = s - 1$ to both sides yields that

$$\rho D_t^\beta D_r^2 \phi = \rho D_t^\beta \rho A_{2n}^{-1} \left[ \rho_1 F^\delta \phi - \sum_{j=1}^{2n-1} A_j D_t^j D_r \phi - \sum_{j,k=1}^{2n-1} A_{j,k} D_t^j D_t^k \phi \right].$$

[Here the terms on the left that arise when $D_t^\beta$ falls on $\rho$ have been absorbed into the last sum on the right.] Thus we see that we can express $\rho D_t^\beta D_r^2 \phi$, $|\beta| = s - 1$, in terms of two types of expressions:

**(a)** $(s-1)$ tangential derivatives of the expression $\rho_1 F^\delta \phi$;
**(b)** $(s+1)$ derivatives, at most one of which is in the normal direction, of $\phi$.

These two types of expressions are both elements of $W_0^{p,q}$. Hence $\rho D_t^\beta D_r^2 \phi \in W_0^{p,q}$. This proves the case $m = 2$.

Proceeding by induction on $m$ we find that $\rho D_t^\beta D_r^2 \phi$ is expressed in terms of **(a)** $(s-1)$ tangential derivatives of $\rho_1 F^\delta \phi$ and **(b)** $(s+1)$ derivatives of $\rho\phi$ of which at most $(m-1)$ are in the normal direction.

This concludes the proof. □

## 10.7 Estimates in the Sobolev $-1/2$ Norm

**Capsule:** As previously noted, the $\overline{\partial}$-Neumann problem on a strongly pseudoconvex domain (indeed, on *any* domain) is not elliptic. There is a loss of smoothness. In the strongly pseudoconvex case the loss of smoothness for the $\overline{\partial}$-problem is of order $1/2$. Thus it follows that (error) estimates in the Sobolev $-1/2$ norm will play a decisive role in our analysis.

We begin this section by doing some calculations in the tangential Sobolev norms. Recall that

$$\|\!|\phi|\!\|_{W^s} \equiv \|\Lambda_t^s \phi\|_{W^0}.$$

If $D$ is any first order linear differential operator then

$$\begin{aligned}\|\!| D\phi |\!\|^2_{W^s} &= \|\!| \sum_{j=1}^{2n} a_j D^j\phi + a_0\phi |\!\|^2_{W^s}\\ &\approx \sum_{j=1}^{2n} \|\!| a_j D^j\phi |\!\|^2_{W^s} + \|\phi\|^2_{W^0}\\ &\approx \|\!| \phi |\!\|^2_{W^{s+1}} + \|\!| D_r\phi |\!\|^2_{W^s}.\end{aligned}$$

Let $A\phi \equiv A_k\phi \equiv \rho_1\Lambda^k_t(\rho\phi)$ $\big($which in turn equals $\Lambda^k_t(\rho_1\rho\phi) + [\rho_1, \Lambda^k_t](\rho\phi)\big)$. Let $A'$ denote the formal adjoint of $A$; that is, if $\phi, \psi \in \bigwedge^{p,q}_c(U \cap \overline{\Omega})$ then $\langle A'\phi, \psi\rangle \equiv \langle \phi, A\psi\rangle$. Fix a special boundary chart $U$.

**Lemma 10.7.1** *For all real $s$ and $\phi \in \bigwedge^{p,q}_c(U \cap \overline{\Omega})$ we have*

**(1)** $\|\!| A\phi |\!\|_{W^s} \lesssim \|\!| \phi |\!\|_{W^{s+k}}$;
**(1')** $\|\!| A'\phi |\!\|_{W^s} \lesssim \|\!| \phi |\!\|_{W^{s+k}}$;
**(2)** $\|\!| (A - A')\phi |\!\|_{W^s} \lesssim \|\!| \phi |\!\|_{W^{s+k-1}}$;
**(3)** *If $D$ is any first order linear differential operator then*

- **(a)** $\|\!| [A, D]\phi |\!\|_{W^s} \lesssim \|\!| D\phi |\!\|_{W^{s+k-1}}$;
- **(b)** $\|\!| [A - A', D]\phi |\!\|_{W^s} \lesssim \|\!| D\phi |\!\|_{W^{s+k-2}}$;
- **(c)** $\|\!| \big[A, [A, D]\big]\phi |\!\|_{W^s} \lesssim \|\!| D\phi |\!\|_{W^{s+2k-2}}$.

**Proof** The proof is straightforward using techniques that we have already presented. We leave the details to the reader. □

Observe that $A$ preserves $\mathcal{D}^{p,q}$.

**Lemma 10.7.2** *It holds that*

$$Q\langle A\phi, A\phi\rangle - Re\, Q\langle \phi, A'A\phi\rangle = \mathcal{O}\big(\|\!| \nabla\phi |\!\|^2_{W^{k-1}}\big),$$

*uniformly in $\phi \in \mathcal{D}^{p,q} \cap \bigwedge^{p,q}_c(U \cap \overline{\Omega})$; here $\nabla$ denotes the gradient of $\phi$.*

**Proof** Recall that $Q(\phi, \psi) = \langle \overline{\partial}\phi, \overline{\partial}\psi\rangle + \langle \vartheta\phi, \vartheta\psi\rangle + \langle \phi, \psi\rangle$. Consider the expression

$$\langle \overline{\partial}A\phi, \overline{\partial}A\phi\rangle - \mathrm{Re}\,\langle \overline{\partial}\phi, \overline{\partial}A'A\phi\rangle = \frac{1}{2}\Big\{2\langle \overline{\partial}A\phi, \overline{\partial}A\phi\rangle - \langle \overline{\partial}\phi, \overline{\partial}A'A\phi\rangle - \langle \overline{\partial}A'A\phi, \overline{\partial}\phi\rangle\Big\}.$$

We write

$$\begin{aligned}\langle \overline{\partial}\phi, \overline{\partial}A'A\phi\rangle &= \langle \overline{\partial}\phi, A'\overline{\partial}A\phi\rangle + \langle \overline{\partial}\phi, [\overline{\partial}, A']A\phi\rangle\\ &= \langle A\overline{\partial}\phi, \overline{\partial}A\phi\rangle + \langle \overline{\partial}\phi, [\overline{\partial}, A']A\phi\rangle\\ &= \langle \overline{\partial}A\phi, \overline{\partial}A\phi\rangle + \langle [A, \overline{\partial}]\phi, \overline{\partial}A\phi\rangle + \langle \overline{\partial}\phi, [\overline{\partial}, A']A\phi\rangle\end{aligned}$$

Also

$$\langle \overline{\partial} A' A\phi, \overline{\partial}\phi\rangle = \langle \overline{\partial} A\phi, \overline{\partial} A\phi\rangle + \langle \overline{\partial} A\phi, [A, \overline{\partial}]\phi\rangle + \langle [\overline{\partial}, A']A\phi, \overline{\partial}\phi\rangle.$$

As a result,

$$\begin{aligned}
&\langle \overline{\partial} A\phi, \overline{\partial} A\phi\rangle - \operatorname{Re}\langle \overline{\partial}\phi, \overline{\partial} A' A\phi\rangle \\
&= -\frac{1}{2}\big\{\langle \overline{\partial}\phi, [\overline{\partial}, A']A\phi\rangle + \langle [A, \overline{\partial}]\phi, \overline{\partial} A\phi\rangle + \langle [\overline{\partial}, A']A\phi, \overline{\partial}\phi\rangle + \langle \overline{\partial} A\phi, [A, \overline{\partial}]\phi\rangle\big\} \\
&= -\frac{1}{2}\big\{I + II + III + IV\big\}.
\end{aligned}$$

We will estimate $II + III$; the corresponding estimate for $I + IV$ will then follow easily.

Notice that

$$\begin{aligned}
&\langle [\overline{\partial}, A' - A]A\phi, \overline{\partial}\phi\rangle + \big\langle \big[[\overline{\partial}, A], A\big]\phi, \overline{\partial}\phi\big\rangle + \langle [\overline{\partial}, A]\phi, (A' - A)\overline{\partial}\phi\rangle + \langle [\overline{\partial}, A]\phi, [A, \overline{\partial}]\phi\rangle \\
&= \langle [\overline{\partial}, A']A\phi, \overline{\partial}\phi\rangle - \langle [\overline{\partial}, A]A\phi, \overline{\partial}\phi\rangle + \langle [\overline{\partial}, A]A\phi, \overline{\partial}\phi\rangle - \langle A[\overline{\partial}, A]\phi, \overline{\partial}\phi\rangle \\
&\quad + \langle [\overline{\partial}, A]\phi, A'\overline{\partial}\phi\rangle - \langle [\overline{\partial}, A]\phi, A\overline{\partial}\phi\rangle + \langle [\overline{\partial}, A]\phi, A\overline{\partial}\phi\rangle - \langle [\overline{\partial}, A]\phi, \overline{\partial} A\phi\rangle \\
&= \langle [\overline{\partial}, A']A\phi, \overline{\partial}\phi\rangle - \langle [\overline{\partial}, A]\phi, \overline{\partial} A\phi\rangle \\
&= \big\langle [\overline{\partial}, A']A\phi, \overline{\partial}\phi\big\rangle + \big\langle [A, \overline{\partial}]\phi, \overline{\partial} A\phi\big\rangle. \qquad (10.7.2.1)
\end{aligned}$$

Therefore

$$\begin{aligned}
\big|II + III\big| &\equiv \big|\langle [A, \overline{\partial}]\phi, \overline{\partial} A\phi\rangle + \langle [\overline{\partial}, A']A\phi, \overline{\partial}\phi\rangle\big| \\
&= \big|\langle [\overline{\partial}, A' - A]A\phi, \overline{\partial}\phi\rangle + \big\langle \big[[\overline{\partial}, A], A\big]\phi, \overline{\partial}\phi\big\rangle \\
&\qquad \langle [\overline{\partial}, A]\phi, (A' - A)\overline{\partial}\phi\rangle \\
&\qquad + \langle [\overline{\partial}, A]\phi, [A, \overline{\partial}]\phi\rangle\big|. \qquad (10.7.2.2)
\end{aligned}$$

Now, using (10.7.2.2) and 10.7.1, we have the estimates

$$\begin{aligned}
|II + III| &\lesssim \|[\overline{\partial}, A' - A]A\phi\|_{W^{1-k}} \cdot \|\overline{\partial}\phi\|_{W^{k-1}} + \big\|\big[[\overline{\partial}, A], A\big]\phi\big\|_{W^{1-k}} \|\overline{\partial}\phi\|_{W^{k-1}} \\
&\quad + \|[\overline{\partial}, A]\phi\|_{W^0} \cdot \|(A' - A)\overline{\partial}\phi\|_{W^0} + \|[\overline{\partial}, A]\phi\|_{W^0}^2 \\
&\lesssim \|\nabla A\phi\|_{W^{-1}} \|\nabla\phi\|_{W^{k-1}} + \|\nabla\phi\|_{W^{k-1}}^2.
\end{aligned}$$

But, by 10.7.1 again,

$$\begin{aligned}\|\nabla A\phi\|^2_{W^{-1}} &\lesssim \sum_j \|AD^j\phi\|^2_{W^{-1}} + \sum_j \|[D^j, A]\phi\|^2_{W^{-1}} + \|A\phi\|^2_{W^{-1}} \\ &\lesssim \sum_j \|D^j\phi\|^2_{W^{k-1}} + \sum_j \|D^j\phi\|^2_{W^{k-2}} + \|\phi\|^2_{W^{k-1}} \\ &\lesssim \|\nabla\phi\|^2_{W^{k-1}}.\end{aligned}$$

Therefore

$$|II + III| \lesssim \|\nabla\phi\|^2_{W^{k-1}}.$$

Since $I + IV = \overline{II + III}$, we may also conclude that

$$|I + IV| \lesssim \|\nabla\phi\|^2_{W^{k-1}}.$$

Together these yield

$$\left|\langle\overline{\partial}A\phi, \overline{\partial}A\phi\rangle - \mathrm{Re}\,\langle\overline{\partial}\phi, \overline{\partial}A'A\phi\rangle\right| \lesssim \|\nabla\phi\|^2_{W^{k-1}}.$$

Similarly, it may be shown that

$$\langle\vartheta A\phi, \vartheta A\phi\rangle - \mathrm{Re}\,\langle\vartheta\phi, \vartheta A'A\phi\rangle \lesssim \|\nabla\phi\|^2_{W^{k-1}}$$

and

$$\langle A\phi, A\phi\rangle - \mathrm{Re}\,\langle\phi, A'A\phi\rangle = 0.$$

This concludes the proof of the lemma. □

It is convenient to think of the preceding lemma as a sophisticated exercise in integration by parts. In the case $k = 0$ we shall now derive a slightly strengthened result:

**Lemma 10.7.3** *We have the estimate*

$$Q\langle\rho\phi, \rho\phi\rangle - Re\,Q(\phi, \rho^2\phi) = \mathcal{O}(\|\phi\|^2_{W^0}).$$

**Proof** We take $A$ in the preceding lemma to be the operator corresponding to multiplication by $\rho$ (this is just the case $k = 0$). Assuming as we may that $\rho$ is real-valued, we know that $A = A'$. By (10.7.2.1) in the proof of 10.7.2 we have

$$\begin{aligned}&\langle[\overline{\partial}, A']A\phi, \overline{\partial}\phi\rangle + \langle[A, \overline{\partial}]\phi, \overline{\partial}A\phi\rangle \\ &= \langle[\overline{\partial}, A - A']A\phi, \overline{\partial}\phi\rangle + \left\langle\left[[\overline{\partial}, A], A\right]\phi, \overline{\partial}\phi\right\rangle \\ &\qquad + \langle[\overline{\partial}, A]\phi, (A - A')\overline{\partial}\phi\rangle + \langle[\overline{\partial}, A]\phi, [A, \overline{\partial}]\phi\rangle.\end{aligned}$$

Since $A = \rho$, the operator $[\overline{\partial}, A]$ is simply multiplication by a matrix of functions. Thus $[\overline{\partial}, A]$ and $A$ commute and $\left[[\overline{\partial}, A], A\right] = 0$ and of course $A - A' = 0$. The result follows. □

Recall the quadratic form

$$E(\phi)^2 \equiv \sum_{j,k} \left\| \frac{\partial \phi_j}{\partial \overline{z}_k} \right\|^2_{W^0} + \int_{\partial\Omega} |\phi|^2 + \|\phi\|^2_{W^0}.$$

The basic estimate is

$$E(\phi)^2 \lesssim Q(\phi, \phi),$$

and it is a standing hypothesis that this estimate holds on the domain under study. *We know that the basic estimate holds, for instance, on any strongly pseudoconvex domain.* The next result contains the key estimate in our proof of regularity for the $\overline{\partial}$-Neumann problem up to the boundary.

**Theorem 10.7.4** *For each $P \in \partial\Omega$ there exists a special boundary chart $V$ about $P$ such that*

$$\|\!|\nabla \phi|\!\|^2_{W^{-1/2}} \lesssim E(\phi)^2$$

*for all $\phi \in \bigwedge^{p,q}_c (V \cap \overline{\Omega})$.*

The result follows quickly from the following lemma:

**Lemma 10.7.5** *Let $U$ be a special boundary chart for $\Omega$ and let $M_1, \ldots, M_N$ be first order, homogeneous differential operators on $U$. Write*

$$M_k = \sum_{j=1}^{2n} a_{jk} D^j.$$

*Assume that there exists no $0 \neq \eta \in T^*(U)$ such that $\sigma(M_k, \eta) = 0$ for all $k \in \{1, \ldots, N\}$. Then for all $P \in \partial\Omega \cap U$ there exists a neighborhood $V \subseteq U$ of $P$ such that*

$$\sum_{j=1}^{2n} \|\!|D^j \phi|\!\|^2_{W^{-1/2}} \leq \sum_{k=1}^{N} \|\!|M_k \phi|\!\|^2_{W^{-1/2}} + \int_{\partial\Omega} |\phi|^2 \tag{10.7.5.1}$$

*for all $\phi \in \bigwedge^{p,q}_c (V \cap \overline{\Omega})$.*

Assuming the lemma for the moment, let us prove the theorem.

**Proof of Theorem 10.7.4** The vector fields

$$\frac{\partial}{\partial \overline{z}_1}, \cdots, \frac{\partial}{\partial \overline{z}_n}$$

satisfy the conditions of the lemma so that

$$\begin{aligned}
\sum_{j=1}^{2n} \|D^j \phi\|^2_{W^{-1/2}} &= \|\nabla \phi\|^2_{W^{-1/2}} \\
&\lesssim \sum_{j,k=1}^{n} \left\| \frac{\partial \phi_k}{\partial \overline{z}_j} \right\|^2_{W^{-1/2}} + \int_{\partial\Omega} |\phi|^2 \\
&\leq \sum_{j,k=1}^{n} \left\| \frac{\partial \phi_k}{\partial \overline{z}_j} \right\|^2_{W^0} + \int_{\partial\Omega} |\phi|^2 + \|\phi\|^2_{W^0} \\
&= E(\phi)^2 .
\end{aligned}$$

□

Note that the $M_k$'s in the lemma cannot be too few. For they have to span all possible directions. Elementary considerations of dimensionality show that $N \geq n$ (the base field is the complex numbers). The $\{D^j\}$ are the simplest example of a collection of vector fields that satisfies the symbol condition in the Lemma.

Before proving Lemma 10.7.5, we need a preliminary result:

**Lemma 10.7.6** *Let $s > t$. Then for any $\epsilon > 0$ there is a neighborhood $V$ of $P \in \Omega$ such that*

$$\|u\|_{W^t} \leq \epsilon \|u\|_{W^s}$$

*whenever $u$ is supported in $V \cap \overline{\Omega}$.*

**Remark 10.7.7** Results of this sort are commonly used in elliptic theory. They are a form of the Rellich lemma. Rellich's lemma says that, on a compact set $K \subseteq \mathbb{R}^N$, bounded sets in the Sobolev space $W^s(K)$ have compact closure in the Sobolev space $W^r(K)$ when $s > r$. □

**Proof of Lemma 10.7.6** We first prove the statement in the case when there is no intervention of the boundary, that is when $u \in C_c^\infty(V)$ and $V \subset\subset \Omega$. Also assume at first that $s > t \geq 0$. If the assertion were false then there would exist an $\epsilon > 0$ and a sequence $\{u_k\}$ of functions with $\operatorname{supp} u_k \searrow \{P\}$, $\|u_k\|_{W^t} = 1$, and $\|u_k\|_{W^s} < 1/\epsilon$. By Rellich's lemma there is a subsequence converging in $W_t$ to a function $u$ (it is a function since we have assumed that $t \geq 0$). But $\operatorname{supp} u = \{P\}$, which is impossible since $\|u\|_{W^t} = 1$.

If $s \leq 0$ but $P$ is still in the interior of $\Omega$, we let $V'$ be a neighborhood of $P$ for which $\|u\|_{W^{-s}} \leq \epsilon \|u\|_{W^{-t}}$ for all $u \in C_c^\infty(V')$. Let $V \subseteq \overline{V} \subseteq V'$. Then we may exploit the duality between $W_t$ and $W_{-t}$ to see that if $u \in C_c^\infty(V)$ then

$$\begin{aligned}\|u\|_{W^t} &= \sup_{\substack{v\in C_c^\infty(V)\\ v\neq 0}} \frac{|\langle u, v\rangle|}{\|v\|_{W^{-t}}}\\ &\le \epsilon \sup_{\substack{v\in C_c^\infty(V)\\ v\neq 0}} \frac{|\langle u, v\rangle|}{\|v\|_{W^{-s}}}\\ &= \epsilon\|u\|_{W^s}.\end{aligned}$$

Notice that the case $t < 0 \le s$ follows from these first two cases.

Now let us consider the case that $P \in \partial\Omega$. It is enough to consider the problem on the half space in $\mathbb{R}^{2n}$ with boundary $\mathbb{R}^{2n-1}$. Let $V' \subseteq \mathbb{R}^{2n-1}$ be a relative neighborhood of $P \in \mathbb{R}^{2n-1}$ such that

$$\|u\|_{W^t} \le \epsilon\|u\|_{W^s}$$

for all $u$ supported in $V'$ (note here that $P$ is in the *interior* of $\mathbb{R}^{2n-1}$ so our preceding result applies). Let $V \equiv V' \times I$, where $I$ is any interval in $(0, \infty)$. Then

$$\begin{aligned}|||u|||_{W^t}^2 &= \int_{-\infty}^0 \|u(\cdot, r)\|_{W^t}^2\, dr\\ &\le \epsilon^2 \int_{-\infty}^0 \|u(\cdot, r)\|_{W^s}^2\, dr\\ &= \epsilon^2 |||u|||_{W^s}^2.\end{aligned}$$

□

**Proof of Lemma 10.7.5** The first step is to reduce to the case in which the $M_k$'s are operators with constant coefficients. Let $V \subset\subset U$ be a neighborhood of $P$, $\rho \in C_c^\infty, 0 \le \rho \le 1$. Suppose also that $\rho \equiv 1$ on $V$ and $W \equiv \operatorname{supp}\rho \subseteq U$. It suffices to prove our result for functions since the Sobolev norm on forms is defined componentwise.

Let $u \in \bigwedge_c^{0,0}(V \cap \overline{\Omega})$. Now we freeze the coefficients of the $M_k$; that is, we set

$$N_k = \sum_{j=1}^{2n} a_{j,k}(P)D^j,$$

$k = 1, \ldots, N$. Let $b_{j,k}(x) \equiv a_{j,k}(x) - a_{j,k}(P)$. Then

$$\begin{aligned}|||(M_k - N_k)u|||_{W^{-1/2}} &= |||\sum_j \big(a_{j,k}(x) - a_{j,k}(P)\big)D^j u|||_{W^{-1/2}}\\ &\lesssim \sum_j |||b_{j,k}D^j u|||_{W^{-1/2}}\end{aligned}$$

$$
\begin{aligned}
&= \sum_j \|\rho b_{j,k} D^j u\|_{W^{-1/2}} \\
&= \sum_j \left\| \Lambda_t^{-1/2} \left( \rho b_{j,k} D^j u \right) \right\|_{W^0} \\
&= \sum_j \left\| \rho b_{j,k} \Lambda_t^{-1/2} D^j u + [\Lambda_t^{-1/2}, \rho b_{j,k}] D^j u \right\|_{W^0} \\
&\lesssim \sum_j \sup_W |\rho b_{j,k}| \cdot \|\Lambda_t^{-1/2} D^j u\|_{W^0} \\
&\qquad + \sum_j \left\| [\Lambda_t^{-1/2}, \rho b_{j,k}] D^j u \right\|_{W^0}.
\end{aligned}
$$

Note that if $W$ is small enough then $\sup_W |\rho b_{j,k}| < \epsilon$ since $|b_{j,k}(P)| = 0$. Now the commutator in the second sum is of tangential order $-3/2$. Thus the last line is

$$
\begin{aligned}
&\lesssim \epsilon \sum_j \|D^j u\|_{W^{-1/2}} + \sum_j \|D^j u\|_{W^{-3/2}} \\
&\lesssim \epsilon \|D^j u\|_{W^{-1/2}},
\end{aligned}
$$

where (shrinking $V$ if necessary) we have applied Lemma 10.7.6. As a result,

$$
\begin{aligned}
\|N_k u\|_{W^{-1/2}} &\leq \|M_k u\|_{W^{-1/2}} + \|(M_k - N_k) u\|_{W^{-1/2}} \\
&\lesssim \|M_k u\|_{W^{-1/2}} + \epsilon \sum_j \|D^j u\|_{W^{-1/2}}. \qquad (10.7.5.1)
\end{aligned}
$$

If we can prove the constant coefficient case of our inequality then we would have

$$
\sum_j \|D^j u\|^2_{W^{-1/2}} \lesssim \sum_{k=1}^N \|N_k u\|^2_{W^{-1/2}} + \int_{\partial\Omega} |u|^2.
$$

Coupling this with (10.7.5.1) would yield

$$
\begin{aligned}
\sum_{j=1}^{2n} \|D^j u\|^2_{W^{-1/2}} &\lesssim \sum_{k=1}^N \left( \|M_k u\|_{W^{-1/2}} + \epsilon \sum_j \|D^j u\|_{W^{-1/2}} \right)^2 \\
&\qquad + \int_{\partial\Omega} |u|^2 \\
&\lesssim \sum_{k=1}^N \|M_k u\|^2_{W^{-1/2}} + \epsilon \sum_j \|D^j u\|^2_{W^{-1/2}} + \int_{\partial\Omega} |u|^2.
\end{aligned}
$$

The full result then follows.

So we have reduced matters to the case of the $M_k$'s having constant coefficients. Assume for the moment that $u\big|_{\partial\Omega} \equiv 0$. Then we can extend $u$ to be zero outside $\Omega$. The extended function will be continuous on all of space. We may suppose that $V$ is a special boundary chart. Notice that, on $V$, the $D_t^j u$ are continuous and $D_r u$ has only a jump discontinuity. Therefore all of these first derivative functions are square integrable.

Write $\xi = (\tau, \zeta)$ with $\tau \in \mathbb{R}^{2n-1}, \zeta \in \mathbb{R}$. We use the symbol condition on the $M_k$'s to see that

$$\begin{aligned}
\sum_{k=1}^N |||M_k u|||_{W^{-1/2}}^2 &\equiv \sum_{k=1}^N \left\| (1+|\tau|^2)^{-1/4} \widetilde{M_k u}(\tau, r) \right\|_{W^0}^2 \\
&\overset{\left(\substack{\text{Plancherel in} \\ \text{last var.}}\right)}{=} \sum_{k=1}^N \left\| (1+|\tau|^2)^{-1/4} \left[ \sum_{j=1}^{2n-1} a_{j,k}(P)\tau_j \right. \right. \\
&\qquad \left. \left. + a_{2n,k}(P)\zeta \right] \widehat{u}(\xi) \right\|_{W^0}^2 \\
&= \sum_{k=1}^N \int_{\mathbb{R}^{2n}} (1+|\tau|^2)^{-1/2} |\sigma(N_k, \xi)|^2 |\widehat{u}(\xi)|^2 \, d\xi \\
&\gtrsim \int_{\mathbb{R}^{2n}} (1+|\tau|^2)^{-1/2} |\xi|^2 |\widehat{u}(\xi)|^2 \, d\xi \\
&\gtrsim \int_{\mathbb{R}^{2n}} (1+|\tau|^2)^{-1/2} \left( \sum_j |D^j u|^2 \right)\widehat{\ } (\xi) \, d\xi
\end{aligned}$$

$$\begin{aligned}
&\overset{\left(\substack{\text{Plancherel in} \\ \text{last var.}}\right)}{=} \int_{\mathbb{R}^{2n-1}} (1+|\tau|^2)^{-1/2} \int_{\mathbb{R}} \sum_j \left| \widetilde{D^j u}(\tau, r) \right|^2 dr d\tau \\
&\equiv \sum_j |||D^j u|||_{W^{-1/2}}^2 .
\end{aligned}$$

Therefore

$$\sum_{k=1}^N |||M_k u|||_{W^{-1/2}}^2 \gtrsim \sum_{j=1}^{2n} |||D^j u|||_{W^{-1/2}}^2$$

in the constant coefficient case provided that $u$ vanishes on $\partial\Omega$.

Next suppose that $u$ may or may not vanish at the boundary. Let

$$\widetilde{w}(\tau, r) \equiv \exp\left[(1 + |\tau|^2)^{1/2} r\right] \widetilde{u}(\tau, 0).$$

This is a regular extension of $\widetilde{u}(\tau, 0)$ to $(\tau, r), r \leq 0$. By the Fourier inversion theorem we have

$$w(t, 0) = u(t, 0).$$

Set $v = u - w$. Then $v$ vanishes on the boundary so that the previous result applies to $v$. We then have

$$\sum_{j=1}^{2n} \|D^j v\|^2_{W^{-1/2}} \lesssim \sum_{k=1}^{N} \|M_k v\|^2_{W^{-1/2}}.$$

Therefore

$$\begin{aligned}
\sum_{j=1}^{2n} \|D^j u\|^2_{W^{-1/2}} &\lesssim \sum_{j=1}^{2n} \|D^j v\|^2_{W^{-1/2}} + \|D^j w\|^2_{W^{-1/2}} \\
&\lesssim \sum_{k=1}^{N} \|M_k v\|^2_{W^{-1/2}} + \sum_{j=1}^{2n} \|D^j w\|^2_{W^{-1/2}} \\
&\lesssim \sum_{k=1}^{N} \left[ \|M_k u\|^2_{W^{-1/2}} + \|M_k w\|^2_{W^{-1/2}} \right] + \sum_{j=1}^{2n} \|D^j w\|^2_{W^{-1/2}} \\
&\lesssim \sum_{k=1}^{N} \|M_k u\|^2_{W^{-1/2}} + \sum_{j=1}^{2n} \|D^j w\|^2_{W^{-1/2}},
\end{aligned}$$

since the $M_k$'s are linear combinations of the $D^j$'s. Observe that we are finished if we can show that

$$\|D^j w\|^2_{W^{-1/2}} \lesssim \int_{\partial\Omega} |u|^2 \, d\sigma,$$

$j = 1, \ldots, 2n$.

First suppose that $j \in \{1, \ldots, 2n - 1\}$. Then

$$\begin{aligned}
\|D^j w\|^2_{W^{-1/2}} = \int_{\mathbb{R}^{2n-1}} \int_{-\infty}^{0} &(1 + |\tau|^2)^{-1/2} |\tau_j|^2 \exp\left[2(1 + |\tau|^2)^{1/2} r\right] \\
&\times |\widetilde{u}(\tau, 0)|^2 \, dr d\tau
\end{aligned}$$

$$\begin{aligned}
&\lesssim \int_{\mathbb{R}^{2n-1}} \int_{-\infty}^{0} (1+|\tau|^2)^{-1/2}(1+|\tau|^2)\exp\left[2(1+|\tau|^2)^{1/2}r\right] \\
&\qquad \times |\widetilde{u}(\tau,0)|^2 \, dr d\tau \\
&= \int_{\mathbb{R}^{2n-1}} |\widetilde{u}(\tau,0)|^2 \left( \frac{1}{2}\exp\left[2(1+|\tau|^2)^{1/2}r\right]\Big|_{-\infty}^{0} \right) d\tau \\
&= \frac{1}{2}\int_{\mathbb{R}^{2n-1}} |\widetilde{u}(\tau,0)|^2 \, d\tau \\
&= \frac{1}{2}\int_{\partial\Omega} |u|^2 \, d\sigma.
\end{aligned}$$

If instead $j = 2n$, so that $D^j$ is the usual normal derivative, then

$$\|\!| D^{2n} w |\!\|^2_{W^{-1/2}} = \int_{\mathbb{R}^{2n}_+} (1+|\tau|^2)^{-1/2} |\widetilde{D_r w}(\tau, r)|^2 \, d\tau dr.$$

Since the derivative does not affect the variables in which we take the Fourier transform, the two operations commute. Hence

$$\widetilde{D_r w}(\tau, r) = (1+|\tau|^2)^{1/2}\exp\left[(1+|\tau|^2)^{1/2}r\right]\widetilde{u}(\tau,0).$$

Therefore

$$\begin{aligned}
\|\!| D_r w |\!\|^2_{W^{-1/2}} &= \int_{\mathbb{R}^{2n-1}} \int_{-\infty}^{0} (1+|\tau|^2)^{-1/2}(1+|\tau|^2)\exp\left[2(1+|\tau|^2)^{1/2}r\right] \cdot |\widetilde{u}(\tau,r)|^2 \, dr d\tau \\
&= \int_{\mathbb{R}^{2n-1}} \int_{-\infty}^{0} (1+|\tau|^2)^{1/2}\exp\left[2(1+|\tau|^2)^{1/2}r\right] |\widetilde{u}(\tau,0)|^2 \, dr d\tau \\
&= \frac{1}{2}\int_{\mathbb{R}^{2n-1}} |\widetilde{u}(\tau,0)|^2 \, d\tau \\
&= \frac{1}{2}\int_{\partial\Omega} |u|^2 \, d\sigma.
\end{aligned}$$

This concludes the second case and the proof. □

## 10.8 Proof of the Main Theorem

**Capsule:** Now we draw together all of our threads of reasoning to prove the Main Theorem. In particular, we use a bootstrapping argument to pass from the $-1/2$ Sobolev norm to higher order norms.

We pass from the Sobolev $-1/2$ norm to higher order norms.

**Lemma 10.8.1** *Suppose that the basic estimate $E(\phi)^2 \lesssim Q(\phi,\phi), \phi \in \mathcal{D}^{p,q}$, holds on a special boundary chart $V$. Let $\{\rho_k\}$ be a sequence of cutoff functions in $\bigwedge_c^{0,0}(V \cap \overline{\Omega})$ such that*

$$\rho_k \equiv 1 \qquad on \quad supp\, \rho_{k+1}.$$

*Then for each $k > 0$ we have an a priori estimate*

$$|||\nabla \rho_k \phi|||^2_{W^{(k-2)/2}} \lesssim |||\rho_1 F\phi|||^2_{W^{(k-2)/2}} + \|F\phi\|^2_{W^0}.$$

**Proof** We proceed by induction on $k$. For $k = 1$ we have

$$\begin{aligned}
|||\nabla \rho_1 \phi|||^2_{W^{-1/2}} &\overset{\text{Theorem 10.7.4}}{\lesssim} E(\rho_1\phi)^2 \\
&\overset{\text{(basic est.)}}{\lesssim} Q(\rho_1\phi, \rho_1\phi) \\
&\overset{\text{Lemma 10.7.3}}{=} \operatorname{Re} Q(\phi, \rho_1^2\phi) + \mathcal{O}(\|\phi\|^2_{W^0}) \\
&\overset{\text{(Friedrichs)}}{=} \operatorname{Re}\langle F\phi, \rho_1^2\phi\rangle + \mathcal{O}(\|\phi\|^2_{W^0}) \\
&\lesssim \|F\phi\|_{W^0}\, \|\rho_1^2\phi\|_{W^0} + \mathcal{O}(\|\phi\|^2_{W^0}) \\
&\lesssim \|F\phi\|_{W^0} \|\phi\|_{W^0} + \mathcal{O}(\|\phi\|^2_{W^0}) \\
&\lesssim \|F\phi\|^2_{W^0} + \mathcal{O}(\|\phi\|^2_{W^0}) \\
&\lesssim \|F\phi\|^2_{W^0} \\
&\lesssim |||\rho_1 F\phi|||^2_{W^{-1/2}} + \|F\phi\|^2_{W^0}.
\end{aligned}$$

In the last two lines but one we have used the fact that $T = F^{-1}$ is bounded on $L^2$. This is the first step of the induction.

Now take $k > 1$ and assume that we have proved the result for $k-1$. Set $\Lambda = \Lambda_t^{(k-1)/2}$. Note that $\Lambda$ commutes with $D^j, j = 1, \ldots, 2n-1$, because $\Lambda$ is a convolution operator in the tangential variables. If $j = 2n$ then it is even easier to see that $D^j$ commutes with $\Lambda$ since $\Lambda$ does not act in the $r$ variable. Thus

$$\begin{aligned}
|||\nabla \rho_k \phi|||^2_{W^{(k-2)/2}} &= |||\Lambda(\nabla \rho_k \phi)|||^2_{W^{-1/2}} \\
&= |||\nabla \Lambda(\rho_k \phi)|||^2_{W^{-1/2}} \\
&= |||\nabla \Lambda(\rho_1 \rho_k \phi)|||^2_{W^{-1/2}}. \qquad (10.8.1.1)
\end{aligned}$$

Observe that

$$\begin{aligned}
&\big[\Lambda,[D^j,\rho_1]\big]\rho_k\rho_{k-1}\phi+[\Lambda,\rho_1][D^j,\rho_k]\rho_{k-1}\phi+[\Lambda,\rho_1]\rho_kD^j\rho_{k-1}\phi\\
&\quad= \big[\Lambda,[D^j,\rho_1]\big]\rho_k\rho_{k-1}\phi+[\Lambda,\rho_1]D^j\rho_k\rho_{k-1}\phi\\
&\overset{\text{(Jacobi identity)}}{=} -\big[\rho_1,[\Lambda,D^j]\big]\rho_k\rho_{k-1}\phi-\big[D^j,[\rho_1,\Lambda]\big]\rho_k\rho_{k-1}\phi\\
&\qquad\qquad +[\Lambda,\rho_1]D^j\rho_k\rho_{k-1}\phi\\
&\overset{(\Lambda,D^j\text{ commute})}{=} -0-D^j[\rho_1,\Lambda]\rho_k\rho_{k-1}\phi\\
&\quad= D^j[\Lambda,\rho_1]\rho_k\phi.
\end{aligned}$$

As a result,

$$\begin{aligned}
D^j\Lambda\rho_1\rho_k\phi &= D^j\rho_1\Lambda\rho_k\phi+D^j[\Lambda,\rho_1]\rho_k\phi\\
&= D^j\rho_1\Lambda\rho_k\phi+\big[\Lambda,[D^j,\rho_1]\big]\rho_k\rho_{k-1}\phi+[\Lambda,\rho_1][D^j,\rho_k]\rho_{k-1}\phi\\
&\quad +[\Lambda,\rho_1]\rho_kD^j\rho_{k-1}\phi.
\end{aligned}$$

Therefore, using (10.8.1.1),

$$\begin{aligned}
\|\nabla\rho_k\phi\|^2_{W^{(k-2)/2}} &= \|\nabla\Lambda\rho_1\rho_k\phi\|^2_{W^{-1/2}}\\
&\lesssim \|\nabla\rho_1\Lambda\rho_k\phi\|^2_{W^{-1/2}}+\sum_j\|\big[\Lambda,[D^j,\rho_1]\big]\rho_k\rho_{k-1}\phi\|^2_{W^{-1/2}}\\
&\quad+\sum_j\|[\Lambda,\rho_1][D^j,\rho_k]\rho_{k-1}\phi\|^2_{W^{-1/2}}\\
&\quad+\sum_j\|[\Lambda,\rho_1]\rho_kD^j\rho_{k-1}\phi\|^2_{W^{-1/2}}\\
&\lesssim \|\nabla\rho_1\Lambda\rho_k\phi\|^2_{W^{-1/2}}+\|\rho_{k-1}\phi\|^2_{W^{-1/2+(k-3)/2}}\\
&\quad\|\rho_{k-1}\phi\|^2_{W^{-1/2+(k-3)/2}}+\|\nabla\rho_{k-1}\phi\|^2_{W^{-1/2+(k-3)/2}}\\
&\lesssim \|\nabla\rho_1\Lambda\rho_k\phi\|^2_{W^{-1/2}}+\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}. \qquad (10.8.1.2)
\end{aligned}$$

Consider the term $\|\nabla\rho_1\Lambda\rho_k\phi\|^2_{W^{-1/2}}$ on the righthand side. Set

$$\rho_1\Lambda\rho_k\equiv\rho_1\Lambda_t^{(k-1)/2}\rho_k\equiv A.$$

Then

$$\begin{aligned}
\|\nabla\rho_1\Lambda\rho_k\phi\|^2_{W^{-1/2}} &= \|\nabla A\rho_{k-1}\phi\|^2_{W^{-1/2}}\\
&\lesssim Q(A\rho_{k-1}\phi,A\rho_{k-1}\phi)\\
&= \operatorname{Re}Q(\rho_{k-1}\phi,A'A\rho_{k-1}\phi)
\end{aligned}$$

$$
\begin{aligned}
&\qquad\qquad +\mathcal{O}\big(\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}\big)\\
&\overset{(\rho_{k-1}A'=A')}{=} \operatorname{Re} Q(\phi, A'A\rho_{k-1}\phi) + \mathcal{O}\big(\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}\big)\\
&= \qquad \operatorname{Re}(F\phi, A'A\rho_{k-1}\phi) + \mathcal{O}(\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}
\end{aligned}
$$

$$
\begin{aligned}
&= \operatorname{Re}(AF\phi, A\rho_{k-1}\phi) + \mathcal{O}\big(\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}\big)\\
&\lesssim \|A\rho_1 F\phi\|_{W^{-1/2}}\|A\phi\|_{W^{1/2}} + \mathcal{O}\big(\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}\\
&= \|\rho_1\Lambda_t^{(k-1)/2}\rho_k\rho_1 F\phi\|_{W^{-1/2}}\|\rho_1\Lambda_t^{(k-1)/2}\rho_k\phi\|_{W^{1/2}}\\
&\qquad +\mathcal{O}\big(\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}\\
&\lesssim \|\rho_1 F\phi\|_{W^{(k-1)/2-1/2}}\|\rho_k\phi\|_{W^{(k-1)/2+1/2}}\\
&\qquad +\mathcal{O}\big(\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}\\
&\lesssim \frac{1}{\epsilon}\|\rho_1 F\phi\|^2_{W^{(k-2)/2}} + \epsilon\|\nabla\rho_k\phi\|^2_{W^{(k-2)/2}}\\
&\qquad +\|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}.
\end{aligned}
$$

Substituting this last inequality into (10.8.1.2) we obtain

$$
\begin{aligned}
\|\nabla\rho_k\phi\|^2_{W^{(k-2)/2}} &\le \|\nabla\rho_1\Lambda\rho_k\phi\|^2_{W^{-1/2}} + \|\nabla\rho_{k-1}\phi\|^2_{W^{(k-3)/2}}\\
&\lesssim \frac{1}{\epsilon}\|\rho_1 F\phi\|^2_{(k-2)/2} + \epsilon\|\nabla\rho_k\phi\|^2_{W^{(k-2)/2}} + \|\nabla\rho_{k-1}\phi\|_{(k-3)/2}.
\end{aligned}
$$

Therefore

$$
\begin{aligned}
\|\nabla\rho_k\phi\|^2_{W^{(k-2)/2}} &\lesssim \frac{1}{\epsilon}\|\rho_1 F\phi\|^2_{W^{(k-2)/2}} + \|\nabla\rho_{k-1}\phi\|^2_{(k-3)/2}\\
&\overset{\text{(induction)}}{\lesssim} \frac{1}{\epsilon}\|\rho_1 F\phi\|^2_{(k-2)/2} + \|\rho_1 F\phi\|^2_{W^{(k-3)/2}} + \|F\phi\|^2_{W^0}\\
&\lesssim \|\rho_1 F\phi\|^2_{W^{(k-2)/2}} + \|F\phi\|^2_{W^0}.
\end{aligned}
$$

This completes the proof. □

**Theorem 10.8.2** *Assume that the basic estimate holds in $\mathcal{D}^{p,q}$. Let $V$ be a special boundary neighborhood on which $\|\nabla\phi\|^2_{W^{-1/2}} \le E(\phi)^2$. Let $U \subset\subset V$ and choose a cutoff function $\rho_1 \in \bigwedge_c^{0,0}(V \cap \overline{\Omega})$ such that $\rho_1 \equiv 1$ on $U$.*

*Then for each $\rho \in \bigwedge_c^{0,0}(U \cap \overline{\Omega})$ and each non-negative integer $s$ it holds that*

$$
\|\rho\phi\|^2_{W^{s+1}} \lesssim \|\rho_1 F\phi\|^2_{W^s} + \|F\phi\|^2_{W^0}
$$

*for all* $\phi \in dom\,(F \cap \mathcal{D}^{p,q})$.

**Proof** We proceed by induction on $s$. We apply the preceding lemma with $k = 2, \rho_2 = \rho$, and $0 = \rho_3 = \rho_4 = \cdots$. It tells us that

$$\|\!|\nabla \rho_2 \phi|\!\|^2_{W^0} \leq \|\!|\rho_1 F\phi|\!\|^2_{W^0} + \|\!|F\phi|\!\|^2_{W^0},$$

that is

$$\|\nabla \rho\phi\|^2_{W^0} \leq \|\rho_1 F\phi\|^2_{W^0} + \|\!|F\phi|\!\|^2_{W^0}.$$

Therefore

$$\begin{aligned}\|\rho\phi\|^2_{W^1} &\approx \|\nabla\rho\phi\|^2_{W^0} + \|\rho\phi\|^2_{W^0} \\ &\lesssim \|\rho_1 F\phi\|^2_{W^0} + \|\!|F\phi|\!\|^2_{W^0},\end{aligned}$$

which is the statement that we wish to prove for $s = 0$.

Now suppose the statement to be true for $s$, some $s \geq 0$. Then

$$\begin{aligned}\|\rho\phi\|^2_{W^{s+1}} &\approx \sum_{|\alpha|\leq s+1} \|D^\alpha(\rho\phi)\|^2_{W^0} \\ &= \sum_{|\alpha|=s+1} \|D^\alpha(\rho\phi)\|^2_{W^0} + \|\rho\phi\|^2_{W^s} \\ &\lesssim \sum_{|\alpha|=s+1} \|D^\alpha(\rho\phi)\|^2_{W^0} + \|\rho_1 F\phi\|^2_{W^{s-1}} + \|\!|F\phi|\!\|^2_{W^0} \\ &\leq \sum_{|\alpha|=s+1} \|D^\alpha(\rho\phi)\|^2_{W^0} + \|\rho_1 F\phi\|^2_{W^s} + \|\!|F\phi|\!\|^2_{W^0}.\end{aligned}$$

It remains to estimate the top order term. Pick a sequence of cutoff functions $\rho_1 \geq \rho_2 \geq \cdots \geq \rho_{2s+2} = \rho$ such that $\operatorname{supp}\rho_j \subset\subset \operatorname{supp}\rho_{j-1}$, $j = 2, \ldots, 2s+2$ and set $\rho_j \equiv 0$ for $j > 2s+2$. We apply 10.8.1 with $k = 2s+2$ to obtain

$$\|\!|\nabla \rho_{2s+2}\phi|\!\|^2_{W^s} \lesssim \|\!|\rho_1 F\phi|\!\|^2_{W^s} + \|\!|F\phi|\!\|^2_{W^0}.$$

If $|\beta| = s+1$ then

$$\begin{aligned}\|D_t^\beta \rho_{2s+2}\phi\|^2_{W^0} &\lesssim \|\!|\nabla\rho_{2s+2}\phi|\!\|^2_{W^s} \\ &\lesssim \|\!|\rho_1 F\phi|\!\|^2_{W^s} + \|\!|F\phi|\!\|^2_{W^0} \\ &\lesssim \|\rho_1 F\phi\|^2_{W^s} + \|\!|F\phi|\!\|^2_{W^0}.\end{aligned}$$

For $|\beta| = s$ we have

$$\begin{aligned} \|D_t^\beta D_r \rho\phi\|_{W^0}^2 &\lesssim \|\!|\nabla \rho\phi|\!\|_{W^s}^2 \\ &\lesssim \|\rho_1 F\phi\|_{W^s}^2 + \|F\phi\|_{W^0}^2 . \end{aligned}$$

Thus it remains to estimate $\|D_t^\beta D_r^m \rho\phi\|_{W^0}^2$ for $m \geq 2$ and $|\beta| + m = s + 1$.

We proceed by induction on $m$ (this is a second induction within the first induction on $s$). We use the differential equation as follows: We know that $\Box + I = F$. On the special boundary chart the function $F\phi$ can be expressed as

$$F\phi = A_{2n} D_r^2 + \left[ \sum_{j=1}^{2n-1} A_j D_t^j D_r \phi + \sum_{j,k=1}^{2n-1} A_{j,k} D_t^j D_t^k \phi \right] .$$

The operator $F$ is strongly elliptic so that $A_{2n}$ is an invertible matrix. Therefore we can express the second derivative $D_r^2 \phi$ in terms of the remaining expressions on the right side of the last equation (each of which involves at most *one* normal derivative) and in terms of $F\phi$ itself. This means that we may estimate expressions of the form $\|D_t^\beta D_r^2 \rho\phi\|_{W^0}^2$ in terms of $\|D_t^\beta D_r \rho\phi\|_{W^0}^2$ and $\|D_t^\beta \rho\phi\|_{W^0}^2$, both of which have already been estimated. Thus we have handled the case $m = 2$.

Inductively, we can handle any term of the form $\|D_t^\beta D_r^m \rho\phi\|_{W^0}^2$.

This concludes the induction on $m$, which in turn concludes the induction on $s$. The proof is therefore complete. □

The final step of the proof of the last theorem is decisive. While the $\overline{\partial}$-Neumann boundary conditions are degenerate, the operator $F = \Box + I$ is strongly ellipic—as nice an operator as you could want. One of the special features of such an operator $L$ is that *any* second derivative of a function $f$ can be controlled by $Lf$, modulo error terms.

Now recall the Main Theorem:

**Theorem 10.8.3 (Main Theorem)** *Let $\Omega \subseteq \mathbb{C}^n$ be a smoothly bounded domain. Assume that the basic estimate holds for elements of $\mathcal{D}^{p,q}$. For $\alpha \in W_0^{p,q}$, we let $\phi$ denote the unique solution to the equation $F\phi = \alpha$. Then we have:*

**(1)** *If $W$ is a relatively open subset of $\overline{\Omega}$ and if $\alpha\big|_W \in \bigwedge^{p,q}(W)$ then $\phi\big|_W \in \bigwedge^{p,q}(W)$;*

**(2)** *Let $\rho, \rho_1$ be smooth functions with $\operatorname{supp}\rho \subseteq \operatorname{supp}\rho_1 \subseteq W$ and $\rho_1 \equiv 1$ on $\operatorname{supp}\rho$. Then:*

  **(a)** *If $W \cap \partial\Omega = \emptyset$ then $\forall s \geq 0$ there is a constant $c_s > 0$ (depending on $\rho, \rho_1$ but independent of $\alpha$) such that*

$$\|\rho\phi\|_{W^{s+2}}^2 \leq c_s \big( \|\rho_1 \alpha\|_{W^s}^2 + \|\alpha\|_{W^0}^2 \big). \qquad (10.8.3.2.a)$$

**(b)** *If $W \cap \partial\Omega \neq \emptyset$ then $\forall s \geq 0$ there exists a constant $c_s \geq 0$ such that*

$$\|\rho\phi\|^2_{W^{s+1}} \leq c_s\big(\|\rho_1\alpha\|^2_{W^s} + \|\alpha\|^2_{W^0}\big). \tag{10.8.3.2.b}$$

**Remark 10.8.4** In the theorem that we just proved, we established the *a priori* estimate

$$\|\rho\phi\|^2_{W^{s+1}} \lesssim \|\rho_1 F\phi\|^2_{W^s} + \|F\phi\|^2_{W^0}.$$

That is, we know that this inequality holds for a testing function $\phi$. In the Main Theorem we are now addressing the problem of *existence*: given a $v \in W_0^{p,q}$ we want to know that there exists a $\phi$ which is smooth and satisfies **(10.8.3.2.a)** and **(10.8.3.2.b)**. We *do* know that the $\phi$ given by Friedrichs is in $L^2$, but nothing further. It is in these arguments that the elliptic regularization technique comes to the rescue. In the original paper [KOH2] Kohn had to use deep functional analysis techniques to address the existence issue. □

**Proof of the Main Theorem** Let $\alpha \in W_0^{p,q}$. Let $\phi \in \widetilde{\mathcal{D}}^{p,q}$ be the unique solution to the equation $F\phi = \alpha$. This will of course be the $\phi$ that we seek, but we must see that it is smooth. Observe that this assertion, namely part **(1)** of the theorem, follows from part **(2)**.

Let $U$ be a subregion of $\mathbb{C}^n$. Assume that

$$\alpha\big|_{U\cap\Omega} \in \bigwedge^{p,q}(U \cap \overline{\Omega}).$$

We need to see that

$$\phi\big|_{U\cap\Omega} \in \bigwedge^{p,q}(U \cap \overline{\Omega}).$$

The easy case is if $U \cap \partial\Omega = \emptyset$. For then $\phi\big|_U \in \bigwedge^{p,q}(U)$ by the interior elliptic regularity for $F = \Box + I$. This gives us the estimate **(10.8.3.2.a)** as well.

More interesting is the case $U \cap \partial\Omega \neq \emptyset$. Then the last theorem tells us that

$$\|\rho\phi\|^2_{W^{s+1}} \lesssim \|\rho\alpha\|^2_{W^s} + \|\alpha\|^2_{W^0}$$

*provided that we know in advance that $\phi$ is smooth on $U \cap \overline{\Omega}$*. Again we emphasize the importance of this subtlety: we know that $\phi$ exists; we also know that *if $\phi$ is known to be smooth on $U \cap \overline{\Omega}$* then it satisfies the desired regularity estimates. We need to pass from this *a priori* information to a general regularity result. To this task we now turn.

Let $0 < \delta \leq 1$ and let $\phi^\delta$ be the solution of $F^\delta\phi = \alpha$. We know that $F^\delta$ is hypoelliptic up to the boundary (see 10.6.3). Recall that

$$Q^\delta(\phi, \psi) = Q(\phi, \psi) + \delta \sum_{j=1}^{2n} \langle D^j\phi, D^j\psi\rangle$$

and that

$$Q^\delta(\phi,\phi) \geq Q(\phi,\phi) \geq E(\phi)^2.$$

The proof of the last theorem then applies to $F^\delta$ and $\phi^\delta$; thus we have

$$\|\rho\phi^\delta\|^2_{W^{s+1}} \lesssim \|\rho_1\alpha\|^2_{W^s} + \|\alpha\|^2_{W^0}.$$

*The constant in the inequality is independent of* $\delta$ since the estimate depends only on the majorization

$$\|\!|\nabla\rho\phi^\delta|\!\|^2_{W^{-1/2}} \leq Q(\rho\phi^\delta,\rho\phi^\delta) \approx Q^\delta(\rho\phi^\delta,\rho\phi^\delta) - \delta\|\rho\phi^\delta\|^2_{W^1}.$$

[In fact we shall provide the details of this important assertion in the Appendix to this chapter.] Thus $\{\rho\phi^\delta\}_{0\leq\delta\leq 1}$ is uniformly bounded in $\|\cdot\|_{W^{s+1}}$ for every $s$. Fix an $s$. By Rellich's lemma, there is a subsequence $\{\rho\phi^{\delta_n}\}$ which converges in $W^{p,q}_s$ as $n\to\infty$. By diagonalization, we may assume that $\{\rho\phi^{\delta_n}\}$ converges in $W^{p,q}_s$ for *every* $s$—to the same function $\rho\psi$. We wish to show that $\rho\psi = \rho\phi$, where $\phi$ is the function whose existence comes from the Fredholm theory. Then we will know, by the Sobolev theorem, that $\rho\phi$ is smooth and we will be done.

It suffices to show that $\phi^{\delta_n}\to\phi$ in the $W^{p,q}_0$ topology (for of course $\phi^{\delta_n}\to\psi$ in that topology). We know that the interior estimates hold uniformly in $\delta$:

$$\|\rho\phi^\delta\|_{W^2} \lesssim \|\alpha\|_{W^0}$$

for any $\rho$ with compact support in $\Omega$. By 10.7.6, $\phi^\delta\in W^{p,q}_1$. If $\alpha$ is globally smooth, we can apply a partitition of unity argument and the boundary estimate for $s=0$ (see 10.7.2) to see that

$$\|\phi^\delta\|_{W^1} \lesssim \|\alpha\|_{W^0}, \tag{10.8.3.3}$$

uniformly in $\delta$ as $\delta$ as $\delta\to 0$. By the density of smooth forms in the space $W^{p,q}_0$, we conclude that (10.3.2.3) holds for all elements $\alpha\in W^{p,q}_0$.

Next we calculate that

$$\begin{aligned} Q(\phi,\psi) &= \langle\alpha,\psi\rangle = Q^\delta(\phi^\delta,\psi) \\ &= Q(\phi^\delta,\psi) + \mathcal{O}(\delta\|\phi^\delta\|_{W^1}\|\psi\|_{W^1}) \\ &= Q(\phi^\delta,\psi) + \|\alpha\|_{W^0}\|\psi\|_{W^1}\cdot\mathcal{O}(\delta). \end{aligned} \tag{10.8.3.4}$$

Give the equation

$$\langle\alpha,\psi\rangle = Q(\phi^\delta,\psi) + \|\alpha\|_{W^0}\|\psi\|_{W^1}\cdot\mathcal{O}(\delta)$$

the name $R(\delta)$. By subtracting $R(\delta')$ from $R(\delta)$ we find that

$$Q(\phi^\delta - \phi^{\delta'}, \psi) = \|\alpha\|_{W^0} \|\psi\|_{W^1} \cdot \mathcal{O}(\delta + \delta').$$

Lemma 10.6.3 we can find a sequence $\{\psi_n\} \subseteq \mathcal{D}^{p,q}$ converging with respect to both the norms $Q$ and $\| \cdot \|_{W^1}$ to $\phi^\delta - \phi^{\delta'}$. The result is that

$$\begin{aligned} Q(\phi^\delta - \phi^{\delta'}, \phi^\delta - \phi^{\delta'}) &= \mathcal{O}(\delta + \delta') \|\alpha\|_{W^0} \|\phi^\delta - \phi^{\delta'}\|_{W^1} \\ &= \mathcal{O}(\delta + \delta') \cdot \|\alpha\|^2_{W^0} \\ &\to 0 \end{aligned}$$

as $\delta, \delta' \to 0$.

We conclude that $\phi^\delta$ converges in the topology of $\widetilde{\mathcal{D}}^{p,q}$ as $\delta \to 0$ and (10.8.3.4) shows that the limit is $\phi$. Consequently $\phi^\delta \to \phi$ in $W_0^{p,q}$ and we are done. □

In the next section we shall develop the Main Theorem into some useful results about the original $\overline{\partial}$-Neumann problem.

## 10.9 Solution of the $\overline{\partial}$-Neumann Problem

**Capsule:** Our ultimate goal has been to solve the $\overline{\partial}$-Neumann problem. This powerful result was achieved by J. J. Kohn in 1963/1964. It has been intensely studied ever since.

Our ultimate goal is to understand existence and regularity for the $\overline{\partial}$-Neumann problem. We begin with some remarks about the operator $F$. We assume throughout this section that the basic estimate holds on $\Omega$.

**Proposition 10.9.1** *Let $\alpha, \phi, U, \rho_1, \rho$ be as in the Main Theorem. Let $\alpha\big|_U \in W_s^{p,q}(U \cap \overline{\Omega})$. Then $\rho\phi \in W_{s+k}^{p,q}(U \cap \overline{\Omega})$ where $k$ either 1 or 2 according to whether $U \cap \partial\Omega \neq \emptyset$ or $U \cap \partial\Omega = \emptyset$.*

**Proof** Let $\rho_0$ be a smooth function with support in $U$ such that $\rho_0 \equiv 1$ on $\operatorname{supp} \rho_1$. Let $\{\beta_n\}, \{\gamma_n\}$ be sequences of smooth forms such that

$$g\operatorname{supp} \beta_n \subseteq \operatorname{supp} \rho_0,$$

$$\operatorname{supp} \gamma_n \subseteq \operatorname{supp} (1 - \rho_0),$$

and

$$\beta_n \to \rho_0 \alpha \quad \text{in } W_s^{p,q},$$

$$\gamma_n \to (1 - \rho_0)\alpha \quad \text{in } W_0^{p,q}.$$

Then

$$\alpha_n = \beta_n + \gamma_n \to \alpha \quad \text{in } W_0^{p,q}$$

and

$$\rho_1 \alpha_n \to \rho_1 \alpha \quad \text{in } W_s^{p,q}.$$

Let $\phi_n \equiv F^{-1}\alpha_n$. Now $F^{-1}$ is bounded in the Sobolev topology so that $\phi_n \to F^{-1}\alpha = \phi$ in $W_0^{p,q}$. Now we apply the Main Theorem to obtain

$$\|\rho(\phi_n - \phi_m)\|_{W^{s+k}} \lesssim \|\rho_1(\alpha_n - \alpha_m)\|_{W^s} + \|\alpha_n - \alpha_m\|_{W^0}.$$

Therefore

$$\lim_{n\to\infty} \rho\phi_n \in W_{s+k}^{p,q},$$

that is, $\rho\phi \in W_{s+k}^{p,q}$. The desired estimate therefore holds. □

**Proposition 10.9.2** *If $F\phi = \alpha$ and $\alpha \in \bigwedge^{p,q}(\overline{\Omega})$ then $\phi \in \bigwedge^{p,q}(\overline{\Omega})$ and $\|\phi\|_{W^{s+1}}^2 \lesssim \|\alpha\|_{W^s}^2$ for every $s$.*

**Proof** This follows at once from the Main Theorem by taking $U \supseteq \overline{\Omega}$ and noticing that $\|\alpha\|_{W^0} \leq \|\alpha\|_{W^s}$. □

**Corollary 10.9.3** *If $F\phi = \alpha$ and $\alpha \in W_s^{p,q}$ then $\phi \in W_{s+1}^{p,q}$ and $\|\phi\|_{W^{s+1}} \lesssim \|\alpha\|_{W^s}$.*

**Proof** Immediate. □

**Corollary 10.9.4** *The operator $F^{-1}$ is a compact operator on $W_s^{p,q}$.*

**Proof** By the corollary we know that $F^{-1}$ is bounded from $W_s^{p,q}$ to $W_{s+1}^{p,q}$ so we apply Rellich's lemma to obtain the result. □

**Corollary 10.9.5** *The operator $F$ has discrete spectrum with no finite limit point and each eigenvalue occurs with finite multiplicity.*

**Proof** By the theory of compact operators (for which see, for instance [WID, RUW2]), we know that $F^{-1}$ has countable, compact spectrum with 0 as its only possible limit point. Also each eigenvalue has finite multiplicity. Since $\lambda$ is an eigenvalue for $F$ if and only if $\lambda^{-1}$ is an eigenvalue for $F^{-1}$ we have proved the corollary. □

**Proposition 10.9.6** *Let $U, \rho, \rho_1, \alpha$, and $k$ be as in Proposition 10.9.1. Suppose that $\rho_1\alpha \in W_s^{p,q}$ for some integer $s > 0$. If $\phi$ satisfies $(F - \lambda)\phi = \alpha$ for some constant $\lambda$ then $\rho\phi \in W_{s+k}^{p,q}(\Omega)$.*

**Proof** Consider the case $k = 1$. Set $\alpha' = \alpha + \lambda\phi$. Then $\phi$ satisfies the equation $F\phi = \alpha'$. Now $\alpha' \in W_0^{p,q}$ so that $\rho_1\phi \in W_1^{p,q}$ by Proposition 10.9.1. Let $\{\rho_k\}_{k=2}^s$ be

a sequence of smooth functions with $\rho_s = \rho$ and $\rho_j \equiv 1$ on $\operatorname{supp} \rho_{j+1}$. Inductively we may reason that

$$\rho_1 \alpha' \in W_1^{p,q} \Rightarrow \rho_2 \phi \in W_2^{p,q} \Rightarrow \rho_2 \alpha' \in W_2^{p,q} \ldots .$$

The result follows.

The case $k = 2$ is similar. □

**Corollary 10.9.7** *The operator $F - \lambda I$ is hypoelliptic.*

**Proof** Immediate from the proposition and Sobolev's theorem. □

**Corollary 10.9.8** *The eigenforms of $F$ are all smooth.*

**Proof** Obvious. □

**Proposition 10.9.9** *The space $W_0^{p,q}$ has a complete orthonormal basis of eigenforms for the operator $\Box_F$ which are smooth up to the boundary of $\Omega$. The eigenvalues are non-negative, with no finite accumulation point, and occur with finite multiplicity. Moreover, for each $s$,*

$$\|\phi\|_{W^{s+1}}^2 \lesssim \|\Box \phi\|_{W^s}^2 + \|\phi\|$$

*for all $\phi \in dom\,(F) \cap \bigwedge^{p,q}(\overline{\Omega})$.*

**Proof** Recall that $\Box_F = F - I$ is the restriction of $\Box$ to the domain of $F$. We know that $W_0^{p,q}$ has a complete orthonormal basis of eigenforms for $F^{-1}$ (just because it is a compact operator on a Hilbert space). Then the same holds for $F$ and thus for $F - I$. We also have that the eigenvalues are non-negative, with no finite accumulation point and with finite multiplicity.

The desired estimates follow by induction on $s$ and by the global regularity statement for $F$:

$$\begin{aligned} \|\phi\|_{W^1}^2 &\lesssim \|F\phi\|_{W^0}^2 \\ &\lesssim \|\Box\phi\|_{W^0}^2 + \|\phi\|_{W^0}^2; \end{aligned}$$

$$\begin{aligned} \|\phi\|_{W^2}^2 &\lesssim \|F\phi\|_{W^1}^2 \\ &\leq \|\Box\phi\|_{W^1}^2 + \|\phi\|_{W^1}^2 \\ &\leq \|\Box\phi\|_{W^1}^2 + \|\Box\phi\|_{W^0}^2 + \|\phi\|_{W^0}^2 \\ &\lesssim \|\Box\phi\|_{W^1}^2 + \|\phi\|_{W^0}^2; \end{aligned}$$

and so forth. □

**Proposition 10.9.10** *The space $W_0^{p,q}$ admits the strong orthogonal decomposition*

$$\begin{aligned} W_0^{p,q} &= \text{range}\,(\Box_F) \oplus \text{kernel}\,(\Box_F) \\ &= \overline{\partial}\vartheta\text{dom}\,(F) \oplus \vartheta\overline{\partial}\text{dom}\,(F) \oplus \text{kernel}\,(\Box_F). \end{aligned}$$

**Proof** First of all we need to show that range $(\Box_F)$ is closed. Set $\mathcal{H}^{p,q} \equiv$ kernel $(\Box_F)$. Then $\mathcal{H}^{p,q}$ is the eigenspace corresponding to the eigenvalue 0. The orthogonal complement of $\mathcal{H}^{p,q}$ is $\bigoplus_{|\lambda|>0} V_\lambda$, where $V_\lambda$ is the eigenspace corresponding to the eigenvalue $\lambda$. Then $\Box_F$ is bounded away from 0 on $(\mathcal{H}^{p,q})^\perp$ and it is one-to-one on this space. Thus $\Box_F$ restricted to the closure of the range of $\Box_F$ has a continuous inverse which we call $L$.

Let $\Box_F x_n \to y$. Then $L\Box_F x_n \to Ly$, that is, $x_n \to Ly$ and $\Box_F(Ly) = y$. Thus $y \in$ range $\Box_F$ and range $\Box_F$ is closed. Since range $\Box_F = (\mathcal{H}^{p,q})^\perp$, the first equality follows.

For the second equality, notice that $\overline{\partial}^2 = 0$ hence range $\overline{\partial} \perp$ range $\overline{\partial}^*$. Also $\overline{\partial}^* = \vartheta\big|_{\text{dom}\,\overline{\partial}^*}$ and the second equality follows as well. □

**Corollary 10.9.11** *The range of $\overline{\partial}$ on* dom $(\overline{\partial}) \cap W_0^{p,q-1}$ *is closed.*

**Proof** Since range $\overline{\partial} \perp$ kernel $(\overline{\partial}^*)$ and $\overline{\partial}^*(\vartheta\overline{\partial}\text{dom}\,(F) \oplus \mathcal{H}^{p,q}) = 0$, we may conclude that range $\overline{\partial} = \overline{\partial}\vartheta\text{dom}\,(F)$. □

We are engaged in setting up a Hodge theory for the $\overline{\partial}$ operator. For analogous material in the classical setting of the exterior differential operator $d$ we refer the reader to [CON].

Now we define the harmonic projector $H$ to be the orthogonal projection from $W_0^{p,q}$ onto $\mathcal{H}^{p,q}$. We use that operator in turn to define the $\overline{\partial}$-Neumann operator.

**Definition 10.9.12** The Neumann operator $N : W_0^{p,q} \to \text{dom}\,(F)$ is defined by

$$\begin{aligned} N\alpha &= 0 \qquad \text{if } \alpha \in \mathcal{H}^{p,q} \\ N\alpha &= \phi \qquad \text{if } \alpha \in \text{range}\,\Box_F \text{ and } \phi \text{ is the unique} \\ &\qquad\qquad \text{solution of } \Box_F\phi = \alpha \text{ with } \phi \perp \mathcal{H}^{p,q}. \end{aligned}$$

Then we extend $N$ to all of $W_0^{p,q}$ by linearity.

Notice that $N\alpha$ is the unique solution $\phi$ to the equations

$$\begin{aligned} H\phi &= 0 \\ \Box_F\phi &= \alpha - H\alpha. \end{aligned}$$

Finally, we obtain the solution to the $\overline{\partial}$-Neumann problem:

**Theorem 10.9.13**

**(1)** *The operator $N$ is compact.*
**(2)** *For any $\alpha \in W_0^{p,q}$ we have $\alpha = \overline{\partial}\vartheta N\alpha + \vartheta\overline{\partial}N\alpha + H\alpha$.*

**(3)** $NH = HN = 0, N\Box = \Box N = I - H$ *on* $\operatorname{dom} F$, $N\overline{\partial} = \overline{\partial}N$ *on* $\operatorname{dom}\overline{\partial}$, *and* $N\vartheta = \vartheta N$ *on* $\operatorname{dom}\overline{\partial}^*$.

**(4)** $N(\bigwedge^{p,q}(\overline{\Omega})) \subseteq \bigwedge^{p,q}(\overline{\Omega})$ *and for each s the inequality*

$$\|N\alpha\|_{W^{s+1}} \lesssim \|\alpha\|_{W^s}$$

*holds for all* $\alpha \in \bigwedge^{p,q}(\overline{\Omega})$.

**Proof** Statement **(2)** is part of Proposition 10.9.10. It is also immediate from the definitions that $NH = HN = 0$. Next, $N\Box = \Box N = I - H$ follows from part **(2)**.

If $\alpha \in \operatorname{dom}\overline{\partial}$ then, since $\overline{\partial}^2 = 0$ and $\overline{\partial}H = 0$, we have

$$\begin{aligned} N\overline{\partial}\alpha &= N\overline{\partial}(\vartheta\overline{\partial}N\alpha + \overline{\partial}\vartheta N\alpha + H\alpha) \\ &= N\overline{\partial}(\vartheta\overline{\partial}N\alpha) \\ &= N(\overline{\partial}\vartheta + \vartheta\overline{\partial})\overline{\partial}N\alpha \\ &= N\Box\overline{\partial}N\alpha \\ &= (I - H)\overline{\partial}N\alpha \\ &= \overline{\partial}N\alpha. \end{aligned}$$

The same reasoning applies to see that $\vartheta N = N\vartheta$.

The first statement of part **(4)** follows because $H\alpha$ is smooth whenever $\alpha$ is (by part **(2)**) and $\phi = N(\alpha - H\alpha)$ satisfies $\Box_F\phi = \alpha - N\alpha$. Hence this $\phi$ is smooth. Furthermore, we see that

$$\begin{aligned} \|N\alpha\|^2_{W^{s+1}} &\lesssim \|\Box N\alpha\|^2_{W^s} + \|N\alpha\|^2_{W^0} \\ &\leq \|\alpha\|^2_{W^s} + \|H\alpha\|^2_{W^s} + \|N\alpha\|^2_{W^0} \\ &\lesssim \|\alpha\|^2_{W^s} + \|H\alpha\|^2_{W^0} + \|N\alpha\|^2_{W^0} \\ &\lesssim \|\alpha\|^2_{W^s}. \end{aligned}$$

[We use here the fact that all norms on the finite dimensional space $\mathcal{H}^{p,q}$ are equivalent.] That proves the second statement of part **(4)**.

Finally, **(1)** follows from **(4)** and Rellich's lemma. □

Next we want to solve the inhomogeneous Cauchy–Riemann equation $\overline{\partial}\phi = \alpha$. Notice that there is no hope to solve this equation unless $\alpha \perp \operatorname{kernel}(\overline{\partial}^*)$ or equivalently $\overline{\partial}\alpha = 0$ and $H\alpha = 0$.

**Theorem 10.9.14** *Suppose that* $\alpha \in W_0^{p,q}$, $\overline{\partial}\alpha = 0$ *and* $H\alpha = 0$. *Then there exists a unique* $\phi \in W_0^{p,q-1}$ *such that* $\phi \perp$ *kernel* $(\overline{\partial})$ *and* $\overline{\partial}\phi = \alpha$. *If* $\alpha \in \bigwedge^{p,q}(\overline{\Omega})$ *then* $\phi \in \bigwedge^{p,q-1}(\overline{\Omega})$ *and*

$$\|\phi\|_{W^s} \lesssim \|\alpha\|_{W^s}\,.$$

**Proof** By the conditions on $\alpha$ we have that $\alpha = \overline{\partial}\vartheta N\alpha$. Thus we take $\phi = \vartheta N\alpha$ and $\phi \perp \text{kernel}\,(\overline{\partial})$ implies uniqueness. By part **(4)** of the last theorem, we know that $N\alpha \in \bigwedge^{p,q}$ if $\alpha \in \bigwedge^{p,q}$. Hence $\phi \in \bigwedge^{p,q-1}$ and

$$\begin{aligned}\|\phi\|_{W^s} &\equiv \|\vartheta N\alpha\|_{W^s}\\ &\lesssim \|N\alpha\|_{W^{s+1}}\\ &\leq \|\alpha\|_{W^s}\,.\end{aligned}$$

□

It is in fact the case that, on a domain in Euclidean space, the harmonic space $\mathcal{H}^{p,q}$ is zero dimensional. Thus the condition $H\alpha = 0$ is vacuous. There is no known elementary way to see this assertion. It follows from the Kodaira vanishing theorem (see [WEL, Ch. 6]), or from solving the $\overline{\partial}$-Neumann problem with certain weights. A third way to see the assertion appears in [FOK, Ch. 4]. We shall say no more about it here.

In fact it is possible to prove a stronger result than 10.9.14: if $\alpha$ has $W^s$ coefficients then $\phi$ has $W^{s+1/2}$ coefficients. This gain of order $1/2$ is sharp. We refer the reader to [FOK, p. 53] for details.

## Appendix to Section 10.8

## Uniform Estimates for $F^\delta$ and $\phi^\delta$.

Refer to Section 10.8—especially the proof of the Main Theorem—for terminology. The purpose of this appendix is to prove that the estimate

$$\|\rho\phi^\delta\|_{W^{s+1}} \lesssim \|\rho_1 F^\delta\phi^\delta\|_{W^s} + \|\,F^\delta\phi^\delta\|_{W^0} \tag{10.A.1}$$

holds with a constant which is independent of $\delta$. We begin as follows:

**PROPOSITION 10.A.2** We have

$$Q^\delta(A\phi, A\phi) - \text{Re}\,Q^\delta(\phi, A'A\phi) = \mathcal{O}(\|\nabla\phi\|^2_{W^{k-1}}),$$

where the constant in $\mathcal{O}$ is independent of $\delta$.

**Proof** The proof of 10.7.2 goes through, with $\overline{\partial}$ replaced by any first order differential operator $\widetilde{D}$ with constant coefficients, without any change. Thus

$$\langle\widetilde{D}A\phi, \widetilde{D}A\phi\rangle - \text{Re}\,\langle\widetilde{D}\phi, \widetilde{D}A'A\phi\rangle = \mathcal{O}(\|\nabla\phi\|^2_{W^{k-1}}).$$

Then

$$\begin{aligned} Q^\delta(A\phi, A\phi) - \operatorname{Re} Q^\delta(\phi, A'A\phi) &= Q(A\phi, A\phi) + \delta \sum_j (D^j A\phi, D^j A\phi) \\ &\quad - \operatorname{Re}\Big[ Q(\phi, A'A\phi) \\ &\quad + \delta \sum_j \langle D^j\phi, D^j A'A\phi\rangle \Big] \\ &= Q(A\phi, A\phi) - \operatorname{Re} Q(\phi, A'A\phi) \\ &\quad + \delta \sum_j \Big[ \langle D^j A\phi, D^j A\phi\rangle \\ &\quad - \langle D^j\phi, D^j A'A\phi\rangle \Big] \\ &= (1+\delta)\mathcal{O}\big(|||\nabla\phi|||^2_{W^{k-1}}\big) \\ &= \mathcal{O}\big(|||\nabla\phi|||^2_{W^{k-1}}\big), \end{aligned}$$

where the constant in $\mathcal{O}$ is independent of $\delta$. □

We need one more preliminary result:

**PROPOSITION 10.A.3** Let hypotheses be as in 10.8.1 with $\phi \in \widetilde{\mathcal{D}}^{p,q}_\delta$. Then

$$|||\nabla\rho_k\phi^\delta|||_{W^{(k-2)/2}} \lesssim |||\rho_1 F^\delta\phi^\delta|||^2_{W^{(k-2)/2}} + \|F^\delta\phi^\delta\|^2_{W^0}, \tag{10.A.3.1}$$

where the constants are independent of $\delta$.

**Proof** We follow the proof of 10.8.1 closely, checking that all constants that arise are independent of $\delta$. We induct on $k$. First let $k = 1$. Then

$$\begin{aligned} |||\nabla\rho_1\phi^\delta|||^2_{W^{-1/2}} &\overset{(10.7.4)}{\lesssim} E(\rho_1\phi^\delta) \\ &\overset{\text{(basic estimate)}}{\leq} Q(\rho_1\phi^\delta, \rho_1\phi^\delta) \\ &\leq Q^\delta(\rho_1\phi^\delta, \rho_1\phi^\delta) \\ &\overset{(10.A.2)}{=} \operatorname{Re} Q^\delta(\phi^\delta, \rho_1^2\phi^\delta) + \mathcal{O}\big(\|\phi^\delta\|^2_{W^0}\big) \\ &\overset{\text{(Friedrichs)}}{=} \operatorname{Re}\big(F^\delta\phi^\delta, \rho_1^2\phi^\delta\big) + \mathcal{O}\big(\|\phi^\delta\|^2_{W^0}\big) \\ &\leq \|F^\delta\phi^\delta\|_{W^0}\|\phi^\delta\|_{W^0} + \mathcal{O}\big(\|\phi^\delta\|^2_{W^0}\big). \end{aligned}$$

Notice that $T^\delta \equiv (\mathcal{F}^\delta)^{-1}$ is bounded on $L^2$ uniformly in $\delta$. Indeed, by the Friedrichs theorem we have

$$\begin{aligned}\|T^\delta\alpha\|^2_{W^0} &\le Q^\delta(T^\delta\alpha, T^\delta\alpha)\\ &= \langle\alpha, T^\delta\alpha\rangle\\ &\le \|\alpha\|_{W^0}\|T^\delta\alpha\|_{W^0}.\end{aligned}$$

It follows that $\|T^\delta\|_{\mathrm{op}} \le 1$. Therefore

$$\begin{aligned}\|\mathcal{F}^\delta\phi^\delta\|_{W^0}\|\phi^\delta\|_{W^0} + \mathcal{O}\big(\|\phi^\delta\|^2_{W^0}\big) &\le \|\mathcal{F}^\delta\phi^\delta\|^2_{W^0} + \mathcal{O}\big(\|\mathcal{F}^\delta\phi^\delta\|^2_{W^0}\big)\\ &\lesssim \|\mathcal{F}^\delta\phi^\delta\|^2_{W^0}\\ &\lesssim |\!|\!|\rho_1\mathcal{F}^\delta\phi^\delta|\!|\!|^2_{W^{-1/2}} + \|\mathcal{F}^\delta\phi^\delta\|^2_{W^0}.\end{aligned}$$

This proves the case $k = 1$.

Next we have for the tangential Bessel potential $\Lambda \equiv \Lambda_t^{(k-1)/2}$ that

$$|\!|\!|\nabla\rho_k\phi^\delta|\!|\!|_{W^{(k-2)/2}} \le |\!|\!|\nabla\rho_1\Lambda\rho_k\phi^\delta|\!|\!|^2_{W^{-1/2}} + |\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}. \tag{10.A.3.2}$$

Setting $\rho_1\Lambda\rho_k \equiv A$ we now calculate that

$$\begin{aligned}|\!|\!|\nabla A\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{-1/2}} &\le Q^\delta(A\rho_{k-1}\phi^\delta, A\rho_{k-1}\phi^\delta)\\ &= \mathrm{Re}\, Q^\delta(\rho_{k-1}\phi^\delta, A'A\rho_{k-1}\phi^\delta) + \mathcal{O}\big(|\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}\big)\\ &= \mathrm{Re}\, Q^\delta(\phi^\delta, A'A\rho_{k-1}\phi^\delta) + \mathcal{O}\big(|\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}\big)\\ &= \mathrm{Re}\, \langle\mathcal{F}^\delta\phi^\delta, A'A\rho_{k-1}\phi^\delta\rangle + \mathcal{O}\big(|\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}\big)\\ &= \mathrm{Re}\, \langle A\mathcal{F}^\delta\phi^\delta, A\rho_{k-1}\phi^\delta\rangle + \mathcal{O}\big(|\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}\big)\\ &\le |\!|\!|A\mathcal{F}^\delta\phi^\delta|\!|\!|_{W^{-1/2}}\|A\rho_{k-1}\phi^\delta\|_{W^{1/2}} + \mathcal{O}\big(|\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}\big)\\ &\le |\!|\!|\rho_1\mathcal{F}^\delta\phi^\delta|\!|\!|_{W^{(k-1)/2-1/2}}\|\rho_k\phi^\delta\|_{W^{(k-1)/2+1/2}}\\ &\quad + \mathcal{O}\big(|\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}\big)\\ &\le \frac{1}{\epsilon}|\!|\!|\rho_1\mathcal{F}^\delta\phi^\delta|\!|\!|_{W^{(k-2)/2}} + \epsilon|\!|\!|\nabla\rho_k\phi^\delta|\!|\!|^2_{W^{(k-2)/2}}\\ &\quad + \mathcal{O}\big(|\!|\!|\nabla\rho_{k-1}\phi^\delta|\!|\!|^2_{W^{(k-3)/2}}\big).\end{aligned}$$

Substituting into (10.A.3.2.) and using induction we get

$$|\!|\!|\nabla\rho_k\phi^\delta|\!|\!|^2_{W^{(k-2)/2}} \lesssim |\!|\!|\rho_1\mathcal{F}^\delta\phi^\delta|\!|\!|^2_{W^{(k-2)/2}} + \|\mathcal{F}^\delta\phi^\delta\|^2_{W^0},$$

where the constants are independent of $\delta$. That completes the proof of (10.A.3.1). □

Now proving the inequality (10.A.1) is straightforward.

## Exercises

**1.** *Distributions.* Let $D = C_c^\infty(\mathbb{R}^N)$. If $K \subset\subset \mathbb{R}^N$ and $\alpha$ is a multi-index then we let $\rho_{K,\alpha}$ on $D$ be the seminorm

$$\rho_{K,\alpha}(f) = \sup_K \left| \left( \frac{\partial}{\partial x} \right)^\alpha f \right| .$$

A sequence $\{\phi_j\}_{j=1}^\infty \subseteq D$ is said to converge to a limit function $\phi \in D$ if all $\phi_j$ are supported in a common compact set and $\lim_{j\to\infty} \rho_{K,\alpha}(\phi_j - \phi) = 0$ for every $K, \alpha$.

The space of *distributions* is the dual space $D'$ of $D$. Put in other words, a linear functional $\varphi$ on $D$ is a distribution if there are a positive integer $k$ and a positive constant $C$ so that

$$|\varphi(f)| \le C \cdot \sum_{|\alpha|\le k} \sup |\partial^\alpha f| .$$

Prove each of the following statements:

**(a)** If $\beta$ is a multi-index, then the differential operator $f \mapsto (\partial/\partial x)^\beta f(0)$ is a distribution.
**(b)** If $g \in L^1_{\text{loc}}(\mathbb{R}^N)$ then the operator $f \mapsto \int f g dV$ is a distribution.
**(c)** If $\mu$ is a finite Borel measure on $\mathbb{R}^N$ then $f \mapsto \int f d\mu$ is a distribution.
**(d)** In general, $T \in D'$ will not depend only on semi-norms on a fixed compact set.

**2.** *The Sobolev spaces by way of the Fourier transform.* For $f \in L^1(\mathbb{R}^N)$, let $\widehat{f}(\xi) = \int f(t) e^{i\xi t} dt$. Verify each of the following assertions:

**(a)** If $f, g \in L^1(\mathbb{R}^N)$ then $\widehat{f * g}(\xi) = \hat{f} \cdot \hat{g}(\xi)$.
**(b)** If $f \in L^1(\mathbb{R}^N)$, let $\tilde{f}(x) = f(-x)$. Then $\hat{\tilde{f}}(\xi) = \tilde{\hat{f}}(\xi)$ and $\bar{\hat{f}} = \bar{\hat{f}}(\xi)$.
**(c)** If $f \in D(\mathbb{R}^N)$, then $|\hat{f}(\xi)| \le C \cdot (1 + |\xi|)^{-N-1}$. (*Hint:* Integrate by parts.) Therefore $\hat{f} \in L^1(\mathbb{R}^N)$.
**(d)** If $f \in D(\mathbb{R}^N)$ then $f(0) = c_N \int \hat{f}(\xi) d\xi$.
**(e)** Apply **(d)** to $f(x) = g * \bar{\tilde{g}}$, any $g \in D$. Then

$$\int |g(t)|^2 dt = c_N \int |\hat{g}(\xi)|^2 d\xi .$$

**(f)** Extend **(e)** to all of $g \in L^2(\mathbb{R}^N)$.
(g) If $f \in D, \alpha$ is a multi-index, then

$$\left( \left( \frac{\partial}{\partial x} \right)^\alpha f \right)\hat{}\,(\xi) = (-i\xi)^\alpha \hat{f}(\xi).$$

(h) If $r > 0$ then we define the Sobolev space

$$W^r(\mathbb{R}^N) = \left\{ f \in L^2(\mathbb{R}^N) : \int |\hat{f}(\xi)|^2 (1 + |\xi|^2)^{r/2} dV(\xi) \right.$$
$$\left. \equiv \|f\|_{W^r}^2 < \infty \right\}.$$

Let $r \in \{0, 1, \ldots, \}$. Then $f \in W^r$ if and only if for all multi-indices $\alpha$ with $|\alpha| \leq r$, we have that $(\partial/\partial x)^\alpha f$ exists weakly and is in $L^2$.

**3.** This problem is needed in order to make the next one rigorous. Let $\mathcal{S}(\mathbb{R}^N)$, the *Schwartz space*, consist of those $f \in C^\infty(\mathbb{R}^N)$ satisfying

$$\rho_{\alpha,\beta}(f) = \sup_x \left| x^\alpha \left( \frac{\partial}{\partial x} \right)^\beta f \right| < \infty$$

for all multi-indices $\alpha, \beta$. Prove each of the following statements:

**(a)** The seminorms $\rho_{\alpha,\beta}$ make $\mathcal{S}$ into a Frechet space. Its dual $\mathcal{S}'$ is called the *space of Schwartz distributions*. Write down the condition describing the continuity of a Schwartz distribution.

**(b)** If $f \in \mathcal{S}$ then $((ix)^\alpha f)\hat{} = (\partial/\partial\xi)^\alpha \hat{f}(\xi)$ and $((\partial/\partial x)^\alpha f)\hat{} = (-i\xi)^\alpha \hat{f}$ for any multi-index $\alpha$. Use this fact, together with Exercise 2(d) to conclude that $\hat{}$ maps $\mathcal{S}$ to $\mathcal{S}$ both injectively and surjectively.

**(c)** If $\mu \in \mathcal{S}'$ then define $\hat{\mu}$ by $\hat{\mu}(f) = \mu(\hat{f})$ for all $f \in \mathcal{S}$. What property of the Fourier transform can you derive from Exercise 2 that justifies this definition?

**(d)** If $\mu \in \mathcal{S}'$ and $f \in \mathcal{S}$, then define $\mu * f$ to be the function given by $(\mu * f)(x) = \mu(\tau_x \tilde{f})$, where $\tau_x(g) \equiv g(t - x)$ and $\tilde{}$ is as in Exercise 2. What property of convolution of functions motivates this definition?

**(e)** If $\mu \in \mathcal{S}'$ and $f \in \mathcal{S}$, then what can you say about $\widehat{\mu * f}$?

**4.** Discover the fundamental solution for the Laplacian using the Fourier transform:

**(a)** If $\delta > 0, f \in \mathcal{S}(\mathbb{R}^N)$, then let $\alpha_\delta f(x) = f(\delta x)$.

**(b)** $\widehat{\alpha_\delta f}(x) = \delta^{-N} \hat{f}(x/\delta) \equiv \alpha^\delta \hat{f}(x)$.

**(c)** If $\rho$ is a rotation of $\mathbb{R}^N$ then $\widehat{f \circ \rho}(\xi) = \hat{f} \circ \rho(\xi)$, all $f \in \mathcal{S}$.

**(d)** If $f \in D$ then $\widehat{\Delta f} = -|\xi|^2 \hat{f}$.

**(e)** Assuming that $f = \Delta f * \Gamma$ for some function $\Gamma$ to be determined, then $\hat{\Gamma}(\xi) = -1/|\xi|^2$.

**(f)** The function $\Gamma$ is rotationally invariant and $\alpha_\delta \Gamma(x) = \delta^{-N+2} \Gamma(x)$.

**(g)** $\Gamma(x) = c_N |x|^{-N+2}$.

**5.** *The Sobolev embedding theorem on domains.* Let $\Omega \subseteq \mathbb{R}^N$. Let

$$W^s(\Omega) = \left\{ f \in L^2(\Omega) : \left( \frac{\partial}{\partial x} \right)^\alpha f \in L^2(\Omega), \text{ all } |\alpha| \leq s \right\}.$$

Here, as usual, the derivatives are interpreted in the weak sense. We wish to consider a Sobolev embedding theorem for these Sobolev spaces. However geometric conditions must be imposed on $\partial\Omega$.

Optimal conditions on $\partial\Omega$ are rather complicated (see [ADA]). But suppose that $\Omega$ is bounded and that $\partial\Omega$ is $C^k, k > s$. Prove that the usual embedding theorem then prevails for our new Sobolev spaces. [*Hint:* After a partition of unity, the problem is a local one. But then a coordinate change maps $\partial\Omega$ locally to $\{(x_1, \ldots, x_N) : x_N = 0\}$. So it is enough to do the problem on the Euclidean upper half space in a neighborhood of 0, say on $\tilde{\Omega} = B \cap \{x \in \mathbb{R}^N : x_N > 0\}$. Show that the functions that extend to be $C^k$ on $B$ are dense in $W^s(\tilde{\Omega})$—use convolution and dilation. Then compute an *a priori* estimate for $C^\infty$ functions.] See R. Adams [ADA] for more on this matter.

**6.** Refer to Exercise 5. Let $\Omega \subset\subset \mathbb{R}^N$ have $C^1$ boundary. Let $L^p_k(\Omega) = \{f \in L^p(\Omega) : (\partial/\partial x)^\alpha f \in L^p, \text{all } |\alpha| \leq k\}$. Here, as usual, derivatives are all in the weak sense. If $1/q = 1/p - k/N$ and $1 \leq p < N/k$, then $L^p_k(\Omega) \subseteq L^q(\Omega)$ with continuous inclusion. If $p > N/k$, then $L^p_k(\Omega) \subseteq \Lambda^{\text{loc}}_\alpha(\Omega)$, where $\alpha = k - N/p$. See R. Adams [ADA] for a detailed treatment of these assertions. Items **(a)**–**(d)** below show that these hypotheses are necessary.

- **(a)** Let $\Omega = \{|x| < 1\} \subseteq \mathbb{R}^N$. Let $f(x) = |x|^{(-N/p)+1} / (\log(1/|x|))^{N+1}$. Assume that $N > p$. Then $f \in L^p_1(\Omega)$, but $f \notin L^q(\Omega)$ if $q > Np/(N-p)$, $1 \leq p < \infty$.
- **(b)** Generalize (a) to the case $k > 1$. What about $p = \infty$?
- **(c)** Some form of boundary regularity is necessary. Let $\Omega = \{(x_1, x_2) \in \mathbb{R}^2 : |x_1| < |x_2|^{1+\epsilon} < 1, x_2 > 0\}$, and let $f(x) = |x|^{-\epsilon/4}$. Then $f \in L^{2+\epsilon/4}_1(\Omega)$, but $f \notin L^\infty(\Omega)$.
- **(d)** Generalize (c) to $\mathbb{R}^N$.

**7.** Verify that the Laplacian, acting on $C^\infty_c(\mathbb{R}^N)$, is self-adjoint. Verify that $-\Delta$ is a positive operator in the sense that $\langle -\Delta\phi, \phi\rangle \geq 0$ for all $\phi \in C^\infty_c$. Use integration by parts on this inner product to prove the *a priori* estimate

$$\|\nabla\phi\|_{L^2} \leq C \cdot \|\phi\|^2_{L^2} + C \cdot \|\Delta\phi\|^2_{L^2}.$$

Conclude that

$$\|f\|^2_{W^1} \leq C\|f\|^2_{L^2} + C\|\Delta f\|^2_{L^2},$$

all $f \in W^2$.

**8.** Consider Sobolev spaces on a fixed compact set $K$. Prove that if $s_1 > s_2$, then bounded sets in $W^{s_1}$ have compact closure in $W^{s_2}$. Equivalently, if $\{f_j\} \subseteq W^{s_2}$ is bounded in the $W^{s_1}$ topology then it has a convergent subsequence in $W^{s_2}$. Formulate and prove an analogous result for Lipschitz spaces.

**9.** Refer to Exercises 3 and 4. On elements of the Schwartz space, the Laplacian $\Delta$ is equivalent to the Fourier multiplier $-|\xi|^2$. More generally, if $L(D) =$

$\sum_\alpha a_\alpha \cdot (\partial/\partial x)^\alpha$ is a *linear, constant coefficient,* partial differential operator, then $L(D)$ corresponds to the Fourier multiplier $\sum_\alpha a_\alpha(-i\xi)^\alpha$. It is a powerful idea of Mikhlin–Calderón–Zygmund–Kohn–Nirenberg–Hörmander and others to reverse this process and concentrate on the Fourier multiplier as the principal tool. We allow the multiplier to depend both on $x$ (the space variable) and on $\xi$ (the Fourier transform or frequency variable).

Define a smooth function $p(x, \xi)$ on $\mathbb{R}^N \times (\mathbb{R}^N \setminus \{0\})$ to be a *symbol* of class $m \in \mathbb{Z}$ (denoted $p \in S^m$) if $p$ is compactly supported in $x$ and

$$\left|(\partial/\partial x)^\alpha \, (\partial/\partial \xi)^\beta \, p(x, \xi)\right| \leq C_{\alpha,\beta} \cdot (1 + |\xi|)^{m-|\beta|}$$

for any multi-indices $\alpha, \beta$. To such a symbol $p$ is associated a *pseudodifferential operator*

$$T_p\phi(x) = \int_{\mathbb{R}^N} p(x, \xi) e^{-ix\cdot\xi} \hat{\phi}(\xi) d\xi \, .$$

The principal results of an elementary calculus of pseudodifferential operators amount to proving that calculus on the operator level (with the operators $T_p$) is equivalent (up to acceptable error terms) with calculus on the symbol level (with the symbols $p$). More precisely, we have

(i) $(T_p)^* = T_{\bar{p}} +$ (negligible error)
(ii) $T_p \circ T_q = T_{pq} +$ (negligible error)

By the phrase "negligible error" we mean here an operator with symbol that lies in $S^r$ with $r < p$ (case (i)) or $r < p + q$ (case (ii)). This calculus enables one to construct, in a natural fashion, parametrices (approximate inverses) for a large collection of partial differential operators (including elliptic ones). It is hard work to make all this precise, and we do not attempt to do so (good references are M. E. Taylor [TAY] and F. Treves [TRE]). Instead, we include a few simple properties of the symbolic calculus, which the reader may verify, and give an application to the regularity theory of the Laplacian. It is of historical interest to note that Kohn and Nirenberg [KOHN2] introduced the first modern calculus of pseudodifferential operators; they were motivated in their work by considerations connected with the $\overline{\partial}$-problem. See G. B. Folland and J. J. Kohn [FOK, Ch. 1, 3] and S. G. Krantz [KRA5, Ch. 3] for more on these matters.

**(a)** Let $p \in S^m$, some $m \in \mathbb{Z}$. Then, for $x$ fixed, the operator $\phi \mapsto T_p\phi(x)$ is a distribution.

**(b)** If $L = \sum_\alpha a_\alpha(x) \cdot (\partial/\partial x)^\alpha$ is a linear partial differential operator of degree $m$ with coefficients $a_\alpha \in C_c^\infty(\mathbb{R}^N)$, then $L$ is a pseudodifferential operator. What is its symbol? In what symbol class $S^m$ does it lie?

**(c)** Let $p(x, \xi) \in C_c^\infty(\mathbb{R}^N \times \mathbb{R}^N)$. Then $p \in \cap_{m=-\infty}^\infty S^m$.

**(d)** If $L_1$ and $L_2$ are partial differential operators of degrees $m_1$ and $m_2$ respectively (as in **(b)** above) then $[L_1, L_2]$ is of degree not exceeding

$m_1 + m_2 - 1$. A similar result holds for $T_{p_1}, T_{p_2}$ in $S^{m_1}, S^{m_2}$ but this is more difficult to prove. It also holds that $T_{p_1} \circ T_{p_2} \in S^{m_1+m_2}$.

**(e)** Prove that if $p \in S^m$ then $T_p : W^s \to W^{s-m}$, every $s \in \mathbb{Z}$.

**(f)** Fix a $\phi \in C_c^\infty$. Define $L = \phi(x) \cdot \Delta$. The symbol of $L$ is $p(x, \xi) = -\phi(x)|\xi|^2$. Assume that $\phi(x) = 1$ when $|x| \leq 1, \phi(x) = 0$ when $|x| \geq 2$. Now write

$$\begin{aligned} p(x, \xi) &= \phi(\xi) \cdot p(x, \xi) + (1 - \phi(\xi)) \cdot p(x, \xi) \\ &= p_1(x, \xi) + p_2(x, \xi). \end{aligned}$$

Then $p_1 \in S^m$, every $m$, while $p_2 \in S^2$. Define $q(x, \xi) = \phi(2x)(1 - \phi(\xi/2))/p(x, \xi)$. Then the operator $T_q$ is a parametrix for $L$ in the following sense. First write

$$T_p T_q = T_{(p_1+p_2)} T_q = T_{p_1} T_q + T_{p_2} T_q \equiv E + F.$$

Now $E \in S^m$ for every $m$. Also, by (ii) above,

$$\begin{aligned} T_{p_2} T_q &= T_{p_2 q} + \mathcal{E} = T_{\phi(2x)(1-\phi(\xi/2))} + \mathcal{E} \\ &= T_{\phi(2x)} + \left( - T_{\phi(2x)\phi(\xi/2)} \right) + \mathcal{E} \\ &= \phi(2x) \cdot I + \mathcal{E} \end{aligned}$$

Here $\mathcal{E}$ is a "negligible error" whose meaning changes from line to line and that we shall assume, without proof, is in $S^{-1}$. Therefore we have computed that

$$L T_q = T_p T_q = \phi(2x) \cdot I + \mathcal{E},$$

where $\mathcal{E} \in S^{-1}$. Use a similar argument to show that $T_q L = \phi(2x) \cdot I + \mathcal{E}'$, some $\mathcal{E}' \in S^{-1}$. Note that $q \in S^{-2}$.

**(g)** We apply the results of (f) as follows. Let $\eta \in C_c^\infty$ with $\operatorname{supp} \eta \subseteq \{x : \phi(2x) = 1\}$. Assume that $L\eta = \psi$. We wish to estimate the Sobolev norms of $\eta$ in terms of the Sobolev norms of $\psi$. So we write

$$\begin{aligned} \|\eta\|_{W^2} &= \|\phi(2x)\eta\|_{W^2} \\ &= \|\phi(2x) I \eta\|_{W^2} \\ &= \|T_q L \eta - \mathcal{E}' \eta\|_{W^2} \\ &\leq \|T_q \psi\|_{W^2} + \|\mathcal{E}' \eta\|_{W^2} \\ &\leq C\|\psi\|_{W^0} + C\|\eta\|_{W^1} \\ &\leq C\|\psi\|_{W^0} + C\|\eta\|_{W^0} + (1/2)\|\eta\|_{W^2}. \end{aligned}$$

Thus

$$\|\phi(2x)\eta\|_{W^2} \leq C\|\psi\|_{W^0} + C\|\eta\|_{W^0}.$$

Iterate this device to prove

$$\|\eta\|_{W^{s+2}} \leq C\{\|\psi\|_{W^s} + \|\eta\|_{W^0}\}, \text{ any } s \in \mathbb{Z}.$$

Use this result to say something about regularity for $\Delta$.

**(h)** Give two reasons why the naive theory of pseudodifferential operators presented above is useless for solving the $\overline{\partial}$-problem on a pseudoconvex domain.

**10.** J. J. Kohn [KOH3] used the method of weight functions to prove the following result: Let $\Omega \subset\subset \mathbb{C}^n$ be a pseudoconvex domain with $C^\infty$ boundary. Let $f$ be a $\overline{\partial}$-closed $(0, 1)$ form on $\Omega$ with coefficients that extend to be $C^\infty$ on $\bar{\Omega}$. Then there is a $u \in C^\infty(\bar{\Omega})$ such that $\overline{\partial}u = f$. Kohn's theorem is difficult and we shall not prove it. Let us take Kohn's result for granted.

Suppose that $P \in \partial\Omega$ and $U$ is a neighborhood of $P$. Let $g : U \cap \Omega \to \mathbb{C}$ be holomorphic and satisfy (a) $g$ extends to be $C^\infty$ on $\overline{U \cap \Omega}$, (b) $g(P) = 0$, (c) (Image $g$) omits a sector in $\mathbb{C}$. Use Kohn's result to construct a (global) peaking function for $A(\Omega)$ at $P$. Will your peaking function be $C^\infty$ on $\bar{\Omega}$? Will it satisfy a Lipschitz condition?

**11.** Let $\mu$ be the ordinary rotationally invariant length measure on the unit circle in the complex plane. Compute explicitly a solution to $\overline{\partial}u = \mu\, d\overline{z}$. Is $u$ in any Sobolev class near the unit circle? How does $u$ behave away from the circle?

**12.** *The Lewy–Pincuk reflection principle* (H. Lewy [LEW2], S. Pincuk [PIN]). The proof of Schwarz's reflection principle is so simple that it obscures the essential geometry of the reflection. The following generalization is harder to prove but is, in the end, more enlightening.

**Theorem** Let $\Omega_1, \Omega_2 \subset\subset \mathbb{C}^n$ be strongly pseudoconvex domains with real analytic boundaries. Let $P \in \partial\Omega_1$. Let $U \subseteq \mathbb{C}^n$ be a neighborhood of $P$. Suppose that $f : \overline{U \cap \Omega_1} \to \mathbb{C}^n$ is $C^1$ and univalent and that $f|_{U\cap\Omega_1}$ is holomorphic. Finally, suppose that $f(U \cap \partial\Omega_1) \subseteq \partial\Omega_2$. Then $f$ extends holomorphically to a neighborhood of $U \cap \partial\Omega_1$.

Complete the following outline to obtain a proof of this theorem.

**(a)** We may as well suppose from the outset that $\partial\Omega_1 \cap U$ is strongly convex.

**(b)** If $w, \zeta \in \mathbb{C}^n$, then define the complex line

$$\ell^w(\zeta) = \{\zeta + w\lambda : \lambda \in \mathbb{C}\}.$$

Let

$$\gamma^w(\zeta) = \ell^w(\zeta) \cap \partial\Omega_1.$$

For $\epsilon > 0$, $\rho_j$ a defining function for $\Omega_j$, define

$$\Omega^-(\epsilon, w, \zeta) = \{z \in \ell^w(\zeta) : -\epsilon < \rho_1(z) < 0\},$$
$$\Omega^+(\epsilon, w, \zeta) = \{z \in \ell^w(\zeta) : 0 < \rho_1(z) < \epsilon\},$$
$$\Omega(\epsilon, w, \zeta) = \{z \in \ell^w(\zeta) : -\epsilon < \rho_1(z) < \epsilon\}.$$

The idea is to choose $w, \zeta^0$ so that $\gamma^w(\zeta)$ is a closed real analytic curve in $\partial\Omega_1$ for all $\zeta$ sufficiently close to $\zeta^0$. *Suppose hereafter that $\zeta_0 = P$ and $w$ is chosen in this way.* Then we extend $f$ in the classical fashion, across each $\gamma^w(\zeta)$, from $\Omega^-(\epsilon, w, \zeta)$ to $\Omega^+(\epsilon, w, \zeta)$.

**(c)** There is a neighborhood $V \subseteq \mathbb{C}^n$ of $\zeta^0$ such that the following holds: If $\epsilon > 0$ then there is a $\delta > 0$ such that for all $\zeta \in V$ and $g$ holomorphic on $\Omega^-(\epsilon, w, \zeta)$ and continuous up to $\gamma^w(\zeta)$, there is a function $G$ conjugate holomorphic on $\Omega^+(\epsilon, w, \zeta)$ and continuous up to $\gamma^w(\zeta)$ such that $G|_{\gamma^w(\zeta)} = g|_{\gamma^w(\zeta)}$.

**(d)** Let $\phi$ be real analytic on $\partial\Omega_1$. There is an $\epsilon > 0$ and a neighborhood $W \subseteq \mathbb{C}^n$ of $\zeta^0$ such that, for any $\zeta \in W$, it holds that $\phi|_{\gamma^w(\zeta)}$ extends holomorphically to $\Omega(\epsilon, w, \zeta)$.

**(e)** If $z \in U \cap \partial\Omega_1$, then $\rho_2 \circ f(z) = 0$. Asssume that $\partial\rho_1/\partial z_n \neq 0$ on $U \cap \partial\Omega_1$. Define $T_j = (\partial\rho_1/\partial z_n)\partial/\partial z_j - (\partial\rho_1/\partial z_j)\partial/\partial z_n$ on $\partial\Omega_1, j = 1, \ldots, n-1$. Then $T_j(\rho_2 \circ f) = 0$ on $U \cap \partial\Omega_1$.

**(f)** The result of (e) may be rewritten as

$$\sum_{k=1}^{n} \frac{\partial\rho_2}{\partial w_k}(f(z))\, T_j f_k(z) = 0, \quad j = 1, \ldots, n-1,\ z \in \partial\Omega_1.$$

**(g)** Let $w$ be the complex coordinate on $\Omega_2$. Writing $w_k = u_k + iv_k, k = 1, \ldots, n$, and assuming as we may that $P = f(P) = 0$, we may expand

$$\rho_2(u + iv) = \rho_2(w) = \sum_{\alpha,\beta} a_{\alpha\beta} u^\alpha v^\beta$$

in a neighborhood of $f(P)$.

**(h)** Write $u_k = (w_k + \bar{w}_k)/2$, $v_k = (w_k - \bar{w}_k)/(2i)$ and note that this transforms the expansion in **(g)** to one of the form

$$P(w, \bar{w}) = \sum_{\alpha,\beta} a'_{\alpha\beta} w^\alpha \bar{w}^\beta .$$

Now formally define

$$P(w, \mu) = \sum a'_{\alpha\beta} w^\alpha \mu^\beta, \quad w \in \mathbb{C}^n, \quad \mu \in \mathbb{C}^n .$$

**(i)** Rewrite the $n$ equations in **(e)** as

$$P\left(f(z),\overline{f(z)}\right) = 0$$

$$\sum_{k=1}^{n} \frac{\partial P}{\partial w_k}\left(f(z),\overline{f(z)}\right) T_j f_k(z), \quad j = 1,\dots,n-1\,.$$

**(j)** Consider the matrix $\mathcal{C} = (c_{ij})_{i,j=1}^{n}$ given by the condition

$$c_{ij} = \sum_{k=1}^{n} \frac{\partial^2 P}{\partial w_k \partial \mu_j}(0,0) T_i f_k(0), \quad i = 1,\dots,n-1\,,$$

$$c_{nj} = \frac{\partial P}{\partial \mu_j}(0,0)\,.$$

Notice that if $f$ is nonconstant (which we may as well assume), then $\det J_{\mathbb{C}} f(0) \neq 0$. As a result, the vectors $T_i f(0), i = 1,\dots,n-1$ span the complex tangent space to $\partial\Omega_1$ at 0. Since the Levi form is positive definite, the first $(n-1)$ rows of $\mathcal{C}$ are linearly independent at 0. Since the last row is a normal vector, $\mathcal{C}$ has rank $n$.

**(k)** Parts **(i)** and **(j)** imply that we may use the implicit function theorem to solve for $\overline{f_1(z)},\dots,\overline{f_n(z)}$ as holomorphic functions of the $n^2$ variables $f_1(z),\dots,f_n(z)$ and $T_j f_k, j = 1,\dots,n-1, k = 1,\dots n$. More precisely, write

$$Tf(z) = (T_1 f_1(z),\dots,T_1 f_n(z),\dots,T_{n-1} f_1(z),\dots,T_{n-1} f_n(z))\,.$$

Then we can find holomorphic function $h_i$ of $n^2$ complex variables such that

$$\tilde{f}_1(z) = h_1\,(f(z), Tf(z))$$

$$\vdots$$

$$\tilde{f}_n(z) = h_n\,(f(z), Tf(z))$$

whenever $(f(z), Tf(z))$ lies in a neighborhood $N$ of $(0, Tf(0))$.

**(l)** If $w$ is fixed appropriately and $\zeta$ is sufficiently near $P$, then we may apply **(d)** to find $\epsilon > 0$ and a neighborhood $W$ of 0 in $\mathbb{C}^n$ such that for $\zeta \in W$, the $T_i f_k|_{\partial\Omega_1}$ extends holomorphically to $\Omega^-(\epsilon, w, \zeta)$. We may also assume that $h_k(f, Tf)$ extends holomorphically to $\Omega^-(\epsilon, w, \zeta)$.

**(m)** By (c), the $h_k(f, Tf)$ extend conjugate holomorphically to $\Omega^+(\epsilon, w, \zeta)$, some $\epsilon > 0$.

**(n)** Shrinking $W$ if necessary, relation (k) is satisfied on $\partial\Omega_1 \cap U$ so that $f_k|_{\gamma^w(\zeta)}$ can be holomorphically continued to $\Omega(\delta, w, \zeta), k = 1, \ldots, n$, some $\delta > 0$.

**(o)** Now conclude that $f$ holomorphically continues to some small neighborhood of $P$. Complete the proof in an obvious way. (See also Exercise 20 at the end of Chapter 5.)

**(p)** Notice that the injectivity of $f$ was needed only to see that $J_{\mathbb{C}} f(0)$ has rank $n$. This turns out to be true even if $f$ is not univalent. Try to prove this.

**13.** Apply the result of the preceding exercise to prove the following:

**Theorem** If $\Omega_1, \Omega_2$ are strongly pseudoconvex domains with real analytic boundaries and $f : \Omega_1 \to \Omega_2$ is biholomorphic, then $f$ extends holomorphically to a neighborhood of $\bar{\Omega}_1$.

(*Hint:* Assume Fefferman's theorem, which states that $f$ and $f^{-1}$ extend smoothly to $\bar{\Omega}_1, \bar{\Omega}_2$ respectively.)

**14.** Let $\omega \subset\subset \mathbb{C}^n$ have $C^k$ boundary. Let $f \in C^{k+1}(\Omega)$. If $(\partial/\partial x)^\alpha f$ is bounded for every $|\alpha| \leq k + 1$, then $f$ has a $C^k$ extension to $\bar{\Omega}$. Can you obtain the same result with a weaker hypothesis than the existence and boundedness of the $(k + 1)^{\text{st}}$ derivatives?

**15.** *The tangential Cauchy–Riemann equations and the Bochner extension phenomenon.* Let $\Omega \subset\subset \mathbb{C}^n$ be a domain, $n > 1$, with $C^1$ boundary and defining function $\rho$. Let $f$ be a $C^1$ function on $\partial\Omega$. We say that $f$ satisfies the *tangential Cauchy–Riemann equations* if there is a $C^1$ extension $F$ of $f$ to a neighborhood of $\partial\Omega$ such that $\overline{\partial} F \wedge \overline{\partial}\rho = 0$ on $\partial\Omega$. [Note: this *ad hoc* definition is equivalent with several other more intrinsic ones—see G. B. Folland and J. J. Kohn [FOK, Ch. 5] and A. Boggess [BOG].]

**(a)** Show that the preceding definition is unambiguous (i.e. is independent of the choice of $F$).

**(b)** Show that $f$ satisfies the tangential Cauchy–Riemann equations if and only if, on $\partial\Omega$,

$$\sum_j \alpha_j \frac{\partial F}{\partial \overline{z}_j} = 0 \quad \text{whenever} \quad \sum_j \alpha_j \frac{\partial \rho}{\partial \overline{z}_j} = 0 \,.$$

Here $F$ is *any* $C^1$ extension of $f$.

**(c)** Let $\Omega$ now have $C^4$ boundary, and suppose that $f \in C^4(\partial\Omega)$ satisfies the tangential Cauchy–Riemann equations. Let $F$ be a $C^4$ extension of $f$ to $\bar{\Omega}$. Then there exist $\alpha \in C^3(\bar{\Omega}), \beta \in C^2_{(0,1)}(\bar{\Omega})$ with $\overline{\partial} F = \alpha \overline{\partial}\rho + \beta \cdot \rho$ on $\bar{\Omega}$. (Why is $\beta$ only $C^2$?)

**(d)** Notice that $\overline{\partial}(F - \alpha\rho) = \gamma \cdot \rho$, some $\gamma \in C^2_{(0,1)}(\bar{\Omega})$.

**(e)** Compute that $0 = \overline{\partial}(\rho\gamma) = \overline{\partial}\rho \wedge \gamma + \rho\overline{\partial}\gamma$. Therefore $\overline{\partial}\rho \wedge \gamma = 0$ on $\partial\Omega$. So $\gamma = \mu\overline{\partial}\rho + \rho\eta$, some $\mu \in C^2(\bar{\Omega}), \eta \in C^1_{(0,1)}(\bar{\Omega})$.

**(f)** Define $\Phi = f - \alpha\rho - \mu\rho^2/2$. Then $\Phi$ agrees with $f$ on $\partial\Omega$, $\Phi \in C^2(\bar{\Omega})$ and $\overline{\partial}\Phi = O(\rho^2)$.

**Theorem** Let $\Omega \subset\subset \mathbb{C}^n$ have $C^4$ boundary. Assume that ${}^c\bar{\Omega}$ is connected. Let $f \in C^4(\partial\Omega)$ satisfy the tangential Cauchy–Riemann equations. Then there is a function $\hat{f} \in C^1(\overline{\Omega})$ that is holomorphic on $\Omega$ and such that $\hat{f}\Big|_{\partial\Omega} = f$.

**(g)** To prove the theorem, choose $\Phi$ as in (f). Let

$$\omega(z) = \begin{cases} \overline{\partial}\Phi(z) \text{ if } z \in \bar{\Omega} \\ 0 \qquad\quad \text{if } z \notin \bar{\Omega}\,. \end{cases}$$

Then $\omega \in C^1_c(\mathbb{C}^n)$ and $\overline{\partial}\omega = 0$. So there is a function $u \in C^1_c(\mathbb{C}^n)$ with $\overline{\partial}u = \omega$, $u \equiv 0$ off $\Omega$. (Careful—what hypothesis do we need to use here?) Now set $\hat{f} = \Phi - u$.

**(h)** The hypotheses of this theorem may be considerably weakened (see [BOG]). Even in the proof just outlined (due to [HOR1]), the hypothesis may be weakened from $C^4$ to $C^2$. There are local versions of the extension theorem, but they are harder to prove. Many problems remain open.

**(i)** Let $\Omega \subseteq \mathbb{C}^2$ be given by $\Omega = B((0,0),6) \setminus \{\bar{B}((0,0),5) \setminus B((4,0),1)\}$. Show that, on $\Omega$, $\overline{\partial}$ on functions does not have closed range. (*Hint:* Construct a peaking function at $P = (3,0)$.) Details of this construction may be found in [KRA1, Ch. 3].

**(j)** Let $\Omega \subseteq \mathbb{C}^n$ be any domain. Let $f$ be a $\overline{\partial}$-closed $(0,1)$ form on $\Omega$ with $C^k$ coefficients. Assume that $u \in L^2_{\text{loc}}(\Omega)$ is a function on $\Omega$ that satisfies $\overline{\partial}u = f$ in the weak sense. Prove that $u$ is $C^k$ on $\Omega$. (*Hint:* Exploit the trivial solution to the $\overline{\partial}$ equation for compactly supported forms together with the fact proved in Section 4.6 that there are no "fake" holomorphic functions.)

**16.** Let

$$\Omega = \left\{(z_1, z_2) : |z_1|^2 + 2e^{-1/|z_2|^2} < 1\right\}.$$

Show that for *no* $\alpha > 0$ can it always hold that

$$\|u\|_{\text{Lip}_\alpha} \leq C\|f\|_{L^\infty}$$

for solutions to $\overline{\partial}u = f$, $f$ a $\overline{\partial}$-closed $(0,1)$ form (see [RAN2]).

**17.** *Failure of local hypoellipticity for the $\overline{\partial}$-problem* Let $\Omega \subseteq \mathbb{C}^2$ be a smoothly bounded domain. Let $P = 0 \in \partial\Omega$. Suppose that for some $r > 0$ we have $B(P,r) \cap \partial\Omega = \{z \in B(P,r) : \text{Re}\, z_1 = 0\}$. So the analytic disc $\{P + (0, z_2) : |z_2| < r\}$ is contained in $\partial\Omega$. Let $\phi \in C^\infty_c(B(P,r))$ satisfy $\phi \equiv 1$ on $B(P, r/2)$. Let $f = \overline{\partial}\phi/z_1$. Then $f$ is a $\overline{\partial}$-closed $(0,1)$ form on $\Omega$ and $f \in C^\infty_{(0,1)}(\bar{B}(P, r/3) \cap \bar{\Omega})$. Prove that it is impossible to find a $u$ on $\Omega$ such that $\overline{\partial}u = f$ and $u \in$

$C^\infty(\bar{B}(P, r/4) \cap \bar{\Omega})$. This example illustrates the phenomenon of "propagation of singularities" along analytic discs in $\partial\Omega$. That is, the function $u$ can have singularities on the disc even at point where $f$ does not. See [KOH3] for more on these matters.

**18.** Consider the operator on $\mathbb{R}^2$ given by

$$P = \frac{\partial}{\partial x} + ix\frac{\partial}{\partial y}.$$

This is perhaps the simplest example of a partial differential operator with the property that there is an $f \in C^\infty(\mathbb{R}^2)$ such that $Pu = f$ is not locally solvable. More precisely, let $D_j = \bar{D}(j^{-1}, 4^{-j}), j \in \mathbb{N}$. Let $f \in C_c^\infty(\mathbb{R})$ satisfy (i) $f(\cdot, y)$ is even for each $y$, (ii) $\text{supp} f \cap \{x \geq 0\} = \cup_j D_j$, and (iii) $\int_{D_j} f\, dxdy > 0$ for all $j$. Complete the following outline to show that there is no $u \in C^1(\mathbb{R}^2)$ such that $Pu = f$.

**(a)** Suppose that $u$ is a $C^1$ solution; write $u = u_e + u_o$, where $u_e$ is even in $x$ and $u_o$ is odd in $x$. The even part of the equation $Pu = f$ is

$$\frac{\partial u_o}{\partial x} + ix\frac{\partial u_o}{\partial y} = f. \qquad (*)$$

It follows that $u(0, y)$ is constant hence we may suppose that $u_o(0, y) = 0$.

**(b)** Restrict attention to $\{x \geq 0\}$. Let $s = x^2/2, \partial/\partial s = (1/x)(\partial/\partial x)$. Then $(*)$ becomes

$$\frac{\partial u_o}{\partial s} + i\frac{\partial u_o}{\partial y} = \tfrac{1}{\sqrt{2s}} f\left(\sqrt{2s}, y\right), \; s > 0,$$
$$u_o = 0, \qquad s = 0.$$

**(c)** The transformed function $u_o(s, y) = u_o(x, y)$ satisfies the Cauchy–Riemann equations in the complex variable $s + iy$ off of $\cup_j D_j$.

**(d)** It follows that $u \equiv 0$ on $\mathbb{R}^2 \setminus \cup_j D_j$.

**(e)** Apply Stokes's theorem to $u_o dy - ixu_o dx$ on $D_j$ to obtain a contradiction.

**(f)** The same proof shows that $Pu = f$ has no $C^1$ solution in any neighborhood of 0.

The fact that $f$ is not real analytic is crucial here. Indeed, the Cauchy–Kowalewski theory guarantees local real analytic solutions (infinitely many of them) in case $f$ is real analytic. Conversely, for some operators of this type, P. Greiner, J. J. Kohn, and E. M. Stein [GRNKS] have shown that real analyticity of $f$ is necessary.

The first example of an operator that is not locally solvable is due to H. Lewy [LEW1]. It was discovered in the context of extension phenomena for holomorphic functions on $B \subseteq \mathbb{C}^2$. Indeed, the operator is obtained by

restricting the $\overline{\partial}$ operator to $B$ (see G. B. Folland and E. M. Stein [FOS1]). Exercise 25 below has more on this point of view.

More recently, H. Jacobowitz and F. Treves [JAC] have shown that partial differential operators that are not locally solvable are, in a precise sense, generic.

**19.** Let $\Omega = \{z \in \mathbb{C}^3 : \rho(z) = \operatorname{Re} z_3 + |z_1^2 - z_3 z_2|^2 + |z_2^2|^2 < 0\}$. Verify that if $\phi : D \to \mathbb{C}^3$ is univalent, $\phi(0) = 0$, and $\phi'(\zeta) \neq 0$ for all $\zeta \in D$ (so that $\phi$ is an immersed, non-singular, analytic disc) then $|\rho(\phi(\zeta))| \geq C|\zeta|^4$ for all $\zeta \in D$. On the other hand, for $t$ small and real, let $\gamma_t(\zeta) = (\zeta, \zeta^2/(it), it)$ for all $\zeta \in D$. Then $\rho(\gamma_t(\zeta)) = O(|\zeta|^8/t^4)$. So the order of tangency of discs *near* 0 is different from the order of tangency of discs *at* 0. This example, due to D'Angelo, is important in the study of subelliptic estimates for the $\overline{\partial}$-problem.

**20.** Let $\Omega \subset\subset \mathbb{C}^n$ and $A(\Omega)$ as usual. Call $P \in \partial\Omega$ a *local peak point* if there is a neighborhood $U$ of $P$ and a function $f \in A(\Omega \cap U)$ such that $f(P) = 1$ and $|f(z)| < 1$ for $z \in \bar{\Omega} \cap U \setminus \{P\}$. Suppose that the equation $\overline{\partial} u = f$ has a bounded solution on $\Omega$ whenever $f$ is a bounded, $C^1$, $\overline{\partial}$-closed $(0, 1)$ form on $\Omega$. Prove that every local peak point in $\partial\Omega$ is a (global) peak point for $A(\Omega)$. N. Sibony [SIB2] has used this idea to construct a smooth pseudoconvex $\Omega \subset\subset \mathbb{C}^3$ for which the equation $\overline{\partial} u = f$ does not have any bounded solution for some smooth $\overline{\partial}$-closed $(0, 1)$ form $f$ with bounded coefficients.

**21.** It is well known and easy to compute that the operator $\overline{\partial}$ is elliptic on relatively compact subsets of any domain $\Omega \subseteq \mathbb{C}^n$ (see Folland and Kohn [FOK, pp. 10–11] or S. G. Krantz [KRA5, Ch. 7] for details). Let us use an idea of N. Kerzman [KER1] that exploits the interior ellipticity in an elementary manner to give very good estimates for $\overline{\partial}$ in the interior.

Let $\Omega_0 \subset\subset \Omega_1 \subset\subset \Omega \subset\subset \mathbb{C}^n \cong \mathbb{R}^{2n}$. Let $\Gamma(x)$ be the fundamental solution to the Laplacian on $\mathbb{R}^{2n}$ that was constructed in Proposition 1.3.2.

**(a)** Let $u \in C^1(\Omega)$. Let $\phi \in C_c^\infty(\Omega)$ be identically equal to 1 on $\Omega_1$. Then, for $z \in \Omega_0$, we have

$$u(z) = \int_\Omega \Gamma(z - w)\Delta(\phi(w)u(w))dV(w).$$

**(b)** Write $\Delta = 4\sum_j (\partial^2/\partial w_j \partial \bar{w}_j)$. Integrate by parts to obtain

$$u(z) = 4\sum_{j=1}^n \int_\Omega \Gamma_{w_j}(z - w)\frac{\partial}{\partial \bar{w}_j}(\phi(w)u(w))dV(w).$$

**(c)** Obtain the estimate

$$|u(z)| \leq C\int_\Omega |u(w)|dV(w) + C\int_\Omega \frac{1}{|z - w|^{2n-1}}|\overline{\partial} u(w)|dV(w)$$

for $z \in \Omega_0$. You must use the fact that $\nabla\phi \equiv 0$ on $\Omega_1$.

**(d)** Notice that if $K \subset\subset \mathbb{C}^n$ then

$$\int_K \frac{1}{|\zeta|^{2n-1}} dV(\zeta) \leq C(K).$$

**(d)** Conclude that, for $1 \leq p \leq \infty$,

$$\|u\|_{L^p(\Omega_0)} \leq C(\Omega_0, \Omega_1) \left\{ \|u\|_{L^1(\Omega)} + \|\overline{\partial} u\|_{L^p(\Omega)} \right\}.$$

**(e)** The fact that we are estimating only in the interior was used decisively in two places. What are they?

**(f)** When $p = \infty$, refine the estimate to

$$\|u\|_{L^\infty(\Omega_0)} \leq C(\Omega_0, \Omega_1, \epsilon) \left\{ \|u\|_{L^1(\Omega)} + \|\overline{\partial} u\|_{L^{2n+\epsilon}(\Omega)} \right\}, \text{ any } \epsilon > 0.$$

**22.** Modify the argument in the last exercise to show that for any $0 \leq k \in \mathbb{Z}$ it holds that

$$\|u\|_{C^{k+1}(\Omega_0} \leq C(\Omega_0, \Omega_1) \left\{ \|u\|_{L^1(\Omega)} + \|\overline{\partial} u\|_{C^k(\Omega)} \right\}.$$

**23.** Use the outline that follows to prove this proposition about almost analytic extensions:

**PROPOSITION** Let $k \in \mathbb{N}$. Let $M \subseteq \mathbb{C}^n$ be a totally real submanifold, that is, $T_P M$ has no non-trivial complex subspace for any $P \in M$ (see also Exercise 1 at the end of Chapter 8). Let $K \subseteq M$ be a compact set. Let $u \in C^k(M)$. Then there is a function $U \in C^1(\mathbb{C}^n)$ such that $U|_K = u$ and $\overline{\partial} U(z) = \mathcal{O}(\text{dist}\,(z, K)^{k-1})$. In a sense, $U$ is an "almost analytic" extension of $U$. For the proof, we proceed by induction on $k$. The case $k = 1$ is trivial. Now let $k = 2$.

**(a)** If each $x \in K$ has a neighborhood $W_x$ such that $u|_{K \cap W_x}$ has the indicated extension, then passing to a finite subcover and using a partition of unity gives the full result. So it suffices to treat the problem locally.

**(b)** Choose $x \in K$. Choose a neighborhood $W_x$ of $x$ and real functions $\rho_1, \ldots, \rho_m$ such that

$$M \cap U = \{z \in U; \rho_1(z) = \cdots = \rho_m(z) = 0\}.$$

Notice that, since $M$ is totally real, $m \geq n$.

**(c)** Shrinking $W_x$ if necessary, there is a sequence $1 \leq j_1, \ldots, j_n \leq m$ such that

$$\left\{ \left( \frac{\partial \rho_{j_i}}{\partial \overline{z}_1}(z), \ldots \frac{\partial \rho_{j_i}}{\partial \overline{z}_n}(z) \right) \right\}_{i=1}^n$$

is a basis for $\mathbb{C}^n$ over the field $\mathbb{C}$ at each point $z \in W_x$.

**(d)** Shrinking $W_x$ if necessary, we see that $u|_{W_x \cap M}$ has a $C^k$ extension to $\mathbb{C}^n$ (with only bounded increase in norm). Denote this extension also by $u$.

**(e)** Denote the functions $\rho_{j_i}$ chosen in (c) by $\rho_1, \ldots, \rho_n$. Define functions $h_i(z)$ on $W_x$ by

$$\left(\frac{\partial u}{\partial \overline{z}_1}(z), \ldots, \frac{\partial u}{\partial \overline{z}_n}(z)\right) = \sum_{i=1}^{n} h_i(z) \left(\frac{\partial \rho_i}{\partial \overline{z}_1}(z), \ldots, \frac{\partial \rho_i}{\partial \overline{z}_n}(z)\right).$$

Then $h_i \in C^{k-1}(W_x), i = 1, \ldots, n$. In the standard several complex variables shorthand,

$$\overline{\partial} u = \sum_{i=1}^{n} h_i \overline{\partial} \rho_i.$$

**(f)** Define $u_1 = u - \sum_{i=1}^{n} h_i \rho_i$. Then $u_1 = u$ on $M \cap W_x$ and $\overline{\partial} u_1 = -\sum_{i=1}^{n} (\overline{\partial} h_i) \rho_i = \mathcal{O}(\text{dist}\,(\cdot, M))$. This completes the proof when $k = 2$.

**(g)** For $k = 3$, notice that (in the notation of part (e)) $\{\overline{\partial} \rho_j \wedge \overline{\partial} \rho_i\}_{i<j}$ are linearly independent. Define $h_{ij}(z)$ by the equations

$$\overline{\partial} h_i = \sum_{j=1}^{n} h_{ij}(z) \overline{\partial} \rho_j(z).$$

Then

$$0 = \overline{\partial}^2 u = \sum_{i<j} [h_{ij} - h_{ji}] \overline{\partial} \rho_j \wedge \overline{\partial} \rho_i.$$

**(g)** Conclude that $h_{ij} = h_{ji}$ for $i < j$. Define

$$u_2(z) = u_1(z) + \frac{1}{2!} \sum_{i,j} h_{ij} \rho_i \rho_j.$$

Then $u_2 = u$ on $M$ and

$$\overline{\partial} u_2(z) = \frac{1}{2!} \sum_{i,j} \overline{\partial} h_{ij}(z) \rho_i(z) \rho_j(z) = \mathcal{O}\left((\text{dist}\,(z, M))^2\right).$$

**(h)** Iterate this procedure to construct $u_{k-1}$, each $k$.

**24.** Refer to Exercise 10. Let $\Omega \subset\subset \mathbb{C}^n$ be a smoothly bounded pseudoconvex domain. J. J. Kohn [KOH3] has proved that for each $k \in \mathbb{N}$ there is an $m(k) \in \mathbb{N}, m(k) \geq k$, and a constant $C(k)$ such that the following property holds: For

each $\overline{\partial}$-closed $(0,1)$ form $f$ with $C^{m(k)}$ coefficients on $\bar{\Omega}$ there is a $u_k \in C^k(\bar{\Omega})$ such that $\overline{\partial} u_k = f$. Also

$$\|u_k\|_{C^k(\bar{U})} \leq C(k)\|f\|_{C^{m(k)}(\bar{U})}.$$

It should be understood that $m(k), C(k)$ depend on the geometry on $\partial\Omega$ but *not* on $f$. It is not well understood how $m(k)$ depends on $k$. In general, $m(k) >> k$. The proof of Kohn's theorem is too difficult to consider here so we take it for granted.

Prove that, with $\Omega \subset\subset \mathbb{C}^n$ as above and $f$ a $\overline{\partial}$-closed $(0,1)$ form with $C^\infty$ coefficients on $\bar{\Omega}$, there is a $u \in C^\infty(\bar{\Omega})$ such that $\overline{\partial} u = f$. Details of this last assertion appear in J. J. Kohn [KOH3]. Here are some hints:

**(a)** It is enough to construct $v_j \in C^j, j = 1, 2, \ldots,$ such that $\overline{\partial} v_j = f$ and $\|v_j - v_{j+1}\|_{C^j} < 2^{-j}$.

**(b)** Assume inductively that $v_1, \ldots, v_j$ have been constructed. Let $\tilde{v}_{j+1}$ be a solution in $C^{j+1}$ to $\overline{\partial} u = f$. Imitate the proof of Theorem 10.4.2 to find a holomorphic $V$ on $\Omega$ such that $\|(v_j - \tilde{v}_{j+1}) - V\|_{C^j} < 2^{-j}$.

**(c)** Define $v_{j+1} = \tilde{v}_{j+1} + V$.

**25.** Refer to Exercise 18. Provide the details for the following argument which provides an elegant way to think about the local solvability question. Let $(x, y, t)$ be coordinates in $\mathbb{R}^3$. Identify $(x, y)$ with $z = x + iy$. Define

$$L = \frac{\partial}{\partial z} + i\overline{z}\frac{\partial}{\partial t}.$$

Refer to Exercise 10 at the end of Chapter 2. Let $\mathcal{U} \subseteq \mathbb{C}^2$ be given by $\mathcal{U} = \{z \in \mathbb{C}^2 : \operatorname{Im} z_1 > |z_2|^2\}$. Map $\mathbb{R}^3$ to $\partial\mathcal{U}$ by

$$(z, t) \mapsto (t + i|z|^2, z).$$

Then $L$ on $\mathbb{R}^3$ is transformed to

$$L' = \frac{\partial}{\partial z_2} + 2i\overline{z}_2\frac{\partial}{\partial z_1}.$$

Notice that $L'$ is the formal adjoint to

$$A = \frac{\partial}{\partial \overline{z}_2} - 2iz_2\frac{\partial}{\partial \overline{z}_1}.$$

If $f \in L^2(\partial\mathcal{U})$, let $Sf$ be its Szegö integral to $\mathcal{U}$. We have

**Theorem** The equation $L'u = f$ has a solution near $P \in \partial\mathcal{U}$ if and only if $Sf$ is real analytic near $P$.

The "if" part of this theorem exceeds the scope of this exercise. For the "only if" part, proceed as follows. Let $\phi \in C_c^\infty(\partial\mathcal{U})$ satisfy $\phi = 1$ near $P$. Prove that

$$S(f - A^*(\phi u)) = Sf.$$

But $f - A^*(\phi u)$ equals 0 near $P$ hence $S(f - A^*(\phi u))$ extends holomorphically past $P$.

This argument may be found in [KRA5, Ch. 7]. Compare this exercise with Exercise 6 above.

# Chapter 11
# A Few Miscellaneous Topics

## 11.1 Ideas of Christ/Geller

One of the beautiful features of harmonic analysis on $\mathbb{R}^N$ is that we have nice characterizations of the real-variable $H^p$ spaces in terms of the Riesz transforms—see [STW1] and [FES], for instance. It is not at all obvious how to develop an analogous theory on the ball in $\mathbb{C}^n$, for instance. But Christ and Geller [CHSTG] made great strides in that direction. We should like to describe some of their ideas here.

First we note that, in the pioneering work of Stein and Weiss [STW1], the real variable Hardy spaces were *defined* in terms of conjugate systems of harmonic functions. That definition is shown to be equivalent to one in terms of the Riesz transforms. Just as, on the real line, a function $f$ is in $H^1$ if and only if $f \in L^1$ and the Hilbert transform $Hf$ is in $L^1$, so it is that a function on $\mathbb{R}^N$ is in real-variable $H^1$ if and only if $f \in L^1$ and the Riesz transforms $R_j f \in L^1$, where

$$R_j f(x) \equiv \text{P.V.} \int_{\mathbb{R}^N} f(t) \cdot \frac{x_j - t_j}{|x-t|^{N+1}}\, dt\,.$$

A natural question to ask at this point is which other collections of $N$ singular integrals will characterize $H^1$. In an important work [UCH], Akihito Uchiyama gave us the answer. Consider the convolution operators

$$K_j f(x) \equiv \left(\theta_j(\xi/|\xi|)\widehat{f}(\xi)\right)^\vee(x)\,,$$

with $\theta_j$ in $C^\infty$ of the unit sphere and satisfying the condition

$$\text{rank} \begin{pmatrix} \theta_1(\xi) & \cdots & \theta_m(\xi) \\ \theta_1(-\xi) & \cdots & \theta_m(-\xi) \end{pmatrix} \equiv 2 \quad \text{for } |\xi| = 1\,.$$

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Such a system of singular integrals $K_j$ will in fact characterize $H^1_{\text{Re}}(\mathbb{R}^N)$. [It may be noted that the Riesz transform kernels obviously satisfy this condition.]

Now Christ and Geller [CHSTG] built on the ideas of Uchiyama to prove the following result. It is the first of its kind along the lines of characterizing $H^p$ on a homogeneous group using singular integrals. Prior to the work in [CHSTG], the primary ways to approach Hardy spaces on homogeneous groups were by way of maximal functions or the atomic theory—see [COG1, Ch. 1].

Now the result of [CHSTG] is this: A collection $K_j$ of singular integral operators on the Heisenberg group characterizes (in the sense discussed in the preceding paragraph) $H^1$ of the Heisenberg group if, for each $v \in \mathbb{R}^m$, there are singular integral operators $L_j$ such that $\sum K_j * L_j = I$ and $\sum v_j L_j = 0$. We cannot provide here all the technical details of the Christ/Geller argument. Let us just say that this is a foundational result for harmonic analysis on the unit ball in $\mathbb{C}^n$, and much work remains to be done in developing this point of view.

The entire idea of developing a working harmonic analysis on the Heisenberg group was explored by D. Geller in [GEL1, GEL2, GEL3]. See also [GELS].

## 11.2 Square Functions

The idea of square functions goes back at least to work of Lusin. We learn in a basic function theory course that, if $U \subseteq \mathbb{C}$ is a bounded domain and $f : U \to \mathbb{C}$ is a univalent holomorphic mapping, then $f(U)$ is a domain and

$$\int_U |f'(\zeta)|^2 \, dA(\zeta) = \int_{f(U)} 1 \, dA(\zeta) = \text{ area } (f(U)) \, .$$

This is simply the change of variables formula, for $|f'(\zeta)|^2$ is nothing other than the Jacobian determinant of the mapping.

Thanks to work of E. M. Stein and others, the idea of the square function has been generalized to the harmonic analysis of several variables. And, more recently, Stein [STE1, Ch. 3] has proved results about the square function on strongly pseudoconvex domains. Following that work, Krantz and Li [KRL1, KRL3] established results about the square function on finite type domains in $\mathbb{C}^n$.

A detailed history of the square function is given in [STE5]. Its history began with a theorem of Kaczmarz and Zygmund in 1926 (see [KAC] and [ZYG2]). It slowly evolved, through work of Calderón, Zygmund, and others, to a more modern form that we shall emphasize here.

We begin our work on the upper half space $\mathbb{R}^{N+1}_+ \equiv \{x = (x_1, x_2, \ldots, x_N, t) : t > 0\}$. If $x = (x_1, x_2, \ldots, x_N) \in \mathbb{R}^N = \partial \mathbb{R}^{N+1}_+$, then let

$$\Gamma(x) = \{(x, t) : |x| < t\} \, .$$

For a harmonic function $u$ on the upper half space $\mathbb{R}^{N+1}_+$ which is the Poisson integral of an integrable function $f$ on the boundary $\mathbb{R}^N$, and for $x \in \partial\mathbb{R}^{N+1}_+$, we define

$$[Au(x)]^2 = \int_{\Gamma(x)} |\nabla u|^2 y^{1-N}\, dxdy\,.$$

Notice that this is an $(N+1)$-dimensional integral. The presence of the factor $y^{1-N}$ is justified by dimensionality considerations (it trivializes to the zeroth power in the classical setting of the disc). It will be put into a more natural context when we formulate the area integral on a strongly pseudoconvex domain in the language of the Bergman metric.

Now the fundamental theorem about the operator $A$ is as follows:

**Theorem 11.2.1** *For $1 < p < \infty$,*

$$\|Au\|_{L^p} \cong \|f\|_{L^p}\,.$$

In this formulation, the result is due to Stein [STE3]. Stein's original rather complicated proof has been simplified (see, for instance [STE2, Ch. 2]). But we still cannot present the details here.

In the book [STE1, Ch. 3], Stein defined a version of the area integral for strongly pseudoconvex domains. We give a more general version of his idea below in the description of the work of Krantz and Li. Stein's main theorem is this:

**Theorem 11.2.2 (Stein)** *Let $\Omega$ be a smoothly bounded, strongly pseudoconvex domain in $\mathbb{C}^n$. Let $F$ be a holomorphic function on $\Omega$. Then, for almost every $\zeta \in \partial\Omega$, the following are equivalent:*

(1) *$F$ is admissibly bounded at $\zeta$;*
(2) *$F$ has an admissible limit at $\zeta$;*
(3)

$$\int_{\mathcal{A}_\alpha(\zeta)} |\nabla F|^2\, d\Omega(z) < \infty\,.$$

Here of course $\mathcal{A}_\alpha$ is the usual admissible approach region as we studied in Chapter 2.

Without going into all the details, we merely point out that Stein works in what he calls a *preferred metric*. With Fefferman's work of a few years later (see [FEF1]), he could have worked in the Bergman metric on a strongly pseudoconvex domain. Then $|\nabla F|$ is calculated in the metric, and $d\Omega$ is the volume element in the metric.

Stein's proof [STE1] is rather technical, and we cannot reproduce it here. In some sense the argument is a sophisticated application of Stokes's theorem.

In later work, Krantz and Li [KRL1, KRL2] were able to generalize what Stein did in [STE1]. For a fixed domain $\Omega$, $z \in \Omega$, and $\alpha > 0$, define

$$D_\alpha(z) = \{w \in \Omega : \pi(w) \in \beta_2(z, \alpha\delta(w))\}\,.$$

Here $\pi$ is orthogonal projection to the boundary and $\delta$ is distance to the boundary. Also let

$$d\lambda(z) = K(z,z)dV(z)\,,$$

where $K$ is the Bergman kernel and $dV$ is Euclidean volume measure. Now we define the *area integral* of a function $f$ on $\Omega$ to be

$$A_\alpha f(z) = \left(\int_{D_\alpha(z)} |\nabla f(w)|^2 \delta(w)^2\, d\lambda(w)\right)^{1/2}.$$

The maximal function of a given function $f$ on $\Omega$ is given by

$$f_\alpha^*(z) = \sup\{|f(w)| : w \in D_\alpha(a)\}\,.$$

If $\partial\Omega$ is smooth then we may take $U$ to be a tubular neighborhood of the boundary. And we may also choose an $\epsilon_0 > 0$ so that this tubular neighborhood has radius greater than $\epsilon_0$.

The radial maximal function is defined to be

$$f^+(z) = \sup\{|f(z - t\nu(z))| : 0 \le t \le \epsilon_0\}\,.$$

The Littlewood-Paley $g$-function is given by

$$g(f)(z) = \left[\int_0^{\epsilon_0} |\nabla f(z + t\nu(z))|^2 t\, dt\right]^{1/2}.$$

We let $H^p(\Omega)$ denote the Hardy space of $p$th-power integrable holomorphic functions on $\Omega$. Now we have the following theorem:

**Theorem 11.2.3 (Krantz–Li)** *Let $\Omega$ be a smoothly bounded domain which is either strongly pseudoconvex, of finite type in $\mathbb{C}^2$, or convex and of finite type in any dimension. Let $0 < p < \infty$ and let $\Omega$ be a regular domain in $\mathbb{C}^n$. Then the following are equivalent:*

**(1)** $f \in H^p(\Omega)$.
**(2)** $f^+ \in L^p(\partial\Omega)$.
**(3)** $f_\alpha^* \in L^p(\partial\Omega)$.
**(4)** $g(f) \in L^p(\partial\Omega)$.
**(5)** $A_\alpha(f) \in L^p(\partial\Omega)$.

Of course the ideas here originate in [PAW]. Their modern multivariable form was initiated in [STE1].

## 11.3 Ideas of Nagel/Stein and Di Biase

As mentioned earlier, the traditional wisdom has been that, in the context of harmonic functions on the disc, the optimal approach regions for boundary limits are the nontangential approach regions. Well established examples of Littlewood, Rudin, and others reinforce this notion. Thus it was a remarkable observation of Nagel and Stein [NAS1] that what is important about the shape of the approach region is *not* its shape (in the common geometric sense of the word "shape"), but rather the measures of its cross sections. That is, it turns out that what is important about the nontangential approach region

$$\Gamma_\alpha(P) = \{z \in D : |z - P| < \alpha(1 - |z|)\}$$

is that

> The one-dimensional measure of the set of points in $\Gamma_\alpha(P)$ that are distance $\delta$ from $P$ is about $\delta$. (11.3.1)

It is not difficult to see that one can construct a region having property (11.3.1) that is *not* nontangential. Yet, through such a region, there are boundary limits for $p^{\text{th}}$ power integrable harmonic functions. The technique of Nagel and Stein involves a new version of the Hardy–Littlewood maximal function.

Fausto Di Biase—see for instance [DIF] and [DIB]—was able to generalize the ideas of Nagel and Stein to the several complex variable setting. Thus he replaces admissible approach regions by regions that have cross sections with the same area as the cross sections of $\mathcal{A}_\alpha$. And then there are boundary limits for $H^p$ functions through these new approach regions. Di Biase's work is rather complex, involving harmonic analysis on trees.

## 11.4 $H^1$ and *BMO*

The Hilbert transform on the unit circle is defined by the formula

$$Hf(x) = P.V. \int_{-\pi}^{\pi} f(e^{i(x-t)}) \frac{1}{2} \cot(t/2)\, dt\,.$$

Here "P.V." denotes *principal value*, which means that we integrate over the deleted interval $[-\pi, -\epsilon] \cup [\epsilon, \pi]$ and let $\epsilon$ tend to 0. We must use this device because the integrand (particularly the function $\cot(t/2)$) is not Lebesgue integrable.

The Hilbert transform is one of the most important linear operators in all of analysis. It controls the convergence of Fourier series (see particularly the reference [KRA2, Ch. 2]). It also plays a significant role in complex function theory.

Specifically, let $f$ be an $L^1$ function on the unit circle $\mathbb{T}$. Let $F$ be its Poisson integral to the disc. Let $\widetilde{F}$ be the harmonic conjugate of $F$ that vanishes at the origin.

It can be shown, by an elementary argument, that $\widetilde{F}$ has a boundary function $\widetilde{f}$ (again see [KRA2]). The operator

$$\mathcal{H} : f \longmapsto \widetilde{f}$$

is a matter of considerable interest. It turns out to be the Hilbert transform.

Earlier in the book (see particularly Chapters 1 and 2) we studied the Hardy space $H^1$. It can be shown that we can think of $H^1$ as

$$H^1 = \{f \in L^1(\mathbb{T}) : \widetilde{f} \in L^1(\mathbb{T})\}. \tag{11.4.1}$$

Because $\mathcal{H}$ is not bounded on $L^1$ (simply calculate the Hilbert transform of the characteristic function $\chi_{[0,\pi/6]}$), it is natural to seek a substitute space for $L^1$ when studying singular integral operators. It turns out that $H^1$, as realized in equation (11.4.1), is the right substitute. And it can be shown that the Hilbert transform is bounded on $H^1$.

It is natural to want to know the dual of $H^1$. Of course the dual of $L^1$ is $L^\infty$, and that is a useful fact. The dual of $H^1$ is rather more subtle.

One of the seminal results of the early 1970s was C. Fefferman's theorem [FEF3] that the dual of the first Hardy class $H^1$ is *BMO*. Here, on a space of homogeneous type $(X, \mu)$, *BMO* is defined to be

$$BMO(X) = \{f \in L^1_{\text{loc}} : \sup_{B(x,r)} \frac{1}{\mu(B(x,r))} \int_{B(x,r)} |f(t) - f_{B(x,r)}|\, d\mu(t) < \infty\}.$$

Here $f_{B(x,r)}$ simply denotes the average of $f$ over the ball $B(x, r)$ with respect to the measure $\mu$. The space *BMO* was discovered by John and Nirenberg [JON] in the context of partial differential equations. It is fairly straightforward to show that the dual of $H^1$ lies in *BMO*. But Fefferman showed that they are equal.

It turns out, by a straightforward but definitely nontrivial calculation (see [FES]) that $\mathcal{H}$ is bounded on *BMO* and, given Fefferman's theorem, this is the most convenient way to prove that $\mathcal{H}$ is bounded on $H^1$. Of course $\mathcal{H}$ is *not* bounded on $L^\infty$ (simply calculate the Hilbert transform of the characteristic function $\chi_{[0,\pi/6]}$).

This key idea spearheaded a development of the real variable theory of Hardy spaces—see, for instance, [FES]. In turn, there developed an atomic theory of Hardy spaces (see [COG1, Ch. 1]), a maximal function theory of Hardy spaces (see [FES]), and several other approaches as well.

With a view that harmonic analysts are always seeking new venues in which to develop their ideas, it became natural to seek to prove Fefferman's theorem in the several complex variable context. The first paper in this direction is [COFW]. There the authors study, among many other topics, a version of the duality of $H^1$ and *BMO* on the unit ball in $\mathbb{C}^n$. To do so they exploit the atomic theory of Hardy spaces—see Section 11.6 below.

Later, the papers [KRL2, KRL5] and [DAF1, DAF2] treat the duality of $H^1$ and *BMO* on strongly pseudoconvex domains in $\mathbb{C}^n$ or domains of finite type in $\mathbb{C}^2$. The methodologies in these two sets of papers are quite different. In particular, the first of these papers relies on maximal functions, while the latter two use a version of the Calderón reproducing formula.

## 11.5 Factorization of Hardy Space Functions

The celebrated *Riesz factorization* of a Hardy space function (on the unit disc) says this: Every function $f \in H^1(D)$ can be written as $f = g_1 \cdot g_2$, where $g_1, g_2 \in H^2(D)$. It is known (see [RUS]) that such a factorization is not possible in the context of several complex variables.

Thus it was a real breakthrough when Coifman, Rochberg, and Weiss discovered a substitute for the Riesz factorization in several variables. What they proved is that, if $f$ is in the Hardy space $H^1(B)$, then $f$ can be written as

$$f = \sum_j g_1^j \cdot g_2^j ,$$

where $g_1^j, g_2^j$ are elements of the Hardy space $H^2(B)$. Their proof relies on a factorization of real variable Hardy spaces using the Riesz transforms.

In the paper [KRL3], Krantz and Li generalize this last result to strongly pseudoconvex domains, and also to factoring an $H^p$ function as an infinite sum of products of $H^{2p}$ functions, $0 < p < \infty$.

## 11.6 The Atomic Theory of Hardy Spaces

The atomic theory of Hardy spaces is an idea of C. Fefferman. In its simplest form it takes the following form. We say that $f \in H^1(\mathbb{R})$ if $f \in L^1$ and its Hilbert transform $\mathcal{H}f \in L^1$.

A function $a$ on the real line is called a *1-atom* if

**(a)** $a$ is supported in an interval $I$;
**(b)** $|a(x)| \leq 1/|I|$ for all $x$;
**(c)** $\int a(x)\, dx = 0$.

The fundamental theorem is then that, if $f \in H^1(\mathbb{R})$, then

$$f = \sum_j \alpha_j a_j ,$$

where the $a_j$ are atoms and $\sum_k |\alpha_j| < \infty$. The converse is true as well.

There is a corresponding atomic theory for $H^p$ spaces, $0 < p < \infty$, and also a theory on $\mathbb{R}^N$. More generally, there is a well developed theory of atomic Hardy spaces on a space of homogeneous type—see [COG2].

In the paper [KRL3], Krantz and Li define a *holomorphic atom* on the boundary of a domain in $\mathbb{C}^n$ to be the Szegő projection of a real atom on the boundary. They use this idea to study duality and other fundamental questions in that context.

## 11.7 Concluding Remarks

There are many aspects of analysis and several complex variables that we have not considered here. Just as an instance, the Szegő and Bergman kernels have had a powerful impact on Kähler geometry (by way of the recently proved Yau–Tian–Donaldson conjecture). The book [MAM] gives a glimpse of some of these developments.

The theory of harmonic analysis in the several complex variables setting is really quite young. We have only begun to scratch the surface of what is possible. In particular, while the strongly pseudoconvex sitution is fairly well understood, that for finite type domains or even more general domains is mostly an open book. New techniques are required in order to make any meaningful progress.

It is clear that the harmonic analysis of several complex variables will be key to understanding the corona problem and other important problems in the function theory of several complex variables. This is a field in which it is worth investing some effort.

## Exercises

1. Give an explicit example on the disc $D$ of an approach region that is *not* nontangential, but which satisfies the conditions of Nagel and Stein.
2. Formulate a version of the Christ/Geller result on the Siegel upper half space.
3. It is fairly clear that if you differentiate with respect to $r$ the area of the disc with center 0 and radius $r$ then you get the linear measure of the boundary. Formulate this result in the language of the Lusin area integral.
4. Refer to [CIK]. What does the result of Nagel and Stein say about the Lindelöf principle?
5. The results of Di Biase give a way to interpret the ideas of Nagel and Stein in the context of admissible approach regions on strongly pseudoconvex domains. Give an explicit example on the ball $B$ of an approach region that is *not* admissible, but which satisfies the conditions of Nagel and Stein and Di Biase.

**6.** Do the same as for Exercise 5 with the domain

$$\Omega = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^2 + |z_2|^4 < 1\}$$

at the boundary point $(1, 0)$.

**7.** Prove, on the real line, that the Hilbert transform of an atom $a$ is in $L^1$, and the norm of $\mathcal{H}a$ is independent of the choice of $a$.

**8.** Coifman, Rochberg, and Weiss prove their infinite Riesz decomposition of an $H^1$ function by exploiting the duality of $H^1$ and *BMO*. What is the connection?

**9.** A function on $\mathbb{R}$ is said to be of *vanishing mean oscillation VMO* if

$$\lim_{|B|\to 0} \frac{1}{|B|} \int |f - f_B| \, dV = 0 \, .$$

D. Sarason [SAR] proved that the dual of *VMO* is $H^1$. Give an example of a function that is in *VMO*. Give an example of a function that is in *BMO* but not *VMO*.

**10.** Look up Stein's version of the area function as explicated in [STE1]. Show how it is related to the Krantz/Li version described in the text.

# Bibliography


[ADA] Adams, *Sobolev Spaces*, Academic Press, New York, 1975.

[AFR] P. Ahern, M. Flores, and W. Rudin, An invariant volume-mean-value property, *Jour. Functional Analysis* 11(1993), 380–397.

[AHL1] L. Ahlfors, *Complex Analysis*, 3rd. ed., McGraw-Hill, New York, 1978.

[AIZ] L. A. Aĭzenberg, Fatou's theorem, the convergence of sequences, and a uniqueness theorem for analytic functions of several variables, *Moskov. Oblast. Ped. Inst. Učen. Zap.* 77(1959), 111–125.

[ARA] H. Arai, Singular elliptic operators related to harmonic analysis and complex analysis of several variables, *Trends in Probability and Related Analysis* (Taipei, 1998), 1–34, World Sci. Publ., River Edge, NJ, 1999.

[ARE] J. Arazy and M. Engliš, Iterates and the boundary behavior of the Berezin transform, *Ann Inst. Fourier (Grenoble)* 51(2001), 1101–1133.

[ARO] N. Aronszajn, Theory of reproducing kernels, *Trans. Am. Math. Soc.* 68(1950), 337–404.

[BAG] F. Bagemihl and W. Seidel, Some boundary properties of analytic functions, *Math. Zeitschr.* 61(1954), 186–199.

[BAR] S. R. Barker, Two theorems on boundary values of analytic functions, *Proc. A. M. S.* 68(1978), 48–54.

[BARR1] D. Barrett, The behavior of the Bergman projection on the Diederich–Fornaess worm, *Acta Math.* 168(1992), 1–10.

[BARR2] D. Barrett, Irregularity of the Bergman projection on a smooth bounded domain in $\mathbb{C}^2$, *Annals of Math.* 119(1984), 431–436.

[BARR3] D. Barrett, Regularity of the Bergman projection on domains with transverse symmetries, *Math. Ann.* 258(1981/82), 441–446.

[BEA] T. N. Bailey, M. G. Eastwood, and C. R. Graham, Invariant theory for conformal and CR geometry, *Annals of Math.* 139(1994), 491–552.

[BEF1] E. Bedford and J.-E. Fornæss, A construction of peak functions on weakly pseudo-convex domains, *Ann. Math.* 107(1978), 555–568.

[BEF2] E. Bedford and J. E. Fornæss, Counterexamples to regularity for the complex Monge–Ampère equation, *Invent. Math.* 50 (1978/79), 129–134.

[BEGA] D. Békollé, A. Bonami, G. Garrigós, C. Nana, M. M. Peloso, F. Ricci, Lecture notes on Bergman projectors in tube domains over cones: an analytic and geometric viewpoint, *IMHOTEP J. Afr. Math. Pures Appl.* 5(2004), Exp. I, front matter + ii + 75 pp.


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[BE1] S. Bell, Biholomorphic mappings and the $\overline{\partial}$-problem, *Ann. Math.*, 114(1981), 103–113.

[BE2] S. Bell, *The Cauchy Transform, Potential Theory, and Conformal Mapping*, Studies in Advanced Mathematics. CRC Press, Boca Raton, FL, 1992.

[BE3] S. Bell, Local boundary behavior of proper holomorphic mappings, *Proc. Sympos. Pure Math*, vol. 41, American Math. Soc., Providence R.I., 1984, 1–7.

[BEB] S. Bell and H. Boas, Regularity of the Bergman projection in weakly pseudoconvex domains, *Math. Annalen* 257(1981), 23–30.

[BEK] S. Bell and S. G. Krantz, Smoothness to the boundary of conformal maps, *Rocky Mt. Jour. Math.* 17(1987), 23–40.

[BELL] S. Bell and E. Ligocka, A simplification and extension of Fefferman's theorem on biholomorphic mappings, *Invent. Math.* 57(1980), 283–289.

[BER1] C. Berenstein, D.-C. Chang, W. Eby, $L^p$ results for the Pompeiu problem with moments on the Heisenberg group. *J. Fourier Anal. Appl.* 10(2004), 545–571.

[BER2] C. Berenstein, D.-C. Chang, J. Tie, Laguerre calculus and its applications on the Heisenberg group. *AMS/IP Studies in Advanced Mathematics*, 22. American Mathematical Society, Providence, RI; International Press, Somerville, MA, 2001. xii+319 pp.

[BERE] F. A. Berezin, Quantization in complex symmetric spaces, *Math. USSR Izvestia* 9(1975), 341–379.

[BERL] J. Bergh and J. Löfström, *Interpolation Spaces: An Introduction*, Springer-Verlag, Berlin, 1976.

[BERM1] S. Bergman, Über die Entwicklung der harmonischen Funktionen der Ebene und des Raumes nach Orthogonal funktionen, *Math. Annalen* 86(1922), 238–271.

[BERM2] S. Bergman, *The Kernel Function and Conformal Mapping*, Am. Math. Soc., Providence, RI, 1970.

[BERN] S. Bergman and M. Schiffer, *Kernel Functions and Elliptic Differential Equations in Mathematical Physics*, Academic Press, New York, 1953.

[BERNC] B. Berndtsson and P. Charpentier, A Sobolev mapping property of the Bergman kernel, *Math. Z.* 235(2000), 1–10.

[BERS] L. Bers, *Introduction to Several Complex Variables*, New York Univ. Press, New York, 1964.

[BLK] B. Blank and S. G. Krantz, *Calculus*, Key College Press, Emeryville, CA, 2006.

[BLP] Z. Blocki and P. Pflug, Hyperconvexity and Bergman completeness, *Nagoya Math. J.* 151(1998), 221–225.

[BOA1] H. Boas, Counterexample to the Lu Qi-Keng conjecture, *Proc. Am. Math. Soc.* 97(1986), 374–375.

[BOA2] H. Boas, Sobolev space projections in strictly pseudoconvex domains, *Trans. Amer. Math. Soc.* 288(1985), 227–240.

[BOA3] H. Boas, Lu Qi-Keng's problem. Several complex variables (Seoul, 1998), *J. Korean Math. Soc.* 37(2000), 253–267.

[BOA4] H. Boas, The Lu Qi-Keng conjecture fails generically, *Proc. Amer. Math. Soc.* 124(1996), 2021–2027.

[BOAKP] H. Boas, S. G. Krantz, M. M. Peloso, unpublished.

[BOAS1] H. Boas and E. Straube, Equivalence of regularity for the Bergman projection and the $\overline{\partial}$-Neumann operator, *Manuscripta Math.* 67(1990), 25–33.

[BOAS2] H. Boas and E. Straube, Sobolev estimates for the $\overline{\partial}$-Neumann operator on domains in $\mathbb{C}^n$ admitting a defining function that is plurisubharmonic on the boundary, *Math. Z.* **206** (1991), 81–88.

[BOC] S. Bochner, Orthogonal systems of analytic functions, *Math. Z.* 14(1922), 180–207.

[BOG] A. Boggess, *CR Functions and the Tangential Cauchy–Riemann Complex*, CRC Press, Boca Raton, 1991.

[BOK] A. Boussejra, K. Koufany, Characterization of the Poisson integrals for the non-tube bounded symmetric domains, *J. Math. Pures Appl.* 87(2007), 438–451.

[BOUT] L. Boutet de Monvel, Le noyau de Bergman en dimension 2, *Séminaire sur les Équations aux Dérivées Partielles* 1987–1988, Exp. no. XXII, École Polytechnique Palaiseau, 1988, p. 13.

[BOUTS] L. Boutet de Monvel and J. Sjöstrand, Sur la singularité des noyaux de Bergman et Szegő, *Soc. Mat. de France Asterisque* 34–35(1976), 123–164.

[BRE] H. Bremerman, Über die Äquivalenz der pseudoconvex Gebiete und der Holomorphie-gebiete in Raum von $n$ Komplexen Veränderlichen, *Math. Ann.* 128(1954), 63–91.

[BUN] L. Bungart, Holomorphic functions with values in locally convex spaces and applications to integral formulas, *Trans. Am. Math. Soc.* 111(1964), 317–344.

[BUR] D. Burkholder, R. Gundy, and M. Silverstein, A maximal function characterization of the class $H^p$, *Trans. Am. Math. Soc.* 157(1971), 137–153.

[BURS] D. Burns, S. Shnider, and R. O. Wells, On deformations of strictly pseudoconvex domains, *Invent. Math.* 46(1978), 237–253.

[CAF] L. Caffarelli, J. J. Kohn, L. Nirenberg, and J. Spruck, The Dirichlet problem for nonlinear second-order elliptic equations. II. Complex Monge–Ampère, and uniformly elliptic, equations, *Comm. Pure Appl. Math.* 38(1985), 209–252.

[CAL] A. Calderón, On the behavior of harmonic functions near the boundary, *Trans. Amer. Math. Soc.* 68(1950), 47–54.

[CALZ1] A. Calderón and A. Zygmund, Note on the boundary values of functions of several complex variables, *Contributions to Fourier Analysis*, Annals of Mathematics Studies, no. 25, Princeton University Press, Princeton, NJ, 1950, 145–165.

[CALZ2] A. P. Calderón and A. Zygmund, Note on the limit values of analytic functions. (Spanish) *Revista Uni. Mat. Argentina* 14(1949), 16–19.

[CAR] A. Carbery, A version of Cotlar's lemma for $L^p$ spaces and some applications, *Harmonic Analysis and Operator Theory* (Caracas, 1994), 117–134, Contemp. Math., 189, Amer. Math. Soc., Providence, RI, 1995.

[CARR] Carrier, Crook, and Pearson, *Functions of a Complex Variable*, McGraw–Hill, New York, 1966.

[CAT1] D. Catlin, Necessary conditions for subellipticity of the $\overline{\partial}-$Neumann problem, *Ann. Math.* 117(1983), 147–172.

[CAT2] D. Catlin, Subelliptic estimates for the $\overline{\partial}$Neumann problem, *Ann. Math.* 126(1987), 131–192.

[CAT3] D. Catlin, Boundary behavior of holomorphic functions on pseudoconvex domains, *J. Differential Geom.* 15(1980), 605–625.

[CHA1] D.-C. Chang, On the boundary of Fourier and complex analysis: the Pompeiu problem, *Taiwanese J. Math.* 6(2002), 1–37.

[CHA2] D.-C. Chang, Singular integrals: Calderón–Zygmund-Stein theory, preprint.

[CHAE] D.-C. Chang, W. Eby, Pompeiu problem on product of Heisenberg groups, *Complex Anal. Oper. Theory* 4 (2010), 619–683.

[CHE1] S.-C. Chen, Characterization of automorphisms on the Barrett and the Diederich–Fornaess worm domains, *Trans. Amer. Math. Soc.* 338(1993), 431–440.

[CHE2] S.-C. Chen, A counterexample to the differentiability of the Bergman kernel function, *Proc. AMS* 124(1996), 1807–1810.

[CHEF] B.-Y. Chen and S. Fu, Comparison of the Bergman and Szegő kernels, *Advances in Math.* 228(2011), 2366–2384.

[CHS] S.-C. Chen and M.-C. Shaw, *Partial Differential Equations in Several Complex Variables*, AMS/IP Studies in Advanced Mathematics, 19, American Mathematical Society, Providence, RI, 2001.

[CHSG] S.-Y. Cheng, Open problems, *Conference on Nonlinear Problems in Geometry Held in Katata*, September, 1979, Tohoku University, Dept. of Mathematics, Sendai, 1979, p. 2.

[CHSM] S. S. Chern and J. Moser, Real hypersurfaces in complex manifolds, *Acta Math.* 133(1974), 219–271.

[CHSS] K. S. Choi, Notes on Carleson type measures on bounded symmetric domain, *Commun. Korean Math. Soc.* 22(2007), 65–74.

[CHST1] M. Christ, *Singular Integrals*, CBMS, Am. Math. Society, Providence, RI, 1990.

[CHST2] M. Christ, Global $C^\infty$ irregularity of the $\overline{\partial}$-Neumann problem for worm domains, *J. Amer. Math. Soc.* 9(1996), 1171–1185.

[CHSTG] M. Christ and D. Geller, Singular integral characterizations of Hardy spaces on homogeneous groups, *Duke Math. J.* 51(1984), 547–598.

[CIK] J. Cima and S. G. Krantz, A Lindelöf principle and normal functions in several complex variables, *Duke Math. Jour.* 50(1983), 303–328.

[COFW] R. Coifman, R. Rochberg, and G. Weiss, Factorization theorems for Hardy spaces in several variables, *Ann. of Math.* 103(1976), 611–635.

[COG1] R. R. Coifman and G. L. Weiss, *Analyse harmonique non-commutative sur certains espaces homogénes*, Lecture Notes in Mathematics, Vol. 242. Springer–Verlag, Berlin-New York, 1971.

[COG2] R. R. Coifman and G. L. Weiss, Extensions of Hardy spaces and their use in analysis, *Bull. Amer. Math. Soc.* 83(1977), 569–645.

[COL] E. F. Collingwood and A. J. Lohwater, *The Theory of Cluster Sets*, Cambridge University Press, Cambridge, 1966.

[CON] P. E. Conner, *The Neumann's Problem for Differential Forms on Riemannian Manifolds*, Mem. Am. Math. Soc. #20, 1956.

[COR] A. Cordoba and R. Fefferman, On the equivalence between the boundedness of certain classes of maximal and multiplier operators in Fourier analysis, *Proc. Nat. Acad. Sci. U.S.A.* 74(1977), 423–425.

[COT] M. Cotlar, A combinatorial inequality and its applications to $L^2$-spaces, *Rev. Mat. Cuyana* 1(1955), 41–55.

[COU] R. Courant and D. Hilbert, *Methods of Mathematical Physics*, 2nd ed., Interscience, New York, 1966.

[DAF1] G. Dafni, Hardy spaces on strongly pseudoconvex domains in $\mathbb{C}^n$ and domains of finite type in $\mathbb{C}^2$, Thesis (Ph.D.) Princeton University, 1993, 86 pp.

[DAF2] G. Dafni, Hardy spaces on some pseudoconvex domains, *J. Geom. Anal.* 4(1994), 273–316.,

[DAM] S. B. Damelin, A. J. Devaney, Local Paley–Wiener theorems for functions analytic on unit spheres, *Inverse Problems* 23(2007), 463–474.

[DEG] M. de Guzman, *Differentiation of Integrals in* $\mathbb{R}^n$, Springer Lecture Notes, Springer-Verlag, Berlin and New York, 1975.

[DEM] J.-P. Demailly, *Analytic Methods in Algebraic Geometry*, Surveys of Modern Mathematics, 1. International Press, Somerville, MA; Higher Education Press, Beijing, 2012. viii+231 pp.

[DIB] F. Di Biase, *Fatou Type Theorems. Maximal Functions and Approach Regions*, Progress in Mathematics, 147. Birkhäuser Boston, Boston, MA, 1998.

[DIF] F. Di Biase and B. Fischer, Boundary behaviour of $H^p$ functions on convex domains of finite type in $\mathbb{C}^n$, *Pacific J. Math.* 183(1998), 25–38.

[DIK] F. DiBiase and S. G. Krantz, *The Boundary Behavior of Holomorphic Functions*, Birkhäuser Publishing, to appear.

[DIR1] K. Diederich, Das Randverhalten der Bergmanschen Kernfunktion und Metrik in streng pseudo-konvexen Gebieten, *Math. Ann.* 187(1970), 9–36.

[DIR2] K. Diederich, Über die 1. and 2. Ableitungen der Bergmanschen Kernfunktion und ihr Randverhalten, *Math. Ann.* 203(1973), 129–170.

[DIRF1] K. Diederich and J.-E. Fornæss, Pseudoconvex domains: Bounded strictly plurisubharmonic exhaustion functions, *Invent. Math.* 39(1977), 129–141.

[DIRF2] K. Diederich and J.-E. Fornæss, Pseudoconvex domains: An example with nontrivial Nebenhülle, *Math. Ann.* 225(1977), 275–292.

[DIRF3] K. Diederich and J.-E. Fornæss, Pseudoconvex domains with real-analytic boundary, *Annals of Math.* 107(1978), 371–384.

[DIRF4] K. Diederich and J. E. Fornæss, Smooth extendability of proper holomorphic mappings, *Bull. Amer. Math. Soc.* (N.S.) 7(1982), 264–268.

[DOM] C. Domenichino, Converse of spherical mean-value property for invariant harmonic functions, *Complex Variables Theory Appl.* 45(2001), 371–385.

[DOU] E. Doubtsov, An F. and M. Riesz theorem on the complex sphere, *J. Fourier Anal. Appl.* 12(2006), 225–231.

[DUB1] E. S. Dubtsov, Some problems of harmonic analysis on a complex sphere, *Vestnik St. Petersburg Univ. Math.* 28(1995), 12–16.

[DUB2] E. S. Dubtsov, On the radial behavior of functions holomorphic in a ball, *Vestnik St. Petersburg Univ. Math.* 27(1994), 21–27.

[DUR] P. L. Duren, B. W. Romberg, and A. L. Shields, Linear functionals on $H^p$ spaces with $0 < p < 1$, *J. Reine Angew. Math.* 238(1969), 32–60.

[ENG1] M. Engliš, Functions invariant under the Berezin transform, *J. Funct. Anal.* 121(1994), 233–254.

[ENG2] M. Engliš, Asymptotics of the Berezin transform and quantization on planar domains, *Duke Math. J.* 79(1995), 57–76.

[EPS] B. Epstein, *Orthogonal Families of Functions*, Macmillan, New York, 1965.

[FED] H. Federer, *Geometric Measure Theory*, Springer-Verlag, New York, 1969.

[FEF1] C. Fefferman, The Bergman kernel and biholomorphic mappings of pseudoconvex domains, *Invent. Math.* 26(1974), 1–65.

[FEF2] C. Fefferman, Monge–Ampère equations, the Bergman kernel, and geometry of pseudoconvex domains, *Ann. of Math.* 103(1976), 395–416.

[FEF3] C. Fefferman, Characterizations of bounded mean oscillation, *Bull. Amer. Math. Soc.* 77(1971), 587–588.

[FES] C. Fefferman and E. M. Stein, $H^p$ spaces of several variables, *Acta Math.* 129(1972), 137–193.

[FOD] G. B. Folland, Spherical harmonic expansion of the Poisson–Szegö kernel for the ball, *Proc. Am. Math. Soc.* 47(1975), 401–408.

[FOL] G. B. Folland, Some topics in the history of harmonic analysis in the twentieth century, *Indian J. Pure Appl. Math.* 48(2017), 1–58.

[FOK] G. B. Folland and J. J. Kohn, *The Neumann Problem for the Cauchy–Riemann Complex*, Princeton University Press, Princeton, 1972.

[FOS1] G. B. Folland and E. M. Stein, Estimates for the $\overline{\partial}_b$ complex and analysis on the Heisenberg group, *Comm. Pure Appl. Math.* 27(1974), 429–522.

[FOS2] G. B. Folland and E. M. Stein, *Hardy Spaces on Homogeneous Groups*, Mathematical Notes, 28. Princeton University Press, Princeton, N.J.; University of Tokyo Press, Tokyo, 1982.

[FOSP] L. Fontana, S. G. Krantz, and M. M. Peloso, The $\overline{\partial}$-Neumann problem in the Sobolev topology, *Indiana Journal of Math.* 48(1999), 275–293.

[FOST] F. Forstneric, An elementary proof of Fefferman's theorem, *Expositiones Math.*, 10(1992), 136–149.

[FRI] B. Fridman, A universal exhausting domain, *Proc. Am. Math. Soc.* 98(1986), 267–270.

[FUW] S. Fu and B. Wong, On strictly pseudoconvex domains with Kähler–Einstein Bergman metrics, *Math. Res. Letters* 4(1997), 697–703.

[GAM] T. W. Gamelin, *Uniform Algebras*, Prentice Hall, Englewood Cliffs, 1969.

[GAN] T. Gamelin and N. Sibony, Subharmonicity for uniform algebras. *J. Funct. Anal.* 35 (1980), 64–108.

[GARA] P. R. Garabedian, A Green's function in the theory of functions of several complex variables, *Ann. of Math.* 55(1952). 19–33.

[GARN] J. B. Garnett, *Bounded Analytic Functions*, Academic Press, New York, 1981.

[GARNL] J. B. Garnett and R. H. Latter, The atomic decomposition for Hardy spaces in several complex variables, *Duke Math. J.* 45(1978), 815–845.

[GEL1] D. Geller, Fourier analysis on the Heisenberg group, *Proc. Nat. Acad. Sci. USA* 74(1977), 1328–1331.

[GEL2] D. Geller, Fourier analysis on the Heisenberg group, *Lie Theories and their Applications*, Queen's University, Kingston, Ontario, 1978, 434–438.

[GEL3] D. Geller, Fourier analysis on the Heisenberg group. I. Schwarz space, *J. Functional Anal.* 36(1980), 205–254.

[GELS] D. Geller and E. M. Stein, Convolution operators on the Heisenberg group, *Bull. Am. Math. Soc.* 6(1982), 99–103.

[GIN] S. Gindikin, Holomorphic harmonic functions. *Russ. J. Math. Phys.* 15(2008), 243–245.

[GLE] A. Gleason, Finitely generated ideals in Banach algebras, *J. Math. Mech.* 13(1964), 125–132.

[1] The abstract theorem of Cauchy–Weil, *Pac. J. Math.* 12(1962), 511–525.

[GOL] G. M. Goluzin, *Geometric Theory of Functions of a Complex Variable*, American Mathematical Society, Providence, 1969.

[GON] D. D. Gong, Some estimates for singular integrals in the building domain of complex biballs (Chinese), *J. Math. Study* 40(2007), 290–296.

[GRA1] C. R. Graham, The Dirichlet problem for the Bergman Laplacian, I, *Comm. Partial Differential Equations* 8(1983), 433–476.

[GRA2] C. R. Graham, The Dirichlet problem for the Bergman Laplacian, II, *Comm. Partial Differential Equations* 8(1983), 563–641.

[GRA3] C. R. Graham, Scalar boundary invariants and the Bergman kernel, *Complex analysis, II* (College Park, Md., 1985–86), 108–135, Lecture Notes in Math. 1276, Springer, Berlin, 1987.

[GRAI] I. Graham, Boundary behavior of the Carathéodory and Kobayashi metrics on strongly pseudoconvex domains in $\mathbb{C}^n$ with smooth boundary, *Trans. Am. Math. Soc.* 207(1975), 219–240.

[GRAL] H. Grauert and I. Lieb, Das Ramirezsche Integral und die Gleichung $\overline{\partial} u = \alpha$ im Bereich der beschränkten Formen, *Rice University Studies* 56(1970), 29–50.

[GKK] R. E. Greene, K.-T. Kim, and S. G. Krantz, *The Geometry of Complex Domains*, Birkhäuser, Boston, 2011.

[GRK1] R. E. Greene and S. G. Krantz, Stability properties of the Bergman kernel and curvature properties of bounded domains, *Recent Developments in Several Complex Variables* (J. E. Fornæss, ed.), Princeton University Press (1979), 179–198.

[GRK2] R. E. Greene and S. G. Krantz, Deformation of complex structures, estimates for the $\overline{\partial}$ equation, and stability of the Bergman kernel, *Adv. Math.* 43(1982), 1–86.

[GRK3] R. E. Greene and S. G. Krantz, Biholomorphic self-maps of domains, *Complex Analysis II* (C. Berenstein, ed.), Springer Lecture Notes, vol. 1276, 1987, 136–207.

[GRK4] R. E. Greene and S. G. Krantz, *Function Theory of One Complex Variable*, 3rd ed., American Mathematical Society, Providence, RI, 2006.

[GRK5] R. E. Greene and S. G. Krantz, The automorphism groups of strongly pseudoconvex domains, *Math. Annalen* 261(1982), 425–446.

[GRN] P. Greiner, Subelliptic estimates for the $\overline{\partial}$-Neumann problem in $\mathbb{C}^2$, *J. Diff. Geom.* 9(1974), 239–250.

[GRNKS] P. Greiner, J. J. Kohn, and E. M. Stein, Necessary and sufficient conditions for solvability of the Lewy equation, *Proc. Nat. Acad. Sci.* 72(1975), 3287–3289.

[GRS] P. Greiner and E. M. Stein, *Estimates for the $\overline{\partial}$-Neumann Problem*, Princeton University Press, Princeton, NJ, 1977.

[HAN] N. Hanges, Explicit formulas for the Szegö kernel for some domains in $\mathbb{C}^2$. *J. Funct. Anal.* 88 (1990), 153–165.

[HAR1] G. H. Hardy and J. E. Littlewood, Theorems concerning mean values of analytic or harmonic functions, *Q. J. Math. Oxford Ser.* 12 (1941), 221–256.

[HAR2] G. H. Hardy and J. E. Littlewood, Some properties of functional integrals II, *Math. Zeit.* 4(1932), 403–439.

[HARV] R. Harvey and J. Polking, Fundamental solutions in complex analysis I and II, *Duke Math. J.* 46(1979), 253–300 and 46(1979), 301–340.

[HEL] S. Helgason, *Differential Geometry and Symmetric Spaces*, Academic Press, New York, 1962.

[HEN1] G. M. Henkin, Integral representations of functions holomorphic in strictly pseudoconvex domains and applications to the $\overline{\partial}$ problem, *Mat. Sb.* 82(124), 300–308 (1970); *Math. U.S.S.R. Sb.* 11(1970), 273–281.

[HEN2] G. M. Henkin, Solutions with estimates of the H. Lewy and Poincaré–Lelong equations, the construction of functions of a Nevanlinna class with given zeroes in a strictly pseudoconvex domain, *Dok. Akad. Nauk. SSSR* 224(1975), 771–774; *Sov. Math. Dokl.* 16(1975), 1310–1314.

[HIL] E. Hille, *Analytic Function Theory*, Ginn & Co., Boston, MA, 1959.

[HIR1] K. Hirachi, The second variation of the Bergman kernel of ellipsoids, *Osaka J. Math.* 30(1993), 457–473.

[HIR2] K. Hirachi, Scalar pseudo-Hermitian invariants and the Szegő kernel on three-dimensional CR manifolds, *Complex Geometry* (Osaka, 1990), Lecture Notes Pure Appl. Math., v. 143, Marcel Dekker, New York, 1993, 67–76.

[HIR3] K. Hirachi, Construction of boundary invariants and the logarithmic singularity in the Bergman kernel, *Annals of Math.* 151(2000), 151–190.

[HIR] M. Hirsch, *Differential Topology*, Springer–Verlag, New York, 1976.

[HOF] K. Hoffman, *Banach Spaces of Holomorphic Functions*, Prentice-Hall, Englewood Cliffs, NJ. 1962.

[HOR1] L. Hörmander, $L^2$ estimates and existence theorems for the $\overline{\partial}$ operator, *Acta Math.* 113(1965), 89–152.

[HOR2] L. Hörmander, *Linear Partial Differential Operators*, Springer, Berlin, 1963.

[HOR3] L. Hörmander, Fourier integral operators, I, *Acta Math.* 127(1971), 79–183.

[HOR4] L. Hörmander, *Introduction to Complex Analysis in Several Variables*, North Holland, Amsterdam, 1973.

[HOR5] $L^p$ estimates for (pluri-)subharmonic functions, *Math. Scand.* 20(1967), 65–78.

[HOR6] L. Hörmander and J. Sjöstrand, Fourier integral operators, II, *Acta Math.* 128(1972), 183–269.

[HOR7] L. Hörmander, Estimates for translation invariant o perators in $L^p$ spaces, *Acta Math.* 104(1960), 93–140.

[HUA] L. Hua, *Harmonic Analysis of Functions of Several Complex Variables in the Classical Domains*, American Mathematical Society, Providence, 1963.

[ISK] A. Isaev and S. G. Krantz, Domains with non-compact automorphism group: A Survey, *Advances in Math.* 146(1999), 1–38.

[JAC] H. Jacobowitz and F. Treves, Nowhere solvable homogeneous partial differential equations, *Bull. A.M.S.* 8(1983), 467–469.

[JAK] S. Jakobsson, Weighted Bergman kernels and biharmonic Green functions, Ph.D. thesis, Lunds Universitet, 2000, 134 pages.

[JAR] P. Jaming, M. Roginskaya, Boundary behaviour of $M$-harmonic functions and non-isotropic Hausdorff measure, *Monatsh. Math.* 134(2002), 217–226.

[JON] F. John and L. Nirenberg, On functions of bounded mean oscillation, *Comm. Pure and Appl. Math.* 14(1961), 415–426.

[KAC] S. Kaczmarz, Über die Konvergenz der Reihen von Orthogonal-funktionen, *Math. Z.* 23(1925), 263–270.

[KAN] Kaneyuki, *Homogeneous Domains and Siegel Domains*, Springer Lecture Notes #241, Berlin, 1971.

[KAT] Y. Katznelson, *Introduction to Harmonic Analysis*, Wiley, New York, 1968.

[KEL] O. Kellogg, Harmonic functions and Green's integral, *Trans. Am. Math. Soc.* 13(1912), 109–132.

[KER1] N. Kerzman, Hölder and $L^p$ estimates for solutions of $\overline{\partial} u = f$ on strongly pseudoconvex domains, *Comm. Pure Appl. Math.* XXIV(1971), 301–380.

[KER2] N. Kerzman, The Bergman kernel function. Differentiability at the boundary, *Math. Ann.* 195(1972), 149–158.

[KER3] N. Kerzman, A Monge–Ampère equation in complex analysis, Several complex variables (*Proc. Sympos. Pure Math.*, Vol. XXX, Part 1, Williams Coll., Williamstown, Mass., 1975), pp. 161–167. Amer. Math. Soc., Providence, R.I., 1977.

[KER4] N. Kerzman, Topics in Complex Analysis, unpublished notes, MIT, 1978.

[KES] N. Kerzman and E. M. Stein, The Szegő kernel in terms of Cauchy–Fantappiè kernels, *Duke Math. J.* 45(1978), 197–224.

[KIS] C. Kiselman, A study of the Bergman projection in certain Hartogs domains, *Proc. Symposia Pure Math.*, vol. 52 (E. Bedford, J. D'Angelo, R. Greene, and S. Krantz eds.), American Mathematical Society, Providence, 1991.

[KLE] P. Klembeck, Kähler metrics of negative curvature, the Bergman metric near the boundary and the Kobayashi metric on smooth bounded strictly pseudoconvex sets, *Indiana Univ. Math. J.* 27(1978), 275–282.

[KNS] A. Knapp and E. M. Stein, Intertwining operators for semisimple groups, *Ann. of Math.* 93(1971), 489–578.

[KOB] S. Kobayashi, *Hyperbolic Manifolds and Holomorphic Mappings*, Marcel Dekker, New York, 1970.

[KOB1] S. Kobayashi, Geometry of bounded domains, *Trans. AMS* 92(1959), 267–290.

[KOBN] S. Kobayashi and K. Nomizu, *Foundations of Differential Geometry*, Vols. I and II, Interscience, New York, 1963, 1969.

[KOH1] J. J. Kohn, Quantitative estimates for global regularity, *Analysis and geometry in several complex variables* (Katata, 1997), 97–128, Trends Math., Birkhäuser Boston, Boston, MA, 1999.

[KOH2] J. J. Kohn, Harmonic integrals on strongly pseudoconvex manifolds I, *Ann. Math.* 78(1963), 112–148; II, ibid. 79(1964), 450–472.

[KOH3] J. J. Kohn, Global regularity for $\overline{\partial}$ on weakly pseudoconvex manifolds, *Trans. A. M. S.* 181(1973), 273–292.

[KOHN1] J. J. Kohn and L. Nirenberg, Non-coercive boundary value problems, *Comm. Pure Appl. Math.* 18(1965), 443–492.

[KOHN2] J. J. Kohn and L. Nirenberg, On the algebra of pseudo-differential operators, *Comm. Pure Appl. Math.* 18(1965), 269–305.

[KOO] P. Koosis, *Lectures on $H^p$ Spaces*, Cambridge University Press, Cambridge, UK, 1980.

[KOR1] A. Korányi, Harmonic functions on Hermitian hyperbolic space, *Trans. A.M.S.* 135(1969), 507–516.

[KOR2] A. Korányi, Boundary behavior of Poisson integrals on symmetric spaces, *Trans. A.M.S.* 140(1969), 393–409.

[KOV] A. Koranyi and S. Vagi, Singular integrals in homogeneous spaces and some problems of classical analysis, *Ann. Scuola Norm. Sup. Pisa* 25(1971), 575–648.

[KRA1] S. G. Krantz, *Function Theory of Several Complex Variables*, 2nd ed., American Mathematical Society, Providence, RI, 2001.

[KRA2] S. G. Krantz, *A Panorama of Harmonic Analysis*, Mathematical Association of America, Washington, D.C., 1999.

[KRA3] S. G. Krantz, *Real Analysis and Foundations*, CRC Press, Boca Raton, 1992.

[KRA4] S. G. Krantz, Fractional integration on Hardy spaces, *Studia Math.* 73(1982), 87–94.

[KRA5] S. G. Krantz, *Partial Differential Equations and Complex Analysis*, CRC Press, Boca Raton, 1992.

[KRA6] S. G. Krantz, Lipschitz spaces, smoothness of functions, and approximation theory, *Expositiones Math.* 3(1983), 193–260.

[KRA7] S. G. Krantz, Invariant metrics and the boundary behavior of holomorphic functions on domains in $\mathbb{C}^n$, *J. Geom. Anal.* 1(1991), 71–97.

[KRA8] S. G. Krantz, Canonical kernels versus constructible kernels, *Rocky Mountain Journal of Mathemtics*, to appear.

[KRA9] S. G. Krantz, Holomorphic functions of bounded mean oscillation and mapping properties of the Szegő projection, *Duke Math. Jour.* 47(1980), 743–761.

[KRA10] S. G. Krantz, Optimal Lipschitz and $L^p$ regularity for the equation $\overline{\partial} u = f$ on strongly pseudo-convex domains, *Math. Annalen* 219(1976), 233–260.

[KRA11] S. G. Krantz, *Explorations in Harmonic Analysis, with Applications in Complex Function Theory and the Heisenberg Group*, Birkhäuser Publishing, Boston, 2009.

[KRA12] S. G. Krantz, Geometric Lipschitz spaces and applications to complex function theory and nilpotent groups, *J. Funct. Anal.* 34(1980), 456–471.

[KRA13] S. G. Krantz, Calculation and estimation of the Poisson kernel, *J. Math. Anal. Appl.* 302(2005)143–148.

[KRA14] S. G. Krantz, The Lindelöf principle in several complex variables, *Journal of Math. Anal. and Applications* 326(2007), 1190–1198.

[KRA15] S. G. Krantz, *Geometric Analysis and Function Spaces*, CBMS and the American Mathematical Society, Providence, 1993.

[KRA16] S. G. Krantz, *Complex Analysis: The Geometric Viewpoint*, 2nd ed., Mathematical Association of America, Washington, D.C., 2004.

[KRA17] S. G. Krantz, *Cornerstones of Geometric Function Theory: Explorations in Complex Analysis*, Birkhäuser Publishing, Boston, 2006.

[KRA18] S. G. Krantz, *Geometric Analysis of the Bergman Kernel and Metric*, Birkhäuser, Boston, 2013.

[KRA19] S. G. Krantz, Characterization of various domains of holomorphy via $\overline{\partial}$ estimates and applications to a problem of Kohn, *Ill. J. Math.* 23(1979), 267–286.

[KRA20] S. G. Krantz, On a construction of L. Hua for positive reproducing kernels, *Michigan Math. J.* 59(2010), 211–230.

[KRA21] S. G. Krantz, A new proof and a generalization of Ramadanov's theorem, *Complex Variables and Elliptic Eq.* 51(2006), 1125–1128.

[KRA22] S. G. Krantz, Boundary decomposition of the Bergman kernel, *Rocky Mountain Journal of Math.*, 41(2011), 1265–1272.

[KRA23] S. G. Krantz, A direct connection between the Bergman and Szegő kernels, *Complex Analysis and Operator Theory* 8(2014), 571–579.

[KRA24] S. G. Krantz, Lipschitz spaces on stratified groups, *Trans. Am. Math. Soc.* 269(1982), 39–66.

[KRA25] S. G. Krantz, Harmonic analysis of several complex variables: a survey, *Expo. Math.* 31(2013), 215–255.

[KRL1] S. G. Krantz and S.-Y. Li, Area integral characterizations of functions in Hardy spaces on domains in $\mathbb{C}^n$, *Complex Variables Theory Appl.* 32(1997), 373–399.

[KRL2] S. G. Krantz and S.-Y. Li, A Note on Hardy Spaces and Functions of Bounded Mean Oscillation on Domains in $\mathbb{C}^n$, *Michigan Jour. Math.* 41 (1994), 51–72.

[KRL3] S. G. Krantz and S.-Y. Li, On decomposition theorems for Hardy spaces on domains in $\mathbb{C}^n$ and applications, *J. Fourier Anal. Appl.* 2(1995), 65–107.

[KRL4] S. G. Krantz and S.-Y. Li, On the existence of smooth plurisubharmonic solutions for certain degenerate Monge–Ampère equations, *Complex Variables Theory Appl.* 41(2000), 207–219.

[KRL5] S. G. Krantz and S.-Y. Li, Duality theorems for Hardy and Bergman spaces on convex domains of finite type in $\mathbb{C}^n$, *Ann. Inst. Fourier* (Grenoble) 45(1995), 1305–1327.

[KRM] S. G. Krantz and D. Ma, BMO and Duality on pseudoconvex domains, preprint.

[KRPA1] S. G. Krantz and H. R. Parks, *A Primer of Real Analytic Functions*, 2nd ed., Birkhäuser, Boston, 2002.

[KRPA2] S. G. Krantz and H. R. Parks, *The Geometry of Domains in Space*, Birkhäuser, Boston, 1996.

[KRPA3] S. G. Krantz and H. R. Parks, *The Implicit Function Theorem*, Birkhäuser, Boston, 2002.

[KRPA4] S. G. Krantz and H. R. Parks, Distance to $C^k$ manifolds, *Jour. Diff. Equations* 40(1981), 116–120.

[KRPE1] S. G. Krantz and M. M. Peloso, The Bergman kernel and projection on non-smooth worm domains, *Houston J. Math.* 34 (2008), 873–950.

[KRPE2] S. G. Krantz and M. M. Peloso, Analysis and geometry on worm domains, *J. Geom. Anal.* 18(2008), 478–510.

[KRPS1] S. G. Krantz, M. M. Peloso, and C. Stoppato, Bergman kernel and projection on the unbounded Diederich–Fornæss worm domain, *Annali della Scuola Normale Superiore*, to appear.

[KRPS2] S. G. Krantz, M. M. Peloso, and C. Stoppato, Completeness on the worm domain and the Müntz–Szász problem for the Bergman space, preprint.

[LAN] L. Lanzani, Harmonic analysis techniques in several complex variables, Bruno Pini Mathematical Analysis Seminar 2014, 83–110, *Bruno Pini Math. Anal. Semin.*, 2014, Univ. Bologna, Alma Mater Stud., Bologna, 2014.

[LAR] L. Lanzani and E. M. Stein, Cauchy-type integrals in several complex variables. *Bull. Math. Sci.* 3 (2013), 241–285.

[LEM1] L. Lempert, La metrique Kobayashi et la representation des domains sur la boule, *Bull. Soc. Math. France* 109(1981), 427–474.

[LEM2] L. Lempert, Boundary behaviour of meromorphic functions of several variables, *Acta Math.* 144(1980), 1–25.

[LEM3] L. Lempert, A precise result on the boundary regularity of biholomorphic mappings, *Math. Z.* 193(1986), 559–579.

[LEW1] H. Lewy, An example of a smooth linear partial differential equation without solution, *Ann. of Math.* 66(1957), 155–158.

[LEW2] H. Lewy, On the boundary behavior of holomorphic mappings, *Acad. Naz. Lincei* 35(1977), 1–8.

[LI1] S.-Y. Li, On the Neumann problems for complex Monge–Ampère equations, *Indiana University J. of Math*, 43(1994), 1099–1122.

[LI2] S.-Y. Li, D. Wei, On the rigidity theorem for harmonic functions in Kähler metric of Bergman type, *Sci. China Math.* 53(2010), 779–790.

[LIG1] E. Ligocka, On orthogonal projections onto spaces of pluriharmonic functions and duality, *Studia Math.* 84(1986), 279–295.

[LIG2] E. Ligocka, Remarks on the Bergman kernel function of a worm domain, *Studia Mathematica* 130(1998), 109–113.

[LIP1] J.-L. Lions and J. Peetre, Sur une classe d'espaces d'interpolation. (French), *Inst. Hautes Études Sci. Publ. Math.* 19(1964), 5–68.

[LIP2] J.-L. Lions and J. Peetre, Propriétés d'espaces d'interpolation. (French), *C. R. Acad. Sci. Paris* 253(1961), 1747–1749.

[MAL] A. Malaspina, A brothers Riesz theorem in the theory of holomorphic functions of several complex variables, Homage to Gaetano Fichera, 273–288, *Quad. Mat.*, 7, Dept. Math., Seconda Univ. Napoli, Caserta, 2000.

[MAM] X. Ma and G. Marinescu, *Holomorphic Morse Inequalities and Bergman Kernels*, Progress in Mathematics, 254. Birkhäuser Verlag, Basel, 2007. xiv+422 pp.

[MEP] V. A. Menegatto, A. P. Peron, Positive definite kernels on complex spheres, *J. Math. Anal. Appl.* 254(2001), 219–232.

[MIN] B. Min, thesis, Washington University, 2011.

[NARSW] A. Nagel, J.-P. Rosay, E. M. Stein, and S. Wainger, Estimates for the Bergman and Szegő kernels in $\mathbb{C}^2$, *Ann. Math.* 129(1989), 113–149.

[NARU] A. Nagel and W. Rudin, Local boundary behavior of bounded holomorphic functions, *Can. Jour. Math.* 30(1978), 583–592.

[NAS1] A. Nagel and E. M. Stein, On certain maximal functions and approach regions, *Adv. in Math.* 54(1984), 83–106.

[NAS2] A. Nagel and E. M. Stein, *Lectures on Pseudodifferential Operators: Regularity Theorems and Applications to Nonelliptic Problems*, Mathematical Notes, 24, Princeton University Press, Princeton, N.J.; University of Tokyo Press, Tokyo, 1979.

[NASW1] A. Nagel, E. M. Stein, and S. Wainger, Boundary behavior of functions holomorphic in domains of finite type, *Proc. Nat. Acad. Sci. USA* 78(1981), 6596–6599.

[NASW2] A. Nagel, E. M. Stein, and S. Wainger, Balls and metrics defined by vector fields, I. Basic properties. *Acta Math.* 155(1985), 103–147.

[NNA] C. Nana, $L^\infty$-estimates of the Bergman projection in the Lie ball of $\mathbb{C}^n$, *J. Funct. Spaces Appl.* 9(2011), 109–128.

[NRT] Y. A. Neretin, On the separation of spectra in the analysis of Berezin kernels (Russian), *Funktsional. Anal. i Prilozhen* 34(2000), 49–62, 96; translation in *Funct. Anal. Appl.* 34(2000), 197–207.

[NWY] L. Nirenberg, S. Webster, and P. Yang, Local boundary regularity of holomorphic mappings. *Comm. Pure Appl. Math.* 33(1980), 305–338.

[OHS] T. Ohsawa, A remark on the completeness of the Bergman metric, *Proc. Japan Acad.* Ser. A Math. Sci., 57(1981), 238–240.

[OHT] T. Ohsawa and K. Takegoshi, On the extension of $L^2$ holomorphic functions, *Math. Z.* 195(1987), 197–204.

[PAI] P. Painlevé, Sur les lignes singulières des functions analytiques, *Thèse*, Gauthier-Villars, Paris, 1887.

[PAW] R. E. A. C. Paley and N. Wiener, *Fourier Transforms in the Complex Domain*, Reprint of the 1934 original, American Mathematical Society Colloquium Publications, 19, American Mathematical Society, Providence, RI, 1987.

[PEE] J. Peetre, The Berezin transform and Ha-Plitz operators, *J. Operator Theory* 24(1990), 165–186.

[PHS] D. H. Phong and E. M. Stein, Hilbert integrals, singular integrals, and Radon transforms, I, *Acta Math.* 157(1986), 99–157.

[PIN] S. Pinchuk, On the analytic continuation of holomorphic mappings, *Mat. Sb.* 98(140)(1975), 375–392; *Mat. U.S.S.R. Sb.* 27(1975), 416–435.

[PIV] S. Pinchuk and S. V. Hasanov, Asymptotically holomorphic functions (Russian), *Mat. Sb.* 134(176) (1987), 546–555.

[PTS] S. I. Pinchuk and S. I. Tsyganov, Smoothness of CR-mappings between strictly pseudoconvex hypersurfaces, (Russian) *Izv. Akad. Nauk SSSR* Ser. Mat. 53(1989), 1120–1129, 1136; translation in *Math. USSR-Izv.* 35(1990), 457–467.

[PWW] I. Priwalow, *Randeigenschaften Analytischer Funktionen*, Deutsch Verlag der Wissenschaften, Berlin, 1956.

[RAM1] I. Ramadanov, Sur une propriété de la fonction de Bergman, *C. R. Acad. Bulgare Sci.* 20(1967), 759–762.

[RAM2] I. Ramadanov, A characterization of the balls in $\mathbb{C}^n$ by means of the Bergman kernel, *C. R. Acad. Bulgare Sci.* 34(1981), 927–929.

[RAM] E. Ramirez, Divisions problem in der komplexen analysis mit einer Anwendung auf Rand integral darstellung, *Math. Ann.* 184(1970), 172–187.

[RAN1] R. M. Range, A remark on bounded strictly plurisubharmonic exhaustion functions, *Proc. A.M.S.* 81(1981), 220–222.

[RAN2] R. M. Range, *Holomorphic Functions and Integral Representations in Several Complex Variables*, Springer Verlag, Berlin, 1986.

[ROR] B. Rodin and S. Warschawski, Estimates of the Riemann mapping function near a boundary point, in *Romanian-Finnish Seminar on Complex Analysis*, Springer Lecture Notes, vol. 743, 1979, 349–366.

[ROS1] J.-P. Rosay, Sur une characterization de la boule parmi les domains de $\mathbb{C}^n$ par son groupe d'automorphismes, *Ann. Inst. Four. Grenoble* XXIX(1979), 91–97.

[ROST] L. P. Rothschild and E. M. Stein, Hypoelliptic differential operators and nilpotent groups, *Acta Math.* 137 (1976), 247–320.

[RUS] L. A. Rubel and A. Shields, The failure of interior-exterior factorization in the polydisc and the ball, *Tôhoku Math. J.* 24(1972), 409–413.

[RUW1] W. Rudin, Lumer's Hardy spaces, *Michigan Math. J.* 24(1977), 1–5.

[RUW2] W. Rudin, *Functional Analysis*, McGraw-Hill, New York, 1973.

[RUW3] W. Rudin, *Function Theory in the Unit Ball of* $\mathbb{C}^n$, Springer, Berlin, 1980.

[SAR] D. Sarason, Functions of vanishing mean oscillation, *Trans. Amer. Math. Soc.* 207(1975), 391–405.

[SEM] S. Semmes, A generalization of Riemann mappings and geometric structures on a space of domains in $\mathbb{C}^n$, *Memoirs of the American Mathematical Society*, 1991.

[SER] J.-P. Serre, *Lie algebras and Lie groups*, Lectures given at Harvard University, 1964, W. A. Benjamin, Inc., New York-Amsterdam, 1965.

[SHE] G. Sheng, *Singular Integrals in Several Complex Variables*, Oxford University Press, 1991.

[SHI] A. B. Shishkin, Spectral synthesis for systems of differential operators with constant coefficients. The duality theorem (Russian) *Mat. Sb.* 189(1998), no. 9, 143–160; translation in *Sb. Math.* 189(1998), 1423–1440.

[SIB1] N. Sibony, Prolongement analytique des fonctions holomorphes bornées. (French) *C. R. Acad. Sci. Paris* Sér. A-B 275(1972), A973–A976.

[SIB2] N. Sibony, Un exemple de domain pseudoconvexe regulier ou l'equation $\overline{\partial} u = f$ n'admet pas de solution bornee pour $f$ bornee, *Invent. Math.* 62(1980), 235–242.

[SIU] Y.-T. Siu, The $\overline{\partial}$ problem with uniform bounds on derivatives, *Math. Ann.* 207(1974), 163–176.

[SKO] H. Skoda, Valeurs au bord pour les solutions de l'operateur $d''$ et caracterization des zeros des fonctions de la classe Nevanlinna, *Bull. Soc. Math. de France* 104(1976), 225–229.

[SKW] M. Skwarczynski, The distance in the theory of pseudo-conformal transformations and the Lu Qi-King conjecture, *Proc. A.M.S.* 22(1969), 305–310.

[SMI] K. T. Smith, A generalization of an inequality of Hardy and Littlewood, *Canad. J. Math.* 8(1956), 157–170.

[SPE] D. C. Spencer, Overdetermined systems of linear partial differential equations, *Bull. Am. Math. Soc.* 75(1969), 179–239.

[STE1] E. M. Stein, *The Boundary Behavior of Holomorphic Functions of Several Complex Variables*, Princeton University Press, Princeton, NJ, 1972.

[STE2] E. M. Stein, *Harmonic Analysis: Real-Variable Methods, Orthogonality, and Oscillatory Integrals*, with the assistance of Timothy S. Murphy, Princeton University Press, Princeton, NJ, 1993.

[STE3] E. M. Stein, On the functions of Littlewood-Paley, Lusin and Marcinkiewicz, *Trans. Amer. Math. Soc.* 88(1958), 4430–466.

[STE4] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton University Press, Princeton, NJ, 1970.

[STE5] E. M. Stein, The development of square functions in the work of A. Zygmund, *Bull. Amer. Math. Soc.* 7(1982), 359–376.

[STE6] E. M. Stein, Singular integrals and estimates for the Cauchy–Riemann equations, *Bull. Amer. Math. Soc.* 79(1973), 440–445.

[STW1] E. M. Stein and G. Weiss, On the theory of harmonic functions of several variables, *Acta Math.* 103(1960), 25–62.

[STW2] E. M. Stein and G. Weiss, *Introduction to Fourier Analysis on Euclidean Space*, Princeton University Press, Princeton, NJ, 1971.

[SUY] N. Suita and A. Yamada, On the Lu Qi-Keng conjecture, *Proc. A.M.S.* 59(1976), 222–224.

[SZE] G. Szegő, Über orthogonalsysteme von Polynomen, *Math. Z.* 4(1919), 139–151.

[TAI] M. Taibleson, The preservation of Lipschitz spaces under singular integral operators, *Studia Math.* 24(1964), 107–111.

[TAN] Tanaka, On generalized graded Lie algebras and geometric structures, I, *J. Math. Soc. Japan* 19(1967), 215–254.

[TAY] M. E. Taylor, *Pseudodifferential Operators*, Princeton University Press, Princeton, 1981.

[THO] G. B. Thomas, *Calculus and Analytic Geometry*, alternate ed., Addison-Wesley, Reading, MA, 1972.

[TRE] F. Treves, *Introduction to Pseudodifferential and Fourier Integral Operators*, Vol. II, Plenum Press, New York 1982.

[UCH] A. Uchiyama, A constructive proof of the Fefferman–Stein decomposition of $BMO(\mathbb{R}^n)$, *Acta Math.* 148(1982), 215–241.

[VAR] N. T. Varopoulos, Zeros of $H^p$ functions in several complex variables,*Pacific J. Math.* 88(1980), 189–246.

[WAD] R. Wada, Explicit formulas for the reproducing kernels of the space of harmonic polynomials in the case of classical real rank 1, *Sci. Math. Jpn.* 65(2007), 387–406.

[WAR1] S. Warschawski, On differentiability at the boundary in conformal mapping, *Proc. Amer. Math. Soc.* 12(1961), 614–620.

[WAR2] S. Warschawski, On the boundary behavior of conformal maps, *Nagoya Math. Jour.* 30(1967), 83–101.

[WAR3] S. Warschawski, On Hölder continuity at the boundary in conformal maps, *J. Math. Mech.* 18(1968/1969), 423–427.

[WAR4] S. Warschawski, On boundary derivations in conformal mapping, *Ann. Acad. Sci. Fenn.* Ser. A I No. 420 (1968), 22 pp.

[WEB1] S. Webster, Biholomorphic mappings and the Bergman kernel off the diagonal, *Invent. Math.* 51(1979), 155–169.

[WEB2] S. Webster, On the reflection principle in several complex variables, *Proc. Amer. Math. Soc.* 71(1978), 26–28.

[WEL] R. O. Wells, *Differential Analysis on Complex Manifolds*, 2nd ed., Springer Verlag, New York, 1979.

[WHI] H. Whitney, Analytic extensions of differentiable functions defined in closed sets, *Trans. Amer. Math. Soc.* 36(1934), 63–89.

[WHW] E. Whittaker and G. Watson, *A Course of Modern Analysis*, 4th ed., Cambridge Univ. Press, London, 1935.

[WID] K. O. Widman, On the boundary behavior of solutions to a class of elliptic partial differential equations, *Ark. Mat.* 6(1966), 485–533.

[WIE] J. Wiegerinck, Domains with finite dimensional Bergman space, *Math. Z.* 187(1984), 559–562.

[WON] B. Wong, Characterizations of the ball in $\mathbb{C}^n$ by its automorphism group, *Invent. Math.* 41(1977), 253–257.

[YAO] Z. X. Yao, A new proof for boundary behavior of the Poisson-Hua integrals on the unit ball of $\mathbb{C}^n$ (Chinese), *Math. Practice Theory* 32(2002), 523–528.

[YAU] S.-T. Yau, Problem section, *Seminar on Differential Geometry*, S.-T. Yau ed., *Annals of Math. Studies*, vol. 102, Princeton University Press, 1982, 669–706.

[YIN] C. Yin, A note on property of Cauchy integral on classical domains, *Chinese Quart. J. Math.* 13(1998), 41–43.

[ZHU] K. Zhu, *Spaces of Holomorphic Functions in the Unit Ball*, Springer, New York, 2005.

[ZYG1] A. Zygmund, A remark on functions of several complex variables, *Acta Sci. Math. Szeged* 12(1950), 66–88.

[ZYG2] A. Zygmund, Une remarque sur un théorème de M. Kaczmarz, *Math. Z.* 25(1926), 297–298.

[ZYG3] A. Zygmund, *Trigonometric Series*, 3rd. ed., Cambridge University Press, Cambridge, 2002.

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